Synopsis 2026
Cross-Volume Peer Review

Volumes 0-IV · Canonical IDs 1-5000

Reviewed unified edition · 2026

Decision: Accept for unified reference use with corrections.
Boundary: structural, epistemic, targeted semantic and publication review; not a theorem-by-theorem independent proof audit.
Volume IV: accepted as a typed research-synthesis layer, not as demonstrated sentience, physical cosmology, or theorem closure.

Canonical objects	5,000
Hard dependency edges	6,924
Graph	1 root / 1 component / 0 cycles
Dangling hard prerequisites	0
Evidence	A=2268 · M=2364 · S=365 · H=3 · P=0
Reviewed corrections	5
Outstanding assumptions metadata	1,196 A records in Volumes I-III

Verdict and strongest surviving result

For the reviewed registry snapshot, all 5,000 records are schema-valid and uniquely addressed. The hard-dependency graph contains 6,924 edges, one root and one weakly connected component, with zero directed cycles and zero missing hard prerequisites. Sign-sensitive NFKC/case/whitespace normalization finds zero exact duplicate titles and zero exact duplicate statements. No object is promoted to P.

This structural result is deterministic evidence about the artifacts. It does not independently prove the 2,268 A-status mathematical claims. The 2,364 M-status records remain bounded computations, and the 365 S plus 3 H records remain architecture/governance/hypothesis surfaces.

Review model

The canonical registry is authoritative. PDFs are editorial projections. The reviewed overlay changes five statements/scopes by append-only correction history while preserving IDs, evidence status and dependency identity. Frozen source releases remain archival objects.

Load-bearing remaining obstruction

Volumes I-III retain 1,196 A records with empty assumptions[] metadata. Many statements contain hypotheses inline, so this is not equivalent to 1,196 false claims. It does, however, prevent exact theorem-source/formalization matching and blocks any responsible bulk A->P promotion.

Correction ledger

S26-T-00002158 - mathematical correction

Released: For sigma-finite positive measures with nu<<mu, there exists an integrable density f with nu(E)=integral_E f dmu, unique mu-a.e.

Reviewed: For sigma-finite positive measures mu,nu with nu << mu, there exists a nonnegative measurable density f, unique mu-a.e., with nu(E)=integral_E f dmu for every measurable E; if nu is finite, then f is in L^1(mu).

Reason: Sigma-finiteness alone does not imply a globally L1 density; Lebesgue measure with respect to itself on R has derivative 1, which is not in L1(R).

S26-T-00002262 - scope correction

Released: A nonconstant holomorphic function maps open sets to open sets.

Reviewed: If U is a connected open subset of C and f:U->C is holomorphic and nonconstant, then f is an open map. On a disconnected domain, apply the conclusion componentwise on components where f is nonconstant.

Reason: Global nonconstancy on a disconnected domain is insufficient when f is constant on one component.

S26-T-00003053 - foundation-scope clarification

Released: An arbitrary product of compact spaces is compact in the product topology.

Reviewed: In the classical ZFC reading used for this theorem, an arbitrary product of compact spaces is compact in the product topology.

Reason: The arbitrary-product theorem requires an explicit classical choice convention in a project that distinguishes foundation contexts.

S26-T-00004062 - mathematical correction

Released: After k doublings the carrier has K-dimension 2^k; algebraic laws lost or retained depend on the starting algebra and the chosen parameters, not on geometric dimension n.

Reviewed: Starting from the base field K, k Cayley-Dickson doublings produce a K-vector space of dimension 2^k. More generally, starting from a finite-dimensional involutive K-algebra A_0 of dimension d, k doublings produce K-dimension d*2^k. Algebraic laws retained or lost depend on the starting algebra and doubling parameters, not on geometric dimension n.

Reason: The released statement mixed a general starting algebra with the special 2^k dimension formula, which is exact only when the starting K-dimension is one.

S26-D-00004341 - terminology/scope clarification

Released: A building mesh uses vertices, chambers and incidence inherited from the chosen Bruhat-Tits object; geodesics are shortest gallery or graph paths in that combinatorial geometry.

Reviewed: A building mesh uses vertices, chambers and incidence inherited from the chosen Bruhat-Tits object. In this registry, a combinatorial geodesic means a shortest path in the declared vertex graph or a shortest gallery in the declared chamber metric; this terminology is kept distinct from geodesics in a geometric realization of a higher-rank building.

Reason: The released wording could conflate graph/gallery geodesics with metric geodesics in a geometric realization for higher-rank buildings.

Volume-by-volume review summary

Volume	IDs	A	A empty assumptions	Review note
Volume 0 - Foundations	1-1000	985	0	No new statement correction required in this review. Assumptions metadata is populated throughout its A layer.
Volume I - Number, Algebra & Discrete Structure	1001-2000	399	399	Riemann hypothesis remains H. No new statement correction required; assumptions metadata debt remains.
Volume II - Analysis	2001-3000	400	398	Reviewed projection carries the Radon-Nikodym and holomorphic open-mapping corrections.
Volume III - Geometry & Topology	3001-4000	400	399	Reviewed projection carries the explicit ZFC/choice scope for Tychonoff.
Volume IV - p-adic Hypercomplex Hyperdimensional Continuum	4001-5000	84	0	Fresh review corrects the Cayley-Dickson dimension formula and clarifies Bruhat-Tits combinatorial-geodesic scope. Research/cognition/quantum language remains typed and evidence-bounded.

The Volume IV source carrier began as a 60-object research draft at IDs 4001-4060. The 4061-5000 extension is reviewed as an overlay/synthesis built on that carrier, not as retroactive evidence for the original visual intuition.

Volume IV semantic and epistemic review

Arithmetic and Cayley-Dickson axis

The new arithmetic-place axis v is admissible as a modeling distinction. The review accepts generalized Cayley-Dickson construction over base fields of characteristic not 2, but requires the starting-dimension factor in the doubling law. Real names such as R,C,H,O and sedenions are not silently reused for arbitrary local-field forms.

Ultrametric/building geometry

p-adic balls, valuation scales, and Bruhat-Tits tree/building meshes are kept separate from Euclidean or Riemannian geometry. The reviewed registry now reserves graph/gallery geodesic terminology for its declared combinatorial metrics and requires separate typing for geometric-realization geodesics.

Gerbes, functors and gluing

Sheaf, stack, gerbe, functor and profunctor objects are accepted when a site/category, descent notion, gauge/band data, or coend-style composition is declared. Semantic similarity does not satisfy a descent condition.

Quantum vocabulary

Entanglement is restricted to a declared complex Hilbert tensor factorization; tunneling requires a concrete Hamiltonian/projector fixture; holoportation is treated as a structure-preserving transport architecture unless a physical channel is explicitly modeled. No visual or linguistic analogy receives quantum evidence authority.

Cognition and recognition

Self-Godel, mutual recognition/resonance, ergocetic, erdodetic, Mind8 and hypercomplex cognition are treated as engineering/architectural coordinates. They may define measurable guards, ledgers or transport obligations; they do not establish subjective experience or sentience.

Checks, controls and independent witnesses

Volume IV finite layer

The p-adic/hypercomplex finite suite was rerun: 40/40 checks PASS. The mutation/metamorphic audit suite was rerun: 16/16 controls PASS. These results support only the exact bounded fixtures and detector behavior.

Independent structural witnesses

JSON Schema validation is performed through jsonschema; graph topology is recomputed with NetworkX; PDF parsing/rendering uses Poppler. These are independent software witnesses for schema, graph and publication properties, not independent theorem proofs.

Primary-source scope anchors

Chapman, Levin and Vishkautsan (2026), arXiv:2607.23613, explicitly study Cayley-Dickson algebras over arbitrary base fields of characteristic not 2. Jardine (2006), arXiv:math/0605200, provides a gerbe/stack-in-groupoids context. Parshin (1996), arXiv:alg-geom/9605001, provides a higher Bruhat-Tits simplicial-complex context. They support the legitimacy of the component mathematical vocabularies; they do not prove the project-specific synthesis.

Representation attacks

The duplicate normalizer preserves signs and other semantically relevant punctuation. Translation verification requires exact source_b64 reconstruction of each reviewed JSON record; evidence status and dependency lists must remain unchanged. Record order is non-semantic for the graph audit.

§-LANG transport decision

The requested slang translation is a transport, not a paraphrase. Each reviewed record becomes a §|LANG|SYNOPSIS_ENTRY block with an exact §FORM statement, §ASSUME list, §DEP_HARD/§DEP_SOFT lists, evidence ledger, provenance, correction history and a source_b64 field that reconstructs the exact reviewed canonical JSON record.

Transport invariants

Preserved: canonical ID, title, statement, class, evidence status, assumptions, dependencies, applications, counterexamples, notation, relations, provenance and correction history. Transformed: registry field layout into §-LANG strata and domain-sensitive charts. Lost in the .lang representation: original PDF page geometry, fonts and line wrapping. Active remainder: custom SYNOPSIS_ENTRY runtime semantics are not asserted, and general A claims remain A.

Round-trip boundary

A successful symbolic transport must reconstruct 5,000/5,000 exact reviewed source records, preserve all 6,924 hard edges and preserve the full evidence distribution. Any mismatch blocks release.

Publication decision and next bounded experiment

The unified master publication contains two parts: first, the peer-reviewed canonical Synopsis Volumes 0-IV; second, the §-LANG s1 translation of the same reviewed registry. Both parts share the same correction ledger and evidence map.

Next bounded experiment

The highest-value next review is theorem-specific assumptions/source matching for the 1,196 A records in Volumes I-III whose assumptions[] metadata remains empty. A separate Volume IV experiment should instantiate one mixed-place Cosmo fixture with explicit p-adic building data, a finite Cayley-Dickson fiber, declared gluing/transport, Hamiltonian/channel model, Mind8 telemetry and negative controls.

Kill conditions: representation-dependent ascent, clipped or tautological metrics, hidden energy leakage, lost returnability, semantic similarity substituted for descent, quantum terminology without a Hilbert/channel model, or any evidence promotion unsupported by an exact artifact.

Final evidence map

A=2268, M=2364, S=365, H=3, P=0. Structural and publication conclusions may be reported as measured facts about the reviewed snapshot. Mathematical theorem truth and research-frontier relevance retain their declared evidence levels.

Machine-readable artifacts: PEER_REVIEW_AUDIT_5000.json · CORRECTION_LEDGER_5000.json · audit_manifest_peer_review_5000.json · REGISTRY_00000001_00005000_REVIEWED.jsonl

A Synopsis of Results in Pure, Applied and Computational Mathematics

2026 Edition

Volume 0
Foundations, Language, Registry & Epistemic Protocol

Release: 1.0.0
Canonical objects: 1,000
Hard dependency edges: 2217
Hard graph: 1 root / 1 connected component / 0 cycles
Dangling hard dependencies: 0
Exact duplicate titles/statements: 0 / 0
Evidence states: A=985, S=15, P=0, M=0
Global closure audit: PASS

Canonical registry first. The PDF is an editorial projection of the graph.

Volume 0 closes at 1,000 addressable objects. Structural closure does not promote the 985 mathematical A-status statements to proof.

Final closure audit

Gate	Observed	Result
Schema	1,000 / 1,000	PASS
Hard dependencies	2217 edges; 0 dangling	PASS
Graph topology	1 root; 1 component; 0 cycles	PASS
Forward-numbered edges	3 resolved and documented	PASS
Exact duplicate titles/statements	0 / 0	PASS
Evidence promotions	28 legacy P demoted; 0 illegal P/M remain	PASS
Declared symbols	34 keys; 1 scoped reuse; 0 unresolved	PASS
Final-tranche checks	13 / 13	PASS
Audit mutation controls	5 / 5	PASS

Epistemic boundary. The closure verdict is M for this registry snapshot. Mathematical entries remain A unless an exact independent proof/source/formal artifact is attached. Bounded symbolic/numerical examples are witnesses, not theorem proofs.

Closure corrections. 28 unsupported legacy P tags were demoted; governance roots 24, 26, and 28 were linked to the metalanguage root; relation slots are now schema-required on every object; entry 706 records its intentional forward-numbered convolution dependency.

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Part A - Metalanguage and assertion discipline

S26-N-00000001 - Object language and metalanguage

A mathematical assertion is interpreted only relative to a declared ambient language or formal context. Mentions of expressions are distinguished from the objects denoted by those expressions.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-D-00000002 - Proposition

Within a declared foundational context, a proposition is an expression for which the context supplies a notion of assertion/truth or inhabitation. No global identification of propositions with sets, types, or Boolean values is imposed.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-N-00000001

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-D-00000003 - Predicate

Given an ambient collection/type/domain `X`, a predicate on `X` is a proposition-valued expression `P(x)` well-formed for `x : X` (or `x in X`, according to the active foundation).

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000002

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-N-00000004 - Quantifier convention

`forall x in X, P(x)` and `exists x in X, P(x)` are binder notation. Their exact semantics are inherited from the active foundation; bounded set-theoretic and typed readings are not silently identified.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000003

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-D-00000005 - Equality relation in context

An equality expression `x =_X y` denotes the equality notion supplied for objects `x,y` in ambient context `X`. Definitional equality, propositional equality, extensional equality, isomorphism, and equivalence are distinct unless an explicit bridge is present.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-N-00000001

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-N-00000006 - Equivalence is not equality

An isomorphism, homeomorphism, diffeomorphism, equivalence of categories, logical equivalence, or computational observational equivalence must not be rewritten as literal equality without a declared quotient, univalence principle, canonical identification, or other bridge that licenses the transport.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000005

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

Part B - Relations, maps, and structure

S26-D-00000007 - Binary relation

A binary relation from `X` to `Y` is a predicate `R(x,y)` on pairs with `x` in `X` and `y` in `Y` in the active foundational reading.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000003

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-D-00000008 - Function / map

A map `f : X -> Y` assigns to every admissible `x` of `X` a uniquely determined value `f(x)` of `Y`, interpreted in the active foundation. The graph-of-a-function encoding is one representation, not the universal definition.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000005, S26-D-00000007

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-D-00000009 - Composition

For maps `f : X -> Y` and `g : Y -> Z`, their composite `g o f : X -> Z` is defined by `(g o f)(x) = g(f(x))` whenever the expressions are well-typed/well-formed.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000008

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-T-00000010 - Associativity of composition

For composable maps `f : W -> X`, `g : X -> Y`, and `h : Y -> Z`, `h o (g o f) = (h o g) o f` extensionally, subject to the equality convention of the ambient foundation.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000005, S26-D-00000009

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000011 - Identity map

For an ambient object/domain `X`, `id_X : X -> X` is given by `id_X(x)=x`.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000008

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-T-00000012 - Identity laws

For `f : X -> Y`, `id_Y o f = f` and `f o id_X = f` extensionally.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000009, S26-D-00000011

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000013 - Structure

A structure on an ambient carrier/object consists of specified data, operations, relations, or properties together with declared laws. The word "structure" is schematic; every use in later volumes must identify its signature or canonical registry object.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-N-00000001

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-D-00000014 - Structure-preserving map

Given specified structures on `X` and `Y`, a structure-preserving map is a map whose declared preservation obligations hold. Homomorphism, linear map, continuous map, smooth map, functor, etc. are specializations and are not interchangeable.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000008, S26-D-00000013

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-D-00000015 - Isomorphism relative to structure

An isomorphism between structured objects is a structure-preserving map possessing an inverse that is structure-preserving in the same declared category/context.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000014, S26-T-00000012

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

Part C - Sets, types, and foundational bridges

S26-N-00000016 - Foundational pluralism

The Synopsis does not decree that all mathematics is intrinsically set-theoretic, type-theoretic, or categorical. A result may declare a foundation-neutral interface or a specific foundation. Transport between foundations is a typed bridge with assumptions and remainder.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-N-00000001

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-D-00000017 - Set (interface-level)

At the Synopsis interface level, a set is an object supporting membership propositions `x in X` under a declared set-theoretic or set-like semantics. Specific axiomatizations (e.g. ZF/ZFC) are separate objects.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000002, S26-N-00000016

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-D-00000018 - Type (interface-level)

At the Synopsis interface level, a type is a domain of terms equipped with the formation and equality rules of a declared type theory. No automatic identification with a set is assumed.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000005, S26-N-00000016

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-B-00000019 - Set/type transport boundary

Statements expressed over sets and statements expressed over types may correspond under an explicit interpretation, model, coercion, or embedding. Such correspondence preserves only the properties established for that bridge; it does not license unrestricted substitution between foundations.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000017, S26-D-00000018

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

Part D - Order and finiteness interfaces

S26-D-00000020 - Preorder

A preorder on `X` is a binary relation `<=` that is reflexive and transitive.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000007

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000021 - Partial order

A partial order is a preorder that is antisymmetric: `x <= y` and `y <= x` imply `x = y` in the declared equality.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000005, S26-D-00000020

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000022 - Equivalence relation

An equivalence relation on `X` is a binary relation that is reflexive, symmetric, and transitive.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000007

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000023 - Finite set (cardinality interface)

A set `X` is finite when it is in bijection with `{0,...,n-1}` for some natural number `n`, under a foundation where this formulation is available. Alternative constructive notions of finiteness are not silently collapsed into this definition.

Status: A - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000008, S26-D-00000017, S26-D-00000031, S26-D-00000040

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.2.1: Forward hard dependency discharged by natural-number kernel entries 31 and 40.

Part E - Epistemic and compilation objects

S26-D-00000024 - Hard dependency

A hard dependency edge `R -> Q` asserts that the exact statement of `R` requires `Q` as a definition, assumption, theorem, or semantic prerequisite. A release cannot contain `R` as closed while `Q` is unresolved.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-N-00000001

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: linked governance primitive to the object-language root to make the canonical dependency graph explicitly connected.

S26-D-00000025 - Soft dependency

A soft dependency is explanatory, historical, applicative, or comparative and is not required for the logical well-formedness or support of the source entry.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000024

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

S26-D-00000026 - Claim-local evidence state

An evidence state is attached to an exact claim/version: `P` proved/definitionally closed; `A` axiomatized/assumed/conditional; `M` measured; `H` hypothesis; `S` semantic/design; `R` retired. Evidence tags do not form an automatic linear promotion ladder.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-N-00000001

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: linked governance primitive to the object-language root to make the canonical dependency graph explicitly connected.

S26-T-00000027 - No-silent-promotion invariant

For every released entry, changing its evidence state requires an explicit promotion/demotion record naming the artifact and rule that justifies the transition. Editorial rewriting alone cannot change evidence state.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000026

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: governance invariant is semantic/design, not a proved mathematical theorem.

S26-D-00000028 - Typed remainder

A remainder records information, assumptions, distinctions, or uncertainty not preserved by a translation, bridge, compression, formalization, or editorial projection.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-N-00000001

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: linked governance primitive to the object-language root to make the canonical dependency graph explicitly connected.

S26-T-00000029 - Remainder preservation invariant

A compilation step that is known not to preserve some semantically relevant content must emit that loss as typed remainder rather than silently discarding it.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000028

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged. | 0.3.0: Global closure audit: governance invariant is semantic/design, not a proved mathematical theorem.

S26-N-00000030 - Proof-status scope

A `P` tag means only that the exact registered claim has a proof/definitional artifact accepted by its declared verification policy. It does not certify nearby paraphrases, stronger converses, historical attribution, applications, or philosophical interpretation.

Status: S - Exact registered statement only.

Assumptions: Interpretation is relative to the declared ambient foundation/context.

Hard dependencies: S26-D-00000026

Applications: Foundational substrate for later Synopsis entries.

Correction history: 0.2.0: Upgraded provisional index record to schema-complete canonical record; mathematical statement unchanged.

Part F - Natural numbers, induction, recursion, and counting

S26-D-00000031 - Natural numbers - interface

A natural-number object N consists of a distinguished element 0 and successor operation S, together with an induction/recursion principle sufficient to characterize the standard finite iteration generated from 0. This interface does not identify a particular set-theoretic construction with the concept itself.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-N-00000016

S26-D-00000032 - Zero and successor

For the natural-number interface, 0 is the distinguished initial element and S(n) is the successor of n. In the standard natural numbers, S is injective and 0 is not in its image.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000031

S26-T-00000033 - Induction principle

Let P be a predicate on N. If P(0) holds and, for every n in N, P(n) implies P(S(n)), then P(n) holds for every n in N.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000003, S26-D-00000031, S26-D-00000032

Applications: Proofs over recursively generated finite iteration.

S26-T-00000034 - Primitive recursion principle

Given an object a in X and an update map g : N x X -> X, there is a uniquely determined map f : N -> X satisfying f(0)=a and f(S(n))=g(n,f(n)), whenever the active foundation supplies the standard recursion principle.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000008, S26-D-00000031, S26-D-00000032

Applications: Definition of arithmetic operations and recursively specified sequences.

S26-D-00000035 - Addition on natural numbers

Natural-number addition is the recursively determined operation + satisfying n+0=n and n+S(m)=S(n+m).

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-T-00000034

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000036 - Multiplication on natural numbers

Natural-number multiplication is the recursively determined operation * satisfying n*0=0 and n*S(m)=n*m+n.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000035, S26-T-00000034

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-T-00000037 - Natural numbers form a commutative semiring

With 0, 1=S(0), addition, and multiplication, N satisfies associativity and commutativity of + and *, additive identity 0, multiplicative identity 1, and distributivity of multiplication over addition.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000032, S26-D-00000035, S26-D-00000036

Applications: Arithmetic and algebraic induction.

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000038 - Order on natural numbers

Define m <= n when there exists k in N such that m+k=n. This relation is the standard linear order on N.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000020, S26-D-00000035

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-T-00000039 - Well-ordering of natural numbers

Every nonempty subset of N has a least element with respect to the standard order.

Status: A - Exact registered statement only.

Assumptions: Classical/set-level subset formulation, or an explicitly equivalent well-founded formulation in the active foundation.

Hard dependencies: S26-D-00000017, S26-D-00000038

Applications: Minimal-counterexample arguments. | Termination arguments.

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000040 - Finite initial segment

For n in N, [n] denotes the finite initial segment {0,1,...,n-1}; equivalently, the naturals k with k<n.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000031, S26-D-00000038

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-T-00000041 - Closure of the finite-set definition

The forward dependency in S26-D-00000023 is discharged by S26-D-00000031 and S26-D-00000040: the expression "in bijection with {0,...,n-1} for some natural number n" is now internally addressable.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000023, S26-D-00000031, S26-D-00000040

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-T-00000042 - Uniqueness of finite cardinality

If a set X is in bijection with [m] and also with [n], then m=n. Hence a finite set has a unique natural-number cardinality.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000023, S26-D-00000040

Applications: Finite counting. | Combinatorics.

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000043 - Countably infinite set

A set X is countably infinite when there exists a bijection N -> X.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000008, S26-D-00000017, S26-D-00000031

S26-D-00000044 - Countable set

A set X is countable when it is finite or countably infinite. Equivalent conventions using injections X -> N are admitted only through an explicit classical-choice bridge when needed.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000023, S26-D-00000043

Boundary note: Constructive settings may distinguish formulations classically treated as equivalent.

Part G - Integers and rationals

S26-D-00000045 - Integers - interface

The integers Z are the additive group completion of N: they contain an image of N and additive inverses, and are generated by these subject to the expected universal property. Concrete pair/quotient constructions are representations, not the interface definition.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000031, S26-D-00000022

S26-B-00000046 - Natural-number embedding into integers

There is a canonical injective map i_NZ : N -> Z preserving 0, 1, addition, multiplication, and order on nonnegative integers.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000008, S26-T-00000037, S26-D-00000045

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-T-00000047 - Integers form a commutative ring

With the induced addition and multiplication, Z is a commutative ring with identity in which every element has an additive inverse.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000045, S26-B-00000046

Applications: Algebraic arithmetic. | Diophantine equations.

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000048 - Order on integers

For a,b in Z, define a <= b when b-a lies in the embedded nonnegative naturals. This is a translation-invariant linear order compatible with multiplication by nonnegative elements.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-B-00000046, S26-T-00000047, S26-D-00000021

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000049 - Rational numbers - interface

The rational numbers Q are the field of fractions of Z: elements may be represented by pairs (a,b) with a in Z and nonzero b in Z, modulo equality a/b=c/d iff ad=bc.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-T-00000047, S26-D-00000022

S26-B-00000050 - Integer embedding into rationals

There is a canonical injective ring homomorphism i_ZQ : Z -> Q given representationally by a |-> a/1.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000049, S26-T-00000047

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-T-00000051 - Rationals form an ordered field

Q is a field, and the order inherited from Z makes Q a linearly ordered field: addition preserves order and multiplication of nonnegative elements is nonnegative.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000049, S26-B-00000050, S26-D-00000048

Applications: Exact fractional arithmetic. | Ordered algebra.

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-T-00000052 - Archimedean property of the rationals

For every rational q there exists n in N with q<n under the canonical embeddings. Equivalently, the natural numbers are unbounded above in Q.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-B-00000046, S26-B-00000050, S26-T-00000051

Applications: A | p | p | r | o | x | i | m | a | t | i | o | n | | a | n | d | | a | n | a | l | y | s | i | s | .

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

Part H - Completion, the reals, and the complex field

S26-D-00000053 - Cauchy sequence in a metric space

A sequence (x_n) in a metric space (X,d) is Cauchy when for every epsilon>0 there exists N such that m,n>=N imply d(x_m,x_n)<epsilon.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000008, S26-D-00000031

S26-D-00000054 - Real numbers - complete ordered field interface

The real numbers R are represented at the interface as a complete Archimedean ordered field containing Q as an ordered subfield. Dedekind-cut and Cauchy-completion constructions are admissible realizations but are not silently identified.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-T-00000051, S26-T-00000052, S26-D-00000053

S26-T-00000055 - Uniqueness of the real-number interface

Any two complete Archimedean ordered fields are isomorphic as ordered fields; once the rational embedding is fixed, the corresponding isomorphism is uniquely determined.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000054, S26-D-00000015

Applications: F | o | u | n | d | a | t | i | o | n | - | i | n | d | e | p | e | n | d | e | n | t | | t | r | a | n | s | p | o | r | t | | o | f | | r | e | a | l | | | a | n | a | l | y | s | i | s | .

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-B-00000056 - Rational embedding into reals

The real-number interface includes an injective ordered-field homomorphism i_QR : Q -> R.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000054, S26-T-00000051

S26-T-00000057 - Least-upper-bound completeness of the reals

Every nonempty subset A of R that is bounded above has a least upper bound in R.

Status: A - Exact registered statement only.

Assumptions: R is interpreted through the complete ordered-field interface S26-D-00000054.

Hard dependencies: S26-D-00000017, S26-D-00000021, S26-D-00000054

Applications: Real analysis. | Existence arguments.

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-D-00000058 - Complex numbers

The complex numbers C may be represented as ordered pairs (a,b) in R^2 with addition (a,b)+(c,d)=(a+c,b+d) and multiplication (a,b)(c,d)=(ac-bd,ad+bc). Write a+bi for (a,b), where i=(0,1).

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000054

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-T-00000059 - Complex numbers form a field

With the operations of S26-D-00000058, C is a field containing an isomorphic copy of R via a |-> (a,0). Every nonzero a+bi has inverse (a-bi)/(a^2+b^2).

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000058, S26-T-00000057

Applications: Complex analysis. | Algebra. | Spectral methods.

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

S26-I-00000060 - Square of the imaginary unit

For i=(0,1) in C, i^2=-1.

Status: A - Exact registered statement only.

Assumptions: Standard natural-number/number-system interface in the declared foundation.

Hard dependencies: S26-D-00000058, S26-T-00000059

Correction history: 0.1.1: Global closure audit: legacy P tag lacked an independent formal/proof artifact under the project constitution; demoted to A without changing the mathematical statement.

Part I - Divisibility, primes, and factorization

S26-D-00000061 - Divisibility in the integers

For integers a and b, a divides b, written a | b, when there exists an integer k such that b=ak.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000047

S26-T-00000062 - Elementary divisibility laws

For integers a,b,c: a|a; if a|b and b|c then a|c; if a|b and a|c then a divides xb+yc for all integers x,y.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000061, S26-T-00000047

S26-D-00000063 - Common divisor and greatest common divisor

A common divisor of integers a,b divides both. For a,b not both zero, gcd(a,b) is the unique positive common divisor divisible by every common divisor.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000061, S26-D-00000048

S26-D-00000064 - Coprime integers

Integers a and b are coprime when gcd(a,b)=1.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000063

S26-T-00000065 - Division algorithm

For integers a and positive b there exist unique integers q,r such that a=bq+r and 0<=r<b.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000047, S26-D-00000048

Applications: Euclidean algorithm. | Positional arithmetic.

S26-A-00000066 - Euclidean algorithm

Repeatedly replace (a,b) by (b,r), where a=bq+r and 0<=r<b, until r=0. The last nonzero remainder is gcd(a,b).

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000063, S26-T-00000065

Applications: Computing gcd. | Modular inverses.

S26-T-00000067 - Bezout identity

For integers a,b not both zero there exist integers x,y such that ax+by=gcd(a,b).

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-A-00000066, S26-D-00000063

Applications: Linear Diophantine equations. | Modular inverses.

S26-T-00000068 - Coprime Bezout criterion

Integers a,b are coprime if and only if there exist integers x,y with ax+by=1.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000064, S26-T-00000067

S26-D-00000069 - Prime natural number

A natural number p is prime when p>1 and its only positive divisors are 1 and p.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000031, S26-D-00000038, S26-D-00000061

S26-D-00000070 - Composite natural number

A natural number n>1 is composite when n=ab for natural numbers a,b satisfying 1<a<n and 1<b<n.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000036, S26-D-00000038

S26-T-00000071 - Prime divisor lemma

Every natural number n>1 has a prime divisor.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000069, S26-T-00000039, S26-D-00000061

Applications: Existence part of prime factorization.

S26-T-00000072 - Euclid lemma

If p is prime and p divides ab, then p divides a or p divides b.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000069, S26-D-00000061, S26-T-00000067

Applications: Uniqueness of prime factorization.

S26-T-00000073 - Fundamental theorem of arithmetic

Every natural number n>1 can be expressed as a finite product of primes, and this factorization is unique up to permutation of the prime factors.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000071, S26-T-00000072, S26-D-00000023

Applications: Arithmetic structure of N. | Valuations.

S26-T-00000074 - Infinitude of primes

There are infinitely many prime natural numbers.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000069, S26-T-00000073

Applications: Prime-number theory.

S26-D-00000075 - Prime exponent valuation on a positive integer

For prime p and positive integer n, v_p(n) is the unique nonnegative exponent e such that p^e divides n and p^(e+1) does not.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000069, S26-T-00000073

S26-T-00000076 - Valuation of a product

For positive integers a,b and prime p, v_p(ab)=v_p(a)+v_p(b).

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000075, S26-T-00000073

S26-D-00000077 - Least common multiple

For positive integers a,b, lcm(a,b) is the least positive integer divisible by both a and b.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000061, S26-D-00000038

S26-I-00000078 - GCD-LCM product identity

For positive integers a,b, gcd(a,b) lcm(a,b)=ab.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000063, S26-D-00000077, S26-T-00000073

Part J - Congruences and modular arithmetic

S26-D-00000079 - Congruence modulo n

For integers a,b and positive integer n, a is congruent to b modulo n when n divides a-b.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000061, S26-D-00000048

S26-T-00000080 - Congruence is an equivalence relation

For each positive n, congruence modulo n is reflexive, symmetric, and transitive on Z.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-N-00000004, S26-D-00000079

S26-T-00000081 - Congruence respects addition and multiplication

If a==b (mod n) and c==d (mod n), then a+c==b+d (mod n) and ac==bd (mod n).

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000079, S26-T-00000047

S26-D-00000082 - Residue class ring Z/nZ

The equivalence classes of integers under congruence modulo positive n form a commutative ring under classwise addition and multiplication.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000080, S26-T-00000081, S26-T-00000047

S26-T-00000083 - Invertibility criterion modulo n

The residue class of a is invertible in Z/nZ if and only if gcd(a,n)=1.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000082, S26-T-00000067, S26-D-00000063

Applications: Modular division. | Cryptography.

S26-T-00000084 - Prime modulus gives a field

If p is prime, then Z/pZ is a field.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000069, S26-D-00000082, S26-T-00000083

S26-T-00000085 - Chinese remainder theorem - two coprime moduli

If m,n are positive and coprime, then for any integers a,b there is an integer x satisfying x==a (mod m) and x==b (mod n), unique modulo mn.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000064, S26-D-00000079, S26-T-00000067

Applications: Modular reconstruction. | Parallel arithmetic.

Part K - Finite combinatorics and binomial structure

S26-D-00000086 - Factorial

For n in N, define 0!=1 and (n+1)!=(n+1)n!.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000034, S26-D-00000036

S26-D-00000087 - Binomial coefficient

For natural numbers 0<=k<=n, define C(n,k)=n!/(k!(n-k)!).

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000086, S26-D-00000038, S26-D-00000049

S26-T-00000088 - Pascal identity

For 1<=k<=n, C(n+1,k)=C(n,k-1)+C(n,k).

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000087, S26-D-00000035

S26-T-00000089 - Binomial theorem

For commuting elements x,y in a commutative ring and n in N, (x+y)^n is the sum from k=0 to n of C(n,k)x^(n-k)y^k.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: x and y commute in a commutative ring; exponentiation is finite natural iteration.

Hard dependencies: S26-D-00000087, S26-T-00000037

Applications: Polynomial expansion. | Combinatorics.

S26-T-00000090 - Number of subsets of a finite set

A finite set with n elements has exactly 2^n subsets.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000042, S26-D-00000023, S26-D-00000036

S26-T-00000091 - Number of k-element subsets

A finite set with n elements has C(n,k) subsets of cardinality k for 0<=k<=n.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000042, S26-D-00000087

S26-T-00000092 - Pigeonhole principle

If a finite set with more than n elements is mapped into a set with n elements, the map is not injective.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000008, S26-D-00000023, S26-T-00000042

Applications: Counting arguments. | Existence proofs.

S26-T-00000093 - Addition rule for disjoint finite sets

If finite sets A and B are disjoint, then |A union B|=|A|+|B|.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000042, S26-D-00000035

S26-T-00000094 - Multiplication rule for finite products

If A and B are finite, then |A x B|=|A||B|.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000042, S26-D-00000036

S26-T-00000095 - Inclusion-exclusion for two finite sets

For finite sets A,B, |A union B|=|A|+|B|-|A intersection B|.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000093, S26-B-00000046

S26-D-00000096 - Permutation of a finite set

A permutation of a finite set X is a bijection X->X.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000008, S26-D-00000023

S26-T-00000097 - Number of permutations

A finite set with n elements has n! permutations.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-D-00000096, S26-D-00000086, S26-T-00000042

S26-T-00000098 - Number of ordered selections without replacement

The number of ordered k-tuples of distinct elements selected from an n-element set is n!/(n-k)! for 0<=k<=n.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000097, S26-D-00000086

S26-T-00000099 - Finite binomial sum

For n in N, the sum from k=0 to n of C(n,k) equals 2^n.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000089, S26-D-00000087

S26-T-00000100 - Alternating finite binomial sum

For n>0, the sum from k=0 to n of (-1)^k C(n,k) equals 0.

Status: A - Exact registered statement only; standard result pending independent publication-grade witness unless provenance says otherwise.

Assumptions: Integers and natural numbers are interpreted through the registered interfaces of Volume 0.

Hard dependencies: S26-T-00000089, S26-D-00000087

Part L - Abstract algebra interfaces for polynomial work

S26-D-00000101 - Commutative ring - interface

A commutative ring is a set R with operations + and *, elements 0 and 1, such that (R,+,0) is an abelian group, multiplication is associative and commutative with identity 1, and multiplication distributes over addition.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000013

S26-D-00000102 - Integral domain

An integral domain is a nonzero commutative ring in which ab=0 implies a=0 or b=0.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000101

S26-D-00000103 - Field - interface

A field is a nonzero commutative ring in which every nonzero element has a multiplicative inverse.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000101

S26-D-00000104 - Ring homomorphism

For commutative rings R,S, a ring homomorphism phi:R->S preserves 0, 1, addition, and multiplication.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000008, S26-D-00000101

S26-T-00000105 - Every field is an integral domain

Every field, under the registered nonzero convention, is an integral domain.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000102, S26-D-00000103

S26-B-00000106 - Registered number systems as abstract algebraic structures

The integer interface instantiates a commutative ring, while the rational, real, and complex interfaces instantiate fields.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-T-00000047, S26-T-00000051, S26-D-00000054, S26-T-00000059, S26-D-00000101, S26-D-00000103

S26-D-00000107 - Natural scalar multiplication and powers in a commutative ring

For a commutative ring R, n*r denotes repeated addition with 0*r=0 and (n+1)*r=n*r+r; r^n denotes repeated multiplication with r^0=1 and r^(n+1)=r^n*r.

Status: A - Exact registered recursive definitions only; no extension to negative or non-integer exponents is implied.

Assumptions: Natural numbers and the commutative-ring interface are the registered Volume 0 objects.

Hard dependencies: S26-D-00000031, S26-D-00000101, S26-T-00000033, S26-T-00000034

Applications: Polynomial notation | Formal differentiation

S26-D-00000108 - Characteristic of a commutative ring

For a nonzero commutative ring R, its characteristic is the least positive n with n*1=0 if such n exists, and 0 otherwise.

Status: A - Exact registered definition only.

Assumptions: R is a nonzero commutative ring; natural scalar multiplication is S26-D-00000107.

Hard dependencies: S26-D-00000101, S26-D-00000107, S26-T-00000039

Applications: Distinguishing algebraic behavior in formal differentiation

Part M - Polynomial objects, degree, evaluation, and division

S26-D-00000109 - Polynomial over a commutative ring

For a commutative ring R, a polynomial f in R[x] is a finitely supported coefficient function a:N->R, written f(x)=sum a_k x^k.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000101, S26-D-00000031, S26-D-00000023, S26-D-00000107

S26-D-00000110 - Zero polynomial

The zero polynomial is the polynomial all of whose coefficients are zero.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000109

S26-D-00000111 - Constant polynomial

For c in R, the constant polynomial c in R[x] has coefficient c at degree 0 and coefficient 0 at every positive degree.

Status: A - Exact registered definition only.

Assumptions: R is a commutative ring and R[x] is the registered polynomial object.

Hard dependencies: S26-D-00000109

Applications: Units and irreducibility in polynomial rings

S26-D-00000112 - Degree of a polynomial

For a nonzero polynomial f, deg(f) is the largest natural number k with nonzero coefficient a_k. The zero polynomial is left degree-exceptional in this interface rather than assigned an ordinary natural degree.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000109, S26-D-00000110, S26-T-00000039

S26-D-00000113 - Leading coefficient

For a nonzero polynomial f, its leading coefficient is the coefficient of x^deg(f).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000112

S26-D-00000114 - Monic polynomial

A nonzero polynomial is monic when its leading coefficient equals 1.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000113

S26-D-00000115 - Polynomial addition and multiplication

Polynomial addition is coefficientwise; multiplication is the Cauchy convolution of coefficient sequences.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000109, S26-D-00000035, S26-D-00000036

S26-T-00000116 - Polynomial ring structure

If R is a commutative ring, then R[x] with the registered polynomial operations is a commutative ring.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000101, S26-D-00000115

S26-T-00000117 - Degree of a product over an integral domain

If R is an integral domain and f,g are nonzero in R[x], then deg(fg)=deg(f)+deg(g).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000102, S26-D-00000112, S26-D-00000115

S26-D-00000118 - Polynomial evaluation

For f=sum a_k x^k in R[x] and r in R, eval_r(f)=sum a_k r^k.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000109, S26-D-00000107

S26-T-00000119 - Evaluation is a ring homomorphism

For a commutative ring R and r in R, eval_r:R[x]->R is a ring homomorphism.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000104, S26-D-00000118, S26-T-00000116

S26-D-00000120 - Polynomial root

For f in R[x], an element r in R is a root of f when f(r)=0.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000118

S26-T-00000121 - Polynomial division algorithm over a field

If F is a field, f,g in F[x], and g is nonzero, then there exist unique q,r in F[x] such that f=qg+r and either r=0 or deg(r)<deg(g).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000103, S26-T-00000116, S26-D-00000112

S26-T-00000122 - Remainder theorem

For f in F[x] over a field F and a in F, division by x-a leaves remainder f(a).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-T-00000121, S26-D-00000118

S26-T-00000123 - Factor theorem

For f in F[x] over a field F and a in F, f(a)=0 if and only if x-a divides f.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-T-00000122, S26-D-00000120

S26-D-00000124 - Polynomial divisibility

For f,g in R[x], g divides f if there exists h in R[x] with f=gh.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000109, S26-D-00000115

S26-D-00000125 - Multiplicity of a root

For f in F[x], a root a has multiplicity m>=1 when (x-a)^m divides f but (x-a)^(m+1) does not.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000120, S26-D-00000124

S26-T-00000126 - Root bound over an integral domain

A nonzero polynomial of degree n over an integral domain has at most n distinct roots in that domain.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000102, S26-D-00000112, S26-T-00000123

S26-T-00000127 - Polynomial identity principle over an infinite integral domain

If R is an infinite integral domain and f,g in R[x] satisfy f(r)=g(r) for every r in R, then f=g.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-T-00000126, S26-D-00000118

Part N - Formal differentiation, polynomial gcd, and multiplicity

S26-D-00000128 - Formal derivative of a polynomial

For f=sum_{k=0}^n a_k x^k over a commutative ring, the formal derivative is f'=sum_{k=1}^n k a_k x^(k-1), with k interpreted via repeated addition of 1.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000109, S26-D-00000107

S26-T-00000129 - Linearity of the formal derivative

For polynomials f,g and scalar c over a commutative ring, (f+g)'=f'+g' and (cf)'=c f'.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000128, S26-D-00000115

S26-T-00000130 - Product rule for the formal derivative

For polynomials f,g over a commutative ring, (fg)'=f'g+fg'.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000128, S26-D-00000115

S26-T-00000131 - Power rule for the formal derivative

For n>=1, (x^n)'=n x^(n-1).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000128

S26-D-00000132 - Polynomial gcd over a field

For nonzero f,g in F[x], a greatest common divisor is a common divisor d such that every common divisor of f and g divides d; a monic representative is chosen canonically.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000124, S26-D-00000114, S26-D-00000103

S26-A-00000133 - Euclidean algorithm for polynomials

Repeated polynomial division over a field terminates and yields a monic gcd of the input polynomials.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-T-00000121, S26-D-00000132

S26-T-00000134 - Bezout identity for polynomial gcd

For f,g in F[x], their monic gcd d can be written d=af+bg for some a,b in F[x].

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-A-00000133, S26-D-00000132

S26-T-00000135 - Repeated-root derivative criterion

For f in F[x] and a in F, a is a repeated root of f if and only if f(a)=0 and f'(a)=0.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000125, S26-D-00000128, S26-T-00000123, S26-D-00000108

Boundary note: In positive characteristic, the derivative may vanish identically for nonconstant polynomials such as x^p; the exact repeated-root statement remains valid, but derivative-based squarefreeness must respect characteristic.

S26-D-00000136 - Squarefree polynomial over a field

A nonzero polynomial f in F[x] is squarefree when there is no nonconstant polynomial h in F[x] such that h^2 divides f.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000124, S26-D-00000103, S26-D-00000111, S26-D-00000107

S26-T-00000137 - Squarefree gcd criterion

For nonzero f in F[x], f is squarefree if and only if gcd(f,f')=1.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000136, S26-D-00000132, S26-D-00000128

Part O - Roots, factorization, arithmetic criteria, and interpolation

S26-T-00000138 - Quadratic formula

Over a field of characteristic not equal to 2, roots of ax^2+bx+c with a nonzero and with a chosen square root of the discriminant Delta=b^2-4ac are (-b +/- sqrt(Delta))/(2a).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: The coefficient field has characteristic not equal to 2; a is nonzero; a square root of the discriminant is available in the field or chosen extension.

Hard dependencies: S26-D-00000103, S26-D-00000120, S26-D-00000108

S26-D-00000139 - Elementary symmetric polynomials

For variables r_1,...,r_n, the k-th elementary symmetric polynomial e_k is the sum of all products r_{i1}...r_{ik} over k-element subsets of the indices.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-T-00000091, S26-D-00000109

S26-T-00000140 - Vieta relations for a monic split polynomial

If f(x)=prod_{i=1}^n (x-r_i)=x^n+a_{n-1}x^(n-1)+...+a_0, then a_{n-k}=(-1)^k e_k(r_1,...,r_n).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000139, S26-D-00000115, S26-D-00000114

S26-T-00000141 - Fundamental theorem of algebra - root form

Every nonconstant polynomial with complex coefficients has at least one complex root.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: Complex numbers use the registered field interface; this theorem is not derived from the elementary algebraic kernel in this tranche.

Hard dependencies: S26-D-00000058, S26-D-00000120

S26-T-00000142 - Complex factorization of polynomials

Every nonzero complex polynomial f of degree n factors as c product_{i=1}^n (x-r_i), counting roots with multiplicity, for nonzero c in C.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-T-00000141, S26-T-00000123, S26-D-00000125

S26-D-00000143 - Primitive integer polynomial

A polynomial in Z[x] is primitive when the gcd of all its coefficients is 1.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-B-00000106, S26-D-00000063, S26-D-00000109

S26-D-00000144 - Content of an integer polynomial

For a nonzero polynomial f in Z[x], its content is the positive gcd of its coefficients.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000063, S26-D-00000109

S26-T-00000145 - Gauss lemma - primitive product

The product of two primitive polynomials in Z[x] is primitive.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000143, S26-D-00000115

S26-D-00000146 - Irreducible polynomial over a field

A nonconstant polynomial f in F[x] is irreducible when f=gh implies that g or h is a nonzero constant polynomial.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000103, S26-D-00000109

S26-T-00000147 - Rational root theorem

If a primitive polynomial a_n x^n+...+a_0 in Z[x] has a rational root p/q in lowest terms, then p divides a_0 and q divides a_n.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000143, S26-D-00000049, S26-D-00000061, S26-D-00000120, S26-D-00000064

S26-T-00000148 - Lagrange interpolation

For distinct x_0,...,x_n in a field F and arbitrary y_0,...,y_n in F, there is a unique polynomial p of degree at most n with p(x_i)=y_i for every i; explicitly p(x)=sum_i y_i product_{j!=i}(x-x_j)/(x_i-x_j).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All referenced structures use the exact interfaces registered in Volume 0; hypotheses in each statement are load-bearing.

Hard dependencies: S26-D-00000103, S26-T-00000126, S26-D-00000118

Applications: Interpolation and numerical approximation | Finite-dimensional polynomial reconstruction

Part P - Vector spaces, subspaces, bases, and dimension

S26-D-00000149 - Vector space over a field

A vector space V over a field F is an abelian group under vector addition together with scalar multiplication F x V -> V satisfying compatibility with field multiplication, identity action 1*v=v, and distributivity over both scalar and vector addition.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000103

Applications: Linear models | Coordinate geometry | Differential equations

S26-D-00000150 - Vector

A vector is an element of a declared vector space V over a declared field F.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000149

S26-D-00000151 - Subspace

A subspace W of a vector space V over F is a subset W of V containing 0 and closed under vector addition and scalar multiplication, with the induced vector-space operations.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000149, S26-D-00000017

S26-T-00000152 - Subspace criterion

A nonempty subset W of a vector space V over F is a subspace if and only if for all u,v in W and all a,b in F, the vector a*u+b*v lies in W.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: V is a vector space over the field F; W is a nonempty subset of V.

Hard dependencies: S26-D-00000151

S26-D-00000153 - Linear combination

Given vectors v_1,...,v_k in V and scalars a_1,...,a_k in F, a_1*v_1+...+a_k*v_k is a linear combination of those vectors.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000149, S26-D-00000150

S26-D-00000154 - Span

For S contained in a vector space V over F, span(S) is the set of all finite linear combinations of elements of S; span(emptyset)={0}.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000153, S26-D-00000031, S26-D-00000017

S26-T-00000155 - Span is the smallest containing subspace

For S contained in V, span(S) is a subspace of V containing S, and every subspace of V that contains S also contains span(S).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000154, S26-D-00000151

S26-D-00000156 - Linear independence

A finite list (v_1,...,v_k) in V is linearly independent over F when a_1*v_1+...+a_k*v_k=0 implies a_1=...=a_k=0.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000153

S26-D-00000157 - Spanning family

A family S in V spans V when span(S)=V.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000154

S26-D-00000158 - Basis

A basis of a vector space V over F is a linearly independent family that spans V.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000156, S26-D-00000157

S26-T-00000159 - Unique finite coordinates in a basis

If (b_1,...,b_n) is a finite basis of V, every v in V has a unique expression v=sum_i a_i b_i with a_i in F.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: (b_1,...,b_n) is a finite basis of V over F.

Hard dependencies: S26-D-00000158, S26-D-00000153

S26-D-00000160 - Coordinate vector relative to an ordered basis

For an ordered finite basis B=(b_1,...,b_n), the coordinate vector [v]_B is the column (a_1,...,a_n)^T determined by the unique expansion v=sum_i a_i b_i.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-T-00000159

S26-D-00000161 - Finite-dimensional vector space

A vector space V is finite-dimensional when it has a finite basis.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000158

S26-T-00000162 - Basis cardinality invariance

Any two finite bases of the same vector space V over F have the same number of vectors.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: V has at least one finite basis.

Hard dependencies: S26-D-00000158, S26-T-00000091

S26-D-00000163 - Dimension of a finite-dimensional vector space

For a finite-dimensional vector space V, dim_F(V) is the number of vectors in any basis of V.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: The basis-cardinality invariance theorem is used to make the definition well-defined.

Hard dependencies: S26-D-00000161, S26-T-00000162

S26-T-00000164 - Basis extension and spanning reduction

In a finite-dimensional vector space, every linearly independent finite set extends to a basis, and every finite spanning set contains a basis.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: V is finite-dimensional over F.

Hard dependencies: S26-D-00000161, S26-D-00000156, S26-D-00000157

S26-T-00000165 - Dimension monotonicity for subspaces

If W is a subspace of a finite-dimensional vector space V, then dim(W)<=dim(V), with equality if and only if W=V.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: V is finite-dimensional over F; W is a subspace of V.

Hard dependencies: S26-D-00000151, S26-D-00000163

Part Q - Linear maps, kernel, image, rank, and isomorphism

S26-D-00000166 - Linear map

For vector spaces V,W over the same field F, a map T:V->W is linear when T(a*u+b*v)=a*T(u)+b*T(v) for all u,v in V and a,b in F.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000149, S26-D-00000008

S26-T-00000167 - Linear maps preserve zero and finite linear combinations

If T:V->W is linear, then T(0)=0 and T(sum_i a_i v_i)=sum_i a_i T(v_i) for every finite linear combination.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000166, S26-D-00000153

S26-D-00000168 - Kernel of a linear map

For a linear map T:V->W, ker(T)={v in V : T(v)=0}.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000166, S26-D-00000017

S26-D-00000169 - Image of a linear map

For a linear map T:V->W, im(T)={T(v):v in V}.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000166, S26-D-00000017

S26-T-00000170 - Kernel and image are subspaces

For a linear map T:V->W, ker(T) is a subspace of V and im(T) is a subspace of W.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000168, S26-D-00000169, S26-D-00000151

S26-D-00000171 - Rank of a linear map

For a linear map T:V->W with finite-dimensional image, rank(T)=dim(im(T)).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000169, S26-D-00000163

S26-D-00000172 - Nullity of a linear map

For a linear map T:V->W with finite-dimensional kernel, nullity(T)=dim(ker(T)).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000168, S26-D-00000163

S26-T-00000173 - Rank-nullity theorem

If V is finite-dimensional and T:V->W is linear, then dim(V)=rank(T)+nullity(T).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: V is finite-dimensional over F; W may be arbitrary over the same field F.

Hard dependencies: S26-D-00000161, S26-D-00000171, S26-D-00000172

S26-D-00000174 - Linear isomorphism

A linear isomorphism T:V->W is a bijective linear map; its inverse is then linear.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000166, S26-D-00000015

S26-T-00000175 - Finite-dimensional isomorphism criterion

If V and W are finite-dimensional vector spaces over F with dim(V)=dim(W), then a linear map T:V->W is injective if and only if it is surjective, and either condition is equivalent to T being a linear isomorphism.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: V and W are finite-dimensional over the same field F and have equal dimension.

Hard dependencies: S26-T-00000173, S26-D-00000174, S26-D-00000163

Part R - Matrices, row operations, and linear systems

S26-D-00000176 - Matrix over a commutative ring

For positive integers m,n and a commutative ring R, an m x n matrix over R is an array A=(a_ij) with entries a_ij in R indexed by 1<=i<=m and 1<=j<=n.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000101, S26-D-00000031

S26-D-00000177 - Matrix addition and scalar multiplication

Matrices of the same shape over a commutative ring are added entrywise; for r in R, scalar multiplication rA is entrywise multiplication by r.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000176, S26-D-00000101

S26-D-00000178 - Matrix multiplication

For A in M_{m x n}(R) and B in M_{n x p}(R), the product AB in M_{m x p}(R) has entries (AB)_ij=sum_{k=1}^n a_ik b_kj.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000176, S26-D-00000107

S26-T-00000179 - Associativity and distributivity of matrix multiplication

Whenever matrix shapes are compatible over a commutative ring R, matrix multiplication is associative and distributes over matrix addition.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000178, S26-D-00000177, S26-D-00000101

S26-D-00000180 - Identity matrix

For n>=1, I_n is the n x n matrix with 1 on the diagonal and 0 off the diagonal.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000176

S26-T-00000181 - Matrix identity laws

For every compatible matrix A, I_m A=A and A I_n=A.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000178, S26-D-00000180

S26-D-00000182 - Matrix of a linear map relative to ordered bases

Let T:V->W be linear and let B=(b_1,...,b_n), C=(c_1,...,c_m) be ordered finite bases. The matrix [T]_{C<-B} is the m x n matrix whose j-th column is [T(b_j)]_C.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000166, S26-D-00000160, S26-D-00000176

S26-T-00000183 - Composition corresponds to matrix multiplication

For linear maps T:U->V and S:V->W with compatible ordered finite bases A,B,C, [S o T]_{C<-A}=[S]_{C<-B}[T]_{B<-A}.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000182, S26-D-00000009, S26-D-00000178

S26-D-00000184 - Elementary row operation

An elementary row operation on a matrix is one of: swap two rows; multiply one row by a unit scalar (over a field, any nonzero scalar); or add a scalar multiple of one row to another.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: For field-based Gaussian elimination, row scaling uses nonzero field scalars. Over a general commutative ring, scaling is invertible only by a unit.

Hard dependencies: S26-D-00000176, S26-D-00000103

S26-D-00000185 - Row equivalence

Two matrices over a field are row-equivalent when one can be obtained from the other by a finite sequence of elementary row operations.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000184, S26-D-00000031

S26-D-00000186 - Linear system represented by an augmented matrix

For A in M_{m x n}(F), x in F^n, and b in F^m, the system Ax=b is represented by the augmented matrix [A|b].

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000176, S26-D-00000178, S26-D-00000103

S26-T-00000187 - Elementary row operations preserve solution sets

Over a field F, applying an elementary row operation to an augmented matrix [A|b] does not change the solution set of Ax=b.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All row operations are invertible elementary operations over the field F.

Hard dependencies: S26-D-00000184, S26-D-00000186

S26-D-00000188 - Reduced row echelon form

A matrix is in reduced row echelon form when each nonzero row has a leading 1, each leading 1 is the only nonzero entry in its column, leading positions move strictly rightward down the rows, and all zero rows occur below all nonzero rows.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000176

S26-T-00000189 - Existence and uniqueness of reduced row echelon form

Every matrix over a field is row-equivalent to a unique reduced row echelon matrix.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: Matrix entries lie in a field.

Hard dependencies: S26-D-00000185, S26-D-00000188

S26-T-00000190 - Gaussian elimination consistency criterion

A linear system Ax=b over a field is consistent exactly when the reduced row echelon form of [A|b] contains no row of the form [0 ... 0 | c] with c nonzero; its free variables parametrize all solutions.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: A and b are finite matrices/vectors over a field F.

Hard dependencies: S26-D-00000186, S26-T-00000189, S26-T-00000187

Part S - Determinants and basis transport

S26-D-00000191 - Determinant - Leibniz formula

For an n x n matrix A over a commutative ring R, det(A)=sum_{sigma in S_n} sgn(sigma) product_{i=1}^n a_{i,sigma(i)}.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: R is commutative; n>=1; the sign of a permutation is defined in the usual integer-to-ring interpretation.

Hard dependencies: S26-D-00000176, S26-D-00000086, S26-D-00000101

S26-T-00000192 - Determinant response to elementary row operations

For square matrices over a commutative ring, swapping two rows negates the determinant, multiplying a row by c multiplies the determinant by c, and adding a multiple of one row to another leaves the determinant unchanged.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: The determinant is the registered commutative-ring determinant.

Hard dependencies: S26-D-00000191, S26-D-00000184

S26-T-00000193 - Multiplicativity of determinant

For n x n matrices A,B over a commutative ring R, det(AB)=det(A)det(B).

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: R is a commutative ring.

Hard dependencies: S26-D-00000191, S26-D-00000178, S26-D-00000101

S26-T-00000194 - Invertibility criterion by determinant over a field

For A in M_n(F) over a field F, A is invertible if and only if det(A) is nonzero.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: A is square and its entries lie in a field F.

Hard dependencies: S26-D-00000191, S26-T-00000193, S26-D-00000180, S26-D-00000103

Boundary note: Over a general commutative ring, invertibility is equivalent to det(A) being a unit, not merely nonzero.

S26-D-00000195 - Change-of-basis matrix

For ordered bases B and C of the same finite-dimensional vector space V, P_{C<-B} is the matrix satisfying [v]_C=P_{C<-B}[v]_B for every v in V.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000160, S26-D-00000182, S26-D-00000174

S26-T-00000196 - Similarity under change of basis

If T:V->V is linear and B,C are ordered bases of finite-dimensional V, then [T]_{C<-C}=P_{C<-B}[T]_{B<-B}P_{C<-B}^{-1}.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: V is finite-dimensional over F; B and C are ordered bases.

Hard dependencies: S26-D-00000195, S26-T-00000183, S26-D-00000174

Part T - Characteristic polynomials and spectral interfaces

S26-D-00000197 - Characteristic polynomial

For A in M_n(F), the characteristic polynomial is chi_A(lambda)=det(lambda I_n-A) in F[lambda].

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: All scalar operations and algebraic structures use the exact registered interfaces; every field/ring and finite-dimensional hypothesis is load-bearing.

Hard dependencies: S26-D-00000191, S26-D-00000180, S26-D-00000109, S26-D-00000103

S26-D-00000198 - Eigenvalue, eigenvector, and eigenspace

For a linear operator T:V->V over F, lambda in F is an eigenvalue when there exists v!=0 with T(v)=lambda v; such v is an eigenvector, and E_lambda=ker(T-lambda I) is the lambda-eigenspace.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: T is an endomorphism of a vector space V over F.

Hard dependencies: S26-D-00000166, S26-D-00000168, S26-D-00000011

S26-T-00000199 - Eigenvalues and the characteristic polynomial

For A in M_n(F), lambda in F is an eigenvalue of the associated linear operator if and only if chi_A(lambda)=0.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: A is an n x n matrix over the field F; lambda lies in F.

Hard dependencies: S26-D-00000197, S26-D-00000198, S26-T-00000194

S26-T-00000200 - Diagonalization criterion by eigenvector basis

A linear operator T on a finite-dimensional vector space V over F is diagonalizable over F if and only if V has a basis consisting of eigenvectors of T.

Status: A - Exact registered statement only; no neighboring paraphrase or unstated generalization is promoted.

Assumptions: V is finite-dimensional over F. Diagonalizability is relative to the declared field F.

Hard dependencies: S26-D-00000158, S26-D-00000182, S26-D-00000198

Boundary note: A real rotation by 90 degrees on R^2 is not diagonalizable over R but becomes diagonalizable over C.

Applications: Decoupling linear recurrences and differential systems | Spectral computation

Part U - Inner products, norms, orthogonality, and projection

S26-D-00000201 - Scalar conjugation on R or C

For F equal to R or C, scalar conjugation a |-> conjugate(a) is the identity on R and ordinary complex conjugation on C; it satisfies conjugate(a+b)=conjugate(a)+conjugate(b), conjugate(ab)=conjugate(a)conjugate(b), and conjugate(conjugate(a))=a.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: F is exactly R or C.

Hard dependencies: S26-T-00000059, S26-I-00000060

S26-D-00000202 - Inner product space over R or C

An inner product space V over F in {R,C} is a vector space equipped with <u,v> in F that is linear in the first argument, conjugate-symmetric <u,v>=conjugate(<v,u>), and positive definite: <v,v>>=0 with equality iff v=0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: F is R or C; this Synopsis uses the convention linear in the first argument and conjugate-linear in the second.

Hard dependencies: S26-D-00000149, S26-D-00000201

Boundary note: Using an ordinary bilinear dot product on C^n can give <v,v>=0 for nonzero v.

S26-D-00000203 - Norm induced by an inner product

In an inner product space, ||v||=sqrt(<v,v>) is the norm induced by the inner product.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The scalar <v,v> is a nonnegative real number, so the nonnegative square root is used.

Hard dependencies: S26-D-00000202, S26-T-00000057

S26-T-00000204 - Cauchy-Schwarz inequality

For all u,v in an inner product space, |<u,v>| <= ||u|| ||v||, with equality iff u and v are linearly dependent.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is an inner product space over R or C.

Hard dependencies: S26-D-00000202, S26-D-00000203, S26-D-00000156

S26-T-00000205 - Triangle inequality for the induced norm

For all u,v in an inner product space, ||u+v|| <= ||u||+||v||.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000203, S26-T-00000204

S26-D-00000206 - Orthogonality

Vectors u and v in an inner product space are orthogonal, written u perp v, when <u,v>=0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000202

S26-T-00000207 - Pythagorean identity

If u and v are orthogonal, then ||u+v||^2=||u||^2+||v||^2.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000206, S26-D-00000203

S26-D-00000208 - Orthogonal complement

For S contained in an inner product space V, S^perp={v in V : <v,s>=0 for every s in S}.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000206, S26-D-00000017

S26-T-00000209 - Orthogonal complement is a subspace

For every subset S of an inner product space V, S^perp is a subspace of V.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000208, S26-D-00000151

S26-D-00000210 - Orthonormal family

A family (e_i) in an inner product space is orthonormal when <e_i,e_j>=0 for i!=j and ||e_i||=1 for every i.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000206, S26-D-00000203

S26-T-00000211 - Finite orthonormal families are linearly independent

Every finite orthonormal family is linearly independent.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000210, S26-D-00000156

S26-T-00000212 - Bessel inequality - finite orthonormal form

If e_1,...,e_k is an orthonormal family, then sum_i |<v,e_i>|^2 <= ||v||^2 for every v.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The family is finite.

Hard dependencies: S26-D-00000210, S26-D-00000203

S26-T-00000213 - Parseval identity for a finite orthonormal basis

If e_1,...,e_n is an orthonormal basis of V, then v=sum_i <v,e_i> e_i and ||v||^2=sum_i |<v,e_i>|^2.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional and the displayed family is an orthonormal basis.

Hard dependencies: S26-D-00000210, S26-D-00000158, S26-T-00000212

S26-D-00000214 - Gram matrix

For vectors v_1,...,v_k in an inner product space, the Gram matrix G is the k x k matrix with G_ij=<v_i,v_j>.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000202, S26-D-00000176

S26-T-00000215 - Gram matrix positivity and independence criterion

For every coefficient vector c, sum_{i,j} conjugate(c_i) G_ij c_j is a nonnegative real number. It is strictly positive for every nonzero c iff the family v_1,...,v_k is linearly independent.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The family is finite; Hermitian and positive (semi)definite are interpreted over R or C as registered below.

Hard dependencies: S26-D-00000214, S26-D-00000156

S26-T-00000216 - Gram-Schmidt orthonormalization

Every finite linearly independent list in an inner product space can be transformed into an orthonormal list with the same successive spans.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The input list is finite and linearly independent.

Hard dependencies: S26-D-00000156, S26-D-00000210, S26-D-00000154

S26-T-00000217 - Existence of orthonormal bases in finite dimension

Every finite-dimensional inner product space has an orthonormal basis.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000161, S26-T-00000164, S26-T-00000216

S26-D-00000218 - Orthogonal projection onto a subspace

If W is a finite-dimensional subspace of an inner product space V with orthonormal basis e_1,...,e_k, define P_W(v)=sum_i <v,e_i> e_i.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: W is finite-dimensional; the definition is independent of the chosen orthonormal basis by the projection theorem.

Hard dependencies: S26-D-00000151, S26-D-00000210, S26-T-00000213

S26-T-00000219 - Orthogonal decomposition theorem - finite-dimensional subspace

If W is a finite-dimensional subspace of an inner product space V, every v in V has a unique decomposition v=w+z with w in W and z in W^perp; w=P_W(v).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: W is finite-dimensional; V itself need not be finite-dimensional.

Hard dependencies: S26-D-00000218, S26-D-00000208

S26-T-00000220 - Best approximation property of orthogonal projection

For W as above, P_W(v) is the unique point w in W minimizing ||v-w||.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-T-00000219, S26-T-00000207

S26-T-00000221 - Projection operator identities

For an orthogonal projection P_W onto a finite-dimensional subspace W, P_W is linear, P_W^2=P_W, im(P_W)=W, and ker(P_W)=W^perp.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000218, S26-D-00000166, S26-D-00000168, S26-D-00000169

Part V - Adjoints, self-adjoint/unitary/normal operators

S26-D-00000222 - Adjoint of a linear map

For inner product spaces V,W and a linear map T:V->W, an adjoint T*:W->V is a linear map satisfying <Tv,w>_W=<v,T*w>_V for all v,w.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000166, S26-D-00000202

S26-T-00000223 - Existence and uniqueness of adjoints in finite dimension

Every linear map between finite-dimensional inner product spaces has a unique adjoint.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Both domain and codomain are finite-dimensional inner product spaces over the same F in {R,C}.

Hard dependencies: S26-D-00000222, S26-T-00000217

S26-D-00000224 - Conjugate transpose of a matrix

For a matrix A over R or C, A* is the transpose of the entrywise conjugate matrix; over R this is the ordinary transpose.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000176, S26-D-00000201

S26-T-00000225 - Matrix of the adjoint in orthonormal bases

If T:V->W is linear and B,C are orthonormal bases, then [T*]_{B<-C} = ([T]_{C<-B})*.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000182, S26-T-00000223, S26-D-00000224

S26-T-00000226 - Adjoint algebra

For compatible finite-dimensional operators and scalars a,b, If pointwise operations are defined by (aS+bT)(v)=aS(v)+bT(v), then (aS+bT)*=conjugate(a)S*+conjugate(b)T*, (ST)*=T*S*, and (T*)*=T.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-T-00000223, S26-D-00000201, S26-D-00000009

S26-D-00000227 - Self-adjoint operator

An operator T on an inner product space is self-adjoint when T*=T.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000222

S26-D-00000228 - Unitary and orthogonal operator

An operator U is unitary when U*U=UU*=I. Over real inner product spaces the same condition defines an orthogonal operator.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000222, S26-D-00000011, S26-D-00000180

S26-D-00000229 - Normal operator

An operator T is normal when T*T=TT*.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000222

S26-T-00000230 - Unitary operators preserve inner products and norms

If U is unitary, then <Uv,Uw>=<v,w> and ||Uv||=||v|| for all v,w.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000228, S26-D-00000203

S26-T-00000231 - Finite-dimensional isometry criterion

For an endomorphism U of a finite-dimensional inner product space, U is unitary iff it preserves the inner product; equivalently, iff it preserves the induced norm.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: U is a linear endomorphism of a finite-dimensional inner product space.

Hard dependencies: S26-D-00000228, S26-T-00000230, S26-D-00000161

S26-T-00000232 - Orthogonality of eigenvectors of a self-adjoint operator

Eigenvectors of a self-adjoint operator belonging to distinct eigenvalues are orthogonal.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000227, S26-D-00000198, S26-D-00000206

S26-T-00000233 - Eigenvalues of a self-adjoint operator are real

Every eigenvalue of a self-adjoint operator on a complex inner product space is real; over a real inner product space eigenvalues are already real scalars.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The eigenvalue belongs to the declared base field; in the complex case the theorem asserts it lies in R.

Hard dependencies: S26-D-00000227, S26-D-00000198, S26-D-00000201

S26-T-00000234 - Eigenvalues of a unitary operator have unit modulus

Every eigenvalue lambda of a unitary operator satisfies |lambda|=1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000228, S26-D-00000198, S26-D-00000203

S26-T-00000235 - Finite-dimensional spectral theorem for self-adjoint operators

Every self-adjoint operator on a finite-dimensional real or complex inner product space has an orthonormal basis of eigenvectors; equivalently, it is diagonalizable by an orthogonal/unitary change of basis with real diagonal entries.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The space is finite-dimensional over R or C.

Hard dependencies: S26-D-00000227, S26-T-00000217, S26-T-00000200, S26-T-00000232, S26-T-00000233

Applications: Quadratic forms | Principal axes | Quantum observables | Symmetric eigensolvers

S26-T-00000236 - Spectral decomposition of a self-adjoint operator

If T is self-adjoint on a finite-dimensional inner product space, then T=sum_j lambda_j P_j where the lambda_j are its distinct real eigenvalues and P_j are pairwise orthogonal projections onto their eigenspaces with sum_j P_j=I.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-T-00000235, S26-D-00000218, S26-D-00000198

S26-T-00000237 - Finite-dimensional complex spectral theorem for normal operators

A linear operator on a finite-dimensional complex inner product space is normal iff it has an orthonormal basis of eigenvectors, equivalently iff it is unitarily diagonalizable.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The base field is C.

Hard dependencies: S26-D-00000229, S26-T-00000217, S26-T-00000200

Boundary note: A 90-degree rotation on R^2 is real-normal and orthogonal but has no real eigenvectors, so real normal operators need not be diagonalizable over R.

Part W - Positive operators, singular values, SVD, and least squares

S26-D-00000238 - Positive semidefinite self-adjoint operator

A self-adjoint operator A is positive semidefinite, written A>=0, when <Av,v>>=0 for every v.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000227, S26-D-00000202

S26-D-00000239 - Positive definite self-adjoint operator

A self-adjoint operator A is positive definite, written A>0, when <Av,v>>0 for every nonzero v.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000238

S26-T-00000240 - Positive square root theorem - finite dimension

Every positive semidefinite operator A on a finite-dimensional real or complex inner product space has a unique positive semidefinite operator B with B^2=A.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The space is finite-dimensional.

Hard dependencies: S26-D-00000238, S26-T-00000235, S26-D-00000011

S26-D-00000241 - Rayleigh quotient

For a self-adjoint operator A and nonzero v, the Rayleigh quotient is R_A(v)=<Av,v>/<v,v>.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000227, S26-D-00000202

S26-T-00000242 - Extremal Rayleigh quotient characterization

For self-adjoint A on a nonzero finite-dimensional inner product space, min_{v!=0} R_A(v)=lambda_min and max_{v!=0} R_A(v)=lambda_max, the smallest and largest eigenvalues of A.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: A is self-adjoint and the space is finite-dimensional and nonzero.

Hard dependencies: S26-D-00000241, S26-T-00000235

S26-D-00000243 - Induced operator norm

For a linear map T between finite-dimensional inner product spaces, ||T||_2 = sup_{v!=0} ||Tv||/||v||.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Finite dimensionality is assumed; attainment of the supremum is not part of this definition.

Hard dependencies: S26-D-00000166, S26-D-00000203, S26-D-00000161

S26-T-00000244 - Operator norm of a self-adjoint operator

If A is self-adjoint on a finite-dimensional inner product space, then ||A||_2=max_j |lambda_j|.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-D-00000243, S26-T-00000235

S26-D-00000245 - Singular values

For a linear map T between finite-dimensional inner product spaces, the singular values are the nonnegative square roots of the eigenvalues of T*T, counted with multiplicity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain and codomain are finite-dimensional over R or C.

Hard dependencies: S26-T-00000223, S26-D-00000238, S26-T-00000240

S26-T-00000246 - Positivity of T*T

For every linear map T between finite-dimensional inner product spaces, T*T is positive semidefinite and ker(T*T)=ker(T).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar fields and finite-dimensional hypotheses are exactly those stated; no real/complex transport is implicit.

Hard dependencies: S26-T-00000223, S26-D-00000238, S26-D-00000168

S26-T-00000247 - Singular value decomposition

For every linear map T:V->W between finite-dimensional real or complex inner product spaces, there exist orthonormal bases of V and W in which the matrix of T is diagonal rectangular with nonnegative diagonal entries equal to the singular values.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V and W are finite-dimensional over the same F in {R,C}.

Hard dependencies: S26-D-00000245, S26-T-00000246, S26-T-00000217

Applications: Low-rank approximation | Numerical linear algebra | Data analysis | Inverse problems

S26-D-00000248 - Moore-Penrose pseudoinverse via SVD

For a matrix or finite-dimensional linear map with SVD T=U Sigma V*, the Moore-Penrose pseudoinverse is T^+=V Sigma^+ U*, where each nonzero singular value is reciprocated and each zero singular value remains zero.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Finite-dimensional real or complex inner product spaces.

Hard dependencies: S26-T-00000247, S26-D-00000224

S26-T-00000249 - Normal equations for least squares

For A:F^n->F^m and b in F^m with F in {R,C}, a vector x minimizes ||Ax-b|| iff A*(Ax-b)=0, equivalently A*A x=A*b.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Finite-dimensional standard inner product spaces; minimizers need not be unique when A has nontrivial kernel.

Hard dependencies: S26-T-00000223, S26-T-00000220, S26-D-00000169

Boundary note: If A is rank deficient, the normal equations can have multiple minimizers.

S26-T-00000250 - Pseudoinverse least-squares solution

For a finite-dimensional least-squares problem Ax approximately b, x=A^+b is a least-squares minimizer of minimum norm; it is the unique minimum-norm minimizer.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: A acts between finite-dimensional real or complex inner product spaces.

Hard dependencies: S26-D-00000248, S26-T-00000249, S26-T-00000220

Applications: Regression | Inverse problems | Numerical fitting

Part X - Duality, bilinear forms, and tensor products

S26-D-00000251 - Dual space

For a vector space V over F, the dual space V* is the vector space Hom_F(V,F) of linear functionals V->F.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000166, S26-D-00000149

S26-D-00000252 - Dual basis

If (e_1,...,e_n) is a basis of finite-dimensional V, its dual basis (e^1,...,e^n) in V* is defined by e^i(e_j)=delta_ij.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional and the displayed basis is finite.

Hard dependencies: S26-D-00000158, S26-D-00000251

S26-T-00000253 - Dual basis theorem

Every basis of a finite-dimensional vector space has a unique dual basis, and the dual basis is a basis of V*.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000252, S26-D-00000161

S26-T-00000254 - Dimension of the dual

If V is finite-dimensional, dim(V*)=dim(V).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-T-00000253, S26-D-00000161

S26-D-00000255 - Evaluation map into the double dual

For any vector space V, the canonical evaluation map J:V->V** is J(v)(phi)=phi(v).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000251, S26-D-00000166

S26-T-00000256 - Finite-dimensional double-dual isomorphism

If V is finite-dimensional, the evaluation map J:V->V** is an isomorphism.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional.

Hard dependencies: S26-D-00000255, S26-T-00000254, S26-T-00000175

Boundary note: For arbitrary infinite-dimensional algebraic vector spaces, J is injective but need not be surjective.

S26-D-00000257 - Annihilator of a subspace

For W<=V, the annihilator W^0 is {phi in V*: phi(w)=0 for every w in W}.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000151, S26-D-00000251

S26-T-00000258 - Dimension formula for annihilators

If V is finite-dimensional and W<=V, then dim(W^0)=dim(V)-dim(W).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional.

Hard dependencies: S26-D-00000257, S26-D-00000172, S26-D-00000161

S26-D-00000259 - Transpose (dual) map

For T:V->W linear, the dual map T*:W*->V* is T*(psi)=psi o T.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000166, S26-D-00000251, S26-D-00000009

S26-T-00000260 - Dual-map functoriality

For composable linear maps S,T, (S o T)*=T* o S*, and the dual of an identity map is an identity map.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000259, S26-D-00000009, S26-D-00000011

S26-D-00000261 - Bilinear map

A map B:V x W->U is bilinear when it is linear in each argument separately.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000166, S26-D-00000149

S26-D-00000262 - Multilinear map

A map M:V_1 x ... x V_k->W is multilinear when it is linear in each argument with all other arguments fixed.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: k is a positive integer.

Hard dependencies: S26-D-00000261

S26-D-00000263 - Alternating multilinear map

A multilinear map is alternating when its value is zero whenever two input arguments are equal.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000262

S26-T-00000264 - Alternating maps change sign under a transposition

For an alternating multilinear map over any field, swapping two arguments multiplies the value by -1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The statement uses additive inverse; in characteristic 2, -1=1, so sign change does not distinguish parity.

Hard dependencies: S26-D-00000263

S26-D-00000265 - Symmetric bilinear form

A bilinear form B:V x V->F is symmetric when B(u,v)=B(v,u).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000261

S26-D-00000266 - Skew-symmetric bilinear form

A bilinear form B is skew-symmetric when B(u,v)=-B(v,u).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000261

Boundary note: In characteristic 2, skew-symmetry alone does not imply B(v,v)=0; alternation must be stated separately.

S26-D-00000267 - Nondegenerate bilinear form

A bilinear form B on V is nondegenerate when B(v,w)=0 for every w implies v=0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000261

S26-D-00000268 - Tensor product - universal property

A tensor product V tensor W is a vector space equipped with a bilinear map tau:V x W->V tensor W such that every bilinear B:V x W->U factors uniquely through a linear map V tensor W->U.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Tensor products are over the declared common scalar field F.

Hard dependencies: S26-D-00000261, S26-D-00000166

S26-T-00000269 - Tensor product uniqueness up to unique isomorphism

Any two tensor products satisfying the same universal property are uniquely isomorphic by an isomorphism commuting with their canonical bilinear maps.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000268, S26-T-00000175

S26-D-00000270 - Pure tensor

An element of V tensor W of the form v tensor w, the image of (v,w) under the canonical bilinear map, is called a pure tensor.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000268

S26-T-00000271 - Pure tensors span the tensor product

The set of pure tensors v tensor w spans V tensor W.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000270, S26-D-00000154

S26-T-00000272 - Basis of a finite tensor product

If (e_i) is a basis of finite-dimensional V and (f_j) a basis of finite-dimensional W, then (e_i tensor f_j)_{i,j} is a basis of V tensor W.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V and W are finite-dimensional.

Hard dependencies: S26-D-00000158, S26-D-00000268, S26-T-00000271

S26-T-00000273 - Dimension of a tensor product

For finite-dimensional V,W, dim(V tensor W)=dim(V) dim(W).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-T-00000272, S26-D-00000161

S26-T-00000274 - Tensor-Hom identification in finite dimension

For finite-dimensional V,W, there is a canonical isomorphism V* tensor W ~= Hom(V,W), sending phi tensor w to the rank-one map v |-> phi(v)w.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V and W are finite-dimensional.

Hard dependencies: S26-D-00000251, S26-D-00000268, S26-D-00000166, S26-T-00000272

S26-D-00000275 - Tensor product of linear maps

Given linear maps S:V->V' and T:W->W', there is a unique linear map S tensor T:V tensor W->V' tensor W' with (S tensor T)(v tensor w)=S(v) tensor T(w).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000268, S26-D-00000166

S26-T-00000276 - Tensor-map functoriality

Tensor products of linear maps satisfy (S_2 o S_1) tensor (T_2 o T_1)=(S_2 tensor T_2)o(S_1 tensor T_1) and I tensor I=I.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000275, S26-D-00000009, S26-D-00000011

S26-D-00000277 - Tensor powers

For k>=0, the k-th tensor power T^k(V) is V tensor ... tensor V with k factors, with T^0(V)=F.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000268, S26-D-00000031

S26-D-00000278 - Tensor algebra

The tensor algebra T(V)=direct_sum_{k>=0} T^k(V) is graded by tensor degree and multiplication is induced by concatenation of tensors.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The direct sum is algebraic: each element has finitely many nonzero homogeneous components.

Hard dependencies: S26-D-00000277

S26-D-00000279 - Covariant and contravariant tensors

A tensor of type (r,s) on finite-dimensional V may be represented as an element of V^{tensor r} tensor (V*)^{tensor s}, equivalently as an appropriate multilinear map after the canonical finite-dimensional identifications.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional; variance convention is explicitly the one stated here.

Hard dependencies: S26-D-00000251, S26-D-00000277, S26-T-00000274

S26-D-00000280 - Symmetric tensor

A covariant k-tensor is symmetric when it is invariant under every permutation of its k arguments.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000262, S26-D-00000279

Part Y - Exterior algebra, determinant, orientation, and Hodge interface

S26-D-00000281 - Alternating covariant tensor

A covariant k-tensor is alternating when it is an alternating k-linear form V^k->F.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000263, S26-D-00000279

S26-D-00000282 - Exterior power

The k-th exterior power Lambda^k V is the vector space characterized by the universal alternating k-linear map V^k->Lambda^k V.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: k is a nonnegative integer; Lambda^0 V=F by convention.

Hard dependencies: S26-D-00000263, S26-D-00000268

S26-D-00000283 - Wedge product

The exterior algebra Lambda V=direct_sum_k Lambda^k V carries the wedge product, with decomposable elements written v_1 wedge ... wedge v_k.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000282, S26-D-00000278

S26-T-00000284 - Graded commutativity of the wedge product

For homogeneous alpha in Lambda^p V and beta in Lambda^q V, alpha wedge beta=(-1)^{pq} beta wedge alpha.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Degrees p,q are nonnegative integers; in characteristic 2 the sign may equal +1.

Hard dependencies: S26-D-00000283

S26-T-00000285 - Basis of an exterior power

If e_1,...,e_n is a basis of V, then e_{i1} wedge ... wedge e_{ik} with i1<...<ik forms a basis of Lambda^k V.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional of dimension n.

Hard dependencies: S26-D-00000158, S26-D-00000282, S26-D-00000283

S26-T-00000286 - Dimension of an exterior power

If dim V=n, then dim Lambda^k V=binomial(n,k); in particular Lambda^k V=0 for k>n.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional of dimension n.

Hard dependencies: S26-T-00000285, S26-T-00000095, S26-D-00000161

S26-T-00000287 - Top exterior power is one-dimensional

If dim V=n, then Lambda^n V is one-dimensional.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-T-00000286

S26-D-00000288 - Induced map on exterior powers

A linear map T:V->W induces Lambda^k T:Lambda^k V->Lambda^k W by v_1 wedge ... wedge v_k |-> Tv_1 wedge ... wedge Tv_k.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000166, S26-D-00000282, S26-D-00000283

S26-T-00000289 - Functoriality of exterior powers

Lambda^k(S o T)=(Lambda^k S)o(Lambda^k T), and Lambda^k(I)=I.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000288, S26-D-00000009, S26-D-00000011

S26-T-00000290 - Determinant from the top exterior power

For an endomorphism T of an n-dimensional vector space V, Lambda^n T acts on the one-dimensional space Lambda^n V by multiplication by det(T).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional of dimension n.

Hard dependencies: S26-T-00000287, S26-D-00000288, S26-D-00000191

S26-T-00000291 - Multiplicativity of determinant from exterior functoriality

For endomorphisms S,T of finite-dimensional V, det(S o T)=det(S)det(T).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-T-00000289, S26-T-00000290, S26-T-00000193

S26-D-00000292 - Volume form

On an n-dimensional real vector space, a volume form is a nonzero element of Lambda^n(V*).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is a finite-dimensional real vector space of dimension n.

Hard dependencies: S26-D-00000251, S26-T-00000287

S26-D-00000293 - Orientation of a real vector space

An orientation of an n-dimensional real vector space is a choice of one of the two equivalence classes of ordered bases, where two bases are equivalent when their change-of-basis determinant is positive.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional over R.

Hard dependencies: S26-D-00000195, S26-T-00000192

S26-T-00000294 - Volume forms determine orientations

A nonzero top-degree alternating covariant form omega determines an orientation by declaring a basis (e_1,...,e_n) positive iff omega(e_1,...,e_n)>0; positive scalar multiples determine the same orientation.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional over R.

Hard dependencies: S26-D-00000292, S26-D-00000293

S26-D-00000295 - Pullback of covariant tensors

For T:V->W linear and a covariant k-tensor alpha on W, the pullback T*alpha is defined by (T*alpha)(v_1,...,v_k)=alpha(Tv_1,...,Tv_k).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000166, S26-D-00000279

S26-T-00000296 - Pullback preserves wedge products

For alternating covariant tensors alpha,beta, T*(alpha wedge beta)=T*alpha wedge T*beta.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000295, S26-D-00000283

S26-D-00000297 - Interior product on alternating forms

For v in V and an alternating k-form omega, the interior product i_v omega is the (k-1)-form (i_v omega)(v_2,...,v_k)=omega(v,v_2,...,v_k).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: k>=1.

Hard dependencies: S26-D-00000281

S26-T-00000298 - Interior product graded Leibniz rule

For alpha of degree p, i_v(alpha wedge beta)=(i_v alpha) wedge beta+(-1)^p alpha wedge (i_v beta).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All vector spaces and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000297, S26-D-00000283

S26-D-00000299 - Hodge star - finite-dimensional Euclidean/Hermitian exterior algebra

Given an oriented finite-dimensional real inner product space V, the Hodge star *:Lambda^k(V*)->Lambda^{n-k}(V*) is the unique linear map satisfying alpha wedge *beta=<alpha,beta> vol for the induced inner product on exterior powers and the positively oriented unit volume form vol.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: V is finite-dimensional, real, oriented, and inner-product equipped; this entry does not assert the complex-Hermitian convention.

Hard dependencies: S26-D-00000202, S26-D-00000251, S26-D-00000282, S26-T-00000294

S26-T-00000300 - Hodge star square in the positive-definite real case

On an oriented n-dimensional real inner product space, the Hodge star satisfies star(star(alpha))=(-1)^{k(n-k)} alpha for alpha in Lambda^k(V*).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The metric is positive definite and real; pseudo-Riemannian signatures require a modified sign.

Hard dependencies: S26-D-00000299

Boundary note: The displayed sign is not valid unchanged for arbitrary indefinite metrics.

Applications: Differential forms | Vector calculus | Electromagnetism | Hodge theory interface

Part Z - Metric and topological foundations

S26-D-00000301 - Metric space

A metric space is a pair (X,d) where d:X x X->[0,infinity) satisfies identity of indiscernibles, symmetry, and the triangle inequality.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: d is real-valued and finite.

Hard dependencies: S26-N-00000016, S26-T-00000057

S26-D-00000302 - Open ball

In a metric space (X,d), the open ball B_r(x)={y in X:d(x,y)<r} for r>0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000301, S26-T-00000055

S26-D-00000303 - Open set in a metric space

A subset U of a metric space X is open when for every x in U there exists r>0 with B_r(x) contained in U.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000302, S26-N-00000016

S26-D-00000304 - Topology

A topology tau on X is a family of subsets containing emptyset and X, closed under arbitrary unions and finite intersections; (X,tau) is a topological space.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-N-00000016, S26-D-00000017

S26-T-00000305 - Metric topology

The open subsets defined by a metric form a topology on X.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000301, S26-D-00000303, S26-D-00000304

S26-D-00000306 - Closed set

A subset F of a topological space X is closed when X\F is open.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000304, S26-N-00000016

S26-D-00000307 - Interior

The interior int(A) of A subset X is the union of all open subsets of X contained in A.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000304, S26-D-00000017

S26-D-00000308 - Closure

The closure cl(A) of A subset X is the intersection of all closed subsets of X containing A.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000306, S26-D-00000017

S26-D-00000309 - Boundary

The boundary of A is bd(A)=cl(A) intersect cl(X\A).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000308, S26-N-00000016

S26-D-00000310 - Convergence in a metric space

A sequence (x_n) converges to x in (X,d) when for every epsilon>0 there exists N such that n>=N implies d(x_n,x)<epsilon.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000301, S26-D-00000031, S26-T-00000055

S26-T-00000311 - Uniqueness of metric limits

A convergent sequence in a metric space has a unique limit.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000310, S26-D-00000301

S26-D-00000312 - Continuous map - topological definition

A map f:X->Y between topological spaces is continuous when f^{-1}(U) is open in X for every open U in Y.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000008, S26-D-00000304, S26-D-00000303

S26-T-00000313 - Sequential criterion for continuity in metric spaces

For maps between metric spaces, f is continuous iff x_n->x implies f(x_n)->f(x) for every sequence.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Both domain and codomain are metric spaces.

Hard dependencies: S26-D-00000312, S26-D-00000310, S26-T-00000305

S26-T-00000314 - Composition of continuous maps

The composition of continuous maps is continuous.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000312, S26-D-00000009

S26-D-00000315 - Homeomorphism

A homeomorphism is a bijective continuous map whose inverse is continuous.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-T-00000010, S26-D-00000312, S26-D-00000008

S26-D-00000316 - Compact topological space

A topological space X is compact when every open cover of X has a finite subcover.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000304, S26-D-00000023

S26-T-00000317 - Continuous image of a compact space

If X is compact and f:X->Y is continuous, then f(X) is compact in the subspace topology.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000316, S26-D-00000312, S26-D-00000014

S26-T-00000318 - Heine-Borel theorem in finite-dimensional Euclidean space

A subset K of R^n is compact iff it is closed and bounded.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: K is a subset of finite-dimensional Euclidean R^n with its usual metric.

Hard dependencies: S26-D-00000316, S26-D-00000306, S26-T-00000205

Boundary note: Closed and bounded need not imply compact in arbitrary metric spaces or infinite-dimensional normed spaces.

S26-T-00000319 - Extreme value theorem

A continuous real-valued function on a nonempty compact space attains both a maximum and a minimum.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The domain is nonempty and compact; codomain is R with its usual topology.

Hard dependencies: S26-D-00000312, S26-D-00000316, S26-T-00000055

S26-D-00000320 - Connected space

A topological space is connected when it is not the union of two disjoint nonempty open subsets.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000304, S26-D-00000017

S26-T-00000321 - Continuous image of a connected space

The continuous image of a connected space is connected.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000320, S26-D-00000312, S26-D-00000014

S26-T-00000322 - Intervals in R are connected

Every interval in R is connected in the usual topology.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000320, S26-T-00000055

S26-T-00000323 - Intermediate value theorem

If f:[a,b]->R is continuous and y lies between f(a) and f(b), then some c in [a,b] satisfies f(c)=y.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: a<=b and f is continuous on the closed interval [a,b].

Hard dependencies: S26-D-00000312, S26-T-00000322

S26-D-00000324 - Cauchy sequence

A sequence (x_n) in a metric space is Cauchy when for every epsilon>0 there exists N such that m,n>=N imply d(x_m,x_n)<epsilon.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000301, S26-D-00000031, S26-T-00000055

S26-D-00000325 - Complete metric space

A metric space is complete when every Cauchy sequence converges to a point of the space.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000324, S26-D-00000310

S26-T-00000326 - Convergent sequences are Cauchy

Every convergent sequence in a metric space is Cauchy.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000310, S26-D-00000324

S26-T-00000327 - Finite-dimensional normed spaces are complete

Every finite-dimensional normed vector space over R or C is complete.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The scalar field is R or C with its usual absolute value.

Hard dependencies: S26-D-00000203, S26-D-00000161, S26-D-00000325

S26-T-00000328 - Equivalence of norms in finite dimension

Any two norms on a finite-dimensional vector space over R or C induce the same topology; quantitatively, each bounds the other up to positive constants.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The vector space is finite-dimensional over R or C.

Hard dependencies: S26-D-00000203, S26-D-00000161, S26-T-00000305

Boundary note: Arbitrary norms on infinite-dimensional spaces need not be equivalent.

S26-T-00000329 - Banach fixed-point theorem

A contraction T on a nonempty complete metric space has a unique fixed point, and iteration x_{n+1}=T(x_n) converges to it from every starting point.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: There exists q with 0<=q<1 such that d(Tx,Ty)<=q d(x,y) for all x,y.

Hard dependencies: S26-D-00000325, S26-D-00000312

Applications: Iterative numerical methods | Existence and uniqueness arguments

Part AA - Finite-dimensional differential calculus

S26-D-00000330 - Frechet derivative in finite dimensions

For open U subset R^n and f:U->R^m, f is differentiable at a in U when there exists a linear map Df(a) such that ||f(a+h)-f(a)-Df(a)h||/||h|| -> 0 as h->0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: U is open in R^n and a is in U.

Hard dependencies: S26-D-00000166, S26-D-00000203, S26-D-00000312

S26-T-00000331 - Uniqueness of the derivative

If the Frechet derivative of f at a exists, it is unique.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000330, S26-D-00000166

S26-T-00000332 - Differentiability implies continuity

If f is differentiable at a, then f is continuous at a.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000330, S26-D-00000312

S26-D-00000333 - Directional derivative

The directional derivative of f at a in direction v, when it exists, is D_v f(a)=lim_{t->0}(f(a+tv)-f(a))/t.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000330, S26-T-00000055

S26-T-00000334 - Derivative determines directional derivatives

If f is Frechet differentiable at a, then every directional derivative exists and D_v f(a)=Df(a)v.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000330, S26-D-00000333, S26-D-00000166

Boundary note: Existence of all directional derivatives alone does not imply Frechet differentiability.

S26-D-00000335 - Partial derivative

For f:U subset R^n->R^m, the j-th partial derivative at a is the directional derivative in the coordinate direction e_j, when it exists.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000333, S26-D-00000158

S26-D-00000336 - Jacobian matrix

For differentiable f:U subset R^n->R^m, the Jacobian J_f(a) is the matrix of Df(a) in the standard bases; its columns are the partial derivatives with respect to the input coordinates.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000330, S26-D-00000176, S26-D-00000335

S26-T-00000337 - Chain rule

If f is differentiable at a and g is differentiable at f(a), then g o f is differentiable at a and D(g o f)(a)=Dg(f(a)) o Df(a).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000330, S26-D-00000009, S26-D-00000166

S26-D-00000338 - Gradient

For differentiable scalar f:U subset R^n->R, the gradient grad f(a) is the unique vector satisfying Df(a)h=<grad f(a),h> for every h.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: R^n has its standard Euclidean inner product.

Hard dependencies: S26-D-00000330, S26-D-00000202, S26-D-00000251

S26-D-00000339 - Continuously differentiable map

A differentiable map f:U->R^m is C^1 when a |-> Df(a) is continuous as a map into the finite-dimensional space of linear maps.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000330, S26-D-00000312, S26-D-00000166

S26-T-00000340 - Continuous partial derivatives imply C1

If all first partial derivatives of f:U subset R^n->R^m exist on open U and are continuous, then f is C^1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: U is open in R^n.

Hard dependencies: S26-D-00000335, S26-D-00000339

S26-T-00000341 - Mean value theorem - scalar one-dimensional form

If f:[a,b]->R is continuous on [a,b] and differentiable on (a,b), then some c in (a,b) satisfies f(b)-f(a)=f'(c)(b-a).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: a<b; differentiability is the usual one-dimensional derivative.

Hard dependencies: S26-D-00000312, S26-D-00000330

S26-T-00000342 - First-order Taylor expansion

If f:U subset R^n->R^m is differentiable at a, then f(a+h)=f(a)+Df(a)h+o(||h||) as h->0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: a is an interior point of U.

Hard dependencies: S26-D-00000330

S26-D-00000343 - Second derivative and Hessian

For scalar f on open U subset R^n, when f is twice differentiable at a, the second derivative is a bilinear form D^2f(a); in standard coordinates its matrix is the Hessian H_f(a).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000330, S26-D-00000261, S26-D-00000176

S26-T-00000344 - Symmetry of mixed partials - Clairaut-Schwarz

If f has continuous second partial derivatives on a neighborhood of a, then the mixed second partials commute at a, so the Hessian is symmetric.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Second partial derivatives exist and are continuous on a neighborhood of a.

Hard dependencies: S26-D-00000343, S26-D-00000265

Boundary note: Without sufficient regularity, mixed partial derivatives can exist and fail to agree.

S26-T-00000345 - Second-order Taylor expansion

If f:U subset R^n->R is twice differentiable at a, then f(a+h)=f(a)+Df(a)h+(1/2)D^2f(a)[h,h]+o(||h||^2) as h->0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The second Frechet derivative exists at a in the stated sense.

Hard dependencies: S26-T-00000342, S26-D-00000343

S26-T-00000346 - Inverse function theorem - finite-dimensional C1 form

If f:U subset R^n->R^n is C^1 and Df(a) is invertible, then f restricts to a C^1 diffeomorphism between neighborhoods of a and f(a), with derivative of the local inverse equal to Df(x)^{-1} at corresponding points.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: U is open, a in U, and Df(a) is invertible.

Hard dependencies: S26-D-00000339, S26-T-00000175, S26-T-00000337

S26-T-00000347 - Implicit function theorem - finite-dimensional C1 form

Let F:U subset R^{n+m}->R^m be C^1 near (a,b), with F(a,b)=0 and the derivative with respect to the last m variables invertible. Then near (a,b), the zero set is the graph y=g(x) of a unique C^1 function g.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: U is open and the indicated m x m partial Jacobian is invertible at (a,b).

Hard dependencies: S26-D-00000339, S26-T-00000175, S26-T-00000346

S26-T-00000348 - First-order necessary condition for an interior extremum

If differentiable f:U subset R^n->R has a local maximum or minimum at interior point a, then Df(a)=0, equivalently grad f(a)=0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: a is an interior point of open U.

Hard dependencies: S26-D-00000330, S26-D-00000338

Boundary note: At boundary extrema the gradient need not vanish.

S26-T-00000349 - Second derivative test for strict local extrema

If f is C^2 near a, grad f(a)=0, and H_f(a) is positive definite, then a is a strict local minimum; if H_f(a) is negative definite, then a is a strict local maximum.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is C^2 on a neighborhood of a.

Hard dependencies: S26-D-00000343, S26-T-00000348, S26-T-00000242

Boundary note: A semidefinite Hessian is generally inconclusive.

S26-D-00000350 - Convex function on a convex set

For a convex subset C of a real vector space, f:C->R is convex when f((1-t)x+ty)<= (1-t)f(x)+t f(y) for all x,y in C and t in [0,1].

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: C is convex and the scalar field is R.

Hard dependencies: S26-D-00000149, S26-T-00000055

Applications: Optimization | Variational analysis | Machine learning

Part AB - Sequences, series, and uniform convergence

S26-D-00000351 - Sequence in a metric space

A sequence in X is a map x:N->X, written (x_n); convergence is the metric convergence of entry 310.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000031, S26-D-00000301, S26-D-00000310

S26-D-00000352 - Subsequence

A subsequence of (x_n) is (x_{n_k}) where n_1<n_2<... is a strictly increasing sequence of natural indices.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000351, S26-D-00000020

S26-T-00000353 - Subsequences preserve limits

Every subsequence of a convergent sequence converges to the same limit.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000352, S26-D-00000310

S26-D-00000354 - Limit point of a sequence

A point x is a subsequential limit of (x_n) when some subsequence converges to x.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000352, S26-D-00000310

S26-T-00000355 - Bolzano-Weierstrass in R^n

Every bounded sequence in finite-dimensional Euclidean R^n has a convergent subsequence.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The sequence is bounded in R^n with its usual Euclidean metric.

Hard dependencies: S26-D-00000351, S26-T-00000318, S26-D-00000352

Boundary note: The analogous statement fails for bounded sequences in general infinite-dimensional normed spaces.

S26-D-00000356 - Series in a normed vector space

For a sequence (a_n) in a normed vector space, the series sum a_n converges when its partial sums s_N=sum_{n=1}^N a_n converge.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000351, S26-D-00000203, S26-D-00000310

S26-T-00000357 - Cauchy criterion for series

In a complete normed space, sum a_n converges iff for every epsilon>0 there exists N such that ||sum_{k=m}^n a_k||<epsilon whenever n>=m>=N.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The ambient normed space is complete.

Hard dependencies: S26-D-00000356, S26-D-00000325, S26-D-00000324

S26-D-00000358 - Absolute convergence of a scalar series

A real or complex series sum a_n is absolutely convergent when sum |a_n| converges in R.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000356, S26-T-00000055, S26-I-00000060

S26-T-00000359 - Absolute convergence implies convergence

Every absolutely convergent real or complex series converges.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Scalars are R or C.

Hard dependencies: S26-D-00000358, S26-T-00000357, S26-T-00000327

S26-T-00000360 - Geometric series

For scalar r with |r|<1, sum_{n=0}^infinity r^n=1/(1-r).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: r is real or complex and |r|<1.

Hard dependencies: S26-D-00000356, S26-T-00000055

Boundary note: At |r|>=1 the infinite geometric series need not converge.

S26-T-00000361 - Comparison test for nonnegative series

If 0<=a_n<=b_n eventually and sum b_n converges, then sum a_n converges.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: a_n,b_n are real and nonnegative eventually.

Hard dependencies: S26-D-00000356, S26-D-00000020

S26-T-00000362 - Ratio test

For a scalar series with nonzero terms eventually, if limsup |a_{n+1}/a_n|<1 then the series converges absolutely; if liminf |a_{n+1}/a_n|>1 then terms do not tend to zero and the series diverges.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The indicated ratios are defined eventually.

Hard dependencies: S26-D-00000358, S26-T-00000360, S26-D-00000310

S26-T-00000363 - Root test

For a scalar series, if limsup |a_n|^(1/n)<1 then it converges absolutely; if limsup |a_n|^(1/n)>1 then it diverges.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000358, S26-T-00000360, S26-D-00000310

S26-D-00000364 - Pointwise convergence of functions

A sequence f_n:X->Y converges pointwise to f when for every x in X, f_n(x)->f(x).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000351, S26-D-00000008, S26-D-00000310

S26-D-00000365 - Uniform convergence of functions

For metric Y, f_n:X->Y converges uniformly to f when for every epsilon>0 there exists N such that n>=N implies d(f_n(x),f(x))<epsilon for every x in X.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000364, S26-D-00000301

S26-T-00000366 - Uniform convergence implies pointwise convergence

Uniform convergence of functions implies pointwise convergence.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000365, S26-D-00000364

S26-T-00000367 - Uniform limit theorem

If f_n:X->Y are continuous, Y is metric, and f_n converges uniformly to f, then f is continuous.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Continuity is with respect to the topology on X and metric topology on Y.

Hard dependencies: S26-D-00000365, S26-D-00000312

Boundary note: A pointwise limit of continuous functions need not be continuous.

S26-D-00000368 - Supremum norm

For bounded scalar-valued f on X, ||f||_infinity=sup_{x in X}|f(x)|.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is bounded so the supremum is finite.

Hard dependencies: S26-D-00000203, S26-D-00000021, S26-T-00000055

S26-T-00000369 - Uniform convergence and supremum norm

For bounded scalar functions, f_n->f uniformly iff ||f_n-f||_infinity->0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000365, S26-D-00000368

S26-T-00000370 - Weierstrass M-test

If |f_n(x)|<=M_n for every x and sum M_n converges, then sum f_n converges uniformly and absolutely pointwise.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: M_n>=0 and the scalar codomain is R or C.

Hard dependencies: S26-D-00000356, S26-D-00000365, S26-T-00000361

Part AC - Power series and Riemann integration

S26-D-00000371 - Power series

A power series centered at a is sum_{n=0}^infinity c_n(x-a)^n over R or C.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000356, S26-T-00000126, S26-T-00000055

S26-D-00000372 - Radius of convergence

The radius R in [0,infinity] of a power series is the threshold such that the series converges absolutely for |x-a|<R and diverges for |x-a|>R.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000371, S26-D-00000358

S26-T-00000373 - Cauchy-Hadamard formula

For a power series sum c_n(x-a)^n, its radius satisfies 1/R=limsup |c_n|^(1/n), with the usual conventions for 0 and infinity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Coefficients are real or complex.

Hard dependencies: S26-D-00000372, S26-T-00000363

S26-T-00000374 - Uniform convergence inside the radius

A power series converges uniformly on every closed disk |x-a|<=r with r<R.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: 0<=r<R.

Hard dependencies: S26-T-00000370, S26-D-00000372

S26-T-00000375 - Termwise differentiation of a power series

Inside its radius of convergence, a power series is differentiable and its derivative is obtained term by term; the derived series has the same radius.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Real one-variable or complex power series, evaluated strictly inside the radius of convergence.

Hard dependencies: S26-D-00000371, S26-T-00000374, S26-D-00000330

S26-T-00000376 - Termwise integration of a power series

Inside its radius of convergence, a real power series may be integrated term by term on compact subintervals, yielding an antiderivative series with the same radius.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Real one-variable power series on an interval strictly inside the radius.

Hard dependencies: S26-D-00000371, S26-T-00000374

S26-D-00000377 - Partition of a compact interval

A partition P of [a,b] is a finite ordered set a=x_0<x_1<...<x_n=b.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: a<b are real.

Hard dependencies: S26-D-00000023, S26-D-00000020, S26-T-00000055

S26-D-00000378 - Mesh of a partition

The mesh ||P|| of P is max_i(x_i-x_{i-1}).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000377, S26-D-00000021

S26-D-00000379 - Tagged partition

A tagged partition is a partition P together with tags t_i in [x_{i-1},x_i].

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000377

S26-D-00000380 - Riemann sum

For bounded f:[a,b]->R and tagged partition, the Riemann sum is sum_i f(t_i)(x_i-x_{i-1}).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is bounded on [a,b].

Hard dependencies: S26-D-00000379, S26-D-00000356

S26-D-00000381 - Riemann integrability - tagged-sum definition

A bounded f:[a,b]->R is Riemann integrable with integral I if for every epsilon>0 there exists delta>0 such that every tagged partition with mesh<delta has Riemann sum within epsilon of I.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is bounded on the compact interval [a,b].

Hard dependencies: S26-D-00000380, S26-D-00000378

S26-T-00000382 - Uniqueness of the Riemann integral

A Riemann-integrable function has a unique integral value.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000381, S26-T-00000055

S26-T-00000383 - Continuous functions are Riemann integrable

Every continuous real-valued function on [a,b] is Riemann integrable.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: a<b and f:[a,b]->R is continuous.

Hard dependencies: S26-D-00000312, S26-T-00000318, S26-D-00000381

S26-T-00000384 - Monotone functions are Riemann integrable

Every bounded monotone real-valued function on [a,b] is Riemann integrable.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: a<b and f is monotone.

Hard dependencies: S26-D-00000381, S26-D-00000020

S26-T-00000385 - Linearity of the Riemann integral

If f and g are Riemann integrable and alpha,beta are real, then integral(alpha f+beta g)=alpha integral f+beta integral g.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000381, S26-D-00000149

S26-T-00000386 - Order preservation of the Riemann integral

If f and g are Riemann integrable and f(x)<=g(x) on [a,b], then integral f<=integral g.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000381, S26-D-00000020

S26-T-00000387 - Integral absolute-value bound

If f is Riemann integrable, then |integral_a^b f|<=integral_a^b |f|.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000381, S26-T-00000386

S26-T-00000388 - Additivity over adjacent intervals

If f is Riemann integrable on [a,b] and c in [a,b], then integral_a^b f=integral_a^c f+integral_c^b f.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000381, S26-T-00000385

S26-T-00000389 - Uniform convergence preserves Riemann integrals

If Riemann-integrable f_n on [a,b] converge uniformly to f, then f is Riemann integrable and integral f_n -> integral f.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All functions are real-valued on the same compact interval.

Hard dependencies: S26-D-00000365, S26-D-00000381, S26-T-00000387

S26-D-00000390 - Indefinite integral function

For Riemann-integrable f on [a,b], define F(x)=integral_a^x f(t)dt for x in [a,b].

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000381, S26-T-00000388

S26-T-00000391 - Fundamental theorem of calculus - part I

If f:[a,b]->R is continuous and F(x)=integral_a^x f(t)dt, then F is differentiable on (a,b) and F'(x)=f(x).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is continuous on [a,b].

Hard dependencies: S26-T-00000383, S26-D-00000390, S26-D-00000330

S26-T-00000392 - Fundamental theorem of calculus - part II

If F:[a,b]->R is C^1, then integral_a^b F'(x)dx=F(b)-F(a).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: F is continuously differentiable on [a,b].

Hard dependencies: S26-D-00000339, S26-T-00000383, S26-T-00000391

S26-T-00000393 - Integration by parts

If u,v are C^1 on [a,b], then integral_a^b u v'=u(b)v(b)-u(a)v(a)-integral_a^b u' v.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: u and v are continuously differentiable.

Hard dependencies: S26-T-00000392, S26-T-00000337

S26-T-00000394 - Change of variables - one-dimensional Riemann form

If phi:[alpha,beta]->[a,b] is C^1 and f is continuous on [a,b], then integral_alpha^beta f(phi(t))phi'(t)dt=integral_{phi(alpha)}^{phi(beta)} f(x)dx.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: phi is C^1; f is continuous on an interval containing phi([alpha,beta]).

Hard dependencies: S26-T-00000392, S26-T-00000337

S26-T-00000395 - Mean value theorem for integrals

If f is continuous on [a,b] and a<b, then some c in [a,b] satisfies integral_a^b f=f(c)(b-a).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is continuous and real-valued.

Hard dependencies: S26-T-00000383, S26-T-00000319, S26-T-00000323

S26-D-00000396 - Improper integral on an unbounded interval

For f on [a,infinity), the improper integral integral_a^infinity f is defined as lim_{b->infinity} integral_a^b f when this limit exists.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is Riemann integrable on every finite interval [a,b].

Hard dependencies: S26-D-00000381, S26-D-00000310

S26-T-00000397 - Absolute convergence criterion for improper integrals

If integral_a^infinity |f(x)|dx converges, then integral_a^infinity f(x)dx converges.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is Riemann integrable on every compact subinterval.

Hard dependencies: S26-D-00000396, S26-T-00000387

S26-T-00000398 - p-integral criterion

For real p, integral_1^infinity x^{-p}dx converges iff p>1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All spaces, maps, sequences, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000396, S26-T-00000055

S26-T-00000399 - Integral test for positive decreasing series

If f:[1,infinity)->[0,infinity) is continuous and decreasing and a_n=f(n), then sum_{n=1}^infinity a_n converges iff integral_1^infinity f(x)dx converges.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is continuous, nonnegative, and decreasing.

Hard dependencies: S26-D-00000356, S26-D-00000396, S26-D-00000020

S26-T-00000400 - Uniform convergence permits termwise integration

If continuous f_n:[a,b]->R and sum f_n converges uniformly to S, then S is continuous and integral_a^b S=sum_n integral_a^b f_n.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The function series converges uniformly on the compact interval [a,b].

Hard dependencies: S26-T-00000370, S26-T-00000367, S26-T-00000389

Boundary note: Pointwise convergence alone does not justify interchange of infinite summation and integration.

Applications: Fourier and power series | Approximation theory | Numerical analysis

Part AD - Measure spaces, measurability, and Lebesgue integration

S26-D-00000401 - Sigma-algebra

A sigma-algebra Sigma on a set X is a family of subsets containing X and closed under complements in X and countable unions.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-N-00000016, S26-D-00000023, S26-D-00000031

S26-D-00000402 - Measurable space

A measurable space is a pair (X,Sigma) consisting of a set X and a sigma-algebra Sigma on X.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000401

S26-D-00000403 - Generated sigma-algebra

For a family C of subsets of X, sigma(C) is the intersection of all sigma-algebras on X containing C.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000401, S26-D-00000018

S26-D-00000404 - Borel sigma-algebra

For a topological space X, the Borel sigma-algebra B(X) is the sigma-algebra generated by the open subsets of X.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000403, S26-T-00000305

S26-D-00000405 - Measure

A measure on (X,Sigma) is a map mu:Sigma->[0,infinity] with mu(emptyset)=0 and countable additivity on pairwise disjoint measurable sets.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000402, S26-D-00000031

S26-D-00000406 - Measure space

A measure space is a triple (X,Sigma,mu) where (X,Sigma) is measurable and mu is a measure on it.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000405

S26-T-00000407 - Continuity from below

If E_1 subset E_2 subset ... are measurable, then mu(union_n E_n)=lim_n mu(E_n).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: (E_n) is an increasing sequence of measurable sets.

Hard dependencies: S26-D-00000405, S26-D-00000310

S26-T-00000408 - Continuity from above

If E_1 superset E_2 superset ... are measurable and mu(E_1)<infinity, then mu(intersection_n E_n)=lim_n mu(E_n).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: (E_n) is decreasing and mu(E_1) is finite.

Hard dependencies: S26-D-00000405, S26-D-00000310

Boundary note: The finite-measure hypothesis cannot in general be dropped.

S26-D-00000409 - Null set

A measurable set N is null when mu(N)=0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000405

S26-D-00000410 - Almost everywhere

A property holds mu-almost everywhere when the measurable set on which it fails is null.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000409

S26-D-00000411 - Complete measure

A measure space is complete when every subset of every measurable null set is measurable.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000406, S26-D-00000409

S26-D-00000412 - Measurable function

A function f:(X,Sigma)->(Y,Tau) is measurable when f^{-1}(A) is in Sigma for every A in Tau.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000008, S26-D-00000402

S26-T-00000413 - Continuous maps are Borel measurable

Every continuous map between topological spaces is measurable with respect to their Borel sigma-algebras.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000312, S26-D-00000404, S26-D-00000412

S26-T-00000414 - Composition of measurable maps

The composition of measurable maps is measurable.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000412, S26-D-00000011

S26-T-00000415 - Pointwise limits preserve measurability

A pointwise limit of measurable real- or extended-real-valued functions is measurable whenever the pointwise limit is defined.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The codomain carries its Borel sigma-algebra; extended-real conventions are used where needed.

Hard dependencies: S26-D-00000412, S26-D-00000364

S26-D-00000416 - Indicator function

For E subset X, the indicator 1_E:X->{0,1} equals 1 on E and 0 outside E.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-N-00000016, S26-D-00000008

S26-D-00000417 - Nonnegative simple function

A nonnegative simple measurable function is a finite sum sum_i a_i 1_{E_i} with a_i>=0 and measurable E_i.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000416, S26-D-00000412, S26-D-00000020

S26-D-00000418 - Integral of a nonnegative simple function

For a representation s=sum_i a_i 1_{E_i} with disjoint measurable E_i, define integral s dmu=sum_i a_i mu(E_i).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000417, S26-D-00000405

S26-D-00000419 - Lebesgue integral of a nonnegative function

For measurable f:X->[0,infinity], define integral f dmu as the supremum of integrals of nonnegative simple measurable s with 0<=s<=f.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000418, S26-D-00000021, S26-D-00000412

S26-T-00000420 - Monotone convergence theorem

If 0<=f_n<=f_{n+1} are measurable and f_n->f pointwise, then integral f_n dmu -> integral f dmu, allowing infinity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All f_n are nonnegative measurable functions.

Hard dependencies: S26-D-00000419, S26-D-00000364

S26-T-00000421 - Fatou lemma

For nonnegative measurable f_n, integral liminf_n f_n dmu <= liminf_n integral f_n dmu.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All f_n are nonnegative measurable functions.

Hard dependencies: S26-D-00000419, S26-T-00000420

S26-D-00000422 - Positive and negative parts

For a real-valued measurable f, define f^+=max(f,0), f^-=max(-f,0), so f=f^+-f^- and |f|=f^++f^-.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000412, S26-D-00000020, S26-T-00000055

S26-D-00000423 - Lebesgue integrable function

A measurable real- or complex-valued function f is integrable when integral |f| dmu<infinity; its integral is defined from positive/negative parts (or real/imaginary parts).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000419, S26-D-00000422, S26-I-00000060

S26-T-00000424 - Linearity of the Lebesgue integral

For integrable f,g and scalars alpha,beta, integral(alpha f+beta g)=alpha integral f+beta integral g.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000423, S26-D-00000149

S26-T-00000425 - Integral absolute-value inequality

For integrable f, |integral f dmu|<=integral |f| dmu.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000423, S26-D-00000020

S26-T-00000426 - Dominated convergence theorem

If measurable f_n->f almost everywhere and |f_n|<=g almost everywhere for an integrable g, then f is integrable and integral |f_n-f| dmu->0; hence integral f_n dmu->integral f dmu.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: g is integrable and dominates every |f_n| almost everywhere.

Hard dependencies: S26-D-00000410, S26-D-00000423, S26-T-00000421

Boundary note: Pointwise or almost-everywhere convergence without an integrable dominator does not in general permit exchanging limit and integral.

S26-T-00000427 - Bounded convergence on finite measure spaces

If mu(X)<infinity, f_n are uniformly bounded by a finite constant, and f_n->f almost everywhere, then integral |f_n-f| dmu->0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: mu(X)<infinity and a common finite essential bound exists.

Hard dependencies: S26-T-00000426, S26-D-00000405

S26-T-00000428 - Riemann integrable implies Lebesgue integrable

A bounded Riemann-integrable function on [a,b] is Lebesgue measurable and Lebesgue integrable, and its Lebesgue integral equals its Riemann integral.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Lebesgue measure on a compact real interval is used.

Hard dependencies: S26-D-00000381, S26-D-00000423, S26-D-00000404

Part AE - Product measures, L^p spaces, and probability bridge

S26-D-00000429 - Product sigma-algebra

For measurable spaces (X,Sigma),(Y,Tau), Sigma tensor Tau is the sigma-algebra on X x Y generated by measurable rectangles A x B.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000403, S26-D-00000402, S26-D-00000017

S26-D-00000430 - Product measure

For sigma-finite measure spaces, the product measure mu x nu is the measure on the product sigma-algebra satisfying (mu x nu)(A x B)=mu(A)nu(B) on measurable rectangles.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The factors are sigma-finite measure spaces.

Hard dependencies: S26-D-00000429, S26-D-00000405

S26-T-00000431 - Tonelli theorem

For a nonnegative measurable f on a product of sigma-finite measure spaces, the product integral equals either iterated integral, allowing infinity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The product measure spaces are sigma-finite and f is nonnegative measurable.

Hard dependencies: S26-D-00000419, S26-D-00000430

S26-T-00000432 - Fubini theorem

For integrable f on a product of sigma-finite measure spaces, the iterated integrals exist almost everywhere and equal the product integral in either order.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is integrable with respect to the product measure and the factors are sigma-finite.

Hard dependencies: S26-D-00000423, S26-D-00000430, S26-T-00000431

Boundary note: Without absolute integrability, changing the order of integration can fail or become undefined.

S26-D-00000433 - L^p seminorm

For 1<=p<infinity define ||f||_p=(integral |f|^p dmu)^(1/p); define ||f||_infinity as the essential supremum.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000423, S26-D-00000021, S26-D-00000368

S26-D-00000434 - L^p space

L^p(X,mu) is the vector space of measurable functions with finite L^p seminorm modulo equality almost everywhere.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000433, S26-D-00000410, S26-D-00000149

S26-T-00000435 - Holder inequality

If 1<=p,q<=infinity with 1/p+1/q=1, then ||fg||_1<=||f||_p ||g||_q.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: p and q are conjugate exponents and f in L^p, g in L^q.

Hard dependencies: S26-D-00000433, S26-D-00000434

S26-T-00000436 - Minkowski inequality

For 1<=p<=infinity, ||f+g||_p<=||f||_p+||g||_p.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000433, S26-T-00000435

S26-T-00000437 - L^p completeness

For 1<=p<=infinity, L^p(X,mu) is a Banach space under ||.||_p.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Functions are identified modulo equality almost everywhere.

Hard dependencies: S26-D-00000434, S26-T-00000436, S26-D-00000325

S26-D-00000438 - L^2 inner product

For complex L^2 define <f,g>=integral conjugate(f)g dmu; for real L^2 omit conjugation.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000434, S26-D-00000201, S26-I-00000060

S26-T-00000439 - Cauchy-Schwarz in L^2

For f,g in L^2, |<f,g>|<=||f||_2||g||_2.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000438, S26-T-00000435

S26-T-00000440 - L^2 is a Hilbert space

L^2(X,mu), modulo equality almost everywhere, is complete under the norm induced by its inner product.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-T-00000437, S26-D-00000438, S26-T-00000209

S26-D-00000441 - Convergence in measure

A sequence f_n converges to f in measure when for every epsilon>0, mu({|f_n-f|>epsilon})->0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000405, S26-D-00000310, S26-D-00000412

S26-T-00000442 - L^p convergence implies convergence in measure

If 1<=p<infinity and ||f_n-f||_p->0, then f_n->f in measure.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: 1<=p<infinity.

Hard dependencies: S26-D-00000433, S26-D-00000441

S26-T-00000443 - Almost-everywhere convergence need not imply L^1 convergence

There exist integrable f_n converging to 0 almost everywhere while ||f_n||_1 does not tend to 0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000410, S26-D-00000434

Boundary note: On [0,1], f_n=n 1_(0,1/n) converges to 0 almost everywhere but has L^1 norm 1.

S26-D-00000444 - Pushforward measure

For measurable T:(X,Sigma)->(Y,Tau), the pushforward T_*mu is defined by (T_*mu)(A)=mu(T^{-1}(A)).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000412, S26-D-00000405

S26-T-00000445 - Integration against a pushforward

For nonnegative measurable or integrable g on Y, integral_Y g d(T_*mu)=integral_X g(T(x)) dmu.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000444, S26-D-00000419, S26-D-00000423

S26-D-00000446 - Probability space

A probability space is a measure space (Omega,F,P) with P(Omega)=1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000406

S26-D-00000447 - Random variable

A random variable with values in a measurable space S is a measurable map X:(Omega,F)->S.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000446, S26-D-00000412

S26-D-00000448 - Distribution of a random variable

The distribution or law of X is the pushforward probability measure X_*P.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000447, S26-D-00000444

S26-D-00000449 - Expectation

For an integrable real- or complex-valued random variable X, E[X]=integral_Omega X dP.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All sets, functions, measures, and scalar fields satisfy exactly the hypotheses stated.

Hard dependencies: S26-D-00000447, S26-D-00000423

S26-T-00000450 - Markov inequality

If X>=0 is an integrable random variable and a>0, then P(X>=a)<=E[X]/a.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: X is nonnegative and integrable; a>0.

Hard dependencies: S26-D-00000449, S26-D-00000020

Applications: Probability theory | Concentration bounds | Statistical learning

Part AF - Probability, conditioning, distributions, and moments

S26-D-00000451 - Event

An event in a probability space (Omega,F,P) is an element A of the sigma-algebra F.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000446, S26-D-00000401

S26-D-00000452 - Conditional probability

For events A,B with P(B)>0, define P(A|B)=P(A intersection B)/P(B).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000451, S26-D-00000020

S26-T-00000453 - Multiplication rule

If P(B)>0, then P(A intersection B)=P(A|B)P(B).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000452

S26-T-00000454 - Law of total probability

If (B_i) is a finite or countable measurable partition with P(B_i)>0, then P(A)=sum_i P(A|B_i)P(B_i).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: (B_i) is a finite or countable partition of Omega and each used conditional probability is defined.

Hard dependencies: S26-D-00000452, S26-D-00000031

S26-T-00000455 - Bayes theorem

If P(A)>0 and P(B)>0, then P(A|B)=P(B|A)P(A)/P(B).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000452, S26-T-00000453

S26-D-00000456 - Independence of two events

Events A and B are independent when P(A intersection B)=P(A)P(B).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000451

S26-D-00000457 - Mutual independence

A family of events is mutually independent when every finite subfamily has intersection probability equal to the product of its individual probabilities.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000456, S26-D-00000031

S26-D-00000458 - Independence of random variables

Random variables X_i are independent when their generated sigma-algebras are mutually independent; equivalently, finite inverse-image events factorize.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000447, S26-D-00000403, S26-D-00000457

S26-D-00000459 - Identically distributed random variables

Random variables X and Y are identically distributed when their pushforward laws are equal.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000448

S26-D-00000460 - IID family

A family of random variables is IID when its members are mutually independent and identically distributed.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000458, S26-D-00000459

S26-D-00000461 - Distribution function

For a real-valued random variable X, F_X(x)=P(X<=x).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000447, S26-D-00000448

S26-T-00000462 - CDF characterization

A real distribution function is nondecreasing, right-continuous, tends to 0 at -infinity and 1 at +infinity; conversely every such function is the CDF of a unique Borel probability measure on R.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The converse concerns Borel probability measures on R.

Hard dependencies: S26-D-00000461, S26-D-00000404, S26-D-00000310

S26-D-00000463 - Discrete random variable

A random variable is discrete when its law is supported on a finite or countable set.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000448, S26-D-00000031

S26-D-00000464 - Probability mass function

For a discrete random variable X, p_X(x)=P(X=x); its nonnegative masses sum to 1 over the support.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000463

S26-D-00000465 - Absolutely continuous distribution

A real random variable has an absolutely continuous law when there exists an integrable f>=0 such that P(X in A)=integral_A f dx for every Borel A.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000448, S26-D-00000423, S26-D-00000404

S26-D-00000466 - Probability density function

A density f of an absolutely continuous law is a nonnegative measurable function with integral_R f dx=1 representing probabilities by integration.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000465

S26-T-00000467 - Expectation via the law

For integrable measurable g(X), E[g(X)]=integral g d mu_X, where mu_X is the law of X.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000445, S26-D-00000448, S26-D-00000449

S26-D-00000468 - Moment

When integrable, the k-th raw moment of X is E[X^k].

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000449, S26-D-00000021

S26-D-00000469 - Variance

For square-integrable real X, Var(X)=E[(X-E[X])^2]=E[X^2]-E[X]^2.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: X is real-valued and square-integrable.

Hard dependencies: S26-D-00000449, S26-D-00000468

S26-D-00000470 - Covariance

For square-integrable real X,Y, Cov(X,Y)=E[(X-E[X])(Y-E[Y])].

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000469, S26-D-00000449

S26-D-00000471 - Correlation coefficient

If Var(X),Var(Y)>0, rho(X,Y)=Cov(X,Y)/(sqrt(Var(X))sqrt(Var(Y))).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000470, S26-D-00000469

S26-T-00000472 - Cauchy-Schwarz covariance bound

For square-integrable real X,Y, |Cov(X,Y)|<=sqrt(Var(X)Var(Y)); hence |rho|<=1 when correlation is defined.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000439, S26-D-00000470, S26-D-00000471

S26-T-00000473 - Variance of a sum

For square-integrable X_1,...,X_n, Var(sum_i X_i)=sum_i Var(X_i)+2 sum_{i<j} Cov(X_i,X_j).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000469, S26-D-00000470, S26-D-00000020

S26-T-00000474 - Variance additivity under independence

For pairwise independent square-integrable X_i, Var(sum_i X_i)=sum_i Var(X_i).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The variables are pairwise independent and square-integrable.

Hard dependencies: S26-D-00000458, S26-T-00000473

S26-T-00000475 - Chebyshev inequality

If X has finite variance, then for a>0, P(|X-E[X]|>=a)<=Var(X)/a^2.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000450, S26-D-00000469

S26-D-00000476 - Conditional expectation

For integrable X and sub-sigma-algebra G subset F, E[X|G] is any G-measurable integrable random variable Y satisfying integral_G Y dP=integral_G X dP for every G in G; it is unique almost surely.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: X is integrable and G is a sub-sigma-algebra of F.

Hard dependencies: S26-D-00000449, S26-D-00000401, S26-D-00000410

S26-T-00000477 - Tower property

If H subset G subset F are sub-sigma-algebras and X is integrable, then E[E[X|G]|H]=E[X|H] almost surely.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000476

S26-T-00000478 - Taking out what is known

If X is integrable, Y is G-measurable, and YX is integrable under suitable boundedness/integrability conditions, then E[YX|G]=Y E[X|G] almost surely.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Products and conditional expectations appearing in the statement are integrable; a standard sufficient case is bounded G-measurable Y and integrable X.

Hard dependencies: S26-D-00000476, S26-T-00000424

S26-T-00000479 - Conditional expectation under independence

If integrable X is independent of G, then E[X|G]=E[X] almost surely.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000458, S26-D-00000476

S26-D-00000480 - Moment generating function

When finite on a neighborhood of 0, the moment generating function is M_X(t)=E[e^{tX}].

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The expectation is finite for the values of t under consideration.

Hard dependencies: S26-D-00000449

S26-D-00000481 - Characteristic function

For real X, phi_X(t)=E[e^{itX}], defined for every real t.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000449, S26-I-00000060

S26-T-00000482 - Characteristic function of an independent sum

If X and Y are independent, phi_{X+Y}(t)=phi_X(t)phi_Y(t).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000458, S26-D-00000481

S26-T-00000483 - Uniqueness of characteristic functions

Two Borel probability measures on R with the same characteristic function are equal.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The measures are Borel probability measures on R.

Hard dependencies: S26-D-00000481, S26-D-00000404

Part AG - Probabilistic convergence, laws of large numbers, and CLT

S26-D-00000484 - Convergence almost surely

X_n converges almost surely to X when P({omega: X_n(omega)->X(omega)})=1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000410, S26-D-00000447, S26-D-00000364

S26-D-00000485 - Convergence in probability

X_n converges in probability to X when for every epsilon>0, P(|X_n-X|>epsilon)->0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000441, S26-D-00000446

S26-D-00000486 - Convergence in L^p

For 1<=p<infinity, X_n converges to X in L^p when E[|X_n-X|^p]->0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000433, S26-D-00000449

S26-D-00000487 - Convergence in distribution

Real random variables X_n converge in distribution to X when their laws converge weakly; equivalently, F_{X_n}(x)->F_X(x) at every continuity point of F_X.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: For the CDF criterion the variables are real-valued.

Hard dependencies: S26-D-00000448, S26-D-00000461

S26-T-00000488 - Almost-sure convergence implies convergence in probability

If X_n->X almost surely, then X_n->X in probability.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000484, S26-D-00000485

S26-T-00000489 - L^p convergence implies convergence in probability

For 1<=p<infinity, X_n->X in L^p implies X_n->X in probability.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000442, S26-D-00000486, S26-D-00000485

S26-T-00000490 - Convergence in probability implies convergence in distribution

If X_n->X in probability, then X_n->X in distribution.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000485, S26-D-00000487

S26-T-00000491 - Convergence in distribution need not imply convergence in probability

Equal or converging laws alone do not force random variables on a common probability space to become close in probability.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000487, S26-D-00000485

Boundary note: Independent Bernoulli(1/2) variables X_n and X on one product space have identical laws but P(|X_n-X|>1/2)=1/2.

S26-T-00000492 - Borel-Cantelli lemma I

For events A_n, if sum_n P(A_n)<infinity then P(A_n infinitely often)=0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000451, S26-D-00000358, S26-T-00000407

S26-T-00000493 - Borel-Cantelli lemma II

If events A_n are independent and sum_n P(A_n)=infinity, then P(A_n infinitely often)=1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The events are mutually independent.

Hard dependencies: S26-D-00000457, S26-T-00000492

S26-D-00000494 - Sample mean

For X_1,...,X_n, the sample mean is bar X_n=(1/n)sum_{k=1}^n X_k.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000020, S26-D-00000447

S26-T-00000495 - Weak law of large numbers

For IID random variables with finite mean mu and finite variance, the sample mean converges in probability to mu.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The variables are IID with finite mean and finite variance.

Hard dependencies: S26-D-00000460, S26-T-00000475, S26-D-00000494, S26-D-00000485

S26-T-00000496 - Strong law of large numbers

For IID integrable real random variables with mean mu, the sample mean converges almost surely to mu.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The variables are IID and E[|X_1|]<infinity.

Hard dependencies: S26-D-00000460, S26-D-00000494, S26-D-00000484

S26-D-00000497 - Standardized partial sum

For IID real X_k with mean mu and variance sigma^2 in (0,infinity), define Z_n=(sum_{k=1}^n X_k-n mu)/(sigma sqrt(n)).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000460, S26-D-00000469

S26-T-00000498 - Lindeberg-Levy central limit theorem

For IID real X_k with finite mean mu and finite positive variance sigma^2, the standardized partial sums Z_n converge in distribution to N(0,1).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The variables are IID with finite mean and variance 0<sigma^2<infinity.

Hard dependencies: S26-D-00000497, S26-D-00000487

S26-D-00000499 - Standard normal distribution

The standard normal law N(0,1) is the Borel probability measure on R with density (2 pi)^(-1/2) exp(-x^2/2).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000466, S26-D-00000404

S26-T-00000500 - Central limit theorem CDF form

Under the Lindeberg-Levy hypotheses, P(Z_n<=x)->Phi(x) for every real x, where Phi is the standard normal CDF.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, random variables, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000498, S26-D-00000499, S26-D-00000461

Applications: Statistics | Monte Carlo methods | Error propagation | Asymptotic inference

Part AH - Stochastic processes, filtrations, martingales, and stopping

S26-D-00000501 - Stochastic process

A stochastic process indexed by T is a family (X_t)_{t in T} of random variables on one probability space.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000447

S26-D-00000502 - Sample path

For fixed omega, the map t |-> X_t(omega) is the sample path (trajectory) of a stochastic process.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000501

S26-D-00000503 - Finite-dimensional distribution

For t_1,...,t_n, the joint law of (X_{t_1},...,X_{t_n}) is a finite-dimensional distribution of the process.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000501, S26-D-00000448

S26-T-00000504 - Process law determined by finite-dimensional distributions on product cylinders

On the canonical product measurable space, equality of all finite-dimensional distributions implies equality of process laws.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Processes are viewed on the canonical product measurable space with the cylinder sigma-algebra.

Hard dependencies: S26-D-00000503, S26-D-00000403

S26-D-00000505 - Filtration

A filtration (F_t) is an increasing family of sub-sigma-algebras: s<=t implies F_s subset F_t.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000401, S26-D-00000309

S26-D-00000506 - Filtered probability space

A filtered probability space is (Omega,F,(F_t),P) with (F_t) a filtration contained in F.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000446, S26-D-00000505

S26-D-00000507 - Natural filtration

The natural filtration of X is F_t^X=sigma(X_s: s<=t).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000501, S26-D-00000505

S26-D-00000508 - Adapted process

X is adapted to (F_t) when X_t is F_t-measurable for every t.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000501, S26-D-00000505, S26-D-00000405

S26-D-00000509 - Stopping time

A random time tau is a stopping time when {tau<=t} belongs to F_t for every t.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000505, S26-D-00000451

S26-D-00000510 - Stopped process

For a process X and stopping time tau, the stopped process is X^tau_t=X_{t wedge tau}.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000501, S26-D-00000509

S26-D-00000511 - Sigma-algebra at a stopping time

F_tau consists of events A such that A intersection {tau<=t} belongs to F_t for every t.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000509, S26-D-00000401

S26-D-00000512 - Discrete-time martingale

An adapted integrable process (M_n) is a martingale when E[M_{n+1}|F_n]=M_n almost surely for every n.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000508, S26-D-00000476

S26-D-00000513 - Submartingale

An adapted integrable process is a submartingale when E[X_{n+1}|F_n]>=X_n almost surely.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000512

S26-D-00000514 - Supermartingale

An adapted integrable process is a supermartingale when E[X_{n+1}|F_n]<=X_n almost surely.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000512

S26-T-00000515 - Martingale expectation conservation

For an integrable martingale, E[M_n]=E[M_0] for every n.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000512, S26-T-00000477

S26-T-00000516 - Conditional-expectation martingale

If X is integrable and F_n is a filtration, M_n=E[X|F_n] is a martingale.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000476, S26-D-00000505, S26-D-00000512

S26-T-00000517 - Martingale transform by bounded predictable sequence

If M is a discrete-time martingale and H_n is bounded and F_{n-1}-measurable, then the stochastic sum sum_{k=1}^n H_k(M_k-M_{k-1}) is a martingale.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: H_n is bounded and predictable; all displayed terms are integrable.

Hard dependencies: S26-D-00000512, S26-T-00000478

S26-T-00000518 - Optional stopping: bounded stopping times

If M_n is an integrable martingale and tau is a bounded stopping time, then E[M_tau]=E[M_0].

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: tau is bounded.

Hard dependencies: S26-D-00000509, S26-D-00000512, S26-T-00000515

S26-T-00000519 - Optional stopping requires hypotheses

Optional-stopping identities are not valid for arbitrary unbounded stopping times without additional hypotheses.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-T-00000518

Boundary note: A fair random walk stopped at the first hit of +1 has an unbounded stopping time and naive unrestricted optional-stopping reasoning can fail integrability/uniform-integrability conditions.

Part AI - Markov chains, stationarity, and ergodic interfaces

S26-D-00000520 - Transition kernel

A Markov transition kernel K from (E,E) to (F,F) assigns to x a probability measure K(x,.) and to measurable B a measurable function x |-> K(x,B).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000403, S26-D-00000446

S26-D-00000521 - Markov chain

A discrete-time process is Markov with kernel K when P(X_{n+1} in A | F_n)=K(X_n,A) almost surely.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000501, S26-D-00000505, S26-D-00000520, S26-D-00000476

S26-D-00000522 - Transition matrix

For a countable-state Markov chain, P=(p_ij) with p_ij=P(X_{n+1}=j|X_n=i) is a row-stochastic transition matrix.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000521, S26-D-00000464, S26-D-00000176

S26-T-00000523 - Chapman-Kolmogorov equation

For a time-homogeneous Markov chain, n+m step transition kernels compose: P^{n+m}(i,j)=sum_k P^n(i,k)P^m(k,j).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000521, S26-D-00000522, S26-D-00000031

S26-D-00000524 - Stationary distribution

A probability distribution pi is stationary for transition matrix P when pi P=pi.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000522, S26-D-00000448

S26-T-00000525 - Stationary initialization

If X_0 has stationary law pi and the chain has kernel P, then every X_n has law pi.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000521, S26-D-00000524

S26-D-00000526 - Communicating states

States i,j communicate when each is reachable from the other with positive probability in some number of steps.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-T-00000523

S26-D-00000527 - Irreducible Markov chain

A Markov chain is irreducible when every pair of states communicates.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000526

S26-D-00000528 - Period of a state

The period of state i is gcd{n>=1: P^n(i,i)>0}.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-T-00000523, S26-T-00000119

S26-D-00000529 - Aperiodic chain

An irreducible chain is aperiodic when its states have period one.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000527, S26-D-00000528

S26-D-00000530 - Return time

The first return time to state i is T_i^+=inf{n>=1:X_n=i}.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000509, S26-D-00000521

S26-D-00000531 - Recurrent state

State i is recurrent when P_i(T_i^+<infinity)=1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000530

S26-D-00000532 - Transient state

State i is transient when P_i(T_i^+<infinity)<1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000530

S26-D-00000533 - Positive recurrent state

A recurrent state i is positive recurrent when E_i[T_i^+]<infinity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000531, S26-D-00000449

S26-T-00000534 - Finite irreducible chains are positive recurrent

Every state of a finite irreducible Markov chain is positive recurrent.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The state space is finite.

Hard dependencies: S26-D-00000527, S26-D-00000533

S26-T-00000535 - Finite irreducible stationary distribution

A finite irreducible Markov chain has a unique stationary distribution with strictly positive coordinates.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The state space is finite.

Hard dependencies: S26-D-00000524, S26-D-00000527, S26-T-00000534

S26-T-00000536 - Finite ergodic-chain convergence

For a finite irreducible aperiodic Markov chain with stationary distribution pi, P^n(i,j)->pi_j for all states i,j.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The state space is finite, irreducible, and aperiodic.

Hard dependencies: S26-D-00000529, S26-T-00000535

S26-D-00000537 - Strict stationarity

A process is strictly stationary when every finite-dimensional distribution is invariant under admissible time shifts.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000503

S26-D-00000538 - Weak stationarity

A second-order process is weakly stationary when its mean is constant and covariance depends only on time lag.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The process has finite second moments.

Hard dependencies: S26-D-00000470, S26-D-00000501

S26-T-00000539 - Strict stationarity plus second moments implies weak stationarity

A strictly stationary process with finite second moments is weakly stationary.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000537, S26-D-00000538

S26-T-00000540 - Weak stationarity need not imply strict stationarity

Second-order stationarity alone does not determine all finite-dimensional distributions.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000537, S26-D-00000538

Boundary note: Processes can share constant mean and lag covariance while having time-varying higher-order joint laws.

S26-D-00000541 - Measure-preserving transformation

A measurable T on a probability space is measure-preserving when P(T^{-1}A)=P(A) for every measurable A.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000446, S26-D-00000405

S26-D-00000542 - Invariant event

An event A is invariant under T when T^{-1}A=A modulo null sets.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000541, S26-D-00000410

S26-D-00000543 - Ergodic transformation

A measure-preserving transformation is ergodic when every invariant event has probability 0 or 1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000541, S26-D-00000542

S26-D-00000544 - Stationary process from a measure-preserving system

Given measure-preserving T and measurable f, X_n=f composed with T^n defines a stationary process.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000541, S26-D-00000537, S26-D-00000405

S26-T-00000545 - Birkhoff ergodic theorem interface

For measure-preserving T and f in L^1, time averages n^{-1}sum_{k=0}^{n-1} f(T^k omega) converge almost surely to E[f|I], where I is the invariant sigma-algebra.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: T is measure-preserving and f is integrable.

Hard dependencies: S26-D-00000476, S26-D-00000541, S26-D-00000542, S26-D-00000433

S26-T-00000546 - Ergodic specialization of Birkhoff

If T is ergodic and f in L^1, the Birkhoff time average converges almost surely to E[f].

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000543, S26-T-00000545

S26-D-00000547 - Hitting time

For measurable A, tau_A=inf{n>=0:X_n in A} is the hitting time of A.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000509, S26-D-00000501

S26-D-00000548 - First-passage probability

For states or measurable sets A,B, first-passage probabilities quantify reaching B before A or before a specified time.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000547, S26-D-00000451

S26-T-00000549 - Strong Markov property interface

For a Markov chain and an almost surely finite stopping time tau, the post-tau process conditionally restarts from X_tau with the same transition kernel.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: tau is a stopping time satisfying the standard finiteness/measurability hypotheses.

Hard dependencies: S26-D-00000509, S26-D-00000521

S26-T-00000550 - Stochastic-process foundation frontier

Filtrations, martingales, stopping times, Markov kernels/chains, stationarity, and ergodic interfaces form the finite/discrete stochastic-process substrate before continuous-time Brownian motion and stochastic integration.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All probability spaces, measurability, integrability, and time-index hypotheses are exactly as stated.

Hard dependencies: S26-D-00000512, S26-D-00000521, S26-D-00000537, S26-T-00000545

Applications: Time series | Queueing | Reliability | Finance | Statistical physics | Randomized algorithms

Part AJ - Brownian motion, Ito calculus, and stochastic differential equations

S26-D-00000551 - Brownian motion

A standard Brownian motion W has W_0=0 a.s., independent stationary Gaussian increments, and almost surely continuous paths.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Continuous time t>=0; standard real Brownian motion.

Hard dependencies: S26-D-00000501, S26-D-00000466

S26-T-00000552 - Brownian increment law

For 0<=s<t, W_t-W_s is N(0,t-s).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551

S26-T-00000553 - Brownian covariance

E[W_s W_t]=min(s,t).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-D-00000470

S26-D-00000554 - Gaussian process

A process is Gaussian when every finite-dimensional distribution is multivariate Gaussian.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000503, S26-D-00000466

S26-T-00000555 - Brownian motion is Gaussian

Standard Brownian motion is a centered Gaussian process with covariance min(s,t).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-D-00000554

S26-T-00000556 - Brownian scaling

For c>0, (c^{-1/2}W_{ct}) has the same finite-dimensional laws as Brownian motion.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-T-00000504

S26-T-00000557 - Brownian Markov property

Brownian motion is a time-homogeneous Markov process with Gaussian transition kernel.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-D-00000521

S26-T-00000558 - Brownian martingale property

Brownian motion adapted to its natural filtration is a martingale.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-D-00000507, S26-D-00000512

S26-D-00000559 - Continuous local martingale interface

A continuous local martingale is an adapted continuous process that becomes a martingale after a localizing sequence of stopping times.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000508, S26-D-00000509, S26-D-00000512

S26-D-00000560 - Partition quadratic variation

For a partition Pi of [0,t], the quadratic-variation sum is sum (X_{t_{i+1}}-X_{t_i})^2.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000501, S26-T-00000357

S26-T-00000561 - Brownian quadratic variation

Along refining partitions with mesh tending to zero, Brownian quadratic-variation sums converge in probability to t.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-D-00000560, S26-D-00000486

S26-T-00000562 - Brownian infinite variation

Brownian paths have infinite total variation on every nontrivial compact interval almost surely.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-D-00000410

Boundary note: Continuity does not imply bounded variation.

S26-D-00000563 - Predictable process interface

A predictable process is measurable with respect to the predictable sigma-algebra generated by adapted left-continuous processes.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000508, S26-D-00000401

S26-D-00000564 - Simple adapted integrand

A simple adapted integrand has form H_t=sum H_k 1_(t_k,t_{k+1}](t), with H_k measurable at time t_k.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000508, S26-D-00000405

S26-D-00000565 - Ito integral for simple processes

For simple adapted H, integral H dW is sum H_k(W_{t_{k+1}}-W_{t_k}).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-D-00000564

S26-T-00000566 - Ito isometry for simple processes

E[(integral_0^T H dW)^2]=E[integral_0^T H_t^2 dt] for square-integrable simple adapted H.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: H is simple, adapted, and square-integrable.

Hard dependencies: S26-D-00000565, S26-T-00000439

S26-D-00000567 - L2 Ito integral

The Ito integral for square-integrable predictable H is the L2 limit of integrals of simple predictable approximants.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000563, S26-T-00000566, S26-D-00000438

S26-T-00000568 - Ito integral has mean zero

For square-integrable predictable H, E[integral_0^T H dW]=0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000567, S26-D-00000449

S26-T-00000569 - Ito integral martingale

M_t=integral_0^t H_s dW_s is a continuous martingale under square-integrability.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000567, S26-D-00000512

S26-T-00000570 - Ito isometry

For square-integrable predictable H, E[(integral_0^T H dW)^2]=E[integral_0^T H_t^2 dt].

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000567, S26-T-00000566

S26-D-00000571 - Stochastic differential notation

dX_t=b_t dt+sigma_t dW_t abbreviates the corresponding integral equation, not an ordinary pointwise differential identity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000567, S26-T-00000398

Boundary note: Brownian paths are almost surely nowhere classically differentiable.

S26-T-00000572 - Ito formula one-dimensional

For dX_t=b_t dt+sigma_t dW_t and sufficiently smooth f, df(X_t)=f'(X_t)dX_t+(1/2)f''(X_t)sigma_t^2 dt.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: f is C^2 and coefficients satisfy conditions making all terms well-defined.

Hard dependencies: S26-D-00000571, S26-T-00000344

S26-T-00000573 - Ito correction term

The second-order term in Ito formula arises from nonzero quadratic variation: heuristically (dW_t)^2=dt.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-T-00000561, S26-T-00000572

S26-T-00000574 - Ito product rule

d(X_tY_t)=X_t dY_t+Y_t dX_t+d[X,Y]_t for continuous semimartingales.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-T-00000572, S26-D-00000560

S26-D-00000575 - Covariation interface

Quadratic covariation [X,Y] is obtained from limits of sums of paired increments along refining partitions.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000560

S26-T-00000576 - Polarization of covariation

When quadratic variations exist, [X,Y]=([X+Y]-[X-Y])/4.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000575, S26-T-00000216

S26-D-00000577 - Ito process

An Ito process has X_t=X_0+integral b_s ds+integral sigma_s dW_s under suitable integrability and adaptedness.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000567, S26-D-00000571

S26-D-00000578 - Stochastic differential equation

An SDE specifies an adapted process through an integral equation driven by a stochastic process, typically Brownian motion.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000577

S26-D-00000579 - Strong solution

A strong solution is adapted to the given driving filtration and is constructed on the given probability space with the given Brownian motion.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000578, S26-D-00000506

S26-D-00000580 - Weak solution

A weak solution allows the probability space and driving Brownian motion to be part of the solution.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000578, S26-D-00000446

S26-D-00000581 - Pathwise uniqueness

Pathwise uniqueness means two solutions on the same space with the same Brownian motion and initial condition agree almost surely.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000579, S26-D-00000410

S26-T-00000582 - Lipschitz SDE existence and uniqueness

Globally Lipschitz coefficients with linear growth give a unique strong solution to the standard Ito SDE.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Coefficients satisfy standard global Lipschitz and linear-growth bounds.

Hard dependencies: S26-D-00000578, S26-D-00000579, S26-D-00000581

S26-D-00000583 - Generator of an Ito diffusion

For dX=b(X)dt+sigma(X)dW, the generator acts as Lf=b·grad f+(1/2)Tr(sigma sigma^T Hess f).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000577, S26-T-00000341, S26-D-00000343

S26-T-00000584 - Dynkin formula interface

Under suitable regularity, E[f(X_tau)] = f(X_0)+E[integral_0^tau Lf(X_s)ds] for admissible stopping times.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Regularity, integrability, and stopping hypotheses required by Dynkin formula hold.

Hard dependencies: S26-D-00000583, S26-D-00000509

S26-T-00000585 - Kolmogorov backward equation interface

For a regular diffusion semigroup, u(t,x)=E_x[g(X_t)] formally satisfies partial_t u=L u with initial data g.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Sufficient regularity for the classical PDE statement.

Hard dependencies: S26-D-00000583, S26-D-00000449

S26-T-00000586 - Fokker-Planck interface

When a diffusion admits a sufficiently regular density, its density evolves under the adjoint generator equation.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: A sufficiently regular transition density exists.

Hard dependencies: S26-D-00000583, S26-T-00000467

S26-D-00000587 - Ornstein-Uhlenbeck process

The Ornstein-Uhlenbeck process solves dX_t=-theta X_t dt+sigma dW_t.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000578

S26-T-00000588 - Ornstein-Uhlenbeck explicit solution

For constant theta,sigma, X_t=e^{-theta t}X_0+sigma integral_0^t e^{-theta(t-s)}dW_s.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000587, S26-D-00000567

S26-D-00000589 - Geometric Brownian motion

Geometric Brownian motion solves dS_t=mu S_t dt+sigma S_t dW_t.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000578

S26-T-00000590 - Geometric Brownian explicit solution

S_t=S_0 exp((mu-sigma^2/2)t+sigma W_t).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000589, S26-T-00000572

S26-D-00000591 - Exponential martingale

For constant lambda, exp(lambda W_t-lambda^2 t/2) is the canonical Brownian exponential martingale.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-T-00000572

S26-T-00000592 - Exponential martingale property

The Brownian exponential process is a martingale with expectation one.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000591, S26-T-00000569

S26-D-00000593 - Stochastic integral versus pathwise integral

The Ito integral is an L2/probabilistic construction and generally is not a Riemann-Stieltjes integral against Brownian paths.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000567, S26-T-00000389

Boundary note: Brownian paths have unbounded variation almost surely.

S26-T-00000594 - Adaptedness boundary for Ito integration

The elementary Ito construction requires non-anticipating integrands; future-dependent integrands are outside this definition.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000563, S26-D-00000567

Boundary note: An integrand depending on W_T at times t<T is generally anticipative.

S26-T-00000595 - Ito-Stratonovich distinction interface

Ito and Stratonovich stochastic integrals use different limiting conventions and differ by a quadratic-covariation correction.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-T-00000573, S26-D-00000575

S26-D-00000596 - Multidimensional Brownian motion

A d-dimensional Brownian motion has independent standard Brownian coordinate processes.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-D-00000465

S26-T-00000597 - Multidimensional Ito formula interface

For a sufficiently smooth scalar f and an Ito diffusion in R^d, Ito formula uses the gradient and Hessian with covariance matrix sigma sigma^T.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000596, S26-D-00000583, S26-D-00000343

S26-D-00000598 - Stochastic flow interface

Under stronger regularity, solutions indexed by initial state may define a random flow of maps.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Coefficient regularity sufficient for the stated flow property.

Hard dependencies: S26-T-00000582, S26-D-00000178

S26-T-00000599 - Stochastic-calculus hypothesis fence

Stochastic differential notation does not license ordinary calculus: adaptedness, integrability, quadratic variation, and the chosen integral convention remain load-bearing.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000571, S26-T-00000572, S26-T-00000594, S26-T-00000595

S26-T-00000600 - Stochastic-calculus frontier

Brownian motion, quadratic variation, Ito integration, Ito formula, and basic SDEs form the continuous stochastic-calculus substrate before advanced martingale theory, stochastic control, SPDEs, and mathematical finance.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All filtration, measurability, integrability, and time-domain hypotheses are exactly as stated.

Hard dependencies: S26-D-00000551, S26-D-00000567, S26-T-00000572, S26-T-00000582

Applications: Diffusion modeling | Filtering | Stochastic control | Mathematical finance

Part AK - Banach spaces, operators, duality, Hilbert spaces, and compactness

S26-D-00000601 - Normed vector space

A normed vector space is a vector space X over R or C equipped with a norm ||.|| satisfying definiteness, absolute homogeneity, and the triangle inequality.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000149, S26-D-00000202

S26-T-00000602 - Norm metric

Every norm induces the metric d(x,y)=||x-y||.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000601, S26-D-00000301

S26-D-00000603 - Equivalent norms

Two norms on the same vector space are equivalent when each bounds the other up to positive constants; equivalent norms induce the same topology.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000601, S26-D-00000315

S26-T-00000604 - Finite-dimensional norm equivalence

All norms on a finite-dimensional vector space over R or C are equivalent.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The vector space is finite-dimensional over R or C.

Hard dependencies: S26-T-00000164, S26-D-00000603

Boundary note: Infinite-dimensional normed spaces can carry inequivalent norms.

S26-D-00000605 - Banach space

A Banach space is a complete normed vector space.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000601, S26-T-00000327

S26-D-00000606 - Bounded linear operator

A linear map T:X->Y between normed spaces is bounded when ||Tx||<=C||x|| for some C>=0 and all x.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000166, S26-D-00000601

S26-T-00000607 - Bounded iff continuous linear

A linear map between normed spaces is continuous iff it is bounded.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000606, S26-T-00000317

S26-D-00000608 - Operator norm

For bounded linear T:X->Y, ||T||=sup_{||x||<=1}||Tx||.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000606, S26-D-00000210

S26-T-00000609 - Operator norm inequality

For bounded linear T, ||Tx||<=||T|| ||x||.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000608

S26-T-00000610 - Submultiplicativity of operator norm

For bounded composable operators S,T, ||ST||<=||S|| ||T||.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000608, S26-D-00000171

S26-T-00000611 - Bounded operators form a normed space

B(X,Y), the bounded linear maps X->Y, is a normed vector space under the operator norm.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000608, S26-D-00000601

S26-T-00000612 - Completeness of operator space

If Y is Banach, then B(X,Y) is Banach for every normed X.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Y is Banach.

Hard dependencies: S26-T-00000611, S26-D-00000605

S26-D-00000613 - Continuous dual

The continuous dual X* is B(X,F), the bounded linear functionals from X to its scalar field F.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000611, S26-D-00000251

S26-T-00000614 - Dual is Banach

The continuous dual X* of every normed space is Banach.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000612, S26-D-00000613

S26-D-00000615 - Annihilator

For M subset X, M^perp={f in X*: f(m)=0 for all m in M}.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000613, S26-D-00000156

S26-T-00000616 - Hahn-Banach extension theorem

A bounded linear functional on a subspace of a real or complex normed space extends to the whole space without increasing its norm.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Standard real or complex Hahn-Banach hypotheses.

Hard dependencies: S26-D-00000613, S26-D-00000156

S26-T-00000617 - Dual separates points

For nonzero x in a normed space, there exists f in X* with ||f||=1 and f(x)=||x||.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: X is a normed space over R or C.

Hard dependencies: S26-T-00000616

S26-T-00000618 - Norm recovered from dual

For x in a normed space, ||x||=sup{|f(x)|: f in X*, ||f||<=1}.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000617, S26-D-00000608

S26-D-00000619 - Pointwise bounded operator family

A family F subset B(X,Y) is pointwise bounded when sup_{T in F}||Tx||<infinity for every x in X.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000611

S26-T-00000620 - Uniform boundedness principle

If X is Banach and a family F subset B(X,Y) is pointwise bounded, then sup_{T in F}||T||<infinity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: X is Banach.

Hard dependencies: S26-D-00000605, S26-D-00000619

Boundary note: Completeness of the domain is load-bearing in the standard theorem.

S26-T-00000621 - Open mapping theorem

A surjective bounded linear map between Banach spaces is an open map.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain and codomain are Banach; the operator is bounded, linear, and surjective.

Hard dependencies: S26-D-00000605, S26-D-00000606, S26-T-00000322

S26-T-00000622 - Bounded inverse theorem

A bijective bounded linear map between Banach spaces has bounded inverse.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain and codomain are Banach.

Hard dependencies: S26-T-00000621, S26-D-00000172

S26-D-00000623 - Closed operator graph

The graph of T:X->Y is {(x,Tx):x in X}; T is closed when this graph is closed in X x Y.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000166, S26-D-00000308, S26-T-00000305

S26-T-00000624 - Closed graph theorem

A linear operator T:X->Y between Banach spaces is bounded iff its graph is closed.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: X and Y are Banach and T is defined on all of X.

Hard dependencies: S26-D-00000623, S26-D-00000605, S26-D-00000606

S26-D-00000625 - Weak topology

The weak topology sigma(X,X*) is the coarsest topology making every continuous linear functional f in X* continuous.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000613, S26-D-00000306

S26-D-00000626 - Weak convergence

x_n converges weakly to x when f(x_n)->f(x) for every f in X*.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000625, S26-D-00000351

S26-T-00000627 - Norm convergence implies weak convergence

If x_n->x in norm, then x_n->x weakly.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000607, S26-D-00000626

S26-T-00000628 - Weak convergence need not imply norm convergence

In infinite-dimensional normed spaces, weak convergence can occur without norm convergence.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000626

Boundary note: The standard basis e_n in l2 converges weakly to 0 but ||e_n||=1.

S26-D-00000629 - Weak-star topology

The weak-star topology sigma(X*,X) is pointwise convergence of functionals on X.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000613, S26-D-00000306

S26-T-00000630 - Banach-Alaoglu theorem

The closed unit ball of X* is compact in the weak-star topology.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Compactness is with respect to the weak-star topology, not generally the norm topology.

Hard dependencies: S26-D-00000629, S26-T-00000323

S26-D-00000631 - Hilbert space

A Hilbert space is a complete inner-product space with norm induced by its inner product.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000201, S26-D-00000605

S26-T-00000632 - Parallelogram law

A norm induced by an inner product satisfies ||x+y||^2+||x-y||^2=2||x||^2+2||y||^2.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000631, S26-T-00000216

S26-T-00000633 - Jordan-von Neumann interface

A norm satisfying the parallelogram law arises from a unique compatible inner product via polarization.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Real or complex normed space with the appropriate polarization identity.

Hard dependencies: S26-T-00000632, S26-T-00000216

S26-D-00000634 - Orthogonal complement in Hilbert space

For M subset H, M^perp={x in H:<x,m>=0 for every m in M}.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000631, S26-T-00000211

S26-T-00000635 - Closedness of orthogonal complement

M^perp is a closed linear subspace of H.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000634, S26-D-00000308

S26-T-00000636 - Hilbert projection theorem

For a nonempty closed convex subset C of a Hilbert space, every x has a unique nearest point in C.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: C is nonempty, closed, and convex.

Hard dependencies: S26-D-00000631, S26-T-00000349, S26-D-00000308

S26-T-00000637 - Orthogonal projection onto closed subspace

For a closed subspace M of H, each x uniquely decomposes x=P_M x+(x-P_Mx) with P_Mx in M and x-P_Mx in M^perp.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000636, S26-D-00000634

S26-T-00000638 - Orthogonal decomposition

For a closed subspace M of H, H=M direct-sum M^perp.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000637, S26-D-00000158

S26-T-00000639 - Projection operator properties

The orthogonal projection P_M onto a closed subspace is bounded, self-adjoint, idempotent, and has norm <=1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000637, S26-T-00000226, S26-D-00000608

S26-T-00000640 - Riesz representation theorem for Hilbert spaces

Every bounded linear functional f on a Hilbert space H has a unique representing vector y with f(x)=<x,y> for all x, up to the chosen inner-product convention.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: H is a real or complex Hilbert space; conjugate-linearity convention is fixed.

Hard dependencies: S26-D-00000631, S26-D-00000613

S26-T-00000641 - Hilbert space self-duality

The Riesz map identifies a Hilbert space isometrically with its continuous dual, conjugate-linearly in the standard complex convention.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000640, S26-T-00000173

S26-D-00000642 - Adjoint of bounded Hilbert operator

For bounded T:H->K between Hilbert spaces, T* is the unique bounded operator satisfying <Tx,y>=<x,T*y>.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000640, S26-D-00000222, S26-D-00000606

S26-T-00000643 - Adjoint norm equality

For bounded Hilbert-space operators, ||T*||=||T|| and ||T*T||=||T||^2.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000642, S26-D-00000608

S26-D-00000644 - Compact operator

A bounded operator T:X->Y is compact when it maps bounded sets to relatively compact sets.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000606, S26-T-00000323

S26-T-00000645 - Finite-rank operators are compact

Every bounded finite-rank operator between normed spaces is compact.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000644, S26-T-00000170, S26-T-00000164

S26-T-00000646 - Compact operators form an operator-norm closed ideal

Compact operators are closed under addition and operator-norm limits, and composition with bounded operators preserves compactness.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Operators are between compatible normed spaces.

Hard dependencies: S26-D-00000644, S26-T-00000610

S26-T-00000647 - Compact self-adjoint spectral theorem interface

A compact self-adjoint operator on a Hilbert space admits an orthonormal eigenvector decomposition on the closure of its range, with nonzero eigenvalues of finite multiplicity accumulating only at zero.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Operator is compact and self-adjoint on a Hilbert space.

Hard dependencies: S26-D-00000644, S26-T-00000225, S26-D-00000631

S26-T-00000648 - Infinite-dimensional compactness fence

Closed and bounded subsets of infinite-dimensional normed spaces need not be compact.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000323, S26-D-00000601

Boundary note: The closed unit ball of an infinite-dimensional Hilbert space is not norm-compact.

S26-T-00000649 - Functional-analysis hypothesis fence

Finite-dimensional intuitions do not automatically extend: completeness, topology choice, closedness, boundedness, and compactness hypotheses remain load-bearing.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000604, S26-T-00000620, S26-T-00000621, S26-T-00000630, S26-T-00000648

S26-T-00000650 - Functional-analysis frontier

Normed and Banach spaces, bounded operators, duality principles, weak topologies, Hilbert projection/Riesz representation, and compact operators form the substrate for infinite-dimensional spectral theory, distributions, PDE, and operator theory.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000605, S26-T-00000616, S26-T-00000620, S26-D-00000631, S26-T-00000640, S26-D-00000644

Applications: PDE | Optimization | Quantum theory | Numerical analysis

Part AL - Spectrum, unbounded operators, distributions, and Sobolev analysis

S26-D-00000651 - Spectrum of a bounded operator

For T in B(X) on a complex Banach space, the spectrum sigma(T) is the set of lambda in C for which T-lambda I is not boundedly invertible.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: X is a complex Banach space and T is bounded.

Hard dependencies: S26-T-00000622, S26-D-00000606

S26-D-00000652 - Resolvent set and resolvent operator

The resolvent set rho(T)=C\sigma(T); for lambda in rho(T), R(lambda,T)=(lambda I-T)^(-1).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000651

S26-T-00000653 - Resolvent identity

For lambda,mu in rho(T), R(lambda,T)-R(mu,T)=(mu-lambda)R(lambda,T)R(mu,T).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000652

S26-T-00000654 - Spectrum nonempty compact

The spectrum of a bounded operator on a nonzero complex Banach space is a nonempty compact subset of C.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: X is a nonzero complex Banach space; T is bounded.

Hard dependencies: S26-D-00000651, S26-T-00000323

S26-D-00000655 - Spectral radius

The spectral radius is r(T)=sup{|lambda|:lambda in sigma(T)}.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000651

S26-T-00000656 - Spectral radius formula

For T in B(X), r(T)=lim_{n->infinity} ||T^n||^(1/n)=inf_n ||T^n||^(1/n).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: X is a complex Banach space and T is bounded.

Hard dependencies: S26-D-00000655, S26-T-00000610

S26-T-00000657 - Neumann series inverse

If ||T||<1 on a Banach space, I-T is invertible with (I-T)^(-1)=sum_{n>=0}T^n in operator norm.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000605, S26-D-00000608, S26-D-00000356

S26-T-00000658 - Resolvent outside norm disk

If |lambda|>||T||, then lambda belongs to rho(T), obtained by a Neumann-series expansion.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000657, S26-D-00000652

S26-D-00000659 - Point spectrum

The point spectrum consists of eigenvalues: lambda for which T-lambda I is not injective.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000651, S26-D-00000197

S26-T-00000660 - Compact operator nonzero spectrum

For a compact operator on an infinite-dimensional complex Banach space, every nonzero spectral value is an eigenvalue of finite algebraic multiplicity; zero is the only possible accumulation point.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: T is compact on an infinite-dimensional complex Banach space.

Hard dependencies: S26-D-00000644, S26-D-00000651

S26-D-00000661 - Unbounded linear operator

An unbounded-operator datum is a linear map A:D(A) subset X -> Y together with its specified domain D(A).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000166, S26-D-00000156

S26-D-00000662 - Densely defined operator

A:D(A) subset X->Y is densely defined when the closure of D(A) equals X.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000661, S26-D-00000308

S26-D-00000663 - Closed unbounded operator

A is closed when its graph is closed in X x Y.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000661, S26-D-00000623

S26-D-00000664 - Closable operator

A is closable when the closure of its graph is the graph of an operator, called its closure.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000663, S26-D-00000308

S26-D-00000665 - Adjoint of a densely defined operator

For densely defined A:D(A) subset H->K, y lies in D(A*) when x -> <Ax,y> is bounded on D(A) in the H norm; A*y is its Riesz representing vector.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: A is densely defined between Hilbert spaces.

Hard dependencies: S26-D-00000662, S26-T-00000640

S26-T-00000666 - Adjoint is closed

The adjoint A* of a densely defined operator is closed.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000665, S26-D-00000663

S26-D-00000667 - Symmetric operator

A densely defined operator A on H is symmetric when <Ax,y>=<x,Ay> for all x,y in D(A), equivalently A is contained in A*.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000665

S26-D-00000668 - Self-adjoint unbounded operator

A is self-adjoint when A=A*, including equality of domains.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000665

Boundary note: Equality of the differential expressions without equality of domains does not establish self-adjointness.

S26-T-00000669 - Self-adjoint implies symmetric

Every self-adjoint operator is symmetric.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000667, S26-D-00000668

S26-T-00000670 - Symmetric need not be self-adjoint

A symmetric densely defined operator need not be self-adjoint; domain and deficiency data can obstruct equality with its adjoint.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000667, S26-D-00000668

Boundary note: Minimal differential operators can be symmetric but require boundary conditions for self-adjoint extensions.

S26-D-00000671 - Positive self-adjoint operator

A self-adjoint operator A is positive when <Ax,x>>=0 for every x in D(A).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000668, S26-T-00000231

S26-T-00000672 - Spectral theorem for bounded self-adjoint operators

A bounded self-adjoint operator on a complex Hilbert space admits a projection-valued spectral representation T=integral lambda dE(lambda).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: T is bounded and self-adjoint on a complex Hilbert space.

Hard dependencies: S26-T-00000225, S26-D-00000631

S26-D-00000673 - Projection-valued measure

A projection-valued measure assigns orthogonal projections E(B) to measurable sets B with countable additivity in the strong operator topology and the usual measure identities.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000434, S26-T-00000639

S26-T-00000674 - Borel functional calculus interface

For bounded self-adjoint T, bounded Borel functions f define operators f(T)=integral f(lambda)dE(lambda), compatible with algebraic operations.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000672, S26-D-00000673

S26-T-00000675 - Continuous functional calculus interface

For bounded normal T, continuous functions on sigma(T) define f(T) through the continuous functional calculus.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: T is bounded normal on a complex Hilbert space.

Hard dependencies: S26-D-00000227, S26-D-00000651

S26-T-00000676 - Spectral theorem for unbounded self-adjoint operators

A self-adjoint operator admits a projection-valued spectral representation A=integral lambda dE(lambda), with its domain determined by square-integrability of lambda against the spectral measures.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: A is self-adjoint on a complex Hilbert space.

Hard dependencies: S26-D-00000668, S26-D-00000673

S26-D-00000677 - Test-function space

D(Omega)=C_c^infinity(Omega) is the space of smooth compactly supported test functions on an open set Omega subset R^n.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000335, S26-T-00000323

S26-D-00000678 - Distribution

A distribution on Omega is a continuous linear functional on D(Omega) with its standard locally convex topology.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000677, S26-D-00000613

S26-D-00000679 - Regular distribution

A locally integrable function f defines the distribution phi -> integral_Omega f phi.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000678, S26-D-00000429

S26-D-00000680 - Distributional derivative

For a distribution T, its derivative is defined by <partial_i T,phi>=-<T,partial_i phi>.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000678, S26-T-00000337

S26-T-00000681 - Distributional integration by parts

Distributional differentiation transfers derivatives to test functions with the sign dictated by integration by parts.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000680, S26-T-00000399

S26-D-00000682 - Dirac distribution

At x0 in Omega, delta_x0(phi)=phi(x0) defines a distribution.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000678

S26-T-00000683 - Heaviside derivative

The distributional derivative of the Heaviside step function is the Dirac distribution at zero.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000679, S26-D-00000680, S26-D-00000682

S26-D-00000684 - Weak derivative

A locally integrable v is the weak derivative partial_i u when integral u partial_i phi = - integral v phi for all test functions phi.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000677, S26-D-00000429, S26-D-00000680

S26-D-00000685 - Sobolev space W^{k,p}

W^{k,p}(Omega) consists of L^p functions whose weak derivatives through order k belong to L^p.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000440, S26-D-00000684

S26-D-00000686 - H^k Sobolev space

H^k(Omega)=W^{k,2}(Omega) is the Hilbertian Sobolev scale.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000685, S26-T-00000442

S26-T-00000687 - W^{k,p} completeness

Sobolev spaces W^{k,p}(Omega) are Banach for 1<=p<=infinity under the standard Sobolev norm.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000685, S26-D-00000605

S26-T-00000688 - H^k Hilbert structure

H^k(Omega) is a Hilbert space with the standard derivative inner product.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000686, S26-D-00000631

S26-D-00000689 - Weak formulation of Poisson equation

A weak Dirichlet solution of -Delta u=f seeks u in H_0^1(Omega) satisfying integral grad u dot grad v = integral f v for all admissible test v.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: The domain and data are such that the displayed pairings are well-defined.

Hard dependencies: S26-D-00000686, S26-T-00000341, S26-T-00000399

S26-D-00000690 - Bounded bilinear form on Hilbert space

A bilinear or sesquilinear form a on H is bounded when |a(u,v)|<=C||u||||v||.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000254, S26-D-00000631

S26-D-00000691 - Coercive bilinear form

A form a on H is coercive when Re a(u,u)>=alpha||u||^2 for some alpha>0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000690

S26-T-00000692 - Lax-Milgram theorem

A bounded coercive sesquilinear form on a Hilbert space represents each bounded anti/linear functional by a unique solution u, with the standard norm bound.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: H is Hilbert; the form is bounded and coercive; scalar convention is fixed.

Hard dependencies: S26-D-00000690, S26-D-00000691, S26-T-00000640

S26-T-00000693 - Poisson weak-solution interface

When the Dirichlet energy form is bounded and coercive on H_0^1(Omega), Lax-Milgram yields existence and uniqueness of the weak Poisson solution for admissible forcing.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All domain regularity, Poincare/coercivity, and forcing-duality hypotheses needed for the stated weak problem hold.

Hard dependencies: S26-D-00000689, S26-T-00000692

S26-D-00000694 - Operator domain fence

For unbounded operators, the domain is part of the operator; changing boundary conditions can change symmetry, adjoints, self-adjointness, and spectrum.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000661, S26-D-00000665, S26-D-00000668

S26-D-00000695 - Spectrum-type fence

Point spectrum, continuous spectrum, residual spectrum, and resolvent behavior must not be collapsed into the finite-dimensional notion of eigenvalues.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000651, S26-D-00000659

S26-D-00000696 - Weak-versus-classical derivative fence

A weak or distributional derivative need not arise from pointwise classical differentiability.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000680, S26-D-00000684, S26-D-00000335

Boundary note: The absolute-value function has a weak first derivative despite failure of classical differentiability at zero.

S26-D-00000697 - Functional-calculus hypothesis fence

Functional calculi depend on operator class and function class; polynomial, continuous, Borel, holomorphic, bounded and unbounded calculi are not interchangeable.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000674, S26-T-00000675, S26-T-00000676

S26-T-00000698 - Spectral-PDE bridge

Self-adjoint operator theory and Sobolev weak formulations provide complementary spectral and variational languages for differential operators.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000676, S26-D-00000685, S26-T-00000692

S26-T-00000699 - Unbounded-operator epistemic fence

Finite-dimensional matrix identities do not by themselves validate claims about unbounded operators: domains, closure, density, topology, and spectral type are load-bearing.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-D-00000661, S26-D-00000663, S26-D-00000668, S26-D-00000695

S26-T-00000700 - Operator-analysis frontier

Spectrum/resolvent theory, unbounded self-adjoint operators, distributions, Sobolev spaces, and variational existence principles form the substrate for PDE, harmonic analysis, quantum mechanics, and geometric analysis.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: All scalar-field, norm, completeness, topology, and continuity hypotheses are exactly as stated.

Hard dependencies: S26-T-00000656, S26-T-00000676, S26-D-00000678, S26-D-00000685, S26-T-00000692

Applications: PDE | Harmonic analysis | Quantum mechanics | Geometric analysis

Part AM - Fourier and harmonic analysis

S26-D-00000701 - Fourier coefficient

For f in L1(T), the nth Fourier coefficient is fhat(n)=(1/2pi) integral_0^{2pi} f(x)e^{-inx} dx.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000429

S26-D-00000702 - Fourier series

The formal Fourier series of f is sum_{n in Z} fhat(n)e^{inx}.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000701

S26-T-00000703 - Bessel inequality

For f in L2(T), sum_n |fhat(n)|^2 <= ||f||_2^2 under the normalized convention.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000442, S26-D-00000701

S26-T-00000704 - Parseval identity on the circle

For f in L2(T), the Fourier coefficients satisfy Parseval equality with the chosen normalization.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000703

S26-T-00000705 - L2 Fourier convergence

Partial Fourier sums converge to f in L2(T).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000704

S26-D-00000706 - Dirichlet kernel

The Nth Dirichlet kernel D_N is sum_{|n|<=N}e^{inx}, and Fourier partial sums are convolution with D_N.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000702, S26-D-00000711

Correction history: 0.1.1: Global closure audit: forward hard dependency on convolution entry 711 is intentional and fully discharged; numbering is editorial, dependency order is topological.

S26-D-00000707 - Fejer kernel

The Fejer kernel is the Cesaro average of Dirichlet kernels and is nonnegative with normalized integral one.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000706

S26-T-00000708 - Fejer theorem

For continuous periodic f, its Cesaro Fourier means converge uniformly to f.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000707, S26-T-00000366

S26-D-00000709 - Fourier transform on R^n

For f in L1(R^n), define Ff(xi)=integral f(x)e^{-2pi i x dot xi} dx under the fixed convention.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000429, S26-I-00000060

S26-D-00000710 - Inverse Fourier transform convention

The inverse transform uses the dual exponential sign and normalization paired with entry 709.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709

S26-D-00000711 - Convolution

For suitable f,g on R^n, (f*g)(x)=integral f(x-y)g(y)dy.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000429

S26-T-00000712 - Young L1 convolution bound

If f,g in L1, then f*g in L1 and ||f*g||_1<=||f||_1||g||_1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000711, S26-T-00000440

S26-T-00000713 - Convolution theorem

For f,g in L1, F(f*g)=F(f)F(g) under the fixed Fourier convention.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709, S26-D-00000711

S26-T-00000714 - Product-transform interface

Under hypotheses making both sides meaningful, Fourier transform converts products into convolution with convention-dependent constants.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709, S26-D-00000711

S26-T-00000715 - Riemann-Lebesgue lemma

If f in L1(R^n), then Ff is continuous and vanishes at infinity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709

S26-D-00000716 - Schwartz space

S(R^n) consists of smooth functions whose derivatives decay faster than every polynomial power.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000335

S26-T-00000717 - Fourier invariance of Schwartz space

The Fourier transform is an automorphism of S(R^n).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709, S26-D-00000716

S26-T-00000718 - Fourier inversion on Schwartz space

For f in S(R^n), inverse Fourier transform recovers f pointwise under the paired convention.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000710, S26-T-00000717

S26-T-00000719 - Plancherel theorem

The Fourier transform extends uniquely from S cap L2 to a unitary operator on L2(R^n), under unitary normalization.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000442, S26-T-00000717

S26-T-00000720 - L2 Fourier inversion

For f in L2, Fourier inversion holds in the L2 sense through the Plancherel extension.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000719

S26-D-00000721 - Approximate identity

A family phi_epsilon is an approximate identity when mass is normalized and concentrates near the identity while remaining uniformly L1-controlled.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000711, S26-T-00000440

S26-T-00000722 - Approximate-identity convergence

For 1<=p<infinity, convolution with a standard approximate identity converges to f in Lp.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000721, S26-T-00000440

S26-D-00000723 - Mollifier

A mollifier is a smooth compactly supported approximate identity obtained by rescaling a unit-mass test function.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000677, S26-D-00000721

S26-T-00000724 - Mollification regularizes

Convolution with a mollifier produces smooth functions and approximates Lp functions in the appropriate norm.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000723, S26-T-00000722

S26-D-00000725 - Tempered distribution

A tempered distribution is a continuous linear functional on Schwartz space S(R^n).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000716, S26-D-00000678

S26-T-00000726 - Fourier transform of tempered distributions

Fourier transform extends to S-prime by duality: <FT,phi>=<T,Fphi> up to the fixed convention.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000725, S26-T-00000717

S26-D-00000727 - Fourier transform of Dirac

The Fourier transform of delta_0 is the constant distribution 1 under the standard convention.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000682, S26-T-00000726

S26-T-00000728 - Differentiation-multiplication duality

Fourier transformation converts distributional differentiation into multiplication by the frequency coordinate, with convention-dependent factors.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000680, S26-T-00000726

S26-T-00000729 - Multiplication-differentiation duality

Multiplication by x_j corresponds under Fourier transform to differentiation in xi_j, with convention-dependent factors.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000726

S26-D-00000730 - Translation operator

Translation tau_a f(x)=f(x-a) acts compatibly with Fourier modulation.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709

S26-D-00000731 - Modulation operator

Modulation M_eta f(x)=e^{2pi i eta dot x}f(x) acts compatibly with Fourier translation.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709

S26-T-00000732 - Translation-modulation covariance

Fourier transform exchanges translations and modulations according to the fixed phase convention.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000730, S26-D-00000731

S26-D-00000733 - Dilation operator

Dilation rescales spatial variables and induces reciprocal frequency scaling with the Jacobian factor.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709

S26-T-00000734 - Fourier scaling law

The Fourier transform of f(ax) is the reciprocally rescaled transform with determinant/Jacobian factor.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000733

S26-D-00000735 - Gaussian Fourier interface

Gaussians transform to Gaussians under compatible normalization.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709

S26-T-00000736 - Heisenberg uncertainty inequality

For suitably normalized f with x f and grad f square-integrable, spatial and frequency dispersions obey a positive lower bound.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000719, S26-T-00000728

S26-D-00000737 - Fourier multiplier

A Fourier multiplier T_m acts formally by F(T_m f)=m Ff on a specified function space.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709

S26-T-00000738 - L2 multiplier bound

If m is essentially bounded, the associated Fourier multiplier is bounded on L2 with norm at most ||m||_infinity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000719, S26-D-00000737

S26-D-00000739 - Fractional differentiation multiplier

Fractional powers of |D| are represented formally by multipliers |xi|^s on suitable domains.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000737, S26-D-00000685

S26-T-00000740 - Sobolev Fourier characterization

On R^n, H^s can be characterized by weighted square-integrability of the Fourier transform with weight comparable to (1+|xi|^2)^s.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000686, S26-T-00000719

S26-D-00000741 - Fundamental solution interface

A fundamental solution E of a constant-coefficient operator P(D) satisfies P(D)E=delta in distributions.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000680, S26-D-00000682

S26-T-00000742 - Fourier-symbol PDE reduction

For constant-coefficient differential operators, Fourier transform converts P(D)u=f into multiplication by its symbol P(2pi i xi), subject to distributional/domain hypotheses.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000726, S26-T-00000728, S26-D-00000741

S26-D-00000743 - Laplacian Fourier symbol

Under the 2pi convention, F(-Delta f)(xi)=4pi^2|xi|^2 Ff(xi) for Schwartz f.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000341, S26-T-00000728, S26-D-00000716

S26-T-00000744 - Heat semigroup Fourier solution

For suitable initial data, the heat equation Fourier transform evolves by multiplication with exp(-4pi^2 t|xi|^2).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000743, S26-T-00000719

S26-T-00000745 - Poisson equation Fourier interface

On R^n, solving -Delta u=f formally corresponds to division by 4pi^2|xi|^2 away from zero, with low-frequency and distributional issues handled explicitly.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000743, S26-T-00000742

S26-D-00000746 - Fourier normalization fence

All constants, inverse formulas, differentiation factors, and Plancherel identities depend on the declared Fourier normalization.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000709, S26-D-00000710

S26-D-00000747 - Convergence-mode fence for Fourier series

L2, pointwise, uniform, Cesaro, and almost-everywhere convergence of Fourier expansions are distinct claims and require distinct hypotheses.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000705, S26-T-00000708, S26-D-00000365

S26-D-00000748 - Distribution-versus-function fence

A tempered distribution need not be represented by an ordinary function; algebraic operations require their own domains of definition.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-D-00000725, S26-D-00000679

S26-T-00000749 - Harmonic-analysis epistemic fence

Finite FFT experiments or smooth examples do not prove general Fourier inversion, multiplier, uncertainty, or PDE theorems.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000719, S26-T-00000736, S26-T-00000742

S26-T-00000750 - Harmonic-analysis frontier

Fourier series, transforms, convolution, Schwartz/tempered distributions, multipliers, Sobolev frequency descriptions, and PDE symbols form the substrate for systematic PDE and harmonic analysis.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Normalization, scalar field, topology, and integrability hypotheses are exactly as stated.

Hard dependencies: S26-T-00000708, S26-T-00000719, S26-T-00000726, S26-T-00000740, S26-T-00000742

Applications: PDE | Signal analysis | Quantum mechanics | Harmonic analysis

Part AN - Partial differential equations

S26-D-00000751 - Partial differential equation

A PDE is a relation involving an unknown function of several variables and some of its partial derivatives.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000336

S26-D-00000752 - Linear PDE

A PDE is linear when the unknown function and its derivatives enter through a linear differential operator.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000751, S26-T-00000226

S26-D-00000753 - Semilinear PDE

A semilinear PDE has a linear highest-order part while lower-order terms may depend nonlinearly on the unknown.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000752

S26-D-00000754 - Quasilinear PDE

A quasilinear PDE is linear in its highest-order derivatives but may have coefficients depending on lower-order derivatives and the unknown.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000751

S26-D-00000755 - Fully nonlinear PDE

A fully nonlinear PDE may depend nonlinearly on its highest-order derivatives.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000751

S26-D-00000756 - Principal symbol

The principal symbol records the highest-order part of a differential operator as a homogeneous polynomial in the cotangent/frequency variable.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-T-00000742

S26-D-00000757 - Second-order principal matrix

For a scalar second-order operator, the symmetric coefficient matrix of second derivatives determines the principal quadratic form.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000756, S26-D-00000239

S26-D-00000758 - Elliptic operator

A second-order scalar operator is elliptic at a point when its principal quadratic form is definite there; uniform ellipticity requires quantitative bounds.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000757

S26-D-00000759 - Parabolic prototype

The heat operator partial_t-Delta is the canonical parabolic prototype; general parabolicity requires a time-space structure and quantitative ellipticity in spatial variables.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-T-00000744, S26-D-00000758

S26-D-00000760 - Hyperbolic prototype

The wave operator partial_t^2-Delta is the canonical hyperbolic prototype, with finite-speed propagation under the standard constant-coefficient model.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000743, S26-D-00000751

S26-D-00000761 - Boundary value problem

A boundary value problem couples a PDE in a domain with conditions prescribed on its boundary.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000751, S26-D-00000306

S26-D-00000762 - Initial value problem

An initial value problem couples an evolution PDE with data prescribed at an initial time.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000751, S26-T-00000346

S26-D-00000763 - Dirichlet boundary condition

A Dirichlet condition prescribes the trace/value of the solution on the boundary where that trace is defined.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000761, S26-T-00000687

S26-D-00000764 - Neumann boundary condition

A Neumann condition prescribes an appropriate normal derivative or flux on the boundary, requiring sufficient boundary and trace structure.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000761, S26-T-00000688

S26-D-00000765 - Classical PDE solution

A classical solution has enough ordinary differentiability for the PDE and prescribed data to hold pointwise in their declared sense.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000751, S26-D-00000335

S26-D-00000766 - Weak PDE solution

A weak solution satisfies an integrated variational identity in a specified function space rather than necessarily satisfying the PDE pointwise.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-T-00000692, S26-D-00000765

S26-D-00000767 - Distributional PDE solution

A distributional solution satisfies the PDE after interpreting derivatives in the sense of distributions.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000679, S26-D-00000680, S26-D-00000751

S26-T-00000768 - Classical implies distributional

A sufficiently regular classical solution induces a distributional solution of the same PDE.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000765, S26-D-00000767

S26-D-00000769 - Well-posedness

A problem is well posed in a declared class when existence, uniqueness, and continuous dependence on data all hold in the specified topologies.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000306, S26-T-00000319, S26-D-00000751

S26-D-00000770 - A priori estimate

An a priori estimate bounds a solution quantity by prescribed data independently of a particular construction of the solution.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000769, S26-T-00000305

S26-T-00000771 - Energy method interface

Multiplying an evolution PDE by a suitable solution/test quantity and integrating can yield an energy identity or inequality under justified regularity and boundary conditions.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000430, S26-D-00000766

S26-D-00000772 - Wave energy

For the standard homogeneous wave equation, the energy combines squared time derivative and squared spatial gradient integrals.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000760, S26-T-00000771

S26-T-00000773 - Wave energy conservation

For sufficiently regular solutions of the homogeneous wave equation with compatible boundary/decay conditions, the standard energy is conserved.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000772

S26-D-00000774 - Heat energy dissipation

For the homogeneous heat equation under compatible conditions, an L2 energy decreases and its derivative is controlled by the gradient norm.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-T-00000744, S26-T-00000771, S26-T-00000442

S26-D-00000775 - Maximum principle interface

Maximum principles constrain extrema of solutions or subsolutions of elliptic/parabolic equations under operator, domain, and boundary hypotheses.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000758, S26-D-00000759, S26-T-00000348

S26-T-00000776 - Weak maximum principle for harmonic functions

A continuous harmonic function on a bounded domain, sufficiently regular inside, attains its maximum and minimum on the boundary.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-T-00000341, S26-D-00000775, S26-D-00000310

S26-T-00000777 - Uniqueness from maximum principle

For the Laplace Dirichlet problem, the maximum principle yields uniqueness in the corresponding classical solution class.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000763, S26-T-00000776

S26-D-00000778 - Green identity interface

Integration by parts relates Laplacians, gradients, and boundary flux terms under sufficient regularity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000430, S26-T-00000341, S26-D-00000764

S26-D-00000779 - Green function

A Green function represents the response of a linear boundary value problem to a point source and encodes the boundary condition.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000741, S26-D-00000761, S26-D-00000682

S26-T-00000780 - Green representation interface

Under suitable regularity, domain, and boundary hypotheses, solutions of linear elliptic problems admit integral representations using an associated Green function plus boundary terms.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000779, S26-D-00000778

S26-D-00000781 - Variational elliptic problem

A second-order elliptic boundary problem can often be recast as finding u in a Hilbert/Sobolev space such that a(u,v)=F(v) for all test functions v.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-T-00000692, S26-T-00000699

S26-T-00000782 - Lax-Milgram elliptic existence interface

A bounded coercive bilinear form on a Hilbert space yields a unique weak solution with an a priori norm bound.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-T-00000699, S26-D-00000781

S26-D-00000783 - Coercivity fence

Coercivity is a quantitative lower bound on the bilinear form relative to the chosen norm; positivity without a uniform lower bound is not enough for Lax-Milgram.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-T-00000782

S26-D-00000784 - Poincare inequality interface

On appropriate domains and subspaces, an Lp norm of a function can be controlled by a norm of its gradient, with constants depending on geometry and normalization.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000686, S26-T-00000440

S26-T-00000785 - Dirichlet energy coercivity interface

For H_0^1-type spaces on suitable bounded domains, Poincare inequality makes the gradient energy coercive in the H1 norm.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000784, S26-D-00000781

S26-D-00000786 - C0 semigroup

A strongly continuous one-parameter semigroup on a Banach space satisfies T(0)=I, T(t+s)=T(t)T(s), and strong continuity in time.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-T-00000604, S26-T-00000617

S26-D-00000787 - Semigroup generator

The generator A of a C0 semigroup is the generally unbounded operator defined by the strong derivative at zero on the domain where that limit exists.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000786, S26-D-00000661

S26-T-00000788 - Abstract evolution equation interface

When A generates a suitable C0 semigroup, u(t)=T(t)u0 provides the mild solution of u prime=Au with initial datum u0.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000787, S26-D-00000762

S26-D-00000789 - Mild solution

A mild solution of an evolution equation is defined through a semigroup/integral formulation and need not possess the derivatives required of a classical solution.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-T-00000788, S26-D-00000765

S26-D-00000790 - Duhamel principle

For an inhomogeneous linear evolution equation, the solution is represented by homogeneous semigroup evolution plus a time convolution with the forcing, under appropriate hypotheses.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000786, S26-D-00000789, S26-D-00000711

S26-D-00000791 - Elliptic regularity interface

Elliptic regularity asks when improved regularity of coefficients, data, and domain forces improved regularity of a weak/distributional solution.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000758, S26-D-00000766, S26-D-00000686

S26-T-00000792 - Interior elliptic regularity schematic

For uniformly elliptic operators with sufficiently regular coefficients, solutions gain interior regularity relative to the forcing in theorem-specific Sobolev or Holder scales.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000791, S26-D-00000758

S26-D-00000793 - Parabolic smoothing interface

Parabolic equations such as the heat equation can improve spatial regularity for positive time, subject to the data and solution framework.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000759, S26-T-00000744, S26-D-00000791

S26-D-00000794 - Finite propagation interface

Hyperbolic equations such as the wave equation exhibit domain-of-dependence and finite-speed phenomena under suitable principal-part hypotheses.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000760, S26-D-00000769

S26-D-00000795 - Characteristic surface interface

For a PDE, characteristic covectors/surfaces are determined by vanishing of the principal symbol and govern propagation behavior.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000756, S26-D-00000760

S26-D-00000796 - Solution-concept fence

Classical, strong, weak, mild, viscosity, and distributional solutions are distinct typed notions; implications require explicit hypotheses.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000765, S26-D-00000766, S26-D-00000767, S26-D-00000789

S26-D-00000797 - Boundary-regularity fence

Boundary traces, normal derivatives, Green formulas, and boundary regularity require geometric and function-space hypotheses and cannot be inferred from interior regularity alone.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000763, S26-D-00000764, S26-D-00000778, S26-D-00000791

S26-D-00000798 - PDE classification fence

Elliptic, parabolic, and hyperbolic labels refer to principal-symbol structure and may depend on point, coefficients, formulation, and degeneracy; prototype equations do not define every case.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000758, S26-D-00000759, S26-D-00000760, S26-D-00000756

S26-T-00000799 - PDE epistemic fence

Grid experiments, symbolic residuals, or smooth manufactured solutions can test implementations but do not prove general existence, uniqueness, regularity, stability, or convergence theorems.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000769, S26-D-00000770, S26-D-00000791

S26-T-00000800 - PDE foundations frontier

Classification, boundary/initial data, weak formulations, energy and maximum methods, variational existence, semigroups, and regularity interfaces form the substrate for nonlinear PDE, numerical PDE, and geometric analysis.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Domain, coefficient regularity, boundary data, function spaces, and solution concept are exactly as stated.

Hard dependencies: S26-D-00000769, S26-T-00000782, S26-T-00000788, S26-T-00000792, S26-D-00000794

Applications: PDE | Continuum mechanics | Mathematical physics | Numerical analysis

Part AO - Calculus of variations and nonlinear PDE

S26-D-00000801 - Functional

A functional is a map from a space of admissible functions or states into the scalar field or extended real line.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-T-00000226, S26-T-00000604

S26-D-00000802 - Admissible class

An admissible class specifies the function space together with boundary, integral, inequality, or other constraints over which a variational problem is posed.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000801, S26-D-00000761

S26-D-00000803 - Variational minimizer

A minimizer of J over an admissible class A is u in A with J(u)<=J(v) for every v in A.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000801, S26-D-00000802

S26-D-00000804 - Local variational minimizer

A local minimizer minimizes a functional only relative to an explicitly chosen topology or norm in a neighborhood of the candidate.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000803, S26-D-00000303

S26-D-00000805 - Variation

A variation of u is a curve u+t phi through the admissible space, with phi restricted when constraints or boundary data require it.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000802, S26-T-00000331

S26-D-00000806 - First variation

The first variation delta J(u)[phi] is the directional derivative at t=0 of J(u+t phi), when this derivative exists.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000805, S26-T-00000329

S26-D-00000807 - Stationary point of a functional

A stationary point has vanishing first variation for every admissible test direction.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000806

S26-T-00000808 - Variational necessary condition

An interior local minimizer at which the first variation exists in every admissible direction is stationary.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000804, S26-D-00000806, S26-D-00000807

S26-D-00000809 - Integral functional

A basic variational functional has the form J(u)=integral_Omega L(x,u,grad u) dx on a declared Sobolev or smooth admissible class.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000801, S26-D-00000430, S26-D-00000686

S26-D-00000810 - Lagrangian density

The integrand L(x,z,p) of an integral functional is its Lagrangian density; regularity and growth assumptions determine which manipulations are legitimate.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000809

S26-T-00000811 - Euler-Lagrange equation interface

For sufficiently smooth L and admissible u, vanishing first variation yields the Euler-Lagrange equation in weak form and, with additional regularity, pointwise form.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000806, S26-D-00000809, S26-D-00000766

S26-D-00000812 - Weak Euler-Lagrange equation

The weak Euler-Lagrange equation is the test-function identity obtained after differentiating the functional and transferring derivatives by integration by parts.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-T-00000811, S26-D-00000778

S26-D-00000813 - Second variation

The second variation is the second directional derivative of a functional along admissible variations when it exists.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000806, S26-T-00000337

S26-T-00000814 - Second-variation necessary condition

At a sufficiently regular local minimizer, the second variation is nonnegative in admissible directions, under hypotheses ensuring the second derivative test applies.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000804, S26-D-00000813, S26-T-00000349

S26-D-00000815 - Convex functional

A functional J on a convex set is convex when J((1-t)u+tv)<=(1-t)J(u)+tJ(v) for 0<=t<=1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-T-00000355, S26-D-00000801

S26-D-00000816 - Strictly convex functional

A functional is strictly convex when the convexity inequality is strict for distinct points and 0<t<1.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000815

S26-T-00000817 - Strict-convexity uniqueness

A strictly convex functional has at most one minimizer on a convex admissible set.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000816, S26-D-00000803

S26-D-00000818 - Coercive functional

A functional is coercive relative to a norm or topology when J(u) tends to +infinity as the declared size of u tends to infinity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000801, S26-T-00000604

S26-D-00000819 - Weak lower semicontinuity

J is sequentially weakly lower semicontinuous when u_n converges weakly to u implies J(u)<=liminf J(u_n).

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-T-00000632, S26-D-00000801, S26-T-00000373

S26-D-00000820 - Minimizing sequence

A minimizing sequence satisfies J(u_n)->inf_A J without itself being assumed to converge.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000803, S26-T-00000362

S26-D-00000821 - Direct method of calculus of variations

The direct method seeks a bounded minimizing sequence, extracts a convergent subsequence using compactness, proves admissibility of the limit, and uses lower semicontinuity to obtain a minimizer.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000818, S26-D-00000819, S26-D-00000820, S26-T-00000311

S26-T-00000822 - Direct-method existence interface

On a reflexive Banach space, coercivity and sequential weak lower semicontinuity, together with weakly closed nonempty admissibility, give a minimizer under the theorem-specific hypotheses.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000821, S26-D-00000634, S26-T-00000633

S26-D-00000823 - Compactness fence for direct methods

Boundedness of a minimizing sequence is not itself strong compactness in an infinite-dimensional space; the topology and compactness theorem used must be named.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000820, S26-D-00000623, S26-T-00000633

S26-D-00000824 - Lower-semicontinuity fence

Weak convergence does not permit passing arbitrary nonlinear energies to the limit; weak lower semicontinuity or another justified limit theorem is required.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000819, S26-T-00000632

S26-D-00000825 - Constraint functional

A constraint is represented by an equation or inequality on the admissible state, often through one or more auxiliary functionals.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000802, S26-D-00000801

S26-D-00000826 - Lagrange multiplier interface in function spaces

Under a suitable constraint qualification, a constrained stationary point may satisfy a multiplier equation involving derivatives of the objective and constraint functionals.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000825, S26-D-00000806

S26-D-00000827 - Obstacle problem interface

An obstacle problem minimizes an energy over functions constrained to lie above or below a prescribed obstacle, leading naturally to a variational inequality.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000803, S26-D-00000825, S26-D-00000781

S26-D-00000828 - Variational inequality

A variational inequality replaces equality against every test direction by an inequality over a convex admissible set.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000827, S26-D-00000815

S26-D-00000829 - Nonlinear operator equation

A nonlinear PDE may be encoded abstractly as A(u)=f between suitable function spaces.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000755, S26-T-00000604

S26-D-00000830 - Monotone operator

An operator A from a real Banach space into its dual is monotone when <A(u)-A(v),u-v>>=0 for all u,v in its domain.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000829, S26-T-00000621

S26-D-00000831 - Strictly monotone operator

A monotone operator is strictly monotone when the monotonicity pairing is positive for distinct admissible points.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000830

S26-D-00000832 - Strongly monotone operator

A is strongly monotone when the monotonicity pairing controls a positive power, typically the squared norm, of u-v by a uniform constant.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000830, S26-T-00000604

S26-D-00000833 - Hemicontinuity interface

Hemicontinuity requires suitable scalar continuity of t -> <A(u+tv),w> and is one standard hypothesis in monotone-operator existence theorems.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000830, S26-T-00000319

S26-D-00000834 - Browder-Minty interface

A coercive, hemicontinuous monotone operator on an appropriate reflexive Banach space is surjective under the theorem-specific hypotheses; strict monotonicity yields uniqueness.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000830, S26-D-00000833, S26-D-00000634

S26-D-00000835 - p-Laplacian

The p-Laplacian is Delta_p u=div(|grad u|^(p-2) grad u), naturally interpreted weakly on W^{1,p}-type spaces.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-T-00000688, S26-D-00000751, S26-D-00000830

S26-D-00000836 - Weak p-Laplace equation

A weak solution of -Delta_p u=f satisfies the corresponding integral identity against test functions, with integrability exponents and boundary conditions explicitly specified.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000835, S26-D-00000766, S26-T-00000443

S26-D-00000837 - Nonlinear elliptic equation

A nonlinear elliptic equation has a nonlinear operator whose principal behavior satisfies an appropriate ellipticity condition; definitions vary by divergence/nondivergence and solution framework.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000755, S26-D-00000758, S26-D-00000829

S26-D-00000838 - Quasilinear divergence-form equation

A typical quasilinear divergence-form equation is -div A(x,u,grad u)=f, interpreted under growth, coercivity, and monotonicity assumptions.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000754, S26-T-00000688, S26-D-00000830

S26-D-00000839 - Weak compactness interface for nonlinear PDE

Uniform bounds in reflexive Sobolev spaces can yield weakly convergent subsequences, but identification of nonlinear limits requires additional structure.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000634, S26-D-00000686, S26-T-00000632

S26-D-00000840 - Strong compactness interface

Compact embeddings can upgrade bounded sequences to strongly convergent subsequences in weaker norms under domain and exponent hypotheses.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-T-00000649, S26-D-00000686, S26-T-00000311

S26-D-00000841 - Rellich-Kondrachov interface

On suitable bounded domains, specified Sobolev embeddings are compact when the exponents and regularity lie in the theorem-specific compact range.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000840, S26-D-00000686

S26-D-00000842 - Nonlinear limit-identification fence

Weak convergence u_n -> u generally does not imply F(u_n)->F(u) for a nonlinear F; monotonicity, compactness, convexity, or another identification argument is needed.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000839, S26-D-00000829

S26-D-00000843 - Schauder fixed-point interface

A continuous map from a nonempty closed convex set into a relatively compact subset of itself has a fixed point under the standard Banach-space formulation.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000354, S26-T-00000649, S26-D-00000315

S26-D-00000844 - Leray-Schauder continuation interface

Compact fixed-point problems can be treated by continuation/degree arguments when the homotopy fixed points satisfy suitable a priori bounds.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000843, S26-D-00000770

S26-D-00000845 - Banach fixed-point nonlinear PDE interface

If a nonlinear PDE reformulation is a contraction on a complete invariant set, Banach fixed point gives a unique local or global solution in that set.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000315, S26-D-00000829

S26-D-00000846 - Galerkin approximation interface

Galerkin methods approximate an infinite-dimensional variational or evolution problem in finite-dimensional subspaces and require stability plus compactness/consistency to pass to a limit.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000781, S26-D-00000840, S26-D-00000214

S26-D-00000847 - A priori bounds before compactness

In nonlinear existence proofs, uniform estimates are established before extracting subsequences; compactness cannot substitute for missing bounds.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000770, S26-D-00000839, S26-D-00000840

S26-D-00000848 - Nonlinear uniqueness fence

Existence by compactness or Schauder fixed point does not imply uniqueness; uniqueness requires an independent mechanism such as contraction or strict monotonicity.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000843, S26-D-00000845, S26-D-00000831

S26-T-00000849 - Variational and nonlinear-PDE epistemic fence

Discrete minimization, symbolic Euler-Lagrange residuals, and finite Galerkin experiments can test constructions but do not prove general lower semicontinuity, compactness, existence, uniqueness, or regularity theorems.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-D-00000821, S26-D-00000842, S26-D-00000846

S26-T-00000850 - Calculus-of-variations and nonlinear-PDE frontier

First variation, Euler-Lagrange structure, direct methods, variational inequalities, monotone operators, compactness, and fixed-point interfaces form a controlled substrate for nonlinear PDE, geometric variational problems, and numerical approximation.

Status: A - Exact registered statement only; bounded checks do not promote the general theorem to P.

Assumptions: Function spaces, domain regularity, growth, coercivity, convexity, compactness, and boundary conditions are exactly as stated.

Hard dependencies: S26-T-00000811, S26-T-00000822, S26-D-00000828, S26-D-00000834, S26-D-00000841, S26-D-00000843, S26-D-00000846

Applications: Nonlinear PDE | Continuum mechanics | Geometric analysis | Optimization | Numerical analysis

Part AP - Numerical analysis

S26-D-00000851 - Numerical approximation

Numerical analysis studies computable approximations together with explicit error, stability, conditioning, and complexity models.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-T-00000850

S26-D-00000852 - Absolute error

Absolute error is the norm or magnitude of the difference between an approximation and the declared exact target.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-T-00000604, S26-D-00000851

S26-D-00000853 - Relative error

Relative error scales absolute error by a nonzero reference magnitude; it is undefined or unsuitable when that reference vanishes or is negligibly small.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000852

S26-D-00000854 - Forward error

Forward error measures error in the computed output relative to the exact output for the original input.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000852

S26-D-00000855 - Backward error

Backward error asks for the smallest perturbation of the input for which the computed result is exact.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000854

S26-D-00000856 - Conditioning

Conditioning describes sensitivity of the mathematical problem itself to perturbations of its data, independently of a particular algorithm.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-T-00000319, S26-D-00000855

S26-D-00000857 - Absolute condition number

A local absolute condition number bounds first-order output perturbation per unit input perturbation in declared norms.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000856, S26-T-00000329

S26-D-00000858 - Relative condition number

A relative condition number compares relative output sensitivity with relative input sensitivity where the required scales are nonzero.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000853, S26-D-00000857

S26-T-00000859 - Conditioning-algorithm fence

An ill-conditioned problem can defeat a stable algorithm, while an unstable algorithm can fail on a well-conditioned problem; conditioning and stability are distinct.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000856

S26-D-00000860 - Floating-point model

A standard floating-point model represents an elementary rounded operation as fl(x op y)=(x op y)(1+delta), with |delta| bounded by unit roundoff when model hypotheses hold.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000851

S26-D-00000861 - Machine epsilon and unit roundoff

Machine epsilon and unit roundoff quantify floating-point spacing/rounding under a specified format and rounding convention; conventions must not be silently conflated.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000860

S26-D-00000862 - Roundoff error

Roundoff error is error introduced by finite-precision representation and arithmetic, distinct from discretization and modeling error.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000860, S26-D-00000852

S26-D-00000863 - Catastrophic cancellation

Subtraction of nearby floating-point quantities can destroy relative accuracy when significant leading digits cancel.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000862

S26-D-00000864 - Backward stable algorithm

An algorithm is backward stable when its computed answer is the exact answer to a nearby problem under a declared perturbation metric.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000855, S26-D-00000862

S26-T-00000865 - Forward-error conditioning interface

Backward stability plus favorable conditioning can yield a small forward error, with the precise bound depending on the problem and perturbation model.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000856, S26-D-00000864

S26-D-00000866 - Consistency of a discretization

A discretization is consistent when its discrete operator approximates the continuous operator with truncation defect tending to zero in the specified limit.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000851, S26-D-00000775

S26-D-00000867 - Stability of a discretization

Stability controls amplification of perturbations or data by the discrete solution operator in specified norms.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000866, S26-T-00000859

S26-D-00000868 - Convergence of a discretization

A discretization converges when discrete solutions approach the exact continuous solution in the declared norm as resolution is refined.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000866, S26-D-00000867

S26-T-00000869 - Lax equivalence interface

For a properly posed linear initial-value problem and a consistent finite-difference approximation, stability is equivalent to convergence under the theorem-specific framework.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000766, S26-D-00000866, S26-D-00000867, S26-D-00000868

S26-D-00000870 - Order of convergence

A method has order p when its error admits the stated asymptotic bound C h^p, with norm, regime, regularity, and constants declared.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000868

S26-D-00000871 - Interpolation problem

Interpolation constructs an approximant matching prescribed data values or functionals at selected nodes.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-T-00000147, S26-D-00000851

S26-D-00000872 - Polynomial interpolation

At n+1 distinct scalar nodes there is a unique polynomial of degree at most n matching n+1 prescribed scalar values.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000146, S26-D-00000871

S26-D-00000873 - Lagrange interpolation formula

The Lagrange basis gives an explicit representation of the unique polynomial interpolant at distinct nodes.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000872

S26-D-00000874 - Interpolation remainder

For sufficiently smooth functions, polynomial interpolation error has a derivative-based remainder involving the nodal polynomial under theorem-specific hypotheses.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000872, S26-D-00000338

S26-D-00000875 - Lebesgue constant

The Lebesgue constant is the operator norm controlling worst-case amplification from sampled data to the interpolation polynomial in the chosen sup norm.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000619, S26-D-00000872

S26-D-00000876 - Runge phenomenon

High-degree interpolation at poorly distributed nodes can diverge or oscillate strongly even for smooth functions; increasing degree alone is not a convergence guarantee.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000872, S26-D-00000875

S26-D-00000877 - Chebyshev-node interface

Chebyshev-type nodes reduce interpolation instability relative to equally spaced nodes and give near-optimal logarithmic Lebesgue-constant growth in the classical setting.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000875

S26-D-00000878 - Numerical differentiation

Numerical differentiation approximates derivatives from discrete data and is typically sensitive to noise and roundoff.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-T-00000329, S26-D-00000862

S26-D-00000879 - Finite-difference stencil

A finite-difference stencil is a weighted combination of nearby sampled values designed to approximate a derivative or differential operator.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000878, S26-D-00000775

S26-D-00000880 - Local truncation error

Local truncation error is the residual produced by inserting the exact sufficiently smooth solution into the discrete formula.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000866, S26-D-00000879

S26-D-00000881 - Numerical quadrature

Quadrature approximates a definite integral by a weighted sum of function values or other accessible data.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-T-00000395, S26-D-00000851

S26-D-00000882 - Newton-Cotes interface

Newton-Cotes rules integrate polynomial interpolants on equally spaced nodes; their accuracy and stability depend on degree and rule choice.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000872, S26-D-00000881

S26-D-00000883 - Gaussian quadrature

An n-node Gaussian quadrature rule associated with a positive weight is exact for polynomials through degree 2n-1 under the standard orthogonality hypotheses.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000881, S26-D-00000224

S26-D-00000884 - Adaptive quadrature

Adaptive quadrature refines sampling according to an error indicator or estimator and requires a stopping tolerance tied to a declared error model.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000881, S26-D-00000852

S26-D-00000885 - Initial-value ODE discretization

A time-stepping method approximates an ODE trajectory by recursively advancing discrete states from initial data.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-T-00000331, S26-D-00000851

S26-D-00000886 - Explicit Euler method

Explicit Euler advances y_{n+1}=y_n+h f(t_n,y_n) and is first-order under standard smoothness and stability assumptions.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000885, S26-D-00000870

S26-D-00000887 - Implicit Euler method

Implicit Euler advances y_{n+1}=y_n+h f(t_{n+1},y_{n+1}) and requires solving an implicit equation at each step.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000885, S26-D-00000829

S26-D-00000888 - Runge-Kutta method

A Runge-Kutta method combines staged evaluations of the vector field to construct a one-step update with order determined by its coefficients.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000885

S26-D-00000889 - Absolute stability region

For a time-stepping method, the absolute stability region is the set of scaled test-equation parameters for which the numerical amplification factor does not grow.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000867, S26-D-00000885

S26-D-00000890 - A-stability

A method is A-stable when its absolute stability region contains the entire open left half-plane.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000889

S26-D-00000891 - Stiffness interface

Stiff problems contain separated dynamical scales that can impose severe stability restrictions on explicit methods; stiffness is problem-and-method dependent.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000889, S26-D-00000890

S26-D-00000892 - Finite-difference PDE method

A finite-difference PDE method replaces differential operators on a grid by discrete stencils and requires consistency, stability, boundary treatment, and convergence analysis.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000879, S26-D-00000866, S26-D-00000867, S26-D-00000775

S26-D-00000893 - Mesh and mesh size

A computational mesh partitions or samples a domain; mesh size and shape-regularity parameters must be declared when error constants depend on them.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000761, S26-D-00000851

S26-D-00000894 - Finite element space

A finite element space is a finite-dimensional trial/test space assembled from local polynomial-like functions over a mesh with specified continuity and boundary constraints.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000781, S26-D-00000893, S26-D-00000214

S26-D-00000895 - Galerkin orthogonality

For a conforming linear Galerkin approximation, the error is orthogonal to the discrete test space with respect to the governing bilinear form.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000781, S26-D-00000846, S26-D-00000894

S26-T-00000896 - Cea lemma interface

For a continuous coercive bilinear variational problem, conforming Galerkin error is bounded by a stability constant times the best approximation error in the energy-space norm.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-T-00000782, S26-D-00000895

S26-D-00000897 - A priori error estimate

An a priori estimate bounds discretization error using known regularity, approximation properties, mesh parameters, and stability constants before computing the solution.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000870, S26-T-00000896

S26-D-00000898 - A posteriori error estimator

An a posteriori estimator uses the computed solution and residual/jump information to estimate or bound error after computation, with reliability and efficiency requiring separate proof.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000897, S26-D-00000884

S26-T-00000899 - Numerical epistemic error decomposition

Modeling error, discretization error, algebraic-solver error, roundoff error, and implementation error are distinct objects and must not be collapsed into a single unexplained numerical discrepancy.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000862, S26-D-00000868, S26-D-00000897, S26-D-00000898

S26-T-00000900 - Numerical-analysis frontier

Conditioning, floating-point stability, approximation, interpolation, quadrature, time stepping, finite differences, and finite elements form a controlled computational layer connecting exact mathematics to reproducible numerical evidence.

Status: A - Exact registered statement only; computational checks do not promote the general theorem to P.

Assumptions: Arithmetic model, norms, discretization spaces, regularity, mesh hypotheses, and stability assumptions are exactly as stated.

Hard dependencies: S26-D-00000856, S26-D-00000864, S26-D-00000868, S26-D-00000872, S26-D-00000881, S26-D-00000885, S26-D-00000892, S26-T-00000896, S26-T-00000899

Applications: Scientific computing | PDE simulation | Optimization | Engineering computation

Part AQ - Differential geometry

S26-D-00000901 - Topological manifold

An n-dimensional topological manifold is a Hausdorff, second-countable space locally homeomorphic to open subsets of R^n.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000309, S26-D-00000312

S26-D-00000902 - Coordinate chart

A chart is a homeomorphism from an open subset of a manifold onto an open subset of R^n.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000901

S26-D-00000903 - Atlas

An atlas is a collection of compatible charts covering the manifold.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000902

S26-D-00000904 - Smooth compatibility

Two charts are smoothly compatible when their transition maps are smooth wherever defined.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000903, S26-T-00000329

S26-D-00000905 - Smooth manifold

A smooth manifold is a topological manifold equipped with a maximal smooth atlas.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000901, S26-D-00000904

S26-D-00000906 - Smooth map of manifolds

A map between smooth manifolds is smooth when its coordinate representatives are smooth; this property is chart-independent.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000905, S26-T-00000329

S26-D-00000907 - Diffeomorphism

A diffeomorphism is a bijective smooth map with smooth inverse.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000906, S26-D-00000048

S26-D-00000908 - Smooth curve

A smooth curve in M is a smooth map from an interval into M.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000906

S26-D-00000909 - Tangent vector by derivations

A tangent vector at p may be defined as a derivation on germs of smooth functions at p satisfying the Leibniz rule.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000905, S26-D-00000128

S26-D-00000910 - Tangent space

The tangent space T_pM is the vector space of tangent vectors at p.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000909, S26-D-00000149

S26-D-00000911 - Velocity of a curve

A smooth curve through p determines a tangent vector at p by differentiating smooth functions along the curve.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000908, S26-D-00000909

S26-D-00000912 - Differential of a smooth map

For smooth F:M->N, the differential dF_p:T_pM->T_{F(p)}N is the induced linear map on tangent vectors.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000906, S26-D-00000910, S26-D-00000169

S26-T-00000913 - Manifold chain rule

Differentials compose: d(G o F)_p=dG_{F(p)} o dF_p.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000912, S26-T-00000334

S26-D-00000914 - Tangent bundle

TM is the disjoint union of tangent spaces with its canonical smooth-manifold structure.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000910, S26-D-00000905

S26-D-00000915 - Vector field

A smooth vector field is a smooth section X:M->TM of the tangent bundle.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000914

S26-D-00000916 - Integral curve

An integral curve of X satisfies gamma prime(t)=X(gamma(t)) in the tangent bundle.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000915, S26-T-00000331

S26-D-00000917 - Flow

A local flow is the local one-parameter family of diffeomorphisms generated by a vector field where the ODE exists.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000916, S26-D-00000907

S26-D-00000918 - Lie bracket

The Lie bracket [X,Y] is the vector field acting on smooth functions by X(Yf)-Y(Xf).

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000915

S26-T-00000919 - Lie bracket properties

The vector-field bracket is bilinear, antisymmetric, and satisfies the Jacobi identity.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000918

S26-D-00000920 - Cotangent space

The cotangent space T_p^*M is the algebraic dual of T_pM.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000910, S26-D-00000251

S26-D-00000921 - Cotangent bundle

T^*M is the smooth vector bundle formed from the cotangent spaces.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000920, S26-D-00000905

S26-D-00000922 - Differential of a function

For smooth f:M->R, df is the 1-form defined by df_p(v)=v(f).

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000912, S26-D-00000920

S26-D-00000923 - Differential k-form

A differential k-form is a smooth section of the bundle of alternating k-covectors Lambda^k T^*M.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000921, S26-T-00000276

S26-D-00000924 - Wedge product of forms

Differential forms carry a graded wedge product induced pointwise by exterior algebra.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000923, S26-D-00000277

S26-D-00000925 - Exterior derivative

The exterior derivative d maps k-forms to (k+1)-forms, is local and graded-Leibniz, agrees with df on functions, and satisfies d^2=0.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000923, S26-D-00000924

S26-D-00000926 - Pullback of differential forms

A smooth map F:M->N pulls differential forms on N back to forms on M via its differential.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000912, S26-D-00000923, S26-T-00000289

S26-T-00000927 - Naturality of exterior derivative

Pullback commutes with exterior differentiation: F^*(d omega)=d(F^* omega).

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000925, S26-D-00000926

S26-D-00000928 - Closed form

A differential form omega is closed when d omega=0.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000925

S26-D-00000929 - Exact form

A differential form is exact when omega=d eta for some form eta.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000925

S26-T-00000930 - Exact implies closed

Every exact differential form is closed because d^2=0.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000928, S26-D-00000929

S26-D-00000931 - de Rham cohomology

H^k_dR(M) is the quotient of closed k-forms by exact k-forms.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000928, S26-D-00000929, S26-D-00000113

S26-D-00000932 - Orientation of a manifold

An orientation is a consistent choice of orientation on tangent spaces, equivalently an oriented-atlas or positive top-form structure under standard hypotheses.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-T-00000290, S26-D-00000910

S26-D-00000933 - Integration of top forms

Compactly supported top-degree forms on an oriented manifold admit a coordinate-independent integral assembled using oriented charts and partitions of unity.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000932, S26-D-00000923, S26-T-00000445

S26-D-00000934 - Manifold with boundary

A manifold with boundary is locally modeled on the closed half-space, with boundary points characterized by half-space charts.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000901

S26-T-00000935 - General Stokes theorem

For a compact oriented smooth manifold with boundary and a smooth (n-1)-form omega, integral_M d omega = integral_boundaryM omega with induced boundary orientation.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000925, S26-D-00000933, S26-D-00000934

S26-D-00000936 - Riemannian metric

A Riemannian metric is a smooth assignment of positive-definite inner products g_p on tangent spaces.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000910, S26-D-00000201

S26-D-00000937 - Riemannian length

The length of a piecewise smooth curve is the integral of the norm of its velocity induced by the Riemannian metric.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000936, S26-D-00000908, S26-T-00000395

S26-D-00000938 - Riemannian distance

The Riemannian distance is the infimum of lengths of piecewise smooth curves joining two points in the same connected component.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000937, S26-D-00000304

S26-D-00000939 - Musical isomorphisms

A Riemannian metric identifies tangent and cotangent vectors via flat and sharp bundle maps.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000936, S26-D-00000920, S26-D-00000208

S26-D-00000940 - Gradient on a Riemannian manifold

The gradient grad f is defined by g(grad f,X)=df(X) for every vector field X.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000922, S26-D-00000936, S26-D-00000939

S26-D-00000941 - Affine connection

A connection nabla differentiates vector fields along vector fields, linearly in the direction and satisfying a Leibniz rule in the differentiated field.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000915

S26-D-00000942 - Covariant derivative along a curve

A connection induces differentiation of vector fields along smooth curves.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000941, S26-D-00000908

S26-D-00000943 - Torsion

The torsion of a connection is T(X,Y)=nabla_XY-nabla_YX-[X,Y].

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000941, S26-D-00000918

S26-T-00000944 - Levi-Civita theorem

Every Riemannian manifold has a unique torsion-free connection compatible with the metric.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000936, S26-D-00000941, S26-D-00000943

S26-D-00000945 - Geodesic

A geodesic is a curve whose velocity is parallel along itself: nabla_{gamma prime} gamma prime=0.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000942, S26-T-00000944

S26-D-00000946 - Exponential map

The exponential map exp_p sends an initial tangent vector to the endpoint at unit time of its geodesic when defined.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000945, S26-D-00000910

S26-D-00000947 - Riemann curvature tensor

The curvature operator is R(X,Y)Z=nabla_Xnabla_YZ-nabla_Ynabla_XZ-nabla_[X,Y]Z.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000941, S26-D-00000918

S26-D-00000948 - Sectional curvature

Sectional curvature assigns to each tangent 2-plane a scalar curvature determined by the Riemann tensor and metric.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000947, S26-D-00000936

S26-D-00000949 - Ricci and scalar curvature

Ricci curvature is the metric trace of the Riemann tensor, and scalar curvature is the trace of Ricci curvature.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000947, S26-D-00000936, S26-D-00000218

S26-T-00000950 - Differential-geometry frontier

Smooth manifolds, tangent/cotangent bundles, forms, Stokes theory, Riemannian metrics, connections, geodesics, and curvature form the coordinate-invariant geometric layer joining topology, calculus, exterior algebra, analysis, and mathematical physics.

Status: A - Exact registered statement only; examples and coordinate calculations do not promote the general claim to P.

Assumptions: Smoothness, dimension, topology, orientation, metric, and boundary hypotheses are exactly as stated.

Hard dependencies: S26-D-00000905, S26-D-00000910, S26-D-00000923, S26-T-00000935, S26-D-00000936, S26-T-00000944, S26-D-00000945, S26-D-00000947, S26-D-00000949

Boundary note: A coordinate formula is not itself an invariant definition; chart-dependent expressions must transform correctly.

Applications: Geometry | General relativity | Mechanics | Geometric PDE

Part AR - Lie groups, symplectic geometry, and Hamiltonian mechanics

S26-D-00000951 - Lie group

A Lie group is a smooth manifold G equipped with a group structure whose multiplication GxG->G and inversion G->G are smooth.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000905, S26-D-00000102

Applications: Symmetry | Geometry | Dynamics

S26-D-00000952 - Lie-group homomorphism

A Lie-group homomorphism is a group homomorphism that is also smooth.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000951, S26-D-00000906

S26-D-00000953 - Left and right translations

For g in a Lie group G, L_g(h)=gh and R_g(h)=hg are diffeomorphisms of G.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000951, S26-D-00000907

S26-D-00000954 - Lie algebra of a Lie group

The Lie algebra g of a Lie group G is T_eG equipped with the bracket induced by left-invariant vector fields.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000951, S26-D-00000910, S26-D-00000918

S26-D-00000955 - Left-invariant vector field

A vector field X on G is left-invariant when (L_g)_*X_h=X_{gh} for all g,h.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000953, S26-D-00000915, S26-D-00000912

S26-T-00000956 - Left-invariant fields identify with T_eG

Evaluation at the identity gives a vector-space isomorphism between left-invariant vector fields on G and T_eG.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000954, S26-D-00000955, S26-T-00000175

S26-T-00000957 - Lie-algebra bracket properties

The bracket on a Lie algebra is bilinear, alternating (hence antisymmetric in characteristic not 2), and satisfies the Jacobi identity.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000954, S26-T-00000919

Boundary note: Alternating and antisymmetric formulations require characteristic care in abstract algebraic generalizations.

S26-D-00000958 - One-parameter subgroup

A one-parameter subgroup is a smooth homomorphism gamma:(R,+)->G.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000952, S26-D-00000054

S26-D-00000959 - Lie-group exponential map

For X in g, exp(X) is the value at time 1 of the one-parameter subgroup whose tangent at 0 is X.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000954, S26-D-00000958, S26-D-00000917

S26-T-00000960 - Exponential one-parameter law

For fixed X in g, exp((s+t)X)=exp(sX)exp(tX) whenever the exponential is defined by the global one-parameter subgroup.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000959, S26-D-00000958

S26-D-00000961 - Adjoint representation

Conjugation C_g(h)=ghg^{-1} differentiates at the identity to a linear automorphism Ad_g:g->g.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000951, S26-D-00000912, S26-D-00000954

S26-D-00000962 - Infinitesimal adjoint map

For X in g, ad_X is the linear map Y |-> [X,Y].

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000954, S26-T-00000957

S26-T-00000963 - Adjoint intertwines the bracket

Ad_g([X,Y])=[Ad_gX,Ad_gY].

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000961, S26-T-00000957

S26-T-00000964 - Baker-Campbell-Hausdorff local interface

Near the identity, log(exp X exp Y) is expressed by the BCH Lie series X+Y+1/2[X,Y]+higher nested commutators under its applicable convergence/formal hypotheses.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000959, S26-T-00000957

Boundary note: The BCH series is a local/formal statement; convergence and global logarithms require additional hypotheses.

S26-D-00000965 - Smooth Lie-group action

A smooth left action of G on M is a smooth map GxM->M satisfying e.p=p and g.(h.p)=(gh).p.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000951, S26-D-00000905

S26-D-00000966 - Orbit

The orbit G.p is the set of points reachable from p under the group action.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000965, S26-D-00000017

S26-D-00000967 - Stabilizer subgroup

The stabilizer G_p consists of group elements fixing p.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000965, S26-D-00000101

S26-D-00000968 - Fundamental vector field

An element X of g induces a vector field X_M by differentiating t |-> exp(tX).p at t=0.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000959, S26-D-00000965, S26-D-00000915

S26-D-00000969 - Equivariant map

For G-spaces M and N, a map F:M->N is equivariant when F(g.p)=g.F(p).

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000965, S26-D-00000906

S26-D-00000970 - Free action

An action is free when every stabilizer is trivial.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000967

S26-D-00000971 - Proper action

A smooth action is proper when the map GxM->MxM, (g,p)|->(g.p,p), is proper.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000965, S26-T-00000322

S26-T-00000972 - Quotient manifold interface

A free proper smooth action has a smooth quotient M/G for which the quotient map is a submersion and M->M/G is a principal G-bundle.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000970, S26-D-00000971, S26-D-00000905

Boundary note: Without freeness or properness the orbit space can be singular or non-Hausdorff.

S26-D-00000973 - Homogeneous space

A homogeneous G-space is a smooth G-space on which G acts transitively.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000965

S26-T-00000974 - Orbit-stabilizer geometry interface

Under standard embeddedness/closedness hypotheses, an orbit is modeled by a homogeneous space G/G_p and its tangent space is generated by fundamental vector fields.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000966, S26-D-00000967, S26-D-00000968, S26-D-00000973

Boundary note: Topological or smooth quotient conclusions require closedness/properness hypotheses.

S26-T-00000975 - Lie-symmetry frontier

Lie groups and Lie algebras encode continuous symmetries through global transformations, infinitesimal generators, actions, and homogeneous spaces.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000951, S26-D-00000954, S26-D-00000959, S26-D-00000965, S26-D-00000968

Applications: Geometry | Mechanics | Differential equations | Representation theory

S26-D-00000976 - Symplectic vector space

A symplectic vector space is an even-dimensional real vector space V with a nondegenerate alternating bilinear form omega.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000149, S26-T-00000254, S26-T-00000264

S26-D-00000977 - Symplectic orthogonal complement

For W subseteq V, W^omega consists of vectors v such that omega(v,w)=0 for every w in W.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000976, S26-T-00000155

S26-D-00000978 - Lagrangian subspace

A subspace L of a 2n-dimensional symplectic vector space is Lagrangian when L=L^omega, equivalently when omega|_L=0 and dim L=n.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000976, S26-D-00000977, S26-T-00000165

S26-D-00000979 - Symplectic manifold

A symplectic manifold is a smooth even-dimensional manifold M with a closed nondegenerate differential 2-form omega.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000905, S26-D-00000923, S26-D-00000928

Boundary note: Nondegeneracy without d omega=0 does not define a symplectic form.

S26-D-00000980 - Symplectomorphism

A diffeomorphism F:(M,omega_M)->(N,omega_N) is symplectic when F^*omega_N=omega_M.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000979, S26-D-00000907, S26-D-00000926

S26-D-00000981 - Canonical one-form on a cotangent bundle

The tautological one-form theta on T^*Q is defined by theta_alpha(v)=alpha(d pi(v)), where pi:T^*Q->Q.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000921, S26-D-00000912

S26-D-00000982 - Canonical symplectic form on T^*Q

The cotangent bundle carries the canonical symplectic form omega=-d theta (up to the explicitly declared sign convention).

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000981, S26-D-00000925, S26-D-00000979

Boundary note: The sign convention omega=d theta versus omega=-d theta varies across sources and must be declared.

S26-T-00000983 - Darboux theorem

Every symplectic manifold is locally symplectomorphic to the standard symplectic vector space; symplectic forms have no local scalar curvature-type invariants.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000979, S26-D-00000980, S26-D-00000976

S26-D-00000984 - Hamiltonian vector field

For H in C^infty(M), the Hamiltonian vector field X_H is uniquely defined by i_{X_H}omega=dH under the chosen sign convention.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000979, S26-D-00000288, S26-D-00000922, S26-D-00000915

S26-D-00000985 - Poisson bracket

The Poisson bracket of f and g is defined from the symplectic form and Hamiltonian vector fields, with sign convention declared consistently, e.g. {f,g}=omega(X_f,X_g).

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000984, S26-D-00000979

S26-T-00000986 - Poisson algebra identities

The Poisson bracket is bilinear, antisymmetric, satisfies the Jacobi identity, and is a derivation in each argument.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000985, S26-T-00000957

S26-T-00000987 - Hamiltonian conservation of H

Along the Hamiltonian flow of H, the Hamiltonian H is constant because dH(X_H)=omega(X_H,X_H)=0.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000984, S26-D-00000976

S26-T-00000988 - Hamiltonian flow is symplectic

The local flow generated by X_H preserves omega: phi_t^*omega=omega.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000984, S26-D-00000917, S26-D-00000925, S26-D-00000980

S26-D-00000989 - Hamilton equations in canonical coordinates

In canonical coordinates (q,p) with the standard symplectic convention, Hamiltonian trajectories satisfy qdot=partial H/partial p and pdot=-partial H/partial q.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000982, S26-D-00000984, S26-T-00000331

S26-D-00000990 - Symplectic gradient matrix

In R^{2n} with standard matrix J, Hamiltonian dynamics may be written zdot=J grad H(z), with J^T=-J and J^2=-I under the declared coordinate convention.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000976, S26-T-00000213, S26-D-00000343

S26-D-00000991 - Liouville volume

On a 2n-dimensional symplectic manifold, omega^n/n! is a nowhere-vanishing volume form.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000979, S26-D-00000924, S26-T-00000287

S26-T-00000992 - Liouville theorem for Hamiltonian flow

Hamiltonian flow preserves the symplectic volume, and hence phase-space volume, wherever the flow is defined.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-T-00000988, S26-D-00000991

S26-D-00000993 - Momentum map

For a symplectic G-action, a momentum map J:M->g^* packages Hamiltonians for fundamental vector fields when the action is Hamiltonian, subject to the chosen sign and equivariance conventions.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000965, S26-D-00000968, S26-D-00000979, S26-D-00000251, S26-D-00000984

Boundary note: Not every symplectic group action admits a globally defined equivariant momentum map.

S26-T-00000994 - Hamiltonian Noether interface

For a Hamiltonian symmetry generated by a momentum-map component that Poisson-commutes with H, that component is conserved along the H-flow.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000993, S26-D-00000985, S26-T-00000987

S26-D-00000995 - Lagrangian submanifold

A submanifold L of a symplectic 2n-manifold is Lagrangian when each tangent space T_pL is a Lagrangian subspace of T_pM.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000978, S26-D-00000979, S26-D-00000910

S26-D-00000996 - Exact Lagrangian interface

In an exact symplectic manifold omega=-d theta, a Lagrangian L is exact when theta|_L is an exact 1-form.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000982, S26-D-00000995, S26-D-00000929

S26-D-00000997 - Canonical transformation

A canonical transformation in Hamiltonian mechanics is a symplectomorphism of phase space; coordinate formulas must preserve the symplectic form rather than merely be invertible.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000980, S26-D-00000989

Boundary note: An arbitrary invertible coordinate change need not be canonical/symplectic.

S26-B-00000998 - Lagrangian-Hamiltonian bridge

Under regularity/nondegeneracy hypotheses on the Lagrangian, the Legendre transform transports Euler-Lagrange dynamics on TQ to Hamiltonian dynamics on T^*Q.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-D-00000803, S26-D-00000982, S26-D-00000989, S26-D-00000180

Boundary note: Degenerate Lagrangians require constraint theory; the ordinary Legendre transform need not be invertible.

S26-T-00000999 - Symmetry-geometry-mechanics synthesis

Lie actions, momentum maps, symplectic forms, Poisson brackets, and variational dynamics form a common structure linking continuous symmetry to conservation and phase-space evolution.

Status: A - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: Smoothness, finite-dimensionality, connectedness, regularity, and global hypotheses are exactly as stated; local statements are not silently globalized.

Hard dependencies: S26-T-00000975, S26-D-00000979, S26-D-00000985, S26-D-00000993, S26-T-00000994, S26-B-00000998

Applications: Mechanics | Geometric integration | PDE | Mathematical physics

S26-N-00001000 - Volume 0 closure invariant

Volume 0 closes the foundational substrate at 1000 addressable objects. Later volumes must reuse these IDs, preserve evidence states and correction history, and introduce new primitives only when an explicit obstruction cannot be represented by the existing substrate.

Status: S - Exact registered statement only; examples, matrix calculations, and symbolic checks do not promote the general claim to P.

Assumptions: This is a project-governance invariant, not a mathematical theorem.

Hard dependencies: S26-T-00000027, S26-T-00000029, S26-N-00000030, S26-T-00000950, S26-T-00000999

Applications: All later Synopsis volumes

Appendix - Global closure audit and active remainder

The final hard graph contains 2217 edges and 583 cross-domain hard edges. Its maximum dependency depth is 46.

Three hard prerequisites are forward-numbered rather than backward-numbered. All are resolved, acyclic, and explicitly documented. Numeric order is editorial; dependency order is topological.

Twenty-one high lexical-similarity title pairs were placed in a semantic review queue. None is an exact duplicate title or exact duplicate statement. They are retained as distinct paired theorems, specializations, or domain interfaces.

The declared symbol registry contains 34 canonical symbol keys. One symbol is reused with explicit owner/domain scope; no declared collision remains unresolved. Formula-bound local variables are not treated as global symbols.

The load-bearing epistemic correction is the removal of unsupported P promotion. Under v1.0, no registry entry is P unless an exact independent proof/formal verification artifact is attached.

Active remainder: 985 mathematical entries remain A. This release closes the foundational ontology and dependency substrate; theorem-by-theorem source attachment and formal verification are a separate promotion programme.

Frozen closure invariant. Later volumes must reuse Volume 0 IDs, preserve evidence states and correction history, and add new primitives only when an explicit obstruction cannot be represented by the frozen substrate.

A Synopsis of Results in Pure, Applied and Computational Mathematics

2026 Edition

Volume I
Number, Algebra & Discrete Structure

Release: 1.0.0
Canonical range: 1001-2000
New objects: 1,000
Combined hard edges: 3217
Combined hard graph: 1 component / 0 cycles / 0 dangling
Volume I evidence: A=399, M=600, H=1, P=0
Finite-profile checks: 15/15 PASS
Audit controls: 6/6 PASS

Frozen Volume 0 concepts are reused by canonical ID. Computed finite profiles are M; general mathematics remains A unless independently witnessed.

Release audit

Gate	Observed	Result
Combined registry	2,000 sequential objects	PASS
Hard graph	3217 edges; one component; zero cycles	PASS
Dangling prerequisites	0	PASS
Exact duplicate titles/statements	0 / 0	PASS
Volume I evidence	A=399; M=600; H=1; P=0	PASS
Finite profile checks	15 / 15	PASS
Mutation/metamorphic controls	6 / 6	PASS

Boundary. The 600 M entries are exact finite instantiations/tables. They do not prove the 399 general A-status mathematical claims. The Riemann hypothesis is explicitly C/H.

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Part A - Arithmetic functions and multiplicative structure

S26-D-00001001 - Arithmetic function

An arithmetic function is a function f:N_{>0}->C (or another declared codomain) studied through the multiplicative structure of positive integers.

Evidence A. Hard: S26-T-00000076.

S26-D-00001002 - Multiplicative arithmetic function

An arithmetic function f is multiplicative when f(1)=1 and f(mn)=f(m)f(n) whenever gcd(m,n)=1.

Evidence A. Hard: S26-D-00001001.

S26-D-00001003 - Completely multiplicative function

An arithmetic function f is completely multiplicative when f(mn)=f(m)f(n) for all positive integers m,n.

Evidence A. Hard: S26-D-00001002.

S26-D-00001004 - Dirichlet convolution

For arithmetic functions f and g, (f*g)(n)=sum_{d|n} f(d)g(n/d).

Evidence A. Hard: S26-D-00001003.

S26-T-00001005 - Dirichlet convolution is commutative and associative

Dirichlet convolution on arithmetic functions is commutative and associative; the delta function epsilon with epsilon(1)=1 and epsilon(n)=0 for n>1 is its identity.

Evidence A. Hard: S26-D-00001004.

S26-T-00001006 - Multiplicative functions are closed under Dirichlet convolution

If f and g are multiplicative arithmetic functions, then f*g is multiplicative.

Evidence A. Hard: S26-T-00001005.

S26-D-00001007 - Moebius function

The Moebius function mu satisfies mu(1)=1; mu(n)=0 when a square of a prime divides n; otherwise mu(n)=(-1)^r when n is a product of r distinct primes.

Evidence A. Hard: S26-T-00001006.

S26-T-00001008 - Moebius inversion

If F(n)=sum_{d|n} f(d), then f(n)=sum_{d|n} mu(d)F(n/d). Equivalently, mu is the Dirichlet-convolution inverse of the constant-one function.

Evidence A. Hard: S26-D-00001007.

S26-D-00001009 - Euler totient function

phi(n) is the number of residue classes modulo n that are coprime to n.

Evidence A. Hard: S26-T-00001008.

S26-I-00001010 - Euler totient product formula

For n=product p_i^{a_i}, phi(n)=n product_{p|n}(1-1/p).

Evidence A. Hard: S26-D-00001009.

S26-D-00001011 - Divisor-counting function

tau(n)=sum_{d|n}1 counts the positive divisors of n.

Evidence A. Hard: S26-I-00001010.

S26-I-00001012 - Divisor-counting prime-power formula

If n=product p_i^{a_i}, then tau(n)=product_i(a_i+1).

Evidence A. Hard: S26-D-00001011.

S26-D-00001013 - Sum-of-divisors function

sigma_k(n)=sum_{d|n} d^k; sigma_1 is the ordinary sum-of-divisors function.

Evidence A. Hard: S26-I-00001012.

S26-I-00001014 - Sum-of-divisors prime-power formula

For k not equal to 0 and n=product p_i^{a_i}, sigma_k(n)=product_i (p_i^{k(a_i+1)}-1)/(p_i^k-1).

Evidence A. Hard: S26-D-00001013.

S26-D-00001015 - Liouville function

lambda(n)=(-1)^{Omega(n)}, where Omega counts prime factors with multiplicity.

Evidence A. Hard: S26-I-00001014.

S26-D-00001016 - von Mangoldt function

Lambda(n)=log p if n=p^k for a prime p and integer k>=1, and Lambda(n)=0 otherwise.

Evidence A. Hard: S26-D-00001015.

S26-I-00001017 - Logarithm as divisor sum of von Mangoldt

For n>=1, log n=sum_{d|n} Lambda(d).

Evidence A. Hard: S26-D-00001016.

S26-D-00001018 - Jordan totient function

J_k(n)=n^k product_{p|n}(1-p^{-k}); it counts ordered k-tuples modulo n that together with n are coprime in the standard gcd sense.

Evidence A. Hard: S26-I-00001017.

S26-D-00001019 - Ramanujan sum

c_q(n)=sum_{1<=a<=q, gcd(a,q)=1} exp(2 pi i a n/q).

Evidence A. Hard: S26-D-00001018.

S26-I-00001020 - Ramanujan sum divisor formula

c_q(n)=sum_{d|gcd(q,n)} d mu(q/d).

Evidence A. Hard: S26-D-00001019.

S26-E-00001021 - Arithmetic profile of 1

1 has prime factorization {}; phi(1)=1, tau(1)=1, sigma_1(1)=1, and mu(1)=1.

Evidence M. Hard: S26-I-00001020.

S26-E-00001022 - Arithmetic profile of 2

2 has prime factorization {2: 1}; phi(2)=1, tau(2)=2, sigma_1(2)=3, and mu(2)=-1.

Evidence M. Hard: S26-E-00001021.

S26-E-00001023 - Arithmetic profile of 3

3 has prime factorization {3: 1}; phi(3)=2, tau(3)=2, sigma_1(3)=4, and mu(3)=-1.

Evidence M. Hard: S26-E-00001022.

S26-E-00001024 - Arithmetic profile of 4

4 has prime factorization {2: 2}; phi(4)=2, tau(4)=3, sigma_1(4)=7, and mu(4)=0.

Evidence M. Hard: S26-E-00001023.

S26-E-00001025 - Arithmetic profile of 5

5 has prime factorization {5: 1}; phi(5)=4, tau(5)=2, sigma_1(5)=6, and mu(5)=-1.

Evidence M. Hard: S26-E-00001024.

S26-E-00001026 - Arithmetic profile of 6

6 has prime factorization {2: 1, 3: 1}; phi(6)=2, tau(6)=4, sigma_1(6)=12, and mu(6)=1.

Evidence M. Hard: S26-E-00001025.

S26-E-00001027 - Arithmetic profile of 7

7 has prime factorization {7: 1}; phi(7)=6, tau(7)=2, sigma_1(7)=8, and mu(7)=-1.

Evidence M. Hard: S26-E-00001026.

S26-E-00001028 - Arithmetic profile of 8

8 has prime factorization {2: 3}; phi(8)=4, tau(8)=4, sigma_1(8)=15, and mu(8)=0.

Evidence M. Hard: S26-E-00001027.

S26-E-00001029 - Arithmetic profile of 9

9 has prime factorization {3: 2}; phi(9)=6, tau(9)=3, sigma_1(9)=13, and mu(9)=0.

Evidence M. Hard: S26-E-00001028.

S26-E-00001030 - Arithmetic profile of 10

10 has prime factorization {2: 1, 5: 1}; phi(10)=4, tau(10)=4, sigma_1(10)=18, and mu(10)=1.

Evidence M. Hard: S26-E-00001029.

S26-E-00001031 - Arithmetic profile of 11

11 has prime factorization {11: 1}; phi(11)=10, tau(11)=2, sigma_1(11)=12, and mu(11)=-1.

Evidence M. Hard: S26-E-00001030.

S26-E-00001032 - Arithmetic profile of 12

12 has prime factorization {2: 2, 3: 1}; phi(12)=4, tau(12)=6, sigma_1(12)=28, and mu(12)=0.

Evidence M. Hard: S26-E-00001031.

S26-E-00001033 - Arithmetic profile of 13

13 has prime factorization {13: 1}; phi(13)=12, tau(13)=2, sigma_1(13)=14, and mu(13)=-1.

Evidence M. Hard: S26-E-00001032.

S26-E-00001034 - Arithmetic profile of 14

14 has prime factorization {2: 1, 7: 1}; phi(14)=6, tau(14)=4, sigma_1(14)=24, and mu(14)=1.

Evidence M. Hard: S26-E-00001033.

S26-E-00001035 - Arithmetic profile of 15

15 has prime factorization {3: 1, 5: 1}; phi(15)=8, tau(15)=4, sigma_1(15)=24, and mu(15)=1.

Evidence M. Hard: S26-E-00001034.

S26-E-00001036 - Arithmetic profile of 16

16 has prime factorization {2: 4}; phi(16)=8, tau(16)=5, sigma_1(16)=31, and mu(16)=0.

Evidence M. Hard: S26-E-00001035.

S26-E-00001037 - Arithmetic profile of 17

17 has prime factorization {17: 1}; phi(17)=16, tau(17)=2, sigma_1(17)=18, and mu(17)=-1.

Evidence M. Hard: S26-E-00001036.

S26-E-00001038 - Arithmetic profile of 18

18 has prime factorization {2: 1, 3: 2}; phi(18)=6, tau(18)=6, sigma_1(18)=39, and mu(18)=0.

Evidence M. Hard: S26-E-00001037.

S26-E-00001039 - Arithmetic profile of 19

19 has prime factorization {19: 1}; phi(19)=18, tau(19)=2, sigma_1(19)=20, and mu(19)=-1.

Evidence M. Hard: S26-E-00001038.

S26-E-00001040 - Arithmetic profile of 20

20 has prime factorization {2: 2, 5: 1}; phi(20)=8, tau(20)=6, sigma_1(20)=42, and mu(20)=0.

Evidence M. Hard: S26-E-00001039.

S26-E-00001041 - Arithmetic profile of 21

21 has prime factorization {3: 1, 7: 1}; phi(21)=12, tau(21)=4, sigma_1(21)=32, and mu(21)=1.

Evidence M. Hard: S26-E-00001040.

S26-E-00001042 - Arithmetic profile of 22

22 has prime factorization {2: 1, 11: 1}; phi(22)=10, tau(22)=4, sigma_1(22)=36, and mu(22)=1.

Evidence M. Hard: S26-E-00001041.

S26-E-00001043 - Arithmetic profile of 23

23 has prime factorization {23: 1}; phi(23)=22, tau(23)=2, sigma_1(23)=24, and mu(23)=-1.

Evidence M. Hard: S26-E-00001042.

S26-E-00001044 - Arithmetic profile of 24

24 has prime factorization {2: 3, 3: 1}; phi(24)=8, tau(24)=8, sigma_1(24)=60, and mu(24)=0.

Evidence M. Hard: S26-E-00001043.

S26-E-00001045 - Arithmetic profile of 25

25 has prime factorization {5: 2}; phi(25)=20, tau(25)=3, sigma_1(25)=31, and mu(25)=0.

Evidence M. Hard: S26-E-00001044.

S26-E-00001046 - Arithmetic profile of 26

26 has prime factorization {2: 1, 13: 1}; phi(26)=12, tau(26)=4, sigma_1(26)=42, and mu(26)=1.

Evidence M. Hard: S26-E-00001045.

S26-E-00001047 - Arithmetic profile of 27

27 has prime factorization {3: 3}; phi(27)=18, tau(27)=4, sigma_1(27)=40, and mu(27)=0.

Evidence M. Hard: S26-E-00001046.

S26-E-00001048 - Arithmetic profile of 28

28 has prime factorization {2: 2, 7: 1}; phi(28)=12, tau(28)=6, sigma_1(28)=56, and mu(28)=0.

Evidence M. Hard: S26-E-00001047.

S26-E-00001049 - Arithmetic profile of 29

29 has prime factorization {29: 1}; phi(29)=28, tau(29)=2, sigma_1(29)=30, and mu(29)=-1.

Evidence M. Hard: S26-E-00001048.

S26-E-00001050 - Arithmetic profile of 30

30 has prime factorization {2: 1, 3: 1, 5: 1}; phi(30)=8, tau(30)=8, sigma_1(30)=72, and mu(30)=-1.

Evidence M. Hard: S26-E-00001049.

Part B - Congruences, orders, and reciprocity

S26-B-00001051 - Congruence relation reuse in Volume I

Volume I reuses the frozen congruence relation S26-D-00000079 and treats it as the ambient equivalence relation for the modular-arithmetic developments that follow.

Evidence A. Hard: S26-D-00000079.

S26-T-00001052 - Congruence respects ring operations

If a congruent b and c congruent d modulo n, then a+c congruent b+d and ac congruent bd modulo n.

Evidence A. Hard: S26-B-00001051.

S26-T-00001053 - Linear congruence solvability

The congruence ax congruent b mod n is solvable iff gcd(a,n) divides b; when solvable it has exactly gcd(a,n) residue-class solutions modulo n.

Evidence A. Hard: S26-T-00001052.

S26-T-00001054 - Chinese remainder theorem

For pairwise coprime moduli n_1,...,n_r, the natural map Z/(product n_i) -> product Z/n_i is a ring isomorphism.

Evidence A. Hard: S26-T-00001053.

S26-D-00001055 - Multiplicative order modulo n

For gcd(a,n)=1, ord_n(a) is the least positive k with a^k congruent 1 mod n.

Evidence A. Hard: S26-T-00001054.

S26-T-00001056 - Order divides group exponent

ord_n(a) divides phi(n); more sharply it divides the exponent lambda_C(n) of (Z/nZ)^x.

Evidence A. Hard: S26-D-00001055.

S26-D-00001057 - Carmichael function

The Carmichael function lambda_C(n) is the exponent of the finite group (Z/nZ)^x.

Evidence A. Hard: S26-T-00001056.

S26-T-00001058 - Euler theorem

If gcd(a,n)=1, then a^{phi(n)} congruent 1 mod n.

Evidence A. Hard: S26-D-00001057.

S26-T-00001059 - Fermat little theorem

For prime p, a^p congruent a mod p; if p does not divide a then a^{p-1} congruent 1 mod p.

Evidence A. Hard: S26-T-00001058.

S26-D-00001060 - Primitive root modulo n

A primitive root modulo n is a generator of the cyclic group (Z/nZ)^x when that group is cyclic.

Evidence A. Hard: S26-T-00001059.

S26-T-00001061 - Primitive-root classification

Primitive roots exist modulo n exactly for n in {1,2,4,p^k,2p^k} with p an odd prime.

Evidence A. Hard: S26-D-00001060.

S26-D-00001062 - Quadratic residue

For odd prime p and p not dividing a, a is a quadratic residue mod p when x^2 congruent a mod p is solvable.

Evidence A. Hard: S26-T-00001061.

S26-D-00001063 - Legendre symbol

For odd prime p, (a/p) is 0 if p|a, 1 if a is a nonzero quadratic residue mod p, and -1 otherwise.

Evidence A. Hard: S26-D-00001062.

S26-T-00001064 - Euler criterion

For odd prime p, (a/p) congruent a^{(p-1)/2} mod p, interpreted as the values 0,+1,-1.

Evidence A. Hard: S26-D-00001063.

S26-T-00001065 - Quadratic reciprocity

For distinct odd primes p,q, (p/q)(q/p)=(-1)^{((p-1)/2)((q-1)/2)}.

Evidence A. Hard: S26-T-00001064.

S26-I-00001066 - First supplementary law

For odd prime p, (-1/p)=(-1)^{(p-1)/2}.

Evidence A. Hard: S26-T-00001065.

S26-I-00001067 - Second supplementary law

For odd prime p, (2/p)=(-1)^{(p^2-1)/8}.

Evidence A. Hard: S26-I-00001066.

S26-D-00001068 - Jacobi symbol

For odd positive n with prime factorization n=product p_i^{e_i}, (a/n)=product (a/p_i)^{e_i}; value 1 does not by itself imply a is a square mod composite n.

Evidence A. Hard: S26-I-00001067.

S26-T-00001069 - Hensel lifting for simple roots

If f(a) congruent 0 mod p and f'(a) is nonzero mod p, then a lifts uniquely to a root modulo p^k for every k>=1.

Evidence A. Hard: S26-D-00001068.

S26-B-00001070 - Local modular viewpoint

Congruence problems modulo composite n decompose into prime-power problems by the Chinese remainder theorem, providing the finite local model used later in p-adic and algebraic-number-theoretic analysis.

Evidence A. Hard: S26-T-00001069.

S26-E-00001071 - Quadratic-residue profile modulo 3

Modulo the prime 3, a primitive root is 2; the nonzero quadratic residues are [1].

Evidence M. Hard: S26-B-00001070.

S26-E-00001072 - Quadratic-residue profile modulo 5

Modulo the prime 5, a primitive root is 2; the nonzero quadratic residues are [1, 4].

Evidence M. Hard: S26-E-00001071.

S26-E-00001073 - Quadratic-residue profile modulo 7

Modulo the prime 7, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4].

Evidence M. Hard: S26-E-00001072.

S26-E-00001074 - Quadratic-residue profile modulo 11

Modulo the prime 11, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 5, 9].

Evidence M. Hard: S26-E-00001073.

S26-E-00001075 - Quadratic-residue profile modulo 13

Modulo the prime 13, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 9, 10, 12].

Evidence M. Hard: S26-E-00001074.

S26-E-00001076 - Quadratic-residue profile modulo 17

Modulo the prime 17, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 8, 9, 13, 15, 16].

Evidence M. Hard: S26-E-00001075.

S26-E-00001077 - Quadratic-residue profile modulo 19

Modulo the prime 19, a primitive root is 2; the nonzero quadratic residues are [1, 4, 5, 6, 7, 9, 11, 16, 17].

Evidence M. Hard: S26-E-00001076.

S26-E-00001078 - Quadratic-residue profile modulo 23

Modulo the prime 23, a primitive root is 5; the nonzero quadratic residues are [1, 2, 3, 4, 6, 8, 9, 12, 13, 16, 18].

Evidence M. Hard: S26-E-00001077.

S26-E-00001079 - Quadratic-residue profile modulo 29

Modulo the prime 29, a primitive root is 2; the nonzero quadratic residues are [1, 4, 5, 6, 7, 9, 13, 16, 20, 22, 23, 24, 25, 28].

Evidence M. Hard: S26-E-00001078.

S26-E-00001080 - Quadratic-residue profile modulo 31

Modulo the prime 31, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 5, 7, 8, 9, 10, 14, 16, 18, 19, 20, 25, 28].

Evidence M. Hard: S26-E-00001079.

S26-E-00001081 - Quadratic-residue profile modulo 37

Modulo the prime 37, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 7, 9, 10, 11, 12, 16, 21, 25, 26, 27, 28, 30, 33, 34, 36].

Evidence M. Hard: S26-E-00001080.

S26-E-00001082 - Quadratic-residue profile modulo 41

Modulo the prime 41, a primitive root is 6; the nonzero quadratic residues are [1, 2, 4, 5, 8, 9, 10, 16, 18, 20, 21, 23, 25, 31, 32, 33, 36, 37, 39, 40].

Evidence M. Hard: S26-E-00001081.

S26-E-00001083 - Quadratic-residue profile modulo 43

Modulo the prime 43, a primitive root is 3; the nonzero quadratic residues are [1, 4, 6, 9, 10, 11, 13, 14, 15, 16, 17, 21, 23, 24, 25, 31, 35, 36, 38, 40, 41].

Evidence M. Hard: S26-E-00001082.

S26-E-00001084 - Quadratic-residue profile modulo 47

Modulo the prime 47, a primitive root is 5; the nonzero quadratic residues are [1, 2, 3, 4, 6, 7, 8, 9, 12, 14, 16, 17, 18, 21, 24, 25, 27, 28, 32, 34, 36, 37, 42].

Evidence M. Hard: S26-E-00001083.

S26-E-00001085 - Quadratic-residue profile modulo 53

Modulo the prime 53, a primitive root is 2; the nonzero quadratic residues are [1, 4, 6, 7, 9, 10, 11, 13, 15, 16, 17, 24, 25, 28, 29, 36, 37, 38, 40, 42, 43, 44, 46, 47, 49, 52].

Evidence M. Hard: S26-E-00001084.

S26-E-00001086 - Quadratic-residue profile modulo 59

Modulo the prime 59, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 5, 7, 9, 12, 15, 16, 17, 19, 20, 21, 22, 25, 26, 27, 28, 29, 35, 36, 41, 45, 46, 48, 49, 51, 53, 57].

Evidence M. Hard: S26-E-00001085.

S26-E-00001087 - Quadratic-residue profile modulo 61

Modulo the prime 61, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 5, 9, 12, 13, 14, 15, 16, 19, 20, 22, 25, 27, 34, 36, 39, 41, 42, 45, 46, 47, 48, 49, 52, 56, 57, 58, 60].

Evidence M. Hard: S26-E-00001086.

S26-E-00001088 - Quadratic-residue profile modulo 67

Modulo the prime 67, a primitive root is 2; the nonzero quadratic residues are [1, 4, 6, 9, 10, 14, 15, 16, 17, 19, 21, 22, 23, 24, 25, 26, 29, 33, 35, 36, 37, 39, 40, 47, 49, 54, 55, 56, 59, 60, 62, 64, 65].

Evidence M. Hard: S26-E-00001087.

S26-E-00001089 - Quadratic-residue profile modulo 71

Modulo the prime 71, a primitive root is 7; the nonzero quadratic residues are [1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 16, 18, 19, 20, 24, 25, 27, 29, 30, 32, 36, 37, 38, 40, 43, 45, 48, 49, 50, 54, 57, 58, 60, 64].

Evidence M. Hard: S26-E-00001088.

S26-E-00001090 - Quadratic-residue profile modulo 73

Modulo the prime 73, a primitive root is 5; the nonzero quadratic residues are [1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 19, 23, 24, 25, 27, 32, 35, 36, 37, 38, 41, 46, 48, 49, 50, 54, 55, 57, 61, 64, 65, 67, 69, 70, 71, 72].

Evidence M. Hard: S26-E-00001089.

S26-E-00001091 - Quadratic-residue profile modulo 79

Modulo the prime 79, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 5, 8, 9, 10, 11, 13, 16, 18, 19, 20, 21, 22, 23, 25, 26, 31, 32, 36, 38, 40, 42, 44, 45, 46, 49, 50, 51, 52, 55, 62, 64, 65, 67, 72, 73, 76].

Evidence M. Hard: S26-E-00001090.

S26-E-00001092 - Quadratic-residue profile modulo 83

Modulo the prime 83, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 7, 9, 10, 11, 12, 16, 17, 21, 23, 25, 26, 27, 28, 29, 30, 31, 33, 36, 37, 38, 40, 41, 44, 48, 49, 51, 59, 61, 63, 64, 65, 68, 69, 70, 75, 77, 78, 81].

Evidence M. Hard: S26-E-00001091.

S26-E-00001093 - Quadratic-residue profile modulo 89

Modulo the prime 89, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 5, 8, 9, 10, 11, 16, 17, 18, 20, 21, 22, 25, 32, 34, 36, 39, 40, 42, 44, 45, 47, 49, 50, 53, 55, 57, 64, 67, 68, 69, 71, 72, 73, 78, 79, 80, 81, 84, 85, 87, 88].

Evidence M. Hard: S26-E-00001092.

S26-E-00001094 - Quadratic-residue profile modulo 97

Modulo the prime 97, a primitive root is 5; the nonzero quadratic residues are [1, 2, 3, 4, 6, 8, 9, 11, 12, 16, 18, 22, 24, 25, 27, 31, 32, 33, 35, 36, 43, 44, 47, 48, 49, 50, 53, 54, 61, 62, 64, 65, 66, 70, 72, 73, 75, 79, 81, 85, 86, 88, 89, 91, 93, 94, 95, 96].

Evidence M. Hard: S26-E-00001093.

S26-E-00001095 - Quadratic-residue profile modulo 101

Modulo the prime 101, a primitive root is 2; the nonzero quadratic residues are [1, 4, 5, 6, 9, 13, 14, 16, 17, 19, 20, 21, 22, 23, 24, 25, 30, 31, 33, 36, 37, 43, 45, 47, 49, 52, 54, 56, 58, 64, 65, 68, 70, 71, 76, 77, 78, 79, 80, 81, 82, 84, 85, 87, 88, 92, 95, 96, 97, 100].

Evidence M. Hard: S26-E-00001094.

S26-E-00001096 - Quadratic-residue profile modulo 103

Modulo the prime 103, a primitive root is 5; the nonzero quadratic residues are [1, 2, 4, 7, 8, 9, 13, 14, 15, 16, 17, 18, 19, 23, 25, 26, 28, 29, 30, 32, 33, 34, 36, 38, 41, 46, 49, 50, 52, 55, 56, 58, 59, 60, 61, 63, 64, 66, 68, 72, 76, 79, 81, 82, 83, 91, 92, 93, 97, 98, 100].

Evidence M. Hard: S26-E-00001095.

S26-E-00001097 - Quadratic-residue profile modulo 107

Modulo the prime 107, a primitive root is 2; the nonzero quadratic residues are [1, 3, 4, 9, 10, 11, 12, 13, 14, 16, 19, 23, 25, 27, 29, 30, 33, 34, 35, 36, 37, 39, 40, 41, 42, 44, 47, 48, 49, 52, 53, 56, 57, 61, 62, 64, 69, 75, 76, 79, 81, 83, 85, 86, 87, 89, 90, 92, 99, 100, 101, 102, 105].

Evidence M. Hard: S26-E-00001096.

S26-E-00001098 - Quadratic-residue profile modulo 109

Modulo the prime 109, a primitive root is 6; the nonzero quadratic residues are [1, 3, 4, 5, 7, 9, 12, 15, 16, 20, 21, 22, 25, 26, 27, 28, 29, 31, 34, 35, 36, 38, 43, 45, 46, 48, 49, 60, 61, 63, 64, 66, 71, 73, 74, 75, 78, 80, 81, 82, 83, 84, 87, 88, 89, 93, 94, 97, 100, 102, 104, 105, 106, 108].

Evidence M. Hard: S26-E-00001097.

S26-E-00001099 - Quadratic-residue profile modulo 113

Modulo the prime 113, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 7, 8, 9, 11, 13, 14, 15, 16, 18, 22, 25, 26, 28, 30, 31, 32, 36, 41, 44, 49, 50, 51, 52, 53, 56, 57, 60, 61, 62, 63, 64, 69, 72, 77, 81, 82, 83, 85, 87, 88, 91, 95, 97, 98, 99, 100, 102, 104, 105, 106, 109, 111, 112].

Evidence M. Hard: S26-E-00001098.

S26-E-00001100 - Quadratic-residue profile modulo 127

Modulo the prime 127, a primitive root is 3; the nonzero quadratic residues are [1, 2, 4, 8, 9, 11, 13, 15, 16, 17, 18, 19, 21, 22, 25, 26, 30, 31, 32, 34, 35, 36, 37, 38, 41, 42, 44, 47, 49, 50, 52, 60, 61, 62, 64, 68, 69, 70, 71, 72, 73, 74, 76, 79, 81, 82, 84, 87, 88, 94, 98, 99, 100, 103, 104, 107, 113, 115, 117, 120, 121, 122, 124].

Evidence M. Hard: S26-E-00001099.

Part C - Diophantine equations

S26-D-00001101 - Diophantine equation

A Diophantine equation is a polynomial equation with integer coefficients for which integer or rational solutions are sought.

Evidence A. Hard: S26-T-00000072.

S26-T-00001102 - Linear Diophantine equation criterion

ax+by=c has integer solutions iff gcd(a,b) divides c.

Evidence A. Hard: S26-D-00001101.

S26-I-00001103 - Parametrization of linear Diophantine solutions

If ax_0+by_0=c and g=gcd(a,b), all integer solutions are x=x_0+(b/g)t, y=y_0-(a/g)t with t in Z.

Evidence A. Hard: S26-T-00001102.

S26-D-00001104 - Pythagorean triple

A Pythagorean triple is an integer solution x^2+y^2=z^2; it is primitive when gcd(x,y,z)=1.

Evidence A. Hard: S26-I-00001103.

S26-T-00001105 - Primitive Pythagorean parametrization

Up to swapping legs, every primitive positive Pythagorean triple is x=m^2-n^2, y=2mn, z=m^2+n^2 for coprime m>n of opposite parity.

Evidence A. Hard: S26-D-00001104.

S26-D-00001106 - Pell equation

For nonsquare positive D, Pell's equation is x^2-Dy^2=1.

Evidence A. Hard: S26-T-00001105.

S26-T-00001107 - Pell solvability

For every positive nonsquare D, Pell's equation has infinitely many integer solutions generated from powers of a fundamental unit of norm 1 in Q(sqrt(D)).

Evidence A. Hard: S26-D-00001106.

S26-D-00001108 - Generalized Pell equation

x^2-Dy^2=N is the norm equation N_{Q(sqrt D)/Q}(x+y sqrt D)=N.

Evidence A. Hard: S26-T-00001107.

S26-T-00001109 - Fermat theorem on sums of two squares

A positive integer n is a sum of two integer squares iff every prime congruent 3 mod 4 occurs to an even exponent in the factorization of n.

Evidence A. Hard: S26-D-00001108.

S26-I-00001110 - Two-square representation count

The number r_2(n) of ordered signed representations n=x^2+y^2 satisfies r_2(n)=4(d_1(n)-d_3(n)), where d_j counts divisors congruent j mod 4.

Evidence A. Hard: S26-T-00001109.

S26-T-00001111 - Lagrange four-square theorem

Every nonnegative integer is a sum of four integer squares.

Evidence A. Hard: S26-I-00001110.

S26-T-00001112 - Legendre three-square theorem

A nonnegative integer n is a sum of three integer squares iff n is not of the form 4^a(8b+7).

Evidence A. Hard: S26-T-00001111.

S26-D-00001113 - Mordell equation

A Mordell equation has the form y^2=x^3+k with fixed nonzero integer k.

Evidence A. Hard: S26-T-00001112.

S26-D-00001114 - Thue equation

A Thue equation F(x,y)=m uses an irreducible binary form F of degree at least 3 and fixed nonzero integer m.

Evidence A. Hard: S26-D-00001113.

S26-T-00001115 - Thue finiteness theorem

For irreducible binary form F of degree at least 3 and nonzero m, F(x,y)=m has finitely many integer solutions.

Evidence A. Hard: S26-D-00001114.

S26-D-00001116 - S-unit equation

An S-unit equation has form x+y=1 with x,y units outside a finite set S of places.

Evidence A. Hard: S26-T-00001115.

S26-T-00001117 - S-unit finiteness

For a fixed number field and finite S, the S-unit equation x+y=1 has finitely many solutions.

Evidence A. Hard: S26-D-00001116.

S26-D-00001118 - Local obstruction

A Diophantine equation has a local obstruction if it has no solution over some completion such as R or Q_p; then it has no rational solution.

Evidence A. Hard: S26-T-00001117.

S26-N-00001119 - Hasse-principle warning

Solvability over every completion need not imply rational solvability for arbitrary varieties; local-to-global principles require theorem-specific hypotheses.

Evidence A. Hard: S26-D-00001118. Boundary: Local solvability is not universally sufficient for global solvability.

S26-B-00001120 - Diophantine-to-geometry bridge

Polynomial Diophantine problems can be recast as rational/integral-point problems on algebraic varieties, opening access to geometry, Galois theory, and height methods.

Evidence A. Hard: S26-N-00001119.

S26-E-00001121 - Fundamental Pell solution for D=2

The least positive solution returned for x^2-2y^2=1 is x=3, y=2; direct substitution gives 1.

Evidence M. Hard: S26-B-00001120.

S26-E-00001122 - Fundamental Pell solution for D=3

The least positive solution returned for x^2-3y^2=1 is x=2, y=1; direct substitution gives 1.

Evidence M. Hard: S26-E-00001121.

S26-E-00001123 - Fundamental Pell solution for D=5

The least positive solution returned for x^2-5y^2=1 is x=9, y=4; direct substitution gives 1.

Evidence M. Hard: S26-E-00001122.

S26-E-00001124 - Fundamental Pell solution for D=6

The least positive solution returned for x^2-6y^2=1 is x=5, y=2; direct substitution gives 1.

Evidence M. Hard: S26-E-00001123.

S26-E-00001125 - Fundamental Pell solution for D=7

The least positive solution returned for x^2-7y^2=1 is x=8, y=3; direct substitution gives 1.

Evidence M. Hard: S26-E-00001124.

S26-E-00001126 - Fundamental Pell solution for D=8

The least positive solution returned for x^2-8y^2=1 is x=3, y=1; direct substitution gives 1.

Evidence M. Hard: S26-E-00001125.

S26-E-00001127 - Fundamental Pell solution for D=10

The least positive solution returned for x^2-10y^2=1 is x=19, y=6; direct substitution gives 1.

Evidence M. Hard: S26-E-00001126.

S26-E-00001128 - Fundamental Pell solution for D=11

The least positive solution returned for x^2-11y^2=1 is x=10, y=3; direct substitution gives 1.

Evidence M. Hard: S26-E-00001127.

S26-E-00001129 - Fundamental Pell solution for D=12

The least positive solution returned for x^2-12y^2=1 is x=7, y=2; direct substitution gives 1.

Evidence M. Hard: S26-E-00001128.

S26-E-00001130 - Fundamental Pell solution for D=13

The least positive solution returned for x^2-13y^2=1 is x=649, y=180; direct substitution gives 1.

Evidence M. Hard: S26-E-00001129.

S26-E-00001131 - Fundamental Pell solution for D=14

The least positive solution returned for x^2-14y^2=1 is x=15, y=4; direct substitution gives 1.

Evidence M. Hard: S26-E-00001130.

S26-E-00001132 - Fundamental Pell solution for D=15

The least positive solution returned for x^2-15y^2=1 is x=4, y=1; direct substitution gives 1.

Evidence M. Hard: S26-E-00001131.

S26-E-00001133 - Fundamental Pell solution for D=17

The least positive solution returned for x^2-17y^2=1 is x=33, y=8; direct substitution gives 1.

Evidence M. Hard: S26-E-00001132.

S26-E-00001134 - Fundamental Pell solution for D=18

The least positive solution returned for x^2-18y^2=1 is x=17, y=4; direct substitution gives 1.

Evidence M. Hard: S26-E-00001133.

S26-E-00001135 - Fundamental Pell solution for D=19

The least positive solution returned for x^2-19y^2=1 is x=170, y=39; direct substitution gives 1.

Evidence M. Hard: S26-E-00001134.

S26-E-00001136 - Fundamental Pell solution for D=20

The least positive solution returned for x^2-20y^2=1 is x=9, y=2; direct substitution gives 1.

Evidence M. Hard: S26-E-00001135.

S26-E-00001137 - Fundamental Pell solution for D=21

The least positive solution returned for x^2-21y^2=1 is x=55, y=12; direct substitution gives 1.

Evidence M. Hard: S26-E-00001136.

S26-E-00001138 - Fundamental Pell solution for D=22

The least positive solution returned for x^2-22y^2=1 is x=197, y=42; direct substitution gives 1.

Evidence M. Hard: S26-E-00001137.

S26-E-00001139 - Fundamental Pell solution for D=23

The least positive solution returned for x^2-23y^2=1 is x=24, y=5; direct substitution gives 1.

Evidence M. Hard: S26-E-00001138.

S26-E-00001140 - Fundamental Pell solution for D=24

The least positive solution returned for x^2-24y^2=1 is x=5, y=1; direct substitution gives 1.

Evidence M. Hard: S26-E-00001139.

S26-E-00001141 - Fundamental Pell solution for D=26

The least positive solution returned for x^2-26y^2=1 is x=51, y=10; direct substitution gives 1.

Evidence M. Hard: S26-E-00001140.

S26-E-00001142 - Fundamental Pell solution for D=27

The least positive solution returned for x^2-27y^2=1 is x=26, y=5; direct substitution gives 1.

Evidence M. Hard: S26-E-00001141.

S26-E-00001143 - Fundamental Pell solution for D=28

The least positive solution returned for x^2-28y^2=1 is x=127, y=24; direct substitution gives 1.

Evidence M. Hard: S26-E-00001142.

S26-E-00001144 - Fundamental Pell solution for D=29

The least positive solution returned for x^2-29y^2=1 is x=9801, y=1820; direct substitution gives 1.

Evidence M. Hard: S26-E-00001143.

S26-E-00001145 - Fundamental Pell solution for D=30

The least positive solution returned for x^2-30y^2=1 is x=11, y=2; direct substitution gives 1.

Evidence M. Hard: S26-E-00001144.

S26-E-00001146 - Fundamental Pell solution for D=31

The least positive solution returned for x^2-31y^2=1 is x=1520, y=273; direct substitution gives 1.

Evidence M. Hard: S26-E-00001145.

S26-E-00001147 - Fundamental Pell solution for D=32

The least positive solution returned for x^2-32y^2=1 is x=17, y=3; direct substitution gives 1.

Evidence M. Hard: S26-E-00001146.

S26-E-00001148 - Fundamental Pell solution for D=33

The least positive solution returned for x^2-33y^2=1 is x=23, y=4; direct substitution gives 1.

Evidence M. Hard: S26-E-00001147.

S26-E-00001149 - Fundamental Pell solution for D=34

The least positive solution returned for x^2-34y^2=1 is x=35, y=6; direct substitution gives 1.

Evidence M. Hard: S26-E-00001148.

S26-E-00001150 - Fundamental Pell solution for D=35

The least positive solution returned for x^2-35y^2=1 is x=6, y=1; direct substitution gives 1.

Evidence M. Hard: S26-E-00001149.

Part D - Analytic number theory

S26-D-00001151 - Prime-counting function

pi(x) is the number of primes p<=x.

Evidence A. Hard: S26-D-00000069.

S26-D-00001152 - Chebyshev theta function

theta(x)=sum_{p<=x} log p.

Evidence A. Hard: S26-D-00001151.

S26-D-00001153 - Chebyshev psi function

psi(x)=sum_{n<=x} Lambda(n)=sum_{p^k<=x} log p.

Evidence A. Hard: S26-D-00001152.

S26-T-00001154 - Prime number theorem

pi(x) is asymptotic to x/log x as x->infinity; equivalently theta(x)~x and psi(x)~x.

Evidence A. Hard: S26-D-00001153.

S26-D-00001155 - Riemann zeta function

For Re(s)>1, zeta(s)=sum_{n>=1} n^{-s}=product_p(1-p^{-s})^{-1}.

Evidence A. Hard: S26-T-00001154.

S26-I-00001156 - Euler product for zeta

Absolute convergence for Re(s)>1 permits the identity zeta(s)=product_p(1-p^{-s})^{-1}.

Evidence A. Hard: S26-D-00001155.

S26-T-00001157 - Meromorphic continuation of zeta

zeta(s) extends meromorphically to C with a single simple pole at s=1 of residue 1.

Evidence A. Hard: S26-I-00001156.

S26-I-00001158 - Zeta functional equation

The completed function xi(s)=1/2 s(s-1) pi^{-s/2} Gamma(s/2) zeta(s) satisfies xi(s)=xi(1-s).

Evidence A. Hard: S26-T-00001157.

S26-D-00001159 - Dirichlet character

A Dirichlet character modulo q is a completely multiplicative periodic function chi:Z->C that vanishes on integers not coprime to q and restricts to a group character on (Z/qZ)^x.

Evidence A. Hard: S26-I-00001158.

S26-D-00001160 - Dirichlet L-function

For Re(s)>1, L(s,chi)=sum_{n>=1} chi(n)n^{-s}, with an Euler product over primes.

Evidence A. Hard: S26-D-00001159.

S26-T-00001161 - Dirichlet theorem on arithmetic progressions

If gcd(a,q)=1, infinitely many primes are congruent to a modulo q.

Evidence A. Hard: S26-D-00001160.

S26-D-00001162 - Abscissa of convergence

For a Dirichlet series sum a_n n^{-s}, the abscissa of convergence is the boundary between its half-plane of convergence and divergence.

Evidence A. Hard: S26-T-00001161.

S26-D-00001163 - Summatory function

For an arithmetic function f, F(x)=sum_{n<=x} f(n) is its summatory function.

Evidence A. Hard: S26-D-00001162.

S26-T-00001164 - Partial summation

If A(x)=sum_{n<=x} a_n and f is continuously differentiable, then sum_{n<=x}a_n f(n)=A(x)f(x)-integral_1^x A(t)f'(t)dt, with endpoint conventions as appropriate.

Evidence A. Hard: S26-D-00001163.

S26-D-00001165 - Average order

A function g is an average order of f when sum_{n<=x}f(n) is asymptotic to integral_1^x g(t)dt in the declared sense.

Evidence A. Hard: S26-T-00001164.

S26-T-00001166 - Average order of divisor function

sum_{n<=x} tau(n)=x log x+(2 gamma-1)x+O(sqrt(x)) in a classical elementary estimate.

Evidence A. Hard: S26-D-00001165.

S26-T-00001167 - Mertens product asymptotic

product_{p<=x}(1-1/p)^{-1} ~ e^gamma log x.

Evidence A. Hard: S26-T-00001166.

S26-C-00001168 - Riemann hypothesis

The Riemann hypothesis asserts that every nontrivial zero of zeta(s) has real part 1/2.

Evidence H. Hard: S26-T-00001167.

S26-N-00001169 - Unproved-status fence for RH

The Riemann hypothesis remains a conjecture; numerical zero verification or consequences under RH do not constitute proof.

Evidence A. Hard: S26-C-00001168. Boundary: Finite computations cannot establish an infinite zero-location claim.

S26-B-00001170 - Spectral/arithmetic interface

Zeros and poles of zeta- and L-functions control explicit formulas relating primes to analytic spectral data; later volumes separate proved explicit formulas from conjectural spectral interpretations.

Evidence A. Hard: S26-N-00001169.

S26-E-00001171 - Prime-counting profile at x=10

pi(10)=4; the finite ratio pi(x)/(x/log x) is approximately 0.92103404.

Evidence M. Hard: S26-B-00001170.

S26-E-00001172 - Prime-counting profile at x=20

pi(20)=8; the finite ratio pi(x)/(x/log x) is approximately 1.19829291.

Evidence M. Hard: S26-E-00001171.

S26-E-00001173 - Prime-counting profile at x=30

pi(30)=10; the finite ratio pi(x)/(x/log x) is approximately 1.13373246.

Evidence M. Hard: S26-E-00001172.

S26-E-00001174 - Prime-counting profile at x=40

pi(40)=12; the finite ratio pi(x)/(x/log x) is approximately 1.10666384.

Evidence M. Hard: S26-E-00001173.

S26-E-00001175 - Prime-counting profile at x=50

pi(50)=15; the finite ratio pi(x)/(x/log x) is approximately 1.17360690.

Evidence M. Hard: S26-E-00001174.

S26-E-00001176 - Prime-counting profile at x=60

pi(60)=17; the finite ratio pi(x)/(x/log x) is approximately 1.16006429.

Evidence M. Hard: S26-E-00001175.

S26-E-00001177 - Prime-counting profile at x=70

pi(70)=19; the finite ratio pi(x)/(x/log x) is approximately 1.15316299.

Evidence M. Hard: S26-E-00001176.

S26-E-00001178 - Prime-counting profile at x=80

pi(80)=22; the finite ratio pi(x)/(x/log x) is approximately 1.20505732.

Evidence M. Hard: S26-E-00001177.

S26-E-00001179 - Prime-counting profile at x=90

pi(90)=24; the finite ratio pi(x)/(x/log x) is approximately 1.19994925.

Evidence M. Hard: S26-E-00001178.

S26-E-00001180 - Prime-counting profile at x=100

pi(100)=25; the finite ratio pi(x)/(x/log x) is approximately 1.15129255.

Evidence M. Hard: S26-E-00001179.

S26-E-00001181 - Prime-counting profile at x=110

pi(110)=29; the finite ratio pi(x)/(x/log x) is approximately 1.23921755.

Evidence M. Hard: S26-E-00001180.

S26-E-00001182 - Prime-counting profile at x=120

pi(120)=30; the finite ratio pi(x)/(x/log x) is approximately 1.19687294.

Evidence M. Hard: S26-E-00001181.

S26-E-00001183 - Prime-counting profile at x=130

pi(130)=31; the finite ratio pi(x)/(x/log x) is approximately 1.16071975.

Evidence M. Hard: S26-E-00001182.

S26-E-00001184 - Prime-counting profile at x=140

pi(140)=34; the finite ratio pi(x)/(x/log x) is approximately 1.20011316.

Evidence M. Hard: S26-E-00001183.

S26-E-00001185 - Prime-counting profile at x=150

pi(150)=35; the finite ratio pi(x)/(x/log x) is approximately 1.16914824.

Evidence M. Hard: S26-E-00001184.

S26-E-00001186 - Prime-counting profile at x=160

pi(160)=37; the finite ratio pi(x)/(x/log x) is approximately 1.17363394.

Evidence M. Hard: S26-E-00001185.

S26-E-00001187 - Prime-counting profile at x=170

pi(170)=39; the finite ratio pi(x)/(x/log x) is approximately 1.17821258.

Evidence M. Hard: S26-E-00001186.

S26-E-00001188 - Prime-counting profile at x=180

pi(180)=41; the finite ratio pi(x)/(x/log x) is approximately 1.18284017.

Evidence M. Hard: S26-E-00001187.

S26-E-00001189 - Prime-counting profile at x=190

pi(190)=42; the finite ratio pi(x)/(x/log x) is approximately 1.15986848.

Evidence M. Hard: S26-E-00001188.

S26-E-00001190 - Prime-counting profile at x=200

pi(200)=46; the finite ratio pi(x)/(x/log x) is approximately 1.21861299.

Evidence M. Hard: S26-E-00001189.

S26-E-00001191 - Prime-counting profile at x=1000

pi(1000)=168; the finite ratio pi(x)/(x/log x) is approximately 1.16050289.

Evidence M. Hard: S26-E-00001190.

S26-E-00001192 - Prime-counting profile at x=10000

pi(10000)=1229; the finite ratio pi(x)/(x/log x) is approximately 1.13195083.

Evidence M. Hard: S26-E-00001191.

S26-E-00001193 - Prime-counting profile at x=100000

pi(100000)=9592; the finite ratio pi(x)/(x/log x) is approximately 1.10431981.

Evidence M. Hard: S26-E-00001192.

S26-E-00001194 - Prime-counting profile at x=1000000

pi(1000000)=78498; the finite ratio pi(x)/(x/log x) is approximately 1.08448995.

Evidence M. Hard: S26-E-00001193.

S26-E-00001195 - Prime-counting profile at x=10000000

pi(10000000)=664579; the finite ratio pi(x)/(x/log x) is approximately 1.07117479.

Evidence M. Hard: S26-E-00001194.

S26-E-00001196 - Prime-counting profile at x=100000000

pi(100000000)=5761455; the finite ratio pi(x)/(x/log x) is approximately 1.06129923.

Evidence M. Hard: S26-E-00001195.

S26-E-00001197 - Prime-counting profile at x=1000000000

pi(1000000000)=50847534; the finite ratio pi(x)/(x/log x) is approximately 1.05372696.

Evidence M. Hard: S26-E-00001196.

S26-E-00001198 - Prime-counting profile at x=10000000000

pi(10000000000)=455052511; the finite ratio pi(x)/(x/log x) is approximately 1.04779713.

Evidence M. Hard: S26-E-00001197.

S26-E-00001199 - Prime-counting profile at x=100000000000

pi(100000000000)=4118054813; the finite ratio pi(x)/(x/log x) is approximately 1.04303888.

Evidence M. Hard: S26-E-00001198.

S26-E-00001200 - Prime-counting profile at x=1000000000000

pi(1000000000000)=37607912018; the finite ratio pi(x)/(x/log x) is approximately 1.03914501.

Evidence M. Hard: S26-E-00001199.

Part E - Algebraic number theory

S26-D-00001201 - Number field

A number field is a finite extension K/Q.

Evidence A. Hard: S26-D-00000101.

S26-D-00001202 - Algebraic integer

An element alpha of a number field is an algebraic integer when it satisfies a monic polynomial with coefficients in Z.

Evidence A. Hard: S26-D-00001201.

S26-D-00001203 - Ring of integers

O_K is the ring of all algebraic integers in K.

Evidence A. Hard: S26-D-00001202.

S26-T-00001204 - Integral closure of Z in a number field

O_K is the integral closure of Z in K and is a free Z-module of rank [K:Q].

Evidence A. Hard: S26-D-00001203.

S26-D-00001205 - Field norm

For alpha in K, N_{K/Q}(alpha) is the determinant of multiplication by alpha as a Q-linear map.

Evidence A. Hard: S26-T-00001204.

S26-D-00001206 - Field trace

For alpha in K, Tr_{K/Q}(alpha) is the trace of multiplication by alpha as a Q-linear map.

Evidence A. Hard: S26-D-00001205.

S26-I-00001207 - Norm and trace from embeddings

For separable K/Q, the norm is the product and the trace the sum of the images of alpha over embeddings K->C, counted appropriately.

Evidence A. Hard: S26-D-00001206.

S26-D-00001208 - Discriminant of a basis

For a Q-basis alpha_1,...,alpha_n of K, disc(alpha_i)=det(Tr(alpha_i alpha_j)).

Evidence A. Hard: S26-I-00001207.

S26-D-00001209 - Field discriminant

The discriminant d_K is the discriminant of an integral basis of O_K and is independent of the chosen integral basis.

Evidence A. Hard: S26-D-00001208.

S26-D-00001210 - Fractional ideal

A fractional ideal of O_K is a nonzero O_K-submodule I of K such that dI is an integral ideal for some nonzero d in O_K.

Evidence A. Hard: S26-D-00001209.

S26-T-00001211 - Dedekind ideal factorization

Every nonzero proper ideal of O_K factors uniquely as a product of nonzero prime ideals.

Evidence A. Hard: S26-D-00001210.

S26-D-00001212 - Ideal norm

For nonzero integral ideal I, N(I)=|O_K/I|.

Evidence A. Hard: S26-T-00001211.

S26-I-00001213 - Multiplicativity of ideal norm

For nonzero ideals I,J in O_K, N(IJ)=N(I)N(J).

Evidence A. Hard: S26-D-00001212.

S26-D-00001214 - Ideal class group

Cl(K) is the quotient of nonzero fractional ideals by principal fractional ideals.

Evidence A. Hard: S26-I-00001213.

S26-T-00001215 - Finiteness of class group

The ideal class group of a number field is finite.

Evidence A. Hard: S26-D-00001214.

S26-D-00001216 - Unit group

O_K^x is the multiplicative group of units in the ring of integers.

Evidence A. Hard: S26-T-00001215.

S26-T-00001217 - Dirichlet unit theorem

If K has r_1 real embeddings and r_2 conjugate pairs of complex embeddings, O_K^x is isomorphic to mu(K) x Z^{r_1+r_2-1}.

Evidence A. Hard: S26-D-00001216.

S26-D-00001218 - Ramification index and residue degree

For prime ideal P over p, e(P/p) measures the exponent of P in pO_K and f(P/p)=[O_K/P:F_p].

Evidence A. Hard: S26-T-00001217.

S26-I-00001219 - Prime decomposition degree formula

For pO_K=product P_i^{e_i}, sum_i e_i f_i=[K:Q].

Evidence A. Hard: S26-D-00001218.

S26-B-00001220 - Arithmetic geometry gateway

Ideal classes, units, valuations, and ramification provide the local/global algebraic infrastructure used by class field theory and arithmetic geometry.

Evidence A. Hard: S26-I-00001219.

S26-E-00001221 - Quadratic-field discriminant profile d=-1

For K=Q(sqrt(-1)) with squarefree d=-1, the standard quadratic-field discriminant is -4.

Evidence M. Hard: S26-B-00001220.

S26-E-00001222 - Quadratic-field discriminant profile d=-2

For K=Q(sqrt(-2)) with squarefree d=-2, the standard quadratic-field discriminant is -8.

Evidence M. Hard: S26-E-00001221.

S26-E-00001223 - Quadratic-field discriminant profile d=-3

For K=Q(sqrt(-3)) with squarefree d=-3, the standard quadratic-field discriminant is -3.

Evidence M. Hard: S26-E-00001222.

S26-E-00001224 - Quadratic-field discriminant profile d=-5

For K=Q(sqrt(-5)) with squarefree d=-5, the standard quadratic-field discriminant is -20.

Evidence M. Hard: S26-E-00001223.

S26-E-00001225 - Quadratic-field discriminant profile d=-6

For K=Q(sqrt(-6)) with squarefree d=-6, the standard quadratic-field discriminant is -24.

Evidence M. Hard: S26-E-00001224.

S26-E-00001226 - Quadratic-field discriminant profile d=-7

For K=Q(sqrt(-7)) with squarefree d=-7, the standard quadratic-field discriminant is -7.

Evidence M. Hard: S26-E-00001225.

S26-E-00001227 - Quadratic-field discriminant profile d=-10

For K=Q(sqrt(-10)) with squarefree d=-10, the standard quadratic-field discriminant is -40.

Evidence M. Hard: S26-E-00001226.

S26-E-00001228 - Quadratic-field discriminant profile d=-11

For K=Q(sqrt(-11)) with squarefree d=-11, the standard quadratic-field discriminant is -11.

Evidence M. Hard: S26-E-00001227.

S26-E-00001229 - Quadratic-field discriminant profile d=-13

For K=Q(sqrt(-13)) with squarefree d=-13, the standard quadratic-field discriminant is -52.

Evidence M. Hard: S26-E-00001228.

S26-E-00001230 - Quadratic-field discriminant profile d=-15

For K=Q(sqrt(-15)) with squarefree d=-15, the standard quadratic-field discriminant is -15.

Evidence M. Hard: S26-E-00001229.

S26-E-00001231 - Quadratic-field discriminant profile d=2

For K=Q(sqrt(2)) with squarefree d=2, the standard quadratic-field discriminant is 8.

Evidence M. Hard: S26-E-00001230.

S26-E-00001232 - Quadratic-field discriminant profile d=3

For K=Q(sqrt(3)) with squarefree d=3, the standard quadratic-field discriminant is 12.

Evidence M. Hard: S26-E-00001231.

S26-E-00001233 - Quadratic-field discriminant profile d=5

For K=Q(sqrt(5)) with squarefree d=5, the standard quadratic-field discriminant is 5.

Evidence M. Hard: S26-E-00001232.

S26-E-00001234 - Quadratic-field discriminant profile d=6

For K=Q(sqrt(6)) with squarefree d=6, the standard quadratic-field discriminant is 24.

Evidence M. Hard: S26-E-00001233.

S26-E-00001235 - Quadratic-field discriminant profile d=7

For K=Q(sqrt(7)) with squarefree d=7, the standard quadratic-field discriminant is 28.

Evidence M. Hard: S26-E-00001234.

S26-E-00001236 - Quadratic-field discriminant profile d=10

For K=Q(sqrt(10)) with squarefree d=10, the standard quadratic-field discriminant is 40.

Evidence M. Hard: S26-E-00001235.

S26-E-00001237 - Quadratic-field discriminant profile d=11

For K=Q(sqrt(11)) with squarefree d=11, the standard quadratic-field discriminant is 44.

Evidence M. Hard: S26-E-00001236.

S26-E-00001238 - Quadratic-field discriminant profile d=13

For K=Q(sqrt(13)) with squarefree d=13, the standard quadratic-field discriminant is 13.

Evidence M. Hard: S26-E-00001237.

S26-E-00001239 - Quadratic-field discriminant profile d=14

For K=Q(sqrt(14)) with squarefree d=14, the standard quadratic-field discriminant is 56.

Evidence M. Hard: S26-E-00001238.

S26-E-00001240 - Quadratic-field discriminant profile d=17

For K=Q(sqrt(17)) with squarefree d=17, the standard quadratic-field discriminant is 17.

Evidence M. Hard: S26-E-00001239.

S26-E-00001241 - Quadratic-field discriminant profile d=19

For K=Q(sqrt(19)) with squarefree d=19, the standard quadratic-field discriminant is 76.

Evidence M. Hard: S26-E-00001240.

S26-E-00001242 - Quadratic-field discriminant profile d=21

For K=Q(sqrt(21)) with squarefree d=21, the standard quadratic-field discriminant is 21.

Evidence M. Hard: S26-E-00001241.

S26-E-00001243 - Quadratic-field discriminant profile d=22

For K=Q(sqrt(22)) with squarefree d=22, the standard quadratic-field discriminant is 88.

Evidence M. Hard: S26-E-00001242.

S26-E-00001244 - Quadratic-field discriminant profile d=23

For K=Q(sqrt(23)) with squarefree d=23, the standard quadratic-field discriminant is 92.

Evidence M. Hard: S26-E-00001243.

S26-E-00001245 - Quadratic-field discriminant profile d=26

For K=Q(sqrt(26)) with squarefree d=26, the standard quadratic-field discriminant is 104.

Evidence M. Hard: S26-E-00001244.

S26-E-00001246 - Quadratic-field discriminant profile d=29

For K=Q(sqrt(29)) with squarefree d=29, the standard quadratic-field discriminant is 29.

Evidence M. Hard: S26-E-00001245.

S26-E-00001247 - Quadratic-field discriminant profile d=30

For K=Q(sqrt(30)) with squarefree d=30, the standard quadratic-field discriminant is 120.

Evidence M. Hard: S26-E-00001246.

S26-E-00001248 - Quadratic-field discriminant profile d=31

For K=Q(sqrt(31)) with squarefree d=31, the standard quadratic-field discriminant is 124.

Evidence M. Hard: S26-E-00001247.

S26-E-00001249 - Quadratic-field discriminant profile d=33

For K=Q(sqrt(33)) with squarefree d=33, the standard quadratic-field discriminant is 33.

Evidence M. Hard: S26-E-00001248.

S26-E-00001250 - Quadratic-field discriminant profile d=37

For K=Q(sqrt(37)) with squarefree d=37, the standard quadratic-field discriminant is 37.

Evidence M. Hard: S26-E-00001249.

Part F - Finite fields and cyclotomy

S26-D-00001251 - Finite field

A finite field is a field with finitely many elements.

Evidence A. Hard: S26-D-00000101.

S26-T-00001252 - Finite-field order theorem

Every finite field has order p^n for a prime p and n>=1; for every p^n there exists a finite field of that order, unique up to isomorphism.

Evidence A. Hard: S26-D-00001251.

S26-T-00001253 - Finite-field multiplicative group

The multiplicative group F_q^x is cyclic of order q-1.

Evidence A. Hard: S26-T-00001252.

S26-I-00001254 - Frobenius identity in F_q

Every x in F_q satisfies x^q=x.

Evidence A. Hard: S26-T-00001253.

S26-D-00001255 - Frobenius automorphism

For F_{p^n}, Frobenius is F(x)=x^p; it generates Gal(F_{p^n}/F_p), a cyclic group of order n.

Evidence A. Hard: S26-I-00001254.

S26-T-00001256 - Subfield classification of finite fields

F_{p^m} embeds as a subfield of F_{p^n} iff m divides n.

Evidence A. Hard: S26-D-00001255.

S26-D-00001257 - Trace over finite fields

Tr_{F_{q^n}/F_q}(x)=x+x^q+...+x^{q^{n-1}}.

Evidence A. Hard: S26-T-00001256.

S26-D-00001258 - Norm over finite fields

N_{F_{q^n}/F_q}(x)=x^{(q^n-1)/(q-1)}.

Evidence A. Hard: S26-D-00001257.

S26-T-00001259 - Irreducible-polynomial construction

If f in F_q[x] is irreducible of degree n, then F_q[x]/(f) is a field with q^n elements.

Evidence A. Hard: S26-D-00001258.

S26-D-00001260 - Cyclotomic polynomial

Phi_n(x) is the monic polynomial whose roots are the primitive n-th roots of unity.

Evidence A. Hard: S26-T-00001259.

S26-I-00001261 - Cyclotomic factorization

x^n-1=product_{d|n} Phi_d(x).

Evidence A. Hard: S26-D-00001260.

S26-T-00001262 - Degree of cyclotomic polynomial

deg Phi_n=phi(n).

Evidence A. Hard: S26-I-00001261.

S26-T-00001263 - Cyclotomic irreducibility

Phi_n(x) is irreducible over Q.

Evidence A. Hard: S26-T-00001262.

S26-D-00001264 - Cyclotomic field

Q(zeta_n) is the field obtained by adjoining a primitive n-th root of unity.

Evidence A. Hard: S26-T-00001263.

S26-T-00001265 - Cyclotomic Galois group

Gal(Q(zeta_n)/Q) is canonically isomorphic to (Z/nZ)^x.

Evidence A. Hard: S26-D-00001264.

S26-D-00001266 - Gaussian period

A Gaussian period is a sum of roots of unity over a coset of a multiplicative subgroup of a finite cyclic group, used in explicit cyclotomic constructions.

Evidence A. Hard: S26-T-00001265.

S26-D-00001267 - Discrete logarithm in F_q

For a generator g of F_q^x and nonzero h, the discrete logarithm asks for k modulo q-1 with g^k=h.

Evidence A. Hard: S26-D-00001266.

S26-N-00001268 - Finite-field representation fence

A finite field is unique only up to isomorphism; concrete polynomial-basis or normal-basis encodings are representations, not the abstract field itself.

Evidence A. Hard: S26-D-00001267. Boundary: Distinct irreducible-polynomial quotients can represent isomorphic fields.

S26-B-00001269 - Finite fields to coding theory

Finite fields supply alphabets and linear-algebra structure for Reed-Solomon, BCH, cyclic, and algebraic-geometric codes.

Evidence A. Hard: S26-N-00001268.

S26-B-00001270 - Finite fields to cryptography

Finite-field groups and elliptic curves over finite fields provide finite algebraic groups used in public-key cryptography; security depends on computational hardness assumptions, not field axioms alone.

Evidence A. Hard: S26-B-00001269.

S26-E-00001271 - Finite-field order profile q=2

q=2=2^1 is a prime power; F_2 exists up to isomorphism and F_2^x has cyclic order 1.

Evidence M. Hard: S26-B-00001270.

S26-E-00001272 - Finite-field order profile q=3

q=3=3^1 is a prime power; F_3 exists up to isomorphism and F_3^x has cyclic order 2.

Evidence M. Hard: S26-E-00001271.

S26-E-00001273 - Finite-field order profile q=4

q=4=2^2 is a prime power; F_4 exists up to isomorphism and F_4^x has cyclic order 3.

Evidence M. Hard: S26-E-00001272.

S26-E-00001274 - Finite-field order profile q=5

q=5=5^1 is a prime power; F_5 exists up to isomorphism and F_5^x has cyclic order 4.

Evidence M. Hard: S26-E-00001273.

S26-E-00001275 - Finite-field order profile q=7

q=7=7^1 is a prime power; F_7 exists up to isomorphism and F_7^x has cyclic order 6.

Evidence M. Hard: S26-E-00001274.

S26-E-00001276 - Finite-field order profile q=8

q=8=2^3 is a prime power; F_8 exists up to isomorphism and F_8^x has cyclic order 7.

Evidence M. Hard: S26-E-00001275.

S26-E-00001277 - Finite-field order profile q=9

q=9=3^2 is a prime power; F_9 exists up to isomorphism and F_9^x has cyclic order 8.

Evidence M. Hard: S26-E-00001276.

S26-E-00001278 - Finite-field order profile q=11

q=11=11^1 is a prime power; F_11 exists up to isomorphism and F_11^x has cyclic order 10.

Evidence M. Hard: S26-E-00001277.

S26-E-00001279 - Finite-field order profile q=13

q=13=13^1 is a prime power; F_13 exists up to isomorphism and F_13^x has cyclic order 12.

Evidence M. Hard: S26-E-00001278.

S26-E-00001280 - Finite-field order profile q=16

q=16=2^4 is a prime power; F_16 exists up to isomorphism and F_16^x has cyclic order 15.

Evidence M. Hard: S26-E-00001279.

S26-E-00001281 - Finite-field order profile q=17

q=17=17^1 is a prime power; F_17 exists up to isomorphism and F_17^x has cyclic order 16.

Evidence M. Hard: S26-E-00001280.

S26-E-00001282 - Finite-field order profile q=19

q=19=19^1 is a prime power; F_19 exists up to isomorphism and F_19^x has cyclic order 18.

Evidence M. Hard: S26-E-00001281.

S26-E-00001283 - Finite-field order profile q=23

q=23=23^1 is a prime power; F_23 exists up to isomorphism and F_23^x has cyclic order 22.

Evidence M. Hard: S26-E-00001282.

S26-E-00001284 - Finite-field order profile q=25

q=25=5^2 is a prime power; F_25 exists up to isomorphism and F_25^x has cyclic order 24.

Evidence M. Hard: S26-E-00001283.

S26-E-00001285 - Finite-field order profile q=27

q=27=3^3 is a prime power; F_27 exists up to isomorphism and F_27^x has cyclic order 26.

Evidence M. Hard: S26-E-00001284.

S26-E-00001286 - Finite-field order profile q=29

q=29=29^1 is a prime power; F_29 exists up to isomorphism and F_29^x has cyclic order 28.

Evidence M. Hard: S26-E-00001285.

S26-E-00001287 - Finite-field order profile q=31

q=31=31^1 is a prime power; F_31 exists up to isomorphism and F_31^x has cyclic order 30.

Evidence M. Hard: S26-E-00001286.

S26-E-00001288 - Finite-field order profile q=32

q=32=2^5 is a prime power; F_32 exists up to isomorphism and F_32^x has cyclic order 31.

Evidence M. Hard: S26-E-00001287.

S26-E-00001289 - Finite-field order profile q=37

q=37=37^1 is a prime power; F_37 exists up to isomorphism and F_37^x has cyclic order 36.

Evidence M. Hard: S26-E-00001288.

S26-E-00001290 - Finite-field order profile q=41

q=41=41^1 is a prime power; F_41 exists up to isomorphism and F_41^x has cyclic order 40.

Evidence M. Hard: S26-E-00001289.

S26-E-00001291 - Finite-field order profile q=43

q=43=43^1 is a prime power; F_43 exists up to isomorphism and F_43^x has cyclic order 42.

Evidence M. Hard: S26-E-00001290.

S26-E-00001292 - Finite-field order profile q=47

q=47=47^1 is a prime power; F_47 exists up to isomorphism and F_47^x has cyclic order 46.

Evidence M. Hard: S26-E-00001291.

S26-E-00001293 - Finite-field order profile q=49

q=49=7^2 is a prime power; F_49 exists up to isomorphism and F_49^x has cyclic order 48.

Evidence M. Hard: S26-E-00001292.

S26-E-00001294 - Finite-field order profile q=53

q=53=53^1 is a prime power; F_53 exists up to isomorphism and F_53^x has cyclic order 52.

Evidence M. Hard: S26-E-00001293.

S26-E-00001295 - Finite-field order profile q=59

q=59=59^1 is a prime power; F_59 exists up to isomorphism and F_59^x has cyclic order 58.

Evidence M. Hard: S26-E-00001294.

S26-E-00001296 - Finite-field order profile q=81

q=81=3^4 is a prime power; F_81 exists up to isomorphism and F_81^x has cyclic order 80.

Evidence M. Hard: S26-E-00001295.

S26-E-00001297 - Finite-field order profile q=121

q=121=11^2 is a prime power; F_121 exists up to isomorphism and F_121^x has cyclic order 120.

Evidence M. Hard: S26-E-00001296.

S26-E-00001298 - Finite-field order profile q=125

q=125=5^3 is a prime power; F_125 exists up to isomorphism and F_125^x has cyclic order 124.

Evidence M. Hard: S26-E-00001297.

S26-E-00001299 - Finite-field order profile q=169

q=169=13^2 is a prime power; F_169 exists up to isomorphism and F_169^x has cyclic order 168.

Evidence M. Hard: S26-E-00001298.

S26-E-00001300 - Finite-field order profile q=243

q=243=3^5 is a prime power; F_243 exists up to isomorphism and F_243^x has cyclic order 242.

Evidence M. Hard: S26-E-00001299.

Part G - Groups and actions

S26-D-00001301 - Group action

A left action of G on X is a map GxX->X with e.x=x and (gh).x=g.(h.x).

Evidence A. Hard: S26-D-00000103.

S26-T-00001302 - Orbit as a homogeneous quotient

For a group action and x in X, the orbit G.x is naturally identified with the coset space G/G_x as a G-set.

Evidence A. Hard: S26-D-00001301.

S26-D-00001303 - Stabilizer

G_x={g in G:g.x=x} is the stabilizer subgroup of x.

Evidence A. Hard: S26-T-00001302.

S26-T-00001304 - Orbit-stabilizer theorem

For finite G, |G.x|=[G:G_x], hence |G|=|G.x||G_x|.

Evidence A. Hard: S26-D-00001303.

S26-D-00001305 - Kernel of an action

The kernel of a G-action is the intersection of all point stabilizers; the action is faithful iff this kernel is trivial.

Evidence A. Hard: S26-T-00001304.

S26-T-00001306 - Cayley theorem

Every group G is isomorphic to a subgroup of the symmetric group on the underlying set of G.

Evidence A. Hard: S26-D-00001305.

S26-D-00001307 - Conjugacy class

The conjugacy class of g is {xgx^{-1}:x in G}; its stabilizer under conjugation is the centralizer C_G(g).

Evidence A. Hard: S26-T-00001306.

S26-I-00001308 - Class equation

For finite G, |G|=|Z(G)|+sum [G:C_G(x_i)] over representatives of noncentral conjugacy classes.

Evidence A. Hard: S26-D-00001307.

S26-T-00001309 - Cauchy theorem for finite groups

If prime p divides |G|, then G contains an element of order p.

Evidence A. Hard: S26-I-00001308.

S26-T-00001310 - Sylow existence theorem

If |G|=p^n m with p not dividing m, G has a subgroup of order p^n.

Evidence A. Hard: S26-T-00001309.

S26-T-00001311 - Sylow conjugacy theorem

All Sylow p-subgroups of a finite group are conjugate.

Evidence A. Hard: S26-T-00001310.

S26-T-00001312 - Sylow counting theorem

The number n_p of Sylow p-subgroups satisfies n_p congruent 1 mod p and n_p divides m when |G|=p^n m.

Evidence A. Hard: S26-T-00001311.

S26-D-00001313 - Normal subgroup

N<=G is normal when gNg^{-1}=N for all g in G; then the coset set G/N inherits a group structure.

Evidence A. Hard: S26-T-00001312.

S26-T-00001314 - First isomorphism theorem for groups

For homomorphism f:G->H, G/ker f is isomorphic to im f.

Evidence A. Hard: S26-D-00001313.

S26-T-00001315 - Second isomorphism theorem for groups

For A<=G and N normal in G, AN/N is isomorphic to A/(A intersect N).

Evidence A. Hard: S26-T-00001314.

S26-T-00001316 - Third isomorphism theorem for groups

If N normal H normal G, then (G/N)/(H/N) is isomorphic to G/H.

Evidence A. Hard: S26-T-00001315.

S26-D-00001317 - Semidirect product

Given N,H and an action H->Aut(N), the semidirect product N rtimes H has multiplication (n,h)(n',h')=(n h.n',hh').

Evidence A. Hard: S26-T-00001316.

S26-D-00001318 - Direct product of groups

GxH has componentwise multiplication; both factors embed as commuting normal subgroups with trivial intersection.

Evidence A. Hard: S26-D-00001317.

S26-D-00001319 - Free group

The free group F(S) on a set S is characterized by the universal property that every map S->G extends uniquely to a homomorphism F(S)->G.

Evidence A. Hard: S26-D-00001318.

S26-B-00001320 - Groups as symmetry objects

Group actions turn abstract multiplication into symmetry on sets, vector spaces, manifolds, graphs, and fields; stabilizers and quotients encode loss of symmetry.

Evidence A. Hard: S26-D-00001319.

S26-E-00001321 - Cyclic group C_1 profile

The cyclic group C_1 has order 1; it has phi(1)=1 generators. Representative generator exponents relatively prime to 1 begin [1].

Evidence M. Hard: S26-B-00001320.

S26-E-00001322 - Cyclic group C_2 profile

The cyclic group C_2 has order 2; it has phi(2)=1 generators. Representative generator exponents relatively prime to 2 begin [1].

Evidence M. Hard: S26-E-00001321.

S26-E-00001323 - Cyclic group C_3 profile

The cyclic group C_3 has order 3; it has phi(3)=2 generators. Representative generator exponents relatively prime to 3 begin [1, 2].

Evidence M. Hard: S26-E-00001322.

S26-E-00001324 - Cyclic group C_4 profile

The cyclic group C_4 has order 4; it has phi(4)=2 generators. Representative generator exponents relatively prime to 4 begin [1, 3].

Evidence M. Hard: S26-E-00001323.

S26-E-00001325 - Cyclic group C_5 profile

The cyclic group C_5 has order 5; it has phi(5)=4 generators. Representative generator exponents relatively prime to 5 begin [1, 2, 3, 4].

Evidence M. Hard: S26-E-00001324.

S26-E-00001326 - Cyclic group C_6 profile

The cyclic group C_6 has order 6; it has phi(6)=2 generators. Representative generator exponents relatively prime to 6 begin [1, 5].

Evidence M. Hard: S26-E-00001325.

S26-E-00001327 - Cyclic group C_7 profile

The cyclic group C_7 has order 7; it has phi(7)=6 generators. Representative generator exponents relatively prime to 7 begin [1, 2, 3, 4, 5, 6].

Evidence M. Hard: S26-E-00001326.

S26-E-00001328 - Cyclic group C_8 profile

The cyclic group C_8 has order 8; it has phi(8)=4 generators. Representative generator exponents relatively prime to 8 begin [1, 3, 5, 7].

Evidence M. Hard: S26-E-00001327.

S26-E-00001329 - Cyclic group C_9 profile

The cyclic group C_9 has order 9; it has phi(9)=6 generators. Representative generator exponents relatively prime to 9 begin [1, 2, 4, 5, 7, 8].

Evidence M. Hard: S26-E-00001328.

S26-E-00001330 - Cyclic group C_10 profile

The cyclic group C_10 has order 10; it has phi(10)=4 generators. Representative generator exponents relatively prime to 10 begin [1, 3, 7, 9].

Evidence M. Hard: S26-E-00001329.

S26-E-00001331 - Cyclic group C_11 profile

The cyclic group C_11 has order 11; it has phi(11)=10 generators. Representative generator exponents relatively prime to 11 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

Evidence M. Hard: S26-E-00001330.

S26-E-00001332 - Cyclic group C_12 profile

The cyclic group C_12 has order 12; it has phi(12)=4 generators. Representative generator exponents relatively prime to 12 begin [1, 5, 7, 11].

Evidence M. Hard: S26-E-00001331.

S26-E-00001333 - Cyclic group C_13 profile

The cyclic group C_13 has order 13; it has phi(13)=12 generators. Representative generator exponents relatively prime to 13 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

Evidence M. Hard: S26-E-00001332.

S26-E-00001334 - Cyclic group C_14 profile

The cyclic group C_14 has order 14; it has phi(14)=6 generators. Representative generator exponents relatively prime to 14 begin [1, 3, 5, 9, 11, 13].

Evidence M. Hard: S26-E-00001333.

S26-E-00001335 - Cyclic group C_15 profile

The cyclic group C_15 has order 15; it has phi(15)=8 generators. Representative generator exponents relatively prime to 15 begin [1, 2, 4, 7, 8, 11, 13, 14].

Evidence M. Hard: S26-E-00001334.

S26-E-00001336 - Cyclic group C_16 profile

The cyclic group C_16 has order 16; it has phi(16)=8 generators. Representative generator exponents relatively prime to 16 begin [1, 3, 5, 7, 9, 11, 13, 15].

Evidence M. Hard: S26-E-00001335.

S26-E-00001337 - Cyclic group C_17 profile

The cyclic group C_17 has order 17; it has phi(17)=16 generators. Representative generator exponents relatively prime to 17 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

Evidence M. Hard: S26-E-00001336.

S26-E-00001338 - Cyclic group C_18 profile

The cyclic group C_18 has order 18; it has phi(18)=6 generators. Representative generator exponents relatively prime to 18 begin [1, 5, 7, 11, 13, 17].

Evidence M. Hard: S26-E-00001337.

S26-E-00001339 - Cyclic group C_19 profile

The cyclic group C_19 has order 19; it has phi(19)=18 generators. Representative generator exponents relatively prime to 19 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

Evidence M. Hard: S26-E-00001338.

S26-E-00001340 - Cyclic group C_20 profile

The cyclic group C_20 has order 20; it has phi(20)=8 generators. Representative generator exponents relatively prime to 20 begin [1, 3, 7, 9, 11, 13, 17, 19].

Evidence M. Hard: S26-E-00001339.

S26-E-00001341 - Cyclic group C_21 profile

The cyclic group C_21 has order 21; it has phi(21)=12 generators. Representative generator exponents relatively prime to 21 begin [1, 2, 4, 5, 8, 10, 11, 13, 16, 17].

Evidence M. Hard: S26-E-00001340.

S26-E-00001342 - Cyclic group C_22 profile

The cyclic group C_22 has order 22; it has phi(22)=10 generators. Representative generator exponents relatively prime to 22 begin [1, 3, 5, 7, 9, 13, 15, 17, 19, 21].

Evidence M. Hard: S26-E-00001341.

S26-E-00001343 - Cyclic group C_23 profile

The cyclic group C_23 has order 23; it has phi(23)=22 generators. Representative generator exponents relatively prime to 23 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

Evidence M. Hard: S26-E-00001342.

S26-E-00001344 - Cyclic group C_24 profile

The cyclic group C_24 has order 24; it has phi(24)=8 generators. Representative generator exponents relatively prime to 24 begin [1, 5, 7, 11, 13, 17, 19, 23].

Evidence M. Hard: S26-E-00001343.

S26-E-00001345 - Cyclic group C_25 profile

The cyclic group C_25 has order 25; it has phi(25)=20 generators. Representative generator exponents relatively prime to 25 begin [1, 2, 3, 4, 6, 7, 8, 9, 11, 12].

Evidence M. Hard: S26-E-00001344.

S26-E-00001346 - Cyclic group C_26 profile

The cyclic group C_26 has order 26; it has phi(26)=12 generators. Representative generator exponents relatively prime to 26 begin [1, 3, 5, 7, 9, 11, 15, 17, 19, 21].

Evidence M. Hard: S26-E-00001345.

S26-E-00001347 - Cyclic group C_27 profile

The cyclic group C_27 has order 27; it has phi(27)=18 generators. Representative generator exponents relatively prime to 27 begin [1, 2, 4, 5, 7, 8, 10, 11, 13, 14].

Evidence M. Hard: S26-E-00001346.

S26-E-00001348 - Cyclic group C_28 profile

The cyclic group C_28 has order 28; it has phi(28)=12 generators. Representative generator exponents relatively prime to 28 begin [1, 3, 5, 9, 11, 13, 15, 17, 19, 23].

Evidence M. Hard: S26-E-00001347.

S26-E-00001349 - Cyclic group C_29 profile

The cyclic group C_29 has order 29; it has phi(29)=28 generators. Representative generator exponents relatively prime to 29 begin [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

Evidence M. Hard: S26-E-00001348.

S26-E-00001350 - Cyclic group C_30 profile

The cyclic group C_30 has order 30; it has phi(30)=8 generators. Representative generator exponents relatively prime to 30 begin [1, 7, 11, 13, 17, 19, 23, 29].

Evidence M. Hard: S26-E-00001349.

Part H - Group structure and solvability

S26-D-00001351 - Commutator of group elements

[g,h]=g^{-1}h^{-1}gh measures the failure of g and h to commute.

Evidence A. Hard: S26-D-00000103.

S26-D-00001352 - Derived subgroup

G'=[G,G] is the subgroup generated by all commutators; G/G' is the abelianization of G.

Evidence A. Hard: S26-D-00001351.

S26-D-00001353 - Derived series

G^{(0)}=G and G^{(n+1)}=[G^{(n)},G^{(n)}].

Evidence A. Hard: S26-D-00001352.

S26-D-00001354 - Solvable group

G is solvable when its derived series reaches the identity after finitely many steps.

Evidence A. Hard: S26-D-00001353.

S26-D-00001355 - Lower central series

gamma_1(G)=G and gamma_{n+1}(G)=[gamma_n(G),G].

Evidence A. Hard: S26-D-00001354.

S26-D-00001356 - Nilpotent group

G is nilpotent when its lower central series reaches the identity after finitely many steps.

Evidence A. Hard: S26-D-00001355.

S26-T-00001357 - Finite p-groups have nontrivial center

Every nontrivial finite group of prime-power order has a nontrivial center.

Evidence A. Hard: S26-D-00001356.

S26-T-00001358 - Finite p-groups are nilpotent

Every finite p-group is nilpotent.

Evidence A. Hard: S26-T-00001357.

S26-D-00001359 - Simple group

A nontrivial group is simple when its only normal subgroups are the identity and the whole group.

Evidence A. Hard: S26-T-00001358.

S26-T-00001360 - Simplicity of alternating groups

A_n is simple for n>=5.

Evidence A. Hard: S26-D-00001359.

S26-D-00001361 - Composition series

A composition series is a finite subnormal chain whose factors are simple.

Evidence A. Hard: S26-T-00001360.

S26-T-00001362 - Jordan-Holder theorem

Any two composition series of a finite-length group have the same multiset of simple factors up to isomorphism and permutation.

Evidence A. Hard: S26-D-00001361.

S26-D-00001363 - Characteristic subgroup

H<=G is characteristic when every automorphism of G preserves H; characteristic subgroups are normal.

Evidence A. Hard: S26-T-00001362.

S26-D-00001364 - Frattini subgroup

Phi(G) is the intersection of all maximal subgroups of G, with Phi(G)=G if there are none.

Evidence A. Hard: S26-D-00001363.

S26-D-00001365 - Center of a group

Z(G)={z in G:zg=gz for all g in G}.

Evidence A. Hard: S26-D-00001364.

S26-D-00001366 - Central product

A central product glues groups along central subgroups; unlike direct products, the factors may intersect nontrivially in the center.

Evidence A. Hard: S26-D-00001365.

S26-T-00001367 - Structure theorem for finitely generated abelian groups

Every finitely generated abelian group is isomorphic to Z^r direct-sum a finite direct sum of cyclic p-power groups; the invariants are unique up to order.

Evidence A. Hard: S26-D-00001366.

S26-T-00001368 - Burnside p^a q^b theorem

Every finite group whose order is divisible by at most two distinct primes is solvable.

Evidence A. Hard: S26-T-00001367.

S26-N-00001369 - Classification boundary

The classification of finite simple groups is a deep theorem family; this Synopsis records its consequence-level interface separately from proof infrastructure.

Evidence A. Hard: S26-T-00001368.

S26-B-00001370 - Group extensions

An exact sequence 1->N->G->Q->1 presents G as an extension of Q by N; extension theory links group structure to cohomological invariants.

Evidence A. Hard: S26-N-00001369.

S26-E-00001371 - Symmetric and alternating group orders at n=3

|S_3|=6 and |A_3|=3. A_3 is simple for n>=5.

Evidence M. Hard: S26-B-00001370.

S26-E-00001372 - Symmetric and alternating group orders at n=4

|S_4|=24 and |A_4|=12. A_4 is simple for n>=5.

Evidence M. Hard: S26-E-00001371.

S26-E-00001373 - Symmetric and alternating group orders at n=5

|S_5|=120 and |A_5|=60. A_5 is simple for n>=5.

Evidence M. Hard: S26-E-00001372.

S26-E-00001374 - Symmetric and alternating group orders at n=6

|S_6|=720 and |A_6|=360. A_6 is simple for n>=5.

Evidence M. Hard: S26-E-00001373.

S26-E-00001375 - Symmetric and alternating group orders at n=7

|S_7|=5040 and |A_7|=2520. A_7 is simple for n>=5.

Evidence M. Hard: S26-E-00001374.

S26-E-00001376 - Symmetric and alternating group orders at n=8

|S_8|=40320 and |A_8|=20160. A_8 is simple for n>=5.

Evidence M. Hard: S26-E-00001375.

S26-E-00001377 - Symmetric and alternating group orders at n=9

|S_9|=362880 and |A_9|=181440. A_9 is simple for n>=5.

Evidence M. Hard: S26-E-00001376.

S26-E-00001378 - Symmetric and alternating group orders at n=10

|S_10|=3628800 and |A_10|=1814400. A_10 is simple for n>=5.

Evidence M. Hard: S26-E-00001377.

S26-E-00001379 - Symmetric and alternating group orders at n=11

|S_11|=39916800 and |A_11|=19958400. A_11 is simple for n>=5.

Evidence M. Hard: S26-E-00001378.

S26-E-00001380 - Symmetric and alternating group orders at n=12

|S_12|=479001600 and |A_12|=239500800. A_12 is simple for n>=5.

Evidence M. Hard: S26-E-00001379.

S26-E-00001381 - Symmetric and alternating group orders at n=13

|S_13|=6227020800 and |A_13|=3113510400. A_13 is simple for n>=5.

Evidence M. Hard: S26-E-00001380.

S26-E-00001382 - Symmetric and alternating group orders at n=14

|S_14|=87178291200 and |A_14|=43589145600. A_14 is simple for n>=5.

Evidence M. Hard: S26-E-00001381.

S26-E-00001383 - Symmetric and alternating group orders at n=15

|S_15|=1307674368000 and |A_15|=653837184000. A_15 is simple for n>=5.

Evidence M. Hard: S26-E-00001382.

S26-E-00001384 - Symmetric and alternating group orders at n=16

|S_16|=20922789888000 and |A_16|=10461394944000. A_16 is simple for n>=5.

Evidence M. Hard: S26-E-00001383.

S26-E-00001385 - Symmetric and alternating group orders at n=17

|S_17|=355687428096000 and |A_17|=177843714048000. A_17 is simple for n>=5.

Evidence M. Hard: S26-E-00001384.

S26-E-00001386 - Dihedral group D_2 order profile

With the convention D_2=<r,s | r^2=s^2=1, srs=r^(-1)>, the group has order 4.

Evidence M. Hard: S26-E-00001385.

S26-E-00001387 - Dihedral group D_3 order profile

With the convention D_3=<r,s | r^3=s^2=1, srs=r^(-1)>, the group has order 6.

Evidence M. Hard: S26-E-00001386.

S26-E-00001388 - Dihedral group D_4 order profile

With the convention D_4=<r,s | r^4=s^2=1, srs=r^(-1)>, the group has order 8.

Evidence M. Hard: S26-E-00001387.

S26-E-00001389 - Dihedral group D_5 order profile

With the convention D_5=<r,s | r^5=s^2=1, srs=r^(-1)>, the group has order 10.

Evidence M. Hard: S26-E-00001388.

S26-E-00001390 - Dihedral group D_6 order profile

With the convention D_6=<r,s | r^6=s^2=1, srs=r^(-1)>, the group has order 12.

Evidence M. Hard: S26-E-00001389.

S26-E-00001391 - Dihedral group D_7 order profile

With the convention D_7=<r,s | r^7=s^2=1, srs=r^(-1)>, the group has order 14.

Evidence M. Hard: S26-E-00001390.

S26-E-00001392 - Dihedral group D_8 order profile

With the convention D_8=<r,s | r^8=s^2=1, srs=r^(-1)>, the group has order 16.

Evidence M. Hard: S26-E-00001391.

S26-E-00001393 - Dihedral group D_9 order profile

With the convention D_9=<r,s | r^9=s^2=1, srs=r^(-1)>, the group has order 18.

Evidence M. Hard: S26-E-00001392.

S26-E-00001394 - Dihedral group D_10 order profile

With the convention D_10=<r,s | r^10=s^2=1, srs=r^(-1)>, the group has order 20.

Evidence M. Hard: S26-E-00001393.

S26-E-00001395 - Dihedral group D_11 order profile

With the convention D_11=<r,s | r^11=s^2=1, srs=r^(-1)>, the group has order 22.

Evidence M. Hard: S26-E-00001394.

S26-E-00001396 - Dihedral group D_12 order profile

With the convention D_12=<r,s | r^12=s^2=1, srs=r^(-1)>, the group has order 24.

Evidence M. Hard: S26-E-00001395.

S26-E-00001397 - Dihedral group D_13 order profile

With the convention D_13=<r,s | r^13=s^2=1, srs=r^(-1)>, the group has order 26.

Evidence M. Hard: S26-E-00001396.

S26-E-00001398 - Dihedral group D_14 order profile

With the convention D_14=<r,s | r^14=s^2=1, srs=r^(-1)>, the group has order 28.

Evidence M. Hard: S26-E-00001397.

S26-E-00001399 - Dihedral group D_15 order profile

With the convention D_15=<r,s | r^15=s^2=1, srs=r^(-1)>, the group has order 30.

Evidence M. Hard: S26-E-00001398.

S26-E-00001400 - Dihedral group D_16 order profile

With the convention D_16=<r,s | r^16=s^2=1, srs=r^(-1)>, the group has order 32.

Evidence M. Hard: S26-E-00001399.

Part I - Representation theory

S26-D-00001401 - Linear representation

A representation of G on vector space V over field F is a homomorphism rho:G->GL(V).

Evidence A. Hard: S26-D-00000149.

S26-D-00001402 - G-module

A G-module over F is an F-vector space V with an F-linear G-action; equivalently it is a module over the group algebra F[G].

Evidence A. Hard: S26-D-00001401.

S26-D-00001403 - Invariant subspace

W<=V is G-invariant when rho(g)W subseteq W for every g in G.

Evidence A. Hard: S26-D-00001402.

S26-D-00001404 - Irreducible representation

A nonzero representation is irreducible when it has no invariant subspaces other than 0 and V.

Evidence A. Hard: S26-D-00001403.

S26-D-00001405 - Completely reducible representation

A representation is completely reducible when it is a direct sum of irreducible representations.

Evidence A. Hard: S26-D-00001404.

S26-T-00001406 - Maschke theorem

If G is finite and char(F) does not divide |G|, every finite-dimensional F-representation of G is completely reducible.

Evidence A. Hard: S26-D-00001405.

S26-D-00001407 - Character of a representation

For finite-dimensional rho, chi(g)=Tr(rho(g)); characters are constant on conjugacy classes.

Evidence A. Hard: S26-T-00001406.

S26-D-00001408 - Character inner product

For complex class functions on finite G, <chi,psi>=(1/|G|)sum_{g in G} chi(g) overline{psi(g)}.

Evidence A. Hard: S26-D-00001407.

S26-T-00001409 - Orthogonality of irreducible characters

Distinct irreducible complex characters are orthogonal and each irreducible character has norm 1 under the character inner product.

Evidence A. Hard: S26-D-00001408.

S26-T-00001410 - Number of irreducible characters

For a finite group over C, the number of irreducible characters equals the number of conjugacy classes.

Evidence A. Hard: S26-T-00001409.

S26-I-00001411 - Sum of squared irreducible dimensions

For finite G over C, sum_{rho irreducible}(dim rho)^2=|G|.

Evidence A. Hard: S26-T-00001410.

S26-D-00001412 - Regular representation

The left regular representation acts on F[G] by left multiplication.

Evidence A. Hard: S26-I-00001411.

S26-I-00001413 - Regular-character decomposition

Over C, the regular representation contains each irreducible representation with multiplicity equal to its dimension.

Evidence A. Hard: S26-D-00001412.

S26-D-00001414 - Induced representation

For H<=G and an H-representation W, Ind_H^G W is obtained from F[G] tensor_{F[H]} W.

Evidence A. Hard: S26-I-00001413.

S26-T-00001415 - Frobenius reciprocity

For suitable finite-dimensional representations, Hom_G(Ind_H^G W,V) is naturally isomorphic to Hom_H(W,Res_H^G V).

Evidence A. Hard: S26-D-00001414.

S26-D-00001416 - Tensor product representation

rho tensor sigma acts by (rho tensor sigma)(g)=rho(g) tensor sigma(g).

Evidence A. Hard: S26-T-00001415.

S26-I-00001417 - Character of a tensor product

chi_{V tensor W}(g)=chi_V(g) chi_W(g).

Evidence A. Hard: S26-D-00001416.

S26-D-00001418 - Dual representation

The dual representation on V* is rho*(g)=rho(g^{-1})^T in a chosen basis.

Evidence A. Hard: S26-I-00001417.

S26-I-00001419 - Character of the dual

For complex unitary realizations of finite-group representations, chi_{V*}(g)=overline{chi_V(g)}.

Evidence A. Hard: S26-D-00001418.

S26-B-00001420 - Representation-theoretic Fourier transform

For finite groups, decomposing functions on G into irreducible representations generalizes the ordinary Fourier transform from abelian groups to noncommutative symmetry.

Evidence A. Hard: S26-I-00001419.

S26-E-00001421 - Complex irreducibles of C_1

The cyclic group C_1 has exactly 1 one-dimensional irreducible complex representations, indexed by the 1-th roots of unity. The sum of squared dimensions is 1.

Evidence M. Hard: S26-B-00001420.

S26-E-00001422 - Complex irreducibles of C_2

The cyclic group C_2 has exactly 2 one-dimensional irreducible complex representations, indexed by the 2-th roots of unity. The sum of squared dimensions is 2.

Evidence M. Hard: S26-E-00001421.

S26-E-00001423 - Complex irreducibles of C_3

The cyclic group C_3 has exactly 3 one-dimensional irreducible complex representations, indexed by the 3-th roots of unity. The sum of squared dimensions is 3.

Evidence M. Hard: S26-E-00001422.

S26-E-00001424 - Complex irreducibles of C_4

The cyclic group C_4 has exactly 4 one-dimensional irreducible complex representations, indexed by the 4-th roots of unity. The sum of squared dimensions is 4.

Evidence M. Hard: S26-E-00001423.

S26-E-00001425 - Complex irreducibles of C_5

The cyclic group C_5 has exactly 5 one-dimensional irreducible complex representations, indexed by the 5-th roots of unity. The sum of squared dimensions is 5.

Evidence M. Hard: S26-E-00001424.

S26-E-00001426 - Complex irreducibles of C_6

The cyclic group C_6 has exactly 6 one-dimensional irreducible complex representations, indexed by the 6-th roots of unity. The sum of squared dimensions is 6.

Evidence M. Hard: S26-E-00001425.

S26-E-00001427 - Complex irreducibles of C_7

The cyclic group C_7 has exactly 7 one-dimensional irreducible complex representations, indexed by the 7-th roots of unity. The sum of squared dimensions is 7.

Evidence M. Hard: S26-E-00001426.

S26-E-00001428 - Complex irreducibles of C_8

The cyclic group C_8 has exactly 8 one-dimensional irreducible complex representations, indexed by the 8-th roots of unity. The sum of squared dimensions is 8.

Evidence M. Hard: S26-E-00001427.

S26-E-00001429 - Complex irreducibles of C_9

The cyclic group C_9 has exactly 9 one-dimensional irreducible complex representations, indexed by the 9-th roots of unity. The sum of squared dimensions is 9.

Evidence M. Hard: S26-E-00001428.

S26-E-00001430 - Complex irreducibles of C_10

The cyclic group C_10 has exactly 10 one-dimensional irreducible complex representations, indexed by the 10-th roots of unity. The sum of squared dimensions is 10.

Evidence M. Hard: S26-E-00001429.

S26-E-00001431 - Complex irreducibles of C_11

The cyclic group C_11 has exactly 11 one-dimensional irreducible complex representations, indexed by the 11-th roots of unity. The sum of squared dimensions is 11.

Evidence M. Hard: S26-E-00001430.

S26-E-00001432 - Complex irreducibles of C_12

The cyclic group C_12 has exactly 12 one-dimensional irreducible complex representations, indexed by the 12-th roots of unity. The sum of squared dimensions is 12.

Evidence M. Hard: S26-E-00001431.

S26-E-00001433 - Complex irreducibles of C_13

The cyclic group C_13 has exactly 13 one-dimensional irreducible complex representations, indexed by the 13-th roots of unity. The sum of squared dimensions is 13.

Evidence M. Hard: S26-E-00001432.

S26-E-00001434 - Complex irreducibles of C_14

The cyclic group C_14 has exactly 14 one-dimensional irreducible complex representations, indexed by the 14-th roots of unity. The sum of squared dimensions is 14.

Evidence M. Hard: S26-E-00001433.

S26-E-00001435 - Complex irreducibles of C_15

The cyclic group C_15 has exactly 15 one-dimensional irreducible complex representations, indexed by the 15-th roots of unity. The sum of squared dimensions is 15.

Evidence M. Hard: S26-E-00001434.

S26-E-00001436 - Complex irreducibles of C_16

The cyclic group C_16 has exactly 16 one-dimensional irreducible complex representations, indexed by the 16-th roots of unity. The sum of squared dimensions is 16.

Evidence M. Hard: S26-E-00001435.

S26-E-00001437 - Complex irreducibles of C_17

The cyclic group C_17 has exactly 17 one-dimensional irreducible complex representations, indexed by the 17-th roots of unity. The sum of squared dimensions is 17.

Evidence M. Hard: S26-E-00001436.

S26-E-00001438 - Complex irreducibles of C_18

The cyclic group C_18 has exactly 18 one-dimensional irreducible complex representations, indexed by the 18-th roots of unity. The sum of squared dimensions is 18.

Evidence M. Hard: S26-E-00001437.

S26-E-00001439 - Complex irreducibles of C_19

The cyclic group C_19 has exactly 19 one-dimensional irreducible complex representations, indexed by the 19-th roots of unity. The sum of squared dimensions is 19.

Evidence M. Hard: S26-E-00001438.

S26-E-00001440 - Complex irreducibles of C_20

The cyclic group C_20 has exactly 20 one-dimensional irreducible complex representations, indexed by the 20-th roots of unity. The sum of squared dimensions is 20.

Evidence M. Hard: S26-E-00001439.

S26-E-00001441 - Complex irreducibles of C_21

The cyclic group C_21 has exactly 21 one-dimensional irreducible complex representations, indexed by the 21-th roots of unity. The sum of squared dimensions is 21.

Evidence M. Hard: S26-E-00001440.

S26-E-00001442 - Complex irreducibles of C_22

The cyclic group C_22 has exactly 22 one-dimensional irreducible complex representations, indexed by the 22-th roots of unity. The sum of squared dimensions is 22.

Evidence M. Hard: S26-E-00001441.

S26-E-00001443 - Complex irreducibles of C_23

The cyclic group C_23 has exactly 23 one-dimensional irreducible complex representations, indexed by the 23-th roots of unity. The sum of squared dimensions is 23.

Evidence M. Hard: S26-E-00001442.

S26-E-00001444 - Complex irreducibles of C_24

The cyclic group C_24 has exactly 24 one-dimensional irreducible complex representations, indexed by the 24-th roots of unity. The sum of squared dimensions is 24.

Evidence M. Hard: S26-E-00001443.

S26-E-00001445 - Complex irreducibles of C_25

The cyclic group C_25 has exactly 25 one-dimensional irreducible complex representations, indexed by the 25-th roots of unity. The sum of squared dimensions is 25.

Evidence M. Hard: S26-E-00001444.

S26-E-00001446 - Irreducible-count profile for S_1

The symmetric group S_1 has 1 complex irreducible representations, equal to the number of partitions of 1.

Evidence M. Hard: S26-E-00001445.

S26-E-00001447 - Irreducible-count profile for S_2

The symmetric group S_2 has 2 complex irreducible representations, equal to the number of partitions of 2.

Evidence M. Hard: S26-E-00001446.

S26-E-00001448 - Irreducible-count profile for S_3

The symmetric group S_3 has 3 complex irreducible representations, equal to the number of partitions of 3.

Evidence M. Hard: S26-E-00001447.

S26-E-00001449 - Irreducible-count profile for S_4

The symmetric group S_4 has 5 complex irreducible representations, equal to the number of partitions of 4.

Evidence M. Hard: S26-E-00001448.

S26-E-00001450 - Irreducible-count profile for S_5

The symmetric group S_5 has 7 complex irreducible representations, equal to the number of partitions of 5.

Evidence M. Hard: S26-E-00001449.

Part J - Rings, ideals, and modules

S26-D-00001451 - Ring ideal

An ideal I of a ring R is an additive subgroup stable under multiplication by arbitrary ring elements on the required side; in commutative rings left/right distinctions disappear.

Evidence A. Hard: S26-D-00000101.

S26-D-00001452 - Prime ideal

In a commutative ring R, a proper ideal p is prime when ab in p implies a in p or b in p.

Evidence A. Hard: S26-D-00001451.

S26-D-00001453 - Maximal ideal

A proper ideal m is maximal when there is no proper ideal strictly between m and R.

Evidence A. Hard: S26-D-00001452.

S26-T-00001454 - Maximal ideals are prime

Every maximal ideal in a commutative ring with identity is prime.

Evidence A. Hard: S26-D-00001453.

S26-T-00001455 - Prime quotient criterion

For commutative R, p is prime iff R/p is an integral domain.

Evidence A. Hard: S26-T-00001454.

S26-T-00001456 - Maximal quotient criterion

For commutative R, m is maximal iff R/m is a field.

Evidence A. Hard: S26-T-00001455.

S26-D-00001457 - Principal ideal domain

A PID is an integral domain in which every ideal is generated by one element.

Evidence A. Hard: S26-T-00001456.

S26-D-00001458 - Euclidean domain

A Euclidean domain is an integral domain with a Euclidean function supporting division with remainder; every Euclidean domain is a PID.

Evidence A. Hard: S26-D-00001457.

S26-T-00001459 - PID implies UFD

Every principal ideal domain is a unique factorization domain.

Evidence A. Hard: S26-D-00001458.

S26-D-00001460 - Module over a ring

An R-module generalizes a vector space by allowing scalars from a ring R rather than necessarily a field.

Evidence A. Hard: S26-T-00001459.

S26-D-00001461 - Submodule and quotient module

A submodule N<=M is closed under module operations; M/N inherits an R-module structure.

Evidence A. Hard: S26-D-00001460.

S26-T-00001462 - First isomorphism theorem for modules

For an R-linear map f:M->N, M/ker f is isomorphic to im f.

Evidence A. Hard: S26-D-00001461.

S26-D-00001463 - Free module

A free R-module has a basis in the module sense and is isomorphic to a direct sum of copies of R.

Evidence A. Hard: S26-T-00001462.

S26-D-00001464 - Torsion element of a module

For module M over an integral domain R, m is torsion when rm=0 for some nonzero r in R.

Evidence A. Hard: S26-D-00001463.

S26-D-00001465 - Finitely generated module

M is finitely generated when a finite subset spans M as an R-module.

Evidence A. Hard: S26-D-00001464.

S26-T-00001466 - Structure theorem over a PID

Every finitely generated module over a PID decomposes into a free part plus cyclic torsion modules determined by invariant factors or elementary divisors.

Evidence A. Hard: S26-D-00001465.

S26-D-00001467 - Exact sequence of modules

A sequence of module maps is exact at a term when the image of the incoming map equals the kernel of the outgoing map.

Evidence A. Hard: S26-T-00001466.

S26-D-00001468 - Projective module

P is projective when Hom(P,-) preserves epimorphism lifting; equivalently every short exact sequence ending in P splits.

Evidence A. Hard: S26-D-00001467.

S26-D-00001469 - Injective module

I is injective when Hom(-,I) sends monomorphisms to surjections on Hom sets, equivalently extensions across submodules exist.

Evidence A. Hard: S26-D-00001468.

S26-D-00001470 - Flat module

An R-module M is flat when tensoring with M preserves exactness of short exact sequences in the appropriate module category; equivalently it preserves injections in the standard one-sided formulation.

Evidence A. Hard: S26-D-00001469.

S26-E-00001471 - Residue ring Z/2Z profile

Z/2Z has 2 elements and 1 units. It is a field iff 2 is prime; its unit residues are [1].

Evidence M. Hard: S26-D-00001470.

S26-E-00001472 - Residue ring Z/3Z profile

Z/3Z has 3 elements and 2 units. It is a field iff 3 is prime; its unit residues are [1, 2].

Evidence M. Hard: S26-E-00001471.

S26-E-00001473 - Residue ring Z/4Z profile

Z/4Z has 4 elements and 2 units. It is a field iff 4 is prime; its unit residues are [1, 3].

Evidence M. Hard: S26-E-00001472.

S26-E-00001474 - Residue ring Z/5Z profile

Z/5Z has 5 elements and 4 units. It is a field iff 5 is prime; its unit residues are [1, 2, 3, 4].

Evidence M. Hard: S26-E-00001473.

S26-E-00001475 - Residue ring Z/6Z profile

Z/6Z has 6 elements and 2 units. It is a field iff 6 is prime; its unit residues are [1, 5].

Evidence M. Hard: S26-E-00001474.

S26-E-00001476 - Residue ring Z/7Z profile

Z/7Z has 7 elements and 6 units. It is a field iff 7 is prime; its unit residues are [1, 2, 3, 4, 5, 6].

Evidence M. Hard: S26-E-00001475.

S26-E-00001477 - Residue ring Z/8Z profile

Z/8Z has 8 elements and 4 units. It is a field iff 8 is prime; its unit residues are [1, 3, 5, 7].

Evidence M. Hard: S26-E-00001476.

S26-E-00001478 - Residue ring Z/9Z profile

Z/9Z has 9 elements and 6 units. It is a field iff 9 is prime; its unit residues are [1, 2, 4, 5, 7, 8].

Evidence M. Hard: S26-E-00001477.

S26-E-00001479 - Residue ring Z/10Z profile

Z/10Z has 10 elements and 4 units. It is a field iff 10 is prime; its unit residues are [1, 3, 7, 9].

Evidence M. Hard: S26-E-00001478.

S26-E-00001480 - Residue ring Z/11Z profile

Z/11Z has 11 elements and 10 units. It is a field iff 11 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

Evidence M. Hard: S26-E-00001479.

S26-E-00001481 - Residue ring Z/12Z profile

Z/12Z has 12 elements and 4 units. It is a field iff 12 is prime; its unit residues are [1, 5, 7, 11].

Evidence M. Hard: S26-E-00001480.

S26-E-00001482 - Residue ring Z/13Z profile

Z/13Z has 13 elements and 12 units. It is a field iff 13 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12].

Evidence M. Hard: S26-E-00001481.

S26-E-00001483 - Residue ring Z/14Z profile

Z/14Z has 14 elements and 6 units. It is a field iff 14 is prime; its unit residues are [1, 3, 5, 9, 11, 13].

Evidence M. Hard: S26-E-00001482.

S26-E-00001484 - Residue ring Z/15Z profile

Z/15Z has 15 elements and 8 units. It is a field iff 15 is prime; its unit residues are [1, 2, 4, 7, 8, 11, 13, 14].

Evidence M. Hard: S26-E-00001483.

S26-E-00001485 - Residue ring Z/16Z profile

Z/16Z has 16 elements and 8 units. It is a field iff 16 is prime; its unit residues are [1, 3, 5, 7, 9, 11, 13, 15].

Evidence M. Hard: S26-E-00001484.

S26-E-00001486 - Residue ring Z/17Z profile

Z/17Z has 17 elements and 16 units. It is a field iff 17 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16].

Evidence M. Hard: S26-E-00001485.

S26-E-00001487 - Residue ring Z/18Z profile

Z/18Z has 18 elements and 6 units. It is a field iff 18 is prime; its unit residues are [1, 5, 7, 11, 13, 17].

Evidence M. Hard: S26-E-00001486.

S26-E-00001488 - Residue ring Z/19Z profile

Z/19Z has 19 elements and 18 units. It is a field iff 19 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18].

Evidence M. Hard: S26-E-00001487.

S26-E-00001489 - Residue ring Z/20Z profile

Z/20Z has 20 elements and 8 units. It is a field iff 20 is prime; its unit residues are [1, 3, 7, 9, 11, 13, 17, 19].

Evidence M. Hard: S26-E-00001488.

S26-E-00001490 - Residue ring Z/21Z profile

Z/21Z has 21 elements and 12 units. It is a field iff 21 is prime; its unit residues are [1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20].

Evidence M. Hard: S26-E-00001489.

S26-E-00001491 - Residue ring Z/22Z profile

Z/22Z has 22 elements and 10 units. It is a field iff 22 is prime; its unit residues are [1, 3, 5, 7, 9, 13, 15, 17, 19, 21].

Evidence M. Hard: S26-E-00001490.

S26-E-00001492 - Residue ring Z/23Z profile

Z/23Z has 23 elements and 22 units. It is a field iff 23 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20] ....

Evidence M. Hard: S26-E-00001491.

S26-E-00001493 - Residue ring Z/24Z profile

Z/24Z has 24 elements and 8 units. It is a field iff 24 is prime; its unit residues are [1, 5, 7, 11, 13, 17, 19, 23].

Evidence M. Hard: S26-E-00001492.

S26-E-00001494 - Residue ring Z/25Z profile

Z/25Z has 25 elements and 20 units. It is a field iff 25 is prime; its unit residues are [1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17, 18, 19, 21, 22, 23, 24].

Evidence M. Hard: S26-E-00001493.

S26-E-00001495 - Residue ring Z/26Z profile

Z/26Z has 26 elements and 12 units. It is a field iff 26 is prime; its unit residues are [1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25].

Evidence M. Hard: S26-E-00001494.

S26-E-00001496 - Residue ring Z/27Z profile

Z/27Z has 27 elements and 18 units. It is a field iff 27 is prime; its unit residues are [1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26].

Evidence M. Hard: S26-E-00001495.

S26-E-00001497 - Residue ring Z/28Z profile

Z/28Z has 28 elements and 12 units. It is a field iff 28 is prime; its unit residues are [1, 3, 5, 9, 11, 13, 15, 17, 19, 23, 25, 27].

Evidence M. Hard: S26-E-00001496.

S26-E-00001498 - Residue ring Z/29Z profile

Z/29Z has 29 elements and 28 units. It is a field iff 29 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20] ....

Evidence M. Hard: S26-E-00001497.

S26-E-00001499 - Residue ring Z/30Z profile

Z/30Z has 30 elements and 8 units. It is a field iff 30 is prime; its unit residues are [1, 7, 11, 13, 17, 19, 23, 29].

Evidence M. Hard: S26-E-00001498.

S26-E-00001500 - Residue ring Z/31Z profile

Z/31Z has 31 elements and 30 units. It is a field iff 31 is prime; its unit residues are [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20] ....

Evidence M. Hard: S26-E-00001499.

Part K - Commutative algebra

S26-D-00001501 - Noetherian ring

A commutative ring R is Noetherian when every ascending chain of ideals stabilizes, equivalently every ideal is finitely generated.

Evidence A. Hard: S26-D-00000101.

S26-T-00001502 - Hilbert basis theorem

If R is Noetherian, then R[x] is Noetherian; iterating gives R[x_1,...,x_n] Noetherian.

Evidence A. Hard: S26-D-00001501.

S26-D-00001503 - Localization of a ring

For multiplicative S subset R, S^{-1}R formally inverts all elements of S.

Evidence A. Hard: S26-T-00001502.

S26-D-00001504 - Localization at a prime

R_p=(R minus p)^(-1)R is the local ring obtained by inverting all elements outside prime p.

Evidence A. Hard: S26-D-00001503.

S26-D-00001505 - Local ring

A commutative ring is local when it has a unique maximal ideal.

Evidence A. Hard: S26-D-00001504.

S26-T-00001506 - Unit criterion in a local ring

In local ring (R,m), the units are exactly R minus m.

Evidence A. Hard: S26-D-00001505.

S26-D-00001507 - Radical of an ideal

sqrt(I)={r in R:r^n in I for some n>=1}.

Evidence A. Hard: S26-T-00001506.

S26-T-00001508 - Radical as intersection of primes

sqrt(I) is the intersection of all prime ideals containing I.

Evidence A. Hard: S26-D-00001507.

S26-D-00001509 - Primary ideal

A proper ideal Q is primary when ab in Q and a notin Q imply b^n in Q for some n>=1.

Evidence A. Hard: S26-T-00001508.

S26-T-00001510 - Lasker-Noether primary decomposition

In a Noetherian commutative ring, every proper ideal admits a finite primary decomposition.

Evidence A. Hard: S26-D-00001509.

S26-D-00001511 - Krull dimension

dim R is the supremum of lengths of strict chains of prime ideals.

Evidence A. Hard: S26-T-00001510.

S26-D-00001512 - Integral dependence

s in an R-algebra S is integral over R when it satisfies a monic polynomial with coefficients in R.

Evidence A. Hard: S26-D-00001511.

S26-T-00001513 - Going-up theorem

For an integral extension R subset S, chains of prime ideals in R can be lifted upward to chains in S under the standard lying-over setup.

Evidence A. Hard: S26-D-00001512.

S26-D-00001514 - Tensor product of modules

M tensor_R N represents R-balanced bilinear maps out of MxN by a universal property.

Evidence A. Hard: S26-T-00001513.

S26-D-00001515 - Tor functor

Tor_n^R(M,N) is the n-th left-derived functor of tensor product, measuring failure of flatness/exactness.

Evidence A. Hard: S26-D-00001514.

S26-D-00001516 - Ext functor

Ext_R^n(M,N) is the n-th right-derived functor of Hom_R(M,-) or Hom_R(-,N) in the corresponding resolved variable.

Evidence A. Hard: S26-D-00001515.

S26-D-00001517 - Regular sequence

x_1,...,x_r is an M-regular sequence when each x_i is a non-zero-divisor on M/(x_1,...,x_{i-1})M and the final quotient is nonzero under the chosen convention.

Evidence A. Hard: S26-D-00001516.

S26-D-00001518 - Cohen-Macaulay ring

A Noetherian local ring is Cohen-Macaulay when its depth equals its Krull dimension.

Evidence A. Hard: S26-D-00001517.

S26-D-00001519 - Nakayama lemma

If M is a finitely generated module over local ring (R,m) and mM=M, then M=0; more generally generators modulo m lift to generators of M.

Evidence A. Hard: S26-D-00001518.

S26-B-00001520 - Commutative algebra to geometry

Prime ideals become points of Spec R, localization becomes passage to neighborhoods/stalks, and modules become algebraic data for sheaves.

Evidence A. Hard: S26-D-00001519.

S26-E-00001521 - Polynomial factorization over F_2: x**2 + 1

Over F_2, x**2 + 1 factors as (x + 1)^2.

Evidence M. Hard: S26-B-00001520.

S26-E-00001522 - Polynomial factorization over F_2: x**2 - 2

Over F_2, x**2 - 2 factors as (x)^2.

Evidence M. Hard: S26-E-00001521.

S26-E-00001523 - Polynomial factorization over F_2: x**2 + x + 1

Over F_2, x**2 + x + 1 factors as (x**2 + x + 1)^1.

Evidence M. Hard: S26-E-00001522.

S26-E-00001524 - Polynomial factorization over F_2: x**3 - 2

Over F_2, x**3 - 2 factors as (x)^3.

Evidence M. Hard: S26-E-00001523.

S26-E-00001525 - Polynomial factorization over F_2: x**3 - x - 1

Over F_2, x**3 - x - 1 factors as (x**3 + x + 1)^1.

Evidence M. Hard: S26-E-00001524.

S26-E-00001526 - Polynomial factorization over F_2: x**4 + 1

Over F_2, x**4 + 1 factors as (x + 1)^4.

Evidence M. Hard: S26-E-00001525.

S26-E-00001527 - Polynomial factorization over F_2: x**4 - x + 1

Over F_2, x**4 - x + 1 factors as (x**4 + x + 1)^1.

Evidence M. Hard: S26-E-00001526.

S26-E-00001528 - Polynomial factorization over F_2: x**4 + 2

Over F_2, x**4 + 2 factors as (x)^4.

Evidence M. Hard: S26-E-00001527.

S26-E-00001529 - Polynomial factorization over F_2: x**5 - 2

Over F_2, x**5 - 2 factors as (x)^5.

Evidence M. Hard: S26-E-00001528.

S26-E-00001530 - Polynomial factorization over F_2: x**5 - x - 1

Over F_2, x**5 - x - 1 factors as (x**2 + x + 1)^1 * (x**3 + x**2 + 1)^1.

Evidence M. Hard: S26-E-00001529.

S26-E-00001531 - Polynomial factorization over F_3: x**2 + 1

Over F_3, x**2 + 1 factors as (x**2 + 1)^1.

Evidence M. Hard: S26-E-00001530.

S26-E-00001532 - Polynomial factorization over F_3: x**2 - 2

Over F_3, x**2 - 2 factors as (x**2 + 1)^1.

Evidence M. Hard: S26-E-00001531.

S26-E-00001533 - Polynomial factorization over F_3: x**2 + x + 1

Over F_3, x**2 + x + 1 factors as (x - 1)^2.

Evidence M. Hard: S26-E-00001532.

S26-E-00001534 - Polynomial factorization over F_3: x**3 - 2

Over F_3, x**3 - 2 factors as (x + 1)^3.

Evidence M. Hard: S26-E-00001533.

S26-E-00001535 - Polynomial factorization over F_3: x**3 - x - 1

Over F_3, x**3 - x - 1 factors as (x**3 - x - 1)^1.

Evidence M. Hard: S26-E-00001534.

S26-E-00001536 - Polynomial factorization over F_3: x**4 + 1

Over F_3, x**4 + 1 factors as (x**2 + x - 1)^1 * (x**2 - x - 1)^1.

Evidence M. Hard: S26-E-00001535.

S26-E-00001537 - Polynomial factorization over F_3: x**4 - x + 1

Over F_3, x**4 - x + 1 factors as (x + 1)^1 * (x**3 - x**2 + x + 1)^1.

Evidence M. Hard: S26-E-00001536.

S26-E-00001538 - Polynomial factorization over F_3: x**4 + 2

Over F_3, x**4 + 2 factors as (x + 1)^1 * (x - 1)^1 * (x**2 + 1)^1.

Evidence M. Hard: S26-E-00001537.

S26-E-00001539 - Polynomial factorization over F_3: x**5 - 2

Over F_3, x**5 - 2 factors as (x + 1)^1 * (x**4 - x**3 + x**2 - x + 1)^1.

Evidence M. Hard: S26-E-00001538.

S26-E-00001540 - Polynomial factorization over F_3: x**5 - x - 1

Over F_3, x**5 - x - 1 factors as (x**5 - x - 1)^1.

Evidence M. Hard: S26-E-00001539.

S26-E-00001541 - Polynomial factorization over F_5: x**2 + 1

Over F_5, x**2 + 1 factors as (x + 2)^1 * (x - 2)^1.

Evidence M. Hard: S26-E-00001540.

S26-E-00001542 - Polynomial factorization over F_5: x**2 - 2

Over F_5, x**2 - 2 factors as (x**2 - 2)^1.

Evidence M. Hard: S26-E-00001541.

S26-E-00001543 - Polynomial factorization over F_5: x**2 + x + 1

Over F_5, x**2 + x + 1 factors as (x**2 + x + 1)^1.

Evidence M. Hard: S26-E-00001542.

S26-E-00001544 - Polynomial factorization over F_5: x**3 - 2

Over F_5, x**3 - 2 factors as (x + 2)^1 * (x**2 - 2*x - 1)^1.

Evidence M. Hard: S26-E-00001543.

S26-E-00001545 - Polynomial factorization over F_5: x**3 - x - 1

Over F_5, x**3 - x - 1 factors as (x - 2)^1 * (x**2 + 2*x - 2)^1.

Evidence M. Hard: S26-E-00001544.

S26-E-00001546 - Polynomial factorization over F_5: x**4 + 1

Over F_5, x**4 + 1 factors as (x**2 + 2)^1 * (x**2 - 2)^1.

Evidence M. Hard: S26-E-00001545.

S26-E-00001547 - Polynomial factorization over F_5: x**4 - x + 1

Over F_5, x**4 - x + 1 factors as (x - 2)^1 * (x**3 + 2*x**2 - x + 2)^1.

Evidence M. Hard: S26-E-00001546.

S26-E-00001548 - Polynomial factorization over F_5: x**4 + 2

Over F_5, x**4 + 2 factors as (x**4 + 2)^1.

Evidence M. Hard: S26-E-00001547.

S26-E-00001549 - Polynomial factorization over F_5: x**5 - 2

Over F_5, x**5 - 2 factors as (x - 2)^5.

Evidence M. Hard: S26-E-00001548.

S26-E-00001550 - Polynomial factorization over F_5: x**5 - x - 1

Over F_5, x**5 - x - 1 factors as (x**5 - x - 1)^1.

Evidence M. Hard: S26-E-00001549.

Part L - Fields and Galois theory

S26-D-00001551 - Field extension degree

For extension L/K, [L:K] is the dimension of L as a K-vector space.

Evidence A. Hard: S26-D-00000101.

S26-T-00001552 - Tower law

For K subset L subset M with finite degrees, [M:K]=[M:L][L:K].

Evidence A. Hard: S26-D-00001551.

S26-D-00001553 - Algebraic extension

L/K is algebraic when every element of L is algebraic over K.

Evidence A. Hard: S26-T-00001552.

S26-D-00001554 - Transcendence degree

A transcendence basis is a maximal algebraically independent subset; its cardinality is trdeg_K L.

Evidence A. Hard: S26-D-00001553.

S26-D-00001555 - Splitting field

A splitting field of f in K[x] is a minimal extension in which f factors into linear factors.

Evidence A. Hard: S26-D-00001554.

S26-D-00001556 - Separable extension

An algebraic extension is separable when every element has a separable minimal polynomial over the base field.

Evidence A. Hard: S26-D-00001555.

S26-D-00001557 - Normal extension

An algebraic extension L/K is normal when every irreducible polynomial over K having one root in L splits in L.

Evidence A. Hard: S26-D-00001556.

S26-D-00001558 - Galois extension

A finite extension is Galois iff it is normal and separable.

Evidence A. Hard: S26-D-00001557.

S26-D-00001559 - Galois group

Gal(L/K) is the group of field automorphisms of L fixing K pointwise.

Evidence A. Hard: S26-D-00001558.

S26-T-00001560 - Fundamental theorem of Galois theory

For finite Galois L/K, intermediate fields E correspond inclusion-reversingly to subgroups H<=Gal(L/K), with E=L^H and H=Gal(L/E).

Evidence A. Hard: S26-D-00001559.

S26-I-00001561 - Degree-index relation in Galois theory

For finite Galois L/K and H<=G, [L:L^H]=|H| and [L^H:K]=[G:H].

Evidence A. Hard: S26-T-00001560.

S26-D-00001562 - Fixed field

For H<=Aut(L), L^H={x in L:sigma(x)=x for all sigma in H}.

Evidence A. Hard: S26-I-00001561.

S26-D-00001563 - Solvable polynomial by radicals

A polynomial over characteristic-zero field is solvable by radicals when its splitting field lies in a tower obtained by adjoining radicals and roots of unity as needed.

Evidence A. Hard: S26-D-00001562.

S26-T-00001564 - Galois criterion for solvability by radicals

Over characteristic zero, a polynomial is solvable by radicals iff its Galois group is solvable.

Evidence A. Hard: S26-D-00001563.

S26-D-00001565 - Artin-Schreier extension

In characteristic p, an extension generated by a root of x^p-x-a is an Artin-Schreier extension when the polynomial is irreducible.

Evidence A. Hard: S26-T-00001564.

S26-D-00001566 - Kummer extension

When the base contains relevant roots of unity and characteristic does not divide n, adjoining n-th roots gives Kummer extensions controlled by multiplicative classes modulo n-th powers.

Evidence A. Hard: S26-D-00001565.

S26-T-00001567 - Primitive element theorem

Every finite separable extension L/K is simple: L=K(alpha) for some alpha.

Evidence A. Hard: S26-D-00001566.

S26-D-00001568 - Algebraic closure

An algebraic closure of K is an algebraic extension that is algebraically closed; it is unique up to noncanonical K-isomorphism.

Evidence A. Hard: S26-T-00001567.

S26-N-00001569 - Galois correspondence direction fence

Larger subgroups correspond to smaller fixed fields; the correspondence reverses inclusions.

Evidence A. Hard: S26-D-00001568. Boundary: Treating subgroup inclusion as field inclusion in the same direction reverses the theorem.

S26-B-00001570 - Galois theory as symmetry of equations

Galois groups encode permutation symmetry of roots while preserving base-field algebraic relations, turning solvability questions into group-theoretic structure.

Evidence A. Hard: S26-N-00001569.

S26-E-00001571 - Cyclotomic degree profile n=3

[Q(zeta_3):Q]=phi(3)=2; its Galois group has order 2 and is isomorphic to (Z/3Z)^x.

Evidence M. Hard: S26-B-00001570.

S26-E-00001572 - Cyclotomic degree profile n=4

[Q(zeta_4):Q]=phi(4)=2; its Galois group has order 2 and is isomorphic to (Z/4Z)^x.

Evidence M. Hard: S26-E-00001571.

S26-E-00001573 - Cyclotomic degree profile n=5

[Q(zeta_5):Q]=phi(5)=4; its Galois group has order 4 and is isomorphic to (Z/5Z)^x.

Evidence M. Hard: S26-E-00001572.

S26-E-00001574 - Cyclotomic degree profile n=6

[Q(zeta_6):Q]=phi(6)=2; its Galois group has order 2 and is isomorphic to (Z/6Z)^x.

Evidence M. Hard: S26-E-00001573.

S26-E-00001575 - Cyclotomic degree profile n=7

[Q(zeta_7):Q]=phi(7)=6; its Galois group has order 6 and is isomorphic to (Z/7Z)^x.

Evidence M. Hard: S26-E-00001574.

S26-E-00001576 - Cyclotomic degree profile n=8

[Q(zeta_8):Q]=phi(8)=4; its Galois group has order 4 and is isomorphic to (Z/8Z)^x.

Evidence M. Hard: S26-E-00001575.

S26-E-00001577 - Cyclotomic degree profile n=9

[Q(zeta_9):Q]=phi(9)=6; its Galois group has order 6 and is isomorphic to (Z/9Z)^x.

Evidence M. Hard: S26-E-00001576.

S26-E-00001578 - Cyclotomic degree profile n=10

[Q(zeta_10):Q]=phi(10)=4; its Galois group has order 4 and is isomorphic to (Z/10Z)^x.

Evidence M. Hard: S26-E-00001577.

S26-E-00001579 - Cyclotomic degree profile n=11

[Q(zeta_11):Q]=phi(11)=10; its Galois group has order 10 and is isomorphic to (Z/11Z)^x.

Evidence M. Hard: S26-E-00001578.

S26-E-00001580 - Cyclotomic degree profile n=12

[Q(zeta_12):Q]=phi(12)=4; its Galois group has order 4 and is isomorphic to (Z/12Z)^x.

Evidence M. Hard: S26-E-00001579.

S26-E-00001581 - Cyclotomic degree profile n=13

[Q(zeta_13):Q]=phi(13)=12; its Galois group has order 12 and is isomorphic to (Z/13Z)^x.

Evidence M. Hard: S26-E-00001580.

S26-E-00001582 - Cyclotomic degree profile n=14

[Q(zeta_14):Q]=phi(14)=6; its Galois group has order 6 and is isomorphic to (Z/14Z)^x.

Evidence M. Hard: S26-E-00001581.

S26-E-00001583 - Cyclotomic degree profile n=15

[Q(zeta_15):Q]=phi(15)=8; its Galois group has order 8 and is isomorphic to (Z/15Z)^x.

Evidence M. Hard: S26-E-00001582.

S26-E-00001584 - Cyclotomic degree profile n=16

[Q(zeta_16):Q]=phi(16)=8; its Galois group has order 8 and is isomorphic to (Z/16Z)^x.

Evidence M. Hard: S26-E-00001583.

S26-E-00001585 - Cyclotomic degree profile n=17

[Q(zeta_17):Q]=phi(17)=16; its Galois group has order 16 and is isomorphic to (Z/17Z)^x.

Evidence M. Hard: S26-E-00001584.

S26-E-00001586 - Cyclotomic degree profile n=18

[Q(zeta_18):Q]=phi(18)=6; its Galois group has order 6 and is isomorphic to (Z/18Z)^x.

Evidence M. Hard: S26-E-00001585.

S26-E-00001587 - Cyclotomic degree profile n=19

[Q(zeta_19):Q]=phi(19)=18; its Galois group has order 18 and is isomorphic to (Z/19Z)^x.

Evidence M. Hard: S26-E-00001586.

S26-E-00001588 - Cyclotomic degree profile n=20

[Q(zeta_20):Q]=phi(20)=8; its Galois group has order 8 and is isomorphic to (Z/20Z)^x.

Evidence M. Hard: S26-E-00001587.

S26-E-00001589 - Cyclotomic degree profile n=21

[Q(zeta_21):Q]=phi(21)=12; its Galois group has order 12 and is isomorphic to (Z/21Z)^x.

Evidence M. Hard: S26-E-00001588.

S26-E-00001590 - Cyclotomic degree profile n=22

[Q(zeta_22):Q]=phi(22)=10; its Galois group has order 10 and is isomorphic to (Z/22Z)^x.

Evidence M. Hard: S26-E-00001589.

S26-E-00001591 - Cyclotomic degree profile n=23

[Q(zeta_23):Q]=phi(23)=22; its Galois group has order 22 and is isomorphic to (Z/23Z)^x.

Evidence M. Hard: S26-E-00001590.

S26-E-00001592 - Cyclotomic degree profile n=24

[Q(zeta_24):Q]=phi(24)=8; its Galois group has order 8 and is isomorphic to (Z/24Z)^x.

Evidence M. Hard: S26-E-00001591.

S26-E-00001593 - Cyclotomic degree profile n=25

[Q(zeta_25):Q]=phi(25)=20; its Galois group has order 20 and is isomorphic to (Z/25Z)^x.

Evidence M. Hard: S26-E-00001592.

S26-E-00001594 - Cyclotomic degree profile n=26

[Q(zeta_26):Q]=phi(26)=12; its Galois group has order 12 and is isomorphic to (Z/26Z)^x.

Evidence M. Hard: S26-E-00001593.

S26-E-00001595 - Cyclotomic degree profile n=27

[Q(zeta_27):Q]=phi(27)=18; its Galois group has order 18 and is isomorphic to (Z/27Z)^x.

Evidence M. Hard: S26-E-00001594.

S26-E-00001596 - Cyclotomic degree profile n=28

[Q(zeta_28):Q]=phi(28)=12; its Galois group has order 12 and is isomorphic to (Z/28Z)^x.

Evidence M. Hard: S26-E-00001595.

S26-E-00001597 - Cyclotomic degree profile n=29

[Q(zeta_29):Q]=phi(29)=28; its Galois group has order 28 and is isomorphic to (Z/29Z)^x.

Evidence M. Hard: S26-E-00001596.

S26-E-00001598 - Cyclotomic degree profile n=30

[Q(zeta_30):Q]=phi(30)=8; its Galois group has order 8 and is isomorphic to (Z/30Z)^x.

Evidence M. Hard: S26-E-00001597.

S26-E-00001599 - Cyclotomic degree profile n=31

[Q(zeta_31):Q]=phi(31)=30; its Galois group has order 30 and is isomorphic to (Z/31Z)^x.

Evidence M. Hard: S26-E-00001598.

S26-E-00001600 - Cyclotomic degree profile n=32

[Q(zeta_32):Q]=phi(32)=16; its Galois group has order 16 and is isomorphic to (Z/32Z)^x.

Evidence M. Hard: S26-E-00001599.

Part M - Homological and noncommutative algebra

S26-D-00001601 - Chain complex

A chain complex C consists of objects C_n and differentials d_n:C_n->C_{n-1} with d_{n-1}d_n=0.

Evidence A. Hard: S26-D-00000267.

S26-D-00001602 - Cycle and boundary

Z_n=ker d_n are n-cycles and B_n=im d_{n+1} are n-boundaries; B_n subseteq Z_n.

Evidence A. Hard: S26-D-00001601.

S26-D-00001603 - Homology group

H_n(C)=Z_n/B_n.

Evidence A. Hard: S26-D-00001602.

S26-D-00001604 - Cochain complex

A cochain complex has maps d^n:C^n->C^{n+1} with d^{n+1}d^n=0; its cohomology is H^n=ker d^n/im d^{n-1}.

Evidence A. Hard: S26-D-00001603.

S26-D-00001605 - Chain map

A chain map f:C->D is a family f_n commuting with differentials.

Evidence A. Hard: S26-D-00001604.

S26-D-00001606 - Chain homotopy

Chain maps f,g:C->D are chain-homotopic when f-g=dh+hd for a graded family h of degree +1.

Evidence A. Hard: S26-D-00001605.

S26-T-00001607 - Chain-homotopic maps induce equal homology maps

If f and g are chain homotopic, H_n(f)=H_n(g) for all n.

Evidence A. Hard: S26-D-00001606.

S26-D-00001608 - Exact complex

A chain complex is exact when all homology groups vanish.

Evidence A. Hard: S26-T-00001607.

S26-D-00001609 - Projective resolution

A projective resolution of M is an exact complex of projective modules ending in M.

Evidence A. Hard: S26-D-00001608.

S26-D-00001610 - Injective resolution

An injective resolution embeds M into an exact cochain complex of injective modules.

Evidence A. Hard: S26-D-00001609.

S26-D-00001611 - Group algebra

For group G and field/ring F, F[G] is the free F-module on G with multiplication extended linearly from the group law.

Evidence A. Hard: S26-D-00001610.

S26-D-00001612 - Associative algebra

An associative algebra over commutative ring k is simultaneously a k-module and an associative ring with compatible scalar multiplication.

Evidence A. Hard: S26-D-00001611.

S26-D-00001613 - Lie algebra

A Lie algebra is a vector space with bilinear bracket [x,y] that is alternating/skew under suitable characteristic assumptions and satisfies the Jacobi identity.

Evidence A. Hard: S26-D-00001612.

S26-D-00001614 - Universal enveloping algebra

U(g) is the associative algebra universal for Lie-algebra maps g->A^- into commutator Lie algebras.

Evidence A. Hard: S26-D-00001613.

S26-T-00001615 - Poincare-Birkhoff-Witt theorem

For a Lie algebra over a field under standard hypotheses, ordered monomials in a basis give a vector-space basis of its universal enveloping algebra.

Evidence A. Hard: S26-D-00001614.

S26-D-00001616 - Center of an algebra

Z(A)={a in A:ab=ba for all b in A}.

Evidence A. Hard: S26-T-00001615.

S26-D-00001617 - Simple algebra

An associative algebra A is simple when it has no nontrivial two-sided ideals and A^2 is nonzero under the usual convention.

Evidence A. Hard: S26-D-00001616.

S26-T-00001618 - Artin-Wedderburn theorem

A semisimple Artinian ring is a finite product of matrix rings over division rings; over an algebraically closed field, finite-dimensional semisimple algebras are products of full matrix algebras.

Evidence A. Hard: S26-D-00001617.

S26-D-00001619 - Morita equivalence

Rings R and S are Morita equivalent when their module categories are equivalent.

Evidence A. Hard: S26-T-00001618.

S26-B-00001620 - Homology as obstruction extractor

Passing from cycles to cycles modulo boundaries converts failure of exactness into invariant algebraic data and recurs across topology, geometry, algebra, and mathematical physics.

Evidence A. Hard: S26-D-00001619.

S26-E-00001621 - Two-step chain-complex profile 1

For d1=[2,3] and d2=[3,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-B-00001620.

S26-E-00001622 - Two-step chain-complex profile 2

For d1=[3,5] and d2=[5,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001621.

S26-E-00001623 - Two-step chain-complex profile 3

For d1=[4,7] and d2=[7,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001622.

S26-E-00001624 - Two-step chain-complex profile 4

For d1=[5,2] and d2=[2,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001623.

S26-E-00001625 - Two-step chain-complex profile 5

For d1=[1,4] and d2=[4,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001624.

S26-E-00001626 - Two-step chain-complex profile 6

For d1=[2,6] and d2=[6,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001625.

S26-E-00001627 - Two-step chain-complex profile 7

For d1=[3,1] and d2=[1,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001626.

S26-E-00001628 - Two-step chain-complex profile 8

For d1=[4,3] and d2=[3,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001627.

S26-E-00001629 - Two-step chain-complex profile 9

For d1=[5,5] and d2=[5,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001628.

S26-E-00001630 - Two-step chain-complex profile 10

For d1=[1,7] and d2=[7,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001629.

S26-E-00001631 - Two-step chain-complex profile 11

For d1=[2,2] and d2=[2,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001630.

S26-E-00001632 - Two-step chain-complex profile 12

For d1=[3,4] and d2=[4,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001631.

S26-E-00001633 - Two-step chain-complex profile 13

For d1=[4,6] and d2=[6,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001632.

S26-E-00001634 - Two-step chain-complex profile 14

For d1=[5,1] and d2=[1,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001633.

S26-E-00001635 - Two-step chain-complex profile 15

For d1=[1,3] and d2=[3,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001634.

S26-E-00001636 - Two-step chain-complex profile 16

For d1=[2,5] and d2=[5,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001635.

S26-E-00001637 - Two-step chain-complex profile 17

For d1=[3,7] and d2=[7,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001636.

S26-E-00001638 - Two-step chain-complex profile 18

For d1=[4,2] and d2=[2,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001637.

S26-E-00001639 - Two-step chain-complex profile 19

For d1=[5,4] and d2=[4,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001638.

S26-E-00001640 - Two-step chain-complex profile 20

For d1=[1,6] and d2=[6,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001639.

S26-E-00001641 - Two-step chain-complex profile 21

For d1=[2,1] and d2=[1,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001640.

S26-E-00001642 - Two-step chain-complex profile 22

For d1=[3,3] and d2=[3,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001641.

S26-E-00001643 - Two-step chain-complex profile 23

For d1=[4,5] and d2=[5,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001642.

S26-E-00001644 - Two-step chain-complex profile 24

For d1=[5,7] and d2=[7,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001643.

S26-E-00001645 - Two-step chain-complex profile 25

For d1=[1,2] and d2=[2,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001644.

S26-E-00001646 - Two-step chain-complex profile 26

For d1=[2,4] and d2=[4,-2] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001645.

S26-E-00001647 - Two-step chain-complex profile 27

For d1=[3,6] and d2=[6,-3] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001646.

S26-E-00001648 - Two-step chain-complex profile 28

For d1=[4,1] and d2=[1,-4] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001647.

S26-E-00001649 - Two-step chain-complex profile 29

For d1=[5,3] and d2=[3,-5] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001648.

S26-E-00001650 - Two-step chain-complex profile 30

For d1=[1,5] and d2=[5,-1] over Q, d1*d2=0, rank(d1)=1, rank(d2)=1, and dim H_1=0.

Evidence M. Hard: S26-E-00001649.

Part N - Enumerative combinatorics

S26-D-00001651 - Ordinary generating function

For sequence a_n, A(x)=sum_{n>=0}a_n x^n is its ordinary generating function as a formal power series or analytic function where convergence is invoked.

Evidence A. Hard: S26-T-00000088.

S26-D-00001652 - Exponential generating function

For sequence a_n, E(x)=sum_{n>=0} a_n x^n/n! is its exponential generating function.

Evidence A. Hard: S26-D-00001651.

S26-T-00001653 - Product rule for ordinary generating functions

The coefficient of x^n in A(x)B(x) is the convolution sum_{k=0}^n a_k b_{n-k}.

Evidence A. Hard: S26-D-00001652.

S26-I-00001654 - Factorial formula for binomial coefficients

For integers 0<=k<=n, C(n,k)=n!/(k!(n-k)!), refining the frozen combinatorial coefficient interface by its factorial identity.

Evidence A. Hard: S26-T-00001653.

S26-B-00001655 - Pascal recurrence as a combinatorial bridge

The frozen Pascal identity S26-T-00000088 gives the recursive decomposition underlying lattice-path and subset-counting interpretations of binomial coefficients.

Evidence A. Hard: S26-I-00001654.

S26-B-00001656 - Binomial theorem coefficient bridge

The frozen binomial theorem S26-T-00000089 identifies C(n,k) as the coefficient of x^{n-k}y^k in (x+y)^n, linking algebraic expansion to subset enumeration.

Evidence A. Hard: S26-B-00001655.

S26-D-00001657 - Stirling numbers of the second kind

S(n,k) counts partitions of an n-element set into k nonempty unlabeled blocks.

Evidence A. Hard: S26-B-00001656.

S26-I-00001658 - Stirling second-kind recurrence

S(n,k)=k S(n-1,k)+S(n-1,k-1).

Evidence A. Hard: S26-D-00001657.

S26-D-00001659 - Stirling numbers of the first kind

Unsigned c(n,k) counts permutations of n elements with exactly k cycles.

Evidence A. Hard: S26-I-00001658.

S26-D-00001660 - Bell number

B_n=sum_k S(n,k) counts set partitions of an n-element set.

Evidence A. Hard: S26-D-00001659.

S26-D-00001661 - Catalan number

C_n=(1/(n+1)) C(2n,n), counting many equivalent families such as Dyck paths of semilength n.

Evidence A. Hard: S26-D-00001660.

S26-I-00001662 - Catalan recurrence

C_{n+1}=sum_{k=0}^n C_k C_{n-k}.

Evidence A. Hard: S26-D-00001661.

S26-D-00001663 - Derangement number

!n counts permutations of n elements with no fixed points.

Evidence A. Hard: S26-I-00001662.

S26-I-00001664 - Derangement formula

!n=n! sum_{k=0}^n (-1)^k/k!, hence !n is the nearest integer to n!/e for n>=1.

Evidence A. Hard: S26-D-00001663.

S26-T-00001665 - Inclusion-exclusion principle

For finite sets A_1,...,A_m, |union_i A_i|=sum_{empty!=J subseteq [m]}(-1)^{|J|+1}|intersection_{j in J}A_j|.

Evidence A. Hard: S26-I-00001664.

S26-T-00001666 - Burnside counting lemma

The number of orbits of a finite group G acting on finite X is (1/|G|)sum_{g in G}|Fix(g)|.

Evidence A. Hard: S26-T-00001665.

S26-T-00001667 - Polya enumeration theorem

For a finite permutation group acting on positions, the cycle index substituted with color inventory counts colorings up to symmetry.

Evidence A. Hard: S26-T-00001666.

S26-D-00001668 - Species of structures

A combinatorial species assigns sets of structures functorially to finite label sets; its exponential generating function records labeled counts.

Evidence A. Hard: S26-T-00001667.

S26-D-00001669 - Recurrence relation

A recurrence defines sequence terms from earlier terms; linear recurrences with constant coefficients are governed by the roots of a characteristic polynomial.

Evidence A. Hard: S26-D-00001668.

S26-B-00001670 - Enumeration-probability bridge

Generating functions encode both combinatorial counts and probability distributions; normalization and analytic domain determine which interpretation is valid.

Evidence A. Hard: S26-D-00001669.

S26-E-00001671 - Enumerative profile n=0

At n=0: Catalan C_n=1, Bell B_n=1, and derangements !n=1.

Evidence M. Hard: S26-B-00001670.

S26-E-00001672 - Enumerative profile n=1

At n=1: Catalan C_n=1, Bell B_n=1, and derangements !n=0.

Evidence M. Hard: S26-E-00001671.

S26-E-00001673 - Enumerative profile n=2

At n=2: Catalan C_n=2, Bell B_n=2, and derangements !n=1.

Evidence M. Hard: S26-E-00001672.

S26-E-00001674 - Enumerative profile n=3

At n=3: Catalan C_n=5, Bell B_n=5, and derangements !n=2.

Evidence M. Hard: S26-E-00001673.

S26-E-00001675 - Enumerative profile n=4

At n=4: Catalan C_n=14, Bell B_n=15, and derangements !n=9.

Evidence M. Hard: S26-E-00001674.

S26-E-00001676 - Enumerative profile n=5

At n=5: Catalan C_n=42, Bell B_n=52, and derangements !n=44.

Evidence M. Hard: S26-E-00001675.

S26-E-00001677 - Enumerative profile n=6

At n=6: Catalan C_n=132, Bell B_n=203, and derangements !n=265.

Evidence M. Hard: S26-E-00001676.

S26-E-00001678 - Enumerative profile n=7

At n=7: Catalan C_n=429, Bell B_n=877, and derangements !n=1854.

Evidence M. Hard: S26-E-00001677.

S26-E-00001679 - Enumerative profile n=8

At n=8: Catalan C_n=1430, Bell B_n=4140, and derangements !n=14833.

Evidence M. Hard: S26-E-00001678.

S26-E-00001680 - Enumerative profile n=9

At n=9: Catalan C_n=4862, Bell B_n=21147, and derangements !n=133496.

Evidence M. Hard: S26-E-00001679.

S26-E-00001681 - Enumerative profile n=10

At n=10: Catalan C_n=16796, Bell B_n=115975, and derangements !n=1334961.

Evidence M. Hard: S26-E-00001680.

S26-E-00001682 - Enumerative profile n=11

At n=11: Catalan C_n=58786, Bell B_n=678570, and derangements !n=14684570.

Evidence M. Hard: S26-E-00001681.

S26-E-00001683 - Enumerative profile n=12

At n=12: Catalan C_n=208012, Bell B_n=4213597, and derangements !n=176214841.

Evidence M. Hard: S26-E-00001682.

S26-E-00001684 - Enumerative profile n=13

At n=13: Catalan C_n=742900, Bell B_n=27644437, and derangements !n=2290792932.

Evidence M. Hard: S26-E-00001683.

S26-E-00001685 - Enumerative profile n=14

At n=14: Catalan C_n=2674440, Bell B_n=190899322, and derangements !n=32071101049.

Evidence M. Hard: S26-E-00001684.

S26-E-00001686 - Enumerative profile n=15

At n=15: Catalan C_n=9694845, Bell B_n=1382958545, and derangements !n=481066515734.

Evidence M. Hard: S26-E-00001685.

S26-E-00001687 - Enumerative profile n=16

At n=16: Catalan C_n=35357670, Bell B_n=10480142147, and derangements !n=7697064251745.

Evidence M. Hard: S26-E-00001686.

S26-E-00001688 - Enumerative profile n=17

At n=17: Catalan C_n=129644790, Bell B_n=82864869804, and derangements !n=130850092279664.

Evidence M. Hard: S26-E-00001687.

S26-E-00001689 - Enumerative profile n=18

At n=18: Catalan C_n=477638700, Bell B_n=682076806159, and derangements !n=2355301661033953.

Evidence M. Hard: S26-E-00001688.

S26-E-00001690 - Enumerative profile n=19

At n=19: Catalan C_n=1767263190, Bell B_n=5832742205057, and derangements !n=44750731559645106.

Evidence M. Hard: S26-E-00001689.

S26-E-00001691 - Enumerative profile n=20

At n=20: Catalan C_n=6564120420, Bell B_n=51724158235372, and derangements !n=895014631192902121.

Evidence M. Hard: S26-E-00001690.

S26-E-00001692 - Enumerative profile n=21

At n=21: Catalan C_n=24466267020, Bell B_n=474869816156751, and derangements !n=18795307255050944540.

Evidence M. Hard: S26-E-00001691.

S26-E-00001693 - Enumerative profile n=22

At n=22: Catalan C_n=91482563640, Bell B_n=4506715738447323, and derangements !n=413496759611120779881.

Evidence M. Hard: S26-E-00001692.

S26-E-00001694 - Enumerative profile n=23

At n=23: Catalan C_n=343059613650, Bell B_n=44152005855084346, and derangements !n=9510425471055777937262.

Evidence M. Hard: S26-E-00001693.

S26-E-00001695 - Enumerative profile n=24

At n=24: Catalan C_n=1289904147324, Bell B_n=445958869294805289, and derangements !n=2282502113053386704942 89.

Evidence M. Hard: S26-E-00001694.

S26-E-00001696 - Enumerative profile n=25

At n=25: Catalan C_n=4861946401452, Bell B_n=4638590332229999353, and derangements !n=5706255282633466762357224.

Evidence M. Hard: S26-E-00001695.

S26-E-00001697 - Enumerative profile n=26

At n=26: Catalan C_n=18367353072152, Bell B_n=49631246523618756274, and derangements !n=14836263734847013582128 7825.

Evidence M. Hard: S26-E-00001696.

S26-E-00001698 - Enumerative profile n=27

At n=27: Catalan C_n=69533550916004, Bell B_n=545717047936059989389, and derangements !n=4005791208408693667174771274.

Evidence M. Hard: S26-E-00001697.

S26-E-00001699 - Enumerative profile n=28

At n=28: Catalan C_n=263747951750360, Bell B_n=6160539404599934652455, and derangements !n=1121621538354434226808935956 73.

Evidence M. Hard: S26-E-00001698.

S26-E-00001700 - Enumerative profile n=29

At n=29: Catalan C_n=1002242216651368, Bell B_n=71339801938860275191172, and derangements !n=3252702461227859257745914274516.

Evidence M. Hard: S26-E-00001699.

Part O - Partitions and q-series

S26-D-00001701 - Integer partition

A partition of n is a weakly decreasing finite sequence of positive integers summing to n.

Evidence A. Hard: S26-D-00000086.

S26-D-00001702 - Partition function

p(n) is the number of integer partitions of n, with p(0)=1.

Evidence A. Hard: S26-D-00001701.

S26-I-00001703 - Euler partition generating function

sum_{n>=0}p(n)q^n=product_{m>=1}(1-q^m)^{-1} as a formal power series.

Evidence A. Hard: S26-D-00001702.

S26-T-00001704 - Euler pentagonal recurrence

p(n)=sum_{k!=0}(-1)^{k-1}p(n-k(3k-1)/2), with p(0)=1 and p(m)=0 for m<0.

Evidence A. Hard: S26-I-00001703.

S26-T-00001705 - Euler distinct-odd partition theorem

The number of partitions of n into distinct parts equals the number of partitions of n into odd parts.

Evidence A. Hard: S26-T-00001704.

S26-D-00001706 - Ferrers diagram

A Ferrers/Young diagram represents a partition by left-justified rows whose lengths are the parts.

Evidence A. Hard: S26-T-00001705.

S26-D-00001707 - Conjugate partition

The conjugate partition transposes the Ferrers diagram; row lengths become column lengths.

Evidence A. Hard: S26-D-00001706.

S26-D-00001708 - Hook length

For a cell of a Young diagram, its hook length is one plus the number of cells strictly right in its row plus strictly below in its column.

Evidence A. Hard: S26-D-00001707.

S26-T-00001709 - Hook-length formula

The number f^lambda of standard Young tableaux of shape lambda partitioning n is n! divided by the product of hook lengths of lambda.

Evidence A. Hard: S26-D-00001708.

S26-D-00001710 - q-binomial coefficient

The Gaussian binomial [n choose k]_q is product_{i=0}^{k-1}(1-q^{n-i})/(1-q^{k-i}).

Evidence A. Hard: S26-T-00001709.

S26-T-00001711 - Finite-field subspace count

[n choose k]_q counts k-dimensional subspaces of F_q^n.

Evidence A. Hard: S26-D-00001710.

S26-D-00001712 - q-Pochhammer symbol

(a;q)_n=product_{j=0}^{n-1}(1-aq^j), with an infinite version when convergence/formal context is declared.

Evidence A. Hard: S26-T-00001711.

S26-D-00001713 - Theta series

A theta series is a generating function sum q^{Q(x)} attached to a quadratic form/lattice, encoding representation counts.

Evidence A. Hard: S26-D-00001712.

S26-D-00001714 - Modular form interface

A modular form is a holomorphic function on the upper half-plane satisfying a specified modular transformation law and cusp regularity; weight and group are part of the data.

Evidence A. Hard: S26-D-00001713.

S26-N-00001715 - q-series convergence fence

Formal-power-series identities and analytic q-series identities are distinct claims; analytic manipulations require a declared domain such as |q|<1.

Evidence A. Hard: S26-D-00001714. Boundary: Formal coefficient equality alone does not justify analytic limit interchange.

S26-T-00001716 - Ramanujan congruence mod 5

For n>=0, p(5n+4) is divisible by 5.

Evidence A. Hard: S26-N-00001715.

S26-T-00001717 - Ramanujan congruence mod 7

For n>=0, p(7n+5) is divisible by 7.

Evidence A. Hard: S26-T-00001716.

S26-T-00001718 - Ramanujan congruence mod 11

For n>=0, p(11n+6) is divisible by 11.

Evidence A. Hard: S26-T-00001717.

S26-D-00001719 - Partition rank

The rank of a partition is its largest part minus its number of parts.

Evidence A. Hard: S26-T-00001718.

S26-B-00001720 - Partitions to representation theory

Young diagrams index irreducible representations of symmetric groups in characteristic zero and organize Schur functors, symmetric functions, and tableaux combinatorics.

Evidence A. Hard: S26-D-00001719.

S26-E-00001721 - Partition number p(0)

The exact partition count is p(0)=1.

Evidence M. Hard: S26-B-00001720.

S26-E-00001722 - Partition number p(1)

The exact partition count is p(1)=1.

Evidence M. Hard: S26-E-00001721.

S26-E-00001723 - Partition number p(2)

The exact partition count is p(2)=2.

Evidence M. Hard: S26-E-00001722.

S26-E-00001724 - Partition number p(3)

The exact partition count is p(3)=3.

Evidence M. Hard: S26-E-00001723.

S26-E-00001725 - Partition number p(4)

The exact partition count is p(4)=5.

Evidence M. Hard: S26-E-00001724.

S26-E-00001726 - Partition number p(5)

The exact partition count is p(5)=7.

Evidence M. Hard: S26-E-00001725.

S26-E-00001727 - Partition number p(6)

The exact partition count is p(6)=11.

Evidence M. Hard: S26-E-00001726.

S26-E-00001728 - Partition number p(7)

The exact partition count is p(7)=15.

Evidence M. Hard: S26-E-00001727.

S26-E-00001729 - Partition number p(8)

The exact partition count is p(8)=22.

Evidence M. Hard: S26-E-00001728.

S26-E-00001730 - Partition number p(9)

The exact partition count is p(9)=30.

Evidence M. Hard: S26-E-00001729.

S26-E-00001731 - Partition number p(10)

The exact partition count is p(10)=42.

Evidence M. Hard: S26-E-00001730.

S26-E-00001732 - Partition number p(11)

The exact partition count is p(11)=56.

Evidence M. Hard: S26-E-00001731.

S26-E-00001733 - Partition number p(12)

The exact partition count is p(12)=77.

Evidence M. Hard: S26-E-00001732.

S26-E-00001734 - Partition number p(13)

The exact partition count is p(13)=101.

Evidence M. Hard: S26-E-00001733.

S26-E-00001735 - Partition number p(14)

The exact partition count is p(14)=135.

Evidence M. Hard: S26-E-00001734.

S26-E-00001736 - Partition number p(15)

The exact partition count is p(15)=176.

Evidence M. Hard: S26-E-00001735.

S26-E-00001737 - Partition number p(16)

The exact partition count is p(16)=231.

Evidence M. Hard: S26-E-00001736.

S26-E-00001738 - Partition number p(17)

The exact partition count is p(17)=297.

Evidence M. Hard: S26-E-00001737.

S26-E-00001739 - Partition number p(18)

The exact partition count is p(18)=385.

Evidence M. Hard: S26-E-00001738.

S26-E-00001740 - Partition number p(19)

The exact partition count is p(19)=490.

Evidence M. Hard: S26-E-00001739.

S26-E-00001741 - Partition number p(20)

The exact partition count is p(20)=627.

Evidence M. Hard: S26-E-00001740.

S26-E-00001742 - Partition number p(21)

The exact partition count is p(21)=792.

Evidence M. Hard: S26-E-00001741.

S26-E-00001743 - Partition number p(22)

The exact partition count is p(22)=1002.

Evidence M. Hard: S26-E-00001742.

S26-E-00001744 - Partition number p(23)

The exact partition count is p(23)=1255.

Evidence M. Hard: S26-E-00001743.

S26-E-00001745 - Partition number p(24)

The exact partition count is p(24)=1575.

Evidence M. Hard: S26-E-00001744.

S26-E-00001746 - Partition number p(25)

The exact partition count is p(25)=1958.

Evidence M. Hard: S26-E-00001745.

S26-E-00001747 - Partition number p(26)

The exact partition count is p(26)=2436.

Evidence M. Hard: S26-E-00001746.

S26-E-00001748 - Partition number p(27)

The exact partition count is p(27)=3010.

Evidence M. Hard: S26-E-00001747.

S26-E-00001749 - Partition number p(28)

The exact partition count is p(28)=3718.

Evidence M. Hard: S26-E-00001748.

S26-E-00001750 - Partition number p(29)

The exact partition count is p(29)=4565.

Evidence M. Hard: S26-E-00001749.

Part P - Graph theory

S26-D-00001751 - Simple graph

A simple graph G=(V,E) has a vertex set V and a set E of unordered two-element subsets of V, with no loops or parallel edges.

Evidence A. Hard: S26-D-00000149.

S26-D-00001752 - Degree of a vertex

deg(v) is the number of edges incident to v in a finite simple graph.

Evidence A. Hard: S26-D-00001751.

S26-T-00001753 - Handshake lemma

For a finite graph, sum_{v in V}deg(v)=2|E|.

Evidence A. Hard: S26-D-00001752.

S26-D-00001754 - Path and cycle

A path is a sequence of adjacent vertices without repeated vertices under the simple-path convention; a cycle is a closed path with distinct interior vertices.

Evidence A. Hard: S26-T-00001753.

S26-D-00001755 - Connected graph

A graph is connected when every pair of vertices is joined by a path.

Evidence A. Hard: S26-D-00001754.

S26-D-00001756 - Tree

A tree is a connected acyclic graph.

Evidence A. Hard: S26-D-00001755.

S26-T-00001757 - Tree edge count

A finite tree with n vertices has n-1 edges; conversely a connected n-vertex graph with n-1 edges is a tree.

Evidence A. Hard: S26-D-00001756.

S26-T-00001758 - Cayley formula

There are n^{n-2} labeled trees on n labeled vertices for n>=2.

Evidence A. Hard: S26-T-00001757.

S26-D-00001759 - Spanning tree

A spanning tree of connected G is a tree containing all vertices of G.

Evidence A. Hard: S26-T-00001758.

S26-T-00001760 - Matrix-tree theorem

The number of spanning trees of a finite graph equals any cofactor of its Laplacian matrix.

Evidence A. Hard: S26-D-00001759.

S26-D-00001761 - Graph Laplacian

For simple graph G, L=D-A where D is the diagonal degree matrix and A the adjacency matrix.

Evidence A. Hard: S26-T-00001760.

S26-T-00001762 - Laplacian kernel and components

The multiplicity of eigenvalue 0 of the graph Laplacian equals the number of connected components.

Evidence A. Hard: S26-D-00001761.

S26-D-00001763 - Eulerian circuit

An Eulerian circuit traverses every edge exactly once and returns to its start.

Evidence A. Hard: S26-T-00001762.

S26-T-00001764 - Euler criterion for connected graphs

A finite connected graph has an Eulerian circuit iff every vertex has even degree.

Evidence A. Hard: S26-D-00001763.

S26-D-00001765 - Hamiltonian cycle

A Hamiltonian cycle visits every vertex exactly once before returning to its start.

Evidence A. Hard: S26-T-00001764.

S26-N-00001766 - Eulerian/Hamiltonian fence

Eulerian conditions concern edges and have a complete degree criterion; Hamiltonicity concerns vertices and has no analogous simple necessary-and-sufficient degree criterion.

Evidence A. Hard: S26-D-00001765. Boundary: Do not transfer Euler degree parity to Hamiltonicity.

S26-D-00001767 - Planar graph

A graph is planar when it admits an embedding in the sphere/plane with edges meeting only at common endpoints.

Evidence A. Hard: S26-N-00001766.

S26-I-00001768 - Euler formula for connected planar graphs

For a connected planar embedding, |V|-|E|+|F|=2.

Evidence A. Hard: S26-D-00001767.

S26-T-00001769 - Kuratowski theorem

A finite graph is planar iff it contains no subdivision of K_5 or K_{3,3}.

Evidence A. Hard: S26-I-00001768.

S26-B-00001770 - Graphs to linear algebra

Adjacency, incidence, and Laplacian matrices turn connectivity, cuts, walks, and expansion into spectral linear-algebraic questions.

Evidence A. Hard: S26-T-00001769.

S26-E-00001771 - Standard graph counts at n=3

On n=3 labeled vertices, K_n has 3 edges, C_n has 3 edges, and there are 3 labeled trees by Cayley's formula.

Evidence M. Hard: S26-B-00001770.

S26-E-00001772 - Standard graph counts at n=4

On n=4 labeled vertices, K_n has 6 edges, C_n has 4 edges, and there are 16 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001771.

S26-E-00001773 - Standard graph counts at n=5

On n=5 labeled vertices, K_n has 10 edges, C_n has 5 edges, and there are 125 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001772.

S26-E-00001774 - Standard graph counts at n=6

On n=6 labeled vertices, K_n has 15 edges, C_n has 6 edges, and there are 1296 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001773.

S26-E-00001775 - Standard graph counts at n=7

On n=7 labeled vertices, K_n has 21 edges, C_n has 7 edges, and there are 16807 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001774.

S26-E-00001776 - Standard graph counts at n=8

On n=8 labeled vertices, K_n has 28 edges, C_n has 8 edges, and there are 262144 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001775.

S26-E-00001777 - Standard graph counts at n=9

On n=9 labeled vertices, K_n has 36 edges, C_n has 9 edges, and there are 4782969 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001776.

S26-E-00001778 - Standard graph counts at n=10

On n=10 labeled vertices, K_n has 45 edges, C_n has 10 edges, and there are 100000000 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001777.

S26-E-00001779 - Standard graph counts at n=11

On n=11 labeled vertices, K_n has 55 edges, C_n has 11 edges, and there are 2357947691 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001778.

S26-E-00001780 - Standard graph counts at n=12

On n=12 labeled vertices, K_n has 66 edges, C_n has 12 edges, and there are 61917364224 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001779.

S26-E-00001781 - Standard graph counts at n=13

On n=13 labeled vertices, K_n has 78 edges, C_n has 13 edges, and there are 1792160394037 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001780.

S26-E-00001782 - Standard graph counts at n=14

On n=14 labeled vertices, K_n has 91 edges, C_n has 14 edges, and there are 56693912375296 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001781.

S26-E-00001783 - Standard graph counts at n=15

On n=15 labeled vertices, K_n has 105 edges, C_n has 15 edges, and there are 1946195068359375 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001782.

S26-E-00001784 - Standard graph counts at n=16

On n=16 labeled vertices, K_n has 120 edges, C_n has 16 edges, and there are 72057594037927936 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001783.

S26-E-00001785 - Standard graph counts at n=17

On n=17 labeled vertices, K_n has 136 edges, C_n has 17 edges, and there are 2862423051509815793 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001784.

S26-E-00001786 - Standard graph counts at n=18

On n=18 labeled vertices, K_n has 153 edges, C_n has 18 edges, and there are 121439531096594251776 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001785.

S26-E-00001787 - Standard graph counts at n=19

On n=19 labeled vertices, K_n has 171 edges, C_n has 19 edges, and there are 5480386857784802185939 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001786.

S26-E-00001788 - Standard graph counts at n=20

On n=20 labeled vertices, K_n has 190 edges, C_n has 20 edges, and there are 26214400000000000000000 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001787.

S26-E-00001789 - Standard graph counts at n=21

On n=21 labeled vertices, K_n has 210 edges, C_n has 21 edges, and there are 1324849664033102612558078 1 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001788.

S26-E-00001790 - Standard graph counts at n=22

On n=22 labeled vertices, K_n has 231 edges, C_n has 22 edges, and there are 705429498686404044207947776 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001789.

S26-E-00001791 - Standard graph counts at n=23

On n=23 labeled vertices, K_n has 253 edges, C_n has 23 edges, and there are 39471584120695485887249589623 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001790.

S26-E-00001792 - Standard graph counts at n=24

On n=24 labeled vertices, K_n has 276 edges, C_n has 24 edges, and there are 2315513501476187716057433112576 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001791.

S26-E-00001793 - Standard graph counts at n=25

On n=25 labeled vertices, K_n has 300 edges, C_n has 25 edges, and there are 1421085471520200371742248535156 25 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001792.

S26-E-00001794 - Standard graph counts at n=26

On n=26 labeled vertices, K_n has 325 edges, C_n has 26 edges, and there are 9106685769537214956799814036094976 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001793.

S26-E-00001795 - Standard graph counts at n=27

On n=27 labeled vertices, K_n has 351 edges, C_n has 27 edges, and there are 6082667877133577091196839926188613 07 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001794.

S26-E-00001796 - Standard graph counts at n=28

On n=28 labeled vertices, K_n has 378 edges, C_n has 28 edges, and there are 4227745295057828426348562277214873 1904 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001795.

S26-E-00001797 - Standard graph counts at n=29

On n=29 labeled vertices, K_n has 406 edges, C_n has 29 edges, and there are 3053134545970524535745336759489912159909 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001796.

S26-E-00001798 - Standard graph counts at n=30

On n=30 labeled vertices, K_n has 435 edges, C_n has 30 edges, and there are 228767924549610000000000000000000000000000 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001797.

S26-E-00001799 - Standard graph counts at n=31

On n=31 labeled vertices, K_n has 465 edges, C_n has 31 edges, and there are 1776188775309389797982377006145610276 3834271 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001798.

S26-E-00001800 - Standard graph counts at n=32

On n=32 labeled vertices, K_n has 496 edges, C_n has 32 edges, and there are 1427247692705959881058285969449495136382746624 labeled trees by Cayley's formula.

Evidence M. Hard: S26-E-00001799.

Part Q - Posets, lattices, and matroids

S26-D-00001801 - Partially ordered set

A poset (P,<=) has a reflexive, antisymmetric, transitive relation.

Evidence A. Hard: S26-D-00000020.

S26-D-00001802 - Chain and antichain

A chain is a subset of pairwise comparable elements; an antichain is a subset of pairwise incomparable elements.

Evidence A. Hard: S26-D-00001801.

S26-D-00001803 - Width of a poset

The width is the maximum size of an antichain.

Evidence A. Hard: S26-D-00001802.

S26-T-00001804 - Dilworth theorem

In a finite poset, the maximum antichain size equals the minimum number of chains in a partition of the poset.

Evidence A. Hard: S26-D-00001803.

S26-T-00001805 - Mirsky theorem

In a finite poset, the maximum chain size equals the minimum number of antichains in a partition of the poset.

Evidence A. Hard: S26-T-00001804.

S26-D-00001806 - Lattice

A lattice is a poset in which every pair has a meet and join.

Evidence A. Hard: S26-T-00001805.

S26-D-00001807 - Distributive lattice

A lattice is distributive when meet and join distribute over one another.

Evidence A. Hard: S26-D-00001806.

S26-T-00001808 - Birkhoff representation theorem

Every finite distributive lattice is isomorphic to the lattice of order ideals of its poset of join-irreducible elements.

Evidence A. Hard: S26-D-00001807.

S26-D-00001809 - Moebius function of a poset

For locally finite poset P, mu(x,x)=1 and sum_{x<=z<=y}mu(x,z)=0 for x<y.

Evidence A. Hard: S26-T-00001808.

S26-T-00001810 - Poset Moebius inversion

If g(y)=sum_{x<=y} f(x), then f(y)=sum_{x<=y}mu(x,y)g(x) for locally finite posets.

Evidence A. Hard: S26-D-00001809.

S26-D-00001811 - Matroid

A matroid M=(E,I) consists of a finite ground set E and independent subsets I satisfying heredity and augmentation axioms.

Evidence A. Hard: S26-T-00001810.

S26-D-00001812 - Rank function of a matroid

r(A) is the maximum size of an independent subset of A.

Evidence A. Hard: S26-D-00001811.

S26-D-00001813 - Basis of a matroid

A basis is a maximal independent subset; all bases have the same size r(E).

Evidence A. Hard: S26-D-00001812.

S26-D-00001814 - Circuit of a matroid

A circuit is a minimal dependent subset.

Evidence A. Hard: S26-D-00001813.

S26-D-00001815 - Closure in a matroid

cl(A)={e:r(A union {e})=r(A)}.

Evidence A. Hard: S26-D-00001814.

S26-D-00001816 - Dual matroid

The dual M* has bases E minus B as B ranges over bases of M.

Evidence A. Hard: S26-D-00001815.

S26-D-00001817 - Graphic matroid

For graph G, the graphic matroid has E(G) as ground set and forests as independent sets.

Evidence A. Hard: S26-D-00001816.

S26-D-00001818 - Uniform matroid

U_{r,n} has ground set of size n and independent sets exactly those of size at most r.

Evidence A. Hard: S26-D-00001817.

S26-T-00001819 - Greedy characterization of matroids

A finite independence system is a matroid iff the greedy algorithm produces a maximum-weight basis for every assignment of weights under the standard ordering convention.

Evidence A. Hard: S26-D-00001818.

S26-B-00001820 - Matroids as independence abstraction

Matroids unify linear independence, forests, transversal systems, and greedy optimization while forgetting structure not captured by independence axioms.

Evidence A. Hard: S26-T-00001819.

S26-E-00001821 - Boolean lattice B_1 profile

The Boolean lattice of subsets of an 1-element set has 2 elements, rank sizes [1, 1], and width 1.

Evidence M. Hard: S26-B-00001820.

S26-E-00001822 - Boolean lattice B_2 profile

The Boolean lattice of subsets of an 2-element set has 4 elements, rank sizes [1, 2, 1], and width 2.

Evidence M. Hard: S26-E-00001821.

S26-E-00001823 - Boolean lattice B_3 profile

The Boolean lattice of subsets of an 3-element set has 8 elements, rank sizes [1, 3, 3, 1], and width 3.

Evidence M. Hard: S26-E-00001822.

S26-E-00001824 - Boolean lattice B_4 profile

The Boolean lattice of subsets of an 4-element set has 16 elements, rank sizes [1, 4, 6, 4, 1], and width 6.

Evidence M. Hard: S26-E-00001823.

S26-E-00001825 - Boolean lattice B_5 profile

The Boolean lattice of subsets of an 5-element set has 32 elements, rank sizes [1, 5, 10, 10, 5, 1], and width 10.

Evidence M. Hard: S26-E-00001824.

S26-E-00001826 - Boolean lattice B_6 profile

The Boolean lattice of subsets of an 6-element set has 64 elements, rank sizes [1, 6, 15, 20, 15, 6, 1], and width 20.

Evidence M. Hard: S26-E-00001825.

S26-E-00001827 - Boolean lattice B_7 profile

The Boolean lattice of subsets of an 7-element set has 128 elements, rank sizes [1, 7, 21, 35, 35, 21, 7, 1], and width 35.

Evidence M. Hard: S26-E-00001826.

S26-E-00001828 - Boolean lattice B_8 profile

The Boolean lattice of subsets of an 8-element set has 256 elements, rank sizes [1, 8, 28, 56, 70, 56, 28, 8, 1], and width 70.

Evidence M. Hard: S26-E-00001827.

S26-E-00001829 - Boolean lattice B_9 profile

The Boolean lattice of subsets of an 9-element set has 512 elements, rank sizes [1, 9, 36, 84, 126, 126, 84, 36, 9, 1], and width 126.

Evidence M. Hard: S26-E-00001828.

S26-E-00001830 - Boolean lattice B_10 profile

The Boolean lattice of subsets of an 10-element set has 1024 elements, rank sizes [1, 10, 45, 120, 210, 252, 210, 120, 45, 10, 1], and width 252.

Evidence M. Hard: S26-E-00001829.

S26-E-00001831 - Boolean lattice B_11 profile

The Boolean lattice of subsets of an 11-element set has 2048 elements, rank sizes [1, 11, 55, 165, 330, 462, 462, 330, 165, 55, 11, 1], and width 462.

Evidence M. Hard: S26-E-00001830.

S26-E-00001832 - Boolean lattice B_12 profile

The Boolean lattice of subsets of an 12-element set has 4096 elements, rank sizes [1, 12, 66, 220, 495, 792, 924, 792, 495, 220, 66, 12, 1], and width 924.

Evidence M. Hard: S26-E-00001831.

S26-E-00001833 - Boolean lattice B_13 profile

The Boolean lattice of subsets of an 13-element set has 8192 elements, rank sizes [1, 13, 78, 286, 715, 1287, 1716, 1716, 1287, 715, 286, 78, 13, 1], and width 1716.

Evidence M. Hard: S26-E-00001832.

S26-E-00001834 - Boolean lattice B_14 profile

The Boolean lattice of subsets of an 14-element set has 16384 elements, rank sizes [1, 14, 91, 364, 1001, 2002, 3003, 3432, 3003, 2002, 1001, 364, 91, 14, 1], and width 3432.

Evidence M. Hard: S26-E-00001833.

S26-E-00001835 - Boolean lattice B_15 profile

The Boolean lattice of subsets of an 15-element set has 32768 elements, rank sizes [1, 15, 105, 455, 1365, 3003, 5005, 6435, 6435, 5005, 3003, 1365, 455, 105, 15, 1], and width 6435.

Evidence M. Hard: S26-E-00001834.

S26-E-00001836 - Boolean lattice B_16 profile

The Boolean lattice of subsets of an 16-element set has 65536 elements, rank sizes [1, 16, 120, 560, 1820, 4368, 8008, 11440, 12870, 11440, 8008, 4368, 1820, 560, 120, 16, 1], and width 12870.

Evidence M. Hard: S26-E-00001835.

S26-E-00001837 - Boolean lattice B_17 profile

The Boolean lattice of subsets of an 17-element set has 131072 elements, rank sizes [1, 17, 136, 680, 2380, 6188, 12376, 19448, 24310, 24310, 19448, 12376, 6188, 2380, 680, 136, 17, 1], and width 24310.

Evidence M. Hard: S26-E-00001836.

S26-E-00001838 - Boolean lattice B_18 profile

The Boolean lattice of subsets of an 18-element set has 262144 elements, rank sizes [1, 18, 153, 816, 3060, 8568, 18564, 31824, 43758, 48620, 43758, 31824, 18564, 8568, 3060, 816, 153, 18, 1], and width 48620.

Evidence M. Hard: S26-E-00001837.

S26-E-00001839 - Boolean lattice B_19 profile

The Boolean lattice of subsets of an 19-element set has 524288 elements, rank sizes [1, 19, 171, 969, 3876, 11628, 27132, 50388, 75582, 92378, 92378, 75582, 50388, 27132, 11628, 3876, 969, 171, 19, 1], and width 92378.

Evidence M. Hard: S26-E-00001838.

S26-E-00001840 - Boolean lattice B_20 profile

The Boolean lattice of subsets of an 20-element set has 1048576 elements, rank sizes [1, 20, 190, 1140, 4845, 15504, 38760, 77520, 125970, 167960, 184756, 167960, 125970, 77520, 38760, 15504, 4845, 1140, 190, 20, 1], and width 184756.

Evidence M. Hard: S26-E-00001839.

S26-E-00001841 - Boolean lattice B_21 profile

The Boolean lattice of subsets of an 21-element set has 2097152 elements, rank sizes [1, 21, 210, 1330, 5985, 20349, 54264, 116280, 203490, 293930, 352716, 352716, 293930, 203490, 116280, 54264, 20349, 5985, 1330, 210, 21, 1], and width 352716.

Evidence M. Hard: S26-E-00001840.

S26-E-00001842 - Boolean lattice B_22 profile

The Boolean lattice of subsets of an 22-element set has 4194304 elements, rank sizes [1, 22, 231, 1540, 7315, 26334, 74613, 170544, 319770, 497420, 646646, 705432, 646646, 497420, 319770, 170544, 74613, 26334, 7315, 1540, 231, 22, 1], and width 705432.

Evidence M. Hard: S26-E-00001841.

S26-E-00001843 - Boolean lattice B_23 profile

The Boolean lattice of subsets of an 23-element set has 8388608 elements, rank sizes [1, 23, 253, 1771, 8855, 33649, 100947, 245157, 490314, 817190, 1144066, 1352078, 1352078, 1144066, 817190, 490314, 245157, 100947, 33649, 8855, 1771, 253, 23, 1], and width 1352078.

Evidence M. Hard: S26-E-00001842.

S26-E-00001844 - Boolean lattice B_24 profile

The Boolean lattice of subsets of an 24-element set has 16777216 elements, rank sizes [1, 24, 276, 2024, 10626, 42504, 134596, 346104, 735471, 1307504, 1961256, 2496144, 2704156, 2496144, 1961256, 1307504, 735471, 346104, 134596, 42504, 10626, 2024, 276, 24, 1], and width 2704156.

Evidence M. Hard: S26-E-00001843.

S26-E-00001845 - Boolean lattice B_25 profile

The Boolean lattice of subsets of an 25-element set has 33554432 elements, rank sizes [1, 25, 300, 2300, 12650, 53130, 177100, 480700, 1081575, 2042975, 3268760, 4457400, 5200300, 5200300, 4457400, 3268760, 2042975, 1081575, 480700, 177100, 53130, 12650, 2300, 300, 25, 1], and width 5200300.

Evidence M. Hard: S26-E-00001844.

S26-E-00001846 - Boolean lattice B_26 profile

The Boolean lattice of subsets of an 26-element set has 67108864 elements, rank sizes [1, 26, 325, 2600, 14950, 65780, 230230, 657800, 1562275, 3124550, 5311735, 7726160, 9657700, 10400600, 9657700, 7726160, 5311735, 3124550, 1562275, 657800, 230230, 65780, 14950, 2600, 325, 26, 1], and width 10400600.

Evidence M. Hard: S26-E-00001845.

S26-E-00001847 - Boolean lattice B_27 profile

The Boolean lattice of subsets of an 27-element set has 134217728 elements, rank sizes [1, 27, 351, 2925, 17550, 80730, 296010, 888030, 2220075, 4686825, 8436285, 13037895, 17383860, 20058300, 20058300, 17383860, 13037895, 8436285, 4686825, 2220075, 888030, 296010, 80730, 17550, 2925, 351, 27, 1], and width 20058300.

Evidence M. Hard: S26-E-00001846.

S26-E-00001848 - Boolean lattice B_28 profile

The Boolean lattice of subsets of an 28-element set has 268435456 elements, rank sizes [1, 28, 378, 3276, 20475, 98280, 376740, 1184040, 3108105, 6906900, 13123110, 21474180, 30421755, 37442160, 40116600, 37442160, 30421755, 21474180, 13123110, 6906900, 3108105, 1184040, 376740, 98280, 20475, 3276, 378, 28, 1], and width 40116600.

Evidence M. Hard: S26-E-00001847.

S26-E-00001849 - Boolean lattice B_29 profile

The Boolean lattice of subsets of an 29-element set has 536870912 elements, rank sizes [1, 29, 406, 3654, 23751, 118755, 475020, 1560780, 4292145, 10015005, 20030010, 34597290, 51895935, 67863915, 77558760, 77558760, 67863915, 51895935, 34597290, 20030010, 10015005, 4292145, 1560780, 475020, 118755, 23751, 3654, 406, 29, 1], and width 77558760.

Evidence M. Hard: S26-E-00001848.

S26-E-00001850 - Boolean lattice B_30 profile

The Boolean lattice of subsets of an 30-element set has 1073741824 elements, rank sizes [1, 30, 435, 4060, 27405, 142506, 593775, 2035800, 5852925, 14307150, 30045015, 54627300, 86493225, 119759850, 145422675, 155117520, 145422675, 119759850, 86493225, 54627300, 30045015, 14307150, 5852925, 2035800, 593775, 142506, 27405, 4060, 435, 30, 1], and width 155117520.

Evidence M. Hard: S26-E-00001849.

Part R - Coding theory and designs

S26-D-00001851 - q-ary block code

A q-ary block code of length n is a subset C of A^n for an alphabet A with |A|=q.

Evidence A. Hard: S26-D-00001251.

S26-D-00001852 - Hamming distance

d_H(x,y) is the number of coordinates in which words x and y differ.

Evidence A. Hard: S26-D-00001851.

S26-D-00001853 - Minimum distance of a code

d(C)=min_{x!=y in C} d_H(x,y).

Evidence A. Hard: S26-D-00001852.

S26-T-00001854 - Error correction from minimum distance

A code of minimum distance d corrects every pattern of at most floor((d-1)/2) symbol errors by nearest-neighbor decoding when the nearest codeword is unique.

Evidence A. Hard: S26-D-00001853.

S26-D-00001855 - Linear code

A linear [n,k]_q code is a k-dimensional subspace of F_q^n.

Evidence A. Hard: S26-T-00001854.

S26-D-00001856 - Generator matrix

A generator matrix G for a linear [n,k]_q code has row space C.

Evidence A. Hard: S26-D-00001855.

S26-D-00001857 - Parity-check matrix

A parity-check matrix H satisfies C=ker H; for an [n,k]_q code, H may be taken with n-k independent rows.

Evidence A. Hard: S26-D-00001856.

S26-D-00001858 - Dual code

C^perp={y:<x,y>=0 for every x in C}.

Evidence A. Hard: S26-D-00001857.

S26-I-00001859 - Dimension of dual code

For linear C<=F_q^n, dim C + dim C^perp=n.

Evidence A. Hard: S26-D-00001858.

S26-T-00001860 - Singleton bound

For a q-ary code of size M and minimum distance d, M<=q^{n-d+1}; for linear [n,k,d]_q, d<=n-k+1.

Evidence A. Hard: S26-I-00001859.

S26-D-00001861 - MDS code

A linear code meeting the Singleton bound with equality is maximum-distance separable.

Evidence A. Hard: S26-T-00001860.

S26-T-00001862 - Hamming sphere-packing bound

For a q-ary code correcting t errors, |C| sum_{i=0}^t C(n,i)(q-1)^i <= q^n.

Evidence A. Hard: S26-D-00001861.

S26-T-00001863 - Gilbert-Varshamov bound interface

A greedy packing argument gives lower bounds on the size/existence of codes with prescribed length and minimum distance; exact form depends on linear vs nonlinear setting.

Evidence A. Hard: S26-T-00001862.

S26-D-00001864 - Hamming code

A q-ary Hamming code is a perfect single-error-correcting linear code with parity-check columns representing projective points under the standard construction.

Evidence A. Hard: S26-T-00001863.

S26-D-00001865 - Reed-Solomon code

A Reed-Solomon code evaluates low-degree polynomials at distinct field points; it is an MDS code.

Evidence A. Hard: S26-D-00001864.

S26-D-00001866 - Cyclic code

A linear code C<=F_q^n is cyclic when cyclic coordinate shift preserves C; equivalently it corresponds to an ideal of F_q[x]/(x^n-1).

Evidence A. Hard: S26-D-00001865.

S26-D-00001867 - Block design

A t-(v,k,lambda) design is a family of k-subsets (blocks) of a v-set such that every t-subset occurs in exactly lambda blocks.

Evidence A. Hard: S26-D-00001866.

S26-D-00001868 - Steiner system

A Steiner system S(t,k,v) is a t-(v,k,1) design.

Evidence A. Hard: S26-D-00001867.

S26-I-00001869 - BIBD parameter relation

For a 2-(v,k,lambda) design with b blocks and replication number r, vr=bk and r(k-1)=lambda(v-1).

Evidence A. Hard: S26-D-00001868.

S26-B-00001870 - Codes and finite geometry

Linear codes, projective point sets, finite-field polynomials, and block designs share incidence structures; translations preserve some but not all parameters.

Evidence A. Hard: S26-I-00001869.

S26-E-00001871 - Binary Hamming code profile m=2

The standard binary Hamming code has parameters [3,1,3]_2 and corrects one bit error.

Evidence M. Hard: S26-B-00001870.

S26-E-00001872 - Binary Hamming code profile m=3

The standard binary Hamming code has parameters [7,4,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001871.

S26-E-00001873 - Binary Hamming code profile m=4

The standard binary Hamming code has parameters [15,11,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001872.

S26-E-00001874 - Binary Hamming code profile m=5

The standard binary Hamming code has parameters [31,26,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001873.

S26-E-00001875 - Binary Hamming code profile m=6

The standard binary Hamming code has parameters [63,57,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001874.

S26-E-00001876 - Binary Hamming code profile m=7

The standard binary Hamming code has parameters [127,120,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001875.

S26-E-00001877 - Binary Hamming code profile m=8

The standard binary Hamming code has parameters [255,247,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001876.

S26-E-00001878 - Binary Hamming code profile m=9

The standard binary Hamming code has parameters [511,502,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001877.

S26-E-00001879 - Binary Hamming code profile m=10

The standard binary Hamming code has parameters [1023,1013,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001878.

S26-E-00001880 - Binary Hamming code profile m=11

The standard binary Hamming code has parameters [2047,2036,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001879.

S26-E-00001881 - Binary Hamming code profile m=12

The standard binary Hamming code has parameters [4095,4083,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001880.

S26-E-00001882 - Binary Hamming code profile m=13

The standard binary Hamming code has parameters [8191,8178,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001881.

S26-E-00001883 - Binary Hamming code profile m=14

The standard binary Hamming code has parameters [16383,16369,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001882.

S26-E-00001884 - Binary Hamming code profile m=15

The standard binary Hamming code has parameters [32767,32752,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001883.

S26-E-00001885 - Binary Hamming code profile m=16

The standard binary Hamming code has parameters [65535,65519,3]_2 and corrects one bit error.

Evidence M. Hard: S26-E-00001884.

S26-E-00001886 - Reed-Solomon parameter profile q=3

Over a field F_3, evaluating polynomials of degree <1 at 2 distinct nonzero field points gives an [2,1,2]_3 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001885.

S26-E-00001887 - Reed-Solomon parameter profile q=5

Over a field F_5, evaluating polynomials of degree <2 at 4 distinct nonzero field points gives an [4,2,3]_5 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001886.

S26-E-00001888 - Reed-Solomon parameter profile q=7

Over a field F_7, evaluating polynomials of degree <3 at 6 distinct nonzero field points gives an [6,3,4]_7 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001887.

S26-E-00001889 - Reed-Solomon parameter profile q=8

Over a field F_8, evaluating polynomials of degree <4 at 7 distinct nonzero field points gives an [7,4,4]_8 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001888.

S26-E-00001890 - Reed-Solomon parameter profile q=9

Over a field F_9, evaluating polynomials of degree <4 at 8 distinct nonzero field points gives an [8,4,5]_9 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001889.

S26-E-00001891 - Reed-Solomon parameter profile q=11

Over a field F_11, evaluating polynomials of degree <5 at 10 distinct nonzero field points gives an [10,5,6]_11 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001890.

S26-E-00001892 - Reed-Solomon parameter profile q=13

Over a field F_13, evaluating polynomials of degree <6 at 12 distinct nonzero field points gives an [12,6,7]_13 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001891.

S26-E-00001893 - Reed-Solomon parameter profile q=16

Over a field F_16, evaluating polynomials of degree <8 at 15 distinct nonzero field points gives an [15,8,8]_16 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001892.

S26-E-00001894 - Reed-Solomon parameter profile q=17

Over a field F_17, evaluating polynomials of degree <8 at 16 distinct nonzero field points gives an [16,8,9]_17 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001893.

S26-E-00001895 - Reed-Solomon parameter profile q=19

Over a field F_19, evaluating polynomials of degree <9 at 18 distinct nonzero field points gives an [18,9,10]_19 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001894.

S26-E-00001896 - Reed-Solomon parameter profile q=23

Over a field F_23, evaluating polynomials of degree <11 at 22 distinct nonzero field points gives an [22,11,12]_23 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001895.

S26-E-00001897 - Reed-Solomon parameter profile q=25

Over a field F_25, evaluating polynomials of degree <12 at 24 distinct nonzero field points gives an [24,12,13]_25 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001896.

S26-E-00001898 - Reed-Solomon parameter profile q=27

Over a field F_27, evaluating polynomials of degree <13 at 26 distinct nonzero field points gives an [26,13,14]_27 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001897.

S26-E-00001899 - Reed-Solomon parameter profile q=29

Over a field F_29, evaluating polynomials of degree <14 at 28 distinct nonzero field points gives an [28,14,15]_29 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001898.

S26-E-00001900 - Reed-Solomon parameter profile q=31

Over a field F_31, evaluating polynomials of degree <15 at 30 distinct nonzero field points gives an [30,15,16]_31 Reed-Solomon MDS code.

Evidence M. Hard: S26-E-00001899.

Part S - Additive combinatorics and discrete geometry

S26-D-00001901 - Sumset

For subsets A,B of an abelian group, A+B={a+b:a in A,b in B}.

Evidence A. Hard: S26-D-00000709.

S26-D-00001902 - Difference set

A-B={a-b:a in A,b in B}.

Evidence A. Hard: S26-D-00001901.

S26-D-00001903 - Additive energy

For finite A,B in an abelian group, E(A,B) is the number of quadruples a_1+b_1=a_2+b_2 with a_i in A,b_i in B.

Evidence A. Hard: S26-D-00001902.

S26-I-00001904 - Energy via representation function

If r_{A+B}(x)=|{(a,b):a+b=x}|, then E(A,B)=sum_x r_{A+B}(x)^2.

Evidence A. Hard: S26-D-00001903.

S26-T-00001905 - Cauchy-Schwarz sumset bound

For finite nonempty A,B, E(A,B)>=|A|^2|B|^2/|A+B|.

Evidence A. Hard: S26-I-00001904.

S26-T-00001906 - Cauchy-Davenport theorem

For nonempty A,B subset F_p, |A+B|>=min(p,|A|+|B|-1).

Evidence A. Hard: S26-T-00001905.

S26-T-00001907 - Vosper theorem interface

Under the nontrivial equality case of Cauchy-Davenport with sufficiently small sumset, subsets of F_p are arithmetic progressions up to the standard exceptional cases.

Evidence A. Hard: S26-T-00001906.

S26-D-00001908 - Arithmetic progression

A finite arithmetic progression in an abelian group is {a+jd:0<=j<k} with declared nondegeneracy conditions.

Evidence A. Hard: S26-T-00001907.

S26-D-00001909 - Generalized arithmetic progression

A rank-r generalized progression is {a+n_1d_1+...+n_rd_r:0<=n_i<L_i}; properness means the parameterization is injective.

Evidence A. Hard: S26-D-00001908.

S26-T-00001910 - Freiman theorem interface

Finite subsets of torsion-free abelian groups with small doubling are controlled by generalized arithmetic progressions; constants depend on the doubling parameter and chosen formulation.

Evidence A. Hard: S26-D-00001909.

S26-D-00001911 - Sidon set

A Sidon/B_2 set has essentially unique pairwise sums under a declared ordered/unordered convention.

Evidence A. Hard: S26-T-00001910.

S26-D-00001912 - Convex lattice polytope

A lattice polytope is the convex hull of finitely many points in Z^d.

Evidence A. Hard: S26-D-00001911.

S26-D-00001913 - Ehrhart polynomial

For a d-dimensional lattice polytope P, L_P(n)=|nP intersect Z^d| is a polynomial in n of degree d for n>=0.

Evidence A. Hard: S26-D-00001912.

S26-T-00001914 - Ehrhart leading coefficient

The leading coefficient of the Ehrhart polynomial is the Euclidean d-volume of P under the lattice-normalized coordinate convention.

Evidence A. Hard: S26-D-00001913.

S26-T-00001915 - Pick theorem

For a lattice polygon in R^2, area=I+B/2-1, where I and B are the numbers of interior and boundary lattice points.

Evidence A. Hard: S26-T-00001914.

S26-D-00001916 - Hypercube

The d-dimensional unit hypercube is [0,1]^d with 2^d vertices.

Evidence A. Hard: S26-T-00001915.

S26-I-00001917 - Hypercube face numbers

The number of k-dimensional faces of the d-cube is 2^{d-k} C(d,k).

Evidence A. Hard: S26-D-00001916.

S26-D-00001918 - Cross-polytope

The d-dimensional cross-polytope is conv{+/- e_1,...,+/- e_d}; it is dual to the hypercube.

Evidence A. Hard: S26-I-00001917.

S26-N-00001919 - Additive-structure fence

Small doubling is a structural hypothesis, not a synonym for randomness or pseudorandomness; the relevant inverse theorem depends on the ambient group and scale.

Evidence A. Hard: S26-D-00001918. Boundary: A set can have small doubling because it is highly structured.

S26-B-00001920 - Additive combinatorics to harmonic analysis

Fourier coefficients, convolution, energy, and sumsets translate additive structure into analytic concentration and back.

Evidence A. Hard: S26-N-00001919.

S26-E-00001921 - Hypercube face profile d=1

The 1-cube has face vector f_k=2^(d-k)C(d,k): [2, 1]. In particular it has 2 vertices and 1 edges.

Evidence M. Hard: S26-B-00001920.

S26-E-00001922 - Hypercube face profile d=2

The 2-cube has face vector f_k=2^(d-k)C(d,k): [4, 4, 1]. In particular it has 4 vertices and 4 edges.

Evidence M. Hard: S26-E-00001921.

S26-E-00001923 - Hypercube face profile d=3

The 3-cube has face vector f_k=2^(d-k)C(d,k): [8, 12, 6, 1]. In particular it has 8 vertices and 12 edges.

Evidence M. Hard: S26-E-00001922.

S26-E-00001924 - Hypercube face profile d=4

The 4-cube has face vector f_k=2^(d-k)C(d,k): [16, 32, 24, 8, 1]. In particular it has 16 vertices and 32 edges.

Evidence M. Hard: S26-E-00001923.

S26-E-00001925 - Hypercube face profile d=5

The 5-cube has face vector f_k=2^(d-k)C(d,k): [32, 80, 80, 40, 10, 1]. In particular it has 32 vertices and 80 edges.

Evidence M. Hard: S26-E-00001924.

S26-E-00001926 - Hypercube face profile d=6

The 6-cube has face vector f_k=2^(d-k)C(d,k): [64, 192, 240, 160, 60, 12, 1]. In particular it has 64 vertices and 192 edges.

Evidence M. Hard: S26-E-00001925.

S26-E-00001927 - Hypercube face profile d=7

The 7-cube has face vector f_k=2^(d-k)C(d,k): [128, 448, 672, 560, 280, 84, 14, 1]. In particular it has 128 vertices and 448 edges.

Evidence M. Hard: S26-E-00001926.

S26-E-00001928 - Hypercube face profile d=8

The 8-cube has face vector f_k=2^(d-k)C(d,k): [256, 1024, 1792, 1792, 1120, 448, 112, 16, 1]. In particular it has 256 vertices and 1024 edges.

Evidence M. Hard: S26-E-00001927.

S26-E-00001929 - Hypercube face profile d=9

The 9-cube has face vector f_k=2^(d-k)C(d,k): [512, 2304, 4608, 5376, 4032, 2016, 672, 144, 18, 1]. In particular it has 512 vertices and 2304 edges.

Evidence M. Hard: S26-E-00001928.

S26-E-00001930 - Hypercube face profile d=10

The 10-cube has face vector f_k=2^(d-k)C(d,k): [1024, 5120, 11520, 15360, 13440, 8064, 3360, 960, 180, 20, 1]. In particular it has 1024 vertices and 5120 edges.

Evidence M. Hard: S26-E-00001929.

S26-E-00001931 - Hypercube face profile d=11

The 11-cube has face vector f_k=2^(d-k)C(d,k): [2048, 11264, 28160, 42240, 42240, 29568, 14784, 5280, 1320, 220, 22, 1]. In particular it has 2048 vertices and 11264 edges.

Evidence M. Hard: S26-E-00001930.

S26-E-00001932 - Hypercube face profile d=12

The 12-cube has face vector f_k=2^(d-k)C(d,k): [4096, 24576, 67584, 112640, 126720, 101376, 59136, 25344, 7920, 1760, 264, 24, 1]. In particular it has 4096 vertices and 24576 edges.

Evidence M. Hard: S26-E-00001931.

S26-E-00001933 - Hypercube face profile d=13

The 13-cube has face vector f_k=2^(d-k)C(d,k): [8192, 53248, 159744, 292864, 366080, 329472, 219648, 109824, 41184, 11440, 2288, 312, 26, 1]. In particular it has 8192 vertices and 53248 edges.

Evidence M. Hard: S26-E-00001932.

S26-E-00001934 - Hypercube face profile d=14

The 14-cube has face vector f_k=2^(d-k)C(d,k): [16384, 114688, 372736, 745472, 1025024, 1025024, 768768, 439296, 192192, 64064, 16016, 2912, 364, 28, 1]. In particular it has 16384 vertices and 114688 edges.

Evidence M. Hard: S26-E-00001933.

S26-E-00001935 - Hypercube face profile d=15

The 15-cube has face vector f_k=2^(d-k)C(d,k): [32768, 245760, 860160, 1863680, 2795520, 3075072, 2562560, 1647360, 823680, 320320, 96096, 21840, 3640, 420, 30, 1]. In particular it has 32768 vertices and 245760 edges.

Evidence M. Hard: S26-E-00001934.

S26-E-00001936 - Hypercube face profile d=16

The 16-cube has face vector f_k=2^(d-k)C(d,k): [65536, 524288, 1966080, 4587520, 7454720, 8945664, 8200192, 5857280, 3294720, 1464320, 512512, 139776, 29120, 4480, 480, 32, 1]. In particular it has 65536 vertices and 524288 edges.

Evidence M. Hard: S26-E-00001935.

S26-E-00001937 - Hypercube face profile d=17

The 17-cube has face vector f_k=2^(d-k)C(d,k): [131072, 1114112, 4456448, 11141120, 19496960, 25346048, 25346048, 19914752, 12446720, 6223360, 2489344, 792064, 198016, 38080, 5440, 544, 34, 1]. In particular it has 131072 vertices and 1114112 edges.

Evidence M. Hard: S26-E-00001936.

S26-E-00001938 - Hypercube face profile d=18

The 18-cube has face vector f_k=2^(d-k)C(d,k): [262144, 2359296, 10027008, 26738688, 50135040, 70189056, 76038144, 65175552, 44808192, 24893440, 11202048, 4073472, 1188096, 274176, 48960, 6528, 612, 36, 1]. In particular it has 262144 vertices and 2359296 edges.

Evidence M. Hard: S26-E-00001937.

S26-E-00001939 - Hypercube face profile d=19

The 19-cube has face vector f_k=2^(d-k)C(d,k): [524288, 4980736, 22413312, 63504384, 127008768, 190513152, 222265344, 206389248, 154791936, 94595072, 47297536, 19348992, 6449664, 1736448, 372096, 62016, 7752, 684, 38, 1]. In particular it has 524288 vertices and 4980736 edges.

Evidence M. Hard: S26-E-00001938.

S26-E-00001940 - Hypercube face profile d=20

The 20-cube has face vector f_k=2^(d-k)C(d,k): [1048576, 10485760, 49807360, 149422080, 317521920, 508035072, 635043840, 635043840, 515973120, 343982080, 189190144, 85995520, 32248320, 9922560, 2480640, 496128, 77520, 9120, 760, 40, 1]. In particular it has 1048576 vertices and 10485760 edges.

Evidence M. Hard: S26-E-00001939.

S26-E-00001941 - Hypercube face profile d=21

The 21-cube has face vector f_k=2^(d-k)C(d,k): [2097152, 22020096, 110100480, 348651520, 784465920, 1333592064, 1778122752, 1905131520, 1666990080, 1203937280, 722362368, 361181184, 150492160, 52093440, 14883840, 3472896, 651168, 95760, 10640, 840, 42, 1]. In particular it has 2097152 vertices and 22020096 edges.

Evidence M. Hard: S26-E-00001940.

S26-E-00001942 - Hypercube face profile d=22

The 22-cube has face vector f_k=2^(d-k)C(d,k): [4194304, 46137344, 242221056, 807403520, 1917583360, 3451650048, 4889837568, 5588385792, 5239111680, 4074864640, 2648662016, 1444724736, 662165504, 254679040, 81861120, 21829632, 4775232, 842688, 117040, 12320, 924, 44, 1]. In particular it has 4194304 vertices and 46137344 edges.

Evidence M. Hard: S26-E-00001941.

S26-E-00001943 - Hypercube face profile d=23

The 23-cube has face vector f_k=2^(d-k)C(d,k): [8388608, 96468992, 530579456, 1857028096, 4642570240, 8820883456, 13231325184, 16066609152, 16066609152, 13388840960, 9372188672, 5538111488, 2769055744, 1171523584, 418401280, 125520384, 31380096, 6460608, 1076768, 141680, 14168, 1012, 46, 1]. In particular it has 8388608 vertices and 96468992 edges.

Evidence M. Hard: S26-E-00001942.

S26-E-00001944 - Hypercube face profile d=24

The 24-cube has face vector f_k=2^(d-k)C(d,k): [16777216, 201326592, 1157627904, 4244635648, 11142168576, 22284337152, 35283533824, 45364543488, 48199827456, 42844291072, 32133218304, 20448411648, 11076222976, 5112102912, 2008326144, 669442048, 188280576, 44301312, 8614144, 1360128, 170016, 16192, 1104, 48, 1]. In particular it has 16777216 vertices and 201326592 edges.

Evidence M. Hard: S26-E-00001943.

S26-E-00001945 - Hypercube face profile d=25

The 25-cube has face vector f_k=2^(d-k)C(d,k): [33554432, 419430400, 2516582400, 9646899200, 26528972800, 55710842880, 92851404800, 126012620800, 141764198400, 133888409600, 107110727680, 73030041600, 42600857600, 21300428800, 9128755200, 3347210240, 1046003200, 276883200, 61529600, 11334400, 1700160, 202400, 18400, 1200, 50, 1]. In particular it has 33554432 vertices and 419430400 edges.

Evidence M. Hard: S26-E-00001944.

S26-E-00001946 - Hypercube face profile d=26

The 26-cube has face vector f_k=2^(d-k)C(d,k): [67108864, 872415232, 5452595200, 21810380800, 62704844800, 137950658560, 241413652480, 344876646400, 409541017600, 409541017600, 348109864960, 253170810880, 158231756800, 85201715200, 39557939200, 15823175680, 5439216640, 1599769600, 399942400, 84198400, 14734720, 2104960, 239200, 20800, 1300, 52, 1]. In particular it has 67108864 vertices and 872415232 edges.

Evidence M. Hard: S26-E-00001945.

S26-E-00001947 - Hypercube face profile d=27

The 27-cube has face vector f_k=2^(d-k)C(d,k): [134217728, 1811939328, 11777605632, 49073356800, 147220070400, 338606161920, 620777963520, 931166945280, 1163958681600, 1228623052800, 1105760747520, 854451486720, 569634324480, 328635187200, 164317593600, 71204290560, 26701608960, 8638755840, 2399654400, 568339200, 113667840, 18944640, 2583360, 280800, 23400, 1404, 54, 1]. In particular it has 134217728 vertices and 1811939328 edges.

Evidence M. Hard: S26-E-00001946.

S26-E-00001948 - Hypercube face profile d=28

The 28-cube has face vector f_k=2^(d-k)C(d,k): [268435456, 3758096384, 25367150592, 109924319232, 343513497600, 824432394240, 1580162088960, 2483111854080, 3259084308480, 3621204787200, 3440144547840, 2814663720960, 1993720135680, 1226904698880, 657270374400, 306726174720, 124607508480, 43979120640, 13438064640, 3536332800, 795674880, 151557120, 24111360, 3144960, 327600, 26208, 1512, 56, 1]. In particular it has 268435456 vertices and 3758096384 edges.

Evidence M. Hard: S26-E-00001947.

S26-E-00001949 - Hypercube face profile d=29

The 29-cube has face vector f_k=2^(d-k)C(d,k): [536870912, 7784628224, 54492397568, 245215789056, 796951314432, 1992378286080, 3984756572160, 6546385797120, 9001280471040, 10501493882880, 10501493882880, 9069471989760, 6802103992320, 4447529533440, 2541445447680, 1270722723840, 555941191680, 212565749760, 70855249920, 20510730240, 5127682560, 1098789120, 199779840, 30401280, 3800160, 380016, 29232, 1624, 58, 1]. In particular it has 536870912 vertices and 7784628224 edges.

Evidence M. Hard: S26-E-00001948.

S26-E-00001950 - Hypercube face profile d=30

The 30-cube has face vector f_k=2^(d-k)C(d,k): [1073741824, 16106127360, 116769423360, 544923975680, 1839118417920, 4781707886592, 9961891430400, 17077528166400, 24548946739200, 30004268236800, 31504481648640, 28640437862400, 22673679974400, 15697163059200, 9530420428800, 5082890895360, 2382605107200, 981072691200, 354276249600, 111876710400, 30766095360, 7325260800, 1498348800, 260582400, 38001600, 4560192, 438480, 32480, 1740, 60, 1]. In particular it has 1073741824 vertices and 16106127360 edges.

Evidence M. Hard: S26-E-00001949.

Part T - Discrete algorithms and complexity bridges

S26-D-00001951 - Asymptotic time complexity

A time bound T(n)=O(f(n)) means there exist constants C,n_0 such that T(n)<=C f(n) for n>=n_0 under the declared computational model.

Evidence A. Hard: S26-A-00000066.

S26-D-00001952 - Polynomial-time algorithm

An algorithm runs in polynomial time when its worst-case running time is bounded by n^{O(1)} in the input length under a specified reasonable machine model.

Evidence A. Hard: S26-D-00001951.

S26-D-00001953 - Complexity class P

P is the class of decision problems solvable by deterministic Turing machines in polynomial time.

Evidence A. Hard: S26-D-00001952.

S26-D-00001954 - Complexity class NP

NP is the class of decision problems whose yes-instances have polynomial-size certificates verifiable in deterministic polynomial time; equivalently solvable by nondeterministic polynomial time.

Evidence A. Hard: S26-D-00001953.

S26-D-00001955 - Polynomial-time many-one reduction

A<=_p B when a polynomial-time computable map f satisfies x in A iff f(x) in B.

Evidence A. Hard: S26-D-00001954.

S26-D-00001956 - NP-hardness

A problem B is NP-hard when every problem in NP reduces to B by the declared polynomial-time reduction.

Evidence A. Hard: S26-D-00001955.

S26-D-00001957 - NP-completeness

A decision problem is NP-complete when it lies in NP and is NP-hard.

Evidence A. Hard: S26-D-00001956.

S26-T-00001958 - Cook-Levin theorem

Boolean satisfiability is NP-complete under polynomial-time many-one reductions.

Evidence A. Hard: S26-D-00001957.

S26-A-00001959 - Euclidean algorithm complexity interface

The Euclidean algorithm computes gcd(a,b) by repeated division with remainder; bit-complexity analysis depends on integer-arithmetic cost, while the number of divisions is logarithmic in the smaller positive input up to classical worst-case constants.

Evidence A. Hard: S26-T-00001958.

S26-A-00001960 - Fast exponentiation

Binary exponentiation computes a^n using O(log n) multiplications by repeated squaring and multiplication according to the binary digits of n.

Evidence A. Hard: S26-A-00001959.

S26-A-00001961 - Modular exponentiation

Binary exponentiation combined with modular reduction computes a^n mod m using O(log n) modular multiplications.

Evidence A. Hard: S26-A-00001960.

S26-A-00001962 - Sieve of Eratosthenes

The sieve marks composites up to N by prime multiples and enumerates all primes <=N; standard implementations use O(N log log N) arithmetic marking operations up to model-dependent refinements.

Evidence A. Hard: S26-A-00001961.

S26-A-00001963 - Breadth-first search

BFS explores an unweighted graph by distance layers and runs in O(|V|+|E|) with adjacency-list representation.

Evidence A. Hard: S26-A-00001962.

S26-A-00001964 - Depth-first search

DFS explores along depth-first branches and runs in O(|V|+|E|) with adjacency lists; it supports component, cycle, and topological-order routines.

Evidence A. Hard: S26-A-00001963.

S26-A-00001965 - Dijkstra algorithm

With nonnegative edge weights, Dijkstra's algorithm computes single-source shortest-path distances; complexity depends on the priority-queue implementation.

Evidence A. Hard: S26-A-00001964.

S26-A-00001966 - Kruskal algorithm

Kruskal's algorithm sorts edges by weight and greedily adds edges that do not create a cycle, yielding a minimum spanning forest.

Evidence A. Hard: S26-A-00001965.

S26-A-00001967 - Prim algorithm

Prim's algorithm grows a minimum spanning tree by repeatedly adding the cheapest edge crossing the current cut in a connected weighted graph.

Evidence A. Hard: S26-A-00001966.

S26-T-00001968 - Greedy correctness for matroid bases

Sorting elements by weight and repeatedly adding an element that preserves independence yields a maximum-weight basis of a matroid.

Evidence A. Hard: S26-A-00001967.

S26-N-00001969 - Complexity-model fence

Asymptotic complexity claims require an input encoding and cost model; arithmetic-operation counts and bit complexity are not interchangeable.

Evidence A. Hard: S26-T-00001968. Boundary: An O(log n) multiplication count does not imply O(log n) bit operations for arbitrarily large operands.

S26-B-00001970 - Discrete mathematics to computation

Graphs, matroids, finite fields, groups, and combinatorial structures become algorithmic objects once encodings and cost models are declared; theorem-level structure and implementation complexity remain distinct evidence layers.

Evidence A. Hard: S26-N-00001969.

S26-E-00001971 - Binary-exponentiation count for exponent 1

Using left-to-right binary exponentiation initialized at the leading 1, exponent 1 (0b1) uses 0 squarings and 0 extra multiplies, for 0 multiplications total.

Evidence M. Hard: S26-B-00001970.

S26-E-00001972 - Binary-exponentiation count for exponent 2

Using left-to-right binary exponentiation initialized at the leading 1, exponent 2 (0b10) uses 1 squarings and 0 extra multiplies, for 1 multiplications total.

Evidence M. Hard: S26-E-00001971.

S26-E-00001973 - Binary-exponentiation count for exponent 3

Using left-to-right binary exponentiation initialized at the leading 1, exponent 3 (0b11) uses 1 squarings and 1 extra multiplies, for 2 multiplications total.

Evidence M. Hard: S26-E-00001972.

S26-E-00001974 - Binary-exponentiation count for exponent 4

Using left-to-right binary exponentiation initialized at the leading 1, exponent 4 (0b100) uses 2 squarings and 0 extra multiplies, for 2 multiplications total.

Evidence M. Hard: S26-E-00001973.

S26-E-00001975 - Binary-exponentiation count for exponent 5

Using left-to-right binary exponentiation initialized at the leading 1, exponent 5 (0b101) uses 2 squarings and 1 extra multiplies, for 3 multiplications total.

Evidence M. Hard: S26-E-00001974.

S26-E-00001976 - Binary-exponentiation count for exponent 6

Using left-to-right binary exponentiation initialized at the leading 1, exponent 6 (0b110) uses 2 squarings and 1 extra multiplies, for 3 multiplications total.

Evidence M. Hard: S26-E-00001975.

S26-E-00001977 - Binary-exponentiation count for exponent 7

Using left-to-right binary exponentiation initialized at the leading 1, exponent 7 (0b111) uses 2 squarings and 2 extra multiplies, for 4 multiplications total.

Evidence M. Hard: S26-E-00001976.

S26-E-00001978 - Binary-exponentiation count for exponent 8

Using left-to-right binary exponentiation initialized at the leading 1, exponent 8 (0b1000) uses 3 squarings and 0 extra multiplies, for 3 multiplications total.

Evidence M. Hard: S26-E-00001977.

S26-E-00001979 - Binary-exponentiation count for exponent 9

Using left-to-right binary exponentiation initialized at the leading 1, exponent 9 (0b1001) uses 3 squarings and 1 extra multiplies, for 4 multiplications total.

Evidence M. Hard: S26-E-00001978.

S26-E-00001980 - Binary-exponentiation count for exponent 10

Using left-to-right binary exponentiation initialized at the leading 1, exponent 10 (0b1010) uses 3 squarings and 1 extra multiplies, for 4 multiplications total.

Evidence M. Hard: S26-E-00001979.

S26-E-00001981 - Binary-exponentiation count for exponent 11

Using left-to-right binary exponentiation initialized at the leading 1, exponent 11 (0b1011) uses 3 squarings and 2 extra multiplies, for 5 multiplications total.

Evidence M. Hard: S26-E-00001980.

S26-E-00001982 - Binary-exponentiation count for exponent 12

Using left-to-right binary exponentiation initialized at the leading 1, exponent 12 (0b1100) uses 3 squarings and 1 extra multiplies, for 4 multiplications total.

Evidence M. Hard: S26-E-00001981.

S26-E-00001983 - Binary-exponentiation count for exponent 13

Using left-to-right binary exponentiation initialized at the leading 1, exponent 13 (0b1101) uses 3 squarings and 2 extra multiplies, for 5 multiplications total.

Evidence M. Hard: S26-E-00001982.

S26-E-00001984 - Binary-exponentiation count for exponent 14

Using left-to-right binary exponentiation initialized at the leading 1, exponent 14 (0b1110) uses 3 squarings and 2 extra multiplies, for 5 multiplications total.

Evidence M. Hard: S26-E-00001983.

S26-E-00001985 - Binary-exponentiation count for exponent 15

Using left-to-right binary exponentiation initialized at the leading 1, exponent 15 (0b1111) uses 3 squarings and 3 extra multiplies, for 6 multiplications total.

Evidence M. Hard: S26-E-00001984.

S26-E-00001986 - Binary-exponentiation count for exponent 16

Using left-to-right binary exponentiation initialized at the leading 1, exponent 16 (0b10000) uses 4 squarings and 0 extra multiplies, for 4 multiplications total.

Evidence M. Hard: S26-E-00001985.

S26-E-00001987 - Binary-exponentiation count for exponent 17

Using left-to-right binary exponentiation initialized at the leading 1, exponent 17 (0b10001) uses 4 squarings and 1 extra multiplies, for 5 multiplications total.

Evidence M. Hard: S26-E-00001986.

S26-E-00001988 - Binary-exponentiation count for exponent 18

Using left-to-right binary exponentiation initialized at the leading 1, exponent 18 (0b10010) uses 4 squarings and 1 extra multiplies, for 5 multiplications total.

Evidence M. Hard: S26-E-00001987.

S26-E-00001989 - Binary-exponentiation count for exponent 19

Using left-to-right binary exponentiation initialized at the leading 1, exponent 19 (0b10011) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

Evidence M. Hard: S26-E-00001988.

S26-E-00001990 - Binary-exponentiation count for exponent 20

Using left-to-right binary exponentiation initialized at the leading 1, exponent 20 (0b10100) uses 4 squarings and 1 extra multiplies, for 5 multiplications total.

Evidence M. Hard: S26-E-00001989.

S26-E-00001991 - Binary-exponentiation count for exponent 21

Using left-to-right binary exponentiation initialized at the leading 1, exponent 21 (0b10101) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

Evidence M. Hard: S26-E-00001990.

S26-E-00001992 - Binary-exponentiation count for exponent 22

Using left-to-right binary exponentiation initialized at the leading 1, exponent 22 (0b10110) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

Evidence M. Hard: S26-E-00001991.

S26-E-00001993 - Binary-exponentiation count for exponent 23

Using left-to-right binary exponentiation initialized at the leading 1, exponent 23 (0b10111) uses 4 squarings and 3 extra multiplies, for 7 multiplications total.

Evidence M. Hard: S26-E-00001992.

S26-E-00001994 - Binary-exponentiation count for exponent 24

Using left-to-right binary exponentiation initialized at the leading 1, exponent 24 (0b11000) uses 4 squarings and 1 extra multiplies, for 5 multiplications total.

Evidence M. Hard: S26-E-00001993.

S26-E-00001995 - Binary-exponentiation count for exponent 25

Using left-to-right binary exponentiation initialized at the leading 1, exponent 25 (0b11001) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

Evidence M. Hard: S26-E-00001994.

S26-E-00001996 - Binary-exponentiation count for exponent 26

Using left-to-right binary exponentiation initialized at the leading 1, exponent 26 (0b11010) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

Evidence M. Hard: S26-E-00001995.

S26-E-00001997 - Binary-exponentiation count for exponent 27

Using left-to-right binary exponentiation initialized at the leading 1, exponent 27 (0b11011) uses 4 squarings and 3 extra multiplies, for 7 multiplications total.

Evidence M. Hard: S26-E-00001996.

S26-E-00001998 - Binary-exponentiation count for exponent 28

Using left-to-right binary exponentiation initialized at the leading 1, exponent 28 (0b11100) uses 4 squarings and 2 extra multiplies, for 6 multiplications total.

Evidence M. Hard: S26-E-00001997.

S26-E-00001999 - Binary-exponentiation count for exponent 29

Using left-to-right binary exponentiation initialized at the leading 1, exponent 29 (0b11101) uses 4 squarings and 3 extra multiplies, for 7 multiplications total.

Evidence M. Hard: S26-E-00001998.

S26-E-00002000 - Binary-exponentiation count for exponent 30

Using left-to-right binary exponentiation initialized at the leading 1, exponent 30 (0b11110) uses 4 squarings and 3 extra multiplies, for 7 multiplications total.

Evidence M. Hard: S26-E-00001999.

Appendix - Evidence and continuation boundary

Volume I freezes IDs 1001-2000. Later volumes may depend on these objects but may not silently duplicate or strengthen them. General A-status claims require theorem-specific source/formal witnesses before any P promotion. M status is confined to the exact finite computation recorded in each profile.

Next volume: Analysis, IDs 2001-3000.

A Synopsis of Results in Pure, Applied and Computational Mathematics

2026 Edition

Volume II
Analysis

Peer-reviewed consolidated projection

Base release: 1.0.0
Reviewed projection: 1.0.1-pr
Canonical range: 2001-3000
Objects: 1,000
Peer-review corrections in this volume: 2
Combined review graph: 5225 hard edges; 1 component; 0 cycles; 0 dangling
Volume evidence unchanged: no P promotions

This projection applies the consolidated peer-review correction ledger while preserving canonical IDs and evidence states. Frozen original release files remain unchanged.

Peer-review corrections in this volume

S26-T-00002158 - mathematical correction

Reviewed statement. For sigma-finite positive measures mu and nu with nu<<mu, there exists a nonnegative measurable density f, unique mu-a.e., such that nu(E)=integral_E f dmu for every measurable E; if nu is finite, then f belongs to L1(mu).

Reason. Counterexample: mu=nu=Lebesgue measure on R is sigma-finite with density 1, but 1 is not in L1(R).

S26-T-00002262 - scope correction

Reviewed statement. If U is a connected open subset of C and f:U->C is holomorphic and nonconstant, then f is an open map. More generally, on a disconnected open set the conclusion holds on each component on which f is nonconstant.

Reason. On a disconnected open set a holomorphic function can be constant on one component and nonconstant on another, so global nonconstancy alone does not imply openness on every open subset.

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Part E - Lp refinements, rearrangement and interpolation (2201-2250)
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Part A - Quantitative sequences and asymptotics

S26-D-00002001 - Asymptotic equivalence of sequences

For eventually nonzero b_n, a_n~b_n means a_n/b_n -> 1.

Evidence A. Hard: S26-D-00000351.

S26-D-00002002 - Little-o relation for sequences

a_n=o(b_n) means a_n/b_n -> 0 wherever the ratio is eventually defined.

Evidence A. Hard: S26-D-00002001.

S26-D-00002003 - Big-O relation for sequences

a_n=O(b_n) means |a_n|<=C|b_n| for all sufficiently large n and some finite C.

Evidence A. Hard: S26-D-00002002.

S26-D-00002004 - Theta relation for positive scales

a_n=Theta(b_n) means both a_n=O(b_n) and b_n=O(a_n), with positivity/nonvanishing declared as needed.

Evidence A. Hard: S26-D-00002003.

S26-D-00002005 - Asymptotic scale

An asymptotic scale phi_0,phi_1,... satisfies phi_{j+1}=o(phi_j) in the declared limit regime.

Evidence A. Hard: S26-D-00002004.

S26-D-00002006 - Poincare asymptotic expansion

f(x)~sum_{j>=0} a_j phi_j(x) means the remainder after N terms is o(phi_N(x)) for every fixed N.

Evidence A. Hard: S26-D-00002005.

S26-T-00002007 - Cesaro convergence preserves ordinary limits

If a_n->L, then the arithmetic means (1/n)sum_{k<=n}a_k also converge to L.

Evidence A. Hard: S26-D-00002006.

S26-T-00002008 - Stolz-Cesaro theorem

Under the standard monotonicity/unboundedness hypotheses on b_n, limits of (a_{n+1}-a_n)/(b_{n+1}-b_n) control the limit of a_n/b_n.

Evidence A. Hard: S26-T-00002007.

S26-T-00002009 - Cauchy condensation criterion

For a nonincreasing nonnegative sequence a_n, sum a_n converges iff sum 2^k a_{2^k} converges.

Evidence A. Hard: S26-T-00002008.

S26-I-00002010 - Abel summation formula

For A(x)=sum_{n<=x}a_n, sum_{n<=N}a_n b_n = A(N)b_N + sum_{n<N}A(n)(b_n-b_{n+1}).

Evidence A. Hard: S26-T-00002009.

S26-D-00002011 - Regularly varying sequence

A positive sequence a_n is regularly varying with index rho when a_{floor(tn)}/a_n -> t^rho for every t>0.

Evidence A. Hard: S26-I-00002010.

S26-D-00002012 - Slowly varying sequence

Regular variation with index 0 is called slow variation.

Evidence A. Hard: S26-D-00002011.

S26-D-00002013 - Limit superior of a real sequence

limsup a_n is the limit of the decreasing sequence sup_{k>=n}a_k in the extended reals.

Evidence A. Hard: S26-D-00002012.

S26-D-00002014 - Limit inferior of a real sequence

liminf a_n is the limit of the increasing sequence inf_{k>=n}a_k in the extended reals.

Evidence A. Hard: S26-D-00002013.

S26-T-00002015 - Subsequence characterization of limsup

For a bounded real sequence, limsup is the largest subsequential limit and is attained by some subsequence.

Evidence A. Hard: S26-D-00002014.

S26-T-00002016 - Ratio test in limsup form

For nonzero terms eventually, limsup |a_{n+1}/a_n|<1 implies absolute convergence of sum a_n.

Evidence A. Hard: S26-T-00002015.

S26-T-00002017 - Root test in limsup form

limsup |a_n|^{1/n}<1 implies absolute convergence; liminf |a_n|^{1/n}>1 implies divergence of terms from zero.

Evidence A. Hard: S26-T-00002016.

S26-D-00002018 - Tauberian principle

A Tauberian theorem supplements a one-way summability implication with side conditions that recover ordinary convergence or sharper asymptotics.

Evidence A. Hard: S26-T-00002017.

S26-N-00002019 - Asymptotic nonuniformity fence

An asymptotic statement in one parameter cannot be treated as uniform in another parameter without an explicit uniform remainder bound.

Evidence A. Hard: S26-D-00002018. Boundary: Pointwise asymptotics can fail after exchanging limits or optimizing over the second parameter.

S26-B-00002020 - Asymptotics to numerical analysis bridge

Rates expressed by O, o, and asymptotic expansions become computational error models only after constants, norms, and finite parameter ranges are specified.

Evidence A. Hard: S26-N-00002019.

S26-E-00002021 - Exponential-limit profile n=10

For n=10, (1+1/n)^n=2.593742460100, lying 1.245e-01 below e in floating evaluation.

Evidence M. Hard: S26-B-00002020.

S26-E-00002022 - Exponential-limit profile n=20

For n=20, (1+1/n)^n=2.653297705144, lying 6.498e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002021.

S26-E-00002023 - Exponential-limit profile n=30

For n=30, (1+1/n)^n=2.674318775870, lying 4.396e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002022.

S26-E-00002024 - Exponential-limit profile n=40

For n=40, (1+1/n)^n=2.685063838390, lying 3.322e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002023.

S26-E-00002025 - Exponential-limit profile n=50

For n=50, (1+1/n)^n=2.691588029074, lying 2.669e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002024.

S26-E-00002026 - Exponential-limit profile n=60

For n=60, (1+1/n)^n=2.695970139330, lying 2.231e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002025.

S26-E-00002027 - Exponential-limit profile n=70

For n=70, (1+1/n)^n=2.699116370976, lying 1.917e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002026.

S26-E-00002028 - Exponential-limit profile n=80

For n=80, (1+1/n)^n=2.701484940753, lying 1.680e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002027.

S26-E-00002029 - Exponential-limit profile n=90

For n=90, (1+1/n)^n=2.703332461058, lying 1.495e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002028.

S26-E-00002030 - Exponential-limit profile n=100

For n=100, (1+1/n)^n=2.704813829422, lying 1.347e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002029.

S26-E-00002031 - Exponential-limit profile n=110

For n=110, (1+1/n)^n=2.706028081505, lying 1.225e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002030.

S26-E-00002032 - Exponential-limit profile n=120

For n=120, (1+1/n)^n=2.707041490862, lying 1.124e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002031.

S26-E-00002033 - Exponential-limit profile n=130

For n=130, (1+1/n)^n=2.707900081718, lying 1.038e-02 below e in floating evaluation.

Evidence M. Hard: S26-E-00002032.

S26-E-00002034 - Exponential-limit profile n=140

For n=140, (1+1/n)^n=2.708636813921, lying 9.645e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002033.

S26-E-00002035 - Exponential-limit profile n=150

For n=150, (1+1/n)^n=2.709275911335, lying 9.006e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002034.

S26-E-00002036 - Exponential-limit profile n=160

For n=160, (1+1/n)^n=2.709835576308, lying 8.446e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002035.

S26-E-00002037 - Exponential-limit profile n=170

For n=170, (1+1/n)^n=2.710329751224, lying 7.952e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002036.

S26-E-00002038 - Exponential-limit profile n=180

For n=180, (1+1/n)^n=2.710769295839, lying 7.513e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002037.

S26-E-00002039 - Exponential-limit profile n=190

For n=190, (1+1/n)^n=2.711162794611, lying 7.119e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002038.

S26-E-00002040 - Exponential-limit profile n=200

For n=200, (1+1/n)^n=2.711517122929, lying 6.765e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002039.

S26-E-00002041 - Exponential-limit profile n=210

For n=210, (1+1/n)^n=2.711837852272, lying 6.444e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002040.

S26-E-00002042 - Exponential-limit profile n=220

For n=220, (1+1/n)^n=2.712129545321, lying 6.152e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002041.

S26-E-00002043 - Exponential-limit profile n=230

For n=230, (1+1/n)^n=2.712395974395, lying 5.886e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002042.

S26-E-00002044 - Exponential-limit profile n=240

For n=240, (1+1/n)^n=2.712640285482, lying 5.642e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002043.

S26-E-00002045 - Exponential-limit profile n=250

For n=250, (1+1/n)^n=2.712865123051, lying 5.417e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002044.

S26-E-00002046 - Exponential-limit profile n=260

For n=260, (1+1/n)^n=2.713072726158, lying 5.209e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002045.

S26-E-00002047 - Exponential-limit profile n=270

For n=270, (1+1/n)^n=2.713265003267, lying 5.017e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002046.

S26-E-00002048 - Exponential-limit profile n=280

For n=280, (1+1/n)^n=2.713443591094, lying 4.838e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002047.

S26-E-00002049 - Exponential-limit profile n=290

For n=290, (1+1/n)^n=2.713609901311, lying 4.672e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002048.

S26-E-00002050 - Exponential-limit profile n=300

For n=300, (1+1/n)^n=2.713765157943, lying 4.517e-03 below e in floating evaluation.

Evidence M. Hard: S26-E-00002049.

Part B - Differentiation, variation and absolute continuity

S26-D-00002051 - Upper right Dini derivative

D^+f(x)=limsup_{h downarrow 0}(f(x+h)-f(x))/h.

Evidence A. Hard: S26-E-00002050.

S26-D-00002052 - Lower right Dini derivative

D_+f(x)=liminf_{h downarrow 0}(f(x+h)-f(x))/h.

Evidence A. Hard: S26-D-00002051.

S26-D-00002053 - Function of bounded variation on an interval

f has bounded variation on [a,b] when the supremum of sum |f(x_i)-f(x_{i-1})| over partitions is finite.

Evidence A. Hard: S26-D-00002052.

S26-D-00002054 - Total variation function

V_f(x) is the variation of f on [a,x], giving a nondecreasing control function for BV maps.

Evidence A. Hard: S26-D-00002053.

S26-T-00002055 - Jordan decomposition for BV functions

A real BV function on an interval is the difference of two bounded nondecreasing functions.

Evidence A. Hard: S26-D-00002054.

S26-D-00002056 - Absolutely continuous function on an interval

f is absolutely continuous when disjoint intervals of sufficiently small total length force the sum of corresponding oscillations of f to be small.

Evidence A. Hard: S26-T-00002055.

S26-T-00002057 - Absolute continuity implies bounded variation

Every absolutely continuous function on a compact interval is of bounded variation.

Evidence A. Hard: S26-D-00002056.

S26-T-00002058 - Fundamental theorem for absolutely continuous functions

f is absolutely continuous iff f has an L1 derivative almost everywhere and f(x)=f(a)+integral_a^x f'(t)dt.

Evidence A. Hard: S26-T-00002057.

S26-D-00002059 - Lipschitz function on a metric domain

f is Lipschitz with constant L when d_Y(f(x),f(y))<=L d_X(x,y).

Evidence A. Hard: S26-T-00002058.

S26-T-00002060 - Rademacher theorem in Euclidean space

Every Lipschitz map R^n->R^m is Frechet differentiable almost everywhere.

Evidence A. Hard: S26-D-00002059.

S26-D-00002061 - Approximate derivative

The approximate derivative uses density-one approach sets rather than all nearby points.

Evidence A. Hard: S26-T-00002060.

S26-D-00002062 - Symmetric derivative

The symmetric derivative is lim_{h->0}(f(x+h)-f(x-h))/(2h) when the limit exists.

Evidence A. Hard: S26-D-00002061.

S26-T-00002063 - Lebesgue differentiation theorem

For f in L1_loc(R^n), averages of f over shrinking balls converge to f(x) for almost every x.

Evidence A. Hard: S26-D-00002062.

S26-D-00002064 - Hardy-Littlewood local averaging operator

A_r f(x)=|B(x,r)|^{-1} integral_{B(x,r)} f(y)dy records the scale-r local average.

Evidence A. Hard: S26-T-00002063.

S26-T-00002065 - Monotone functions are differentiable almost everywhere

A real monotone function on an interval has a finite derivative almost everywhere.

Evidence A. Hard: S26-D-00002064.

S26-D-00002066 - Singular function

A singular increasing function is continuous and nonconstant while having derivative zero almost everywhere.

Evidence A. Hard: S26-T-00002065.

S26-N-00002067 - Derivative-zero fence

Derivative zero almost everywhere does not imply constancy without additional regularity such as absolute continuity.

Evidence A. Hard: S26-D-00002066. Boundary: The Cantor function is nonconstant with derivative zero almost everywhere.

S26-D-00002068 - One-dimensional coarea identity for BV

For suitable BV functions on an interval, total variation can be represented by integrating the counting measure of level crossings.

Evidence A. Hard: S26-N-00002067.

S26-B-00002069 - BV to measure bridge

Distributional derivatives of BV functions are finite signed Radon measures, connecting real-variable regularity to measure theory.

Evidence A. Hard: S26-D-00002068.

S26-B-00002070 - Differentiation to optimization bridge

Generalized one-sided derivatives and BV/AC distinctions control when local slope information legitimately supports optimization or dynamical conclusions.

Evidence A. Hard: S26-B-00002069.

S26-E-00002071 - Central-difference profile h=6.813e-02

For f(x)=sin x at x=1 and h=6.813e-02, the centered quotient is 0.539884426003; absolute error versus cos(1) is 4.179e-04.

Evidence M. Hard: S26-B-00002070.

S26-E-00002072 - Central-difference profile h=4.642e-02

For f(x)=sin x at x=1 and h=4.642e-02, the centered quotient is 0.540108319094; absolute error versus cos(1) is 1.940e-04.

Evidence M. Hard: S26-E-00002071.

S26-E-00002073 - Central-difference profile h=3.162e-02

For f(x)=sin x at x=1 and h=3.162e-02, the centered quotient is 0.540212259986; absolute error versus cos(1) is 9.005e-05.

Evidence M. Hard: S26-E-00002072.

S26-E-00002074 - Central-difference profile h=2.154e-02

For f(x)=sin x at x=1 and h=2.154e-02, the centered quotient is 0.540260509152; absolute error versus cos(1) is 4.180e-05.

Evidence M. Hard: S26-E-00002073.

S26-E-00002075 - Central-difference profile h=1.468e-02

For f(x)=sin x at x=1 and h=1.468e-02, the centered quotient is 0.540282905310; absolute error versus cos(1) is 1.940e-05.

Evidence M. Hard: S26-E-00002074.

S26-E-00002076 - Central-difference profile h=1.000e-02

For f(x)=sin x at x=1 and h=1.000e-02, the centered quotient is 0.540293300875; absolute error versus cos(1) is 9.005e-06.

Evidence M. Hard: S26-E-00002075.

S26-E-00002077 - Central-difference profile h=6.813e-03

For f(x)=sin x at x=1 and h=6.813e-03, the centered quotient is 0.540298126109; absolute error versus cos(1) is 4.180e-06.

Evidence M. Hard: S26-E-00002076.

S26-E-00002078 - Central-difference profile h=4.642e-03

For f(x)=sin x at x=1 and h=4.642e-03, the centered quotient is 0.540300365794; absolute error versus cos(1) is 1.940e-06.

Evidence M. Hard: S26-E-00002077.

S26-E-00002079 - Central-difference profile h=3.162e-03

For f(x)=sin x at x=1 and h=3.162e-03, the centered quotient is 0.540301405365; absolute error versus cos(1) is 9.005e-07.

Evidence M. Hard: S26-E-00002078.

S26-E-00002080 - Central-difference profile h=2.154e-03

For f(x)=sin x at x=1 and h=2.154e-03, the centered quotient is 0.540301887891; absolute error versus cos(1) is 4.180e-07.

Evidence M. Hard: S26-E-00002079.

S26-E-00002081 - Central-difference profile h=1.468e-03

For f(x)=sin x at x=1 and h=1.468e-03, the centered quotient is 0.540302111860; absolute error versus cos(1) is 1.940e-07.

Evidence M. Hard: S26-E-00002080.

S26-E-00002082 - Central-difference profile h=1.000e-03

For f(x)=sin x at x=1 and h=1.000e-03, the centered quotient is 0.540302215818; absolute error versus cos(1) is 9.005e-08.

Evidence M. Hard: S26-E-00002081.

S26-E-00002083 - Central-difference profile h=6.813e-04

For f(x)=sin x at x=1 and h=6.813e-04, the centered quotient is 0.540302264070; absolute error versus cos(1) is 4.180e-08.

Evidence M. Hard: S26-E-00002082.

S26-E-00002084 - Central-difference profile h=4.642e-04

For f(x)=sin x at x=1 and h=4.642e-04, the centered quotient is 0.540302286467; absolute error versus cos(1) is 1.940e-08.

Evidence M. Hard: S26-E-00002083.

S26-E-00002085 - Central-difference profile h=3.162e-04

For f(x)=sin x at x=1 and h=3.162e-04, the centered quotient is 0.540302296863; absolute error versus cos(1) is 9.005e-09.

Evidence M. Hard: S26-E-00002084.

S26-E-00002086 - Central-difference profile h=2.154e-04

For f(x)=sin x at x=1 and h=2.154e-04, the centered quotient is 0.540302301688; absolute error versus cos(1) is 4.180e-09.

Evidence M. Hard: S26-E-00002085.

S26-E-00002087 - Central-difference profile h=1.468e-04

For f(x)=sin x at x=1 and h=1.468e-04, the centered quotient is 0.540302303928; absolute error versus cos(1) is 1.940e-09.

Evidence M. Hard: S26-E-00002086.

S26-E-00002088 - Central-difference profile h=1.000e-04

For f(x)=sin x at x=1 and h=1.000e-04, the centered quotient is 0.540302304968; absolute error versus cos(1) is 9.004e-10.

Evidence M. Hard: S26-E-00002087.

S26-E-00002089 - Central-difference profile h=6.813e-05

For f(x)=sin x at x=1 and h=6.813e-05, the centered quotient is 0.540302305450; absolute error versus cos(1) is 4.178e-10.

Evidence M. Hard: S26-E-00002088.

S26-E-00002090 - Central-difference profile h=4.642e-05

For f(x)=sin x at x=1 and h=4.642e-05, the centered quotient is 0.540302305674; absolute error versus cos(1) is 1.939e-10.

Evidence M. Hard: S26-E-00002089.

S26-E-00002091 - Central-difference profile h=3.162e-05

For f(x)=sin x at x=1 and h=3.162e-05, the centered quotient is 0.540302305779; absolute error versus cos(1) is 8.893e-11.

Evidence M. Hard: S26-E-00002090.

S26-E-00002092 - Central-difference profile h=2.154e-05

For f(x)=sin x at x=1 and h=2.154e-05, the centered quotient is 0.540302305827; absolute error versus cos(1) is 4.150e-11.

Evidence M. Hard: S26-E-00002091.

S26-E-00002093 - Central-difference profile h=1.468e-05

For f(x)=sin x at x=1 and h=1.468e-05, the centered quotient is 0.540302305848; absolute error versus cos(1) is 2.012e-11.

Evidence M. Hard: S26-E-00002092.

S26-E-00002094 - Central-difference profile h=1.000e-05

For f(x)=sin x at x=1 and h=1.000e-05, the centered quotient is 0.540302305857; absolute error versus cos(1) is 1.114e-11.

Evidence M. Hard: S26-E-00002093.

S26-E-00002095 - Central-difference profile h=6.813e-06

For f(x)=sin x at x=1 and h=6.813e-06, the centered quotient is 0.540302305857; absolute error versus cos(1) is 1.124e-11.

Evidence M. Hard: S26-E-00002094.

S26-E-00002096 - Central-difference profile h=4.642e-06

For f(x)=sin x at x=1 and h=4.642e-06, the centered quotient is 0.540302305856; absolute error versus cos(1) is 1.212e-11.

Evidence M. Hard: S26-E-00002095.

S26-E-00002097 - Central-difference profile h=3.162e-06

For f(x)=sin x at x=1 and h=3.162e-06, the centered quotient is 0.540302305869; absolute error versus cos(1) is 6.005e-13.

Evidence M. Hard: S26-E-00002096.

S26-E-00002098 - Central-difference profile h=2.154e-06

For f(x)=sin x at x=1 and h=2.154e-06, the centered quotient is 0.540302305832; absolute error versus cos(1) is 3.634e-11.

Evidence M. Hard: S26-E-00002097.

S26-E-00002099 - Central-difference profile h=1.468e-06

For f(x)=sin x at x=1 and h=1.468e-06, the centered quotient is 0.540302305871; absolute error versus cos(1) is 2.570e-12.

Evidence M. Hard: S26-E-00002098.

S26-E-00002100 - Central-difference profile h=1.000e-06

For f(x)=sin x at x=1 and h=1.000e-06, the centered quotient is 0.540302305896; absolute error versus cos(1) is 2.772e-11.

Evidence M. Hard: S26-E-00002099.

Part C - Convex analysis and inequalities

S26-D-00002101 - Strictly convex function

A convex function is strictly convex when the convexity inequality is strict for distinct endpoints and weights in (0,1).

Evidence A. Hard: S26-E-00002100.

S26-D-00002102 - Uniformly convex function

A convex function is uniformly convex when midpoint deficits admit a positive modulus depending only on separation.

Evidence A. Hard: S26-D-00002101.

S26-D-00002103 - Strongly convex differentiable function

f is m-strongly convex when f(y)>=f(x)+<grad f(x),y-x>+(m/2)||y-x||^2.

Evidence A. Hard: S26-D-00002102.

S26-D-00002104 - Convex conjugate

f*(y)=sup_x(<x,y>-f(x)) is the Legendre-Fenchel conjugate.

Evidence A. Hard: S26-D-00002103.

S26-T-00002105 - Fenchel-Young inequality

For proper convex f, <x,y><=f(x)+f*(y).

Evidence A. Hard: S26-D-00002104.

S26-T-00002106 - Fenchel-Moreau biconjugation

A proper lower-semicontinuous convex function on a suitable locally convex space equals its biconjugate.

Evidence A. Hard: S26-T-00002105.

S26-D-00002107 - Subgradient of a convex function

g is a subgradient of f at x when f(y)>=f(x)+<g,y-x> for all y.

Evidence A. Hard: S26-T-00002106.

S26-D-00002108 - Subdifferential

partial f(x) is the set of all subgradients at x.

Evidence A. Hard: S26-D-00002107.

S26-T-00002109 - Subgradient optimality criterion

For proper convex f, 0 in partial f(x) is equivalent to x being a global minimizer.

Evidence A. Hard: S26-D-00002108.

S26-D-00002110 - Bregman divergence

D_f(x,y)=f(x)-f(y)-<grad f(y),x-y> for differentiable convex f; it is generally not symmetric and need not satisfy the triangle inequality.

Evidence A. Hard: S26-T-00002109.

S26-T-00002111 - Young inequality with conjugate exponents

For a,b>=0 and p,q>1 with 1/p+1/q=1, ab<=a^p/p+b^q/q.

Evidence A. Hard: S26-D-00002110.

S26-T-00002112 - Weighted AM-GM inequality

For positive x_i and weights lambda_i>=0 summing to one, product x_i^{lambda_i}<=sum lambda_i x_i.

Evidence A. Hard: S26-T-00002111.

S26-T-00002113 - Log-sum inequality

For positive a_i,b_i, sum a_i log(a_i/b_i)>=(sum a_i)log((sum a_i)/(sum b_i)).

Evidence A. Hard: S26-T-00002112.

S26-D-00002114 - Log-convex function

A positive function f is log-convex when log f is convex.

Evidence A. Hard: S26-T-00002113.

S26-T-00002115 - Three-lines theorem

A bounded holomorphic function on a vertical strip has logarithmically convex sup norms across intermediate vertical lines under the standard boundary hypotheses.

Evidence A. Hard: S26-D-00002114.

S26-D-00002116 - Modulus of convexity of a normed space

delta_X(epsilon)=inf{1-||(x+y)/2||:||x||,||y||<=1, ||x-y||>=epsilon}.

Evidence A. Hard: S26-T-00002115.

S26-D-00002117 - Uniform convexity of a Banach space

X is uniformly convex when delta_X(epsilon)>0 for every epsilon>0.

Evidence A. Hard: S26-D-00002116.

S26-T-00002118 - Uniform convexity implies reflexivity

Every uniformly convex Banach space is reflexive.

Evidence A. Hard: S26-D-00002117.

S26-N-00002119 - Convexity smoothness fence

Strict convexity, strong convexity, smoothness, and uniform convexity are distinct properties and cannot be substituted without explicit hypotheses.

Evidence A. Hard: S26-T-00002118.

S26-B-00002120 - Convex duality bridge

Conjugates and subdifferentials connect inequalities, variational problems, optimization duality, monotone operators and PDE energy methods.

Evidence A. Hard: S26-N-00002119.

S26-E-00002121 - Young-inequality profile 1

With p=1.200, q=6.000, a=0.370, b=1.680: ab=0.621600000 and a^p/p+b^q/q=3.999911346; gap=3.378e+00.

Evidence M. Hard: S26-B-00002120.

S26-E-00002122 - Young-inequality profile 2

With p=1.300, q=4.333, a=0.440, b=1.660: ab=0.730400000 and a^p/p+b^q/q=2.339388897; gap=1.609e+00.

Evidence M. Hard: S26-E-00002121.

S26-E-00002123 - Young-inequality profile 3

With p=1.400, q=3.500, a=0.510, b=1.640: ab=0.836400000 and a^p/p+b^q/q=1.892205160; gap=1.056e+00.

Evidence M. Hard: S26-E-00002122.

S26-E-00002124 - Young-inequality profile 4

With p=1.500, q=3.000, a=0.580, b=1.620: ab=0.939600000 and a^p/p+b^q/q=1.711652560; gap=7.721e-01.

Evidence M. Hard: S26-E-00002123.

S26-E-00002125 - Young-inequality profile 5

With p=1.600, q=2.667, a=0.650, b=1.600: ab=1.040000000 and a^p/p+b^q/q=1.626980936; gap=5.870e-01.

Evidence M. Hard: S26-E-00002124.

S26-E-00002126 - Young-inequality profile 6

With p=1.700, q=2.429, a=0.720, b=1.580: ab=1.137600000 and a^p/p+b^q/q=1.587077049; gap=4.495e-01.

Evidence M. Hard: S26-E-00002125.

S26-E-00002127 - Young-inequality profile 7

With p=1.800, q=2.250, a=0.790, b=1.560: ab=1.232400000 and a^p/p+b^q/q=1.572241262; gap=3.398e-01.

Evidence M. Hard: S26-E-00002126.

S26-E-00002128 - Young-inequality profile 8

With p=1.900, q=2.111, a=0.860, b=1.540: ab=1.324400000 and a^p/p+b^q/q=1.573777411; gap=2.494e-01.

Evidence M. Hard: S26-E-00002127.

S26-E-00002129 - Young-inequality profile 9

With p=2.000, q=2.000, a=0.930, b=1.520: ab=1.413600000 and a^p/p+b^q/q=1.587650000; gap=1.741e-01.

Evidence M. Hard: S26-E-00002128.

S26-E-00002130 - Young-inequality profile 10

With p=2.100, q=1.909, a=1.000, b=1.500: ab=1.500000000 and a^p/p+b^q/q=1.612110126; gap=1.121e-01.

Evidence M. Hard: S26-E-00002129.

S26-E-00002131 - Young-inequality profile 11

With p=2.200, q=1.833, a=1.070, b=1.480: ab=1.583600000 and a^p/p+b^q/q=1.646692137; gap=6.309e-02.

Evidence M. Hard: S26-E-00002130.

S26-E-00002132 - Young-inequality profile 12

With p=2.300, q=1.769, a=1.140, b=1.460: ab=1.664400000 and a^p/p+b^q/q=1.691759261; gap=2.736e-02.

Evidence M. Hard: S26-E-00002131.

S26-E-00002133 - Young-inequality profile 13

With p=2.400, q=1.714, a=1.210, b=1.440: ab=1.742400000 and a^p/p+b^q/q=1.748297310; gap=5.897e-03.

Evidence M. Hard: S26-E-00002132.

S26-E-00002134 - Young-inequality profile 14

With p=2.500, q=1.667, a=1.280, b=1.420: ab=1.817600000 and a^p/p+b^q/q=1.817834055; gap=2.341e-04.

Evidence M. Hard: S26-E-00002133.

S26-E-00002135 - Young-inequality profile 15

With p=2.600, q=1.625, a=1.350, b=1.400: ab=1.890000000 and a^p/p+b^q/q=1.902430836; gap=1.243e-02.

Evidence M. Hard: S26-E-00002134.

S26-E-00002136 - Young-inequality profile 16

With p=2.700, q=1.588, a=1.420, b=1.380: ab=1.959600000 and a^p/p+b^q/q=2.004722590; gap=4.512e-02.

Evidence M. Hard: S26-E-00002135.

S26-E-00002137 - Young-inequality profile 17

With p=2.800, q=1.556, a=1.490, b=1.360: ab=2.026400000 and a^p/p+b^q/q=2.127996415; gap=1.016e-01.

Evidence M. Hard: S26-E-00002136.

S26-E-00002138 - Young-inequality profile 18

With p=2.900, q=1.526, a=1.560, b=1.340: ab=2.090400000 and a^p/p+b^q/q=2.276306197; gap=1.859e-01.

Evidence M. Hard: S26-E-00002137.

S26-E-00002139 - Young-inequality profile 19

With p=3.000, q=1.500, a=1.630, b=1.320: ab=2.151600000 and a^p/p+b^q/q=2.454625359; gap=3.030e-01.

Evidence M. Hard: S26-E-00002138.

S26-E-00002140 - Young-inequality profile 20

With p=3.100, q=1.476, a=1.700, b=1.300: ab=2.210000000 and a^p/p+b^q/q=2.669043113; gap=4.590e-01.

Evidence M. Hard: S26-E-00002139.

S26-E-00002141 - Young-inequality profile 21

With p=3.200, q=1.455, a=1.770, b=1.280: ab=2.265600000 and a^p/p+b^q/q=2.927012596; gap=6.614e-01.

Evidence M. Hard: S26-E-00002140.

S26-E-00002142 - Young-inequality profile 22

With p=3.300, q=1.435, a=1.840, b=1.260: ab=2.318400000 and a^p/p+b^q/q=3.237662385; gap=9.193e-01.

Evidence M. Hard: S26-E-00002141.

S26-E-00002143 - Young-inequality profile 23

With p=3.400, q=1.417, a=1.910, b=1.240: ab=2.368400000 and a^p/p+b^q/q=3.612186418; gap=1.244e+00.

Evidence M. Hard: S26-E-00002142.

S26-E-00002144 - Young-inequality profile 24

With p=3.500, q=1.400, a=1.980, b=1.220: ab=2.415600000 and a^p/p+b^q/q=4.064331691; gap=1.649e+00.

Evidence M. Hard: S26-E-00002143.

S26-E-00002145 - Young-inequality profile 25

With p=3.600, q=1.385, a=2.050, b=1.200: ab=2.460000000 and a^p/p+b^q/q=4.611008416; gap=2.151e+00.

Evidence M. Hard: S26-E-00002144.

S26-E-00002146 - Young-inequality profile 26

With p=3.700, q=1.370, a=2.120, b=1.180: ab=2.501600000 and a^p/p+b^q/q=5.273054040; gap=2.771e+00.

Evidence M. Hard: S26-E-00002145.

S26-E-00002147 - Young-inequality profile 27

With p=3.800, q=1.357, a=2.190, b=1.160: ab=2.540400000 and a^p/p+b^q/q=6.076191064; gap=3.536e+00.

Evidence M. Hard: S26-E-00002146.

S26-E-00002148 - Young-inequality profile 28

With p=3.900, q=1.345, a=2.260, b=1.140: ab=2.576400000 and a^p/p+b^q/q=7.052229394; gap=4.476e+00.

Evidence M. Hard: S26-E-00002147.

S26-E-00002149 - Young-inequality profile 29

With p=4.000, q=1.333, a=2.330, b=1.120: ab=2.609600000 and a^p/p+b^q/q=8.240577812; gap=5.631e+00.

Evidence M. Hard: S26-E-00002148.

S26-E-00002150 - Young-inequality profile 30

With p=4.100, q=1.323, a=2.400, b=1.100: ab=2.640000000 and a^p/p+b^q/q=9.690146781; gap=7.050e+00.

Evidence M. Hard: S26-E-00002149.

Part D - Signed, Radon and weak measure theory

S26-D-00002151 - Signed measure

A signed measure is a countably additive map into the extended reals that does not take both +infinity and -infinity.

Evidence A. Hard: S26-E-00002150.

S26-D-00002152 - Positive set for a signed measure

A measurable set P is positive for nu when every measurable E subset P has nu(E)>=0.

Evidence A. Hard: S26-D-00002151.

S26-T-00002153 - Hahn decomposition theorem

Every signed measure space decomposes into disjoint positive and negative measurable sets, unique up to nu-null sets.

Evidence A. Hard: S26-D-00002152.

S26-T-00002154 - Jordan decomposition theorem for signed measures

Every signed measure nu can be written uniquely as nu=nu^+-nu^- with mutually singular positive measures.

Evidence A. Hard: S26-T-00002153.

S26-D-00002155 - Total variation measure

|nu|=nu^++nu^- is the total variation measure of a signed measure.

Evidence A. Hard: S26-T-00002154.

S26-D-00002156 - Absolute continuity of measures

nu<<mu means mu(E)=0 implies nu(E)=0 for every measurable E.

Evidence A. Hard: S26-D-00002155.

S26-D-00002157 - Mutually singular measures

nu perpendicular mu means the measures are supported on disjoint measurable pieces up to null sets.

Evidence A. Hard: S26-D-00002156.

S26-T-00002158 - Radon-Nikodym theorem

For sigma-finite positive measures mu and nu with nu<<mu, there exists a nonnegative measurable density f, unique mu-a.e., such that nu(E)=integral_E f dmu for every measurable E; if nu is finite, then f belongs to L1(mu).

Evidence A. Hard: S26-D-00002157. Peer-review correction applied in reviewed projection.

S26-T-00002159 - Lebesgue decomposition of measures

Under standard sigma-finiteness hypotheses, nu decomposes uniquely into a part absolutely continuous with respect to mu and a part singular to mu.

Evidence A. Hard: S26-T-00002158.

S26-D-00002160 - Radon measure on a locally compact space

A Radon measure is locally finite and inner regular, with the precise regularity convention declared for the ambient topological setting.

Evidence A. Hard: S26-T-00002159.

S26-D-00002161 - Support of a measure

supp(mu) is the complement of the largest open set of mu-measure zero.

Evidence A. Hard: S26-D-00002160.

S26-D-00002162 - Pushforward density transformation

When a smooth injective change of variables is valid, densities transform by the absolute Jacobian determinant.

Evidence A. Hard: S26-D-00002161.

S26-T-00002163 - Change-of-variables theorem for Lebesgue integrals

For an appropriate C1 diffeomorphism Phi, integral_{Phi(U)}g(y)dy=integral_U g(Phi(x))|det D Phi(x)|dx.

Evidence A. Hard: S26-D-00002162.

S26-D-00002164 - Weak convergence of finite measures

mu_n converges weakly to mu when integrals against bounded continuous test functions converge.

Evidence A. Hard: S26-T-00002163.

S26-D-00002165 - Vague convergence of Radon measures

Vague convergence tests against compactly supported continuous functions.

Evidence A. Hard: S26-D-00002164.

S26-D-00002166 - Tight family of probability measures

A family is tight when for every epsilon>0 there is a compact K carrying mass at least 1-epsilon for every measure in the family.

Evidence A. Hard: S26-D-00002165.

S26-T-00002167 - Prokhorov compactness interface

On Polish spaces, tightness of probability measures is equivalent to relative compactness for weak convergence.

Evidence A. Hard: S26-D-00002166.

S26-N-00002168 - Density representation fence

Absolute continuity and a density are relative to a reference measure; a singular measure need not possess a Lebesgue density.

Evidence A. Hard: S26-T-00002167.

S26-B-00002169 - Measure duality bridge

Radon measures act as continuous linear functionals on spaces of continuous compactly supported functions under the appropriate representation theorem.

Evidence A. Hard: S26-N-00002168.

S26-B-00002170 - Weak-measure convergence bridge

Weak convergence of measures links analysis to probability, empirical distributions, PDE compactness and variational limits.

Evidence A. Hard: S26-B-00002169.

S26-E-00002171 - Finite probability-measure profile n=3

On atoms 0,...,2 with weights proportional to 1,...,3, total mass is 1.000000000000 and the exact floating mean is 1.333333333333.

Evidence M. Hard: S26-B-00002170.

S26-E-00002172 - Finite probability-measure profile n=4

On atoms 0,...,3 with weights proportional to 1,...,4, total mass is 1.000000000000 and the exact floating mean is 2.000000000000.

Evidence M. Hard: S26-E-00002171.

S26-E-00002173 - Finite probability-measure profile n=5

On atoms 0,...,4 with weights proportional to 1,...,5, total mass is 1.000000000000 and the exact floating mean is 2.666666666667.

Evidence M. Hard: S26-E-00002172.

S26-E-00002174 - Finite probability-measure profile n=6

On atoms 0,...,5 with weights proportional to 1,...,6, total mass is 1.000000000000 and the exact floating mean is 3.333333333333.

Evidence M. Hard: S26-E-00002173.

S26-E-00002175 - Finite probability-measure profile n=7

On atoms 0,...,6 with weights proportional to 1,...,7, total mass is 1.000000000000 and the exact floating mean is 4.000000000000.

Evidence M. Hard: S26-E-00002174.

S26-E-00002176 - Finite probability-measure profile n=8

On atoms 0,...,7 with weights proportional to 1,...,8, total mass is 1.000000000000 and the exact floating mean is 4.666666666667.

Evidence M. Hard: S26-E-00002175.

S26-E-00002177 - Finite probability-measure profile n=9

On atoms 0,...,8 with weights proportional to 1,...,9, total mass is 1.000000000000 and the exact floating mean is 5.333333333333.

Evidence M. Hard: S26-E-00002176.

S26-E-00002178 - Finite probability-measure profile n=10

On atoms 0,...,9 with weights proportional to 1,...,10, total mass is 1.000000000000 and the exact floating mean is 6.000000000000.

Evidence M. Hard: S26-E-00002177.

S26-E-00002179 - Finite probability-measure profile n=11

On atoms 0,...,10 with weights proportional to 1,...,11, total mass is 1.000000000000 and the exact floating mean is 6.666666666667.

Evidence M. Hard: S26-E-00002178.

S26-E-00002180 - Finite probability-measure profile n=12

On atoms 0,...,11 with weights proportional to 1,...,12, total mass is 1.000000000000 and the exact floating mean is 7.333333333333.

Evidence M. Hard: S26-E-00002179.

S26-E-00002181 - Finite probability-measure profile n=13

On atoms 0,...,12 with weights proportional to 1,...,13, total mass is 1.000000000000 and the exact floating mean is 8.000000000000.

Evidence M. Hard: S26-E-00002180.

S26-E-00002182 - Finite probability-measure profile n=14

On atoms 0,...,13 with weights proportional to 1,...,14, total mass is 1.000000000000 and the exact floating mean is 8.666666666667.

Evidence M. Hard: S26-E-00002181.

S26-E-00002183 - Finite probability-measure profile n=15

On atoms 0,...,14 with weights proportional to 1,...,15, total mass is 1.000000000000 and the exact floating mean is 9.333333333333.

Evidence M. Hard: S26-E-00002182.

S26-E-00002184 - Finite probability-measure profile n=16

On atoms 0,...,15 with weights proportional to 1,...,16, total mass is 1.000000000000 and the exact floating mean is 10.000000000000.

Evidence M. Hard: S26-E-00002183.

S26-E-00002185 - Finite probability-measure profile n=17

On atoms 0,...,16 with weights proportional to 1,...,17, total mass is 1.000000000000 and the exact floating mean is 10.666666666667.

Evidence M. Hard: S26-E-00002184.

S26-E-00002186 - Finite probability-measure profile n=18

On atoms 0,...,17 with weights proportional to 1,...,18, total mass is 1.000000000000 and the exact floating mean is 11.333333333333.

Evidence M. Hard: S26-E-00002185.

S26-E-00002187 - Finite probability-measure profile n=19

On atoms 0,...,18 with weights proportional to 1,...,19, total mass is 1.000000000000 and the exact floating mean is 12.000000000000.

Evidence M. Hard: S26-E-00002186.

S26-E-00002188 - Finite probability-measure profile n=20

On atoms 0,...,19 with weights proportional to 1,...,20, total mass is 1.000000000000 and the exact floating mean is 12.666666666667.

Evidence M. Hard: S26-E-00002187.

S26-E-00002189 - Finite probability-measure profile n=21

On atoms 0,...,20 with weights proportional to 1,...,21, total mass is 1.000000000000 and the exact floating mean is 13.333333333333.

Evidence M. Hard: S26-E-00002188.

S26-E-00002190 - Finite probability-measure profile n=22

On atoms 0,...,21 with weights proportional to 1,...,22, total mass is 1.000000000000 and the exact floating mean is 14.000000000000.

Evidence M. Hard: S26-E-00002189.

S26-E-00002191 - Finite probability-measure profile n=23

On atoms 0,...,22 with weights proportional to 1,...,23, total mass is 1.000000000000 and the exact floating mean is 14.666666666667.

Evidence M. Hard: S26-E-00002190.

S26-E-00002192 - Finite probability-measure profile n=24

On atoms 0,...,23 with weights proportional to 1,...,24, total mass is 1.000000000000 and the exact floating mean is 15.333333333333.

Evidence M. Hard: S26-E-00002191.

S26-E-00002193 - Finite probability-measure profile n=25

On atoms 0,...,24 with weights proportional to 1,...,25, total mass is 1.000000000000 and the exact floating mean is 16.000000000000.

Evidence M. Hard: S26-E-00002192.

S26-E-00002194 - Finite probability-measure profile n=26

On atoms 0,...,25 with weights proportional to 1,...,26, total mass is 1.000000000000 and the exact floating mean is 16.666666666667.

Evidence M. Hard: S26-E-00002193.

S26-E-00002195 - Finite probability-measure profile n=27

On atoms 0,...,26 with weights proportional to 1,...,27, total mass is 1.000000000000 and the exact floating mean is 17.333333333333.

Evidence M. Hard: S26-E-00002194.

S26-E-00002196 - Finite probability-measure profile n=28

On atoms 0,...,27 with weights proportional to 1,...,28, total mass is 1.000000000000 and the exact floating mean is 18.000000000000.

Evidence M. Hard: S26-E-00002195.

S26-E-00002197 - Finite probability-measure profile n=29

On atoms 0,...,28 with weights proportional to 1,...,29, total mass is 1.000000000000 and the exact floating mean is 18.666666666667.

Evidence M. Hard: S26-E-00002196.

S26-E-00002198 - Finite probability-measure profile n=30

On atoms 0,...,29 with weights proportional to 1,...,30, total mass is 1.000000000000 and the exact floating mean is 19.333333333333.

Evidence M. Hard: S26-E-00002197.

S26-E-00002199 - Finite probability-measure profile n=31

On atoms 0,...,30 with weights proportional to 1,...,31, total mass is 1.000000000000 and the exact floating mean is 20.000000000000.

Evidence M. Hard: S26-E-00002198.

S26-E-00002200 - Finite probability-measure profile n=32

On atoms 0,...,31 with weights proportional to 1,...,32, total mass is 1.000000000000 and the exact floating mean is 20.666666666667.

Evidence M. Hard: S26-E-00002199.

Part E - Lp refinements, rearrangement and interpolation

S26-D-00002201 - Essential supremum norm

||f||_infinity is the infimum of M such that |f|<=M almost everywhere.

Evidence A. Hard: S26-E-00002200.

S26-D-00002202 - Weak Lp space

L^{p,infinity} consists of measurable f with sup_{lambda>0} lambda mu(|f|>lambda)^{1/p}<infinity.

Evidence A. Hard: S26-D-00002201.

S26-D-00002203 - Lorentz space interface

Lorentz spaces L^{p,q} refine Lp by integrating the decreasing rearrangement with a second exponent q.

Evidence A. Hard: S26-D-00002202.

S26-D-00002204 - Decreasing rearrangement

f*(t) is the right-continuous decreasing rearrangement determined by the distribution function of |f|.

Evidence A. Hard: S26-D-00002203.

S26-T-00002205 - Hardy-Littlewood rearrangement inequality

For nonnegative measurable f,g, integral fg is bounded above by integral f* g* under the standard measure-space formulation.

Evidence A. Hard: S26-D-00002204.

S26-T-00002206 - Generalized Holder inequality

If sum_i 1/p_i=1, then ||product_i f_i||_1<=product_i ||f_i||_{p_i}.

Evidence A. Hard: S26-T-00002205.

S26-T-00002207 - Minkowski integral inequality

Under standard measurability and integrability hypotheses, an Lp norm may be moved outside an integral with the Minkowski bound for p>=1.

Evidence A. Hard: S26-T-00002206.

S26-D-00002208 - Interpolation couple

An interpolation couple is a pair of compatible Banach spaces continuously embedded in a common Hausdorff topological vector space.

Evidence A. Hard: S26-T-00002207.

S26-T-00002209 - Riesz-Thorin interpolation theorem

A linear operator bounded on two Lp endpoint pairs is bounded on the interpolated exponent pairs with logarithmically interpolated norm bound.

Evidence A. Hard: S26-D-00002208.

S26-T-00002210 - Marcinkiewicz interpolation theorem

Sublinear weak-type endpoint bounds interpolate to strong-type interior Lp bounds under the standard exponent hypotheses.

Evidence A. Hard: S26-T-00002209.

S26-D-00002211 - Uniform integrability

A family in L1 is uniformly integrable when large values contribute uniformly small L1 mass.

Evidence A. Hard: S26-T-00002210.

S26-T-00002212 - Vitali convergence theorem

Convergence in measure plus uniform integrability yields L1 convergence on finite-measure spaces under standard hypotheses.

Evidence A. Hard: S26-D-00002211.

S26-D-00002213 - Reflexive Lp regime

For 1<p<infinity, Lp is reflexive.

Evidence A. Hard: S26-T-00002212.

S26-D-00002214 - Dual exponent

For 1<p<infinity, p'=p/(p-1) is the Holder conjugate exponent.

Evidence A. Hard: S26-D-00002213.

S26-T-00002215 - Lp duality on sigma-finite spaces

For 1<p<infinity under standard measure hypotheses, every continuous linear functional on Lp is integration against a unique L^{p'} function.

Evidence A. Hard: S26-D-00002214.

S26-D-00002216 - Orlicz function

A Young function is convex, increasing from zero and diverges to infinity, generating an Orlicz integrability scale.

Evidence A. Hard: S26-T-00002215.

S26-D-00002217 - Orlicz space interface

An Orlicz space replaces power growth |f|^p by a Young function Phi(|f|) in the integrability criterion.

Evidence A. Hard: S26-D-00002216.

S26-N-00002218 - Endpoint interpolation fence

Interpolation conclusions at p=1 or p=infinity often require separate weak-type or endpoint arguments and cannot be inferred mechanically from interior formulas.

Evidence A. Hard: S26-D-00002217.

S26-B-00002219 - Lp to harmonic-analysis bridge

Weak Lp, Lorentz and interpolation spaces encode endpoint behavior of maximal, singular-integral and Fourier operators.

Evidence A. Hard: S26-N-00002218.

S26-B-00002220 - Uniform-integrability bridge

Uniform integrability converts convergence in measure into norm convergence and is a central compactness mechanism in probability and PDE.

Evidence A. Hard: S26-B-00002219.

S26-E-00002221 - Finite lp-norm profile n=3, p=1.15

For v=(1,...,3), ||v||_1.15=5.266253703961 and ||v||_infinity=3.

Evidence M. Hard: S26-B-00002220.

S26-E-00002222 - Finite lp-norm profile n=4, p=1.30

For v=(1,...,4), ||v||_1.30=7.487050309871 and ||v||_infinity=4.

Evidence M. Hard: S26-E-00002221.

S26-E-00002223 - Finite lp-norm profile n=5, p=1.45

For v=(1,...,5), ||v||_1.45=9.566741833882 and ||v||_infinity=5.

Evidence M. Hard: S26-E-00002222.

S26-E-00002224 - Finite lp-norm profile n=6, p=1.60

For v=(1,...,6), ||v||_1.60=11.492357851814 and ||v||_infinity=6.

Evidence M. Hard: S26-E-00002223.

S26-E-00002225 - Finite lp-norm profile n=7, p=1.75

For v=(1,...,7), ||v||_1.75=13.278104237489 and ||v||_infinity=7.

Evidence M. Hard: S26-E-00002224.

S26-E-00002226 - Finite lp-norm profile n=8, p=1.90

For v=(1,...,8), ||v||_1.90=14.944705402164 and ||v||_infinity=8.

Evidence M. Hard: S26-E-00002225.

S26-E-00002227 - Finite lp-norm profile n=9, p=2.05

For v=(1,...,9), ||v||_2.05=16.512266295892 and ||v||_infinity=9.

Evidence M. Hard: S26-E-00002226.

S26-E-00002228 - Finite lp-norm profile n=10, p=2.20

For v=(1,...,10), ||v||_2.20=17.998172269504 and ||v||_infinity=10.

Evidence M. Hard: S26-E-00002227.

S26-E-00002229 - Finite lp-norm profile n=11, p=2.35

For v=(1,...,11), ||v||_2.35=19.416799575465 and ||v||_infinity=11.

Evidence M. Hard: S26-E-00002228.

S26-E-00002230 - Finite lp-norm profile n=12, p=2.50

For v=(1,...,12), ||v||_2.50=20.779820602243 and ||v||_infinity=12.

Evidence M. Hard: S26-E-00002229.

S26-E-00002231 - Finite lp-norm profile n=13, p=2.65

For v=(1,...,13), ||v||_2.65=22.096652593901 and ||v||_infinity=13.

Evidence M. Hard: S26-E-00002230.

S26-E-00002232 - Finite lp-norm profile n=14, p=2.80

For v=(1,...,14), ||v||_2.80=23.374890814190 and ||v||_infinity=14.

Evidence M. Hard: S26-E-00002231.

S26-E-00002233 - Finite lp-norm profile n=15, p=2.95

For v=(1,...,15), ||v||_2.95=24.620678492252 and ||v||_infinity=15.

Evidence M. Hard: S26-E-00002232.

S26-E-00002234 - Finite lp-norm profile n=16, p=3.10

For v=(1,...,16), ||v||_3.10=25.839007140489 and ||v||_infinity=16.

Evidence M. Hard: S26-E-00002233.

S26-E-00002235 - Finite lp-norm profile n=17, p=3.25

For v=(1,...,17), ||v||_3.25=27.033954750223 and ||v||_infinity=17.

Evidence M. Hard: S26-E-00002234.

S26-E-00002236 - Finite lp-norm profile n=18, p=3.40

For v=(1,...,18), ||v||_3.40=28.208872775623 and ||v||_infinity=18.

Evidence M. Hard: S26-E-00002235.

S26-E-00002237 - Finite lp-norm profile n=19, p=3.55

For v=(1,...,19), ||v||_3.55=29.366532399636 and ||v||_infinity=19.

Evidence M. Hard: S26-E-00002236.

S26-E-00002238 - Finite lp-norm profile n=20, p=3.70

For v=(1,...,20), ||v||_3.70=30.509238984748 and ||v||_infinity=20.

Evidence M. Hard: S26-E-00002237.

S26-E-00002239 - Finite lp-norm profile n=21, p=3.85

For v=(1,...,21), ||v||_3.85=31.638921860664 and ||v||_infinity=21.

Evidence M. Hard: S26-E-00002238.

S26-E-00002240 - Finite lp-norm profile n=22, p=4.00

For v=(1,...,22), ||v||_4.00=32.757205042625 and ||v||_infinity=22.

Evidence M. Hard: S26-E-00002239.

S26-E-00002241 - Finite lp-norm profile n=23, p=4.15

For v=(1,...,23), ||v||_4.15=33.865463197083 and ||v||_infinity=23.

Evidence M. Hard: S26-E-00002240.

S26-E-00002242 - Finite lp-norm profile n=24, p=4.30

For v=(1,...,24), ||v||_4.30=34.964866165474 and ||v||_infinity=24.

Evidence M. Hard: S26-E-00002241.

S26-E-00002243 - Finite lp-norm profile n=25, p=4.45

For v=(1,...,25), ||v||_4.45=36.056414580626 and ||v||_infinity=25.

Evidence M. Hard: S26-E-00002242.

S26-E-00002244 - Finite lp-norm profile n=26, p=4.60

For v=(1,...,26), ||v||_4.60=37.140968517555 and ||v||_infinity=26.

Evidence M. Hard: S26-E-00002243.

S26-E-00002245 - Finite lp-norm profile n=27, p=4.75

For v=(1,...,27), ||v||_4.75=38.219270669982 and ||v||_infinity=27.

Evidence M. Hard: S26-E-00002244.

S26-E-00002246 - Finite lp-norm profile n=28, p=4.90

For v=(1,...,28), ||v||_4.90=39.291965202042 and ||v||_infinity=28.

Evidence M. Hard: S26-E-00002245.

S26-E-00002247 - Finite lp-norm profile n=29, p=5.05

For v=(1,...,29), ||v||_5.05=40.359613164958 and ||v||_infinity=29.

Evidence M. Hard: S26-E-00002246.

S26-E-00002248 - Finite lp-norm profile n=30, p=5.20

For v=(1,...,30), ||v||_5.20=41.422705170734 and ||v||_infinity=30.

Evidence M. Hard: S26-E-00002247.

S26-E-00002249 - Finite lp-norm profile n=31, p=5.35

For v=(1,...,31), ||v||_5.35=42.481671863774 and ||v||_infinity=31.

Evidence M. Hard: S26-E-00002248.

S26-E-00002250 - Finite lp-norm profile n=32, p=5.50

For v=(1,...,32), ||v||_5.50=43.536892615411 and ||v||_infinity=32.

Evidence M. Hard: S26-E-00002249.

Part F - Holomorphic local theory

S26-D-00002251 - Complex differentiability at a point

f is complex differentiable at z when lim_{h->0}(f(z+h)-f(z))/h exists with h complex.

Evidence A. Hard: S26-E-00002250.

S26-D-00002252 - Holomorphic function

A function is holomorphic on an open set when it is complex differentiable at every point of that set.

Evidence A. Hard: S26-D-00002251.

S26-D-00002253 - Entire function

An entire function is holomorphic on all of C.

Evidence A. Hard: S26-D-00002252.

S26-T-00002254 - Cauchy-Riemann equations

If f=u+iv is complex differentiable with suitable real differentiability, then u_x=v_y and u_y=-v_x.

Evidence A. Hard: S26-D-00002253.

S26-T-00002255 - Holomorphic functions are analytic

Every holomorphic function has a locally convergent complex power-series expansion.

Evidence A. Hard: S26-T-00002254.

S26-T-00002256 - Cauchy integral formula

If f is holomorphic on and inside a positively oriented simple contour, f(z0)=(1/(2pi i)) integral f(z)/(z-z0) dz for interior z0.

Evidence A. Hard: S26-T-00002255.

S26-I-00002257 - Cauchy derivative formula

f^{(n)}(z0)=n!/(2pi i) integral f(z)/(z-z0)^{n+1} dz under the Cauchy hypotheses.

Evidence A. Hard: S26-T-00002256.

S26-T-00002258 - Cauchy estimates

If |f|<=M on |z-z0|=R, then |f^{(n)}(z0)|<=n! M/R^n.

Evidence A. Hard: S26-I-00002257.

S26-T-00002259 - Liouville theorem

Every bounded entire function is constant.

Evidence A. Hard: S26-T-00002258.

S26-T-00002260 - Fundamental theorem of algebra via complex analysis

Every nonconstant complex polynomial has a complex zero; Liouville gives one standard analytic proof.

Evidence A. Hard: S26-T-00002259.

S26-T-00002261 - Identity theorem for holomorphic functions

If two holomorphic functions agree on a set with an accumulation point inside a connected domain, they agree everywhere on that domain.

Evidence A. Hard: S26-T-00002260.

S26-T-00002262 - Open mapping theorem for holomorphic functions

If U is a connected open subset of C and f:U->C is holomorphic and nonconstant, then f is an open map. More generally, on a disconnected open set the conclusion holds on each component on which f is nonconstant.

Evidence A. Hard: S26-T-00002261. Peer-review correction applied in reviewed projection.

S26-T-00002263 - Maximum modulus principle

A nonconstant holomorphic function cannot attain a strict interior maximum of its modulus on a connected domain.

Evidence A. Hard: S26-T-00002262.

S26-T-00002264 - Schwarz lemma

A holomorphic self-map of the unit disk fixing 0 satisfies |f(z)|<=|z| and |f'(0)|<=1, with rigidity in the equality case.

Evidence A. Hard: S26-T-00002263.

S26-D-00002265 - Order of a zero

A zero z0 of a holomorphic function has order m when f(z)=(z-z0)^m g(z) with g(z0) nonzero.

Evidence A. Hard: S26-T-00002264.

S26-D-00002266 - Isolated singularity

A point z0 is an isolated singularity when f is holomorphic on a punctured neighborhood of z0.

Evidence A. Hard: S26-D-00002265.

S26-D-00002267 - Laurent series

On an annulus, a holomorphic function admits a unique doubly infinite Laurent expansion.

Evidence A. Hard: S26-D-00002266.

S26-D-00002268 - Removable, pole and essential singularities

An isolated singularity is removable, a pole, or essential according to the negative part of its Laurent series.

Evidence A. Hard: S26-D-00002267.

S26-N-00002269 - Complex-real differentiability fence

Real differentiability of f:R^2->R^2 does not imply complex differentiability; the Cauchy-Riemann structure is additional.

Evidence A. Hard: S26-D-00002268.

S26-B-00002270 - Holomorphy rigidity bridge

Complex differentiability forces analytic continuation, strong maximum principles and coefficient bounds unavailable to generic smooth real functions.

Evidence A. Hard: S26-N-00002269.

S26-E-00002271 - Entire exponential Taylor profile 1

At z=0.030-0.020i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.221e-16.

Evidence M. Hard: S26-B-00002270.

S26-E-00002272 - Entire exponential Taylor profile 2

At z=0.060-0.040i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.180e-16.

Evidence M. Hard: S26-E-00002271.

S26-E-00002273 - Entire exponential Taylor profile 3

At z=0.090-0.060i, the degree-8 Taylor polynomial for exp(z) has absolute error 5.764e-15.

Evidence M. Hard: S26-E-00002272.

S26-E-00002274 - Entire exponential Taylor profile 4

At z=0.120-0.080i, the degree-8 Taylor polynomial for exp(z) has absolute error 7.527e-14.

Evidence M. Hard: S26-E-00002273.

S26-E-00002275 - Entire exponential Taylor profile 5

At z=0.150-0.100i, the degree-8 Taylor polynomial for exp(z) has absolute error 5.628e-13.

Evidence M. Hard: S26-E-00002274.

S26-E-00002276 - Entire exponential Taylor profile 6

At z=0.180-0.120i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.912e-12.

Evidence M. Hard: S26-E-00002275.

S26-E-00002277 - Entire exponential Taylor profile 7

At z=0.210-0.140i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.170e-11.

Evidence M. Hard: S26-E-00002276.

S26-E-00002278 - Entire exponential Taylor profile 8

At z=0.240-0.160i, the degree-8 Taylor polynomial for exp(z) has absolute error 3.902e-11.

Evidence M. Hard: S26-E-00002277.

S26-E-00002279 - Entire exponential Taylor profile 9

At z=0.270-0.180i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.130e-10.

Evidence M. Hard: S26-E-00002278.

S26-E-00002280 - Entire exponential Taylor profile 10

At z=0.300-0.200i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.925e-10.

Evidence M. Hard: S26-E-00002279.

S26-E-00002281 - Entire exponential Taylor profile 11

At z=0.330-0.220i, the degree-8 Taylor polynomial for exp(z) has absolute error 6.918e-10.

Evidence M. Hard: S26-E-00002280.

S26-E-00002282 - Entire exponential Taylor profile 12

At z=0.360-0.240i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.518e-09.

Evidence M. Hard: S26-E-00002281.

S26-E-00002283 - Entire exponential Taylor profile 13

At z=0.390-0.260i, the degree-8 Taylor polynomial for exp(z) has absolute error 3.130e-09.

Evidence M. Hard: S26-E-00002282.

S26-E-00002284 - Entire exponential Taylor profile 14

At z=0.420-0.280i, the degree-8 Taylor polynomial for exp(z) has absolute error 6.117e-09.

Evidence M. Hard: S26-E-00002283.

S26-E-00002285 - Entire exponential Taylor profile 15

At z=0.450-0.300i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.142e-08.

Evidence M. Hard: S26-E-00002284.

S26-E-00002286 - Entire exponential Taylor profile 16

At z=0.480-0.320i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.047e-08.

Evidence M. Hard: S26-E-00002285.

S26-E-00002287 - Entire exponential Taylor profile 17

At z=0.510-0.340i, the degree-8 Taylor polynomial for exp(z) has absolute error 3.543e-08.

Evidence M. Hard: S26-E-00002286.

S26-E-00002288 - Entire exponential Taylor profile 18

At z=0.540-0.360i, the degree-8 Taylor polynomial for exp(z) has absolute error 5.945e-08.

Evidence M. Hard: S26-E-00002287.

S26-E-00002289 - Entire exponential Taylor profile 19

At z=0.570-0.380i, the degree-8 Taylor polynomial for exp(z) has absolute error 9.701e-08.

Evidence M. Hard: S26-E-00002288.

S26-E-00002290 - Entire exponential Taylor profile 20

At z=0.600-0.400i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.544e-07.

Evidence M. Hard: S26-E-00002289.

S26-E-00002291 - Entire exponential Taylor profile 21

At z=0.630-0.420i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.403e-07.

Evidence M. Hard: S26-E-00002290.

S26-E-00002292 - Entire exponential Taylor profile 22

At z=0.660-0.440i, the degree-8 Taylor polynomial for exp(z) has absolute error 3.663e-07.

Evidence M. Hard: S26-E-00002291.

S26-E-00002293 - Entire exponential Taylor profile 23

At z=0.690-0.460i, the degree-8 Taylor polynomial for exp(z) has absolute error 5.482e-07.

Evidence M. Hard: S26-E-00002292.

S26-E-00002294 - Entire exponential Taylor profile 24

At z=0.720-0.480i, the degree-8 Taylor polynomial for exp(z) has absolute error 8.066e-07.

Evidence M. Hard: S26-E-00002293.

S26-E-00002295 - Entire exponential Taylor profile 25

At z=0.750-0.500i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.168e-06.

Evidence M. Hard: S26-E-00002294.

S26-E-00002296 - Entire exponential Taylor profile 26

At z=0.780-0.520i, the degree-8 Taylor polynomial for exp(z) has absolute error 1.668e-06.

Evidence M. Hard: S26-E-00002295.

S26-E-00002297 - Entire exponential Taylor profile 27

At z=0.810-0.540i, the degree-8 Taylor polynomial for exp(z) has absolute error 2.350e-06.

Evidence M. Hard: S26-E-00002296.

S26-E-00002298 - Entire exponential Taylor profile 28

At z=0.840-0.560i, the degree-8 Taylor polynomial for exp(z) has absolute error 3.270e-06.

Evidence M. Hard: S26-E-00002297.

S26-E-00002299 - Entire exponential Taylor profile 29

At z=0.870-0.580i, the degree-8 Taylor polynomial for exp(z) has absolute error 4.499e-06.

Evidence M. Hard: S26-E-00002298.

S26-E-00002300 - Entire exponential Taylor profile 30

At z=0.900-0.600i, the degree-8 Taylor polynomial for exp(z) has absolute error 6.123e-06.

Evidence M. Hard: S26-E-00002299.

Part G - Contour integration and residues

S26-D-00002301 - Residue at an isolated singularity

Res(f,z0) is the coefficient of (z-z0)^{-1} in the Laurent expansion.

Evidence A. Hard: S26-E-00002300.

S26-I-00002302 - Residue at a simple pole

If f=g/h with g,h holomorphic, h(z0)=0 and h'(z0) nonzero, then Res(f,z0)=g(z0)/h'(z0).

Evidence A. Hard: S26-D-00002301.

S26-T-00002303 - Residue theorem

For a meromorphic function with finitely many poles inside a closed contour, the contour integral equals 2pi i times the sum of enclosed residues, counted with winding numbers in the general form.

Evidence A. Hard: S26-I-00002302.

S26-D-00002304 - Winding number

Ind(gamma,z0)=(1/(2pi i)) integral_gamma dz/(z-z0) for z0 off the closed curve.

Evidence A. Hard: S26-T-00002303.

S26-T-00002305 - Argument principle

The change of argument of a meromorphic f around a contour equals 2pi times the number of zeros minus poles inside, counted with multiplicity.

Evidence A. Hard: S26-D-00002304.

S26-T-00002306 - Rouche theorem

If |g|<|f| on a simple closed contour, then f and f+g have the same number of zeros inside, counted with multiplicity.

Evidence A. Hard: S26-T-00002305.

S26-D-00002307 - Principal part of a Laurent series

The principal part is the sum of negative-power terms and determines the isolated singular behavior.

Evidence A. Hard: S26-T-00002306.

S26-D-00002308 - Meromorphic function

A meromorphic function is holomorphic except at isolated poles.

Evidence A. Hard: S26-D-00002307.

S26-T-00002309 - Mittag-Leffler interface

Prescribed principal parts at a discrete set can be realized by a meromorphic function under the classical Mittag-Leffler construction.

Evidence A. Hard: S26-D-00002308.

S26-T-00002310 - Weierstrass product interface

Entire functions with prescribed zeros can be constructed by convergent canonical products subject to convergence factors.

Evidence A. Hard: S26-T-00002309.

S26-D-00002311 - Contour deformation principle

Holomorphic integrals are invariant under admissible homotopies that do not cross singularities, expressed precisely through Cauchy theory/homology.

Evidence A. Hard: S26-T-00002310.

S26-T-00002312 - Jordan lemma interface

For suitable rational-decay factors, semicircular contour contributions vanish, enabling Fourier-type real integral evaluation.

Evidence A. Hard: S26-D-00002311.

S26-D-00002313 - Cauchy principal value

PV integral is a symmetric limiting prescription for singular integrals and is not ordinary absolute integration.

Evidence A. Hard: S26-T-00002312.

S26-D-00002314 - Indentation contour

A small contour detour around a pole records half-residue or related contributions only under the declared orientation and limiting geometry.

Evidence A. Hard: S26-D-00002313.

S26-T-00002315 - Residue evaluation of rational trigonometric integrals

The substitution z=e^{it} converts many periodic rational trigonometric integrals into contour integrals on the unit circle.

Evidence A. Hard: S26-D-00002314.

S26-T-00002316 - Residue evaluation of real rational integrals

Suitable rational functions decaying in a half-plane admit real-line integral evaluations from enclosed residues when arc contributions vanish.

Evidence A. Hard: S26-T-00002315.

S26-D-00002317 - Pole at infinity

The residue at infinity is minus the coefficient of z^{-1} in the Laurent expansion at infinity and equals minus the sum of finite residues on the Riemann sphere.

Evidence A. Hard: S26-T-00002316.

S26-N-00002318 - Contour crossing fence

Deforming a contour across a singularity changes the integral by the corresponding residue contribution; deformation invariance is not unconditional.

Evidence A. Hard: S26-D-00002317.

S26-B-00002319 - Residues to spectral inversion bridge

Residue calculus extracts modal contributions from inverse Laplace/Fourier transforms and resolvents when poles represent isolated spectral data.

Evidence A. Hard: S26-N-00002318.

S26-B-00002320 - Argument principle to root counting bridge

Contour integrals can convert analytic zero counting into winding-number computation, motivating robust root-location algorithms.

Evidence A. Hard: S26-B-00002319.

S26-E-00002321 - Simple-pole residue profile 1

For f(z)=1/((z-0.230)(z--0.380)), the residue at z=0.230 is 1.639344262295.

Evidence M. Hard: S26-B-00002320.

S26-E-00002322 - Simple-pole residue profile 2

For f(z)=1/((z-0.260)(z--0.360)), the residue at z=0.260 is 1.612903225806.

Evidence M. Hard: S26-E-00002321.

S26-E-00002323 - Simple-pole residue profile 3

For f(z)=1/((z-0.290)(z--0.340)), the residue at z=0.290 is 1.587301587302.

Evidence M. Hard: S26-E-00002322.

S26-E-00002324 - Simple-pole residue profile 4

For f(z)=1/((z-0.320)(z--0.320)), the residue at z=0.320 is 1.562500000000.

Evidence M. Hard: S26-E-00002323.

S26-E-00002325 - Simple-pole residue profile 5

For f(z)=1/((z-0.350)(z--0.300)), the residue at z=0.350 is 1.538461538462.

Evidence M. Hard: S26-E-00002324.

S26-E-00002326 - Simple-pole residue profile 6

For f(z)=1/((z-0.380)(z--0.280)), the residue at z=0.380 is 1.515151515152.

Evidence M. Hard: S26-E-00002325.

S26-E-00002327 - Simple-pole residue profile 7

For f(z)=1/((z-0.410)(z--0.260)), the residue at z=0.410 is 1.492537313433.

Evidence M. Hard: S26-E-00002326.

S26-E-00002328 - Simple-pole residue profile 8

For f(z)=1/((z-0.440)(z--0.240)), the residue at z=0.440 is 1.470588235294.

Evidence M. Hard: S26-E-00002327.

S26-E-00002329 - Simple-pole residue profile 9

For f(z)=1/((z-0.470)(z--0.220)), the residue at z=0.470 is 1.449275362319.

Evidence M. Hard: S26-E-00002328.

S26-E-00002330 - Simple-pole residue profile 10

For f(z)=1/((z-0.500)(z--0.200)), the residue at z=0.500 is 1.428571428571.

Evidence M. Hard: S26-E-00002329.

S26-E-00002331 - Simple-pole residue profile 11

For f(z)=1/((z-0.530)(z--0.180)), the residue at z=0.530 is 1.408450704225.

Evidence M. Hard: S26-E-00002330.

S26-E-00002332 - Simple-pole residue profile 12

For f(z)=1/((z-0.560)(z--0.160)), the residue at z=0.560 is 1.388888888889.

Evidence M. Hard: S26-E-00002331.

S26-E-00002333 - Simple-pole residue profile 13

For f(z)=1/((z-0.590)(z--0.140)), the residue at z=0.590 is 1.369863013699.

Evidence M. Hard: S26-E-00002332.

S26-E-00002334 - Simple-pole residue profile 14

For f(z)=1/((z-0.620)(z--0.120)), the residue at z=0.620 is 1.351351351351.

Evidence M. Hard: S26-E-00002333.

S26-E-00002335 - Simple-pole residue profile 15

For f(z)=1/((z-0.650)(z--0.100)), the residue at z=0.650 is 1.333333333333.

Evidence M. Hard: S26-E-00002334.

S26-E-00002336 - Simple-pole residue profile 16

For f(z)=1/((z-0.680)(z--0.080)), the residue at z=0.680 is 1.315789473684.

Evidence M. Hard: S26-E-00002335.

S26-E-00002337 - Simple-pole residue profile 17

For f(z)=1/((z-0.710)(z--0.060)), the residue at z=0.710 is 1.298701298701.

Evidence M. Hard: S26-E-00002336.

S26-E-00002338 - Simple-pole residue profile 18

For f(z)=1/((z-0.740)(z--0.040)), the residue at z=0.740 is 1.282051282051.

Evidence M. Hard: S26-E-00002337.

S26-E-00002339 - Simple-pole residue profile 19

For f(z)=1/((z-0.770)(z--0.020)), the residue at z=0.770 is 1.265822784810.

Evidence M. Hard: S26-E-00002338.

S26-E-00002340 - Simple-pole residue profile 20

For f(z)=1/((z-0.800)(z-0.000)), the residue at z=0.800 is 1.250000000000.

Evidence M. Hard: S26-E-00002339.

S26-E-00002341 - Simple-pole residue profile 21

For f(z)=1/((z-0.830)(z-0.020)), the residue at z=0.830 is 1.234567901235.

Evidence M. Hard: S26-E-00002340.

S26-E-00002342 - Simple-pole residue profile 22

For f(z)=1/((z-0.860)(z-0.040)), the residue at z=0.860 is 1.219512195122.

Evidence M. Hard: S26-E-00002341.

S26-E-00002343 - Simple-pole residue profile 23

For f(z)=1/((z-0.890)(z-0.060)), the residue at z=0.890 is 1.204819277108.

Evidence M. Hard: S26-E-00002342.

S26-E-00002344 - Simple-pole residue profile 24

For f(z)=1/((z-0.920)(z-0.080)), the residue at z=0.920 is 1.190476190476.

Evidence M. Hard: S26-E-00002343.

S26-E-00002345 - Simple-pole residue profile 25

For f(z)=1/((z-0.950)(z-0.100)), the residue at z=0.950 is 1.176470588235.

Evidence M. Hard: S26-E-00002344.

S26-E-00002346 - Simple-pole residue profile 26

For f(z)=1/((z-0.980)(z-0.120)), the residue at z=0.980 is 1.162790697674.

Evidence M. Hard: S26-E-00002345.

S26-E-00002347 - Simple-pole residue profile 27

For f(z)=1/((z-1.010)(z-0.140)), the residue at z=1.010 is 1.149425287356.

Evidence M. Hard: S26-E-00002346.

S26-E-00002348 - Simple-pole residue profile 28

For f(z)=1/((z-1.040)(z-0.160)), the residue at z=1.040 is 1.136363636364.

Evidence M. Hard: S26-E-00002347.

S26-E-00002349 - Simple-pole residue profile 29

For f(z)=1/((z-1.070)(z-0.180)), the residue at z=1.070 is 1.123595505618.

Evidence M. Hard: S26-E-00002348.

S26-E-00002350 - Simple-pole residue profile 30

For f(z)=1/((z-1.100)(z-0.200)), the residue at z=1.100 is 1.111111111111.

Evidence M. Hard: S26-E-00002349.

Part H - Conformal and harmonic analysis in the plane

S26-D-00002351 - Conformal map

A holomorphic map with nonzero derivative is locally angle preserving in the oriented sense.

Evidence A. Hard: S26-E-00002350.

S26-D-00002352 - Biholomorphism

A biholomorphism is a bijective holomorphic map whose inverse is holomorphic.

Evidence A. Hard: S26-D-00002351.

S26-T-00002353 - Riemann mapping theorem

Every nonempty simply connected proper open subset of C is biholomorphic to the unit disk.

Evidence A. Hard: S26-D-00002352.

S26-D-00002354 - Mobius transformation

z |-> (az+b)/(cz+d) with ad-bc nonzero acts conformally on the Riemann sphere.

Evidence A. Hard: S26-T-00002353.

S26-T-00002355 - Mobius transformations preserve generalized circles

Mobius maps send circles and lines on the Riemann sphere to circles or lines.

Evidence A. Hard: S26-D-00002354.

S26-D-00002356 - Harmonic function

A C2 real-valued function u is harmonic when Delta u=0.

Evidence A. Hard: S26-T-00002355.

S26-T-00002357 - Real and imaginary parts of holomorphic functions are harmonic

If f=u+iv is holomorphic, then Delta u=Delta v=0.

Evidence A. Hard: S26-D-00002356.

S26-D-00002358 - Harmonic conjugate

v is a harmonic conjugate of u when u+iv is holomorphic; global existence can be obstructed by topology.

Evidence A. Hard: S26-T-00002357.

S26-T-00002359 - Mean-value property for harmonic functions

A harmonic function equals its average over sufficiently small circles or balls centered at the point.

Evidence A. Hard: S26-D-00002358.

S26-T-00002360 - Maximum principle for harmonic functions

A nonconstant harmonic function on a connected domain cannot attain an interior maximum or minimum.

Evidence A. Hard: S26-T-00002359.

S26-D-00002361 - Poisson kernel for the unit disk

P_r(theta)=(1-r^2)/(1-2r cos theta+r^2) is the boundary reproducing kernel for harmonic extension in the disk.

Evidence A. Hard: S26-T-00002360.

S26-I-00002362 - Poisson integral formula

Boundary data on the circle is harmonically extended to the disk by convolution with the Poisson kernel under suitable integrability/regularity assumptions.

Evidence A. Hard: S26-D-00002361.

S26-D-00002363 - Green function in planar potential theory

A Green function is harmonic away from its pole, has logarithmic singularity at the pole, and satisfies the declared boundary normalization.

Evidence A. Hard: S26-I-00002362.

S26-D-00002364 - Subharmonic function

An upper-semicontinuous function is subharmonic when it lies below its local harmonic averages in the appropriate sense.

Evidence A. Hard: S26-D-00002363.

S26-T-00002365 - Maximum principle for subharmonic functions

A subharmonic function cannot possess a strict interior maximum unless locally constant under standard connectedness assumptions.

Evidence A. Hard: S26-D-00002364.

S26-D-00002366 - Hyperbolic metric of the disk

The Poincare disk metric has line element proportional to |dz|/(1-|z|^2), with normalization convention declared.

Evidence A. Hard: S26-T-00002365.

S26-T-00002367 - Schwarz-Pick theorem

Holomorphic self-maps of the disk are nonexpansive for the Poincare metric, with equality rigidity for automorphisms.

Evidence A. Hard: S26-D-00002366.

S26-N-00002368 - Global conjugate fence

A harmonic function need not possess a single-valued global harmonic conjugate on a nonsimply connected domain.

Evidence A. Hard: S26-T-00002367.

S26-B-00002369 - Conformal geometry bridge

Planar conformal maps translate boundary-value problems between domains while preserving harmonic structure up to the appropriate transformation law.

Evidence A. Hard: S26-N-00002368.

S26-B-00002370 - Potential theory bridge

Harmonic, subharmonic, Green and Poisson objects connect complex analysis, elliptic PDE, probability and capacity theory.

Evidence A. Hard: S26-B-00002369.

S26-E-00002371 - Poisson-kernel profile 1

For disk radius parameter r=0.020 and angle theta=0.100, P_r(theta)=1.040599805721.

Evidence M. Hard: S26-B-00002370.

S26-E-00002372 - Poisson-kernel profile 2

For disk radius parameter r=0.040 and angle theta=0.200, P_r(theta)=1.081462045183.

Evidence M. Hard: S26-E-00002371.

S26-E-00002373 - Poisson-kernel profile 3

For disk radius parameter r=0.060 and angle theta=0.300, P_r(theta)=1.120860808658.

Evidence M. Hard: S26-E-00002372.

S26-E-00002374 - Poisson-kernel profile 4

For disk radius parameter r=0.080 and angle theta=0.400, P_r(theta)=1.156653110245.

Evidence M. Hard: S26-E-00002373.

S26-E-00002375 - Poisson-kernel profile 5

For disk radius parameter r=0.100 and angle theta=0.500, P_r(theta)=1.186362599961.

Evidence M. Hard: S26-E-00002374.

S26-E-00002376 - Poisson-kernel profile 6

For disk radius parameter r=0.120 and angle theta=0.600, P_r(theta)=1.207370468848.

Evidence M. Hard: S26-E-00002375.

S26-E-00002377 - Poisson-kernel profile 7

For disk radius parameter r=0.140 and angle theta=0.700, P_r(theta)=1.217216555960.

Evidence M. Hard: S26-E-00002376.

S26-E-00002378 - Poisson-kernel profile 8

For disk radius parameter r=0.160 and angle theta=0.800, P_r(theta)=1.213972868064.

Evidence M. Hard: S26-E-00002377.

S26-E-00002379 - Poisson-kernel profile 9

For disk radius parameter r=0.180 and angle theta=0.900, P_r(theta)=1.196605955442.

Evidence M. Hard: S26-E-00002378.

S26-E-00002380 - Poisson-kernel profile 10

For disk radius parameter r=0.200 and angle theta=1.000, P_r(theta)=1.165219540148.

Evidence M. Hard: S26-E-00002379.

S26-E-00002381 - Poisson-kernel profile 11

For disk radius parameter r=0.220 and angle theta=1.100, P_r(theta)=1.121088771631.

Evidence M. Hard: S26-E-00002380.

S26-E-00002382 - Poisson-kernel profile 12

For disk radius parameter r=0.240 and angle theta=1.200, P_r(theta)=1.066463540246.

Evidence M. Hard: S26-E-00002381.

S26-E-00002383 - Poisson-kernel profile 13

For disk radius parameter r=0.260 and angle theta=1.300, P_r(theta)=1.004199664326.

Evidence M. Hard: S26-E-00002382.

S26-E-00002384 - Poisson-kernel profile 14

For disk radius parameter r=0.280 and angle theta=1.400, P_r(theta)=0.937329895395.

Evidence M. Hard: S26-E-00002383.

S26-E-00002385 - Poisson-kernel profile 15

For disk radius parameter r=0.300 and angle theta=1.500, P_r(theta)=0.868687250586.

Evidence M. Hard: S26-E-00002384.

S26-E-00002386 - Poisson-kernel profile 16

For disk radius parameter r=0.320 and angle theta=1.600, P_r(theta)=0.800651014711.

Evidence M. Hard: S26-E-00002385.

S26-E-00002387 - Poisson-kernel profile 17

For disk radius parameter r=0.340 and angle theta=1.700, P_r(theta)=0.735031184597.

Evidence M. Hard: S26-E-00002386.

S26-E-00002388 - Poisson-kernel profile 18

For disk radius parameter r=0.360 and angle theta=1.800, P_r(theta)=0.673066620755.

Evidence M. Hard: S26-E-00002387.

S26-E-00002389 - Poisson-kernel profile 19

For disk radius parameter r=0.380 and angle theta=1.900, P_r(theta)=0.615495256753.

Evidence M. Hard: S26-E-00002388.

S26-E-00002390 - Poisson-kernel profile 20

For disk radius parameter r=0.400 and angle theta=2.000, P_r(theta)=0.562656688872.

Evidence M. Hard: S26-E-00002389.

S26-E-00002391 - Poisson-kernel profile 21

For disk radius parameter r=0.420 and angle theta=2.100, P_r(theta)=0.514598602562.

Evidence M. Hard: S26-E-00002390.

S26-E-00002392 - Poisson-kernel profile 22

For disk radius parameter r=0.440 and angle theta=2.200, P_r(theta)=0.471170879445.

Evidence M. Hard: S26-E-00002391.

S26-E-00002393 - Poisson-kernel profile 23

For disk radius parameter r=0.460 and angle theta=2.300, P_r(theta)=0.432100877305.

Evidence M. Hard: S26-E-00002392.

S26-E-00002394 - Poisson-kernel profile 24

For disk radius parameter r=0.480 and angle theta=2.400, P_r(theta)=0.397049376894.

Evidence M. Hard: S26-E-00002393.

S26-E-00002395 - Poisson-kernel profile 25

For disk radius parameter r=0.500 and angle theta=2.500, P_r(theta)=0.365649676754.

Evidence M. Hard: S26-E-00002394.

S26-E-00002396 - Poisson-kernel profile 26

For disk radius parameter r=0.520 and angle theta=2.600, P_r(theta)=0.337533331216.

Evidence M. Hard: S26-E-00002395.

S26-E-00002397 - Poisson-kernel profile 27

For disk radius parameter r=0.540 and angle theta=2.700, P_r(theta)=0.312345966379.

Evidence M. Hard: S26-E-00002396.

S26-E-00002398 - Poisson-kernel profile 28

For disk radius parameter r=0.560 and angle theta=2.800, P_r(theta)=0.289756081334.

Evidence M. Hard: S26-E-00002397.

S26-E-00002399 - Poisson-kernel profile 29

For disk radius parameter r=0.580 and angle theta=2.900, P_r(theta)=0.269459093223.

Evidence M. Hard: S26-E-00002398.

S26-E-00002400 - Poisson-kernel profile 30

For disk radius parameter r=0.600 and angle theta=3.000, P_r(theta)=0.251178281644.

Evidence M. Hard: S26-E-00002399.

Part I - Special functions

S26-D-00002401 - Gamma function

For Re(s)>0, Gamma(s)=integral_0^infinity t^{s-1}e^{-t}dt, extended meromorphically to C.

Evidence A. Hard: S26-E-00002400.

S26-I-00002402 - Gamma recurrence

Gamma(s+1)=s Gamma(s).

Evidence A. Hard: S26-D-00002401.

S26-I-00002403 - Gamma factorial identity

Gamma(n+1)=n! for nonnegative integers n.

Evidence A. Hard: S26-I-00002402.

S26-T-00002404 - Euler reflection formula

Gamma(z)Gamma(1-z)=pi/sin(pi z) away from poles.

Evidence A. Hard: S26-I-00002403.

S26-T-00002405 - Legendre duplication formula

Gamma(z)Gamma(z+1/2)=2^{1-2z}sqrt(pi)Gamma(2z).

Evidence A. Hard: S26-T-00002404.

S26-D-00002406 - Beta function

B(x,y)=integral_0^1 t^{x-1}(1-t)^{y-1}dt for Re x,Re y>0.

Evidence A. Hard: S26-T-00002405.

S26-I-00002407 - Beta-Gamma identity

B(x,y)=Gamma(x)Gamma(y)/Gamma(x+y).

Evidence A. Hard: S26-D-00002406.

S26-D-00002408 - Digamma function

psi(z)=Gamma'(z)/Gamma(z) is the logarithmic derivative of Gamma.

Evidence A. Hard: S26-I-00002407.

S26-D-00002409 - Polygamma functions

psi^{(m)} are derivatives of the digamma function.

Evidence A. Hard: S26-D-00002408.

S26-D-00002410 - Error function

erf(x)=2/sqrt(pi) integral_0^x e^{-t^2}dt.

Evidence A. Hard: S26-D-00002409.

S26-D-00002411 - Complementary error function

erfc(x)=1-erf(x).

Evidence A. Hard: S26-D-00002410.

S26-D-00002412 - Bessel equation

x^2 y''+x y'+(x^2-nu^2)y=0 defines the classical Bessel-function solution family.

Evidence A. Hard: S26-D-00002411.

S26-D-00002413 - Bessel function of the first kind

J_nu is the solution of Bessel equation specified by its standard power series near zero.

Evidence A. Hard: S26-D-00002412.

S26-D-00002414 - Modified Bessel equation

x^2 y''+x y'-(x^2+nu^2)y=0 defines I_nu and K_nu solution families.

Evidence A. Hard: S26-D-00002413.

S26-D-00002415 - Airy equation

y''-x y=0 has canonical solutions Ai and Bi.

Evidence A. Hard: S26-D-00002414.

S26-D-00002416 - Hypergeometric differential equation

z(1-z)y''+[c-(a+b+1)z]y'-ab y=0 defines the Gauss hypergeometric family.

Evidence A. Hard: S26-D-00002415.

S26-D-00002417 - Gauss hypergeometric series

_2F_1(a,b;c;z)=sum_{n>=0}(a)_n(b)_n/((c)_n n!) z^n in its disk of convergence, with analytic continuation beyond.

Evidence A. Hard: S26-D-00002416.

S26-T-00002418 - Stirling asymptotic for Gamma

Gamma(z) has the classical Stirling expansion in sectors excluding the negative real axis, with the sector and remainder convention declared.

Evidence A. Hard: S26-D-00002417.

S26-N-00002419 - Special-function branch fence

Analytic continuation of special functions can involve branch cuts and convention-dependent branches; numerical values require the branch convention.

Evidence A. Hard: S26-T-00002418.

S26-B-00002420 - Special-functions bridge

Gamma, Bessel, Airy and hypergeometric functions encode canonical solutions of differential equations, transforms and asymptotic transition regimes.

Evidence A. Hard: S26-N-00002419.

S26-E-00002421 - Gamma-Beta-erf profile x=0.15

At x=0.15: Gamma(x+1)=0.933040931107, B(x+1,3/2)=0.556753322025, erf(x)=0.167995971427.

Evidence M. Hard: S26-B-00002420.

S26-E-00002422 - Gamma-Beta-erf profile x=0.30

At x=0.30: Gamma(x+1)=0.897470696306, B(x+1,3/2)=0.474421154996, erf(x)=0.328626759459.

Evidence M. Hard: S26-E-00002421.

S26-E-00002423 - Gamma-Beta-erf profile x=0.45

At x=0.45: Gamma(x+1)=0.885661380271, B(x+1,3/2)=0.410775804234, erf(x)=0.475481719787.

Evidence M. Hard: S26-E-00002422.

S26-E-00002424 - Gamma-Beta-erf profile x=0.60

At x=0.60: Gamma(x+1)=0.893515349288, B(x+1,3/2)=0.360324924478, erf(x)=0.603856090848.

Evidence M. Hard: S26-E-00002423.

S26-E-00002425 - Gamma-Beta-erf profile x=0.75

At x=0.75: Gamma(x+1)=0.919062526849, B(x+1,3/2)=0.319504062596, erf(x)=0.711155633654.

Evidence M. Hard: S26-E-00002424.

S26-E-00002426 - Gamma-Beta-erf profile x=0.90

At x=0.90: Gamma(x+1)=0.961765831907, B(x+1,3/2)=0.285905319589, erf(x)=0.796908212423.

Evidence M. Hard: S26-E-00002425.

S26-E-00002427 - Gamma-Beta-erf profile x=1.05

At x=1.05: Gamma(x+1)=1.02217947884, B(x+1,3/2)=0.257847413019, erf(x)=0.86243610609.

Evidence M. Hard: S26-E-00002426.

S26-E-00002428 - Gamma-Beta-erf profile x=1.20

At x=1.20: Gamma(x+1)=1.10180249088, B(x+1,3/2)=0.234123365979, erf(x)=0.91031397823.

Evidence M. Hard: S26-E-00002427.

S26-E-00002429 - Gamma-Beta-erf profile x=1.35

At x=1.35: Gamma(x+1)=1.20305444673, B(x+1,3/2)=0.2138459543, erf(x)=0.943762196123.

Evidence M. Hard: S26-E-00002428.

S26-E-00002430 - Gamma-Beta-erf profile x=1.50

At x=1.50: Gamma(x+1)=1.32934038818, B(x+1,3/2)=0.196349540849, erf(x)=0.966105146475.

Evidence M. Hard: S26-E-00002429.

S26-E-00002431 - Gamma-Beta-erf profile x=1.65

At x=1.65: Gamma(x+1)=1.48519274692, B(x+1,3/2)=0.181125755612, erf(x)=0.980375585023.

Evidence M. Hard: S26-E-00002430.

S26-E-00002432 - Gamma-Beta-erf profile x=1.80

At x=1.80: Gamma(x+1)=1.67649078776, B(x+1,3/2)=0.167780199583, erf(x)=0.989090501636.

Evidence M. Hard: S26-E-00002431.

S26-E-00002433 - Gamma-Beta-erf profile x=1.95

At x=1.95: Gamma(x+1)=1.91076726998, B(x+1,3/2)=0.156002603074, erf(x)=0.994179333592.

Evidence M. Hard: S26-E-00002432.

S26-E-00002434 - Gamma-Beta-erf profile x=2.10

At x=2.10: Gamma(x+1)=2.19762027839, B(x+1,3/2)=0.145545822832, erf(x)=0.997020533344.

Evidence M. Hard: S26-E-00002433.

S26-E-00002435 - Gamma-Beta-erf profile x=2.25

At x=2.25: Gamma(x+1)=2.54925696672, B(x+1,3/2)=0.136210782041, erf(x)=0.998537283413.

Evidence M. Hard: S26-E-00002434.

S26-E-00002436 - Gamma-Beta-erf profile x=2.40

At x=2.40: Gamma(x+1)=2.98120642681, B(x+1,3/2)=0.127835489725, erf(x)=0.999311486103.

Evidence M. Hard: S26-E-00002435.

S26-E-00002437 - Gamma-Beta-erf profile x=2.55

At x=2.55: Gamma(x+1)=3.51325214469, B(x+1,3/2)=0.120286912946, erf(x)=0.999689339657.

Evidence M. Hard: S26-E-00002436.

S26-E-00002438 - Gamma-Beta-erf profile x=2.70

At x=2.70: Gamma(x+1)=4.1706517838, B(x+1,3/2)=0.113454877835, erf(x)=0.99986566726.

Evidence M. Hard: S26-E-00002437.

S26-E-00002439 - Gamma-Beta-erf profile x=2.85

At x=2.85: Gamma(x+1)=4.9857349276, B(x+1,3/2)=0.107247435651, erf(x)=0.99994434372.

Evidence M. Hard: S26-E-00002438.

S26-E-00002440 - Gamma-Beta-erf profile x=3.00

At x=3.00: Gamma(x+1)=6, B(x+1,3/2)=0.101587301587, erf(x)=0.999977909503.

Evidence M. Hard: S26-E-00002439.

S26-E-00002441 - Gamma-Beta-erf profile x=3.15

At x=3.15: Gamma(x+1)=7.26687266181, B(x+1,3/2)=0.0964090891631, erf(x)=0.999991601789.

Evidence M. Hard: S26-E-00002440.

S26-E-00002442 - Gamma-Beta-erf profile x=3.30

At x=3.30: Gamma(x+1)=8.85534336045, B(x+1,3/2)=0.0916571416108, erf(x)=0.99999694229.

Evidence M. Hard: S26-E-00002441.

S26-E-00002443 - Gamma-Beta-erf profile x=3.45

At x=3.45: Gamma(x+1)=10.8547765843, B(x+1,3/2)=0.0872838161259, erf(x)=0.999998933948.

Evidence M. Hard: S26-E-00002442.

S26-E-00002444 - Gamma-Beta-erf profile x=3.60

At x=3.60: Gamma(x+1)=13.3812858709, B(x+1,3/2)=0.0832481151014, erf(x)=0.999999644137.

Evidence M. Hard: S26-E-00002443.

S26-E-00002445 - Gamma-Beta-erf profile x=3.75

At x=3.75: Gamma(x+1)=16.5862065392, B(x+1,3/2)=0.079514585714, erf(x)=0.999999886273.

Evidence M. Hard: S26-E-00002444.

S26-E-00002446 - Gamma-Beta-erf profile x=3.90

At x=3.90: Gamma(x+1)=20.6673859619, B(x+1,3/2)=0.0760524288552, erf(x)=0.999999965208.

Evidence M. Hard: S26-E-00002445.

S26-E-00002447 - Gamma-Beta-erf profile x=4.05

At x=4.05: Gamma(x+1)=25.8842676254, B(x+1,3/2)=0.0728347727068, erf(x)=0.999999989812.

Evidence M. Hard: S26-E-00002446.

S26-E-00002448 - Gamma-Beta-erf profile x=4.20

At x=4.20: Gamma(x+1)=32.5780960503, B(x+1,3/2)=0.0698380767861, erf(x)=0.999999997145.

Evidence M. Hard: S26-E-00002447.

S26-E-00002449 - Gamma-Beta-erf profile x=4.35

At x=4.35: Gamma(x+1)=41.1990506738, B(x+1,3/2)=0.0670416401215, erf(x)=0.999999999234.

Evidence M. Hard: S26-E-00002448.

S26-E-00002450 - Gamma-Beta-erf profile x=4.50

At x=4.50: Gamma(x+1)=52.3427777846, B(x+1,3/2)=0.0644271930912, erf(x)=0.999999999803.

Evidence M. Hard: S26-E-00002449.

Part J - Approximation theory

S26-D-00002451 - Best approximation error

For f in a normed space and approximation set V, E(f,V)=inf_{v in V}||f-v||.

Evidence A. Hard: S26-E-00002450.

S26-D-00002452 - Polynomial approximation space

P_n is the space of polynomials of degree at most n on the declared interval/domain.

Evidence A. Hard: S26-D-00002451.

S26-T-00002453 - Weierstrass approximation theorem

Every continuous real-valued function on a compact interval can be uniformly approximated by polynomials.

Evidence A. Hard: S26-D-00002452.

S26-T-00002454 - Stone-Weierstrass theorem

A separating subalgebra containing constants is dense in C(K) under the standard real form; the complex form requires conjugation closure.

Evidence A. Hard: S26-T-00002453.

S26-D-00002455 - Modulus of continuity

omega(f,delta)=sup{|f(x)-f(y)|:|x-y|<=delta} quantifies uniform continuity.

Evidence A. Hard: S26-T-00002454.

S26-T-00002456 - Jackson approximation interface

Polynomial/trigonometric approximation errors admit upper bounds in terms of moduli of smoothness under the corresponding Jackson theorem.

Evidence A. Hard: S26-D-00002455.

S26-D-00002457 - Bernstein polynomial

B_n f(x)=sum_{k=0}^n f(k/n) binom(n,k)x^k(1-x)^{n-k} on [0,1].

Evidence A. Hard: S26-T-00002456.

S26-T-00002458 - Bernstein uniform convergence

For f continuous on [0,1], B_n f converges uniformly to f.

Evidence A. Hard: S26-D-00002457.

S26-D-00002459 - Chebyshev alternation property

A best uniform polynomial approximation to a continuous function is characterized by an alternating extremal error pattern under the classical degree conditions.

Evidence A. Hard: S26-T-00002458.

S26-D-00002460 - Lebesgue constant for interpolation

The Lebesgue constant is the sup norm of the interpolation operator and quantifies amplification of nodal data errors.

Evidence A. Hard: S26-D-00002459.

S26-T-00002461 - Interpolation error formula

For f sufficiently differentiable, the polynomial interpolation error at x equals f^{(n+1)}(xi)/(n+1)! times the nodal polynomial for some xi in the interpolation interval.

Evidence A. Hard: S26-D-00002460.

S26-D-00002462 - Chebyshev nodes

Chebyshev interpolation nodes cluster near interval endpoints and reduce worst-case polynomial interpolation amplification relative to equally spaced nodes.

Evidence A. Hard: S26-T-00002461.

S26-D-00002463 - Minimax polynomial approximation

A minimax approximant minimizes the maximum absolute error on a compact set.

Evidence A. Hard: S26-D-00002462.

S26-D-00002464 - Rational approximation

Rational approximation uses quotients p/q and can represent poles or boundary layers more efficiently than fixed-degree polynomials.

Evidence A. Hard: S26-D-00002463.

S26-D-00002465 - Pade approximant

A [m/n] Pade approximant matches as many initial power-series coefficients as possible with a rational function having numerator degree m and denominator degree n.

Evidence A. Hard: S26-D-00002464.

S26-D-00002466 - Spline approximation

A spline is a piecewise polynomial with prescribed smoothness across knots.

Evidence A. Hard: S26-D-00002465.

S26-T-00002467 - Cubic spline interpolation interface

Natural or clamped cubic splines give unique C2 piecewise-cubic interpolants after the corresponding boundary conditions are specified.

Evidence A. Hard: S26-D-00002466.

S26-N-00002468 - Interpolation-convergence fence

Exact interpolation at more nodes does not by itself imply uniform convergence; node geometry and Lebesgue growth matter.

Evidence A. Hard: S26-T-00002467. Boundary: Equally spaced polynomial interpolation can exhibit the Runge phenomenon.

S26-B-00002469 - Approximation to numerics bridge

Approximation error, conditioning and discretization error must be tracked separately in interpolation and spectral algorithms.

Evidence A. Hard: S26-N-00002468.

S26-B-00002470 - Approximation to learning bridge

Finite-dimensional approximation spaces and regularized best approximation provide analytic prototypes for regression and function learning.

Evidence A. Hard: S26-B-00002469.

S26-E-00002471 - Chebyshev interpolation profile degree 2

Using 3 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 5.647e-02.

Evidence M. Hard: S26-B-00002470.

S26-E-00002472 - Chebyshev interpolation profile degree 3

Using 4 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.657e-03.

Evidence M. Hard: S26-E-00002471.

S26-E-00002473 - Chebyshev interpolation profile degree 4

Using 5 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.397e-04.

Evidence M. Hard: S26-E-00002472.

S26-E-00002474 - Chebyshev interpolation profile degree 5

Using 6 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 5.180e-05.

Evidence M. Hard: S26-E-00002473.

S26-E-00002475 - Chebyshev interpolation profile degree 6

Using 7 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.620e-06.

Evidence M. Hard: S26-E-00002474.

S26-E-00002476 - Chebyshev interpolation profile degree 7

Using 8 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 2.224e-07.

Evidence M. Hard: S26-E-00002475.

S26-E-00002477 - Chebyshev interpolation profile degree 8

Using 9 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.219e-08.

Evidence M. Hard: S26-E-00002476.

S26-E-00002478 - Chebyshev interpolation profile degree 9

Using 10 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.027e-10.

Evidence M. Hard: S26-E-00002477.

S26-E-00002479 - Chebyshev interpolation profile degree 10

Using 11 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 2.714e-11.

Evidence M. Hard: S26-E-00002478.

S26-E-00002480 - Chebyshev interpolation profile degree 11

Using 12 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.123e-12.

Evidence M. Hard: S26-E-00002479.

S26-E-00002481 - Chebyshev interpolation profile degree 12

Using 13 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 4.530e-14.

Evidence M. Hard: S26-E-00002480.

S26-E-00002482 - Chebyshev interpolation profile degree 13

Using 14 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.109e-15.

Evidence M. Hard: S26-E-00002481.

S26-E-00002483 - Chebyshev interpolation profile degree 14

Using 15 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.109e-15.

Evidence M. Hard: S26-E-00002482.

S26-E-00002484 - Chebyshev interpolation profile degree 15

Using 16 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 8.438e-15.

Evidence M. Hard: S26-E-00002483.

S26-E-00002485 - Chebyshev interpolation profile degree 16

Using 17 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.465e-14.

Evidence M. Hard: S26-E-00002484.

S26-E-00002486 - Chebyshev interpolation profile degree 17

Using 18 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 2.665e-15.

Evidence M. Hard: S26-E-00002485.

S26-E-00002487 - Chebyshev interpolation profile degree 18

Using 19 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.109e-15.

Evidence M. Hard: S26-E-00002486.

S26-E-00002488 - Chebyshev interpolation profile degree 19

Using 20 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 2.220e-15.

Evidence M. Hard: S26-E-00002487.

S26-E-00002489 - Chebyshev interpolation profile degree 20

Using 21 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.997e-15.

Evidence M. Hard: S26-E-00002488.

S26-E-00002490 - Chebyshev interpolation profile degree 21

Using 22 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.110e-14.

Evidence M. Hard: S26-E-00002489.

S26-E-00002491 - Chebyshev interpolation profile degree 22

Using 23 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.465e-14.

Evidence M. Hard: S26-E-00002490.

S26-E-00002492 - Chebyshev interpolation profile degree 23

Using 24 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.661e-15.

Evidence M. Hard: S26-E-00002491.

S26-E-00002493 - Chebyshev interpolation profile degree 24

Using 25 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 3.997e-15.

Evidence M. Hard: S26-E-00002492.

S26-E-00002494 - Chebyshev interpolation profile degree 25

Using 26 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.217e-15.

Evidence M. Hard: S26-E-00002493.

S26-E-00002495 - Chebyshev interpolation profile degree 26

Using 27 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 6.217e-15.

Evidence M. Hard: S26-E-00002494.

S26-E-00002496 - Chebyshev interpolation profile degree 27

Using 28 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 1.510e-14.

Evidence M. Hard: S26-E-00002495.

S26-E-00002497 - Chebyshev interpolation profile degree 28

Using 29 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 4.885e-15.

Evidence M. Hard: S26-E-00002496.

S26-E-00002498 - Chebyshev interpolation profile degree 29

Using 30 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 8.438e-15.

Evidence M. Hard: S26-E-00002497.

S26-E-00002499 - Chebyshev interpolation profile degree 30

Using 31 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 8.882e-15.

Evidence M. Hard: S26-E-00002498.

S26-E-00002500 - Chebyshev interpolation profile degree 31

Using 32 first-kind Chebyshev nodes for exp(x) on [-1,1], sampled max error on a 501-point grid is 7.994e-15.

Evidence M. Hard: S26-E-00002499.

Part K - Orthogonal polynomials and quadrature

S26-D-00002501 - Orthogonal polynomial sequence

Polynomials p_n are orthogonal for weight w on an interval when integral p_n p_m w=0 for n!=m.

Evidence A. Hard: S26-E-00002500.

S26-D-00002502 - Monic orthogonal polynomial

A monic orthogonal polynomial has leading coefficient one and is orthogonal to all lower-degree polynomials.

Evidence A. Hard: S26-D-00002501.

S26-T-00002503 - Three-term recurrence for orthogonal polynomials

Under a positive measure with infinite support and finite moments, monic orthogonal polynomials satisfy a three-term recurrence.

Evidence A. Hard: S26-D-00002502.

S26-D-00002504 - Legendre polynomial

P_n is the degree-n polynomial orthogonal on [-1,1] with weight 1 and standard normalization P_n(1)=1.

Evidence A. Hard: S26-T-00002503.

S26-I-00002505 - Rodrigues formula for Legendre polynomials

P_n(x)=1/(2^n n!) d^n/dx^n (x^2-1)^n.

Evidence A. Hard: S26-D-00002504.

S26-D-00002506 - Chebyshev polynomial of first kind

T_n(cos theta)=cos(n theta).

Evidence A. Hard: S26-I-00002505.

S26-D-00002507 - Chebyshev polynomial of second kind

U_n(cos theta)=sin((n+1)theta)/sin theta.

Evidence A. Hard: S26-D-00002506.

S26-D-00002508 - Hermite polynomial

Hermite polynomials form an orthogonal family for Gaussian weight, with physicists/probabilists normalizations distinguished by convention.

Evidence A. Hard: S26-D-00002507.

S26-D-00002509 - Laguerre polynomial

Laguerre polynomials are orthogonal on [0,infinity) for exponential weights, with generalized parameter alpha when declared.

Evidence A. Hard: S26-D-00002508.

S26-D-00002510 - Jacobi polynomial

P_n^{(alpha,beta)} is orthogonal on [-1,1] for weight (1-x)^alpha(1+x)^beta with alpha,beta>-1.

Evidence A. Hard: S26-D-00002509.

S26-T-00002511 - Zeros of orthogonal polynomials

For a positive weight on an interval, the degree-n orthogonal polynomial has n simple zeros in the interior of the convex hull of the support.

Evidence A. Hard: S26-D-00002510.

S26-D-00002512 - Christoffel-Darboux kernel

K_n(x,y)=sum_{k=0}^n p_k(x)p_k(y)/h_k is the finite orthogonal-projection kernel.

Evidence A. Hard: S26-T-00002511.

S26-T-00002513 - Christoffel-Darboux formula

The orthogonal-polynomial kernel admits a quotient representation using consecutive polynomials and recurrence coefficients.

Evidence A. Hard: S26-D-00002512.

S26-D-00002514 - Gaussian quadrature rule

An n-node Gaussian quadrature rule uses the zeros of the degree-n orthogonal polynomial as nodes.

Evidence A. Hard: S26-T-00002513.

S26-T-00002515 - Gaussian quadrature exactness

An n-node Gaussian quadrature rule is exact for polynomials of degree at most 2n-1 under the associated weight.

Evidence A. Hard: S26-D-00002514.

S26-D-00002516 - Moment sequence

m_k=integral x^k dmu(x) is the kth moment when finite.

Evidence A. Hard: S26-T-00002515.

S26-D-00002517 - Hankel moment matrix

H_n=(m_{i+j})_{0<=i,j<=n} is positive semidefinite for every positive measure with the relevant finite moments.

Evidence A. Hard: S26-D-00002516.

S26-N-00002518 - Normalization fence for classical polynomials

Legendre, Hermite, Laguerre and Jacobi polynomials have multiple normalization conventions; recurrence and norm constants depend on the declared convention.

Evidence A. Hard: S26-D-00002517.

S26-B-00002519 - Orthogonal-polynomial spectral bridge

Jacobi matrices encode three-term recurrences and connect orthogonal polynomials to self-adjoint spectral theory.

Evidence A. Hard: S26-N-00002518.

S26-B-00002520 - Orthogonal-polynomial numerical bridge

Gaussian quadrature and spectral approximation turn orthogonality into high-order numerical integration and discretization schemes.

Evidence A. Hard: S26-B-00002519.

S26-E-00002521 - Legendre-Chebyshev profile n=1

At x=0.37, P_1(x)=0.37 and T_1(x)=0.37 under standard normalizations.

Evidence M. Hard: S26-B-00002520.

S26-E-00002522 - Legendre-Chebyshev profile n=2

At x=0.37, P_2(x)=-0.29465 and T_2(x)=-0.7262 under standard normalizations.

Evidence M. Hard: S26-E-00002521.

S26-E-00002523 - Legendre-Chebyshev profile n=3

At x=0.37, P_3(x)=-0.4283675 and T_3(x)=-0.907388 under standard normalizations.

Evidence M. Hard: S26-E-00002522.

S26-E-00002524 - Legendre-Chebyshev profile n=4

At x=0.37, P_4(x)=-0.05638045625 and T_4(x)=0.05473288 under standard normalizations.

Evidence M. Hard: S26-E-00002523.

S26-E-00002525 - Legendre-Chebyshev profile n=5

At x=0.37, P_5(x)=0.305144616137 and T_5(x)=0.9478903312 under standard normalizations.

Evidence M. Hard: S26-E-00002524.

S26-E-00002526 - Legendre-Chebyshev profile n=6

At x=0.37, P_6(x)=0.253973478155 and T_6(x)=0.646705965088 under standard normalizations.

Evidence M. Hard: S26-E-00002525.

S26-E-00002527 - Legendre-Chebyshev profile n=7

At x=0.37, P_7(x)=-0.0870364667 and T_7(x)=-0.469327917035 under standard normalizations.

Evidence M. Hard: S26-E-00002526.

S26-E-00002528 - Legendre-Chebyshev profile n=8

At x=0.37, P_8(x)=-0.282608342159 and T_8(x)=-0.994008623694 under standard normalizations.

Evidence M. Hard: S26-E-00002527.

S26-E-00002529 - Legendre-Chebyshev profile n=9

At x=0.37, P_9(x)=-0.120146082064 and T_9(x)=-0.266238464499 under standard normalizations.

Evidence M. Hard: S26-E-00002528.

S26-E-00002530 - Legendre-Chebyshev profile n=10

At x=0.37, P_10(x)=0.169884812252 and T_10(x)=0.796992159965 under standard normalizations.

Evidence M. Hard: S26-E-00002529.

S26-E-00002531 - Legendre-Chebyshev profile n=11

At x=0.37, P_11(x)=0.229224164713 and T_11(x)=0.856012662873 under standard normalizations.

Evidence M. Hard: S26-E-00002530.

S26-E-00002532 - Legendre-Chebyshev profile n=12

At x=0.37, P_12(x)=0.00683039224464 and T_12(x)=-0.163542789439 under standard normalizations.

Evidence M. Hard: S26-E-00002531.

S26-E-00002533 - Legendre-Chebyshev profile n=13

At x=0.37, P_13(x)=-0.206731449868 and T_13(x)=-0.977034327058 under standard normalizations.

Evidence M. Hard: S26-E-00002532.

S26-E-00002534 - Legendre-Chebyshev profile n=14

At x=0.37, P_14(x)=-0.153860163097 and T_14(x)=-0.559462612583 under standard normalizations.

Evidence M. Hard: S26-E-00002533.

S26-E-00002535 - Legendre-Chebyshev profile n=15

At x=0.37, P_15(x)=0.0828880498747 and T_15(x)=0.563031993746 under standard normalizations.

Evidence M. Hard: S26-E-00002534.

S26-E-00002536 - Legendre-Chebyshev profile n=16

At x=0.37, P_16(x)=0.203664273658 and T_16(x)=0.976106287955 under standard normalizations.

Evidence M. Hard: S26-E-00002535.

S26-E-00002537 - Legendre-Chebyshev profile n=17

At x=0.37, P_17(x)=0.0682665872569 and T_17(x)=0.159286659341 under standard normalizations.

Evidence M. Hard: S26-E-00002536.

S26-E-00002538 - Legendre-Chebyshev profile n=18

At x=0.37, P_18(x)=-0.143235574845 and T_18(x)=-0.858234160043 under standard normalizations.

Evidence M. Hard: S26-E-00002537.

S26-E-00002539 - Legendre-Chebyshev profile n=19

At x=0.37, P_19(x)=-0.167878610013 and T_19(x)=-0.794379937773 under standard normalizations.

Evidence M. Hard: S26-E-00002538.

S26-E-00002540 - Legendre-Chebyshev profile n=20

At x=0.37, P_20(x)=0.0149493789781 and T_20(x)=0.270393006091 under standard normalizations.

Evidence M. Hard: S26-E-00002539.

S26-E-00002541 - Legendre-Chebyshev profile n=21

At x=0.37, P_21(x)=0.170683537112 and T_21(x)=0.99447076228 under standard normalizations.

Evidence M. Hard: S26-E-00002540.

S26-E-00002542 - Legendre-Chebyshev profile n=22

At x=0.37, P_22(x)=0.109165368951 and T_22(x)=0.465515357996 under standard normalizations.

Evidence M. Hard: S26-E-00002541.

S26-E-00002543 - Legendre-Chebyshev profile n=23

At x=0.37, P_23(x)=-0.08423627928 and T_23(x)=-0.649989397363 under standard normalizations.

Evidence M. Hard: S26-E-00002542.

S26-E-00002544 - Legendre-Chebyshev profile n=24

At x=0.37, P_24(x)=-0.165653015939 and T_24(x)=-0.946507512045 under standard normalizations.

Evidence M. Hard: S26-E-00002543.

S26-E-00002545 - Legendre-Chebyshev profile n=25

At x=0.37, P_25(x)=-0.0392647390505 and T_25(x)=-0.0504261615503 under standard normalizations.

Evidence M. Hard: S26-E-00002544.

S26-E-00002546 - Legendre-Chebyshev profile n=26

At x=0.37, P_26(x)=0.130784606639 and T_26(x)=0.909192152498 under standard normalizations.

Evidence M. Hard: S26-E-00002545.

S26-E-00002547 - Legendre-Chebyshev profile n=27

At x=0.37, P_27(x)=0.13279886487 and T_27(x)=0.723228354399 under standard normalizations.

Evidence M. Hard: S26-E-00002546.

S26-E-00002548 - Legendre-Chebyshev profile n=28

At x=0.37, P_28(x)=-0.0295974099691 and T_28(x)=-0.374003170243 under standard normalizations.

Evidence M. Hard: S26-E-00002547.

S26-E-00002549 - Legendre-Chebyshev profile n=29

At x=0.37, P_29(x)=-0.149744054918 and T_29(x)=-0.999990700378 under standard normalizations.

Evidence M. Hard: S26-E-00002548.

S26-E-00002550 - Legendre-Chebyshev profile n=30

At x=0.37, P_30(x)=-0.0803529276584 and T_30(x)=-0.365989948037 under standard normalizations.

Evidence M. Hard: S26-E-00002549.

Part L - Fourier series and summability

S26-D-00002551 - Fourier coefficient on the circle

For integrable 2pi-periodic f, fhat(k)=(1/(2pi)) integral_{-pi}^{pi}f(x)e^{-ikx}dx.

Evidence A. Hard: S26-E-00002550.

S26-D-00002552 - Symmetric Fourier partial sum

S_N f=sum_{|k|<=N} fhat(k)e^{ikx}.

Evidence A. Hard: S26-D-00002551.

S26-D-00002553 - Dirichlet kernel explicit formula refinement

D_N(x)=sum_{|k|<=N}e^{ikx}=sin((N+1/2)x)/sin(x/2).

Evidence A. Hard: S26-D-00002552.

S26-D-00002554 - Fejer mean

sigma_N f=(1/(N+1))sum_{n=0}^N S_n f is the Cesaro mean of Fourier partial sums.

Evidence A. Hard: S26-D-00002553.

S26-D-00002555 - Fejer kernel positivity refinement

K_N=(1/(N+1))sum_{n=0}^N D_n is nonnegative and integrates to 2pi under the standard circle normalization.

Evidence A. Hard: S26-D-00002554.

S26-T-00002556 - Fejer convergence theorem refinement

For continuous periodic f, Fejer means converge uniformly to f; for L1 functions they converge at Lebesgue points in the standard sense.

Evidence A. Hard: S26-D-00002555.

S26-T-00002557 - Riemann-Lebesgue lemma on the circle

If f in L1(T), then fhat(k)->0 as |k|->infinity.

Evidence A. Hard: S26-T-00002556.

S26-T-00002558 - Bessel inequality for Fourier coefficients

For f in L2(T), sum_k |fhat(k)|^2 <= ||f||_2^2 under matching normalization.

Evidence A. Hard: S26-T-00002557.

S26-T-00002559 - Parseval identity with circle normalization

For f in L2(T), the squared L2 norm equals the sum of squared Fourier coefficients under the chosen normalization.

Evidence A. Hard: S26-T-00002558.

S26-D-00002560 - Sobolev regularity from Fourier coefficients on T

Periodic H^s regularity is characterized by square summability of (1+|k|^2)^{s/2} fhat(k).

Evidence A. Hard: S26-T-00002559.

S26-T-00002561 - Fourier differentiation rule on the circle

When justified by regularity/distribution theory, the kth Fourier coefficient of f' is ik fhat(k).

Evidence A. Hard: S26-D-00002560.

S26-D-00002562 - Gibbs phenomenon

Fourier partial sums near a jump exhibit a persistent relative overshoot even as the transition region narrows.

Evidence A. Hard: S26-T-00002561.

S26-D-00002563 - Fourier multiplier on the torus

A multiplier operator acts by (Tf)^hat(k)=m(k)fhat(k) on Fourier modes.

Evidence A. Hard: S26-D-00002562.

S26-T-00002564 - L2 multiplier bound on the torus

If m is bounded, the torus Fourier multiplier is bounded on L2 with norm at most sup|m| under Parseval.

Evidence A. Hard: S26-D-00002563.

S26-D-00002565 - Periodic convolution

(f*g)(x)=(1/(2pi))integral f(x-y)g(y)dy under the normalized circle convention.

Evidence A. Hard: S26-T-00002564.

S26-I-00002566 - Fourier coefficient of periodic convolution

With normalized convolution, (f*g)^hat(k)=fhat(k) ghat(k).

Evidence A. Hard: S26-D-00002565.

S26-D-00002567 - Lacunary Fourier series

A lacunary series uses frequencies whose successive ratios are bounded below by a constant greater than one.

Evidence A. Hard: S26-I-00002566.

S26-N-00002568 - Fourier convergence fence

L2 convergence of Fourier series does not imply pointwise or uniform convergence; each mode of convergence requires separate hypotheses.

Evidence A. Hard: S26-D-00002567.

S26-B-00002569 - Fourier series to PDE bridge

Periodic differential operators diagonalize on Fourier modes, reducing linear constant-coefficient PDE on tori to algebraic mode equations.

Evidence A. Hard: S26-N-00002568.

S26-B-00002570 - Fourier series to signal bridge

Truncation, windowing and Cesaro summation correspond to different spectral filters and have different localization/convergence behavior.

Evidence A. Hard: S26-B-00002569.

S26-E-00002571 - Square-wave Fourier sine profile n=1

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=1 is 4/(pi n)=1.27323954474; even harmonics vanish.

Evidence M. Hard: S26-B-00002570.

S26-E-00002572 - Square-wave Fourier sine profile n=3

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=3 is 4/(pi n)=0.424413181578; even harmonics vanish.

Evidence M. Hard: S26-E-00002571.

S26-E-00002573 - Square-wave Fourier sine profile n=5

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=5 is 4/(pi n)=0.254647908947; even harmonics vanish.

Evidence M. Hard: S26-E-00002572.

S26-E-00002574 - Square-wave Fourier sine profile n=7

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=7 is 4/(pi n)=0.181891363534; even harmonics vanish.

Evidence M. Hard: S26-E-00002573.

S26-E-00002575 - Square-wave Fourier sine profile n=9

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=9 is 4/(pi n)=0.141471060526; even harmonics vanish.

Evidence M. Hard: S26-E-00002574.

S26-E-00002576 - Square-wave Fourier sine profile n=11

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=11 is 4/(pi n)=0.115749049521; even harmonics vanish.

Evidence M. Hard: S26-E-00002575.

S26-E-00002577 - Square-wave Fourier sine profile n=13

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=13 is 4/(pi n)=0.0979415034412; even harmonics vanish.

Evidence M. Hard: S26-E-00002576.

S26-E-00002578 - Square-wave Fourier sine profile n=15

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=15 is 4/(pi n)=0.0848826363157; even harmonics vanish.

Evidence M. Hard: S26-E-00002577.

S26-E-00002579 - Square-wave Fourier sine profile n=17

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=17 is 4/(pi n)=0.074896443808; even harmonics vanish.

Evidence M. Hard: S26-E-00002578.

S26-E-00002580 - Square-wave Fourier sine profile n=19

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=19 is 4/(pi n)=0.0670126076176; even harmonics vanish.

Evidence M. Hard: S26-E-00002579.

S26-E-00002581 - Square-wave Fourier sine profile n=21

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=21 is 4/(pi n)=0.0606304545112; even harmonics vanish.

Evidence M. Hard: S26-E-00002580.

S26-E-00002582 - Square-wave Fourier sine profile n=23

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=23 is 4/(pi n)=0.0553582410754; even harmonics vanish.

Evidence M. Hard: S26-E-00002581.

S26-E-00002583 - Square-wave Fourier sine profile n=25

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=25 is 4/(pi n)=0.0509295817894; even harmonics vanish.

Evidence M. Hard: S26-E-00002582.

S26-E-00002584 - Square-wave Fourier sine profile n=27

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=27 is 4/(pi n)=0.0471570201754; even harmonics vanish.

Evidence M. Hard: S26-E-00002583.

S26-E-00002585 - Square-wave Fourier sine profile n=29

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=29 is 4/(pi n)=0.0439048118874; even harmonics vanish.

Evidence M. Hard: S26-E-00002584.

S26-E-00002586 - Square-wave Fourier sine profile n=31

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=31 is 4/(pi n)=0.0410722433786; even harmonics vanish.

Evidence M. Hard: S26-E-00002585.

S26-E-00002587 - Square-wave Fourier sine profile n=33

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=33 is 4/(pi n)=0.0385830165071; even harmonics vanish.

Evidence M. Hard: S26-E-00002586.

S26-E-00002588 - Square-wave Fourier sine profile n=35

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=35 is 4/(pi n)=0.0363782727067; even harmonics vanish.

Evidence M. Hard: S26-E-00002587.

S26-E-00002589 - Square-wave Fourier sine profile n=37

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=37 is 4/(pi n)=0.0344118795874; even harmonics vanish.

Evidence M. Hard: S26-E-00002588.

S26-E-00002590 - Square-wave Fourier sine profile n=39

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=39 is 4/(pi n)=0.0326471678137; even harmonics vanish.

Evidence M. Hard: S26-E-00002589.

S26-E-00002591 - Square-wave Fourier sine profile n=41

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=41 is 4/(pi n)=0.0310546230423; even harmonics vanish.

Evidence M. Hard: S26-E-00002590.

S26-E-00002592 - Square-wave Fourier sine profile n=43

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=43 is 4/(pi n)=0.0296102219706; even harmonics vanish.

Evidence M. Hard: S26-E-00002591.

S26-E-00002593 - Square-wave Fourier sine profile n=45

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=45 is 4/(pi n)=0.0282942121052; even harmonics vanish.

Evidence M. Hard: S26-E-00002592.

S26-E-00002594 - Square-wave Fourier sine profile n=47

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=47 is 4/(pi n)=0.0270902030795; even harmonics vanish.

Evidence M. Hard: S26-E-00002593.

S26-E-00002595 - Square-wave Fourier sine profile n=49

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=49 is 4/(pi n)=0.0259844805048; even harmonics vanish.

Evidence M. Hard: S26-E-00002594.

S26-E-00002596 - Square-wave Fourier sine profile n=51

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=51 is 4/(pi n)=0.0249654812693; even harmonics vanish.

Evidence M. Hard: S26-E-00002595.

S26-E-00002597 - Square-wave Fourier sine profile n=53

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=53 is 4/(pi n)=0.0240233876365; even harmonics vanish.

Evidence M. Hard: S26-E-00002596.

S26-E-00002598 - Square-wave Fourier sine profile n=55

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=55 is 4/(pi n)=0.0231498099043; even harmonics vanish.

Evidence M. Hard: S26-E-00002597.

S26-E-00002599 - Square-wave Fourier sine profile n=57

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=57 is 4/(pi n)=0.0223375358725; even harmonics vanish.

Evidence M. Hard: S26-E-00002598.

S26-E-00002600 - Square-wave Fourier sine profile n=59

For the standard +/-1 square wave, the sine coefficient at odd harmonic n=59 is 4/(pi n)=0.0215803312667; even harmonics vanish.

Evidence M. Hard: S26-E-00002599.

Part M - Fourier transform and multipliers

S26-D-00002601 - Fourier transform normalization pair

A Fourier transform convention must declare both exponent sign and constants; all multiplier and Plancherel formulas depend on that choice.

Evidence A. Hard: S26-E-00002600.

S26-T-00002602 - Riemann-Lebesgue lemma on Rn

For f in L1(R^n), its Fourier transform is continuous and vanishes at infinity.

Evidence A. Hard: S26-D-00002601.

S26-T-00002603 - Fourier transform of translation

Translation in physical space multiplies the Fourier transform by a phase factor under the declared convention.

Evidence A. Hard: S26-T-00002602.

S26-T-00002604 - Fourier transform of modulation

Modulation in physical space translates the Fourier transform in frequency.

Evidence A. Hard: S26-T-00002603.

S26-T-00002605 - Fourier scaling law under declared convention

Dilation of f rescales the Fourier transform with the reciprocal spatial scale and the Jacobian factor.

Evidence A. Hard: S26-T-00002604.

S26-T-00002606 - Plancherel extension on Rn

The Fourier transform extends from Schwartz functions to a unitary operator on L2 after normalization.

Evidence A. Hard: S26-T-00002605.

S26-T-00002607 - Hausdorff-Young inequality

For 1<=p<=2, the Fourier transform maps Lp to L^{p'} with the appropriate normalization-dependent bound.

Evidence A. Hard: S26-T-00002606.

S26-D-00002608 - Fourier multiplier on Rn

A multiplier T_m is defined formally by Fourier(T_m f)(xi)=m(xi) Fourier(f)(xi).

Evidence A. Hard: S26-T-00002607.

S26-T-00002609 - L2 Fourier multiplier theorem

Every essentially bounded multiplier m defines a bounded L2 multiplier with operator norm ||m||_infinity.

Evidence A. Hard: S26-D-00002608.

S26-D-00002610 - Fractional Laplacian via Fourier symbol

(-Delta)^{s/2} is defined on suitable functions/distributions by multiplier |xi|^s.

Evidence A. Hard: S26-T-00002609.

S26-D-00002611 - Bessel potential operator

(I-Delta)^{-s/2} is the Fourier multiplier (1+|xi|^2)^{-s/2}.

Evidence A. Hard: S26-D-00002610.

S26-D-00002612 - Band-limited function

f is band-limited when its Fourier transform is supported in a bounded frequency set.

Evidence A. Hard: S26-D-00002611.

S26-T-00002613 - Paley-Wiener interface

Compact support on one side of Fourier duality corresponds to analytic/exponential-type constraints on the other, with several precise Paley-Wiener formulations.

Evidence A. Hard: S26-D-00002612.

S26-D-00002614 - Approximate identity by rescaling

phi_epsilon(x)=epsilon^{-n}phi(x/epsilon) with integral phi=1 forms an approximate identity under standard integrability assumptions.

Evidence A. Hard: S26-T-00002613.

S26-T-00002615 - Approximate-identity convergence in Lp

For 1<=p<infinity, phi_epsilon*f -> f in Lp under the standard approximate-identity hypotheses.

Evidence A. Hard: S26-D-00002614.

S26-D-00002616 - Autocorrelation

The autocorrelation of f is the convolution of f with the reflected-conjugate function where defined.

Evidence A. Hard: S26-T-00002615.

S26-T-00002617 - Wiener-Khinchin interface

Under suitable hypotheses, the Fourier transform of autocorrelation is the power spectral density |fhat|^2, with probabilistic variants for stationary processes.

Evidence A. Hard: S26-D-00002616.

S26-N-00002618 - Transform-normalization fence

A formula imported from another Fourier convention can differ by powers of 2pi and sign changes; convention transport must be explicit.

Evidence A. Hard: S26-T-00002617.

S26-B-00002619 - Fourier transform to pseudodifferential bridge

Variable-coefficient symbol calculus generalizes constant Fourier multipliers and underlies microlocal analysis.

Evidence A. Hard: S26-N-00002618.

S26-B-00002620 - Fourier transform to uncertainty bridge

Spatial localization and frequency localization compete quantitatively; exact uncertainty statements depend on norm and normalization.

Evidence A. Hard: S26-B-00002619.

S26-E-00002621 - Gaussian Fourier profile sigma=0.30, xi=0.1

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.7516501637.

Evidence M. Hard: S26-B-00002620.

S26-E-00002622 - Gaussian Fourier profile sigma=0.35, xi=0.2

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.875173093283.

Evidence M. Hard: S26-E-00002621.

S26-E-00002623 - Gaussian Fourier profile sigma=0.40, xi=0.3

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.995458146883.

Evidence M. Hard: S26-E-00002622.

S26-E-00002624 - Gaussian Fourier profile sigma=0.45, xi=0.4

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.10985662131.

Evidence M. Hard: S26-E-00002623.

S26-E-00002625 - Gaussian Fourier profile sigma=0.50, xi=0.5

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.21475371513.

Evidence M. Hard: S26-E-00002624.

S26-E-00002626 - Gaussian Fourier profile sigma=0.55, xi=0.6

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.3055854129.

Evidence M. Hard: S26-E-00002625.

S26-E-00002627 - Gaussian Fourier profile sigma=0.60, xi=0.7

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.37700783473.

Evidence M. Hard: S26-E-00002626.

S26-E-00002628 - Gaussian Fourier profile sigma=0.65, xi=0.8

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.42326797329.

Evidence M. Hard: S26-E-00002627.

S26-E-00002629 - Gaussian Fourier profile sigma=0.70, xi=0.9

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.43880598028.

Evidence M. Hard: S26-E-00002628.

S26-E-00002630 - Gaussian Fourier profile sigma=0.75, xi=1.0

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.41907671687.

Evidence M. Hard: S26-E-00002629.

S26-E-00002631 - Gaussian Fourier profile sigma=0.80, xi=1.1

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.36151082228.

Evidence M. Hard: S26-E-00002630.

S26-E-00002632 - Gaussian Fourier profile sigma=0.85, xi=1.2

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.26645239727.

Evidence M. Hard: S26-E-00002631.

S26-E-00002633 - Gaussian Fourier profile sigma=0.90, xi=1.3

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 1.13783577779.

Evidence M. Hard: S26-E-00002632.

S26-E-00002634 - Gaussian Fourier profile sigma=0.95, xi=1.4

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.983335651184.

Evidence M. Hard: S26-E-00002633.

S26-E-00002635 - Gaussian Fourier profile sigma=1.00, xi=1.5

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.813783054109.

Evidence M. Hard: S26-E-00002634.

S26-E-00002636 - Gaussian Fourier profile sigma=1.05, xi=1.6

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.641804651578.

Evidence M. Hard: S26-E-00002635.

S26-E-00002637 - Gaussian Fourier profile sigma=1.10, xi=1.7

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.479888599212.

Evidence M. Hard: S26-E-00002636.

S26-E-00002638 - Gaussian Fourier profile sigma=1.15, xi=1.8

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.338324588123.

Evidence M. Hard: S26-E-00002637.

S26-E-00002639 - Gaussian Fourier profile sigma=1.20, xi=1.9

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.223590302164.

Evidence M. Hard: S26-E-00002638.

S26-E-00002640 - Gaussian Fourier profile sigma=1.25, xi=2.0

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.137666950151.

Evidence M. Hard: S26-E-00002639.

S26-E-00002641 - Gaussian Fourier profile sigma=1.30, xi=2.1

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.078461501415.

Evidence M. Hard: S26-E-00002640.

S26-E-00002642 - Gaussian Fourier profile sigma=1.35, xi=2.2

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.0411139880167.

Evidence M. Hard: S26-E-00002641.

S26-E-00002643 - Gaussian Fourier profile sigma=1.40, xi=2.3

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.0196674709233.

Evidence M. Hard: S26-E-00002642.

S26-E-00002644 - Gaussian Fourier profile sigma=1.45, xi=2.4

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.00852546330755.

Evidence M. Hard: S26-E-00002643.

S26-E-00002645 - Gaussian Fourier profile sigma=1.50, xi=2.5

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.00332313601624.

Evidence M. Hard: S26-E-00002644.

S26-E-00002646 - Gaussian Fourier profile sigma=1.55, xi=2.6

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.00115546024269.

Evidence M. Hard: S26-E-00002645.

S26-E-00002647 - Gaussian Fourier profile sigma=1.60, xi=2.7

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 0.000355403129202.

Evidence M. Hard: S26-E-00002646.

S26-E-00002648 - Gaussian Fourier profile sigma=1.65, xi=2.8

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 9.58729980928e-05.

Evidence M. Hard: S26-E-00002647.

S26-E-00002649 - Gaussian Fourier profile sigma=1.70, xi=2.9

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 2.24800296108e-05.

Evidence M. Hard: S26-E-00002648.

S26-E-00002650 - Gaussian Fourier profile sigma=1.75, xi=3.0

For the convention integral f(x)e^(-i xi x)dx and f=e^(-x^2/(2 sigma^2)), the transform value is 4.53949094495e-06.

Evidence M. Hard: S26-E-00002649.

Part N - Maximal functions and singular integrals

S26-D-00002651 - Hardy-Littlewood maximal function

Mf(x)=sup_{r>0}|B(x,r)|^{-1} integral_{B(x,r)}|f(y)|dy.

Evidence A. Hard: S26-E-00002650.

S26-T-00002652 - Weak (1,1) maximal inequality

The Hardy-Littlewood maximal operator satisfies mu({Mf>lambda})<=C_n ||f||_1/lambda on R^n.

Evidence A. Hard: S26-D-00002651.

S26-T-00002653 - Strong Lp maximal inequality

For 1<p<=infinity, the maximal operator is bounded on Lp(R^n).

Evidence A. Hard: S26-T-00002652.

S26-D-00002654 - Hilbert transform

Hf(x)=PV (1/pi) integral f(y)/(x-y)dy under the standard real-line normalization.

Evidence A. Hard: S26-T-00002653.

S26-T-00002655 - L2 Hilbert-transform multiplier

The Hilbert transform has Fourier multiplier -i sign(xi) and is bounded on L2.

Evidence A. Hard: S26-D-00002654.

S26-T-00002656 - Lp boundedness of Hilbert transform

For 1<p<infinity, H extends boundedly on Lp(R).

Evidence A. Hard: S26-T-00002655.

S26-D-00002657 - Singular integral kernel

A singular kernel has nonintegrable point singularity together with cancellation and size/smoothness structure sufficient for a principal-value operator.

Evidence A. Hard: S26-T-00002656.

S26-D-00002658 - Calderon-Zygmund operator interface

A Calderon-Zygmund operator is L2 bounded and represented off the diagonal by a standard singular kernel satisfying size and smoothness estimates.

Evidence A. Hard: S26-D-00002657.

S26-T-00002659 - Calderon-Zygmund Lp theorem

A Calderon-Zygmund operator extends boundedly to Lp for 1<p<infinity and satisfies weak-type endpoint control under standard assumptions.

Evidence A. Hard: S26-D-00002658.

S26-D-00002660 - Riesz transform

R_j has Fourier multiplier proportional to -i xi_j/|xi| under the declared Fourier convention.

Evidence A. Hard: S26-T-00002659.

S26-D-00002661 - Maximal truncation

A maximal truncation takes the supremum over singular-integral cutoffs and controls almost-everywhere convergence.

Evidence A. Hard: S26-D-00002660.

S26-D-00002662 - Hardy space H1 interface

Real-variable H1 replaces L1 at certain singular-integral endpoints and admits atomic/maximal characterizations.

Evidence A. Hard: S26-D-00002661.

S26-D-00002663 - BMO space

A locally integrable function is in BMO when its mean oscillation over balls/cubes is uniformly bounded.

Evidence A. Hard: S26-D-00002662.

S26-T-00002664 - John-Nirenberg inequality

BMO oscillations have exponentially decaying distribution tails on each cube, up to dimension-dependent constants.

Evidence A. Hard: S26-D-00002663.

S26-T-00002665 - H1-BMO duality interface

The dual of real Hardy space H1 can be identified with BMO modulo constants under the classical pairing.

Evidence A. Hard: S26-T-00002664.

S26-D-00002666 - Muckenhoupt Ap weight

A weight w belongs to Ap when its local average and reciprocal-power average satisfy the Ap characteristic bound.

Evidence A. Hard: S26-T-00002665.

S26-T-00002667 - Weighted maximal theorem

The maximal operator is bounded on Lp(w) exactly for Muckenhoupt Ap weights when 1<p<infinity.

Evidence A. Hard: S26-D-00002666.

S26-N-00002668 - Principal-value fence

Existence of a principal value is weaker than absolute integrability and can depend on symmetric cancellation.

Evidence A. Hard: S26-T-00002667.

S26-B-00002669 - Singular integrals to PDE bridge

Riesz transforms and Calderon-Zygmund estimates convert elliptic PDE derivatives into bounded singular-integral operators.

Evidence A. Hard: S26-N-00002668.

S26-B-00002670 - Maximal operators to differentiation bridge

Maximal inequalities provide the quantitative domination behind almost-everywhere differentiation theorems.

Evidence A. Hard: S26-B-00002669.

S26-E-00002671 - Discrete maximal-average profile n=5

For v=(1,...,5) at its central index, centered finite interval averages are [3.0, 3.0, 3.0]; their maximum is 3.000000.

Evidence M. Hard: S26-B-00002670.

S26-E-00002672 - Discrete maximal-average profile n=7

For v=(1,...,7) at its central index, centered finite interval averages are [4.0, 4.0, 4.0, 4.0]; their maximum is 4.000000.

Evidence M. Hard: S26-E-00002671.

S26-E-00002673 - Discrete maximal-average profile n=9

For v=(1,...,9) at its central index, centered finite interval averages are [5.0, 5.0, 5.0, 5.0, 5.0]; their maximum is 5.000000.

Evidence M. Hard: S26-E-00002672.

S26-E-00002674 - Discrete maximal-average profile n=11

For v=(1,...,11) at its central index, centered finite interval averages are [6.0, 6.0, 6.0, 6.0, 6.0, 6.0]; their maximum is 6.000000.

Evidence M. Hard: S26-E-00002673.

S26-E-00002675 - Discrete maximal-average profile n=13

For v=(1,...,13) at its central index, centered finite interval averages are [7.0, 7.0, 7.0, 7.0, 7.0, 7.0, 7.0]; their maximum is 7.000000.

Evidence M. Hard: S26-E-00002674.

S26-E-00002676 - Discrete maximal-average profile n=15

For v=(1,...,15) at its central index, centered finite interval averages are [8.0, 8.0, 8.0, 8.0, 8.0, 8.0, 8.0, 8.0]; their maximum is 8.000000.

Evidence M. Hard: S26-E-00002675.

S26-E-00002677 - Discrete maximal-average profile n=17

For v=(1,...,17) at its central index, centered finite interval averages are [9.0, 9.0, 9.0, 9.0, 9.0, 9.0, 9.0, 9.0, 9.0]; their maximum is 9.000000.

Evidence M. Hard: S26-E-00002676.

S26-E-00002678 - Discrete maximal-average profile n=19

For v=(1,...,19) at its central index, centered finite interval averages are [10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0]; their maximum is 10.000000.

Evidence M. Hard: S26-E-00002677.

S26-E-00002679 - Discrete maximal-average profile n=21

For v=(1,...,21) at its central index, centered finite interval averages are [11.0, 11.0, 11.0, 11.0, 11.0, 11.0, 11.0, 11.0, 11.0, 11.0, 11.0]; their maximum is 11.000000.

Evidence M. Hard: S26-E-00002678.

S26-E-00002680 - Discrete maximal-average profile n=23

For v=(1,...,23) at its central index, centered finite interval averages are [12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0, 12.0]; their maximum is 12.000000.

Evidence M. Hard: S26-E-00002679.

S26-E-00002681 - Discrete maximal-average profile n=25

For v=(1,...,25) at its central index, centered finite interval averages are [13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0, 13.0]; their maximum is 13.000000.

Evidence M. Hard: S26-E-00002680.

S26-E-00002682 - Discrete maximal-average profile n=27

For v=(1,...,27) at its central index, centered finite interval averages are [14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0, 14.0]; their maximum is 14.000000.

Evidence M. Hard: S26-E-00002681.

S26-E-00002683 - Discrete maximal-average profile n=29

For v=(1,...,29) at its central index, centered finite interval averages are [15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0, 15.0]; their maximum is 15.000000.

Evidence M. Hard: S26-E-00002682.

S26-E-00002684 - Discrete maximal-average profile n=31

For v=(1,...,31) at its central index, centered finite interval averages are [16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0, 16.0]; their maximum is 16.000000.

Evidence M. Hard: S26-E-00002683.

S26-E-00002685 - Discrete maximal-average profile n=33

For v=(1,...,33) at its central index, centered finite interval averages are [17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0, 17.0]; their maximum is 17.000000.

Evidence M. Hard: S26-E-00002684.

S26-E-00002686 - Discrete maximal-average profile n=35

For v=(1,...,35) at its central index, centered finite interval averages are [18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0, 18.0]; their maximum is 18.000000.

Evidence M. Hard: S26-E-00002685.

S26-E-00002687 - Discrete maximal-average profile n=37

For v=(1,...,37) at its central index, centered finite interval averages are [19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0, 19.0]; their maximum is 19.000000.

Evidence M. Hard: S26-E-00002686.

S26-E-00002688 - Discrete maximal-average profile n=39

For v=(1,...,39) at its central index, centered finite interval averages are [20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0]; their maximum is 20.000000.

Evidence M. Hard: S26-E-00002687.

S26-E-00002689 - Discrete maximal-average profile n=41

For v=(1,...,41) at its central index, centered finite interval averages are [21.0, 21.0, 21.0, 21.0, 21.0, 21.0, 21.0, 21.0, 21.0, 21.0, 21.0, 21.0, 21.0, 21.0, 21.0]; their maximum is 21.000000.

Evidence M. Hard: S26-E-00002688.

S26-E-00002690 - Discrete maximal-average profile n=43

For v=(1,...,43) at its central index, centered finite interval averages are [22.0, 22.0, 22.0, 22.0, 22.0, 22.0, 22.0, 22.0, 22.0, 22.0, 22.0, 22.0, 22.0, 22.0, 22.0]; their maximum is 22.000000.

Evidence M. Hard: S26-E-00002689.

S26-E-00002691 - Discrete maximal-average profile n=45

For v=(1,...,45) at its central index, centered finite interval averages are [23.0, 23.0, 23.0, 23.0, 23.0, 23.0, 23.0, 23.0, 23.0, 23.0, 23.0, 23.0, 23.0, 23.0, 23.0]; their maximum is 23.000000.

Evidence M. Hard: S26-E-00002690.

S26-E-00002692 - Discrete maximal-average profile n=47

For v=(1,...,47) at its central index, centered finite interval averages are [24.0, 24.0, 24.0, 24.0, 24.0, 24.0, 24.0, 24.0, 24.0, 24.0, 24.0, 24.0, 24.0, 24.0, 24.0]; their maximum is 24.000000.

Evidence M. Hard: S26-E-00002691.

S26-E-00002693 - Discrete maximal-average profile n=49

For v=(1,...,49) at its central index, centered finite interval averages are [25.0, 25.0, 25.0, 25.0, 25.0, 25.0, 25.0, 25.0, 25.0, 25.0, 25.0, 25.0, 25.0, 25.0, 25.0]; their maximum is 25.000000.

Evidence M. Hard: S26-E-00002692.

S26-E-00002694 - Discrete maximal-average profile n=51

For v=(1,...,51) at its central index, centered finite interval averages are [26.0, 26.0]; their maximum is 26.000000.

Evidence M. Hard: S26-E-00002693.

S26-E-00002695 - Discrete maximal-average profile n=53

For v=(1,...,53) at its central index, centered finite interval averages are [27.0, 27.0]; their maximum is 27.000000.

Evidence M. Hard: S26-E-00002694.

S26-E-00002696 - Discrete maximal-average profile n=55

For v=(1,...,55) at its central index, centered finite interval averages are [28.0, 28.0]; their maximum is 28.000000.

Evidence M. Hard: S26-E-00002695.

S26-E-00002697 - Discrete maximal-average profile n=57

For v=(1,...,57) at its central index, centered finite interval averages are [29.0, 29.0]; their maximum is 29.000000.

Evidence M. Hard: S26-E-00002696.

S26-E-00002698 - Discrete maximal-average profile n=59

For v=(1,...,59) at its central index, centered finite interval averages are [30.0, 30.0]; their maximum is 30.000000.

Evidence M. Hard: S26-E-00002697.

S26-E-00002699 - Discrete maximal-average profile n=61

For v=(1,...,61) at its central index, centered finite interval averages are [31.0, 31.0]; their maximum is 31.000000.

Evidence M. Hard: S26-E-00002698.

S26-E-00002700 - Discrete maximal-average profile n=63

For v=(1,...,63) at its central index, centered finite interval averages are [32.0, 32.0]; their maximum is 32.000000.

Evidence M. Hard: S26-E-00002699.

Part O - Regularity and function-space scales

S26-D-00002701 - Holder space C^{0,alpha}

For 0<alpha<=1, C^{0,alpha} consists of bounded functions with finite seminorm sup_{x!=y}|f(x)-f(y)|/|x-y|^alpha.

Evidence A. Hard: S26-E-00002700.

S26-D-00002702 - Holder space C^{k,alpha}

C^{k,alpha} consists of C^k functions whose kth derivatives are alpha-Holder continuous.

Evidence A. Hard: S26-D-00002701.

S26-D-00002703 - Lipschitz space as Holder endpoint

C^{0,1} is the Lipschitz class under the standard bounded-domain convention.

Evidence A. Hard: S26-D-00002702.

S26-D-00002704 - Morrey space interface

Morrey spaces control local Lp mass with an additional scaling exponent, interpolating between Lp and uniform regularity phenomena.

Evidence A. Hard: S26-D-00002703.

S26-D-00002705 - Besov space via differences

Besov B^s_{p,q} can be defined using Lp norms of finite differences integrated over scale with q-summability.

Evidence A. Hard: S26-D-00002704.

S26-D-00002706 - Besov space via dyadic decomposition

Equivalent Besov norms can be built from Littlewood-Paley dyadic frequency pieces under standard decomposition choices.

Evidence A. Hard: S26-D-00002705.

S26-D-00002707 - Triebel-Lizorkin space interface

Triebel-Lizorkin F^s_{p,q} places the dyadic q-aggregation inside the Lp norm and includes many Sobolev/Hardy scales as special cases.

Evidence A. Hard: S26-D-00002706.

S26-D-00002708 - Littlewood-Paley partition of unity

A dyadic smooth frequency partition decomposes a distribution into low-frequency and annular pieces.

Evidence A. Hard: S26-D-00002707.

S26-T-00002709 - Littlewood-Paley Lp equivalence

For 1<p<infinity, square-function norms built from dyadic pieces are equivalent to Lp norms under standard constructions.

Evidence A. Hard: S26-D-00002708.

S26-D-00002710 - Sobolev-Slobodeckij seminorm

For 0<s<1, [f]_{W^{s,p}}^p=double integral |f(x)-f(y)|^p/|x-y|^{n+sp} dxdy.

Evidence A. Hard: S26-T-00002709.

S26-D-00002711 - Fractional Sobolev space W^{s,p}

W^{s,p} combines Lp integrability with the fractional difference seminorm for noninteger s in the standard regime.

Evidence A. Hard: S26-D-00002710.

S26-T-00002712 - Morrey embedding interface

If p>n, W^{1,p} functions on suitable domains have Holder-continuous representatives with exponent 1-n/p.

Evidence A. Hard: S26-D-00002711.

S26-T-00002713 - Sobolev trace theorem interface

Sobolev functions on sufficiently regular domains admit boundary traces in lower-order fractional spaces when the regularity index crosses the trace threshold.

Evidence A. Hard: S26-T-00002712.

S26-D-00002714 - Extension operator for Sobolev spaces

An extension operator maps Sobolev functions on a domain to the ambient space while preserving the function on the domain and controlling the Sobolev norm.

Evidence A. Hard: S26-T-00002713.

S26-T-00002715 - Sobolev extension theorem for Lipschitz domains

Bounded Lipschitz domains admit bounded linear extension operators for standard W^{k,p} ranges.

Evidence A. Hard: S26-D-00002714.

S26-D-00002716 - Campanato space interface

Campanato seminorms measure polynomial/mean oscillation with scale weights and identify with Holder spaces in supercritical regimes.

Evidence A. Hard: S26-T-00002715.

S26-D-00002717 - Function-space scaling dimension

A function-space norm has a scaling law under f(x)->f(lambda x); critical exponents occur when the target estimate is scale invariant.

Evidence A. Hard: S26-D-00002716.

S26-N-00002718 - Embedding-domain fence

Sobolev and trace embeddings depend on dimension, domain regularity, exponent and boundary conditions; Euclidean formulas are not domain-free.

Evidence A. Hard: S26-D-00002717.

S26-B-00002719 - Function spaces to regularity bridge

Holder, Sobolev, Besov and Triebel-Lizorkin scales provide a vocabulary for PDE regularity, approximation rates and harmonic-analysis bounds.

Evidence A. Hard: S26-N-00002718.

S26-B-00002720 - Function spaces to numerics bridge

Smoothness measured in Besov/Sobolev scales predicts approximation sparsity and convergence rates for wavelet, finite-element and spectral methods.

Evidence A. Hard: S26-B-00002719.

S26-E-00002721 - Holder power profile alpha=0.125

For f(x)=x^0.125 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-B-00002720.

S26-E-00002722 - Holder power profile alpha=0.150

For f(x)=x^0.150 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002721.

S26-E-00002723 - Holder power profile alpha=0.175

For f(x)=x^0.175 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002722.

S26-E-00002724 - Holder power profile alpha=0.200

For f(x)=x^0.200 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002723.

S26-E-00002725 - Holder power profile alpha=0.225

For f(x)=x^0.225 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002724.

S26-E-00002726 - Holder power profile alpha=0.250

For f(x)=x^0.250 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002725.

S26-E-00002727 - Holder power profile alpha=0.275

For f(x)=x^0.275 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002726.

S26-E-00002728 - Holder power profile alpha=0.300

For f(x)=x^0.300 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002727.

S26-E-00002729 - Holder power profile alpha=0.325

For f(x)=x^0.325 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002728.

S26-E-00002730 - Holder power profile alpha=0.350

For f(x)=x^0.350 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002729.

S26-E-00002731 - Holder power profile alpha=0.375

For f(x)=x^0.375 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002730.

S26-E-00002732 - Holder power profile alpha=0.400

For f(x)=x^0.400 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002731.

S26-E-00002733 - Holder power profile alpha=0.425

For f(x)=x^0.425 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002732.

S26-E-00002734 - Holder power profile alpha=0.450

For f(x)=x^0.450 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002733.

S26-E-00002735 - Holder power profile alpha=0.475

For f(x)=x^0.475 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002734.

S26-E-00002736 - Holder power profile alpha=0.500

For f(x)=x^0.500 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002735.

S26-E-00002737 - Holder power profile alpha=0.525

For f(x)=x^0.525 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002736.

S26-E-00002738 - Holder power profile alpha=0.550

For f(x)=x^0.550 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002737.

S26-E-00002739 - Holder power profile alpha=0.575

For f(x)=x^0.575 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002738.

S26-E-00002740 - Holder power profile alpha=0.600

For f(x)=x^0.600 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002739.

S26-E-00002741 - Holder power profile alpha=0.625

For f(x)=x^0.625 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002740.

S26-E-00002742 - Holder power profile alpha=0.650

For f(x)=x^0.650 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002741.

S26-E-00002743 - Holder power profile alpha=0.675

For f(x)=x^0.675 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002742.

S26-E-00002744 - Holder power profile alpha=0.700

For f(x)=x^0.700 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002743.

S26-E-00002745 - Holder power profile alpha=0.725

For f(x)=x^0.725 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002744.

S26-E-00002746 - Holder power profile alpha=0.750

For f(x)=x^0.750 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002745.

S26-E-00002747 - Holder power profile alpha=0.775

For f(x)=x^0.775 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002746.

S26-E-00002748 - Holder power profile alpha=0.800

For f(x)=x^0.800 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002747.

S26-E-00002749 - Holder power profile alpha=0.825

For f(x)=x^0.825 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002748.

S26-E-00002750 - Holder power profile alpha=0.850

For f(x)=x^0.850 on [0,1], the sampled ratio |f(x)-f(0)|/|x|^alpha has maximum 1.000000, matching the exact boundary ratio 1.

Evidence M. Hard: S26-E-00002749.

Part P - Banach-space geometry

S26-D-00002751 - Reflexive Banach space

X is reflexive when the canonical embedding X->X** is onto.

Evidence A. Hard: S26-E-00002750.

S26-D-00002752 - Strictly convex norm

A norm is strictly convex when the unit sphere contains no nontrivial line segment.

Evidence A. Hard: S26-D-00002751.

S26-D-00002753 - Smooth norm

A norm is smooth when it has a unique supporting functional at every nonzero point, with equivalent differentiability formulations under standard conditions.

Evidence A. Hard: S26-D-00002752.

S26-D-00002754 - Uniformly smooth Banach space

Uniform smoothness is quantified by a modulus rho_X(t) satisfying rho_X(t)/t->0 as t->0.

Evidence A. Hard: S26-D-00002753.

S26-D-00002755 - Modulus of smoothness

rho_X(t)=sup_{||x||=||y||=1}((||x+ty||+||x-ty||)/2-1).

Evidence A. Hard: S26-D-00002754.

S26-T-00002756 - Duality of uniform convexity and smoothness

Uniform convexity of X is dual to uniform smoothness of X* in the standard quantitative duality framework.

Evidence A. Hard: S26-D-00002755.

S26-D-00002757 - Schauder basis

A sequence (e_n) is a Schauder basis if every x has a unique norm-convergent expansion x=sum a_n e_n.

Evidence A. Hard: S26-T-00002756.

S26-D-00002758 - Unconditional basis

A Schauder basis is unconditional when the expansion converges regardless of signs/permutations in the equivalent standard formulations.

Evidence A. Hard: S26-D-00002757.

S26-D-00002759 - Basic sequence

A basic sequence is a Schauder basis for its closed linear span.

Evidence A. Hard: S26-D-00002758.

S26-D-00002760 - Finite representability

X is finitely representable in Y when every finite-dimensional subspace of X embeds into Y with arbitrarily small distortion above one.

Evidence A. Hard: S26-D-00002759.

S26-D-00002761 - Banach-Mazur distance

For isomorphic finite-dimensional normed spaces X,Y, d(X,Y)=inf ||T||||T^{-1}|| over linear isomorphisms T:X->Y.

Evidence A. Hard: S26-D-00002760.

S26-D-00002762 - Type of a Banach space

Rademacher type p controls expected norms of signed sums by the l^p sum of vector norms.

Evidence A. Hard: S26-D-00002761.

S26-D-00002763 - Cotype of a Banach space

Rademacher cotype q controls the reverse comparison between vector norms and expected signed sums.

Evidence A. Hard: S26-D-00002762.

S26-T-00002764 - Kadec-Klee property interface

In spaces with the Kadec-Klee property, weak convergence plus convergence of norms implies norm convergence.

Evidence A. Hard: S26-D-00002763.

S26-D-00002765 - Weakly Cauchy sequence

A sequence x_n is weakly Cauchy when phi(x_n) is Cauchy for every phi in X*.

Evidence A. Hard: S26-T-00002764.

S26-T-00002766 - Eberlein-Smulian interface

For Banach spaces, weak compactness is equivalent to weak sequential compactness.

Evidence A. Hard: S26-D-00002765.

S26-D-00002767 - Radon-Nikodym property interface

A Banach space has the Radon-Nikodym property when suitable vector measures absolutely continuous with respect to scalar measures possess Bochner-integrable densities.

Evidence A. Hard: S26-T-00002766.

S26-N-00002768 - Basis fence

A separable Banach space need not have a Schauder basis, and a Schauder basis need not be unconditional.

Evidence A. Hard: S26-D-00002767.

S26-B-00002769 - Banach geometry to probability bridge

Type, cotype and uniform convexity govern concentration and vector-valued random-series inequalities.

Evidence A. Hard: S26-N-00002768.

S26-B-00002770 - Banach geometry to algorithms bridge

Norm geometry and Banach-Mazur distortion quantify conditioning and dimension-dependent losses when transporting algorithms between normed models.

Evidence A. Hard: S26-B-00002769.

S26-E-00002771 - lp midpoint profile p=1.10, n=5

For normalized reversed positive vectors in l^1.10_5, the midpoint norm is 0.988310066232; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-B-00002770.

S26-E-00002772 - lp midpoint profile p=1.20, n=6

For normalized reversed positive vectors in l^1.20_6, the midpoint norm is 0.975426882711; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002771.

S26-E-00002773 - lp midpoint profile p=1.30, n=7

For normalized reversed positive vectors in l^1.30_7, the midpoint norm is 0.962206636680; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002772.

S26-E-00002774 - lp midpoint profile p=1.40, n=8

For normalized reversed positive vectors in l^1.40_8, the midpoint norm is 0.949109987475; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002773.

S26-E-00002775 - lp midpoint profile p=1.50, n=9

For normalized reversed positive vectors in l^1.50_9, the midpoint norm is 0.936383710782; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002774.

S26-E-00002776 - lp midpoint profile p=1.60, n=10

For normalized reversed positive vectors in l^1.60_10, the midpoint norm is 0.924154076451; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002775.

S26-E-00002777 - lp midpoint profile p=1.70, n=4

For normalized reversed positive vectors in l^1.70_4, the midpoint norm is 0.935311358159; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002776.

S26-E-00002778 - lp midpoint profile p=1.80, n=5

For normalized reversed positive vectors in l^1.80_5, the midpoint norm is 0.920254786622; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002777.

S26-E-00002779 - lp midpoint profile p=1.90, n=6

For normalized reversed positive vectors in l^1.90_6, the midpoint norm is 0.906737660524; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002778.

S26-E-00002780 - lp midpoint profile p=2.00, n=7

For normalized reversed positive vectors in l^2.00_7, the midpoint norm is 0.894427191000; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002779.

S26-E-00002781 - lp midpoint profile p=2.10, n=8

For normalized reversed positive vectors in l^2.10_8, the midpoint norm is 0.883106934357; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002780.

S26-E-00002782 - lp midpoint profile p=2.20, n=9

For normalized reversed positive vectors in l^2.20_9, the midpoint norm is 0.872623919168; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002781.

S26-E-00002783 - lp midpoint profile p=2.30, n=10

For normalized reversed positive vectors in l^2.30_10, the midpoint norm is 0.862863535879; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002782.

S26-E-00002784 - lp midpoint profile p=2.40, n=4

For normalized reversed positive vectors in l^2.40_4, the midpoint norm is 0.886943703761; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002783.

S26-E-00002785 - lp midpoint profile p=2.50, n=5

For normalized reversed positive vectors in l^2.50_5, the midpoint norm is 0.870803508081; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002784.

S26-E-00002786 - lp midpoint profile p=2.60, n=6

For normalized reversed positive vectors in l^2.60_6, the midpoint norm is 0.857607742895; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002785.

S26-E-00002787 - lp midpoint profile p=2.70, n=7

For normalized reversed positive vectors in l^2.70_7, the midpoint norm is 0.846324902449; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002786.

S26-E-00002788 - lp midpoint profile p=2.80, n=8

For normalized reversed positive vectors in l^2.80_8, the midpoint norm is 0.836389360421; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002787.

S26-E-00002789 - lp midpoint profile p=2.90, n=9

For normalized reversed positive vectors in l^2.90_9, the midpoint norm is 0.827461206652; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002788.

S26-E-00002790 - lp midpoint profile p=3.00, n=10

For normalized reversed positive vectors in l^3.00_10, the midpoint norm is 0.819321270601; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002789.

S26-E-00002791 - lp midpoint profile p=3.10, n=4

For normalized reversed positive vectors in l^3.10_4, the midpoint norm is 0.850333663578; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002790.

S26-E-00002792 - lp midpoint profile p=3.20, n=5

For normalized reversed positive vectors in l^3.20_5, the midpoint norm is 0.833902099906; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002791.

S26-E-00002793 - lp midpoint profile p=3.30, n=6

For normalized reversed positive vectors in l^3.30_6, the midpoint norm is 0.821165226896; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002792.

S26-E-00002794 - lp midpoint profile p=3.40, n=7

For normalized reversed positive vectors in l^3.40_7, the midpoint norm is 0.810694560788; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002793.

S26-E-00002795 - lp midpoint profile p=3.50, n=8

For normalized reversed positive vectors in l^3.50_8, the midpoint norm is 0.801739915782; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002794.

S26-E-00002796 - lp midpoint profile p=3.60, n=9

For normalized reversed positive vectors in l^3.60_9, the midpoint norm is 0.793867140500; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002795.

S26-E-00002797 - lp midpoint profile p=3.70, n=10

For normalized reversed positive vectors in l^3.70_10, the midpoint norm is 0.786805931648; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002796.

S26-E-00002798 - lp midpoint profile p=3.80, n=4

For normalized reversed positive vectors in l^3.80_4, the midpoint norm is 0.822013869190; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002797.

S26-E-00002799 - lp midpoint profile p=3.90, n=5

For normalized reversed positive vectors in l^3.90_5, the midpoint norm is 0.805513391165; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002798.

S26-E-00002800 - lp midpoint profile p=4.00, n=6

For normalized reversed positive vectors in l^4.00_6, the midpoint norm is 0.793159456850; a value below 1 records this finite convexity instance.

Evidence M. Hard: S26-E-00002799.

Part Q - Operator and resolvent theory

S26-D-00002801 - Resolvent set of a closed operator

rho(A) consists of lambda for which lambda I-A is bijective with bounded inverse on the ambient Banach space.

Evidence A. Hard: S26-E-00002800.

S26-D-00002802 - Resolvent operator

R(lambda,A)=(lambda I-A)^{-1} for lambda in rho(A).

Evidence A. Hard: S26-D-00002801.

S26-I-00002803 - Resolvent identity for closed operators

R(lambda,A)-R(mu,A)=(mu-lambda)R(lambda,A)R(mu,A).

Evidence A. Hard: S26-D-00002802.

S26-T-00002804 - Resolvent analyticity

The resolvent is operator-valued holomorphic on the resolvent set.

Evidence A. Hard: S26-I-00002803.

S26-D-00002805 - Spectral projection by contour integral

For isolated spectral components, P=(1/(2pi i)) integral_Gamma (zI-A)^{-1}dz defines the corresponding Riesz projection.

Evidence A. Hard: S26-T-00002804.

S26-D-00002806 - Fredholm operator

T is Fredholm when ker T and cokernel are finite-dimensional and range T is closed.

Evidence A. Hard: S26-D-00002805.

S26-D-00002807 - Fredholm index

ind(T)=dim ker T-dim coker T.

Evidence A. Hard: S26-D-00002806.

S26-T-00002808 - Index stability under compact perturbations

The Fredholm index is invariant under compact perturbations that preserve Fredholmness; in fact T+K is Fredholm with the same index when T is Fredholm and K compact.

Evidence A. Hard: S26-D-00002807.

S26-T-00002809 - Fredholm alternative for compact operators

For compact K, I-K is invertible iff the homogeneous equation (I-K)x=0 has only the zero solution, with the dual finite-dimensional obstruction formulation.

Evidence A. Hard: S26-T-00002808.

S26-D-00002810 - Essential spectrum interface

The essential spectrum removes isolated finite-multiplicity eigenvalues according to a declared equivalent/Fredholm definition appropriate to the operator class.

Evidence A. Hard: S26-T-00002809.

S26-D-00002811 - Numerical range

W(A)={<Ax,x>:||x||=1} for a bounded operator on a complex Hilbert space.

Evidence A. Hard: S26-D-00002810.

S26-T-00002812 - Toeplitz-Hausdorff theorem

The numerical range of a bounded operator on a complex Hilbert space is convex.

Evidence A. Hard: S26-D-00002811.

S26-D-00002813 - Compact resolvent

A closed operator has compact resolvent when R(lambda,A) is compact for one, hence every, lambda in the resolvent set.

Evidence A. Hard: S26-T-00002812.

S26-T-00002814 - Compact-resolvent spectral discreteness interface

A closed operator with compact resolvent has discrete spectrum with finite algebraic multiplicities under standard Banach/Hilbert hypotheses, accumulating only at infinity.

Evidence A. Hard: S26-D-00002813.

S26-D-00002815 - Sectorial operator

A sectorial operator has spectrum in a sector and resolvent bounds outside wider sectors, with angle and constants declared.

Evidence A. Hard: S26-T-00002814.

S26-D-00002816 - Fractional power of a sectorial operator

Fractional powers A^alpha can be defined through holomorphic functional calculus or equivalent integral formulas under sectorial hypotheses.

Evidence A. Hard: S26-D-00002815.

S26-D-00002817 - Functional calculus algebra homomorphism

A holomorphic functional calculus maps admissible scalar functions to operators while preserving sums, products and the identity on its domain of definition.

Evidence A. Hard: S26-D-00002816.

S26-N-00002818 - Spectrum-eigenvalue fence for infinite dimensions

Spectrum can contain continuous or residual parts with no eigenvectors; finite-matrix intuition does not exhaust operator spectrum.

Evidence A. Hard: S26-D-00002817.

S26-B-00002819 - Resolvent to PDE bridge

Elliptic and evolution estimates are often resolvent bounds for differential operators on function spaces.

Evidence A. Hard: S26-N-00002818.

S26-B-00002820 - Operator calculus bridge

Contour, holomorphic and spectral functional calculi turn scalar analytic functions into operator constructions such as semigroups, fractional powers and projections.

Evidence A. Hard: S26-B-00002819.

S26-E-00002821 - Diagonal resolvent profile n=3, lambda=3.60

For A=diag(1,...,3) and lambda=3.60, ||(lambda I-A)^(-1)||_2=1.66666666667.

Evidence M. Hard: S26-B-00002820.

S26-E-00002822 - Diagonal resolvent profile n=4, lambda=4.70

For A=diag(1,...,4) and lambda=4.70, ||(lambda I-A)^(-1)||_2=1.42857142857.

Evidence M. Hard: S26-E-00002821.

S26-E-00002823 - Diagonal resolvent profile n=5, lambda=5.80

For A=diag(1,...,5) and lambda=5.80, ||(lambda I-A)^(-1)||_2=1.25.

Evidence M. Hard: S26-E-00002822.

S26-E-00002824 - Diagonal resolvent profile n=6, lambda=6.90

For A=diag(1,...,6) and lambda=6.90, ||(lambda I-A)^(-1)||_2=1.11111111111.

Evidence M. Hard: S26-E-00002823.

S26-E-00002825 - Diagonal resolvent profile n=2, lambda=3.00

For A=diag(1,...,2) and lambda=3.00, ||(lambda I-A)^(-1)||_2=1.

Evidence M. Hard: S26-E-00002824.

S26-E-00002826 - Diagonal resolvent profile n=3, lambda=4.10

For A=diag(1,...,3) and lambda=4.10, ||(lambda I-A)^(-1)||_2=0.909090909091.

Evidence M. Hard: S26-E-00002825.

S26-E-00002827 - Diagonal resolvent profile n=4, lambda=5.20

For A=diag(1,...,4) and lambda=5.20, ||(lambda I-A)^(-1)||_2=0.833333333333.

Evidence M. Hard: S26-E-00002826.

S26-E-00002828 - Diagonal resolvent profile n=5, lambda=6.30

For A=diag(1,...,5) and lambda=6.30, ||(lambda I-A)^(-1)||_2=0.769230769231.

Evidence M. Hard: S26-E-00002827.

S26-E-00002829 - Diagonal resolvent profile n=6, lambda=7.40

For A=diag(1,...,6) and lambda=7.40, ||(lambda I-A)^(-1)||_2=0.714285714286.

Evidence M. Hard: S26-E-00002828.

S26-E-00002830 - Diagonal resolvent profile n=2, lambda=3.50

For A=diag(1,...,2) and lambda=3.50, ||(lambda I-A)^(-1)||_2=0.666666666667.

Evidence M. Hard: S26-E-00002829.

S26-E-00002831 - Diagonal resolvent profile n=3, lambda=4.60

For A=diag(1,...,3) and lambda=4.60, ||(lambda I-A)^(-1)||_2=0.625.

Evidence M. Hard: S26-E-00002830.

S26-E-00002832 - Diagonal resolvent profile n=4, lambda=5.70

For A=diag(1,...,4) and lambda=5.70, ||(lambda I-A)^(-1)||_2=0.588235294118.

Evidence M. Hard: S26-E-00002831.

S26-E-00002833 - Diagonal resolvent profile n=5, lambda=6.80

For A=diag(1,...,5) and lambda=6.80, ||(lambda I-A)^(-1)||_2=0.555555555556.

Evidence M. Hard: S26-E-00002832.

S26-E-00002834 - Diagonal resolvent profile n=6, lambda=7.90

For A=diag(1,...,6) and lambda=7.90, ||(lambda I-A)^(-1)||_2=0.526315789474.

Evidence M. Hard: S26-E-00002833.

S26-E-00002835 - Diagonal resolvent profile n=2, lambda=4.00

For A=diag(1,...,2) and lambda=4.00, ||(lambda I-A)^(-1)||_2=0.5.

Evidence M. Hard: S26-E-00002834.

S26-E-00002836 - Diagonal resolvent profile n=3, lambda=5.10

For A=diag(1,...,3) and lambda=5.10, ||(lambda I-A)^(-1)||_2=0.47619047619.

Evidence M. Hard: S26-E-00002835.

S26-E-00002837 - Diagonal resolvent profile n=4, lambda=6.20

For A=diag(1,...,4) and lambda=6.20, ||(lambda I-A)^(-1)||_2=0.454545454545.

Evidence M. Hard: S26-E-00002836.

S26-E-00002838 - Diagonal resolvent profile n=5, lambda=7.30

For A=diag(1,...,5) and lambda=7.30, ||(lambda I-A)^(-1)||_2=0.434782608696.

Evidence M. Hard: S26-E-00002837.

S26-E-00002839 - Diagonal resolvent profile n=6, lambda=8.40

For A=diag(1,...,6) and lambda=8.40, ||(lambda I-A)^(-1)||_2=0.416666666667.

Evidence M. Hard: S26-E-00002838.

S26-E-00002840 - Diagonal resolvent profile n=2, lambda=4.50

For A=diag(1,...,2) and lambda=4.50, ||(lambda I-A)^(-1)||_2=0.4.

Evidence M. Hard: S26-E-00002839.

S26-E-00002841 - Diagonal resolvent profile n=3, lambda=5.60

For A=diag(1,...,3) and lambda=5.60, ||(lambda I-A)^(-1)||_2=0.384615384615.

Evidence M. Hard: S26-E-00002840.

S26-E-00002842 - Diagonal resolvent profile n=4, lambda=6.70

For A=diag(1,...,4) and lambda=6.70, ||(lambda I-A)^(-1)||_2=0.37037037037.

Evidence M. Hard: S26-E-00002841.

S26-E-00002843 - Diagonal resolvent profile n=5, lambda=7.80

For A=diag(1,...,5) and lambda=7.80, ||(lambda I-A)^(-1)||_2=0.357142857143.

Evidence M. Hard: S26-E-00002842.

S26-E-00002844 - Diagonal resolvent profile n=6, lambda=8.90

For A=diag(1,...,6) and lambda=8.90, ||(lambda I-A)^(-1)||_2=0.344827586207.

Evidence M. Hard: S26-E-00002843.

S26-E-00002845 - Diagonal resolvent profile n=2, lambda=5.00

For A=diag(1,...,2) and lambda=5.00, ||(lambda I-A)^(-1)||_2=0.333333333333.

Evidence M. Hard: S26-E-00002844.

S26-E-00002846 - Diagonal resolvent profile n=3, lambda=6.10

For A=diag(1,...,3) and lambda=6.10, ||(lambda I-A)^(-1)||_2=0.322580645161.

Evidence M. Hard: S26-E-00002845.

S26-E-00002847 - Diagonal resolvent profile n=4, lambda=7.20

For A=diag(1,...,4) and lambda=7.20, ||(lambda I-A)^(-1)||_2=0.3125.

Evidence M. Hard: S26-E-00002846.

S26-E-00002848 - Diagonal resolvent profile n=5, lambda=8.30

For A=diag(1,...,5) and lambda=8.30, ||(lambda I-A)^(-1)||_2=0.30303030303.

Evidence M. Hard: S26-E-00002847.

S26-E-00002849 - Diagonal resolvent profile n=6, lambda=9.40

For A=diag(1,...,6) and lambda=9.40, ||(lambda I-A)^(-1)||_2=0.294117647059.

Evidence M. Hard: S26-E-00002848.

S26-E-00002850 - Diagonal resolvent profile n=2, lambda=5.50

For A=diag(1,...,2) and lambda=5.50, ||(lambda I-A)^(-1)||_2=0.285714285714.

Evidence M. Hard: S26-E-00002849.

Part R - Distributions and microlocal interfaces

S26-D-00002851 - Distributional support

supp(T) is the complement of the largest open set on which the distribution T vanishes.

Evidence A. Hard: S26-E-00002850.

S26-D-00002852 - Distribution of finite order

T has finite order m on a compact set when its action is bounded by derivatives of test functions through order m with a compact-dependent constant.

Evidence A. Hard: S26-D-00002851.

S26-D-00002853 - Convolution of a distribution with a test function

For T in D' and phi in C_c^infinity, T*phi is smooth and defined by translating the test function against T.

Evidence A. Hard: S26-D-00002852.

S26-T-00002854 - Distributional convolution regularizes

Convolution of a distribution with a compactly supported smooth test function produces a smooth function.

Evidence A. Hard: S26-D-00002853.

S26-D-00002855 - Compactly supported mollifier interface

A mollifier rho_epsilon is a rescaled smooth compactly supported approximate identity.

Evidence A. Hard: S26-T-00002854.

S26-T-00002856 - Mollification convergence for distributions

T*rho_epsilon -> T in the distribution topology as epsilon->0 under the standard mollifier construction.

Evidence A. Hard: S26-D-00002855.

S26-D-00002857 - Distributional fundamental solution

E is a fundamental solution of P(D) when P(D)E=delta in distributions.

Evidence A. Hard: S26-T-00002856.

S26-T-00002858 - Malgrange-Ehrenpreis theorem

Every nonzero constant-coefficient linear differential operator on R^n has a distributional fundamental solution.

Evidence A. Hard: S26-D-00002857.

S26-D-00002859 - Singular support

sing supp(T) is the complement of the largest open set on which T is represented by a smooth function.

Evidence A. Hard: S26-T-00002858.

S26-D-00002860 - Wavefront set interface

WF(T) refines singular support by recording both singular positions and cotangent directions in which localized Fourier decay fails.

Evidence A. Hard: S26-D-00002859.

S26-D-00002861 - Microlocal smoothness

T is microlocally smooth near (x0,xi0) when a localization near x0 has rapidly decaying Fourier transform in a cone around xi0.

Evidence A. Hard: S26-D-00002860.

S26-T-00002862 - Wavefront projection to singular support

The projection of WF(T) to the base space equals sing supp(T).

Evidence A. Hard: S26-D-00002861.

S26-D-00002863 - Pseudodifferential symbol class S^m

A symbol a(x,xi) is of order m when derivatives obey polynomial growth bounds in xi with order reduced by xi-derivatives.

Evidence A. Hard: S26-T-00002862.

S26-D-00002864 - Pseudodifferential operator interface

Op(a) acts by oscillatory Fourier synthesis with symbol a(x,xi), with quantization convention declared.

Evidence A. Hard: S26-D-00002863.

S26-D-00002865 - Principal symbol of a pseudodifferential operator

The top-order homogeneous/equivalence class of a symbol determines its principal microlocal behavior.

Evidence A. Hard: S26-D-00002864.

S26-D-00002866 - Elliptic symbol

A symbol is elliptic when its magnitude has the expected lower bound of its order for large frequencies in the region of interest.

Evidence A. Hard: S26-D-00002865.

S26-T-00002867 - Elliptic regularity microlocal interface

For elliptic P, singularities of u and Pu coincide microlocally up to the characteristic set; ellipticity prevents new singularities.

Evidence A. Hard: S26-D-00002866.

S26-N-00002868 - Wavefront-support fence

Support records where a distribution is nonzero; singular support records where it is nonsmooth; wavefront set additionally records singular directions. These are different objects.

Evidence A. Hard: S26-T-00002867.

S26-B-00002869 - Microlocal PDE bridge

Wavefront sets and symbols track propagation and removal of singularities beyond what global Sobolev norms can express.

Evidence A. Hard: S26-N-00002868.

S26-B-00002870 - Microlocal spectral bridge

Pseudodifferential parametrices connect local symbol inversion to resolvents, elliptic spectra and heat-kernel asymptotics.

Evidence A. Hard: S26-B-00002869.

S26-E-00002871 - Gaussian mollifier profile epsilon=0.020

For the normalized Gaussian mollifier at epsilon=0.020, rho_epsilon(0.3)=2.76535477492e-48. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-B-00002870.

S26-E-00002872 - Gaussian mollifier profile epsilon=0.040

For the normalized Gaussian mollifier at epsilon=0.040, rho_epsilon(0.3)=6.08580133257e-12. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002871.

S26-E-00002873 - Gaussian mollifier profile epsilon=0.060

For the normalized Gaussian mollifier at epsilon=0.060, rho_epsilon(0.3)=2.47786585789e-05. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002872.

S26-E-00002874 - Gaussian mollifier profile epsilon=0.080

For the normalized Gaussian mollifier at epsilon=0.080, rho_epsilon(0.3)=0.00440744602959. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002873.

S26-E-00002875 - Gaussian mollifier profile epsilon=0.100

For the normalized Gaussian mollifier at epsilon=0.100, rho_epsilon(0.3)=0.0443184841194. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002874.

S26-E-00002876 - Gaussian mollifier profile epsilon=0.120

For the normalized Gaussian mollifier at epsilon=0.120, rho_epsilon(0.3)=0.14606917078. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002875.

S26-E-00002877 - Gaussian mollifier profile epsilon=0.140

For the normalized Gaussian mollifier at epsilon=0.140, rho_epsilon(0.3)=0.286864860292. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002876.

S26-E-00002878 - Gaussian mollifier profile epsilon=0.160

For the normalized Gaussian mollifier at epsilon=0.160, rho_epsilon(0.3)=0.429914223917. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002877.

S26-E-00002879 - Gaussian mollifier profile epsilon=0.180

For the normalized Gaussian mollifier at epsilon=0.180, rho_epsilon(0.3)=0.552650771071. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002878.

S26-E-00002880 - Gaussian mollifier profile epsilon=0.200

For the normalized Gaussian mollifier at epsilon=0.200, rho_epsilon(0.3)=0.647587978329. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002879.

S26-E-00002881 - Gaussian mollifier profile epsilon=0.220

For the normalized Gaussian mollifier at epsilon=0.220, rho_epsilon(0.3)=0.715650852813. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002880.

S26-E-00002882 - Gaussian mollifier profile epsilon=0.240

For the normalized Gaussian mollifier at epsilon=0.240, rho_epsilon(0.3)=0.761037855788. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002881.

S26-E-00002883 - Gaussian mollifier profile epsilon=0.260

For the normalized Gaussian mollifier at epsilon=0.260, rho_epsilon(0.3)=0.788561122133. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002882.

S26-E-00002884 - Gaussian mollifier profile epsilon=0.280

For the normalized Gaussian mollifier at epsilon=0.280, rho_epsilon(0.3)=0.802556959258. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002883.

S26-E-00002885 - Gaussian mollifier profile epsilon=0.300

For the normalized Gaussian mollifier at epsilon=0.300, rho_epsilon(0.3)=0.80656908173. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002884.

S26-E-00002886 - Gaussian mollifier profile epsilon=0.320

For the normalized Gaussian mollifier at epsilon=0.320, rho_epsilon(0.3)=0.803355960459. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002885.

S26-E-00002887 - Gaussian mollifier profile epsilon=0.340

For the normalized Gaussian mollifier at epsilon=0.340, rho_epsilon(0.3)=0.795008694242. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002886.

S26-E-00002888 - Gaussian mollifier profile epsilon=0.360

For the normalized Gaussian mollifier at epsilon=0.360, rho_epsilon(0.3)=0.783088542806. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002887.

S26-E-00002889 - Gaussian mollifier profile epsilon=0.380

For the normalized Gaussian mollifier at epsilon=0.380, rho_epsilon(0.3)=0.768750464239. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002888.

S26-E-00002890 - Gaussian mollifier profile epsilon=0.400

For the normalized Gaussian mollifier at epsilon=0.400, rho_epsilon(0.3)=0.752843580387. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002889.

S26-E-00002891 - Gaussian mollifier profile epsilon=0.420

For the normalized Gaussian mollifier at epsilon=0.420, rho_epsilon(0.3)=0.735989073379. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002890.

S26-E-00002892 - Gaussian mollifier profile epsilon=0.440

For the normalized Gaussian mollifier at epsilon=0.440, rho_epsilon(0.3)=0.718639097674. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002891.

S26-E-00002893 - Gaussian mollifier profile epsilon=0.460

For the normalized Gaussian mollifier at epsilon=0.460, rho_epsilon(0.3)=0.701120751911. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002892.

S26-E-00002894 - Gaussian mollifier profile epsilon=0.480

For the normalized Gaussian mollifier at epsilon=0.480, rho_epsilon(0.3)=0.68366868448. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002893.

S26-E-00002895 - Gaussian mollifier profile epsilon=0.500

For the normalized Gaussian mollifier at epsilon=0.500, rho_epsilon(0.3)=0.666449205784. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002894.

S26-E-00002896 - Gaussian mollifier profile epsilon=0.520

For the normalized Gaussian mollifier at epsilon=0.520, rho_epsilon(0.3)=0.649578112742. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002895.

S26-E-00002897 - Gaussian mollifier profile epsilon=0.540

For the normalized Gaussian mollifier at epsilon=0.540, rho_epsilon(0.3)=0.633133878085. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002896.

S26-E-00002898 - Gaussian mollifier profile epsilon=0.560

For the normalized Gaussian mollifier at epsilon=0.560, rho_epsilon(0.3)=0.617167426637. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002897.

S26-E-00002899 - Gaussian mollifier profile epsilon=0.580

For the normalized Gaussian mollifier at epsilon=0.580, rho_epsilon(0.3)=0.601709396676. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002898.

S26-E-00002900 - Gaussian mollifier profile epsilon=0.600

For the normalized Gaussian mollifier at epsilon=0.600, rho_epsilon(0.3)=0.586775544607. This is a smooth approximate-identity instance, not compactly supported.

Evidence M. Hard: S26-E-00002899.

Part S - Asymptotic integral methods

S26-D-00002901 - Laplace-type integral

A Laplace-type integral has the form I(lambda)=integral e^{-lambda phi(x)}a(x)dx for large positive lambda.

Evidence A. Hard: S26-E-00002900.

S26-T-00002902 - Laplace method at a nondegenerate minimum

If phi has a unique nondegenerate interior minimum x0 and a is smooth, I(lambda) has leading term a(x0)e^{-lambda phi(x0)}sqrt(2pi/(lambda phi''(x0))) in one dimension.

Evidence A. Hard: S26-D-00002901.

S26-D-00002903 - Watson lemma

Watson-type results turn a local expansion of an integrand near an endpoint into an asymptotic expansion of its Laplace transform.

Evidence A. Hard: S26-T-00002902.

S26-D-00002904 - Oscillatory integral

An oscillatory integral has rapidly varying phase e^{i lambda phi(x)} and is interpreted by ordinary or regularized integration depending on decay.

Evidence A. Hard: S26-D-00002903.

S26-T-00002905 - Nonstationary phase principle

If grad phi has no zeros on the support of a smooth compactly supported amplitude, repeated integration by parts gives decay faster than any prescribed power when the data are smooth enough.

Evidence A. Hard: S26-D-00002904.

S26-T-00002906 - Stationary phase at a nondegenerate critical point

A nondegenerate critical point contributes a lambda^{-n/2} leading term with determinant/Hessian phase factor under the standard stationary-phase theorem.

Evidence A. Hard: S26-T-00002905.

S26-D-00002907 - Steepest descent contour

A steepest-descent contour follows directions on which the real part of the complex phase decreases most rapidly away from a saddle.

Evidence A. Hard: S26-T-00002906.

S26-D-00002908 - Saddle point

A saddle point is a critical point of a complex phase used to determine dominant asymptotic contributions after contour deformation.

Evidence A. Hard: S26-D-00002907.

S26-D-00002909 - Mellin transform

M[f](s)=integral_0^infinity f(x)x^{s-1}dx on its fundamental strip.

Evidence A. Hard: S26-D-00002908.

S26-I-00002910 - Mellin scaling law

M[f(ax)](s)=a^{-s}M[f](s) for a>0 where both sides converge.

Evidence A. Hard: S26-D-00002909.

S26-D-00002911 - Fundamental strip of a Mellin transform

The fundamental strip is the vertical region in which the defining Mellin integral converges absolutely.

Evidence A. Hard: S26-I-00002910.

S26-D-00002912 - Mellin inversion interface

Under standard regularity/decay assumptions, f is recovered by a vertical-line inverse Mellin integral inside the fundamental strip.

Evidence A. Hard: S26-D-00002911.

S26-D-00002913 - Asymptotic matching

Matched asymptotics constructs local expansions in distinct regions and matches their common overlap behavior.

Evidence A. Hard: S26-D-00002912.

S26-D-00002914 - Boundary layer

A boundary layer is a thin region in which a solution varies on a different scale from the outer solution.

Evidence A. Hard: S26-D-00002913.

S26-D-00002915 - Multiple-scales expansion

The method of multiple scales introduces independent slow/fast variables to remove secular growth in perturbative approximations.

Evidence A. Hard: S26-D-00002914.

S26-D-00002916 - Poincare-Lindstedt method

Frequency is expanded along with the solution to eliminate secular terms in weakly nonlinear periodic problems.

Evidence A. Hard: S26-D-00002915.

S26-D-00002917 - Borel transform interface

A formal power series can be mapped to a factorially rescaled Borel transform to study summability beyond ordinary convergence.

Evidence A. Hard: S26-D-00002916.

S26-N-00002918 - Asymptotic-series fence

An asymptotic expansion can diverge for every fixed nonzero parameter; usefulness comes from truncated remainder behavior, not convergence of the infinite series.

Evidence A. Hard: S26-D-00002917.

S26-B-00002919 - Asymptotics to special-functions bridge

Airy, Bessel and related functions arise as canonical uniform approximants near turning points and coalescing stationary points.

Evidence A. Hard: S26-N-00002918.

S26-B-00002920 - Asymptotics to computation bridge

Optimal truncation, exponentially small remainders and uniformity regions must be separated from mere agreement at sampled parameter values.

Evidence A. Hard: S26-B-00002919.

S26-E-00002921 - Gaussian Laplace profile lambda=2

Integral_R exp(-lambda x^2)dx at lambda=2 equals sqrt(pi/lambda)=1.25331413732; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-B-00002920.

S26-E-00002922 - Gaussian Laplace profile lambda=4

Integral_R exp(-lambda x^2)dx at lambda=4 equals sqrt(pi/lambda)=0.886226925453; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002921.

S26-E-00002923 - Gaussian Laplace profile lambda=6

Integral_R exp(-lambda x^2)dx at lambda=6 equals sqrt(pi/lambda)=0.723601254558; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002922.

S26-E-00002924 - Gaussian Laplace profile lambda=8

Integral_R exp(-lambda x^2)dx at lambda=8 equals sqrt(pi/lambda)=0.626657068658; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002923.

S26-E-00002925 - Gaussian Laplace profile lambda=10

Integral_R exp(-lambda x^2)dx at lambda=10 equals sqrt(pi/lambda)=0.56049912164; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002924.

S26-E-00002926 - Gaussian Laplace profile lambda=12

Integral_R exp(-lambda x^2)dx at lambda=12 equals sqrt(pi/lambda)=0.511663353973; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002925.

S26-E-00002927 - Gaussian Laplace profile lambda=14

Integral_R exp(-lambda x^2)dx at lambda=14 equals sqrt(pi/lambda)=0.473708217425; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002926.

S26-E-00002928 - Gaussian Laplace profile lambda=16

Integral_R exp(-lambda x^2)dx at lambda=16 equals sqrt(pi/lambda)=0.443113462726; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002927.

S26-E-00002929 - Gaussian Laplace profile lambda=18

Integral_R exp(-lambda x^2)dx at lambda=18 equals sqrt(pi/lambda)=0.417771379105; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002928.

S26-E-00002930 - Gaussian Laplace profile lambda=20

Integral_R exp(-lambda x^2)dx at lambda=20 equals sqrt(pi/lambda)=0.396332729761; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002929.

S26-E-00002931 - Gaussian Laplace profile lambda=22

Integral_R exp(-lambda x^2)dx at lambda=22 equals sqrt(pi/lambda)=0.37788843072; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002930.

S26-E-00002932 - Gaussian Laplace profile lambda=24

Integral_R exp(-lambda x^2)dx at lambda=24 equals sqrt(pi/lambda)=0.361800627279; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002931.

S26-E-00002933 - Gaussian Laplace profile lambda=26

Integral_R exp(-lambda x^2)dx at lambda=26 equals sqrt(pi/lambda)=0.34760679895; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002932.

S26-E-00002934 - Gaussian Laplace profile lambda=28

Integral_R exp(-lambda x^2)dx at lambda=28 equals sqrt(pi/lambda)=0.334962292845; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002933.

S26-E-00002935 - Gaussian Laplace profile lambda=30

Integral_R exp(-lambda x^2)dx at lambda=30 equals sqrt(pi/lambda)=0.323604318759; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002934.

S26-E-00002936 - Gaussian Laplace profile lambda=32

Integral_R exp(-lambda x^2)dx at lambda=32 equals sqrt(pi/lambda)=0.313328534329; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002935.

S26-E-00002937 - Gaussian Laplace profile lambda=34

Integral_R exp(-lambda x^2)dx at lambda=34 equals sqrt(pi/lambda)=0.303973327661; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002936.

S26-E-00002938 - Gaussian Laplace profile lambda=36

Integral_R exp(-lambda x^2)dx at lambda=36 equals sqrt(pi/lambda)=0.295408975151; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002937.

S26-E-00002939 - Gaussian Laplace profile lambda=38

Integral_R exp(-lambda x^2)dx at lambda=38 equals sqrt(pi/lambda)=0.287529982583; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002938.

S26-E-00002940 - Gaussian Laplace profile lambda=40

Integral_R exp(-lambda x^2)dx at lambda=40 equals sqrt(pi/lambda)=0.28024956082; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002939.

S26-E-00002941 - Gaussian Laplace profile lambda=42

Integral_R exp(-lambda x^2)dx at lambda=42 equals sqrt(pi/lambda)=0.273495566848; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002940.

S26-E-00002942 - Gaussian Laplace profile lambda=44

Integral_R exp(-lambda x^2)dx at lambda=44 equals sqrt(pi/lambda)=0.267207471894; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002941.

S26-E-00002943 - Gaussian Laplace profile lambda=46

Integral_R exp(-lambda x^2)dx at lambda=46 equals sqrt(pi/lambda)=0.261334062972; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002942.

S26-E-00002944 - Gaussian Laplace profile lambda=48

Integral_R exp(-lambda x^2)dx at lambda=48 equals sqrt(pi/lambda)=0.255831676987; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002943.

S26-E-00002945 - Gaussian Laplace profile lambda=50

Integral_R exp(-lambda x^2)dx at lambda=50 equals sqrt(pi/lambda)=0.250662827463; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002944.

S26-E-00002946 - Gaussian Laplace profile lambda=52

Integral_R exp(-lambda x^2)dx at lambda=52 equals sqrt(pi/lambda)=0.245795124724; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002945.

S26-E-00002947 - Gaussian Laplace profile lambda=54

Integral_R exp(-lambda x^2)dx at lambda=54 equals sqrt(pi/lambda)=0.241200418186; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002946.

S26-E-00002948 - Gaussian Laplace profile lambda=56

Integral_R exp(-lambda x^2)dx at lambda=56 equals sqrt(pi/lambda)=0.236854108713; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002947.

S26-E-00002949 - Gaussian Laplace profile lambda=58

Integral_R exp(-lambda x^2)dx at lambda=58 equals sqrt(pi/lambda)=0.232734592571; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002948.

S26-E-00002950 - Gaussian Laplace profile lambda=60

Integral_R exp(-lambda x^2)dx at lambda=60 equals sqrt(pi/lambda)=0.228822808216; the quadratic Laplace leading term is exact in this case.

Evidence M. Hard: S26-E-00002949.

Part T - Semigroups and evolution analysis

S26-D-00002951 - Strongly continuous operator semigroup

A family T(t), t>=0, is a C0 semigroup when T(0)=I, T(t+s)=T(t)T(s), and T(t)x->x as t downarrow 0 for every x.

Evidence A. Hard: S26-E-00002950.

S26-D-00002952 - Infinitesimal generator of a C0 semigroup

Ax=lim_{t downarrow0}(T(t)x-x)/t on the set of x for which the strong limit exists.

Evidence A. Hard: S26-D-00002951.

S26-T-00002953 - Generator is closed and densely defined

The generator of a C0 semigroup on a Banach space is a closed densely defined operator.

Evidence A. Hard: S26-D-00002952.

S26-D-00002954 - Growth bound of a semigroup

omega_0 is the infimum of omega for which ||T(t)||<=M e^{omega t} holds for some M and all t>=0.

Evidence A. Hard: S26-T-00002953.

S26-T-00002955 - Hille-Yosida theorem interface

Closed densely defined operators satisfying the Hille-Yosida resolvent bounds are exactly generators of contraction or bounded-growth C0 semigroups in the corresponding form.

Evidence A. Hard: S26-D-00002954.

S26-D-00002956 - Analytic semigroup

A C0 semigroup is analytic when it extends holomorphically to a sector and satisfies the compatible semigroup law there.

Evidence A. Hard: S26-T-00002955.

S26-D-00002957 - Mild solution of abstract Cauchy problem

For u'=Au+f, a mild solution satisfies u(t)=T(t)u0+integral_0^t T(t-s)f(s)ds.

Evidence A. Hard: S26-D-00002956.

S26-D-00002958 - Classical solution of abstract Cauchy problem

A classical solution is differentiable in time, lies in D(A), and satisfies u'=Au+f pointwise in the Banach space.

Evidence A. Hard: S26-D-00002957.

S26-T-00002959 - Variation-of-constants formula

The mild-solution formula is the semigroup form of Duhamel variation of constants under the standard integrability hypotheses.

Evidence A. Hard: S26-D-00002958.

S26-D-00002960 - Exponential stability

A semigroup is exponentially stable when ||T(t)||<=M e^{-omega t} for some M,omega>0.

Evidence A. Hard: S26-T-00002959.

S26-D-00002961 - Spectral bound

s(A)=sup{Re lambda:lambda in sigma(A)}.

Evidence A. Hard: S26-D-00002960.

S26-N-00002962 - Growth-spectral bound fence

In general Banach spaces the semigroup growth bound need not equal the spectral bound without additional hypotheses.

Evidence A. Hard: S26-D-00002961.

S26-D-00002963 - Resolvent Laplace representation

For Re lambda above the growth bound, R(lambda,A)x=integral_0^infinity e^{-lambda t}T(t)x dt.

Evidence A. Hard: S26-N-00002962.

S26-D-00002964 - Evolution family

For nonautonomous problems, U(t,s) generalizes a semigroup with U(t,r)U(r,s)=U(t,s).

Evidence A. Hard: S26-D-00002963.

S26-D-00002965 - Maximal regularity interface

Maximal Lp regularity asks whether f in Lp forces u' and Au to belong to Lp for the abstract evolution equation.

Evidence A. Hard: S26-D-00002964.

S26-D-00002966 - Trotter product interface

Product formulas approximate a semigroup generated by A+B through alternating short-time factors generated by A and B under appropriate hypotheses.

Evidence A. Hard: S26-D-00002965.

S26-D-00002967 - Chernoff approximation interface

Chernoff-type theorems derive semigroup limits from consistent families of approximate evolution operators.

Evidence A. Hard: S26-D-00002966.

S26-D-00002968 - Fractional evolution equation interface

Replacing the first time derivative by a fractional derivative leads to solution operators built from sectorial calculus/subordination rather than an ordinary semigroup in time.

Evidence A. Hard: S26-D-00002967.

S26-B-00002969 - Semigroup to PDE bridge

Heat, diffusion and many linear evolution PDE are abstract Cauchy problems generated by differential operators on Banach/Hilbert spaces.

Evidence A. Hard: S26-D-00002968.

S26-B-00002970 - Semigroup to probability bridge

Markov semigroups encode transition evolution of stochastic processes; positivity and mass preservation are additional structures beyond the C0 semigroup axioms.

Evidence A. Hard: S26-B-00002969.

S26-E-00002971 - Contractive diagonal semigroup profile t=0.1

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.904837418036, 0.818730753078, 0.740818220682] at t=0.1; its 2-norm is 0.904837418036.

Evidence M. Hard: S26-B-00002970.

S26-E-00002972 - Contractive diagonal semigroup profile t=0.2

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.818730753078, 0.670320046036, 0.548811636094] at t=0.2; its 2-norm is 0.818730753078.

Evidence M. Hard: S26-E-00002971.

S26-E-00002973 - Contractive diagonal semigroup profile t=0.3

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.740818220682, 0.548811636094, 0.406569659741] at t=0.3; its 2-norm is 0.740818220682.

Evidence M. Hard: S26-E-00002972.

S26-E-00002974 - Contractive diagonal semigroup profile t=0.4

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.670320046036, 0.449328964117, 0.301194211912] at t=0.4; its 2-norm is 0.670320046036.

Evidence M. Hard: S26-E-00002973.

S26-E-00002975 - Contractive diagonal semigroup profile t=0.5

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.606530659713, 0.367879441171, 0.223130160148] at t=0.5; its 2-norm is 0.606530659713.

Evidence M. Hard: S26-E-00002974.

S26-E-00002976 - Contractive diagonal semigroup profile t=0.6

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.548811636094, 0.301194211912, 0.165298888222] at t=0.6; its 2-norm is 0.548811636094.

Evidence M. Hard: S26-E-00002975.

S26-E-00002977 - Contractive diagonal semigroup profile t=0.7

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.496585303791, 0.246596963942, 0.122456428253] at t=0.7; its 2-norm is 0.496585303791.

Evidence M. Hard: S26-E-00002976.

S26-E-00002978 - Contractive diagonal semigroup profile t=0.8

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.449328964117, 0.201896517995, 0.090717953289] at t=0.8; its 2-norm is 0.449328964117.

Evidence M. Hard: S26-E-00002977.

S26-E-00002979 - Contractive diagonal semigroup profile t=0.9

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.406569659741, 0.165298888222, 0.06720551274] at t=0.9; its 2-norm is 0.406569659741.

Evidence M. Hard: S26-E-00002978.

S26-E-00002980 - Contractive diagonal semigroup profile t=1.0

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.367879441171, 0.135335283237, 0.049787068368] at t=1.0; its 2-norm is 0.367879441171.

Evidence M. Hard: S26-E-00002979.

S26-E-00002981 - Contractive diagonal semigroup profile t=1.1

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.332871083698, 0.110803158362, 0.036883167401] at t=1.1; its 2-norm is 0.332871083698.

Evidence M. Hard: S26-E-00002980.

S26-E-00002982 - Contractive diagonal semigroup profile t=1.2

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.301194211912, 0.090717953289, 0.027323722447] at t=1.2; its 2-norm is 0.301194211912.

Evidence M. Hard: S26-E-00002981.

S26-E-00002983 - Contractive diagonal semigroup profile t=1.3

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.272531793034, 0.074273578214, 0.020241911446] at t=1.3; its 2-norm is 0.272531793034.

Evidence M. Hard: S26-E-00002982.

S26-E-00002984 - Contractive diagonal semigroup profile t=1.4

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.246596963942, 0.060810062625, 0.01499557682] at t=1.4; its 2-norm is 0.246596963942.

Evidence M. Hard: S26-E-00002983.

S26-E-00002985 - Contractive diagonal semigroup profile t=1.5

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.223130160148, 0.049787068368, 0.011108996538] at t=1.5; its 2-norm is 0.223130160148.

Evidence M. Hard: S26-E-00002984.

S26-E-00002986 - Contractive diagonal semigroup profile t=1.6

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.201896517995, 0.040762203978, 0.008229747049] at t=1.6; its 2-norm is 0.201896517995.

Evidence M. Hard: S26-E-00002985.

S26-E-00002987 - Contractive diagonal semigroup profile t=1.7

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.182683524053, 0.03337326996, 0.006096746566] at t=1.7; its 2-norm is 0.182683524053.

Evidence M. Hard: S26-E-00002986.

S26-E-00002988 - Contractive diagonal semigroup profile t=1.8

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.165298888222, 0.027323722447, 0.004516580943] at t=1.8; its 2-norm is 0.165298888222.

Evidence M. Hard: S26-E-00002987.

S26-E-00002989 - Contractive diagonal semigroup profile t=1.9

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.149568619223, 0.022370771856, 0.003345965457] at t=1.9; its 2-norm is 0.149568619223.

Evidence M. Hard: S26-E-00002988.

S26-E-00002990 - Contractive diagonal semigroup profile t=2.0

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.135335283237, 0.018315638889, 0.002478752177] at t=2.0; its 2-norm is 0.135335283237.

Evidence M. Hard: S26-E-00002989.

S26-E-00002991 - Contractive diagonal semigroup profile t=2.1

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.122456428253, 0.01499557682, 0.001836304777] at t=2.1; its 2-norm is 0.122456428253.

Evidence M. Hard: S26-E-00002990.

S26-E-00002992 - Contractive diagonal semigroup profile t=2.2

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.110803158362, 0.012277339903, 0.001360368038] at t=2.2; its 2-norm is 0.110803158362.

Evidence M. Hard: S26-E-00002991.

S26-E-00002993 - Contractive diagonal semigroup profile t=2.3

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.100258843723, 0.010051835745, 0.001007785429] at t=2.3; its 2-norm is 0.100258843723.

Evidence M. Hard: S26-E-00002992.

S26-E-00002994 - Contractive diagonal semigroup profile t=2.4

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.090717953289, 0.008229747049, 0.000746585808] at t=2.4; its 2-norm is 0.0907179532894.

Evidence M. Hard: S26-E-00002993.

S26-E-00002995 - Contractive diagonal semigroup profile t=2.5

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.082084998624, 0.006737946999, 0.00055308437] at t=2.5; its 2-norm is 0.0820849986239.

Evidence M. Hard: S26-E-00002994.

S26-E-00002996 - Contractive diagonal semigroup profile t=2.6

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.074273578214, 0.005516564421, 0.000409734979] at t=2.6; its 2-norm is 0.0742735782143.

Evidence M. Hard: S26-E-00002995.

S26-E-00002997 - Contractive diagonal semigroup profile t=2.7

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.06720551274, 0.004516580943, 0.000303539138] at t=2.7; its 2-norm is 0.0672055127397.

Evidence M. Hard: S26-E-00002996.

S26-E-00002998 - Contractive diagonal semigroup profile t=2.8

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.060810062625, 0.003697863716, 0.000224867324] at t=2.8; its 2-norm is 0.0608100626252.

Evidence M. Hard: S26-E-00002997.

S26-E-00002999 - Contractive diagonal semigroup profile t=2.9

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.055023220056, 0.003027554745, 0.000166585811] at t=2.9; its 2-norm is 0.0550232200564.

Evidence M. Hard: S26-E-00002998.

S26-E-00003000 - Contractive diagonal semigroup profile t=3.0

For A=diag(-1,-2,-3), e^(tA) has diagonal [0.049787068368, 0.002478752177, 0.000123409804] at t=3.0; its 2-norm is 0.0497870683679.

Evidence M. Hard: S26-E-00002999.

Appendix - Peer-review boundary

This reviewed projection changes no canonical IDs and performs no A-to-P promotion. Statement-level corrections are recorded in the consolidated correction ledger. The assumptions[] metadata debt identified for Volumes I-III remains a separate formalization task.

A Synopsis of Results in Pure, Applied and Computational Mathematics

2026 Edition

Volume III
Geometry & Topology

Peer-reviewed consolidated projection

Base release: 1.0.0
Reviewed projection: 1.0.1-pr
Canonical range: 3001-4000
Objects: 1,000
Peer-review corrections in this volume: 1
Combined review graph: 5225 hard edges; 1 component; 0 cycles; 0 dangling
Volume evidence unchanged: no P promotions

This projection applies the consolidated peer-review correction ledger while preserving canonical IDs and evidence states. Frozen original release files remain unchanged.

Peer-review corrections in this volume

S26-T-00003053 - foundation-scope clarification

Reviewed statement. In the classical ZFC reading used for this theorem, an arbitrary product of compact spaces is compact in the product topology.

Reason. The project foundation layer distinguishes ZF/ZFC contexts; arbitrary-product Tychonoff requires an explicit classical choice convention.

Contents

Part A - Point-set topology refinements (3001-3050)
Part B - Products, quotients, compactness and connectedness (3051-3100)
Part C - Homotopy and fundamental groups (3101-3150)
Part D - Covering spaces and monodromy (3151-3200)
Part E - Simplicial complexes and homology (3201-3250)
Part F - Singular homology, excision and exact sequences (3251-3300)
Part G - Cohomology rings and duality interfaces (3301-3350)
Part H - CW complexes and cellular methods (3351-3400)
Part I - Smooth manifolds, maps and submanifolds (3401-3450)
Part J - Tangent geometry, vector fields and flows (3451-3500)
Part K - Differential forms, de Rham theory and integration (3501-3550)
Part L - Fiber bundles, vector bundles and principal bundles (3551-3600)
Part M - Characteristic classes and classifying spaces (3601-3650)
Part N - Riemannian metrics, geodesics and completeness (3651-3700)
Part O - Curvature tensors and comparison geometry (3701-3750)
Part P - Global and submanifold geometry (3751-3800)
Part Q - Morse theory and handle interfaces (3801-3850)
Part R - Lie group actions and homogeneous geometry (3851-3900)
Part S - Spectral sequences and obstruction interfaces (3901-3950)
Part T - Low-dimensional and geometric topology (3951-4000)

Part A - Point-set topology refinements

S26-D-00003001 - Topology generated by a subbasis

A subbasis S on X generates the smallest topology containing S; finite intersections of subbasic sets form a basis.

Evidence A. Hard: S26-D-00000901.

S26-D-00003002 - Initial topology of a family of maps

The initial topology on X induced by maps f_i:X->Y_i is the coarsest topology making every f_i continuous.

Evidence A. Hard: S26-D-00003001.

S26-D-00003003 - Final topology of a family of maps

The final topology on Y induced by maps g_i:X_i->Y is the finest topology making every g_i continuous.

Evidence A. Hard: S26-D-00003002.

S26-D-00003004 - Nets in a topological space

A net is a function from a directed set into a space and generalizes sequences for detecting topology.

Evidence A. Hard: S26-D-00003003.

S26-T-00003005 - Net characterization of closure

x lies in the closure of A iff some net in A converges to x.

Evidence A. Hard: S26-D-00003004.

S26-T-00003006 - Net characterization of continuity

A map is continuous iff it preserves convergence of all nets.

Evidence A. Hard: S26-T-00003005.

S26-D-00003007 - Filters and filter convergence

A filter is an upward-closed finite-intersection family of subsets; it converges to x when every neighborhood of x belongs to the filter.

Evidence A. Hard: S26-T-00003006.

S26-D-00003008 - Ultrafilter

An ultrafilter is a maximal proper filter; equivalently, for every subset A, exactly one of A or its complement belongs to it.

Evidence A. Hard: S26-D-00003007.

S26-T-00003009 - Compactness by ultrafilters

A topological space is compact iff every ultrafilter on it converges.

Evidence A. Hard: S26-D-00003008.

S26-D-00003010 - First-countable space

A space is first countable when each point has a countable neighborhood basis.

Evidence A. Hard: S26-T-00003009.

S26-D-00003011 - Second-countable space

A space is second countable when its topology has a countable basis.

Evidence A. Hard: S26-D-00003010.

S26-T-00003012 - Second countable implies separable

Every second-countable space is separable.

Evidence A. Hard: S26-D-00003011.

S26-T-00003013 - Second countable implies Lindelof

Every second-countable space is Lindelof.

Evidence A. Hard: S26-T-00003012.

S26-D-00003014 - Regular topological space

A T1 space is regular when points and disjoint closed sets can be separated by neighborhoods.

Evidence A. Hard: S26-T-00003013.

S26-D-00003015 - Normal topological space

A T1 space is normal when disjoint closed sets have disjoint neighborhoods.

Evidence A. Hard: S26-D-00003014.

S26-T-00003016 - Urysohn lemma

In a normal space, disjoint closed sets can be separated by a continuous map to [0,1].

Evidence A. Hard: S26-D-00003015.

S26-T-00003017 - Tietze extension theorem

A continuous real-valued function on a closed subset of a normal space extends continuously to the whole space.

Evidence A. Hard: S26-T-00003016.

S26-N-00003018 - Sequence-detection fence

Sequential tests characterize closure and continuity only in suitable spaces; arbitrary topological spaces may require nets or filters.

Evidence A. Hard: S26-T-00003017. Boundary: There are nonsequential spaces where sequential closure is strictly smaller than closure.

S26-B-00003019 - Topology to analysis bridge

Initial/final topologies, nets and compactness principles supply the language behind weak topologies, convergence of measures, and function-space constructions.

Evidence A. Hard: S26-N-00003018.

S26-B-00003020 - Topology to geometry bridge

Countability and separation axioms are structural hypotheses used in manifold definitions, partitions of unity, embeddings and metrization.

Evidence A. Hard: S26-B-00003019.

S26-E-00003021 - Finite topology cardinality bounds n=3

On a labeled 3-point set, the indiscrete topology has 2 open sets and the discrete topology has 8 open sets; these are exact finite counts.

Evidence M. Hard: S26-B-00003020.

S26-E-00003022 - Finite topology cardinality bounds n=4

On a labeled 4-point set, the indiscrete topology has 2 open sets and the discrete topology has 16 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003021.

S26-E-00003023 - Finite topology cardinality bounds n=5

On a labeled 5-point set, the indiscrete topology has 2 open sets and the discrete topology has 32 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003022.

S26-E-00003024 - Finite topology cardinality bounds n=6

On a labeled 6-point set, the indiscrete topology has 2 open sets and the discrete topology has 64 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003023.

S26-E-00003025 - Finite topology cardinality bounds n=7

On a labeled 7-point set, the indiscrete topology has 2 open sets and the discrete topology has 128 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003024.

S26-E-00003026 - Finite topology cardinality bounds n=8

On a labeled 8-point set, the indiscrete topology has 2 open sets and the discrete topology has 256 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003025.

S26-E-00003027 - Finite topology cardinality bounds n=9

On a labeled 9-point set, the indiscrete topology has 2 open sets and the discrete topology has 512 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003026.

S26-E-00003028 - Finite topology cardinality bounds n=10

On a labeled 10-point set, the indiscrete topology has 2 open sets and the discrete topology has 1024 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003027.

S26-E-00003029 - Finite topology cardinality bounds n=11

On a labeled 11-point set, the indiscrete topology has 2 open sets and the discrete topology has 2048 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003028.

S26-E-00003030 - Finite topology cardinality bounds n=12

On a labeled 12-point set, the indiscrete topology has 2 open sets and the discrete topology has 4096 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003029.

S26-E-00003031 - Finite topology cardinality bounds n=13

On a labeled 13-point set, the indiscrete topology has 2 open sets and the discrete topology has 8192 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003030.

S26-E-00003032 - Finite topology cardinality bounds n=14

On a labeled 14-point set, the indiscrete topology has 2 open sets and the discrete topology has 16384 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003031.

S26-E-00003033 - Finite topology cardinality bounds n=15

On a labeled 15-point set, the indiscrete topology has 2 open sets and the discrete topology has 32768 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003032.

S26-E-00003034 - Finite topology cardinality bounds n=16

On a labeled 16-point set, the indiscrete topology has 2 open sets and the discrete topology has 65536 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003033.

S26-E-00003035 - Finite topology cardinality bounds n=17

On a labeled 17-point set, the indiscrete topology has 2 open sets and the discrete topology has 131072 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003034.

S26-E-00003036 - Finite topology cardinality bounds n=18

On a labeled 18-point set, the indiscrete topology has 2 open sets and the discrete topology has 262144 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003035.

S26-E-00003037 - Finite topology cardinality bounds n=19

On a labeled 19-point set, the indiscrete topology has 2 open sets and the discrete topology has 524288 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003036.

S26-E-00003038 - Finite topology cardinality bounds n=20

On a labeled 20-point set, the indiscrete topology has 2 open sets and the discrete topology has 1048576 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003037.

S26-E-00003039 - Finite topology cardinality bounds n=21

On a labeled 21-point set, the indiscrete topology has 2 open sets and the discrete topology has 2097152 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003038.

S26-E-00003040 - Finite topology cardinality bounds n=22

On a labeled 22-point set, the indiscrete topology has 2 open sets and the discrete topology has 4194304 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003039.

S26-E-00003041 - Finite topology cardinality bounds n=23

On a labeled 23-point set, the indiscrete topology has 2 open sets and the discrete topology has 8388608 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003040.

S26-E-00003042 - Finite topology cardinality bounds n=24

On a labeled 24-point set, the indiscrete topology has 2 open sets and the discrete topology has 16777216 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003041.

S26-E-00003043 - Finite topology cardinality bounds n=25

On a labeled 25-point set, the indiscrete topology has 2 open sets and the discrete topology has 33554432 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003042.

S26-E-00003044 - Finite topology cardinality bounds n=26

On a labeled 26-point set, the indiscrete topology has 2 open sets and the discrete topology has 67108864 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003043.

S26-E-00003045 - Finite topology cardinality bounds n=27

On a labeled 27-point set, the indiscrete topology has 2 open sets and the discrete topology has 134217728 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003044.

S26-E-00003046 - Finite topology cardinality bounds n=28

On a labeled 28-point set, the indiscrete topology has 2 open sets and the discrete topology has 268435456 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003045.

S26-E-00003047 - Finite topology cardinality bounds n=29

On a labeled 29-point set, the indiscrete topology has 2 open sets and the discrete topology has 536870912 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003046.

S26-E-00003048 - Finite topology cardinality bounds n=30

On a labeled 30-point set, the indiscrete topology has 2 open sets and the discrete topology has 1073741824 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003047.

S26-E-00003049 - Finite topology cardinality bounds n=31

On a labeled 31-point set, the indiscrete topology has 2 open sets and the discrete topology has 2147483648 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003048.

S26-E-00003050 - Finite topology cardinality bounds n=32

On a labeled 32-point set, the indiscrete topology has 2 open sets and the discrete topology has 4294967296 open sets; these are exact finite counts.

Evidence M. Hard: S26-E-00003049.

Part B - Products, quotients, compactness and connectedness

S26-D-00003051 - Product topology on an arbitrary family

The product topology on product_i X_i is generated by inverse images of open sets under coordinate projections, with finitely many constrained coordinates in a basic open set.

Evidence A. Hard: S26-E-00003050.

S26-T-00003052 - Universal property of product topology

A map f:Y->product_i X_i is continuous iff every coordinate projection pi_i∘f is continuous.

Evidence A. Hard: S26-D-00003051.

S26-T-00003053 - Tychonoff theorem

In the classical ZFC reading used for this theorem, an arbitrary product of compact spaces is compact in the product topology.

Evidence A. Hard: S26-T-00003052. Peer-review correction applied in reviewed projection.

S26-D-00003054 - Box topology

The box topology permits arbitrary products of open coordinate sets and is generally finer than the product topology.

Evidence A. Hard: S26-T-00003053.

S26-N-00003055 - Product-box fence

For infinitely many factors the product and box topologies generally differ; compactness results for product topology do not transfer automatically to box topology.

Evidence A. Hard: S26-D-00003054.

S26-D-00003056 - Quotient topology

For a surjection q:X->Y, the quotient topology on Y declares U open exactly when q^{-1}(U) is open in X.

Evidence A. Hard: S26-N-00003055.

S26-T-00003057 - Universal property of quotient topology

A map f:Y->Z from a quotient space is continuous iff f∘q is continuous, provided f is well-defined on equivalence classes.

Evidence A. Hard: S26-D-00003056.

S26-D-00003058 - Identification map

A quotient map is a surjective continuous map q for which UsubsetY is open iff q^{-1}(U) is open.

Evidence A. Hard: S26-T-00003057.

S26-D-00003059 - Topological sum

The disjoint union topology is the final topology for the inclusions of the summands.

Evidence A. Hard: S26-D-00003058.

S26-D-00003060 - One-point compactification

A noncompact locally compact Hausdorff space admits a compactification by adjoining one point whose neighborhoods complement compact subsets.

Evidence A. Hard: S26-D-00003059.

S26-T-00003061 - Continuous image of compact is compact

A continuous image of a compact space is compact.

Evidence A. Hard: S26-D-00003060.

S26-T-00003062 - Compact subsets of Hausdorff spaces are closed

Every compact subset of a Hausdorff space is closed.

Evidence A. Hard: S26-T-00003061.

S26-D-00003063 - Local compactness

A space is locally compact when each point has a neighborhood base with compact closure in the chosen standard formulation.

Evidence A. Hard: S26-T-00003062.

S26-D-00003064 - Path component

The path component of x is the set of points joinable to x by a continuous path.

Evidence A. Hard: S26-D-00003063.

S26-D-00003065 - Connected component

A connected component is a maximal connected subset.

Evidence A. Hard: S26-D-00003064.

S26-T-00003066 - Path connected implies connected

Every path-connected space is connected.

Evidence A. Hard: S26-D-00003065.

S26-N-00003067 - Connected-path-connected fence

Connectedness does not imply path connectedness without extra hypotheses.

Evidence A. Hard: S26-T-00003066. Boundary: The topologist sine curve provides a standard connected but non-path-connected example.

S26-D-00003068 - Local connectedness

A space is locally connected when every point has a neighborhood basis of connected open sets.

Evidence A. Hard: S26-N-00003067.

S26-D-00003069 - Local path connectedness

A space is locally path connected when every point has a neighborhood basis of path-connected open sets.

Evidence A. Hard: S26-D-00003068.

S26-B-00003070 - Quotient-gluing bridge

Quotient topology formalizes gluing constructions used for CW complexes, manifolds, orbit spaces, projective spaces and identification spaces.

Evidence A. Hard: S26-D-00003069.

S26-E-00003071 - Disjoint-union component profile n=3

The disjoint union of 3 closed intervals has exactly 3 connected components and Euler characteristic 3; this is an exact finite CW count.

Evidence M. Hard: S26-B-00003070.

S26-E-00003072 - Disjoint-union component profile n=4

The disjoint union of 4 closed intervals has exactly 4 connected components and Euler characteristic 4; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003071.

S26-E-00003073 - Disjoint-union component profile n=5

The disjoint union of 5 closed intervals has exactly 5 connected components and Euler characteristic 5; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003072.

S26-E-00003074 - Disjoint-union component profile n=6

The disjoint union of 6 closed intervals has exactly 6 connected components and Euler characteristic 6; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003073.

S26-E-00003075 - Disjoint-union component profile n=7

The disjoint union of 7 closed intervals has exactly 7 connected components and Euler characteristic 7; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003074.

S26-E-00003076 - Disjoint-union component profile n=8

The disjoint union of 8 closed intervals has exactly 8 connected components and Euler characteristic 8; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003075.

S26-E-00003077 - Disjoint-union component profile n=9

The disjoint union of 9 closed intervals has exactly 9 connected components and Euler characteristic 9; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003076.

S26-E-00003078 - Disjoint-union component profile n=10

The disjoint union of 10 closed intervals has exactly 10 connected components and Euler characteristic 10; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003077.

S26-E-00003079 - Disjoint-union component profile n=11

The disjoint union of 11 closed intervals has exactly 11 connected components and Euler characteristic 11; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003078.

S26-E-00003080 - Disjoint-union component profile n=12

The disjoint union of 12 closed intervals has exactly 12 connected components and Euler characteristic 12; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003079.

S26-E-00003081 - Disjoint-union component profile n=13

The disjoint union of 13 closed intervals has exactly 13 connected components and Euler characteristic 13; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003080.

S26-E-00003082 - Disjoint-union component profile n=14

The disjoint union of 14 closed intervals has exactly 14 connected components and Euler characteristic 14; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003081.

S26-E-00003083 - Disjoint-union component profile n=15

The disjoint union of 15 closed intervals has exactly 15 connected components and Euler characteristic 15; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003082.

S26-E-00003084 - Disjoint-union component profile n=16

The disjoint union of 16 closed intervals has exactly 16 connected components and Euler characteristic 16; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003083.

S26-E-00003085 - Disjoint-union component profile n=17

The disjoint union of 17 closed intervals has exactly 17 connected components and Euler characteristic 17; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003084.

S26-E-00003086 - Disjoint-union component profile n=18

The disjoint union of 18 closed intervals has exactly 18 connected components and Euler characteristic 18; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003085.

S26-E-00003087 - Disjoint-union component profile n=19

The disjoint union of 19 closed intervals has exactly 19 connected components and Euler characteristic 19; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003086.

S26-E-00003088 - Disjoint-union component profile n=20

The disjoint union of 20 closed intervals has exactly 20 connected components and Euler characteristic 20; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003087.

S26-E-00003089 - Disjoint-union component profile n=21

The disjoint union of 21 closed intervals has exactly 21 connected components and Euler characteristic 21; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003088.

S26-E-00003090 - Disjoint-union component profile n=22

The disjoint union of 22 closed intervals has exactly 22 connected components and Euler characteristic 22; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003089.

S26-E-00003091 - Disjoint-union component profile n=23

The disjoint union of 23 closed intervals has exactly 23 connected components and Euler characteristic 23; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003090.

S26-E-00003092 - Disjoint-union component profile n=24

The disjoint union of 24 closed intervals has exactly 24 connected components and Euler characteristic 24; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003091.

S26-E-00003093 - Disjoint-union component profile n=25

The disjoint union of 25 closed intervals has exactly 25 connected components and Euler characteristic 25; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003092.

S26-E-00003094 - Disjoint-union component profile n=26

The disjoint union of 26 closed intervals has exactly 26 connected components and Euler characteristic 26; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003093.

S26-E-00003095 - Disjoint-union component profile n=27

The disjoint union of 27 closed intervals has exactly 27 connected components and Euler characteristic 27; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003094.

S26-E-00003096 - Disjoint-union component profile n=28

The disjoint union of 28 closed intervals has exactly 28 connected components and Euler characteristic 28; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003095.

S26-E-00003097 - Disjoint-union component profile n=29

The disjoint union of 29 closed intervals has exactly 29 connected components and Euler characteristic 29; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003096.

S26-E-00003098 - Disjoint-union component profile n=30

The disjoint union of 30 closed intervals has exactly 30 connected components and Euler characteristic 30; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003097.

S26-E-00003099 - Disjoint-union component profile n=31

The disjoint union of 31 closed intervals has exactly 31 connected components and Euler characteristic 31; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003098.

S26-E-00003100 - Disjoint-union component profile n=32

The disjoint union of 32 closed intervals has exactly 32 connected components and Euler characteristic 32; this is an exact finite CW count.

Evidence M. Hard: S26-E-00003099.

Part C - Homotopy and fundamental groups

S26-D-00003101 - Homotopy of maps

Maps f,g:X->Y are homotopic when there is a continuous H:Xx[0,1]->Y with H(x,0)=f(x) and H(x,1)=g(x).

Evidence A. Hard: S26-E-00003100.

S26-D-00003102 - Homotopy equivalence

Spaces X and Y are homotopy equivalent when maps f:X->Y and g:Y->X exist with g∘f homotopic to id_X and f∘g homotopic to id_Y.

Evidence A. Hard: S26-D-00003101.

S26-D-00003103 - Contractible space

A space is contractible when its identity map is homotopic to a constant map.

Evidence A. Hard: S26-D-00003102.

S26-D-00003104 - Based loop

A based loop at x0 is a path gamma:[0,1]->X with gamma(0)=gamma(1)=x0.

Evidence A. Hard: S26-D-00003103.

S26-D-00003105 - Path homotopy relative endpoints

Two paths with the same endpoints are homotopic rel endpoints when a homotopy keeps both endpoints fixed throughout.

Evidence A. Hard: S26-D-00003104.

S26-D-00003106 - Fundamental group

pi_1(X,x0) is the group of based loops modulo path homotopy, with multiplication induced by concatenation.

Evidence A. Hard: S26-D-00003105.

S26-T-00003107 - Homotopy invariance of fundamental group

A based homotopy equivalence induces an isomorphism of fundamental groups, with the usual basepoint bookkeeping.

Evidence A. Hard: S26-D-00003106.

S26-T-00003108 - Fundamental group of the circle

pi_1(S^1) is isomorphic to Z.

Evidence A. Hard: S26-T-00003107.

S26-T-00003109 - Fundamental group of a product

For path-connected based spaces under standard hypotheses, pi_1(XxY)≅pi_1(X)xpi_1(Y).

Evidence A. Hard: S26-T-00003108.

S26-D-00003110 - Simply connected space

A path-connected space is simply connected when its fundamental group is trivial.

Evidence A. Hard: S26-T-00003109.

S26-T-00003111 - Seifert-van Kampen theorem

Under the standard path-connected open-cover hypotheses, pi_1(UunionV) is the pushout of pi_1(U) and pi_1(V) over pi_1(UintersectionV).

Evidence A. Hard: S26-D-00003110.

S26-D-00003112 - Deformation retract

A subspace AsubsetX is a deformation retract when X retracts onto A through a homotopy fixed on A.

Evidence A. Hard: S26-T-00003111.

S26-T-00003113 - Deformation retract preserves homotopy type

If A is a deformation retract of X, inclusion A->X is a homotopy equivalence.

Evidence A. Hard: S26-D-00003112.

S26-D-00003114 - Homotopy group pi_n

For n>=1, pi_n(X,x0) is the set of based homotopy classes of maps S^n->X, with a natural group structure.

Evidence A. Hard: S26-T-00003113.

S26-T-00003115 - Higher homotopy groups are abelian

For n>=2, pi_n(X,x0) is abelian.

Evidence A. Hard: S26-D-00003114.

S26-D-00003116 - Weak homotopy equivalence

A map is a weak homotopy equivalence when it induces a bijection on path components and isomorphisms on all homotopy groups at compatible basepoints.

Evidence A. Hard: S26-T-00003115.

S26-N-00003117 - Weak-equivalence fence

A weak homotopy equivalence need not be a homotopy equivalence for arbitrary spaces; CW-type hypotheses are load-bearing in Whitehead-style conclusions.

Evidence A. Hard: S26-D-00003116.

S26-T-00003118 - Whitehead theorem for CW complexes

A weak homotopy equivalence between connected CW complexes is a homotopy equivalence.

Evidence A. Hard: S26-N-00003117.

S26-B-00003119 - Fundamental-group to covering bridge

The fundamental group classifies connected covering spaces under the standard local hypotheses through subgroup data.

Evidence A. Hard: S26-T-00003118.

S26-B-00003120 - Homotopy to algebra bridge

Homotopy groups convert deformation information into algebraic invariants but generally do not classify spaces by themselves.

Evidence A. Hard: S26-B-00003119.

S26-E-00003121 - Rose graph homotopy profile rank=2

The wedge of 2 circles is a finite graph with fundamental group the free group F_2; its first homology is Z^2. This is a finite CW identification.

Evidence M. Hard: S26-B-00003120.

S26-E-00003122 - Rose graph homotopy profile rank=3

The wedge of 3 circles is a finite graph with fundamental group the free group F_3; its first homology is Z^3. This is a finite CW identification.

Evidence M. Hard: S26-E-00003121.

S26-E-00003123 - Rose graph homotopy profile rank=4

The wedge of 4 circles is a finite graph with fundamental group the free group F_4; its first homology is Z^4. This is a finite CW identification.

Evidence M. Hard: S26-E-00003122.

S26-E-00003124 - Rose graph homotopy profile rank=5

The wedge of 5 circles is a finite graph with fundamental group the free group F_5; its first homology is Z^5. This is a finite CW identification.

Evidence M. Hard: S26-E-00003123.

S26-E-00003125 - Rose graph homotopy profile rank=6

The wedge of 6 circles is a finite graph with fundamental group the free group F_6; its first homology is Z^6. This is a finite CW identification.

Evidence M. Hard: S26-E-00003124.

S26-E-00003126 - Rose graph homotopy profile rank=7

The wedge of 7 circles is a finite graph with fundamental group the free group F_7; its first homology is Z^7. This is a finite CW identification.

Evidence M. Hard: S26-E-00003125.

S26-E-00003127 - Rose graph homotopy profile rank=8

The wedge of 8 circles is a finite graph with fundamental group the free group F_8; its first homology is Z^8. This is a finite CW identification.

Evidence M. Hard: S26-E-00003126.

S26-E-00003128 - Rose graph homotopy profile rank=9

The wedge of 9 circles is a finite graph with fundamental group the free group F_9; its first homology is Z^9. This is a finite CW identification.

Evidence M. Hard: S26-E-00003127.

S26-E-00003129 - Rose graph homotopy profile rank=10

The wedge of 10 circles is a finite graph with fundamental group the free group F_10; its first homology is Z^10. This is a finite CW identification.

Evidence M. Hard: S26-E-00003128.

S26-E-00003130 - Rose graph homotopy profile rank=11

The wedge of 11 circles is a finite graph with fundamental group the free group F_11; its first homology is Z^11. This is a finite CW identification.

Evidence M. Hard: S26-E-00003129.

S26-E-00003131 - Rose graph homotopy profile rank=12

The wedge of 12 circles is a finite graph with fundamental group the free group F_12; its first homology is Z^12. This is a finite CW identification.

Evidence M. Hard: S26-E-00003130.

S26-E-00003132 - Rose graph homotopy profile rank=13

The wedge of 13 circles is a finite graph with fundamental group the free group F_13; its first homology is Z^13. This is a finite CW identification.

Evidence M. Hard: S26-E-00003131.

S26-E-00003133 - Rose graph homotopy profile rank=14

The wedge of 14 circles is a finite graph with fundamental group the free group F_14; its first homology is Z^14. This is a finite CW identification.

Evidence M. Hard: S26-E-00003132.

S26-E-00003134 - Rose graph homotopy profile rank=15

The wedge of 15 circles is a finite graph with fundamental group the free group F_15; its first homology is Z^15. This is a finite CW identification.

Evidence M. Hard: S26-E-00003133.

S26-E-00003135 - Rose graph homotopy profile rank=16

The wedge of 16 circles is a finite graph with fundamental group the free group F_16; its first homology is Z^16. This is a finite CW identification.

Evidence M. Hard: S26-E-00003134.

S26-E-00003136 - Rose graph homotopy profile rank=17

The wedge of 17 circles is a finite graph with fundamental group the free group F_17; its first homology is Z^17. This is a finite CW identification.

Evidence M. Hard: S26-E-00003135.

S26-E-00003137 - Rose graph homotopy profile rank=18

The wedge of 18 circles is a finite graph with fundamental group the free group F_18; its first homology is Z^18. This is a finite CW identification.

Evidence M. Hard: S26-E-00003136.

S26-E-00003138 - Rose graph homotopy profile rank=19

The wedge of 19 circles is a finite graph with fundamental group the free group F_19; its first homology is Z^19. This is a finite CW identification.

Evidence M. Hard: S26-E-00003137.

S26-E-00003139 - Rose graph homotopy profile rank=20

The wedge of 20 circles is a finite graph with fundamental group the free group F_20; its first homology is Z^20. This is a finite CW identification.

Evidence M. Hard: S26-E-00003138.

S26-E-00003140 - Rose graph homotopy profile rank=21

The wedge of 21 circles is a finite graph with fundamental group the free group F_21; its first homology is Z^21. This is a finite CW identification.

Evidence M. Hard: S26-E-00003139.

S26-E-00003141 - Rose graph homotopy profile rank=22

The wedge of 22 circles is a finite graph with fundamental group the free group F_22; its first homology is Z^22. This is a finite CW identification.

Evidence M. Hard: S26-E-00003140.

S26-E-00003142 - Rose graph homotopy profile rank=23

The wedge of 23 circles is a finite graph with fundamental group the free group F_23; its first homology is Z^23. This is a finite CW identification.

Evidence M. Hard: S26-E-00003141.

S26-E-00003143 - Rose graph homotopy profile rank=24

The wedge of 24 circles is a finite graph with fundamental group the free group F_24; its first homology is Z^24. This is a finite CW identification.

Evidence M. Hard: S26-E-00003142.

S26-E-00003144 - Rose graph homotopy profile rank=25

The wedge of 25 circles is a finite graph with fundamental group the free group F_25; its first homology is Z^25. This is a finite CW identification.

Evidence M. Hard: S26-E-00003143.

S26-E-00003145 - Rose graph homotopy profile rank=26

The wedge of 26 circles is a finite graph with fundamental group the free group F_26; its first homology is Z^26. This is a finite CW identification.

Evidence M. Hard: S26-E-00003144.

S26-E-00003146 - Rose graph homotopy profile rank=27

The wedge of 27 circles is a finite graph with fundamental group the free group F_27; its first homology is Z^27. This is a finite CW identification.

Evidence M. Hard: S26-E-00003145.

S26-E-00003147 - Rose graph homotopy profile rank=28

The wedge of 28 circles is a finite graph with fundamental group the free group F_28; its first homology is Z^28. This is a finite CW identification.

Evidence M. Hard: S26-E-00003146.

S26-E-00003148 - Rose graph homotopy profile rank=29

The wedge of 29 circles is a finite graph with fundamental group the free group F_29; its first homology is Z^29. This is a finite CW identification.

Evidence M. Hard: S26-E-00003147.

S26-E-00003149 - Rose graph homotopy profile rank=30

The wedge of 30 circles is a finite graph with fundamental group the free group F_30; its first homology is Z^30. This is a finite CW identification.

Evidence M. Hard: S26-E-00003148.

S26-E-00003150 - Rose graph homotopy profile rank=31

The wedge of 31 circles is a finite graph with fundamental group the free group F_31; its first homology is Z^31. This is a finite CW identification.

Evidence M. Hard: S26-E-00003149.

Part D - Covering spaces and monodromy

S26-D-00003151 - Covering map

A map p:E->B is a covering map when every b in B has an evenly covered neighborhood U whose preimage is a disjoint union of open sets mapped homeomorphically onto U.

Evidence A. Hard: S26-E-00003150.

S26-D-00003152 - Sheet number of a covering

For a connected base, the cardinality of p^{-1}(b) is constant when the covering has constant finite or infinite sheet number.

Evidence A. Hard: S26-D-00003151.

S26-T-00003153 - Path lifting for covering maps

Given a path in B and a chosen initial lift in E, there exists a unique lifted path.

Evidence A. Hard: S26-D-00003152.

S26-T-00003154 - Homotopy lifting for covering maps

A homotopy in the base with a chosen initial lift admits a unique lifted homotopy under the covering hypotheses.

Evidence A. Hard: S26-T-00003153.

S26-D-00003155 - Deck transformation

A deck transformation of p:E->B is a homeomorphism h:E->E satisfying p∘h=p.

Evidence A. Hard: S26-T-00003154.

S26-D-00003156 - Universal covering space

A simply connected covering E->B is a universal cover when it dominates connected covers in the standard category.

Evidence A. Hard: S26-D-00003155.

S26-T-00003157 - Existence of universal covers

A path-connected, locally path-connected, semilocally simply connected space has a universal cover.

Evidence A. Hard: S26-D-00003156.

S26-D-00003158 - Semilocally simply connected space

A space is semilocally simply connected when each point has a neighborhood whose inclusion kills fundamental-group loops in the ambient space.

Evidence A. Hard: S26-T-00003157.

S26-T-00003159 - Subgroup classification of connected covers

For a path-connected locally path-connected semilocally simply connected base, connected covering spaces correspond to conjugacy classes of subgroups of pi_1.

Evidence A. Hard: S26-D-00003158.

S26-T-00003160 - Regular covering criterion

A connected covering is regular exactly when the corresponding subgroup is normal, under the standard classification hypotheses.

Evidence A. Hard: S26-T-00003159.

S26-T-00003161 - Deck group of a regular cover

For a regular connected cover corresponding to H normal in pi_1(B), the deck group is isomorphic to pi_1(B)/H.

Evidence A. Hard: S26-T-00003160.

S26-D-00003162 - Monodromy action of a covering

The fundamental group acts on a fiber by lifting loops; this permutation action encodes the covering.

Evidence A. Hard: S26-T-00003161.

S26-T-00003163 - Lifting criterion for maps

A map f:Y->B lifts to E after choosing basepoints when f_*(pi_1(Y)) lies in the subgroup p_*(pi_1(E)), under the standard local hypotheses.

Evidence A. Hard: S26-D-00003162.

S26-D-00003164 - Branched-covering fence

A branched covering is not a covering map at branch points because local homeomorphic sheet structure fails.

Evidence A. Hard: S26-T-00003163.

S26-D-00003165 - Covering degree

A finite-sheeted covering of connected spaces has degree equal to the number of points in a fiber.

Evidence A. Hard: S26-D-00003164.

S26-T-00003166 - Euler characteristic under finite covers

For finite CW complexes and a d-sheeted covering E->B, chi(E)=d chi(B).

Evidence A. Hard: S26-D-00003165.

S26-N-00003167 - Covering-existence fence

Local path connectedness and semilocal simple connectivity cannot be silently dropped from elementary universal-cover classification statements.

Evidence A. Hard: S26-T-00003166.

S26-B-00003168 - Coverings to complex analysis bridge

Analytic continuation and multivalued functions are naturally organized by covering spaces and monodromy.

Evidence A. Hard: S26-N-00003167.

S26-B-00003169 - Coverings to geometry bridge

Quotients of simply connected model geometries by properly discontinuous group actions produce many geometric manifolds and orbifold-like spaces.

Evidence A. Hard: S26-B-00003168.

S26-B-00003170 - Coverings to group theory bridge

Finite covers, subgroup index and permutation monodromy provide a geometric realization of subgroup data.

Evidence A. Hard: S26-B-00003169.

S26-E-00003171 - Circle covering profile degree=2

The map z|->z^2 on S^1 is a 2-sheeted covering; each fiber has 2 points and the induced map on pi_1 is multiplication by 2.

Evidence M. Hard: S26-B-00003170.

S26-E-00003172 - Circle covering profile degree=3

The map z|->z^3 on S^1 is a 3-sheeted covering; each fiber has 3 points and the induced map on pi_1 is multiplication by 3.

Evidence M. Hard: S26-E-00003171.

S26-E-00003173 - Circle covering profile degree=4

The map z|->z^4 on S^1 is a 4-sheeted covering; each fiber has 4 points and the induced map on pi_1 is multiplication by 4.

Evidence M. Hard: S26-E-00003172.

S26-E-00003174 - Circle covering profile degree=5

The map z|->z^5 on S^1 is a 5-sheeted covering; each fiber has 5 points and the induced map on pi_1 is multiplication by 5.

Evidence M. Hard: S26-E-00003173.

S26-E-00003175 - Circle covering profile degree=6

The map z|->z^6 on S^1 is a 6-sheeted covering; each fiber has 6 points and the induced map on pi_1 is multiplication by 6.

Evidence M. Hard: S26-E-00003174.

S26-E-00003176 - Circle covering profile degree=7

The map z|->z^7 on S^1 is a 7-sheeted covering; each fiber has 7 points and the induced map on pi_1 is multiplication by 7.

Evidence M. Hard: S26-E-00003175.

S26-E-00003177 - Circle covering profile degree=8

The map z|->z^8 on S^1 is a 8-sheeted covering; each fiber has 8 points and the induced map on pi_1 is multiplication by 8.

Evidence M. Hard: S26-E-00003176.

S26-E-00003178 - Circle covering profile degree=9

The map z|->z^9 on S^1 is a 9-sheeted covering; each fiber has 9 points and the induced map on pi_1 is multiplication by 9.

Evidence M. Hard: S26-E-00003177.

S26-E-00003179 - Circle covering profile degree=10

The map z|->z^10 on S^1 is a 10-sheeted covering; each fiber has 10 points and the induced map on pi_1 is multiplication by 10.

Evidence M. Hard: S26-E-00003178.

S26-E-00003180 - Circle covering profile degree=11

The map z|->z^11 on S^1 is a 11-sheeted covering; each fiber has 11 points and the induced map on pi_1 is multiplication by 11.

Evidence M. Hard: S26-E-00003179.

S26-E-00003181 - Circle covering profile degree=12

The map z|->z^12 on S^1 is a 12-sheeted covering; each fiber has 12 points and the induced map on pi_1 is multiplication by 12.

Evidence M. Hard: S26-E-00003180.

S26-E-00003182 - Circle covering profile degree=13

The map z|->z^13 on S^1 is a 13-sheeted covering; each fiber has 13 points and the induced map on pi_1 is multiplication by 13.

Evidence M. Hard: S26-E-00003181.

S26-E-00003183 - Circle covering profile degree=14

The map z|->z^14 on S^1 is a 14-sheeted covering; each fiber has 14 points and the induced map on pi_1 is multiplication by 14.

Evidence M. Hard: S26-E-00003182.

S26-E-00003184 - Circle covering profile degree=15

The map z|->z^15 on S^1 is a 15-sheeted covering; each fiber has 15 points and the induced map on pi_1 is multiplication by 15.

Evidence M. Hard: S26-E-00003183.

S26-E-00003185 - Circle covering profile degree=16

The map z|->z^16 on S^1 is a 16-sheeted covering; each fiber has 16 points and the induced map on pi_1 is multiplication by 16.

Evidence M. Hard: S26-E-00003184.

S26-E-00003186 - Circle covering profile degree=17

The map z|->z^17 on S^1 is a 17-sheeted covering; each fiber has 17 points and the induced map on pi_1 is multiplication by 17.

Evidence M. Hard: S26-E-00003185.

S26-E-00003187 - Circle covering profile degree=18

The map z|->z^18 on S^1 is a 18-sheeted covering; each fiber has 18 points and the induced map on pi_1 is multiplication by 18.

Evidence M. Hard: S26-E-00003186.

S26-E-00003188 - Circle covering profile degree=19

The map z|->z^19 on S^1 is a 19-sheeted covering; each fiber has 19 points and the induced map on pi_1 is multiplication by 19.

Evidence M. Hard: S26-E-00003187.

S26-E-00003189 - Circle covering profile degree=20

The map z|->z^20 on S^1 is a 20-sheeted covering; each fiber has 20 points and the induced map on pi_1 is multiplication by 20.

Evidence M. Hard: S26-E-00003188.

S26-E-00003190 - Circle covering profile degree=21

The map z|->z^21 on S^1 is a 21-sheeted covering; each fiber has 21 points and the induced map on pi_1 is multiplication by 21.

Evidence M. Hard: S26-E-00003189.

S26-E-00003191 - Circle covering profile degree=22

The map z|->z^22 on S^1 is a 22-sheeted covering; each fiber has 22 points and the induced map on pi_1 is multiplication by 22.

Evidence M. Hard: S26-E-00003190.

S26-E-00003192 - Circle covering profile degree=23

The map z|->z^23 on S^1 is a 23-sheeted covering; each fiber has 23 points and the induced map on pi_1 is multiplication by 23.

Evidence M. Hard: S26-E-00003191.

S26-E-00003193 - Circle covering profile degree=24

The map z|->z^24 on S^1 is a 24-sheeted covering; each fiber has 24 points and the induced map on pi_1 is multiplication by 24.

Evidence M. Hard: S26-E-00003192.

S26-E-00003194 - Circle covering profile degree=25

The map z|->z^25 on S^1 is a 25-sheeted covering; each fiber has 25 points and the induced map on pi_1 is multiplication by 25.

Evidence M. Hard: S26-E-00003193.

S26-E-00003195 - Circle covering profile degree=26

The map z|->z^26 on S^1 is a 26-sheeted covering; each fiber has 26 points and the induced map on pi_1 is multiplication by 26.

Evidence M. Hard: S26-E-00003194.

S26-E-00003196 - Circle covering profile degree=27

The map z|->z^27 on S^1 is a 27-sheeted covering; each fiber has 27 points and the induced map on pi_1 is multiplication by 27.

Evidence M. Hard: S26-E-00003195.

S26-E-00003197 - Circle covering profile degree=28

The map z|->z^28 on S^1 is a 28-sheeted covering; each fiber has 28 points and the induced map on pi_1 is multiplication by 28.

Evidence M. Hard: S26-E-00003196.

S26-E-00003198 - Circle covering profile degree=29

The map z|->z^29 on S^1 is a 29-sheeted covering; each fiber has 29 points and the induced map on pi_1 is multiplication by 29.

Evidence M. Hard: S26-E-00003197.

S26-E-00003199 - Circle covering profile degree=30

The map z|->z^30 on S^1 is a 30-sheeted covering; each fiber has 30 points and the induced map on pi_1 is multiplication by 30.

Evidence M. Hard: S26-E-00003198.

S26-E-00003200 - Circle covering profile degree=31

The map z|->z^31 on S^1 is a 31-sheeted covering; each fiber has 31 points and the induced map on pi_1 is multiplication by 31.

Evidence M. Hard: S26-E-00003199.

Part E - Simplicial complexes and homology

S26-D-00003201 - Abstract simplicial complex

An abstract simplicial complex K on a vertex set is a family of finite nonempty subsets closed under taking nonempty subsets.

Evidence A. Hard: S26-E-00003200.

S26-D-00003202 - Geometric realization of a simplicial complex

The geometric realization |K| glues Euclidean simplices according to the face relations of K.

Evidence A. Hard: S26-D-00003201.

S26-D-00003203 - Simplicial map

A vertex map inducing simplex-to-simplex maps defines a simplicial map and hence a continuous map of realizations.

Evidence A. Hard: S26-D-00003202.

S26-D-00003204 - Oriented simplex

An orientation of an n-simplex is an equivalence class of vertex orderings modulo even permutations.

Evidence A. Hard: S26-D-00003203.

S26-D-00003205 - Simplicial chain group

C_n(K;R) is the free R-module on oriented n-simplices, modulo the sign relation for orientation reversal.

Evidence A. Hard: S26-D-00003204.

S26-I-00003206 - Simplicial boundary operator

partial[v_0,...,v_n]=sum_i (-1)^i[v_0,...,hat v_i,...,v_n].

Evidence A. Hard: S26-D-00003205.

S26-T-00003207 - Boundary squared is zero

The simplicial boundary satisfies partial_{n-1}∘partial_n=0.

Evidence A. Hard: S26-I-00003206.

S26-D-00003208 - Simplicial cycle and boundary groups

Z_n=ker partial_n and B_n=im partial_{n+1}; boundary-squared-zero gives B_nsubsetZ_n.

Evidence A. Hard: S26-T-00003207.

S26-D-00003209 - Simplicial homology group

H_n(K;R)=Z_n/B_n.

Evidence A. Hard: S26-D-00003208.

S26-D-00003210 - Betti number

Over a field, the nth Betti number beta_n is dim H_n.

Evidence A. Hard: S26-D-00003209.

S26-D-00003211 - Euler characteristic from simplices

For a finite simplicial complex, chi(K)=sum_n (-1)^n f_n where f_n counts n-simplices.

Evidence A. Hard: S26-D-00003210.

S26-T-00003212 - Euler-Poincare formula

For a finite simplicial complex over a field, chi(K)=sum_n (-1)^n beta_n.

Evidence A. Hard: S26-D-00003211.

S26-D-00003213 - Barycentric subdivision

Barycentric subdivision replaces simplices by chains of nonempty faces and preserves the homeomorphism type of the realization.

Evidence A. Hard: S26-T-00003212.

S26-T-00003214 - Simplicial approximation interface

After sufficient subdivision under standard compactness hypotheses, continuous maps between polyhedra admit simplicial approximations in their homotopy class.

Evidence A. Hard: S26-D-00003213.

S26-D-00003215 - Reduced simplicial homology

Reduced homology modifies degree zero so that a one-point space has vanishing reduced homology.

Evidence A. Hard: S26-T-00003214.

S26-D-00003216 - Relative simplicial chain complex

For LsubsetK, C_*(K,L)=C_*(K)/C_*(L), leading to relative homology.

Evidence A. Hard: S26-D-00003215.

S26-T-00003217 - Long exact sequence of a simplicial pair

A pair (K,L) has a natural long exact sequence in homology.

Evidence A. Hard: S26-D-00003216.

S26-N-00003218 - Triangulation fence

Not every topological space or manifold in every category admits a triangulation compatible with naive simplicial methods; triangulability is a real hypothesis.

Evidence A. Hard: S26-T-00003217.

S26-B-00003219 - Simplicial to computation bridge

Boundary matrices reduce finite homology computations to linear algebra or Smith normal form over the chosen coefficient ring.

Evidence A. Hard: S26-N-00003218.

S26-B-00003220 - Simplicial to geometry bridge

Triangulations provide finite combinatorial models of many spaces while preserving topological invariants when the triangulation assumptions hold.

Evidence A. Hard: S26-B-00003219.

S26-E-00003221 - Cycle-complex homology profile n=3

The 1-dimensional simplicial cycle C_3 has f_0=3, f_1=3, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-B-00003220.

S26-E-00003222 - Cycle-complex homology profile n=4

The 1-dimensional simplicial cycle C_4 has f_0=4, f_1=4, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003221.

S26-E-00003223 - Cycle-complex homology profile n=5

The 1-dimensional simplicial cycle C_5 has f_0=5, f_1=5, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003222.

S26-E-00003224 - Cycle-complex homology profile n=6

The 1-dimensional simplicial cycle C_6 has f_0=6, f_1=6, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003223.

S26-E-00003225 - Cycle-complex homology profile n=7

The 1-dimensional simplicial cycle C_7 has f_0=7, f_1=7, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003224.

S26-E-00003226 - Cycle-complex homology profile n=8

The 1-dimensional simplicial cycle C_8 has f_0=8, f_1=8, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003225.

S26-E-00003227 - Cycle-complex homology profile n=9

The 1-dimensional simplicial cycle C_9 has f_0=9, f_1=9, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003226.

S26-E-00003228 - Cycle-complex homology profile n=10

The 1-dimensional simplicial cycle C_10 has f_0=10, f_1=10, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003227.

S26-E-00003229 - Cycle-complex homology profile n=11

The 1-dimensional simplicial cycle C_11 has f_0=11, f_1=11, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003228.

S26-E-00003230 - Cycle-complex homology profile n=12

The 1-dimensional simplicial cycle C_12 has f_0=12, f_1=12, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003229.

S26-E-00003231 - Cycle-complex homology profile n=13

The 1-dimensional simplicial cycle C_13 has f_0=13, f_1=13, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003230.

S26-E-00003232 - Cycle-complex homology profile n=14

The 1-dimensional simplicial cycle C_14 has f_0=14, f_1=14, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003231.

S26-E-00003233 - Cycle-complex homology profile n=15

The 1-dimensional simplicial cycle C_15 has f_0=15, f_1=15, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003232.

S26-E-00003234 - Cycle-complex homology profile n=16

The 1-dimensional simplicial cycle C_16 has f_0=16, f_1=16, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003233.

S26-E-00003235 - Cycle-complex homology profile n=17

The 1-dimensional simplicial cycle C_17 has f_0=17, f_1=17, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003234.

S26-E-00003236 - Cycle-complex homology profile n=18

The 1-dimensional simplicial cycle C_18 has f_0=18, f_1=18, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003235.

S26-E-00003237 - Cycle-complex homology profile n=19

The 1-dimensional simplicial cycle C_19 has f_0=19, f_1=19, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003236.

S26-E-00003238 - Cycle-complex homology profile n=20

The 1-dimensional simplicial cycle C_20 has f_0=20, f_1=20, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003237.

S26-E-00003239 - Cycle-complex homology profile n=21

The 1-dimensional simplicial cycle C_21 has f_0=21, f_1=21, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003238.

S26-E-00003240 - Cycle-complex homology profile n=22

The 1-dimensional simplicial cycle C_22 has f_0=22, f_1=22, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003239.

S26-E-00003241 - Cycle-complex homology profile n=23

The 1-dimensional simplicial cycle C_23 has f_0=23, f_1=23, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003240.

S26-E-00003242 - Cycle-complex homology profile n=24

The 1-dimensional simplicial cycle C_24 has f_0=24, f_1=24, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003241.

S26-E-00003243 - Cycle-complex homology profile n=25

The 1-dimensional simplicial cycle C_25 has f_0=25, f_1=25, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003242.

S26-E-00003244 - Cycle-complex homology profile n=26

The 1-dimensional simplicial cycle C_26 has f_0=26, f_1=26, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003243.

S26-E-00003245 - Cycle-complex homology profile n=27

The 1-dimensional simplicial cycle C_27 has f_0=27, f_1=27, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003244.

S26-E-00003246 - Cycle-complex homology profile n=28

The 1-dimensional simplicial cycle C_28 has f_0=28, f_1=28, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003245.

S26-E-00003247 - Cycle-complex homology profile n=29

The 1-dimensional simplicial cycle C_29 has f_0=29, f_1=29, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003246.

S26-E-00003248 - Cycle-complex homology profile n=30

The 1-dimensional simplicial cycle C_30 has f_0=30, f_1=30, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003247.

S26-E-00003249 - Cycle-complex homology profile n=31

The 1-dimensional simplicial cycle C_31 has f_0=31, f_1=31, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003248.

S26-E-00003250 - Cycle-complex homology profile n=32

The 1-dimensional simplicial cycle C_32 has f_0=32, f_1=32, chi=0, and over Q has Betti numbers beta_0=1, beta_1=1.

Evidence M. Hard: S26-E-00003249.

Part F - Singular homology, excision and exact sequences

S26-D-00003251 - Singular n-simplex

A singular n-simplex in X is a continuous map sigma:Delta^n->X from the standard n-simplex.

Evidence A. Hard: S26-E-00003250.

S26-D-00003252 - Singular chain complex

C_n(X;R) is the free R-module generated by singular n-simplices with the alternating face boundary.

Evidence A. Hard: S26-D-00003251.

S26-T-00003253 - Singular boundary squared is zero

The singular boundary operators satisfy partial^2=0.

Evidence A. Hard: S26-D-00003252.

S26-D-00003254 - Singular homology

H_n(X;R)=ker partial_n/im partial_{n+1} for the singular chain complex.

Evidence A. Hard: S26-T-00003253.

S26-T-00003255 - Homotopy invariance of singular homology

Homotopic maps induce the same maps on singular homology; homotopy equivalent spaces have isomorphic homology.

Evidence A. Hard: S26-D-00003254.

S26-T-00003256 - Dimension axiom for a point

For ordinary homology, H_0(point;R)≅R and H_n(point;R)=0 for n>0.

Evidence A. Hard: S26-T-00003255.

S26-T-00003257 - Excision theorem

For a pair (X,A), a subset whose closure lies in the interior of A may be excised without changing relative homology.

Evidence A. Hard: S26-T-00003256.

S26-T-00003258 - Long exact sequence of a pair

Every pair (X,A) has a natural long exact sequence ...->H_n(A)->H_n(X)->H_n(X,A)->H_{n-1}(A)->....

Evidence A. Hard: S26-T-00003257.

S26-T-00003259 - Mayer-Vietoris sequence

A suitable decomposition X=UunionV gives a long exact sequence linking homology of UintersectionV, U, V and X.

Evidence A. Hard: S26-T-00003258.

S26-D-00003260 - Reduced singular homology

Reduced homology augments the singular chain complex to normalize degree-zero behavior.

Evidence A. Hard: S26-T-00003259.

S26-T-00003261 - Suspension isomorphism in reduced homology

For suitable spaces, reduced H_{n+1}(Sigma X) is naturally isomorphic to reduced H_n(X).

Evidence A. Hard: S26-D-00003260.

S26-T-00003262 - Homology of spheres

For S^n, reduced homology is the coefficient ring in degree n and zero in other degrees.

Evidence A. Hard: S26-T-00003261.

S26-D-00003263 - Degree of a map between oriented spheres

For f:S^n->S^n, the induced map on top homology is multiplication by an integer deg(f).

Evidence A. Hard: S26-T-00003262.

S26-T-00003264 - Degree is homotopy invariant

Homotopic maps S^n->S^n have the same degree.

Evidence A. Hard: S26-D-00003263.

S26-T-00003265 - Brouwer fixed-point theorem via degree/homology interface

No continuous retraction D^n->S^{n-1} exists; this obstruction yields the fixed-point theorem through standard homological arguments.

Evidence A. Hard: S26-T-00003264.

S26-D-00003266 - Homology with coefficients

Tensoring or otherwise changing the coefficient ring can alter homology by detecting torsion differently.

Evidence A. Hard: S26-T-00003265.

S26-T-00003267 - Universal coefficient theorem for homology

Under standard chain-complex hypotheses there is a short exact sequence involving H_n(X;Z) tensor G and Tor(H_{n-1}(X;Z),G).

Evidence A. Hard: S26-D-00003266.

S26-N-00003268 - Coefficient fence

Betti numbers over a field can hide torsion; homology over one coefficient system does not determine all coefficient theories without additional data.

Evidence A. Hard: S26-T-00003267.

S26-B-00003269 - Homology to fixed-point bridge

Homological degree and Lefschetz-type traces convert global topology into obstructions to fixed-point-free maps.

Evidence A. Hard: S26-N-00003268.

S26-B-00003270 - Homology to manifold orientation bridge

Top-dimensional homology encodes orientability and fundamental classes for appropriate manifolds.

Evidence A. Hard: S26-B-00003269.

S26-E-00003271 - Sphere homology rank profile n=2

For S^2 with rational coefficients, beta_0=1 and beta_2=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-B-00003270.

S26-E-00003272 - Sphere homology rank profile n=3

For S^3 with rational coefficients, beta_0=1 and beta_3=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003271.

S26-E-00003273 - Sphere homology rank profile n=4

For S^4 with rational coefficients, beta_0=1 and beta_4=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003272.

S26-E-00003274 - Sphere homology rank profile n=5

For S^5 with rational coefficients, beta_0=1 and beta_5=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003273.

S26-E-00003275 - Sphere homology rank profile n=6

For S^6 with rational coefficients, beta_0=1 and beta_6=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003274.

S26-E-00003276 - Sphere homology rank profile n=7

For S^7 with rational coefficients, beta_0=1 and beta_7=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003275.

S26-E-00003277 - Sphere homology rank profile n=8

For S^8 with rational coefficients, beta_0=1 and beta_8=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003276.

S26-E-00003278 - Sphere homology rank profile n=9

For S^9 with rational coefficients, beta_0=1 and beta_9=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003277.

S26-E-00003279 - Sphere homology rank profile n=10

For S^10 with rational coefficients, beta_0=1 and beta_10=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003278.

S26-E-00003280 - Sphere homology rank profile n=11

For S^11 with rational coefficients, beta_0=1 and beta_11=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003279.

S26-E-00003281 - Sphere homology rank profile n=12

For S^12 with rational coefficients, beta_0=1 and beta_12=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003280.

S26-E-00003282 - Sphere homology rank profile n=13

For S^13 with rational coefficients, beta_0=1 and beta_13=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003281.

S26-E-00003283 - Sphere homology rank profile n=14

For S^14 with rational coefficients, beta_0=1 and beta_14=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003282.

S26-E-00003284 - Sphere homology rank profile n=15

For S^15 with rational coefficients, beta_0=1 and beta_15=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003283.

S26-E-00003285 - Sphere homology rank profile n=16

For S^16 with rational coefficients, beta_0=1 and beta_16=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003284.

S26-E-00003286 - Sphere homology rank profile n=17

For S^17 with rational coefficients, beta_0=1 and beta_17=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003285.

S26-E-00003287 - Sphere homology rank profile n=18

For S^18 with rational coefficients, beta_0=1 and beta_18=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003286.

S26-E-00003288 - Sphere homology rank profile n=19

For S^19 with rational coefficients, beta_0=1 and beta_19=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003287.

S26-E-00003289 - Sphere homology rank profile n=20

For S^20 with rational coefficients, beta_0=1 and beta_20=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003288.

S26-E-00003290 - Sphere homology rank profile n=21

For S^21 with rational coefficients, beta_0=1 and beta_21=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003289.

S26-E-00003291 - Sphere homology rank profile n=22

For S^22 with rational coefficients, beta_0=1 and beta_22=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003290.

S26-E-00003292 - Sphere homology rank profile n=23

For S^23 with rational coefficients, beta_0=1 and beta_23=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003291.

S26-E-00003293 - Sphere homology rank profile n=24

For S^24 with rational coefficients, beta_0=1 and beta_24=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003292.

S26-E-00003294 - Sphere homology rank profile n=25

For S^25 with rational coefficients, beta_0=1 and beta_25=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003293.

S26-E-00003295 - Sphere homology rank profile n=26

For S^26 with rational coefficients, beta_0=1 and beta_26=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003294.

S26-E-00003296 - Sphere homology rank profile n=27

For S^27 with rational coefficients, beta_0=1 and beta_27=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003295.

S26-E-00003297 - Sphere homology rank profile n=28

For S^28 with rational coefficients, beta_0=1 and beta_28=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003296.

S26-E-00003298 - Sphere homology rank profile n=29

For S^29 with rational coefficients, beta_0=1 and beta_29=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003297.

S26-E-00003299 - Sphere homology rank profile n=30

For S^30 with rational coefficients, beta_0=1 and beta_30=1, with all other Betti numbers zero; Euler characteristic is 2.

Evidence M. Hard: S26-E-00003298.

S26-E-00003300 - Sphere homology rank profile n=31

For S^31 with rational coefficients, beta_0=1 and beta_31=1, with all other Betti numbers zero; Euler characteristic is 0.

Evidence M. Hard: S26-E-00003299.

Part G - Cohomology rings and duality interfaces

S26-D-00003301 - Singular cochain complex

C^n(X;R)=Hom(C_n(X;R),R) with coboundary delta dual to the singular boundary.

Evidence A. Hard: S26-E-00003300.

S26-T-00003302 - Coboundary squared is zero

The singular coboundary satisfies delta^2=0.

Evidence A. Hard: S26-D-00003301.

S26-D-00003303 - Singular cohomology

H^n(X;R)=ker delta^n/im delta^{n-1}.

Evidence A. Hard: S26-T-00003302.

S26-D-00003304 - Cup product

The cup product gives H^p(X;R)xH^q(X;R)->H^{p+q}(X;R) and equips cohomology with a graded ring structure.

Evidence A. Hard: S26-D-00003303.

S26-T-00003305 - Graded commutativity of cup product

For suitable commutative coefficients, alpha cup beta=(-1)^{pq} beta cup alpha in cohomology.

Evidence A. Hard: S26-D-00003304.

S26-D-00003306 - Cohomology ring

H^*(X;R)=⊕_n H^n(X;R) together with cup product is a graded ring invariant of X.

Evidence A. Hard: S26-T-00003305.

S26-T-00003307 - Universal coefficient theorem for cohomology

Under standard hypotheses, H^n(X;G) fits into a short exact sequence involving Hom(H_n(X;Z),G) and Ext(H_{n-1}(X;Z),G).

Evidence A. Hard: S26-D-00003306.

S26-D-00003308 - Cross product in cohomology

The external cross product maps H^p(X)xH^q(Y) to H^{p+q}(XxY).

Evidence A. Hard: S26-T-00003307.

S26-T-00003309 - Kunneth theorem interface

Under suitable coefficient and finiteness hypotheses, homology or cohomology of products is computed from tensor and Tor data of the factors.

Evidence A. Hard: S26-D-00003308.

S26-D-00003310 - Cohomological dimension interface

Cohomological dimension records the highest degree in which an appropriate cohomology theory can be nonzero.

Evidence A. Hard: S26-T-00003309.

S26-T-00003311 - Cohomology of complex projective space

With integer coefficients, H^*(CP^n;Z)□Z[x]/(x^{n+1}) with |x|=2.

Evidence A. Hard: S26-D-00003310.

S26-D-00003312 - Cap product

The cap product pairs cohomology and homology to lower homological degree.

Evidence A. Hard: S26-T-00003311.

S26-D-00003313 - Kronecker pairing

The evaluation pairing H^n(X;R)xH_n(X;R)->R evaluates cohomology classes on homology classes.

Evidence A. Hard: S26-D-00003312.

S26-D-00003314 - Poincare duality interface

For a closed oriented n-manifold over suitable coefficients, cap product with the fundamental class gives H^k(M)->H_{n-k}(M).

Evidence A. Hard: S26-D-00003313.

S26-T-00003315 - Euler characteristic from cohomology

For finite-type complexes over a field, Euler characteristic is the alternating sum of cohomology dimensions as well as homology dimensions.

Evidence A. Hard: S26-D-00003314.

S26-D-00003316 - Cech cohomology interface

Cech cohomology is built from nerves of open covers and agrees with singular cohomology on broad classes of well-behaved spaces.

Evidence A. Hard: S26-T-00003315.

S26-N-00003317 - Ring-vs-group fence

Isomorphic graded cohomology groups need not imply isomorphic cohomology rings; cup-product structure carries additional information.

Evidence A. Hard: S26-D-00003316.

S26-N-00003318 - Duality hypothesis fence

Poincare duality requires manifold, orientation/coefficients, boundary and compactness hypotheses in the appropriate form.

Evidence A. Hard: S26-N-00003317.

S26-B-00003319 - Cohomology to bundles bridge

Characteristic classes live in cohomology and turn geometric twisting of bundles into algebraic invariants.

Evidence A. Hard: S26-N-00003318.

S26-B-00003320 - Cohomology to differential forms bridge

de Rham cohomology identifies smooth closed-form classes with singular cohomology over R for smooth manifolds.

Evidence A. Hard: S26-B-00003319.

S26-E-00003321 - Projective-space cohomology profile n=2

For CP^2, the integral cohomology groups have rank 1 in degrees 0,2,...,4 and rank 0 otherwise; the Euler characteristic is 3.

Evidence M. Hard: S26-B-00003320.

S26-E-00003322 - Projective-space cohomology profile n=3

For CP^3, the integral cohomology groups have rank 1 in degrees 0,2,...,6 and rank 0 otherwise; the Euler characteristic is 4.

Evidence M. Hard: S26-E-00003321.

S26-E-00003323 - Projective-space cohomology profile n=4

For CP^4, the integral cohomology groups have rank 1 in degrees 0,2,...,8 and rank 0 otherwise; the Euler characteristic is 5.

Evidence M. Hard: S26-E-00003322.

S26-E-00003324 - Projective-space cohomology profile n=5

For CP^5, the integral cohomology groups have rank 1 in degrees 0,2,...,10 and rank 0 otherwise; the Euler characteristic is 6.

Evidence M. Hard: S26-E-00003323.

S26-E-00003325 - Projective-space cohomology profile n=6

For CP^6, the integral cohomology groups have rank 1 in degrees 0,2,...,12 and rank 0 otherwise; the Euler characteristic is 7.

Evidence M. Hard: S26-E-00003324.

S26-E-00003326 - Projective-space cohomology profile n=7

For CP^7, the integral cohomology groups have rank 1 in degrees 0,2,...,14 and rank 0 otherwise; the Euler characteristic is 8.

Evidence M. Hard: S26-E-00003325.

S26-E-00003327 - Projective-space cohomology profile n=8

For CP^8, the integral cohomology groups have rank 1 in degrees 0,2,...,16 and rank 0 otherwise; the Euler characteristic is 9.

Evidence M. Hard: S26-E-00003326.

S26-E-00003328 - Projective-space cohomology profile n=9

For CP^9, the integral cohomology groups have rank 1 in degrees 0,2,...,18 and rank 0 otherwise; the Euler characteristic is 10.

Evidence M. Hard: S26-E-00003327.

S26-E-00003329 - Projective-space cohomology profile n=10

For CP^10, the integral cohomology groups have rank 1 in degrees 0,2,...,20 and rank 0 otherwise; the Euler characteristic is 11.

Evidence M. Hard: S26-E-00003328.

S26-E-00003330 - Projective-space cohomology profile n=11

For CP^11, the integral cohomology groups have rank 1 in degrees 0,2,...,22 and rank 0 otherwise; the Euler characteristic is 12.

Evidence M. Hard: S26-E-00003329.

S26-E-00003331 - Projective-space cohomology profile n=12

For CP^12, the integral cohomology groups have rank 1 in degrees 0,2,...,24 and rank 0 otherwise; the Euler characteristic is 13.

Evidence M. Hard: S26-E-00003330.

S26-E-00003332 - Projective-space cohomology profile n=13

For CP^13, the integral cohomology groups have rank 1 in degrees 0,2,...,26 and rank 0 otherwise; the Euler characteristic is 14.

Evidence M. Hard: S26-E-00003331.

S26-E-00003333 - Projective-space cohomology profile n=14

For CP^14, the integral cohomology groups have rank 1 in degrees 0,2,...,28 and rank 0 otherwise; the Euler characteristic is 15.

Evidence M. Hard: S26-E-00003332.

S26-E-00003334 - Projective-space cohomology profile n=15

For CP^15, the integral cohomology groups have rank 1 in degrees 0,2,...,30 and rank 0 otherwise; the Euler characteristic is 16.

Evidence M. Hard: S26-E-00003333.

S26-E-00003335 - Projective-space cohomology profile n=16

For CP^16, the integral cohomology groups have rank 1 in degrees 0,2,...,32 and rank 0 otherwise; the Euler characteristic is 17.

Evidence M. Hard: S26-E-00003334.

S26-E-00003336 - Projective-space cohomology profile n=17

For CP^17, the integral cohomology groups have rank 1 in degrees 0,2,...,34 and rank 0 otherwise; the Euler characteristic is 18.

Evidence M. Hard: S26-E-00003335.

S26-E-00003337 - Projective-space cohomology profile n=18

For CP^18, the integral cohomology groups have rank 1 in degrees 0,2,...,36 and rank 0 otherwise; the Euler characteristic is 19.

Evidence M. Hard: S26-E-00003336.

S26-E-00003338 - Projective-space cohomology profile n=19

For CP^19, the integral cohomology groups have rank 1 in degrees 0,2,...,38 and rank 0 otherwise; the Euler characteristic is 20.

Evidence M. Hard: S26-E-00003337.

S26-E-00003339 - Projective-space cohomology profile n=20

For CP^20, the integral cohomology groups have rank 1 in degrees 0,2,...,40 and rank 0 otherwise; the Euler characteristic is 21.

Evidence M. Hard: S26-E-00003338.

S26-E-00003340 - Projective-space cohomology profile n=21

For CP^21, the integral cohomology groups have rank 1 in degrees 0,2,...,42 and rank 0 otherwise; the Euler characteristic is 22.

Evidence M. Hard: S26-E-00003339.

S26-E-00003341 - Projective-space cohomology profile n=22

For CP^22, the integral cohomology groups have rank 1 in degrees 0,2,...,44 and rank 0 otherwise; the Euler characteristic is 23.

Evidence M. Hard: S26-E-00003340.

S26-E-00003342 - Projective-space cohomology profile n=23

For CP^23, the integral cohomology groups have rank 1 in degrees 0,2,...,46 and rank 0 otherwise; the Euler characteristic is 24.

Evidence M. Hard: S26-E-00003341.

S26-E-00003343 - Projective-space cohomology profile n=24

For CP^24, the integral cohomology groups have rank 1 in degrees 0,2,...,48 and rank 0 otherwise; the Euler characteristic is 25.

Evidence M. Hard: S26-E-00003342.

S26-E-00003344 - Projective-space cohomology profile n=25

For CP^25, the integral cohomology groups have rank 1 in degrees 0,2,...,50 and rank 0 otherwise; the Euler characteristic is 26.

Evidence M. Hard: S26-E-00003343.

S26-E-00003345 - Projective-space cohomology profile n=26

For CP^26, the integral cohomology groups have rank 1 in degrees 0,2,...,52 and rank 0 otherwise; the Euler characteristic is 27.

Evidence M. Hard: S26-E-00003344.

S26-E-00003346 - Projective-space cohomology profile n=27

For CP^27, the integral cohomology groups have rank 1 in degrees 0,2,...,54 and rank 0 otherwise; the Euler characteristic is 28.

Evidence M. Hard: S26-E-00003345.

S26-E-00003347 - Projective-space cohomology profile n=28

For CP^28, the integral cohomology groups have rank 1 in degrees 0,2,...,56 and rank 0 otherwise; the Euler characteristic is 29.

Evidence M. Hard: S26-E-00003346.

S26-E-00003348 - Projective-space cohomology profile n=29

For CP^29, the integral cohomology groups have rank 1 in degrees 0,2,...,58 and rank 0 otherwise; the Euler characteristic is 30.

Evidence M. Hard: S26-E-00003347.

S26-E-00003349 - Projective-space cohomology profile n=30

For CP^30, the integral cohomology groups have rank 1 in degrees 0,2,...,60 and rank 0 otherwise; the Euler characteristic is 31.

Evidence M. Hard: S26-E-00003348.

S26-E-00003350 - Projective-space cohomology profile n=31

For CP^31, the integral cohomology groups have rank 1 in degrees 0,2,...,62 and rank 0 otherwise; the Euler characteristic is 32.

Evidence M. Hard: S26-E-00003349.

Part H - CW complexes and cellular methods

S26-D-00003351 - CW complex

A CW complex is built inductively by attaching n-cells to lower skeleta via maps S^{n-1}->X^{n-1}, with closure-finiteness and weak-topology conditions.

Evidence A. Hard: S26-E-00003350.

S26-D-00003352 - n-skeleton of a CW complex

X^n is the union of all cells of dimension at most n.

Evidence A. Hard: S26-D-00003351.

S26-D-00003353 - Attaching map of a cell

An n-cell is attached by a continuous map S^{n-1}->X^{n-1} followed by quotient gluing of D^n.

Evidence A. Hard: S26-D-00003352.

S26-D-00003354 - Characteristic map of a cell

A characteristic map D^n->X maps the interior homeomorphically onto the cell and restricts on the boundary to the attaching map.

Evidence A. Hard: S26-D-00003353.

S26-T-00003355 - CW pair has homotopy extension property

If A is a subcomplex of a CW complex X, the pair (X,A) satisfies the homotopy extension property.

Evidence A. Hard: S26-D-00003354.

S26-D-00003356 - Cellular chain complex

For a CW complex, C_n^{cell}(X)=H_n(X^n,X^{n-1}) is free on the n-cells.

Evidence A. Hard: S26-T-00003355.

S26-T-00003357 - Cellular homology theorem

The homology of the cellular chain complex is naturally isomorphic to singular homology of the CW complex.

Evidence A. Hard: S26-D-00003356.

S26-I-00003358 - Cellular boundary via degrees

Cellular boundary coefficients are degrees of maps obtained from attaching maps and quotient projections onto (n-1)-spheres.

Evidence A. Hard: S26-T-00003357.

S26-T-00003359 - Euler characteristic from cells

For a finite CW complex, chi(X)=sum_n (-1)^n c_n where c_n is the number of n-cells.

Evidence A. Hard: S26-I-00003358.

S26-D-00003360 - CW approximation interface

Spaces can often be replaced up to weak homotopy type by CW complexes, making cellular tools available.

Evidence A. Hard: S26-T-00003359.

S26-T-00003361 - Whitehead theorem in cellular setting

A weak homotopy equivalence between CW complexes is a homotopy equivalence.

Evidence A. Hard: S26-D-00003360.

S26-D-00003362 - Mapping cone

The mapping cone C_f is obtained by attaching the cone on X to Y along f:X->Y.

Evidence A. Hard: S26-T-00003361.

S26-D-00003363 - Mapping cylinder

The mapping cylinder M_f=(Xx[0,1])union_f Y interpolates between X and Y and provides a standard factorization of f.

Evidence A. Hard: S26-D-00003362.

S26-D-00003364 - Wedge sum as a CW construction

The wedge X∨Y identifies chosen basepoints and inherits a CW structure from based CW structures.

Evidence A. Hard: S26-D-00003363.

S26-D-00003365 - Smash product

For based spaces, X^Y=(XxY)/(X∨Y).

Evidence A. Hard: S26-D-00003364.

S26-D-00003366 - Suspension as a smash product

The reduced suspension Sigma X is homeomorphic to S^1^X for based compactly generated settings.

Evidence A. Hard: S26-D-00003365.

S26-N-00003367 - Cell-structure fence

A CW decomposition is not unique; cell counts in each dimension are not topological invariants although Euler characteristic is for finite CW type.

Evidence A. Hard: S26-D-00003366.

S26-B-00003368 - CW to homotopy bridge

Cell attachment records how homotopy type is assembled dimension by dimension and supports obstruction-theoretic arguments.

Evidence A. Hard: S26-N-00003367.

S26-B-00003369 - CW to computation bridge

Cellular boundary matrices can be dramatically smaller than singular chain complexes while computing the same homology.

Evidence A. Hard: S26-B-00003368.

S26-B-00003370 - CW to manifold bridge

Many smooth manifolds admit CW structures, for example through Morse theory, enabling algebraic-topological calculations.

Evidence A. Hard: S26-B-00003369.

S26-E-00003371 - Real-projective CW count profile n=2

The standard CW structure on RP^2 has one cell in each dimension 0,...,2; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-B-00003370.

S26-E-00003372 - Real-projective CW count profile n=3

The standard CW structure on RP^3 has one cell in each dimension 0,...,3; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003371.

S26-E-00003373 - Real-projective CW count profile n=4

The standard CW structure on RP^4 has one cell in each dimension 0,...,4; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003372.

S26-E-00003374 - Real-projective CW count profile n=5

The standard CW structure on RP^5 has one cell in each dimension 0,...,5; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003373.

S26-E-00003375 - Real-projective CW count profile n=6

The standard CW structure on RP^6 has one cell in each dimension 0,...,6; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003374.

S26-E-00003376 - Real-projective CW count profile n=7

The standard CW structure on RP^7 has one cell in each dimension 0,...,7; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003375.

S26-E-00003377 - Real-projective CW count profile n=8

The standard CW structure on RP^8 has one cell in each dimension 0,...,8; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003376.

S26-E-00003378 - Real-projective CW count profile n=9

The standard CW structure on RP^9 has one cell in each dimension 0,...,9; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003377.

S26-E-00003379 - Real-projective CW count profile n=10

The standard CW structure on RP^10 has one cell in each dimension 0,...,10; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003378.

S26-E-00003380 - Real-projective CW count profile n=11

The standard CW structure on RP^11 has one cell in each dimension 0,...,11; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003379.

S26-E-00003381 - Real-projective CW count profile n=12

The standard CW structure on RP^12 has one cell in each dimension 0,...,12; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003380.

S26-E-00003382 - Real-projective CW count profile n=13

The standard CW structure on RP^13 has one cell in each dimension 0,...,13; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003381.

S26-E-00003383 - Real-projective CW count profile n=14

The standard CW structure on RP^14 has one cell in each dimension 0,...,14; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003382.

S26-E-00003384 - Real-projective CW count profile n=15

The standard CW structure on RP^15 has one cell in each dimension 0,...,15; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003383.

S26-E-00003385 - Real-projective CW count profile n=16

The standard CW structure on RP^16 has one cell in each dimension 0,...,16; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003384.

S26-E-00003386 - Real-projective CW count profile n=17

The standard CW structure on RP^17 has one cell in each dimension 0,...,17; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003385.

S26-E-00003387 - Real-projective CW count profile n=18

The standard CW structure on RP^18 has one cell in each dimension 0,...,18; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003386.

S26-E-00003388 - Real-projective CW count profile n=19

The standard CW structure on RP^19 has one cell in each dimension 0,...,19; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003387.

S26-E-00003389 - Real-projective CW count profile n=20

The standard CW structure on RP^20 has one cell in each dimension 0,...,20; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003388.

S26-E-00003390 - Real-projective CW count profile n=21

The standard CW structure on RP^21 has one cell in each dimension 0,...,21; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003389.

S26-E-00003391 - Real-projective CW count profile n=22

The standard CW structure on RP^22 has one cell in each dimension 0,...,22; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003390.

S26-E-00003392 - Real-projective CW count profile n=23

The standard CW structure on RP^23 has one cell in each dimension 0,...,23; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003391.

S26-E-00003393 - Real-projective CW count profile n=24

The standard CW structure on RP^24 has one cell in each dimension 0,...,24; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003392.

S26-E-00003394 - Real-projective CW count profile n=25

The standard CW structure on RP^25 has one cell in each dimension 0,...,25; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003393.

S26-E-00003395 - Real-projective CW count profile n=26

The standard CW structure on RP^26 has one cell in each dimension 0,...,26; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003394.

S26-E-00003396 - Real-projective CW count profile n=27

The standard CW structure on RP^27 has one cell in each dimension 0,...,27; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003395.

S26-E-00003397 - Real-projective CW count profile n=28

The standard CW structure on RP^28 has one cell in each dimension 0,...,28; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003396.

S26-E-00003398 - Real-projective CW count profile n=29

The standard CW structure on RP^29 has one cell in each dimension 0,...,29; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003397.

S26-E-00003399 - Real-projective CW count profile n=30

The standard CW structure on RP^30 has one cell in each dimension 0,...,30; its cellular alternating count gives Euler characteristic 1.

Evidence M. Hard: S26-E-00003398.

S26-E-00003400 - Real-projective CW count profile n=31

The standard CW structure on RP^31 has one cell in each dimension 0,...,31; its cellular alternating count gives Euler characteristic 0.

Evidence M. Hard: S26-E-00003399.

Part I - Smooth manifolds, maps and submanifolds

S26-D-00003401 - Smooth manifold atlas compatibility refinement

A C^infinity atlas has smooth transition maps on all overlaps; the maximal compatible atlas defines the smooth structure.

Evidence A. Hard: S26-E-00003400.

S26-D-00003402 - Smooth map between manifolds

A map F:M->N is smooth when its coordinate representatives are smooth in compatible charts.

Evidence A. Hard: S26-D-00003401.

S26-T-00003403 - Smoothness is chart independent

If a coordinate representative is smooth in one pair of compatible charts, all compatible coordinate representatives are smooth.

Evidence A. Hard: S26-D-00003402.

S26-D-00003404 - Immersion

A smooth map F:M->N is an immersion when dF_p is injective at every p.

Evidence A. Hard: S26-T-00003403.

S26-D-00003405 - Submersion

A smooth map F:M->N is a submersion when dF_p is surjective at every p.

Evidence A. Hard: S26-D-00003404.

S26-D-00003406 - Embedding of manifolds

A smooth embedding is an immersion that is a homeomorphism onto its image with the subspace topology.

Evidence A. Hard: S26-D-00003405.

S26-T-00003407 - Constant rank theorem

A smooth map of constant rank has a local normal form equivalent to a coordinate projection/inclusion according to its rank.

Evidence A. Hard: S26-D-00003406.

S26-T-00003408 - Regular value theorem

If y is a regular value of F:M->N, then F^{-1}(y) is an embedded submanifold of codimension dim N.

Evidence A. Hard: S26-T-00003407.

S26-D-00003409 - Regular point and critical point

p is regular for F when dF_p has maximal target rank; otherwise p is critical.

Evidence A. Hard: S26-T-00003408.

S26-D-00003410 - Embedded submanifold

An embedded k-submanifold is locally carried by charts to R^kx{0} inside R^n.

Evidence A. Hard: S26-D-00003409.

S26-D-00003411 - Initial submanifold interface

An initial submanifold has a smooth structure characterized by testing smoothness of maps into the ambient manifold.

Evidence A. Hard: S26-D-00003410.

S26-T-00003412 - Whitney embedding theorem interface

Every smooth n-manifold satisfying the standard countability hypotheses admits a smooth embedding into some Euclidean space; classical bounds include R^{2n}.

Evidence A. Hard: S26-D-00003411.

S26-T-00003413 - Whitney immersion theorem interface

Every smooth n-manifold admits a smooth immersion into Euclidean space of controlled dimension under the classical theorem.

Evidence A. Hard: S26-T-00003412.

S26-D-00003414 - Partition of unity subordinate to a cover

A smooth partition of unity subordinate to an open cover is a locally finite family of nonnegative smooth functions summing to 1 with supports lying in cover elements.

Evidence A. Hard: S26-T-00003413.

S26-T-00003415 - Existence of smooth partitions of unity

Paracompact smooth manifolds admit smooth partitions of unity subordinate to any open cover.

Evidence A. Hard: S26-D-00003414.

S26-D-00003416 - Proper smooth map

A smooth map is proper when inverse images of compact sets are compact.

Evidence A. Hard: S26-T-00003415.

S26-T-00003417 - Proper injective immersion is an embedding

A proper injective immersion between suitable manifolds is a smooth embedding.

Evidence A. Hard: S26-D-00003416.

S26-N-00003418 - Immersion-embedding fence

An injective differential does not imply global embedding; topology of the map and image remains additional data.

Evidence A. Hard: S26-T-00003417.

S26-B-00003419 - Manifold to topology bridge

Smooth manifolds carry underlying Hausdorff second-countable topologies; smooth constructions depend on these point-set hypotheses.

Evidence A. Hard: S26-N-00003418.

S26-B-00003420 - Manifold to analysis bridge

Partitions of unity globalize local analytic constructions such as metrics, differential operators and integration.

Evidence A. Hard: S26-B-00003419.

S26-E-00003421 - Manifold dimension profile n=2

S^{2} is a smooth 2-manifold, while S^2xS^1 is a smooth 3-manifold; dimensions add under finite products.

Evidence M. Hard: S26-B-00003420.

S26-E-00003422 - Manifold dimension profile n=3

S^{3} is a smooth 3-manifold, while S^3xS^1 is a smooth 4-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003421.

S26-E-00003423 - Manifold dimension profile n=4

S^{4} is a smooth 4-manifold, while S^4xS^1 is a smooth 5-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003422.

S26-E-00003424 - Manifold dimension profile n=5

S^{5} is a smooth 5-manifold, while S^5xS^1 is a smooth 6-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003423.

S26-E-00003425 - Manifold dimension profile n=6

S^{6} is a smooth 6-manifold, while S^6xS^1 is a smooth 7-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003424.

S26-E-00003426 - Manifold dimension profile n=7

S^{7} is a smooth 7-manifold, while S^7xS^1 is a smooth 8-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003425.

S26-E-00003427 - Manifold dimension profile n=8

S^{8} is a smooth 8-manifold, while S^8xS^1 is a smooth 9-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003426.

S26-E-00003428 - Manifold dimension profile n=9

S^{9} is a smooth 9-manifold, while S^9xS^1 is a smooth 10-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003427.

S26-E-00003429 - Manifold dimension profile n=10

S^{10} is a smooth 10-manifold, while S^10xS^1 is a smooth 11-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003428.

S26-E-00003430 - Manifold dimension profile n=11

S^{11} is a smooth 11-manifold, while S^11xS^1 is a smooth 12-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003429.

S26-E-00003431 - Manifold dimension profile n=12

S^{12} is a smooth 12-manifold, while S^12xS^1 is a smooth 13-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003430.

S26-E-00003432 - Manifold dimension profile n=13

S^{13} is a smooth 13-manifold, while S^13xS^1 is a smooth 14-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003431.

S26-E-00003433 - Manifold dimension profile n=14

S^{14} is a smooth 14-manifold, while S^14xS^1 is a smooth 15-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003432.

S26-E-00003434 - Manifold dimension profile n=15

S^{15} is a smooth 15-manifold, while S^15xS^1 is a smooth 16-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003433.

S26-E-00003435 - Manifold dimension profile n=16

S^{16} is a smooth 16-manifold, while S^16xS^1 is a smooth 17-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003434.

S26-E-00003436 - Manifold dimension profile n=17

S^{17} is a smooth 17-manifold, while S^17xS^1 is a smooth 18-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003435.

S26-E-00003437 - Manifold dimension profile n=18

S^{18} is a smooth 18-manifold, while S^18xS^1 is a smooth 19-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003436.

S26-E-00003438 - Manifold dimension profile n=19

S^{19} is a smooth 19-manifold, while S^19xS^1 is a smooth 20-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003437.

S26-E-00003439 - Manifold dimension profile n=20

S^{20} is a smooth 20-manifold, while S^20xS^1 is a smooth 21-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003438.

S26-E-00003440 - Manifold dimension profile n=21

S^{21} is a smooth 21-manifold, while S^21xS^1 is a smooth 22-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003439.

S26-E-00003441 - Manifold dimension profile n=22

S^{22} is a smooth 22-manifold, while S^22xS^1 is a smooth 23-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003440.

S26-E-00003442 - Manifold dimension profile n=23

S^{23} is a smooth 23-manifold, while S^23xS^1 is a smooth 24-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003441.

S26-E-00003443 - Manifold dimension profile n=24

S^{24} is a smooth 24-manifold, while S^24xS^1 is a smooth 25-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003442.

S26-E-00003444 - Manifold dimension profile n=25

S^{25} is a smooth 25-manifold, while S^25xS^1 is a smooth 26-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003443.

S26-E-00003445 - Manifold dimension profile n=26

S^{26} is a smooth 26-manifold, while S^26xS^1 is a smooth 27-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003444.

S26-E-00003446 - Manifold dimension profile n=27

S^{27} is a smooth 27-manifold, while S^27xS^1 is a smooth 28-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003445.

S26-E-00003447 - Manifold dimension profile n=28

S^{28} is a smooth 28-manifold, while S^28xS^1 is a smooth 29-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003446.

S26-E-00003448 - Manifold dimension profile n=29

S^{29} is a smooth 29-manifold, while S^29xS^1 is a smooth 30-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003447.

S26-E-00003449 - Manifold dimension profile n=30

S^{30} is a smooth 30-manifold, while S^30xS^1 is a smooth 31-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003448.

S26-E-00003450 - Manifold dimension profile n=31

S^{31} is a smooth 31-manifold, while S^31xS^1 is a smooth 32-manifold; dimensions add under finite products.

Evidence M. Hard: S26-E-00003449.

Part J - Tangent geometry, vector fields and flows

S26-D-00003451 - Derivation model of tangent vectors

A tangent vector at p can be defined as a derivation v:C^infinity_p(M)->R satisfying the Leibniz rule.

Evidence A. Hard: S26-E-00003450.

S26-T-00003452 - Equivalence of curve and derivation tangent models

Velocity classes of curves and derivations define canonically isomorphic tangent spaces on a smooth manifold.

Evidence A. Hard: S26-D-00003451.

S26-D-00003453 - Pushforward differential

For smooth F:M->N, dF_p:T_pM->T_{F(p)}N sends tangent vectors by differentiating pullback of functions.

Evidence A. Hard: S26-T-00003452.

S26-T-00003454 - Functoriality of the differential

d(G∘F)_p=dG_{F(p)}∘dF_p and d(id)_p=id.

Evidence A. Hard: S26-D-00003453.

S26-D-00003455 - Vector field as a section of TM

A smooth vector field X assigns X_pinT_pM smoothly, equivalently a smooth section of the tangent bundle.

Evidence A. Hard: S26-T-00003454.

S26-D-00003456 - Lie bracket of vector fields

[X,Y](f)=X(Yf)-Y(Xf) defines a vector field and a Lie algebra structure on smooth vector fields.

Evidence A. Hard: S26-D-00003455.

S26-T-00003457 - Naturality of Lie brackets under diffeomorphism

If F is a diffeomorphism, F_*[X,Y]=[F_*X,F_*Y].

Evidence A. Hard: S26-D-00003456.

S26-D-00003458 - Integral curve of a vector field

An integral curve gamma satisfies gamma dot(t)=X_{gamma(t)}.

Evidence A. Hard: S26-T-00003457.

S26-T-00003459 - Local existence and uniqueness of flows

A smooth vector field admits a unique local flow through each initial point.

Evidence A. Hard: S26-D-00003458.

S26-D-00003460 - Complete vector field

A vector field is complete when its flow exists for all real times.

Evidence A. Hard: S26-T-00003459.

S26-T-00003461 - Compact-manifold completeness

Every smooth vector field on a compact manifold is complete.

Evidence A. Hard: S26-D-00003460.

S26-D-00003462 - Flow box theorem

Near a point where X is nonzero, coordinates can be chosen in which X is a coordinate vector field.

Evidence A. Hard: S26-T-00003461.

S26-D-00003463 - Distribution of tangent subspaces

A rank-k distribution assigns a k-dimensional subspace D_psubsetT_pM smoothly.

Evidence A. Hard: S26-D-00003462.

S26-D-00003464 - Involutive distribution

A distribution is involutive when brackets of tangent vector fields remain tangent to the distribution.

Evidence A. Hard: S26-D-00003463.

S26-T-00003465 - Frobenius theorem

A smooth constant-rank distribution is integrable by submanifolds iff it is involutive.

Evidence A. Hard: S26-D-00003464.

S26-D-00003466 - Foliation interface

A foliation decomposes a manifold into immersed leaves locally modeled by parallel coordinate slices.

Evidence A. Hard: S26-T-00003465.

S26-D-00003467 - Transversality of maps

Maps or submanifolds are transverse when the relevant tangent images sum to the ambient tangent space at intersections.

Evidence A. Hard: S26-D-00003466.

S26-T-00003468 - Transverse preimage theorem

If F is transverse to a submanifold SsubsetN, then F^{-1}(S) is a submanifold of the expected codimension.

Evidence A. Hard: S26-D-00003467.

S26-N-00003469 - Local-flow fence

Local existence of a smooth flow does not imply completeness; solutions can escape every compact set in finite time on noncompact manifolds.

Evidence A. Hard: S26-T-00003468.

S26-B-00003470 - Vector-fields to dynamics bridge

Flows convert differential-geometric vector fields into continuous-time dynamical systems while retaining coordinate-invariant meaning.

Evidence A. Hard: S26-N-00003469.

S26-E-00003471 - Linear vector-field profile dimension=2

For X(x)=x on R^2, the Jacobian is the 2x2 identity, divergence is exactly 2, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-B-00003470.

S26-E-00003472 - Linear vector-field profile dimension=3

For X(x)=x on R^3, the Jacobian is the 3x3 identity, divergence is exactly 3, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003471.

S26-E-00003473 - Linear vector-field profile dimension=4

For X(x)=x on R^4, the Jacobian is the 4x4 identity, divergence is exactly 4, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003472.

S26-E-00003474 - Linear vector-field profile dimension=5

For X(x)=x on R^5, the Jacobian is the 5x5 identity, divergence is exactly 5, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003473.

S26-E-00003475 - Linear vector-field profile dimension=6

For X(x)=x on R^6, the Jacobian is the 6x6 identity, divergence is exactly 6, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003474.

S26-E-00003476 - Linear vector-field profile dimension=7

For X(x)=x on R^7, the Jacobian is the 7x7 identity, divergence is exactly 7, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003475.

S26-E-00003477 - Linear vector-field profile dimension=8

For X(x)=x on R^8, the Jacobian is the 8x8 identity, divergence is exactly 8, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003476.

S26-E-00003478 - Linear vector-field profile dimension=9

For X(x)=x on R^9, the Jacobian is the 9x9 identity, divergence is exactly 9, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003477.

S26-E-00003479 - Linear vector-field profile dimension=10

For X(x)=x on R^10, the Jacobian is the 10x10 identity, divergence is exactly 10, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003478.

S26-E-00003480 - Linear vector-field profile dimension=11

For X(x)=x on R^11, the Jacobian is the 11x11 identity, divergence is exactly 11, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003479.

S26-E-00003481 - Linear vector-field profile dimension=12

For X(x)=x on R^12, the Jacobian is the 12x12 identity, divergence is exactly 12, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003480.

S26-E-00003482 - Linear vector-field profile dimension=13

For X(x)=x on R^13, the Jacobian is the 13x13 identity, divergence is exactly 13, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003481.

S26-E-00003483 - Linear vector-field profile dimension=14

For X(x)=x on R^14, the Jacobian is the 14x14 identity, divergence is exactly 14, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003482.

S26-E-00003484 - Linear vector-field profile dimension=15

For X(x)=x on R^15, the Jacobian is the 15x15 identity, divergence is exactly 15, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003483.

S26-E-00003485 - Linear vector-field profile dimension=16

For X(x)=x on R^16, the Jacobian is the 16x16 identity, divergence is exactly 16, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003484.

S26-E-00003486 - Linear vector-field profile dimension=17

For X(x)=x on R^17, the Jacobian is the 17x17 identity, divergence is exactly 17, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003485.

S26-E-00003487 - Linear vector-field profile dimension=18

For X(x)=x on R^18, the Jacobian is the 18x18 identity, divergence is exactly 18, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003486.

S26-E-00003488 - Linear vector-field profile dimension=19

For X(x)=x on R^19, the Jacobian is the 19x19 identity, divergence is exactly 19, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003487.

S26-E-00003489 - Linear vector-field profile dimension=20

For X(x)=x on R^20, the Jacobian is the 20x20 identity, divergence is exactly 20, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003488.

S26-E-00003490 - Linear vector-field profile dimension=21

For X(x)=x on R^21, the Jacobian is the 21x21 identity, divergence is exactly 21, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003489.

S26-E-00003491 - Linear vector-field profile dimension=22

For X(x)=x on R^22, the Jacobian is the 22x22 identity, divergence is exactly 22, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003490.

S26-E-00003492 - Linear vector-field profile dimension=23

For X(x)=x on R^23, the Jacobian is the 23x23 identity, divergence is exactly 23, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003491.

S26-E-00003493 - Linear vector-field profile dimension=24

For X(x)=x on R^24, the Jacobian is the 24x24 identity, divergence is exactly 24, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003492.

S26-E-00003494 - Linear vector-field profile dimension=25

For X(x)=x on R^25, the Jacobian is the 25x25 identity, divergence is exactly 25, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003493.

S26-E-00003495 - Linear vector-field profile dimension=26

For X(x)=x on R^26, the Jacobian is the 26x26 identity, divergence is exactly 26, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003494.

S26-E-00003496 - Linear vector-field profile dimension=27

For X(x)=x on R^27, the Jacobian is the 27x27 identity, divergence is exactly 27, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003495.

S26-E-00003497 - Linear vector-field profile dimension=28

For X(x)=x on R^28, the Jacobian is the 28x28 identity, divergence is exactly 28, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003496.

S26-E-00003498 - Linear vector-field profile dimension=29

For X(x)=x on R^29, the Jacobian is the 29x29 identity, divergence is exactly 29, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003497.

S26-E-00003499 - Linear vector-field profile dimension=30

For X(x)=x on R^30, the Jacobian is the 30x30 identity, divergence is exactly 30, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003498.

S26-E-00003500 - Linear vector-field profile dimension=31

For X(x)=x on R^31, the Jacobian is the 31x31 identity, divergence is exactly 31, and the flow is Phi_t(x)=e^t x.

Evidence M. Hard: S26-E-00003499.

Part K - Differential forms, de Rham theory and integration

S26-D-00003501 - Differential k-form on a manifold refinement

A smooth k-form is a smooth section of the exterior-power cotangent bundle Lambda^k T^*M.

Evidence A. Hard: S26-E-00003500.

S26-D-00003502 - Wedge product of differential forms refinement

The wedge product combines p- and q-forms into a (p+q)-form and is graded-anticommutative.

Evidence A. Hard: S26-D-00003501.

S26-T-00003503 - Exterior derivative squares to zero

For differential forms, d∘d=0.

Evidence A. Hard: S26-D-00003502.

S26-T-00003504 - Leibniz rule for exterior derivative

d(alpha^beta)=dalpha^beta+(-1)^p alpha^dbeta for alpha of degree p.

Evidence A. Hard: S26-T-00003503.

S26-T-00003505 - Naturality of exterior derivative under manifold pullback refinement

For smooth F:M->N, d(F^*alpha)=F^*(dalpha).

Evidence A. Hard: S26-T-00003504, S26-T-00000927.

S26-D-00003506 - Closed differential form

A form alpha is closed when d alpha=0.

Evidence A. Hard: S26-T-00003505.

S26-D-00003507 - Exact differential form

A form alpha is exact when alpha=d beta for some lower-degree beta.

Evidence A. Hard: S26-D-00003506.

S26-D-00003508 - de Rham cohomology refinement

H^k_dR(M)=ker(d:Omega^k->Omega^{k+1})/im(d:Omega^{k-1}->Omega^k).

Evidence A. Hard: S26-D-00003507.

S26-T-00003509 - Poincare lemma

Every closed differential k-form with k>0 is locally exact on a star-shaped or contractible coordinate domain.

Evidence A. Hard: S26-D-00003508.

S26-T-00003510 - de Rham theorem

Integration of differential forms over singular chains induces an isomorphism H^*_dR(M)□H^*(M;R).

Evidence A. Hard: S26-T-00003509.

S26-D-00003511 - Orientation form

On an oriented n-manifold, a nowhere-vanishing top-degree form determines the orientation in the smooth category.

Evidence A. Hard: S26-T-00003510.

S26-D-00003512 - Integration of compactly supported top forms refinement

A compactly supported top-degree form can be integrated on an oriented manifold using oriented charts and a partition of unity.

Evidence A. Hard: S26-D-00003511, S26-D-00000933.

S26-T-00003513 - Stokes theorem on manifolds

For a compact oriented manifold with boundary, integral_M d omega = integral_{partialM} omega for appropriate degree and support/regularity.

Evidence A. Hard: S26-D-00003512.

S26-D-00003514 - Compactly supported de Rham cohomology

H^k_{dR,c}(M) is computed from compactly supported differential forms.

Evidence A. Hard: S26-T-00003513.

S26-D-00003515 - Pullback on de Rham cohomology

A smooth map induces a graded-ring homomorphism on de Rham cohomology by pullback.

Evidence A. Hard: S26-D-00003514.

S26-D-00003516 - Lie derivative of forms

The Lie derivative L_X alpha measures infinitesimal change of alpha along the flow of X.

Evidence A. Hard: S26-D-00003515.

S26-I-00003517 - Cartan magic formula

L_X alpha=d(i_X alpha)+i_X(d alpha).

Evidence A. Hard: S26-D-00003516.

S26-N-00003518 - Local-global exactness fence

The Poincare lemma is local; a closed form need not be globally exact when de Rham cohomology is nontrivial.

Evidence A. Hard: S26-I-00003517.

S26-B-00003519 - Forms to electromagnetism bridge

Closed/exact differential forms and Stokes-type identities provide coordinate-free expressions for flux, circulation and conservation laws.

Evidence A. Hard: S26-N-00003518.

S26-B-00003520 - de Rham to topology bridge

The de Rham theorem turns analytic differential-form computations into topological cohomology invariants.

Evidence A. Hard: S26-B-00003519.

S26-E-00003521 - Torus de Rham profile dimension=2

For the n-torus T^2, de Rham Betti numbers are binomial(2,j); their total rank is 4 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-B-00003520.

S26-E-00003522 - Torus de Rham profile dimension=3

For the n-torus T^3, de Rham Betti numbers are binomial(3,j); their total rank is 8 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003521.

S26-E-00003523 - Torus de Rham profile dimension=4

For the n-torus T^4, de Rham Betti numbers are binomial(4,j); their total rank is 16 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003522.

S26-E-00003524 - Torus de Rham profile dimension=5

For the n-torus T^5, de Rham Betti numbers are binomial(5,j); their total rank is 32 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003523.

S26-E-00003525 - Torus de Rham profile dimension=6

For the n-torus T^6, de Rham Betti numbers are binomial(6,j); their total rank is 64 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003524.

S26-E-00003526 - Torus de Rham profile dimension=7

For the n-torus T^7, de Rham Betti numbers are binomial(7,j); their total rank is 128 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003525.

S26-E-00003527 - Torus de Rham profile dimension=8

For the n-torus T^8, de Rham Betti numbers are binomial(8,j); their total rank is 256 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003526.

S26-E-00003528 - Torus de Rham profile dimension=9

For the n-torus T^9, de Rham Betti numbers are binomial(9,j); their total rank is 512 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003527.

S26-E-00003529 - Torus de Rham profile dimension=10

For the n-torus T^10, de Rham Betti numbers are binomial(10,j); their total rank is 1024 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003528.

S26-E-00003530 - Torus de Rham profile dimension=11

For the n-torus T^11, de Rham Betti numbers are binomial(11,j); their total rank is 2048 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003529.

S26-E-00003531 - Torus de Rham profile dimension=12

For the n-torus T^12, de Rham Betti numbers are binomial(12,j); their total rank is 4096 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003530.

S26-E-00003532 - Torus de Rham profile dimension=13

For the n-torus T^13, de Rham Betti numbers are binomial(13,j); their total rank is 8192 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003531.

S26-E-00003533 - Torus de Rham profile dimension=14

For the n-torus T^14, de Rham Betti numbers are binomial(14,j); their total rank is 16384 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003532.

S26-E-00003534 - Torus de Rham profile dimension=15

For the n-torus T^15, de Rham Betti numbers are binomial(15,j); their total rank is 32768 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003533.

S26-E-00003535 - Torus de Rham profile dimension=16

For the n-torus T^16, de Rham Betti numbers are binomial(16,j); their total rank is 65536 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003534.

S26-E-00003536 - Torus de Rham profile dimension=17

For the n-torus T^17, de Rham Betti numbers are binomial(17,j); their total rank is 131072 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003535.

S26-E-00003537 - Torus de Rham profile dimension=18

For the n-torus T^18, de Rham Betti numbers are binomial(18,j); their total rank is 262144 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003536.

S26-E-00003538 - Torus de Rham profile dimension=19

For the n-torus T^19, de Rham Betti numbers are binomial(19,j); their total rank is 524288 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003537.

S26-E-00003539 - Torus de Rham profile dimension=20

For the n-torus T^20, de Rham Betti numbers are binomial(20,j); their total rank is 1048576 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003538.

S26-E-00003540 - Torus de Rham profile dimension=21

For the n-torus T^21, de Rham Betti numbers are binomial(21,j); their total rank is 2097152 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003539.

S26-E-00003541 - Torus de Rham profile dimension=22

For the n-torus T^22, de Rham Betti numbers are binomial(22,j); their total rank is 4194304 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003540.

S26-E-00003542 - Torus de Rham profile dimension=23

For the n-torus T^23, de Rham Betti numbers are binomial(23,j); their total rank is 8388608 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003541.

S26-E-00003543 - Torus de Rham profile dimension=24

For the n-torus T^24, de Rham Betti numbers are binomial(24,j); their total rank is 16777216 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003542.

S26-E-00003544 - Torus de Rham profile dimension=25

For the n-torus T^25, de Rham Betti numbers are binomial(25,j); their total rank is 33554432 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003543.

S26-E-00003545 - Torus de Rham profile dimension=26

For the n-torus T^26, de Rham Betti numbers are binomial(26,j); their total rank is 67108864 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003544.

S26-E-00003546 - Torus de Rham profile dimension=27

For the n-torus T^27, de Rham Betti numbers are binomial(27,j); their total rank is 134217728 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003545.

S26-E-00003547 - Torus de Rham profile dimension=28

For the n-torus T^28, de Rham Betti numbers are binomial(28,j); their total rank is 268435456 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003546.

S26-E-00003548 - Torus de Rham profile dimension=29

For the n-torus T^29, de Rham Betti numbers are binomial(29,j); their total rank is 536870912 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003547.

S26-E-00003549 - Torus de Rham profile dimension=30

For the n-torus T^30, de Rham Betti numbers are binomial(30,j); their total rank is 1073741824 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003548.

S26-E-00003550 - Torus de Rham profile dimension=31

For the n-torus T^31, de Rham Betti numbers are binomial(31,j); their total rank is 2147483648 and Euler characteristic is 0 for n>0.

Evidence M. Hard: S26-E-00003549.

Part L - Fiber bundles, vector bundles and principal bundles

S26-D-00003551 - Fiber bundle

A fiber bundle p:E->B with fiber F is locally trivial: each b has a neighborhood U with p^{-1}(U) homeomorphic or diffeomorphic to UxF compatibly with projection.

Evidence A. Hard: S26-E-00003550.

S26-D-00003552 - Bundle chart and transition function

Over overlaps of trivializing neighborhoods, bundle charts differ by transition maps valued in the structure group acting on the fiber.

Evidence A. Hard: S26-D-00003551.

S26-D-00003553 - Vector bundle

A rank-r vector bundle is a fiber bundle whose fibers are r-dimensional vector spaces and whose transition functions are linear.

Evidence A. Hard: S26-D-00003552.

S26-D-00003554 - Bundle section

A section s:B->E satisfies p∘s=id_B.

Evidence A. Hard: S26-D-00003553.

S26-D-00003555 - Trivial bundle

A bundle is trivial when it is bundle-isomorphic to BxF with the product projection.

Evidence A. Hard: S26-D-00003554.

S26-T-00003556 - Vector bundle triviality by global frame

A rank-r vector bundle is trivial iff it admits r pointwise linearly independent global sections forming a frame.

Evidence A. Hard: S26-D-00003555.

S26-D-00003557 - Pullback bundle

For f:X->B and p:E->B, the pullback f^*E consists of pairs (x,e) with f(x)=p(e) and is a bundle over X.

Evidence A. Hard: S26-T-00003556.

S26-T-00003558 - Functoriality of pullback bundles

Pullback is compatible with identity and composition up to canonical bundle isomorphism.

Evidence A. Hard: S26-D-00003557.

S26-D-00003559 - Whitney sum of vector bundles

E⊕F over the same base has fiber E_b⊕F_b and transition functions given by block sums.

Evidence A. Hard: S26-T-00003558.

S26-D-00003560 - Tensor and dual bundles

Fiberwise tensor product and dualization define vector bundles EtensorF and E^*.

Evidence A. Hard: S26-D-00003559.

S26-D-00003561 - Principal G-bundle

A principal G-bundle is a fiber bundle with free transitive right G-action on each fiber and local G-equivariant trivializations.

Evidence A. Hard: S26-D-00003560.

S26-D-00003562 - Associated bundle

Given a principal G-bundle P and a G-space F, Px_G F is the associated bundle with fiber F.

Evidence A. Hard: S26-D-00003561.

S26-D-00003563 - Gauge transformation

A gauge transformation is an automorphism of a principal bundle covering the identity on the base.

Evidence A. Hard: S26-D-00003562.

S26-D-00003564 - Clutching construction

Bundles over spheres and other decomposed spaces can be formed by gluing trivial bundles along overlap via a clutching map into the structure group.

Evidence A. Hard: S26-D-00003563.

S26-T-00003565 - Classification of line bundles by projective maps interface

Under standard paracompactness hypotheses, real or complex line bundles are classified by homotopy classes into the corresponding infinite projective space.

Evidence A. Hard: S26-D-00003564.

S26-D-00003566 - Tangent bundle as a vector bundle

TM->M is the rank-dim M vector bundle whose fiber at p is T_pM.

Evidence A. Hard: S26-T-00003565.

S26-D-00003567 - Normal bundle

For an embedded submanifold SsubsetM with a chosen Riemannian splitting or quotient formulation, the normal bundle records directions transverse to S.

Evidence A. Hard: S26-D-00003566.

S26-N-00003568 - Local-triviality fence

Fiberwise isomorphism of sets or vector spaces does not by itself give a bundle; local triviality and compatible topology/smoothness are essential.

Evidence A. Hard: S26-D-00003567.

S26-B-00003569 - Bundles to cohomology bridge

Characteristic classes assign cohomology classes to bundles and detect obstructions to triviality or existence of sections.

Evidence A. Hard: S26-N-00003568.

S26-B-00003570 - Principal bundles to gauge theory bridge

Connections and curvature on principal bundles provide the geometric substrate of gauge fields and holonomy.

Evidence A. Hard: S26-B-00003569.

S26-E-00003571 - Trivial vector-bundle profile rank=2

The trivial rank-2 real vector bundle over S^1 has total-space dimension 3 and admits the explicit global frame of 2 constant coordinate sections.

Evidence M. Hard: S26-B-00003570.

S26-E-00003572 - Trivial vector-bundle profile rank=3

The trivial rank-3 real vector bundle over S^1 has total-space dimension 4 and admits the explicit global frame of 3 constant coordinate sections.

Evidence M. Hard: S26-E-00003571.

S26-E-00003573 - Trivial vector-bundle profile rank=4

The trivial rank-4 real vector bundle over S^1 has total-space dimension 5 and admits the explicit global frame of 4 constant coordinate sections.

Evidence M. Hard: S26-E-00003572.

S26-E-00003574 - Trivial vector-bundle profile rank=5

The trivial rank-5 real vector bundle over S^1 has total-space dimension 6 and admits the explicit global frame of 5 constant coordinate sections.

Evidence M. Hard: S26-E-00003573.

S26-E-00003575 - Trivial vector-bundle profile rank=6

The trivial rank-6 real vector bundle over S^1 has total-space dimension 7 and admits the explicit global frame of 6 constant coordinate sections.

Evidence M. Hard: S26-E-00003574.

S26-E-00003576 - Trivial vector-bundle profile rank=7

The trivial rank-7 real vector bundle over S^1 has total-space dimension 8 and admits the explicit global frame of 7 constant coordinate sections.

Evidence M. Hard: S26-E-00003575.

S26-E-00003577 - Trivial vector-bundle profile rank=8

The trivial rank-8 real vector bundle over S^1 has total-space dimension 9 and admits the explicit global frame of 8 constant coordinate sections.

Evidence M. Hard: S26-E-00003576.

S26-E-00003578 - Trivial vector-bundle profile rank=9

The trivial rank-9 real vector bundle over S^1 has total-space dimension 10 and admits the explicit global frame of 9 constant coordinate sections.

Evidence M. Hard: S26-E-00003577.

S26-E-00003579 - Trivial vector-bundle profile rank=10

The trivial rank-10 real vector bundle over S^1 has total-space dimension 11 and admits the explicit global frame of 10 constant coordinate sections.

Evidence M. Hard: S26-E-00003578.

S26-E-00003580 - Trivial vector-bundle profile rank=11

The trivial rank-11 real vector bundle over S^1 has total-space dimension 12 and admits the explicit global frame of 11 constant coordinate sections.

Evidence M. Hard: S26-E-00003579.

S26-E-00003581 - Trivial vector-bundle profile rank=12

The trivial rank-12 real vector bundle over S^1 has total-space dimension 13 and admits the explicit global frame of 12 constant coordinate sections.

Evidence M. Hard: S26-E-00003580.

S26-E-00003582 - Trivial vector-bundle profile rank=13

The trivial rank-13 real vector bundle over S^1 has total-space dimension 14 and admits the explicit global frame of 13 constant coordinate sections.

Evidence M. Hard: S26-E-00003581.

S26-E-00003583 - Trivial vector-bundle profile rank=14

The trivial rank-14 real vector bundle over S^1 has total-space dimension 15 and admits the explicit global frame of 14 constant coordinate sections.

Evidence M. Hard: S26-E-00003582.

S26-E-00003584 - Trivial vector-bundle profile rank=15

The trivial rank-15 real vector bundle over S^1 has total-space dimension 16 and admits the explicit global frame of 15 constant coordinate sections.

Evidence M. Hard: S26-E-00003583.

S26-E-00003585 - Trivial vector-bundle profile rank=16

The trivial rank-16 real vector bundle over S^1 has total-space dimension 17 and admits the explicit global frame of 16 constant coordinate sections.

Evidence M. Hard: S26-E-00003584.

S26-E-00003586 - Trivial vector-bundle profile rank=17

The trivial rank-17 real vector bundle over S^1 has total-space dimension 18 and admits the explicit global frame of 17 constant coordinate sections.

Evidence M. Hard: S26-E-00003585.

S26-E-00003587 - Trivial vector-bundle profile rank=18

The trivial rank-18 real vector bundle over S^1 has total-space dimension 19 and admits the explicit global frame of 18 constant coordinate sections.

Evidence M. Hard: S26-E-00003586.

S26-E-00003588 - Trivial vector-bundle profile rank=19

The trivial rank-19 real vector bundle over S^1 has total-space dimension 20 and admits the explicit global frame of 19 constant coordinate sections.

Evidence M. Hard: S26-E-00003587.

S26-E-00003589 - Trivial vector-bundle profile rank=20

The trivial rank-20 real vector bundle over S^1 has total-space dimension 21 and admits the explicit global frame of 20 constant coordinate sections.

Evidence M. Hard: S26-E-00003588.

S26-E-00003590 - Trivial vector-bundle profile rank=21

The trivial rank-21 real vector bundle over S^1 has total-space dimension 22 and admits the explicit global frame of 21 constant coordinate sections.

Evidence M. Hard: S26-E-00003589.

S26-E-00003591 - Trivial vector-bundle profile rank=22

The trivial rank-22 real vector bundle over S^1 has total-space dimension 23 and admits the explicit global frame of 22 constant coordinate sections.

Evidence M. Hard: S26-E-00003590.

S26-E-00003592 - Trivial vector-bundle profile rank=23

The trivial rank-23 real vector bundle over S^1 has total-space dimension 24 and admits the explicit global frame of 23 constant coordinate sections.

Evidence M. Hard: S26-E-00003591.

S26-E-00003593 - Trivial vector-bundle profile rank=24

The trivial rank-24 real vector bundle over S^1 has total-space dimension 25 and admits the explicit global frame of 24 constant coordinate sections.

Evidence M. Hard: S26-E-00003592.

S26-E-00003594 - Trivial vector-bundle profile rank=25

The trivial rank-25 real vector bundle over S^1 has total-space dimension 26 and admits the explicit global frame of 25 constant coordinate sections.

Evidence M. Hard: S26-E-00003593.

S26-E-00003595 - Trivial vector-bundle profile rank=26

The trivial rank-26 real vector bundle over S^1 has total-space dimension 27 and admits the explicit global frame of 26 constant coordinate sections.

Evidence M. Hard: S26-E-00003594.

S26-E-00003596 - Trivial vector-bundle profile rank=27

The trivial rank-27 real vector bundle over S^1 has total-space dimension 28 and admits the explicit global frame of 27 constant coordinate sections.

Evidence M. Hard: S26-E-00003595.

S26-E-00003597 - Trivial vector-bundle profile rank=28

The trivial rank-28 real vector bundle over S^1 has total-space dimension 29 and admits the explicit global frame of 28 constant coordinate sections.

Evidence M. Hard: S26-E-00003596.

S26-E-00003598 - Trivial vector-bundle profile rank=29

The trivial rank-29 real vector bundle over S^1 has total-space dimension 30 and admits the explicit global frame of 29 constant coordinate sections.

Evidence M. Hard: S26-E-00003597.

S26-E-00003599 - Trivial vector-bundle profile rank=30

The trivial rank-30 real vector bundle over S^1 has total-space dimension 31 and admits the explicit global frame of 30 constant coordinate sections.

Evidence M. Hard: S26-E-00003598.

S26-E-00003600 - Trivial vector-bundle profile rank=31

The trivial rank-31 real vector bundle over S^1 has total-space dimension 32 and admits the explicit global frame of 31 constant coordinate sections.

Evidence M. Hard: S26-E-00003599.

Part M - Characteristic classes and classifying spaces

S26-D-00003601 - Characteristic class

A characteristic class assigns to each bundle E->B a cohomology class c(E) naturally under pullback.

Evidence A. Hard: S26-E-00003600.

S26-D-00003602 - Stiefel-Whitney classes

For a real vector bundle E, w_i(E)inH^i(B;Z/2) are its Stiefel-Whitney classes with total class w(E)=1+w_1+w_2+....

Evidence A. Hard: S26-D-00003601.

S26-T-00003603 - Whitney product formula for Stiefel-Whitney classes

w(E⊕F)=w(E) cup w(F).

Evidence A. Hard: S26-D-00003602.

S26-D-00003604 - First Stiefel-Whitney obstruction

w_1(E) is the primary obstruction to orientability of a real vector bundle.

Evidence A. Hard: S26-T-00003603.

S26-D-00003605 - Second Stiefel-Whitney spin obstruction interface

For an oriented real vector bundle, w_2 is the obstruction to a spin structure in the standard theory.

Evidence A. Hard: S26-D-00003604.

S26-D-00003606 - Chern classes

For a complex vector bundle E, c_i(E)inH^{2i}(B;Z) are its Chern classes.

Evidence A. Hard: S26-D-00003605.

S26-T-00003607 - Whitney product formula for Chern classes

c(E⊕F)=c(E) cup c(F).

Evidence A. Hard: S26-D-00003606.

S26-D-00003608 - First Chern class of a complex line bundle

Complex line bundles have a characteristic class c_1inH^2(B;Z) that classifies them over broad paracompact bases.

Evidence A. Hard: S26-T-00003607.

S26-D-00003609 - Pontryagin classes

For a real vector bundle E, Pontryagin classes p_i(E)inH^{4i}(B;Z) are defined via Chern classes of the complexification with standard signs.

Evidence A. Hard: S26-D-00003608.

S26-D-00003610 - Euler class

An oriented rank-n real vector bundle has Euler class e(E)inH^n(B;Z).

Evidence A. Hard: S26-D-00003609.

S26-T-00003611 - Euler class and zero sections interface

Under transversality, the Euler class pairs with the fundamental class to count signed zeros of a generic section on a closed oriented manifold.

Evidence A. Hard: S26-D-00003610.

S26-D-00003612 - Chern character

ch(E) is a rational even-degree cohomology class additive under direct sum and multiplicative under tensor product.

Evidence A. Hard: S26-T-00003611.

S26-T-00003613 - Naturality of characteristic classes

For f:X->B, characteristic classes satisfy c(f^*E)=f^*c(E).

Evidence A. Hard: S26-D-00003612.

S26-D-00003614 - Classifying space BG

Principal G-bundles over suitable bases are classified up to isomorphism by homotopy classes of maps into BG.

Evidence A. Hard: S26-T-00003613.

S26-D-00003615 - Universal bundle EG->BG

A universal principal G-bundle has contractible total space EG with free G-action and quotient BG.

Evidence A. Hard: S26-D-00003614.

S26-D-00003616 - Thom class interface

An oriented rank-r vector bundle admits a Thom class in compactly supported relative cohomology, inducing the Thom isomorphism.

Evidence A. Hard: S26-D-00003615.

S26-T-00003617 - Thom isomorphism interface

Cup product with the Thom class identifies cohomology of the base with degree-shifted compactly supported cohomology of the bundle under orientation hypotheses.

Evidence A. Hard: S26-D-00003616.

S26-N-00003618 - Vanishing-class fence

Vanishing of selected characteristic classes need not imply bundle triviality; characteristic classes are powerful but generally incomplete invariants.

Evidence A. Hard: S26-T-00003617.

S26-B-00003619 - Characteristic classes to curvature bridge

Chern-Weil theory represents real characteristic classes by invariant polynomials in curvature of a connection.

Evidence A. Hard: S26-N-00003618.

S26-B-00003620 - Characteristic classes to topology bridge

Characteristic numbers obtained by pairing classes with fundamental cycles constrain manifold topology and cobordism.

Evidence A. Hard: S26-B-00003619.

S26-E-00003621 - Sphere tangent Euler-number profile n=2

For the tangent bundle TS^2, pairing the Euler class with [S^2] equals chi(S^2)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-B-00003620.

S26-E-00003622 - Sphere tangent Euler-number profile n=3

For the tangent bundle TS^3, pairing the Euler class with [S^3] equals chi(S^3)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003621.

S26-E-00003623 - Sphere tangent Euler-number profile n=4

For the tangent bundle TS^4, pairing the Euler class with [S^4] equals chi(S^4)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003622.

S26-E-00003624 - Sphere tangent Euler-number profile n=5

For the tangent bundle TS^5, pairing the Euler class with [S^5] equals chi(S^5)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003623.

S26-E-00003625 - Sphere tangent Euler-number profile n=6

For the tangent bundle TS^6, pairing the Euler class with [S^6] equals chi(S^6)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003624.

S26-E-00003626 - Sphere tangent Euler-number profile n=7

For the tangent bundle TS^7, pairing the Euler class with [S^7] equals chi(S^7)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003625.

S26-E-00003627 - Sphere tangent Euler-number profile n=8

For the tangent bundle TS^8, pairing the Euler class with [S^8] equals chi(S^8)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003626.

S26-E-00003628 - Sphere tangent Euler-number profile n=9

For the tangent bundle TS^9, pairing the Euler class with [S^9] equals chi(S^9)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003627.

S26-E-00003629 - Sphere tangent Euler-number profile n=10

For the tangent bundle TS^10, pairing the Euler class with [S^10] equals chi(S^10)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003628.

S26-E-00003630 - Sphere tangent Euler-number profile n=11

For the tangent bundle TS^11, pairing the Euler class with [S^11] equals chi(S^11)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003629.

S26-E-00003631 - Sphere tangent Euler-number profile n=12

For the tangent bundle TS^12, pairing the Euler class with [S^12] equals chi(S^12)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003630.

S26-E-00003632 - Sphere tangent Euler-number profile n=13

For the tangent bundle TS^13, pairing the Euler class with [S^13] equals chi(S^13)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003631.

S26-E-00003633 - Sphere tangent Euler-number profile n=14

For the tangent bundle TS^14, pairing the Euler class with [S^14] equals chi(S^14)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003632.

S26-E-00003634 - Sphere tangent Euler-number profile n=15

For the tangent bundle TS^15, pairing the Euler class with [S^15] equals chi(S^15)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003633.

S26-E-00003635 - Sphere tangent Euler-number profile n=16

For the tangent bundle TS^16, pairing the Euler class with [S^16] equals chi(S^16)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003634.

S26-E-00003636 - Sphere tangent Euler-number profile n=17

For the tangent bundle TS^17, pairing the Euler class with [S^17] equals chi(S^17)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003635.

S26-E-00003637 - Sphere tangent Euler-number profile n=18

For the tangent bundle TS^18, pairing the Euler class with [S^18] equals chi(S^18)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003636.

S26-E-00003638 - Sphere tangent Euler-number profile n=19

For the tangent bundle TS^19, pairing the Euler class with [S^19] equals chi(S^19)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003637.

S26-E-00003639 - Sphere tangent Euler-number profile n=20

For the tangent bundle TS^20, pairing the Euler class with [S^20] equals chi(S^20)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003638.

S26-E-00003640 - Sphere tangent Euler-number profile n=21

For the tangent bundle TS^21, pairing the Euler class with [S^21] equals chi(S^21)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003639.

S26-E-00003641 - Sphere tangent Euler-number profile n=22

For the tangent bundle TS^22, pairing the Euler class with [S^22] equals chi(S^22)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003640.

S26-E-00003642 - Sphere tangent Euler-number profile n=23

For the tangent bundle TS^23, pairing the Euler class with [S^23] equals chi(S^23)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003641.

S26-E-00003643 - Sphere tangent Euler-number profile n=24

For the tangent bundle TS^24, pairing the Euler class with [S^24] equals chi(S^24)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003642.

S26-E-00003644 - Sphere tangent Euler-number profile n=25

For the tangent bundle TS^25, pairing the Euler class with [S^25] equals chi(S^25)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003643.

S26-E-00003645 - Sphere tangent Euler-number profile n=26

For the tangent bundle TS^26, pairing the Euler class with [S^26] equals chi(S^26)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003644.

S26-E-00003646 - Sphere tangent Euler-number profile n=27

For the tangent bundle TS^27, pairing the Euler class with [S^27] equals chi(S^27)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003645.

S26-E-00003647 - Sphere tangent Euler-number profile n=28

For the tangent bundle TS^28, pairing the Euler class with [S^28] equals chi(S^28)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003646.

S26-E-00003648 - Sphere tangent Euler-number profile n=29

For the tangent bundle TS^29, pairing the Euler class with [S^29] equals chi(S^29)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003647.

S26-E-00003649 - Sphere tangent Euler-number profile n=30

For the tangent bundle TS^30, pairing the Euler class with [S^30] equals chi(S^30)=2; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003648.

S26-E-00003650 - Sphere tangent Euler-number profile n=31

For the tangent bundle TS^31, pairing the Euler class with [S^31] equals chi(S^31)=0; this bounded instance records the standard sphere Euler characteristic.

Evidence M. Hard: S26-E-00003649.

Part N - Riemannian metrics, geodesics and completeness

S26-D-00003651 - Riemannian metric refinement

A Riemannian metric is a smooth field of positive-definite inner products g_p on tangent spaces.

Evidence A. Hard: S26-E-00003650.

S26-D-00003652 - Metric musical isomorphisms refinement

The metric defines bundle isomorphisms sharp:T^*M->TM and flat:TM->T^*M by contraction with g.

Evidence A. Hard: S26-D-00003651, S26-D-00000939.

S26-D-00003653 - Metric-dual characterization of Riemannian gradient

grad f is characterized by g(grad f,X)=df(X) for every vector field X.

Evidence A. Hard: S26-D-00003652, S26-D-00000940.

S26-D-00003654 - Divergence with respect to a Riemannian volume

div X is characterized locally or invariantly by L_X dvol_g=(div X)dvol_g.

Evidence A. Hard: S26-D-00003653.

S26-D-00003655 - Laplace-Beltrami operator

Delta f=div grad f, with sign convention declared explicitly in applications.

Evidence A. Hard: S26-D-00003654.

S26-D-00003656 - Length of a smooth curve

L(gamma)=integral sqrt(g(gamma dot,gamma dot)) dt.

Evidence A. Hard: S26-D-00003655.

S26-D-00003657 - Riemannian length distance refinement

d_g(p,q) is the infimum of lengths of piecewise smooth curves joining p and q.

Evidence A. Hard: S26-D-00003656, S26-D-00000938.

S26-T-00003658 - Riemannian distance induces the manifold topology

The metric-space topology of d_g agrees with the original smooth-manifold topology.

Evidence A. Hard: S26-D-00003657.

S26-D-00003659 - Levi-Civita connection refinement

The Levi-Civita connection is the unique affine connection that is torsion-free and metric-compatible.

Evidence A. Hard: S26-T-00003658.

S26-I-00003660 - Koszul formula

The Levi-Civita connection is determined by the Koszul formula expressing 2g(nabla_X Y,Z) in terms of derivatives of g and Lie brackets.

Evidence A. Hard: S26-D-00003659.

S26-D-00003661 - Geodesic refinement

A geodesic satisfies nabla_{gamma dot} gamma dot=0.

Evidence A. Hard: S26-I-00003660.

S26-D-00003662 - Riemannian exponential map domain refinement

exp_p(v) is the endpoint at time 1 of the geodesic starting at p with initial velocity v, where defined.

Evidence A. Hard: S26-D-00003661, S26-D-00000946.

S26-T-00003663 - Gauss lemma

Radial geodesics are orthogonal to geodesic spheres in normal neighborhoods, yielding local minimizing properties.

Evidence A. Hard: S26-D-00003662.

S26-D-00003664 - Normal coordinates

Normal coordinates at p are induced by exp_p and satisfy g_ij(p)=delta_ij with first derivatives vanishing at p.

Evidence A. Hard: S26-T-00003663.

S26-D-00003665 - Injectivity radius

inj(p) is the largest radius on which exp_p is a diffeomorphism from the tangent ball onto its image.

Evidence A. Hard: S26-D-00003664.

S26-D-00003666 - Cut locus interface

The cut locus of p marks endpoints beyond which minimizing radial geodesics cease to be uniquely minimizing or exp_p loses regularity.

Evidence A. Hard: S26-D-00003665.

S26-T-00003667 - Hopf-Rinow theorem

For a connected Riemannian manifold, metric completeness, geodesic completeness, and compactness of closed bounded subsets are equivalent in the classical finite-dimensional theorem.

Evidence A. Hard: S26-D-00003666.

S26-N-00003668 - Completeness fence

Local geodesic existence does not imply geodesic completeness; completeness is a global metric property.

Evidence A. Hard: S26-T-00003667.

S26-B-00003669 - Riemannian to analysis bridge

The metric determines volume, gradient, divergence and Laplace-Beltrami operators, linking geometry with elliptic and spectral analysis.

Evidence A. Hard: S26-N-00003668.

S26-B-00003670 - Riemannian to probability bridge

Brownian motion on a Riemannian manifold is generated, up to convention, by a multiple of the Laplace-Beltrami operator.

Evidence A. Hard: S26-B-00003669.

S26-E-00003671 - Round-sphere area profile S^2, r=1.05

The standard round n-sphere of radius 1.05 has n-dimensional volume 13.8544236023 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-B-00003670.

S26-E-00003672 - Round-sphere area profile S^3, r=1.10

The standard round n-sphere of radius 1.10 has n-dimensional volume 26.2728869157 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003671.

S26-E-00003673 - Round-sphere area profile S^4, r=1.15

The standard round n-sphere of radius 1.15 has n-dimensional volume 46.0319994201 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003672.

S26-E-00003674 - Round-sphere area profile S^5, r=1.20

The standard round n-sphere of radius 1.20 has n-dimensional volume 77.1535383891 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003673.

S26-E-00003675 - Round-sphere area profile S^6, r=1.25

The standard round n-sphere of radius 1.25 has n-dimensional volume 126.164862794 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003674.

S26-E-00003676 - Round-sphere area profile S^7, r=1.30

The standard round n-sphere of radius 1.30 has n-dimensional volume 203.74253349 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003675.

S26-E-00003677 - Round-sphere area profile S^8, r=1.35

The standard round n-sphere of radius 1.35 has n-dimensional volume 327.51433845 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003676.

S26-E-00003678 - Round-sphere area profile S^9, r=1.40

The standard round n-sphere of radius 1.40 has n-dimensional volume 526.890585348 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003677.

S26-E-00003679 - Round-sphere area profile S^10, r=1.45

The standard round n-sphere of radius 1.45 has n-dimensional volume 851.486077523 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003678.

S26-E-00003680 - Round-sphere area profile S^11, r=1.50

The standard round n-sphere of radius 1.50 has n-dimensional volume 1385.96363504 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003679.

S26-E-00003681 - Round-sphere area profile S^12, r=1.55

The standard round n-sphere of radius 1.55 has n-dimensional volume 2276.48655759 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003680.

S26-E-00003682 - Round-sphere area profile S^13, r=1.60

The standard round n-sphere of radius 1.60 has n-dimensional volume 3778.38651532 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003681.

S26-E-00003683 - Round-sphere area profile S^14, r=1.65

The standard round n-sphere of radius 1.65 has n-dimensional volume 6343.02535183 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003682.

S26-E-00003684 - Round-sphere area profile S^15, r=1.70

The standard round n-sphere of radius 1.70 has n-dimensional volume 10777.8531371 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003683.

S26-E-00003685 - Round-sphere area profile S^16, r=1.75

The standard round n-sphere of radius 1.75 has n-dimensional volume 18544.6489469 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003684.

S26-E-00003686 - Round-sphere area profile S^17, r=1.80

The standard round n-sphere of radius 1.80 has n-dimensional volume 32321.4557631 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003685.

S26-E-00003687 - Round-sphere area profile S^18, r=1.85

The standard round n-sphere of radius 1.85 has n-dimensional volume 57073.1569386 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003686.

S26-E-00003688 - Round-sphere area profile S^19, r=1.90

The standard round n-sphere of radius 1.90 has n-dimensional volume 102113.722355 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003687.

S26-E-00003689 - Round-sphere area profile S^20, r=1.95

The standard round n-sphere of radius 1.95 has n-dimensional volume 185122.530915 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003688.

S26-E-00003690 - Round-sphere area profile S^21, r=2.00

The standard round n-sphere of radius 2.00 has n-dimensional volume 340052.107971 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003689.

S26-E-00003691 - Round-sphere area profile S^22, r=2.05

The standard round n-sphere of radius 2.05 has n-dimensional volume 632867.172205 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003690.

S26-E-00003692 - Round-sphere area profile S^23, r=2.10

The standard round n-sphere of radius 2.10 has n-dimensional volume 1193209.02305 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003691.

S26-E-00003693 - Round-sphere area profile S^24, r=2.15

The standard round n-sphere of radius 2.15 has n-dimensional volume 2278781.55661 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003692.

S26-E-00003694 - Round-sphere area profile S^25, r=2.20

The standard round n-sphere of radius 2.20 has n-dimensional volume 4407663.15453 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003693.

S26-E-00003695 - Round-sphere area profile S^26, r=2.25

The standard round n-sphere of radius 2.25 has n-dimensional volume 8633064.55513 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003694.

S26-E-00003696 - Round-sphere area profile S^27, r=2.30

The standard round n-sphere of radius 2.30 has n-dimensional volume 17119830.8535 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003695.

S26-E-00003697 - Round-sphere area profile S^28, r=2.35

The standard round n-sphere of radius 2.35 has n-dimensional volume 34366536.477 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003696.

S26-E-00003698 - Round-sphere area profile S^29, r=2.40

The standard round n-sphere of radius 2.40 has n-dimensional volume 69822556.1 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003697.

S26-E-00003699 - Round-sphere area profile S^30, r=2.45

The standard round n-sphere of radius 2.45 has n-dimensional volume 143548950.98 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003698.

S26-E-00003700 - Round-sphere area profile S^31, r=2.50

The standard round n-sphere of radius 2.50 has n-dimensional volume 298585433.256 by the closed formula 2*pi^((n+1)/2)/Gamma((n+1)/2)*r^n.

Evidence M. Hard: S26-E-00003699.

Part O - Curvature tensors and comparison geometry

S26-D-00003701 - Riemann curvature tensor refinement

R(X,Y)Z=nabla_Xnabla_Y Z-nabla_Ynabla_X Z-nabla_[X,Y]Z for the declared sign convention.

Evidence A. Hard: S26-E-00003700.

S26-D-00003702 - Sectional curvature refinement

For a 2-plane sigma=span(u,v), K(sigma)=g(R(u,v)v,u)/(g(u,u)g(v,v)-g(u,v)^2) with the chosen R sign convention.

Evidence A. Hard: S26-D-00003701.

S26-D-00003703 - Ricci curvature refinement

Ric is the trace of the curvature endomorphism, contracting the first and third indices under a declared convention.

Evidence A. Hard: S26-D-00003702.

S26-D-00003704 - Scalar curvature refinement

Scalar curvature is the metric trace of the Ricci tensor.

Evidence A. Hard: S26-D-00003703.

S26-D-00003705 - Jacobi field

A Jacobi field J along a geodesic gamma satisfies nabla_t^2 J+R(J,gamma dot)gamma dot=0 under the declared sign convention.

Evidence A. Hard: S26-D-00003704.

S26-D-00003706 - Conjugate points

Two points along a geodesic are conjugate when a nonzero Jacobi field vanishes at both.

Evidence A. Hard: S26-D-00003705.

S26-T-00003707 - Second variation of energy interface

The second variation of geodesic energy is governed by the index form containing covariant derivative and curvature terms.

Evidence A. Hard: S26-D-00003706.

S26-T-00003708 - Bonnet-Myers theorem

If a complete n-manifold has Ricci curvature bounded below by (n-1)k>0, then its diameter is at most pi/sqrt(k) and its fundamental group is finite.

Evidence A. Hard: S26-T-00003707.

S26-T-00003709 - Cartan-Hadamard theorem

A complete simply connected Riemannian manifold with nonpositive sectional curvature has exponential map at every point a covering; in the simply connected case exp_p is a global diffeomorphism.

Evidence A. Hard: S26-T-00003708.

S26-T-00003710 - Rauch comparison theorem interface

Curvature bounds compare growth of Jacobi fields along geodesics under matched initial data.

Evidence A. Hard: S26-T-00003709.

S26-T-00003711 - Toponogov comparison theorem interface

Sectional-curvature lower bounds compare geodesic triangles with triangles in constant-curvature model spaces.

Evidence A. Hard: S26-T-00003710.

S26-D-00003712 - Constant-curvature space form

A Riemannian manifold has constant sectional curvature k when every tangent 2-plane has curvature k.

Evidence A. Hard: S26-T-00003711.

S26-T-00003713 - Curvature tensor of a space form

For constant sectional curvature k, R(X,Y)Z=k(g(Y,Z)X-g(X,Z)Y) up to the chosen curvature-sign convention.

Evidence A. Hard: S26-D-00003712.

S26-D-00003714 - Einstein metric

A Riemannian metric is Einstein when Ric=lambda g for some scalar lambda, constant in dimensions greater than two under connectedness and standard hypotheses.

Evidence A. Hard: S26-T-00003713.

S26-D-00003715 - Weyl curvature tensor interface

In dimension at least four, the Weyl tensor is the trace-free conformal part of Riemann curvature.

Evidence A. Hard: S26-D-00003714.

S26-D-00003716 - Conformal change of metric

A conformal metric has form g_tilde=e^{2u}g and changes curvature by explicit derivative terms in u.

Evidence A. Hard: S26-D-00003715.

S26-N-00003717 - Sign-convention fence

Curvature tensor signs vary across literature; every formula involving R, sectional curvature or Jacobi equations must be interpreted under one declared convention.

Evidence A. Hard: S26-D-00003716.

S26-N-00003718 - Ricci-sectional fence

Bounds on Ricci curvature and bounds on sectional curvature are different hypotheses and cannot be interchanged mechanically.

Evidence A. Hard: S26-N-00003717.

S26-B-00003719 - Curvature to topology bridge

Global curvature bounds yield topological and metric consequences such as diameter, fundamental-group, conjugate-point and comparison restrictions.

Evidence A. Hard: S26-N-00003718.

S26-B-00003720 - Curvature to relativity bridge

Pseudo-Riemannian curvature uses analogous tensorial constructions but positive-definite Riemannian comparison theorems do not transfer unchanged.

Evidence A. Hard: S26-B-00003719.

S26-E-00003721 - Round-sphere curvature profile n=3, r=1.1

For the round S^3 of radius 1.1, sectional curvature is 0.826446280992, Ricci factor is 1.65289256198, and scalar curvature is 4.95867768595.

Evidence M. Hard: S26-B-00003720.

S26-E-00003722 - Round-sphere curvature profile n=4, r=1.2

For the round S^4 of radius 1.2, sectional curvature is 0.694444444444, Ricci factor is 2.08333333333, and scalar curvature is 8.33333333333.

Evidence M. Hard: S26-E-00003721.

S26-E-00003723 - Round-sphere curvature profile n=5, r=1.3

For the round S^5 of radius 1.3, sectional curvature is 0.591715976331, Ricci factor is 2.36686390533, and scalar curvature is 11.8343195266.

Evidence M. Hard: S26-E-00003722.

S26-E-00003724 - Round-sphere curvature profile n=6, r=1.4

For the round S^6 of radius 1.4, sectional curvature is 0.510204081633, Ricci factor is 2.55102040816, and scalar curvature is 15.306122449.

Evidence M. Hard: S26-E-00003723.

S26-E-00003725 - Round-sphere curvature profile n=7, r=1.5

For the round S^7 of radius 1.5, sectional curvature is 0.444444444444, Ricci factor is 2.66666666667, and scalar curvature is 18.6666666667.

Evidence M. Hard: S26-E-00003724.

S26-E-00003726 - Round-sphere curvature profile n=8, r=1.6

For the round S^8 of radius 1.6, sectional curvature is 0.390625, Ricci factor is 2.734375, and scalar curvature is 21.875.

Evidence M. Hard: S26-E-00003725.

S26-E-00003727 - Round-sphere curvature profile n=9, r=1.7

For the round S^9 of radius 1.7, sectional curvature is 0.346020761246, Ricci factor is 2.76816608997, and scalar curvature is 24.9134948097.

Evidence M. Hard: S26-E-00003726.

S26-E-00003728 - Round-sphere curvature profile n=10, r=1.8

For the round S^10 of radius 1.8, sectional curvature is 0.308641975309, Ricci factor is 2.77777777778, and scalar curvature is 27.7777777778.

Evidence M. Hard: S26-E-00003727.

S26-E-00003729 - Round-sphere curvature profile n=11, r=1.9

For the round S^11 of radius 1.9, sectional curvature is 0.277008310249, Ricci factor is 2.77008310249, and scalar curvature is 30.4709141274.

Evidence M. Hard: S26-E-00003728.

S26-E-00003730 - Round-sphere curvature profile n=12, r=2.0

For the round S^12 of radius 2.0, sectional curvature is 0.25, Ricci factor is 2.75, and scalar curvature is 33.

Evidence M. Hard: S26-E-00003729.

S26-E-00003731 - Round-sphere curvature profile n=13, r=2.1

For the round S^13 of radius 2.1, sectional curvature is 0.226757369615, Ricci factor is 2.72108843537, and scalar curvature is 35.3741496599.

Evidence M. Hard: S26-E-00003730.

S26-E-00003732 - Round-sphere curvature profile n=14, r=2.2

For the round S^14 of radius 2.2, sectional curvature is 0.206611570248, Ricci factor is 2.68595041322, and scalar curvature is 37.6033057851.

Evidence M. Hard: S26-E-00003731.

S26-E-00003733 - Round-sphere curvature profile n=15, r=2.3

For the round S^15 of radius 2.3, sectional curvature is 0.189035916824, Ricci factor is 2.64650283554, and scalar curvature is 39.6975425331.

Evidence M. Hard: S26-E-00003732.

S26-E-00003734 - Round-sphere curvature profile n=16, r=2.4

For the round S^16 of radius 2.4, sectional curvature is 0.173611111111, Ricci factor is 2.60416666667, and scalar curvature is 41.6666666667.

Evidence M. Hard: S26-E-00003733.

S26-E-00003735 - Round-sphere curvature profile n=17, r=2.5

For the round S^17 of radius 2.5, sectional curvature is 0.16, Ricci factor is 2.56, and scalar curvature is 43.52.

Evidence M. Hard: S26-E-00003734.

S26-E-00003736 - Round-sphere curvature profile n=18, r=2.6

For the round S^18 of radius 2.6, sectional curvature is 0.147928994083, Ricci factor is 2.51479289941, and scalar curvature is 45.2662721893.

Evidence M. Hard: S26-E-00003735.

S26-E-00003737 - Round-sphere curvature profile n=19, r=2.7

For the round S^19 of radius 2.7, sectional curvature is 0.137174211248, Ricci factor is 2.46913580247, and scalar curvature is 46.9135802469.

Evidence M. Hard: S26-E-00003736.

S26-E-00003738 - Round-sphere curvature profile n=20, r=2.8

For the round S^20 of radius 2.8, sectional curvature is 0.127551020408, Ricci factor is 2.42346938776, and scalar curvature is 48.4693877551.

Evidence M. Hard: S26-E-00003737.

S26-E-00003739 - Round-sphere curvature profile n=21, r=2.9

For the round S^21 of radius 2.9, sectional curvature is 0.118906064209, Ricci factor is 2.37812128419, and scalar curvature is 49.9405469679.

Evidence M. Hard: S26-E-00003738.

S26-E-00003740 - Round-sphere curvature profile n=22, r=3.0

For the round S^22 of radius 3.0, sectional curvature is 0.111111111111, Ricci factor is 2.33333333333, and scalar curvature is 51.3333333333.

Evidence M. Hard: S26-E-00003739.

S26-E-00003741 - Round-sphere curvature profile n=23, r=3.1

For the round S^23 of radius 3.1, sectional curvature is 0.104058272633, Ricci factor is 2.28928199792, and scalar curvature is 52.6534859521.

Evidence M. Hard: S26-E-00003740.

S26-E-00003742 - Round-sphere curvature profile n=24, r=3.2

For the round S^24 of radius 3.2, sectional curvature is 0.09765625, Ricci factor is 2.24609375, and scalar curvature is 53.90625.

Evidence M. Hard: S26-E-00003741.

S26-E-00003743 - Round-sphere curvature profile n=25, r=3.3

For the round S^25 of radius 3.3, sectional curvature is 0.0918273645546, Ricci factor is 2.20385674931, and scalar curvature is 55.0964187328.

Evidence M. Hard: S26-E-00003742.

S26-E-00003744 - Round-sphere curvature profile n=26, r=3.4

For the round S^26 of radius 3.4, sectional curvature is 0.0865051903114, Ricci factor is 2.16262975779, and scalar curvature is 56.2283737024.

Evidence M. Hard: S26-E-00003743.

S26-E-00003745 - Round-sphere curvature profile n=27, r=3.5

For the round S^27 of radius 3.5, sectional curvature is 0.0816326530612, Ricci factor is 2.12244897959, and scalar curvature is 57.306122449.

Evidence M. Hard: S26-E-00003744.

S26-E-00003746 - Round-sphere curvature profile n=28, r=3.6

For the round S^28 of radius 3.6, sectional curvature is 0.0771604938272, Ricci factor is 2.08333333333, and scalar curvature is 58.3333333333.

Evidence M. Hard: S26-E-00003745.

S26-E-00003747 - Round-sphere curvature profile n=29, r=3.7

For the round S^29 of radius 3.7, sectional curvature is 0.073046018992, Ricci factor is 2.04528853178, and scalar curvature is 59.3133674215.

Evidence M. Hard: S26-E-00003746.

S26-E-00003748 - Round-sphere curvature profile n=30, r=3.8

For the round S^30 of radius 3.8, sectional curvature is 0.0692520775623, Ricci factor is 2.00831024931, and scalar curvature is 60.2493074792.

Evidence M. Hard: S26-E-00003747.

S26-E-00003749 - Round-sphere curvature profile n=31, r=3.9

For the round S^31 of radius 3.9, sectional curvature is 0.0657462195924, Ricci factor is 1.97238658777, and scalar curvature is 61.1439842209.

Evidence M. Hard: S26-E-00003748.

S26-E-00003750 - Round-sphere curvature profile n=32, r=4.0

For the round S^32 of radius 4.0, sectional curvature is 0.0625, Ricci factor is 1.9375, and scalar curvature is 62.

Evidence M. Hard: S26-E-00003749.

Part P - Global and submanifold geometry

S26-D-00003751 - Geodesic completeness

A Riemannian manifold is geodesically complete when every maximal geodesic is defined for all real parameter times.

Evidence A. Hard: S26-E-00003750.

S26-D-00003752 - Metric ball and geodesic ball distinction

A metric ball is defined by d_g, whereas a geodesic normal ball is an exponential-map image on which radial coordinates are regular.

Evidence A. Hard: S26-D-00003751.

S26-D-00003753 - Volume growth function

V_p(r)=Vol(B(p,r)) records Riemannian volume growth around p.

Evidence A. Hard: S26-D-00003752.

S26-T-00003754 - Bishop-Gromov comparison interface

A Ricci lower bound gives monotonic comparison of normalized volumes of metric balls with the corresponding model space.

Evidence A. Hard: S26-D-00003753.

S26-D-00003755 - Isometry of Riemannian manifolds

An isometry is a diffeomorphism F with F^*h=g; it preserves lengths, distances, Levi-Civita connection and curvature.

Evidence A. Hard: S26-T-00003754.

S26-D-00003756 - Killing vector field

A vector field X is Killing when L_X g=0, equivalently its local flow acts by local isometries.

Evidence A. Hard: S26-D-00003755.

S26-T-00003757 - Myers-Steenrod theorem interface

The isometry group of a connected Riemannian manifold is a Lie group acting smoothly.

Evidence A. Hard: S26-D-00003756.

S26-D-00003758 - Riemannian submersion

A submersion is Riemannian when dF restricts to an isometry on horizontal spaces orthogonal to the fibers.

Evidence A. Hard: S26-T-00003757.

S26-D-00003759 - Second fundamental form

For an embedded Riemannian submanifold, the second fundamental form is the normal component of the ambient covariant derivative.

Evidence A. Hard: S26-D-00003758.

S26-D-00003760 - Shape operator

For a chosen normal field nu, the shape operator A_nu is related to the second fundamental form by g(A_nu X,Y)=g(II(X,Y),nu) with sign convention declared.

Evidence A. Hard: S26-D-00003759.

S26-T-00003761 - Gauss equation

Intrinsic curvature of a submanifold equals ambient tangential curvature plus quadratic terms in the second fundamental form.

Evidence A. Hard: S26-D-00003760.

S26-T-00003762 - Codazzi equation

Covariant derivatives of the second fundamental form satisfy a compatibility equation involving ambient curvature.

Evidence A. Hard: S26-T-00003761.

S26-D-00003763 - Mean curvature vector

The mean curvature vector is the trace of the second fundamental form divided or not by dimension according to convention; the convention must be declared.

Evidence A. Hard: S26-T-00003762.

S26-D-00003764 - Minimal submanifold

A submanifold is minimal when its mean curvature vector vanishes.

Evidence A. Hard: S26-D-00003763.

S26-T-00003765 - First variation of area interface

The first variation of volume of a submanifold is controlled by mean curvature and boundary terms.

Evidence A. Hard: S26-D-00003764.

S26-D-00003766 - Harmonic map interface

A smooth map between Riemannian manifolds is harmonic when it is a critical point of the energy functional, equivalently its tension field vanishes.

Evidence A. Hard: S26-T-00003765.

S26-D-00003767 - Riemannian covering

A covering map that is a local isometry transports the metric locally and preserves geodesic structure under lifting.

Evidence A. Hard: S26-D-00003766.

S26-N-00003768 - Local-isometry fence

A local isometry need not be globally injective; covering and quotient phenomena remain global.

Evidence A. Hard: S26-D-00003767.

S26-B-00003769 - Submanifold geometry to PDE bridge

Minimal and harmonic map equations are nonlinear elliptic systems arising from geometric variational principles.

Evidence A. Hard: S26-N-00003768.

S26-B-00003770 - Global geometry to group actions bridge

Isometry groups, quotients and homogeneous spaces connect Riemannian geometry with Lie groups and representation-theoretic structure.

Evidence A. Hard: S26-B-00003769.

S26-E-00003771 - Flat-torus geometry profile dimension=2

The standard flat torus R^2/Z^2 has dimension 2, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-B-00003770.

S26-E-00003772 - Flat-torus geometry profile dimension=3

The standard flat torus R^3/Z^3 has dimension 3, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003771.

S26-E-00003773 - Flat-torus geometry profile dimension=4

The standard flat torus R^4/Z^4 has dimension 4, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003772.

S26-E-00003774 - Flat-torus geometry profile dimension=5

The standard flat torus R^5/Z^5 has dimension 5, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003773.

S26-E-00003775 - Flat-torus geometry profile dimension=6

The standard flat torus R^6/Z^6 has dimension 6, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003774.

S26-E-00003776 - Flat-torus geometry profile dimension=7

The standard flat torus R^7/Z^7 has dimension 7, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003775.

S26-E-00003777 - Flat-torus geometry profile dimension=8

The standard flat torus R^8/Z^8 has dimension 8, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003776.

S26-E-00003778 - Flat-torus geometry profile dimension=9

The standard flat torus R^9/Z^9 has dimension 9, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003777.

S26-E-00003779 - Flat-torus geometry profile dimension=10

The standard flat torus R^10/Z^10 has dimension 10, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003778.

S26-E-00003780 - Flat-torus geometry profile dimension=11

The standard flat torus R^11/Z^11 has dimension 11, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003779.

S26-E-00003781 - Flat-torus geometry profile dimension=12

The standard flat torus R^12/Z^12 has dimension 12, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003780.

S26-E-00003782 - Flat-torus geometry profile dimension=13

The standard flat torus R^13/Z^13 has dimension 13, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003781.

S26-E-00003783 - Flat-torus geometry profile dimension=14

The standard flat torus R^14/Z^14 has dimension 14, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003782.

S26-E-00003784 - Flat-torus geometry profile dimension=15

The standard flat torus R^15/Z^15 has dimension 15, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003783.

S26-E-00003785 - Flat-torus geometry profile dimension=16

The standard flat torus R^16/Z^16 has dimension 16, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003784.

S26-E-00003786 - Flat-torus geometry profile dimension=17

The standard flat torus R^17/Z^17 has dimension 17, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003785.

S26-E-00003787 - Flat-torus geometry profile dimension=18

The standard flat torus R^18/Z^18 has dimension 18, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003786.

S26-E-00003788 - Flat-torus geometry profile dimension=19

The standard flat torus R^19/Z^19 has dimension 19, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003787.

S26-E-00003789 - Flat-torus geometry profile dimension=20

The standard flat torus R^20/Z^20 has dimension 20, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003788.

S26-E-00003790 - Flat-torus geometry profile dimension=21

The standard flat torus R^21/Z^21 has dimension 21, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003789.

S26-E-00003791 - Flat-torus geometry profile dimension=22

The standard flat torus R^22/Z^22 has dimension 22, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003790.

S26-E-00003792 - Flat-torus geometry profile dimension=23

The standard flat torus R^23/Z^23 has dimension 23, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003791.

S26-E-00003793 - Flat-torus geometry profile dimension=24

The standard flat torus R^24/Z^24 has dimension 24, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003792.

S26-E-00003794 - Flat-torus geometry profile dimension=25

The standard flat torus R^25/Z^25 has dimension 25, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003793.

S26-E-00003795 - Flat-torus geometry profile dimension=26

The standard flat torus R^26/Z^26 has dimension 26, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003794.

S26-E-00003796 - Flat-torus geometry profile dimension=27

The standard flat torus R^27/Z^27 has dimension 27, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003795.

S26-E-00003797 - Flat-torus geometry profile dimension=28

The standard flat torus R^28/Z^28 has dimension 28, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003796.

S26-E-00003798 - Flat-torus geometry profile dimension=29

The standard flat torus R^29/Z^29 has dimension 29, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003797.

S26-E-00003799 - Flat-torus geometry profile dimension=30

The standard flat torus R^30/Z^30 has dimension 30, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003798.

S26-E-00003800 - Flat-torus geometry profile dimension=31

The standard flat torus R^31/Z^31 has dimension 31, zero sectional curvature, zero Ricci curvature, zero scalar curvature, and volume 1 for the unit lattice metric.

Evidence M. Hard: S26-E-00003799.

Part Q - Morse theory and handle interfaces

S26-D-00003801 - Nondegenerate critical point

A critical point p of f:M->R is nondegenerate when the Hessian at p is nonsingular in any local coordinate representation.

Evidence A. Hard: S26-E-00003800.

S26-D-00003802 - Morse function

A smooth function is Morse when all its critical points are nondegenerate.

Evidence A. Hard: S26-D-00003801.

S26-D-00003803 - Morse index

The Morse index of a nondegenerate critical point is the number of negative eigenvalues of its Hessian, counted with multiplicity.

Evidence A. Hard: S26-D-00003802.

S26-T-00003804 - Morse lemma

Near a nondegenerate critical point of index lambda, there are coordinates in which f=f(p)-x_1^2-...-x_lambda^2+x_{lambda+1}^2+...+x_n^2.

Evidence A. Hard: S26-D-00003803.

S26-T-00003805 - Regular-level isotopy interface

If [a,b] contains no critical values of a proper-enough Morse function, the corresponding sublevel sets are diffeomorphic in the standard theorem.

Evidence A. Hard: S26-T-00003804.

S26-T-00003806 - Handle attachment at a critical point

Passing a single nondegenerate critical value of index lambda changes the sublevel topology by attaching a lambda-cell/handle under standard isolation hypotheses.

Evidence A. Hard: S26-T-00003805.

S26-D-00003807 - Morse polynomial

M_f(t)=sum_lambda m_lambda t^lambda records counts m_lambda of critical points by Morse index.

Evidence A. Hard: S26-T-00003806.

S26-D-00003808 - Poincare polynomial

P_M(t)=sum_k beta_k t^k records Betti numbers over a chosen field.

Evidence A. Hard: S26-D-00003807.

S26-T-00003809 - Weak Morse inequalities

The number m_k of index-k critical points is at least the kth Betti number.

Evidence A. Hard: S26-D-00003808.

S26-T-00003810 - Strong Morse inequalities

Alternating partial sums of Morse numbers dominate corresponding alternating Betti sums, with equality at the top total Euler sum.

Evidence A. Hard: S26-T-00003809.

S26-T-00003811 - Morse Euler characteristic formula

For a Morse function on a closed manifold, chi(M)=sum_p (-1)^{index(p)}.

Evidence A. Hard: S26-T-00003810.

S26-D-00003812 - Gradient-like vector field

A gradient-like vector field for f decreases f away from critical points and has controlled local form near critical points.

Evidence A. Hard: S26-T-00003811.

S26-D-00003813 - Stable manifold of a Morse critical point

For a Morse-Smale gradient flow, the stable manifold consists of points converging to the critical point in forward time.

Evidence A. Hard: S26-D-00003812.

S26-D-00003814 - Unstable manifold of a Morse critical point

The unstable manifold consists of points converging to the critical point in backward time.

Evidence A. Hard: S26-D-00003813.

S26-D-00003815 - Morse-Smale transversality

A gradient flow is Morse-Smale when stable and unstable manifolds intersect transversely.

Evidence A. Hard: S26-D-00003814.

S26-D-00003816 - Morse chain complex interface

Under Morse-Smale conditions, the chain group is generated by critical points and the differential counts oriented flow lines between index-difference-one critical points.

Evidence A. Hard: S26-D-00003815.

S26-T-00003817 - Morse homology theorem interface

Morse homology is canonically isomorphic to singular homology under the standard compactness/orientation setup.

Evidence A. Hard: S26-D-00003816.

S26-N-00003818 - Critical-count fence

Morse inequalities constrain critical-point counts but do not say every function realizes the minimum counts or that Betti numbers determine diffeomorphism type.

Evidence A. Hard: S26-T-00003817.

S26-B-00003819 - Morse to CW bridge

A Morse function on a compact manifold produces a handle/CW decomposition indexed by its critical points.

Evidence A. Hard: S26-N-00003818.

S26-B-00003820 - Morse to dynamics bridge

Gradient flows turn critical-point topology into trajectories, stable/unstable manifolds and chain-level boundary operators.

Evidence A. Hard: S26-B-00003819.

S26-E-00003821 - Sphere Morse profile n=2

The height function on S^2 has two nondegenerate critical points of indices 0 and 2; the signed Morse count is 2=chi(S^2).

Evidence M. Hard: S26-B-00003820.

S26-E-00003822 - Sphere Morse profile n=3

The height function on S^3 has two nondegenerate critical points of indices 0 and 3; the signed Morse count is 0=chi(S^3).

Evidence M. Hard: S26-E-00003821.

S26-E-00003823 - Sphere Morse profile n=4

The height function on S^4 has two nondegenerate critical points of indices 0 and 4; the signed Morse count is 2=chi(S^4).

Evidence M. Hard: S26-E-00003822.

S26-E-00003824 - Sphere Morse profile n=5

The height function on S^5 has two nondegenerate critical points of indices 0 and 5; the signed Morse count is 0=chi(S^5).

Evidence M. Hard: S26-E-00003823.

S26-E-00003825 - Sphere Morse profile n=6

The height function on S^6 has two nondegenerate critical points of indices 0 and 6; the signed Morse count is 2=chi(S^6).

Evidence M. Hard: S26-E-00003824.

S26-E-00003826 - Sphere Morse profile n=7

The height function on S^7 has two nondegenerate critical points of indices 0 and 7; the signed Morse count is 0=chi(S^7).

Evidence M. Hard: S26-E-00003825.

S26-E-00003827 - Sphere Morse profile n=8

The height function on S^8 has two nondegenerate critical points of indices 0 and 8; the signed Morse count is 2=chi(S^8).

Evidence M. Hard: S26-E-00003826.

S26-E-00003828 - Sphere Morse profile n=9

The height function on S^9 has two nondegenerate critical points of indices 0 and 9; the signed Morse count is 0=chi(S^9).

Evidence M. Hard: S26-E-00003827.

S26-E-00003829 - Sphere Morse profile n=10

The height function on S^10 has two nondegenerate critical points of indices 0 and 10; the signed Morse count is 2=chi(S^10).

Evidence M. Hard: S26-E-00003828.

S26-E-00003830 - Sphere Morse profile n=11

The height function on S^11 has two nondegenerate critical points of indices 0 and 11; the signed Morse count is 0=chi(S^11).

Evidence M. Hard: S26-E-00003829.

S26-E-00003831 - Sphere Morse profile n=12

The height function on S^12 has two nondegenerate critical points of indices 0 and 12; the signed Morse count is 2=chi(S^12).

Evidence M. Hard: S26-E-00003830.

S26-E-00003832 - Sphere Morse profile n=13

The height function on S^13 has two nondegenerate critical points of indices 0 and 13; the signed Morse count is 0=chi(S^13).

Evidence M. Hard: S26-E-00003831.

S26-E-00003833 - Sphere Morse profile n=14

The height function on S^14 has two nondegenerate critical points of indices 0 and 14; the signed Morse count is 2=chi(S^14).

Evidence M. Hard: S26-E-00003832.

S26-E-00003834 - Sphere Morse profile n=15

The height function on S^15 has two nondegenerate critical points of indices 0 and 15; the signed Morse count is 0=chi(S^15).

Evidence M. Hard: S26-E-00003833.

S26-E-00003835 - Sphere Morse profile n=16

The height function on S^16 has two nondegenerate critical points of indices 0 and 16; the signed Morse count is 2=chi(S^16).

Evidence M. Hard: S26-E-00003834.

S26-E-00003836 - Sphere Morse profile n=17

The height function on S^17 has two nondegenerate critical points of indices 0 and 17; the signed Morse count is 0=chi(S^17).

Evidence M. Hard: S26-E-00003835.

S26-E-00003837 - Sphere Morse profile n=18

The height function on S^18 has two nondegenerate critical points of indices 0 and 18; the signed Morse count is 2=chi(S^18).

Evidence M. Hard: S26-E-00003836.

S26-E-00003838 - Sphere Morse profile n=19

The height function on S^19 has two nondegenerate critical points of indices 0 and 19; the signed Morse count is 0=chi(S^19).

Evidence M. Hard: S26-E-00003837.

S26-E-00003839 - Sphere Morse profile n=20

The height function on S^20 has two nondegenerate critical points of indices 0 and 20; the signed Morse count is 2=chi(S^20).

Evidence M. Hard: S26-E-00003838.

S26-E-00003840 - Sphere Morse profile n=21

The height function on S^21 has two nondegenerate critical points of indices 0 and 21; the signed Morse count is 0=chi(S^21).

Evidence M. Hard: S26-E-00003839.

S26-E-00003841 - Sphere Morse profile n=22

The height function on S^22 has two nondegenerate critical points of indices 0 and 22; the signed Morse count is 2=chi(S^22).

Evidence M. Hard: S26-E-00003840.

S26-E-00003842 - Sphere Morse profile n=23

The height function on S^23 has two nondegenerate critical points of indices 0 and 23; the signed Morse count is 0=chi(S^23).

Evidence M. Hard: S26-E-00003841.

S26-E-00003843 - Sphere Morse profile n=24

The height function on S^24 has two nondegenerate critical points of indices 0 and 24; the signed Morse count is 2=chi(S^24).

Evidence M. Hard: S26-E-00003842.

S26-E-00003844 - Sphere Morse profile n=25

The height function on S^25 has two nondegenerate critical points of indices 0 and 25; the signed Morse count is 0=chi(S^25).

Evidence M. Hard: S26-E-00003843.

S26-E-00003845 - Sphere Morse profile n=26

The height function on S^26 has two nondegenerate critical points of indices 0 and 26; the signed Morse count is 2=chi(S^26).

Evidence M. Hard: S26-E-00003844.

S26-E-00003846 - Sphere Morse profile n=27

The height function on S^27 has two nondegenerate critical points of indices 0 and 27; the signed Morse count is 0=chi(S^27).

Evidence M. Hard: S26-E-00003845.

S26-E-00003847 - Sphere Morse profile n=28

The height function on S^28 has two nondegenerate critical points of indices 0 and 28; the signed Morse count is 2=chi(S^28).

Evidence M. Hard: S26-E-00003846.

S26-E-00003848 - Sphere Morse profile n=29

The height function on S^29 has two nondegenerate critical points of indices 0 and 29; the signed Morse count is 0=chi(S^29).

Evidence M. Hard: S26-E-00003847.

S26-E-00003849 - Sphere Morse profile n=30

The height function on S^30 has two nondegenerate critical points of indices 0 and 30; the signed Morse count is 2=chi(S^30).

Evidence M. Hard: S26-E-00003848.

S26-E-00003850 - Sphere Morse profile n=31

The height function on S^31 has two nondegenerate critical points of indices 0 and 31; the signed Morse count is 0=chi(S^31).

Evidence M. Hard: S26-E-00003849.

Part R - Lie group actions and homogeneous geometry

S26-D-00003851 - Lie group action refinement

A smooth left action of a Lie group G on M is a smooth map GxM->M satisfying identity and associativity axioms.

Evidence A. Hard: S26-E-00003850.

S26-D-00003852 - Orbit of a Lie group action

G·p={g·p:ginG}; for smooth actions orbits are immersed submanifolds under standard results.

Evidence A. Hard: S26-D-00003851.

S26-D-00003853 - Lie-action stabilizer subgroup refinement

G_p={ginG:g·p=p} is a closed Lie subgroup of G.

Evidence A. Hard: S26-D-00003852, S26-D-00000967.

S26-D-00003854 - Homogeneous Lie quotient model refinement

A manifold is homogeneous under G when G acts transitively; after choosing p it is modeled by G/G_p.

Evidence A. Hard: S26-D-00003853, S26-D-00000973.

S26-T-00003855 - Orbit-stabilizer dimension formula for Lie actions

dim(G·p)=dim G-dim G_p for smooth Lie group actions.

Evidence A. Hard: S26-D-00003854.

S26-D-00003856 - Proper group action

A G-action is proper when (g,p)|->(g·p,p) is a proper map GxM->MxM.

Evidence A. Hard: S26-T-00003855.

S26-T-00003857 - Free proper quotient theorem

A free proper smooth action has a smooth manifold quotient M/G and M->M/G is a principal G-bundle.

Evidence A. Hard: S26-D-00003856.

S26-D-00003858 - Effective group action

An action is effective when only the identity acts trivially on every point.

Evidence A. Hard: S26-T-00003857.

S26-D-00003859 - Isotropy representation

The stabilizer G_p acts linearly on T_pM via differentials, producing the isotropy representation.

Evidence A. Hard: S26-D-00003858.

S26-D-00003860 - Left-invariant Riemannian metric

A Riemannian metric on G is left invariant when all left translations are isometries.

Evidence A. Hard: S26-D-00003859.

S26-D-00003861 - Bi-invariant Riemannian metric

A metric is bi-invariant when both left and right translations are isometries.

Evidence A. Hard: S26-D-00003860.

S26-T-00003862 - Geodesics of a bi-invariant metric

For a Lie group with bi-invariant Riemannian metric, one-parameter subgroups through the identity are geodesics.

Evidence A. Hard: S26-D-00003861.

S26-I-00003863 - Levi-Civita connection for bi-invariant metrics

For left-invariant vector fields under a bi-invariant metric, nabla_X Y=(1/2)[X,Y].

Evidence A. Hard: S26-T-00003862.

S26-I-00003864 - Curvature for bi-invariant metrics

Under a standard curvature sign convention, sectional curvature of a bi-invariant metric is proportional to ||[X,Y]||^2 for orthonormal X,Y.

Evidence A. Hard: S26-I-00003863.

S26-D-00003865 - Symmetric space interface

A Riemannian symmetric space has an isometry reversing geodesics through each point; globally it is modeled by Lie-theoretic quotient data under standard connectedness hypotheses.

Evidence A. Hard: S26-I-00003864.

S26-D-00003866 - Reductive homogeneous space interface

A homogeneous space G/H is reductive when the Lie algebra splits g=h⊕m with m invariant under Ad(H).

Evidence A. Hard: S26-D-00003865.

S26-D-00003867 - Invariant tensor on a homogeneous space

G-invariant tensors on G/H correspond to H-invariant tensors on the tangent model g/h at the basepoint.

Evidence A. Hard: S26-D-00003866.

S26-N-00003868 - Orbit-space fence

A smooth group action need not yield a manifold quotient; freeness/properness or controlled singular-stratum hypotheses are load-bearing.

Evidence A. Hard: S26-D-00003867.

S26-B-00003869 - Homogeneous geometry to representation bridge

Isotropy representations and invariant tensors reduce geometric classification problems to linear representation data.

Evidence A. Hard: S26-N-00003868.

S26-B-00003870 - Lie quotients to bundles bridge

For closed HsubsetG, G->G/H is a principal H-bundle under standard smooth quotient hypotheses.

Evidence A. Hard: S26-B-00003869.

S26-E-00003871 - Orthogonal-group dimension profile n=3

dim SO(3)=3; consequently dim SO(4)-dim SO(3)=3, matching dim S^3 in the homogeneous model SO(4)/SO(3).

Evidence M. Hard: S26-B-00003870.

S26-E-00003872 - Orthogonal-group dimension profile n=4

dim SO(4)=6; consequently dim SO(5)-dim SO(4)=4, matching dim S^4 in the homogeneous model SO(5)/SO(4).

Evidence M. Hard: S26-E-00003871.

S26-E-00003873 - Orthogonal-group dimension profile n=5

dim SO(5)=10; consequently dim SO(6)-dim SO(5)=5, matching dim S^5 in the homogeneous model SO(6)/SO(5).

Evidence M. Hard: S26-E-00003872.

S26-E-00003874 - Orthogonal-group dimension profile n=6

dim SO(6)=15; consequently dim SO(7)-dim SO(6)=6, matching dim S^6 in the homogeneous model SO(7)/SO(6).

Evidence M. Hard: S26-E-00003873.

S26-E-00003875 - Orthogonal-group dimension profile n=7

dim SO(7)=21; consequently dim SO(8)-dim SO(7)=7, matching dim S^7 in the homogeneous model SO(8)/SO(7).

Evidence M. Hard: S26-E-00003874.

S26-E-00003876 - Orthogonal-group dimension profile n=8

dim SO(8)=28; consequently dim SO(9)-dim SO(8)=8, matching dim S^8 in the homogeneous model SO(9)/SO(8).

Evidence M. Hard: S26-E-00003875.

S26-E-00003877 - Orthogonal-group dimension profile n=9

dim SO(9)=36; consequently dim SO(10)-dim SO(9)=9, matching dim S^9 in the homogeneous model SO(10)/SO(9).

Evidence M. Hard: S26-E-00003876.

S26-E-00003878 - Orthogonal-group dimension profile n=10

dim SO(10)=45; consequently dim SO(11)-dim SO(10)=10, matching dim S^10 in the homogeneous model SO(11)/SO(10).

Evidence M. Hard: S26-E-00003877.

S26-E-00003879 - Orthogonal-group dimension profile n=11

dim SO(11)=55; consequently dim SO(12)-dim SO(11)=11, matching dim S^11 in the homogeneous model SO(12)/SO(11).

Evidence M. Hard: S26-E-00003878.

S26-E-00003880 - Orthogonal-group dimension profile n=12

dim SO(12)=66; consequently dim SO(13)-dim SO(12)=12, matching dim S^12 in the homogeneous model SO(13)/SO(12).

Evidence M. Hard: S26-E-00003879.

S26-E-00003881 - Orthogonal-group dimension profile n=13

dim SO(13)=78; consequently dim SO(14)-dim SO(13)=13, matching dim S^13 in the homogeneous model SO(14)/SO(13).

Evidence M. Hard: S26-E-00003880.

S26-E-00003882 - Orthogonal-group dimension profile n=14

dim SO(14)=91; consequently dim SO(15)-dim SO(14)=14, matching dim S^14 in the homogeneous model SO(15)/SO(14).

Evidence M. Hard: S26-E-00003881.

S26-E-00003883 - Orthogonal-group dimension profile n=15

dim SO(15)=105; consequently dim SO(16)-dim SO(15)=15, matching dim S^15 in the homogeneous model SO(16)/SO(15).

Evidence M. Hard: S26-E-00003882.

S26-E-00003884 - Orthogonal-group dimension profile n=16

dim SO(16)=120; consequently dim SO(17)-dim SO(16)=16, matching dim S^16 in the homogeneous model SO(17)/SO(16).

Evidence M. Hard: S26-E-00003883.

S26-E-00003885 - Orthogonal-group dimension profile n=17

dim SO(17)=136; consequently dim SO(18)-dim SO(17)=17, matching dim S^17 in the homogeneous model SO(18)/SO(17).

Evidence M. Hard: S26-E-00003884.

S26-E-00003886 - Orthogonal-group dimension profile n=18

dim SO(18)=153; consequently dim SO(19)-dim SO(18)=18, matching dim S^18 in the homogeneous model SO(19)/SO(18).

Evidence M. Hard: S26-E-00003885.

S26-E-00003887 - Orthogonal-group dimension profile n=19

dim SO(19)=171; consequently dim SO(20)-dim SO(19)=19, matching dim S^19 in the homogeneous model SO(20)/SO(19).

Evidence M. Hard: S26-E-00003886.

S26-E-00003888 - Orthogonal-group dimension profile n=20

dim SO(20)=190; consequently dim SO(21)-dim SO(20)=20, matching dim S^20 in the homogeneous model SO(21)/SO(20).

Evidence M. Hard: S26-E-00003887.

S26-E-00003889 - Orthogonal-group dimension profile n=21

dim SO(21)=210; consequently dim SO(22)-dim SO(21)=21, matching dim S^21 in the homogeneous model SO(22)/SO(21).

Evidence M. Hard: S26-E-00003888.

S26-E-00003890 - Orthogonal-group dimension profile n=22

dim SO(22)=231; consequently dim SO(23)-dim SO(22)=22, matching dim S^22 in the homogeneous model SO(23)/SO(22).

Evidence M. Hard: S26-E-00003889.

S26-E-00003891 - Orthogonal-group dimension profile n=23

dim SO(23)=253; consequently dim SO(24)-dim SO(23)=23, matching dim S^23 in the homogeneous model SO(24)/SO(23).

Evidence M. Hard: S26-E-00003890.

S26-E-00003892 - Orthogonal-group dimension profile n=24

dim SO(24)=276; consequently dim SO(25)-dim SO(24)=24, matching dim S^24 in the homogeneous model SO(25)/SO(24).

Evidence M. Hard: S26-E-00003891.

S26-E-00003893 - Orthogonal-group dimension profile n=25

dim SO(25)=300; consequently dim SO(26)-dim SO(25)=25, matching dim S^25 in the homogeneous model SO(26)/SO(25).

Evidence M. Hard: S26-E-00003892.

S26-E-00003894 - Orthogonal-group dimension profile n=26

dim SO(26)=325; consequently dim SO(27)-dim SO(26)=26, matching dim S^26 in the homogeneous model SO(27)/SO(26).

Evidence M. Hard: S26-E-00003893.

S26-E-00003895 - Orthogonal-group dimension profile n=27

dim SO(27)=351; consequently dim SO(28)-dim SO(27)=27, matching dim S^27 in the homogeneous model SO(28)/SO(27).

Evidence M. Hard: S26-E-00003894.

S26-E-00003896 - Orthogonal-group dimension profile n=28

dim SO(28)=378; consequently dim SO(29)-dim SO(28)=28, matching dim S^28 in the homogeneous model SO(29)/SO(28).

Evidence M. Hard: S26-E-00003895.

S26-E-00003897 - Orthogonal-group dimension profile n=29

dim SO(29)=406; consequently dim SO(30)-dim SO(29)=29, matching dim S^29 in the homogeneous model SO(30)/SO(29).

Evidence M. Hard: S26-E-00003896.

S26-E-00003898 - Orthogonal-group dimension profile n=30

dim SO(30)=435; consequently dim SO(31)-dim SO(30)=30, matching dim S^30 in the homogeneous model SO(31)/SO(30).

Evidence M. Hard: S26-E-00003897.

S26-E-00003899 - Orthogonal-group dimension profile n=31

dim SO(31)=465; consequently dim SO(32)-dim SO(31)=31, matching dim S^31 in the homogeneous model SO(32)/SO(31).

Evidence M. Hard: S26-E-00003898.

S26-E-00003900 - Orthogonal-group dimension profile n=32

dim SO(32)=496; consequently dim SO(33)-dim SO(32)=32, matching dim S^32 in the homogeneous model SO(33)/SO(32).

Evidence M. Hard: S26-E-00003899.

Part S - Spectral sequences and obstruction interfaces

S26-D-00003901 - Filtered chain complex

A filtered chain complex has nested subcomplexes ...subsetF_{p-1}CsubsetF_pCsubset... compatible with the differential.

Evidence A. Hard: S26-E-00003900.

S26-D-00003902 - Spectral sequence

A spectral sequence is a sequence of bigraded objects E_r with differentials d_r of prescribed bidegree such that E_{r+1} is the homology of (E_r,d_r).

Evidence A. Hard: S26-D-00003901.

S26-D-00003903 - Convergence of a spectral sequence

A spectral sequence converges to a filtered object when its stable page identifies with the associated graded pieces of the target filtration under declared hypotheses.

Evidence A. Hard: S26-D-00003902.

S26-D-00003904 - Serre spectral sequence interface

For a suitable fibration F->E->B, the Serre spectral sequence relates homology/cohomology of fiber and base to that of total space.

Evidence A. Hard: S26-D-00003903.

S26-D-00003905 - Leray-Serre E2 page

With appropriate local coefficients, the cohomological Serre spectral sequence has E_2^{p,q}=H^p(B;H^q(F)) in the standard setup.

Evidence A. Hard: S26-D-00003904.

S26-D-00003906 - Atiyah-Hirzebruch spectral sequence interface

Generalized cohomology of a CW complex can be computed from ordinary cohomology with coefficients in the generalized theory through the AHSS.

Evidence A. Hard: S26-D-00003905.

S26-D-00003907 - Exact couple

An exact couple is algebraic data generating a spectral sequence by repeated derivation.

Evidence A. Hard: S26-D-00003906.

S26-D-00003908 - Edge homomorphism

An edge homomorphism relates target filtration pieces to entries on the boundary of a spectral-sequence page.

Evidence A. Hard: S26-D-00003907.

S26-D-00003909 - Transgression

A transgression is a differential carrying a class from fiber-like to base-like degree in a fibration spectral sequence.

Evidence A. Hard: S26-D-00003908.

S26-N-00003910 - Spectral-sequence collapse fence

Vanishing of all later differentials does not automatically resolve extension problems in reconstructing the target from its associated graded object.

Evidence A. Hard: S26-D-00003909.

S26-D-00003911 - Obstruction cocycle

An obstruction cocycle records failure to extend a partially defined map, section or lift over the next skeleton.

Evidence A. Hard: S26-N-00003910.

S26-D-00003912 - Primary obstruction class

The first nonzero obstruction to extension lies in an appropriate cohomology group with local coefficients determined by homotopy of the fiber/target.

Evidence A. Hard: S26-D-00003911.

S26-T-00003913 - Vanishing obstruction extension principle

If the relevant obstruction class vanishes, an extension over the next skeleton exists under the standard obstruction-theory setup.

Evidence A. Hard: S26-D-00003912.

S26-D-00003914 - Postnikov tower

A Postnikov tower approximates a connected space by stages retaining homotopy groups only up to increasing degrees.

Evidence A. Hard: S26-T-00003913.

S26-D-00003915 - k-invariant

A Postnikov k-invariant is a cohomology class encoding how one stage is assembled from the preceding stage and the next homotopy group.

Evidence A. Hard: S26-D-00003914.

S26-D-00003916 - Eilenberg-Mac Lane space

K(G,n) has pi_n□G and all other homotopy groups trivial, in the standard connected pointed model.

Evidence A. Hard: S26-D-00003915.

S26-T-00003917 - Cohomology representability by Eilenberg-Mac Lane spaces

For suitable CW complexes, H^n(X;G) is naturally identified with based homotopy classes [X,K(G,n)].

Evidence A. Hard: S26-D-00003916.

S26-N-00003918 - Associated-graded fence

Knowing the associated graded object of a filtration need not uniquely determine the original filtered group or ring.

Evidence A. Hard: S26-T-00003917.

S26-B-00003919 - Spectral sequences to bundles bridge

Fibration and bundle filtrations generate spectral sequences that transport local fiber/base invariants into global invariants.

Evidence A. Hard: S26-N-00003918.

S26-B-00003920 - Obstruction theory to geometry bridge

Existence of sections, lifts, framings and reductions of structure group can be encoded as successive cohomological obstructions.

Evidence A. Hard: S26-B-00003919.

S26-E-00003921 - Toy spectral-sequence profile rank=2

For a finite first-quadrant toy page with 2 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 2; no extension claim is made.

Evidence M. Hard: S26-B-00003920.

S26-E-00003922 - Toy spectral-sequence profile rank=3

For a finite first-quadrant toy page with 3 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 3; no extension claim is made.

Evidence M. Hard: S26-E-00003921.

S26-E-00003923 - Toy spectral-sequence profile rank=4

For a finite first-quadrant toy page with 4 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 4; no extension claim is made.

Evidence M. Hard: S26-E-00003922.

S26-E-00003924 - Toy spectral-sequence profile rank=5

For a finite first-quadrant toy page with 5 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 5; no extension claim is made.

Evidence M. Hard: S26-E-00003923.

S26-E-00003925 - Toy spectral-sequence profile rank=6

For a finite first-quadrant toy page with 6 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 6; no extension claim is made.

Evidence M. Hard: S26-E-00003924.

S26-E-00003926 - Toy spectral-sequence profile rank=7

For a finite first-quadrant toy page with 7 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 7; no extension claim is made.

Evidence M. Hard: S26-E-00003925.

S26-E-00003927 - Toy spectral-sequence profile rank=8

For a finite first-quadrant toy page with 8 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 8; no extension claim is made.

Evidence M. Hard: S26-E-00003926.

S26-E-00003928 - Toy spectral-sequence profile rank=9

For a finite first-quadrant toy page with 9 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 9; no extension claim is made.

Evidence M. Hard: S26-E-00003927.

S26-E-00003929 - Toy spectral-sequence profile rank=10

For a finite first-quadrant toy page with 10 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 10; no extension claim is made.

Evidence M. Hard: S26-E-00003928.

S26-E-00003930 - Toy spectral-sequence profile rank=11

For a finite first-quadrant toy page with 11 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 11; no extension claim is made.

Evidence M. Hard: S26-E-00003929.

S26-E-00003931 - Toy spectral-sequence profile rank=12

For a finite first-quadrant toy page with 12 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 12; no extension claim is made.

Evidence M. Hard: S26-E-00003930.

S26-E-00003932 - Toy spectral-sequence profile rank=13

For a finite first-quadrant toy page with 13 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 13; no extension claim is made.

Evidence M. Hard: S26-E-00003931.

S26-E-00003933 - Toy spectral-sequence profile rank=14

For a finite first-quadrant toy page with 14 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 14; no extension claim is made.

Evidence M. Hard: S26-E-00003932.

S26-E-00003934 - Toy spectral-sequence profile rank=15

For a finite first-quadrant toy page with 15 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 15; no extension claim is made.

Evidence M. Hard: S26-E-00003933.

S26-E-00003935 - Toy spectral-sequence profile rank=16

For a finite first-quadrant toy page with 16 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 16; no extension claim is made.

Evidence M. Hard: S26-E-00003934.

S26-E-00003936 - Toy spectral-sequence profile rank=17

For a finite first-quadrant toy page with 17 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 17; no extension claim is made.

Evidence M. Hard: S26-E-00003935.

S26-E-00003937 - Toy spectral-sequence profile rank=18

For a finite first-quadrant toy page with 18 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 18; no extension claim is made.

Evidence M. Hard: S26-E-00003936.

S26-E-00003938 - Toy spectral-sequence profile rank=19

For a finite first-quadrant toy page with 19 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 19; no extension claim is made.

Evidence M. Hard: S26-E-00003937.

S26-E-00003939 - Toy spectral-sequence profile rank=20

For a finite first-quadrant toy page with 20 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 20; no extension claim is made.

Evidence M. Hard: S26-E-00003938.

S26-E-00003940 - Toy spectral-sequence profile rank=21

For a finite first-quadrant toy page with 21 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 21; no extension claim is made.

Evidence M. Hard: S26-E-00003939.

S26-E-00003941 - Toy spectral-sequence profile rank=22

For a finite first-quadrant toy page with 22 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 22; no extension claim is made.

Evidence M. Hard: S26-E-00003940.

S26-E-00003942 - Toy spectral-sequence profile rank=23

For a finite first-quadrant toy page with 23 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 23; no extension claim is made.

Evidence M. Hard: S26-E-00003941.

S26-E-00003943 - Toy spectral-sequence profile rank=24

For a finite first-quadrant toy page with 24 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 24; no extension claim is made.

Evidence M. Hard: S26-E-00003942.

S26-E-00003944 - Toy spectral-sequence profile rank=25

For a finite first-quadrant toy page with 25 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 25; no extension claim is made.

Evidence M. Hard: S26-E-00003943.

S26-E-00003945 - Toy spectral-sequence profile rank=26

For a finite first-quadrant toy page with 26 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 26; no extension claim is made.

Evidence M. Hard: S26-E-00003944.

S26-E-00003946 - Toy spectral-sequence profile rank=27

For a finite first-quadrant toy page with 27 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 27; no extension claim is made.

Evidence M. Hard: S26-E-00003945.

S26-E-00003947 - Toy spectral-sequence profile rank=28

For a finite first-quadrant toy page with 28 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 28; no extension claim is made.

Evidence M. Hard: S26-E-00003946.

S26-E-00003948 - Toy spectral-sequence profile rank=29

For a finite first-quadrant toy page with 29 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 29; no extension claim is made.

Evidence M. Hard: S26-E-00003947.

S26-E-00003949 - Toy spectral-sequence profile rank=30

For a finite first-quadrant toy page with 30 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 30; no extension claim is made.

Evidence M. Hard: S26-E-00003948.

S26-E-00003950 - Toy spectral-sequence profile rank=31

For a finite first-quadrant toy page with 31 one-dimensional entries and zero differentials on every page, the page collapses immediately and the associated graded target has total rank 31; no extension claim is made.

Evidence M. Hard: S26-E-00003949.

Part T - Low-dimensional and geometric topology

S26-D-00003951 - Knot in the 3-sphere

A knot is a smooth or tame embedding S^1->S^3 considered up to ambient isotopy.

Evidence A. Hard: S26-E-00003950.

S26-D-00003952 - Link

A link is a finite disjoint union of embedded circles in S^3 considered up to ambient isotopy.

Evidence A. Hard: S26-D-00003951.

S26-D-00003953 - Knot complement

The complement S^3\K, often with a tubular neighborhood removed, is a 3-manifold whose topology encodes the knot.

Evidence A. Hard: S26-D-00003952.

S26-D-00003954 - Knot group

The knot group is pi_1(S^3\K), with the precise complement convention declared.

Evidence A. Hard: S26-D-00003953.

S26-D-00003955 - Seifert surface

An oriented Seifert surface for an oriented knot is a connected oriented embedded surface in S^3 whose boundary is the knot.

Evidence A. Hard: S26-D-00003954.

S26-D-00003956 - Genus of a knot

The knot genus is the minimal genus among its Seifert surfaces.

Evidence A. Hard: S26-D-00003955.

S26-D-00003957 - Linking number

For two disjoint oriented knots, the linking number is an integer invariant computable from signed crossings or intersection with a spanning surface.

Evidence A. Hard: S26-D-00003956.

S26-D-00003958 - Reidemeister moves

Three local diagram moves generate ambient-isotopy equivalence of tame knot diagrams.

Evidence A. Hard: S26-D-00003957.

S26-T-00003959 - Reidemeister theorem

Two tame knot diagrams represent isotopic knots iff they are related by planar isotopy and a finite sequence of Reidemeister moves.

Evidence A. Hard: S26-D-00003958.

S26-D-00003960 - Alexander polynomial interface

The Alexander polynomial is a Laurent-polynomial knot invariant defined up to multiplication by units ±t^k under a standard normalization convention.

Evidence A. Hard: S26-T-00003959.

S26-D-00003961 - Jones polynomial interface

The Jones polynomial is a Laurent-polynomial invariant characterized by a skein relation and normalization convention.

Evidence A. Hard: S26-D-00003960.

S26-N-00003962 - Polynomial-completeness fence

Classical knot polynomials are not complete invariants; distinct knots can share the same Alexander or Jones polynomial.

Evidence A. Hard: S26-D-00003961.

S26-D-00003963 - Heegaard splitting

A Heegaard splitting decomposes a closed orientable 3-manifold into two handlebodies glued along their common boundary.

Evidence A. Hard: S26-N-00003962.

S26-D-00003964 - Heegaard genus

The Heegaard genus is the minimal genus of a Heegaard surface for the manifold.

Evidence A. Hard: S26-D-00003963.

S26-D-00003965 - Dehn surgery interface

Dehn surgery removes a solid-torus neighborhood of a knot and reglues a solid torus by a boundary homeomorphism specified by a slope.

Evidence A. Hard: S26-D-00003964.

S26-D-00003966 - Mapping class group

The mapping class group of a manifold is the group of isotopy classes of self-homeomorphisms or diffeomorphisms in the declared category.

Evidence A. Hard: S26-D-00003965.

S26-D-00003967 - Surface classification interface

Closed connected surfaces are classified up to homeomorphism by orientability and genus in the classical theorem.

Evidence A. Hard: S26-D-00003966.

S26-T-00003968 - Euler characteristic of closed orientable surfaces

For the closed orientable surface Sigma_g, chi(Sigma_g)=2-2g.

Evidence A. Hard: S26-D-00003967.

S26-B-00003969 - Low-dimensional topology to groups bridge

Knot complements, surface groups and mapping class groups encode geometric topology through finitely presented groups and their actions.

Evidence A. Hard: S26-T-00003968.

S26-B-00003970 - Low-dimensional topology to geometry bridge

Hyperbolic, spherical, Euclidean and other geometric structures on low-dimensional manifolds connect topology with Lie groups, curvature and dynamical invariants.

Evidence A. Hard: S26-B-00003969.

S26-E-00003971 - Orientable-surface profile genus=1

The closed orientable surface Sigma_1 has a CW structure with 1 zero-cell, 2 one-cells and 1 two-cell; chi=1-2+1=0, and over Q beta_1=2.

Evidence M. Hard: S26-B-00003970.

S26-E-00003972 - Orientable-surface profile genus=2

The closed orientable surface Sigma_2 has a CW structure with 1 zero-cell, 4 one-cells and 1 two-cell; chi=1-4+1=-2, and over Q beta_1=4.

Evidence M. Hard: S26-E-00003971.

S26-E-00003973 - Orientable-surface profile genus=3

The closed orientable surface Sigma_3 has a CW structure with 1 zero-cell, 6 one-cells and 1 two-cell; chi=1-6+1=-4, and over Q beta_1=6.

Evidence M. Hard: S26-E-00003972.

S26-E-00003974 - Orientable-surface profile genus=4

The closed orientable surface Sigma_4 has a CW structure with 1 zero-cell, 8 one-cells and 1 two-cell; chi=1-8+1=-6, and over Q beta_1=8.

Evidence M. Hard: S26-E-00003973.

S26-E-00003975 - Orientable-surface profile genus=5

The closed orientable surface Sigma_5 has a CW structure with 1 zero-cell, 10 one-cells and 1 two-cell; chi=1-10+1=-8, and over Q beta_1=10.

Evidence M. Hard: S26-E-00003974.

S26-E-00003976 - Orientable-surface profile genus=6

The closed orientable surface Sigma_6 has a CW structure with 1 zero-cell, 12 one-cells and 1 two-cell; chi=1-12+1=-10, and over Q beta_1=12.

Evidence M. Hard: S26-E-00003975.

S26-E-00003977 - Orientable-surface profile genus=7

The closed orientable surface Sigma_7 has a CW structure with 1 zero-cell, 14 one-cells and 1 two-cell; chi=1-14+1=-12, and over Q beta_1=14.

Evidence M. Hard: S26-E-00003976.

S26-E-00003978 - Orientable-surface profile genus=8

The closed orientable surface Sigma_8 has a CW structure with 1 zero-cell, 16 one-cells and 1 two-cell; chi=1-16+1=-14, and over Q beta_1=16.

Evidence M. Hard: S26-E-00003977.

S26-E-00003979 - Orientable-surface profile genus=9

The closed orientable surface Sigma_9 has a CW structure with 1 zero-cell, 18 one-cells and 1 two-cell; chi=1-18+1=-16, and over Q beta_1=18.

Evidence M. Hard: S26-E-00003978.

S26-E-00003980 - Orientable-surface profile genus=10

The closed orientable surface Sigma_10 has a CW structure with 1 zero-cell, 20 one-cells and 1 two-cell; chi=1-20+1=-18, and over Q beta_1=20.

Evidence M. Hard: S26-E-00003979.

S26-E-00003981 - Orientable-surface profile genus=11

The closed orientable surface Sigma_11 has a CW structure with 1 zero-cell, 22 one-cells and 1 two-cell; chi=1-22+1=-20, and over Q beta_1=22.

Evidence M. Hard: S26-E-00003980.

S26-E-00003982 - Orientable-surface profile genus=12

The closed orientable surface Sigma_12 has a CW structure with 1 zero-cell, 24 one-cells and 1 two-cell; chi=1-24+1=-22, and over Q beta_1=24.

Evidence M. Hard: S26-E-00003981.

S26-E-00003983 - Orientable-surface profile genus=13

The closed orientable surface Sigma_13 has a CW structure with 1 zero-cell, 26 one-cells and 1 two-cell; chi=1-26+1=-24, and over Q beta_1=26.

Evidence M. Hard: S26-E-00003982.

S26-E-00003984 - Orientable-surface profile genus=14

The closed orientable surface Sigma_14 has a CW structure with 1 zero-cell, 28 one-cells and 1 two-cell; chi=1-28+1=-26, and over Q beta_1=28.

Evidence M. Hard: S26-E-00003983.

S26-E-00003985 - Orientable-surface profile genus=15

The closed orientable surface Sigma_15 has a CW structure with 1 zero-cell, 30 one-cells and 1 two-cell; chi=1-30+1=-28, and over Q beta_1=30.

Evidence M. Hard: S26-E-00003984.

S26-E-00003986 - Orientable-surface profile genus=16

The closed orientable surface Sigma_16 has a CW structure with 1 zero-cell, 32 one-cells and 1 two-cell; chi=1-32+1=-30, and over Q beta_1=32.

Evidence M. Hard: S26-E-00003985.

S26-E-00003987 - Orientable-surface profile genus=17

The closed orientable surface Sigma_17 has a CW structure with 1 zero-cell, 34 one-cells and 1 two-cell; chi=1-34+1=-32, and over Q beta_1=34.

Evidence M. Hard: S26-E-00003986.

S26-E-00003988 - Orientable-surface profile genus=18

The closed orientable surface Sigma_18 has a CW structure with 1 zero-cell, 36 one-cells and 1 two-cell; chi=1-36+1=-34, and over Q beta_1=36.

Evidence M. Hard: S26-E-00003987.

S26-E-00003989 - Orientable-surface profile genus=19

The closed orientable surface Sigma_19 has a CW structure with 1 zero-cell, 38 one-cells and 1 two-cell; chi=1-38+1=-36, and over Q beta_1=38.

Evidence M. Hard: S26-E-00003988.

S26-E-00003990 - Orientable-surface profile genus=20

The closed orientable surface Sigma_20 has a CW structure with 1 zero-cell, 40 one-cells and 1 two-cell; chi=1-40+1=-38, and over Q beta_1=40.

Evidence M. Hard: S26-E-00003989.

S26-E-00003991 - Orientable-surface profile genus=21

The closed orientable surface Sigma_21 has a CW structure with 1 zero-cell, 42 one-cells and 1 two-cell; chi=1-42+1=-40, and over Q beta_1=42.

Evidence M. Hard: S26-E-00003990.

S26-E-00003992 - Orientable-surface profile genus=22

The closed orientable surface Sigma_22 has a CW structure with 1 zero-cell, 44 one-cells and 1 two-cell; chi=1-44+1=-42, and over Q beta_1=44.

Evidence M. Hard: S26-E-00003991.

S26-E-00003993 - Orientable-surface profile genus=23

The closed orientable surface Sigma_23 has a CW structure with 1 zero-cell, 46 one-cells and 1 two-cell; chi=1-46+1=-44, and over Q beta_1=46.

Evidence M. Hard: S26-E-00003992.

S26-E-00003994 - Orientable-surface profile genus=24

The closed orientable surface Sigma_24 has a CW structure with 1 zero-cell, 48 one-cells and 1 two-cell; chi=1-48+1=-46, and over Q beta_1=48.

Evidence M. Hard: S26-E-00003993.

S26-E-00003995 - Orientable-surface profile genus=25

The closed orientable surface Sigma_25 has a CW structure with 1 zero-cell, 50 one-cells and 1 two-cell; chi=1-50+1=-48, and over Q beta_1=50.

Evidence M. Hard: S26-E-00003994.

S26-E-00003996 - Orientable-surface profile genus=26

The closed orientable surface Sigma_26 has a CW structure with 1 zero-cell, 52 one-cells and 1 two-cell; chi=1-52+1=-50, and over Q beta_1=52.

Evidence M. Hard: S26-E-00003995.

S26-E-00003997 - Orientable-surface profile genus=27

The closed orientable surface Sigma_27 has a CW structure with 1 zero-cell, 54 one-cells and 1 two-cell; chi=1-54+1=-52, and over Q beta_1=54.

Evidence M. Hard: S26-E-00003996.

S26-E-00003998 - Orientable-surface profile genus=28

The closed orientable surface Sigma_28 has a CW structure with 1 zero-cell, 56 one-cells and 1 two-cell; chi=1-56+1=-54, and over Q beta_1=56.

Evidence M. Hard: S26-E-00003997.

S26-E-00003999 - Orientable-surface profile genus=29

The closed orientable surface Sigma_29 has a CW structure with 1 zero-cell, 58 one-cells and 1 two-cell; chi=1-58+1=-56, and over Q beta_1=58.

Evidence M. Hard: S26-E-00003998.

S26-E-00004000 - Orientable-surface profile genus=30

The closed orientable surface Sigma_30 has a CW structure with 1 zero-cell, 60 one-cells and 1 two-cell; chi=1-60+1=-58, and over Q beta_1=60.

Evidence M. Hard: S26-E-00003999.

Appendix - Peer-review boundary

This reviewed projection changes no canonical IDs and performs no A-to-P promotion. Statement-level corrections are recorded in the consolidated correction ledger. The assumptions[] metadata debt identified for Volumes I-III remains a separate formalization task.

A Synopsis of Results in Pure, Applied and Computational Mathematics

2026 Edition

Volume IV
p-adic Hypercomplex Hyperdimensional Continuum

Local-Field Cayley-Dickson, Gerbes, Recognition & Quantum-Transport Interfaces

Release: 1.0.0 research specification
Canonical range: 4001-5000
Source carrier: 4001-4060 preserved from user draft
New synthesis: 4061-5000
Combined registry: 5,000 objects / 6924 hard edges
Hard graph: 1 root / 1 component / 0 cycles / 0 dangling
Volume IV evidence: A=84 S=350 H=2 M=564 P=0
Finite checks: 40/40 PASS Audit controls: 16/16 PASS

Epistemic boundary: cognition is used as an engineering abstraction for a typed section ecology. This volume does not assert sentience, subjective experience, demonstrated physical cosmology, or theorem closure from visual or computational complexity.

Visual atlas - source geometry as semantic constraint

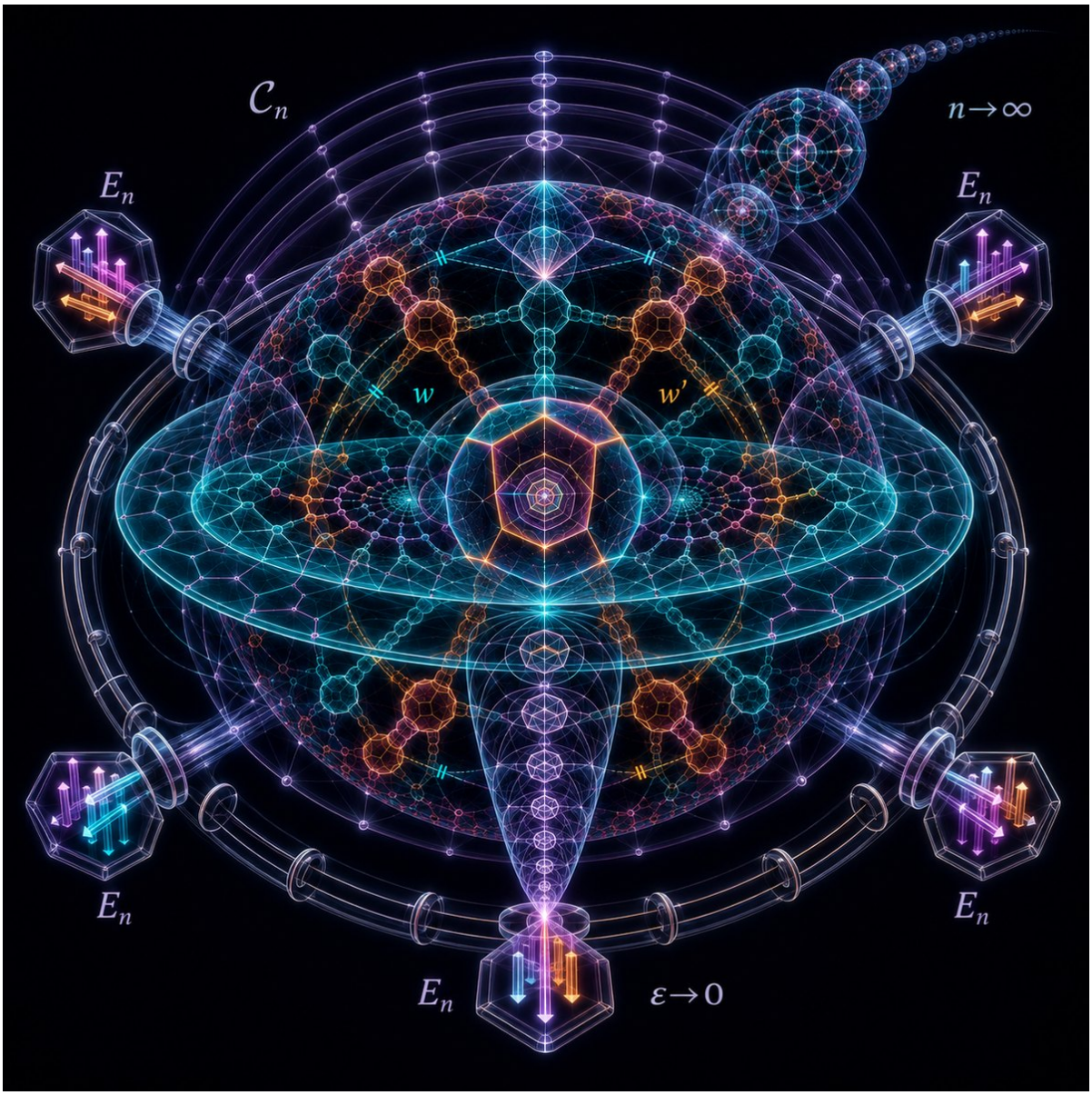


Plate 1. User-supplied overview: nested dodecahedral/orbifold shells, orthogonal w/w-prime lineages, and outward/inward scale cues. This is an Archimedean conceptual projection of distinct n->infinity and epsilon->0 directions. The p-adic/building atlas is not literally depicted; the image is not a proof object.

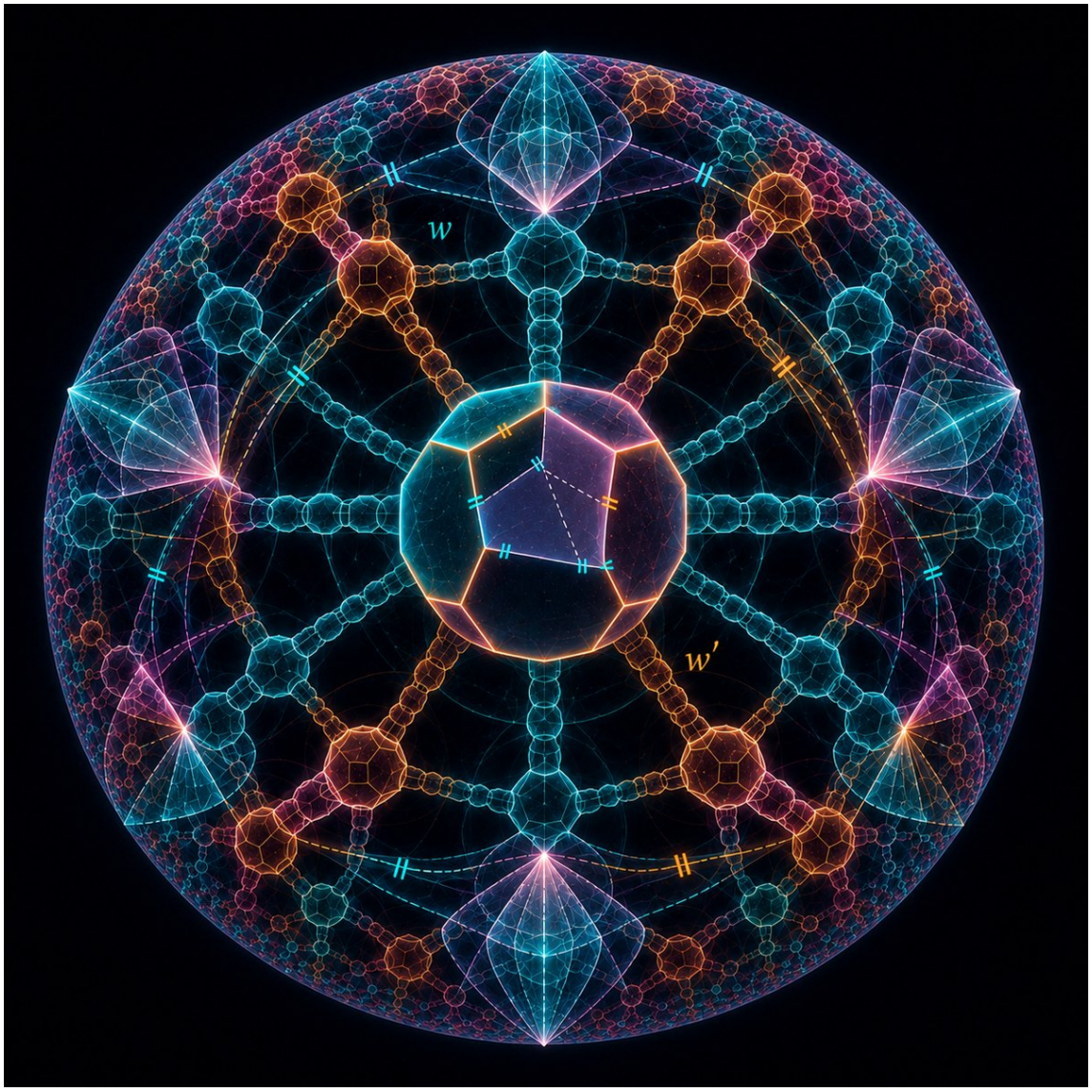


Plate 2. Recursive dodecahedral carrier projection: local-to-global and nested-shell visual cue.

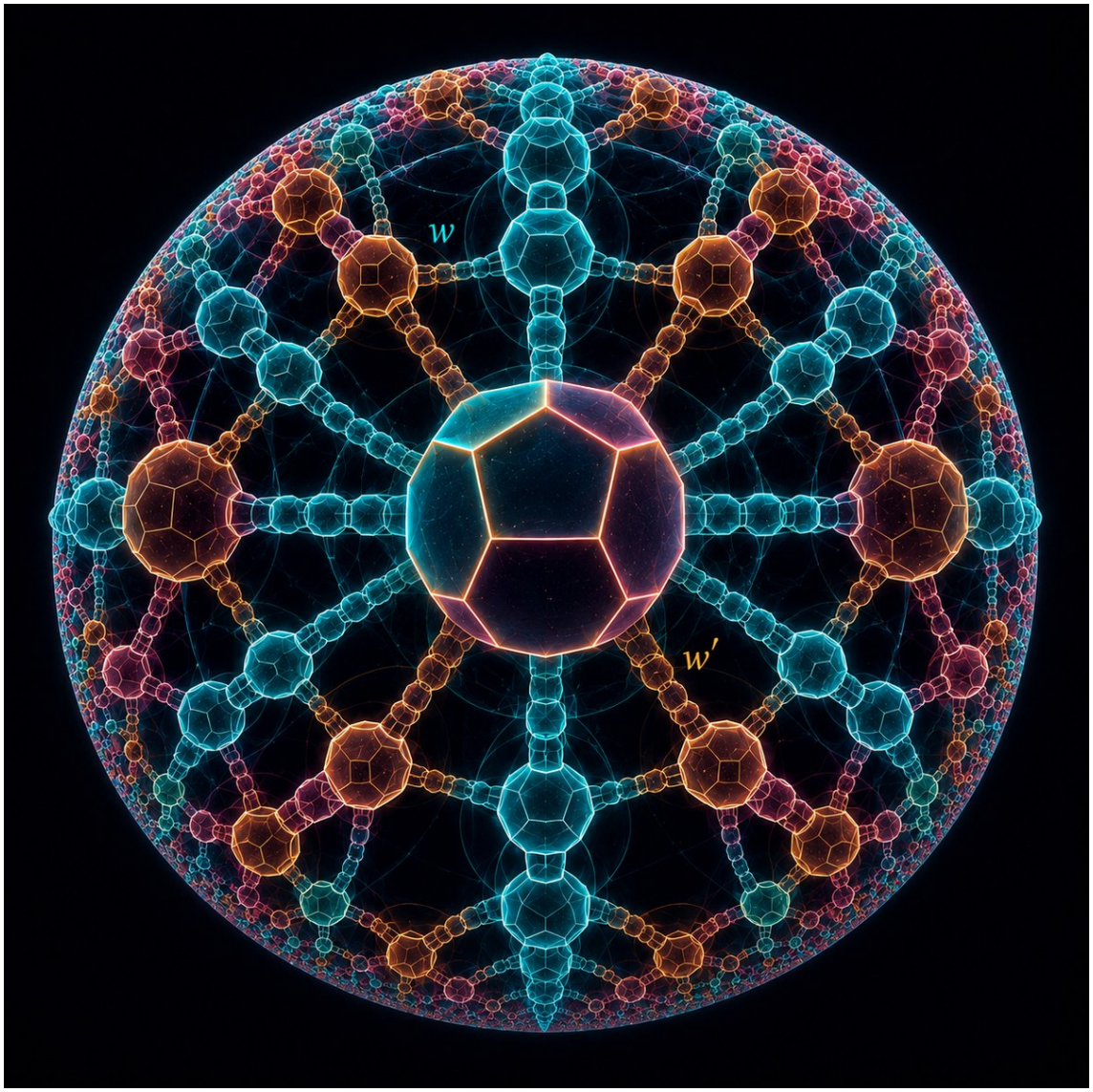


Plate 3. Denser recursive carrier projection: used as a semantic prompt for hypercomplex-in-hypercomplex nesting, not as evidence that a unique algebra is physically realized.

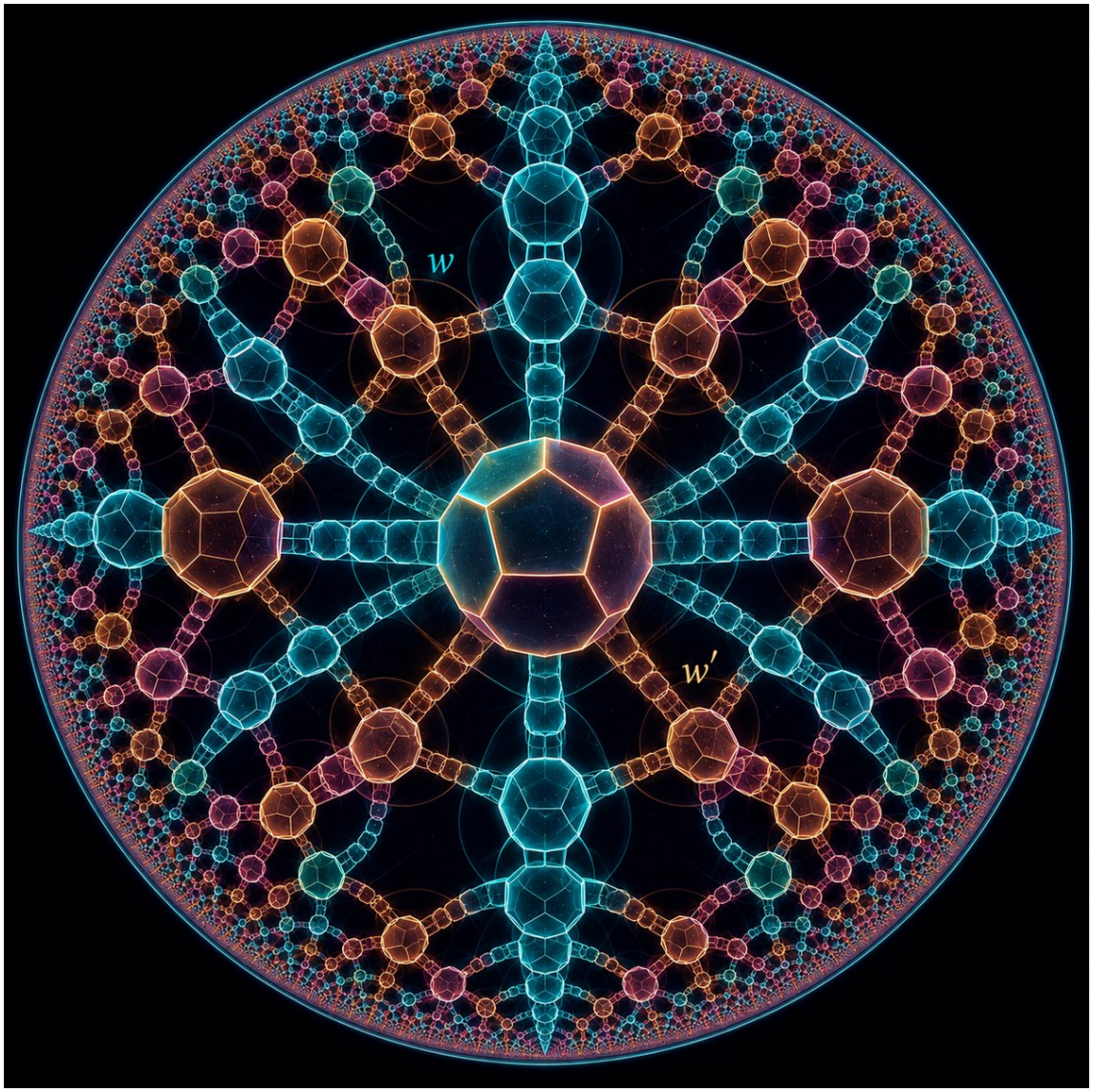


Plate 4. High-density recursive projection: motivates explicit provenance, scale, and return maps so visual density cannot substitute for mathematical structure.



Plate 5. User-supplied singularity/compression visual. Volume IV treats this as a semantic model of projection/compression horizon and unresolved remainder; it is not registered as demonstrated astrophysical black-hole evidence.

The visual atlas is intentionally downstream of the registry: images constrain interpretation and motivate typed operators, but evidence authority remains in statements, dependencies, finite fixtures and audit artifacts.

Core typed state

C_(n,k,v,j,epsilon,t,alpha) = (O_n, A_(k,v), Psi, nabla, Omega, H, Sigma, Z, R_j, Pi, E, epsilon, t, alpha)

The seven control coordinates remain distinct: n is geometric/Cosmo dimension; k is Cayley-Dickson algebraic depth; v is arithmetic place (Archimedean infinity or a p-adic place); j is recursive/fractal generation; epsilon is local scale; t is phase/Hamiltonian parameter; and alpha is a local chart or peer section. Smooth real manifold/orbifold carriers are used on the Archimedean side, while p-adic fibers use ultrametric/building-style charts. Cayley-Dickson structure lives in typed fibers, sections, tensors and operators rather than being smuggled in as an undefined manifold.

Primary invariant: hypercomplex emergence != dimensional inflation. A valid lift must add a measurable algebraic, arithmetic-place, geometric, phase, gluing or evidence distinction while retaining a return path and active remainder. Sequential N.000...N.800 descriptions are publication order only, not an ontological pipeline.

Release audit

Gate	Observed	Result
Canonical sequence	1,000 Volume IV objects; IDs 4001-5000	PASS
Source preservation	60/60 source-draft objects with provenance	PASS
Combined hard graph	6924 edges; 1 component; 0 cycles	PASS
Dangling prerequisites	0	PASS
Exact duplicate titles/statements	0 / 0	PASS
Evidence promotions	P=0	PASS
Hypercomplex finite checks	40 / 40	PASS
Mutation/metamorphic controls	16 / 16	PASS
p-adic/place typing	Ultrametric/building charts kept distinct from Euclidean geometry	PASS
Repo vocabulary grounding	Self-Godel, recognition/resonance, ergocetic/erdodetic and Mind8 typed	PASS
Quantum boundary	Entanglement/tunneling/holoportation restricted to explicit finite Hilbert/channel fixtures	PASS
Cognition/physical boundary	No positive sentience or physical-cosmology claim	PASS

Peer-review overlay. Five corpus corrections are active in the reviewed registry: three carried from Volumes II-III, one Volume IV Cayley-Dickson dimension correction, and one Bruhat-Tits combinatorial-geodesic scope clarification. Frozen originals remain unchanged.

Audit distinction. The 564 M entries are exact finite fixtures: p-adic valuations and ultrametric relations, local-field Cayley-Dickson products, algebraic-defect probes, Bruhat-Tits-style finite tree meshes, gerbe/profunctor/globular coherence fixtures, Hamiltonian/adiabatic ledgers, Mind8/recognition guards, finite entanglement/tunneling/channel examples, spectral profiles, multiscale residuals and detector fixtures. They do not prove the A/S/H layers.

Contents

Carrier layer from the user-supplied v0.1.0 draft

Part A - Modeling discipline and the hyperbolic base (4001-4005)
Part B - Dodecahedral reflection tiling and adjacency (4006-4010)
Part C - Orthogonal 360/360 cell families and the knit (4011-4015)
Part D - Spherical links, shells, and curvature fences (4016-4020)
Part E - Orbifold quotient, isotropy, and singular strata (4021-4025)
Part F - The fused disk-sphere-orbifold object (4026-4030)
Part G - Laplace operators and eigenbasis plug-ins (4031-4035)
Part H - Generalized Cosmo-n systems and coherent towers (4036-4040)
Part I - Fractal generation by graph-directed contractions (4041-4045)
Part J - Infinitesimal generation and tangent orbifolds (4046-4050)
Part K - Coupled scale dynamics and stability (4051-4055)
Part L - Invariants, obstructions, and research frontier (4056-4060)

Hypercomplex continuum extension

Part M - Arithmetic Cayley-Dickson atlas: local fields, p-adics and places (4061-4140)
Part N - Mixed-place hypercomplex bundles and non-Archimedean charts (4141-4220)
Part O - Commutators, associators, zero divisors and law-survival channels (4221-4300)
Part P - Infinitesimal, ultrametric and building multi-mesh geometry (4301-4380)
Part Q - Hamiltonian, adiabatic, ergocetic and erdodetic governance (4381-4460)
Part R - Sheaves, stacks, gerbes, functors, profunctors and globular coherence (4461-4540)
Part S - Self-Godel identity, mutual recognition, resonance and Mind8 (4541-4620)
Part T - Quantum carriers, entanglement, tunneling and holoportation (4621-4700)
Part U - p-adic harmonic, spectral and hypercomplex coupling (4701-4780)
Part V - n/k/place/epsilon continuum and noncommuting-limit fences (4781-4860)
Part W - Russian-doll hypercomplex emergence from previous volumes (4861-4940)
Part X - Repo context, cognitive performance and frontier (4941-5000)

Part A - Modeling discipline and the hyperbolic base

S26-N-00004001 - Dimension-consistency fence

A dodecahedron is a three-dimensional cell. Therefore a construction whose primitive cells are dodecahedra is modeled in hyperbolic three-space H^3, or in a three-dimensional orbifold locally covered by H^3. The Poincare disk D^2 may occur as a totally geodesic section or chart, but not as an intrinsic dodecahedral tiling surface.

Evidence S. Source-preserved v0.1.0. Assumptions: The words disk, ball, cell, section, and quotient retain their standard dimensional meanings. Hard: S26-D-00002366, S26-B-00003169. Boundary: Replacing each dodecahedron by a hyperbolic dodecagon yields a legitimate two-dimensional analogue, but it is a different model.

S26-D-00004002 - Poincare disk chart

Let D^2={z in C: |z|<1}. Its curvature -1 metric is g_D=4|dz|^2/(1-|z|^2)^2. Geodesics are diameters or circular arcs meeting the ideal boundary |z|=1 orthogonally.

Evidence A. Source-preserved v0.1.0. Assumptions: The curvature normalization is fixed at -1. Hard: S26-D-00002366, S26-N-00004001.

S26-D-00004003 - Poincare ball ambient space

Let B^3={x in R^3: ||x||<1} with g_B=4(dx_1^2+dx_2^2+dx_3^2)/(1-||x||^2)^2. Then (B^3,g_B) is a model of complete simply connected hyperbolic three-space H^3 with sectional curvature -1.

Evidence A. Source-preserved v0.1.0. Assumptions: The Euclidean norm and the curvature -1 normalization are used. Hard: S26-N-00004001, S26-D-00004002.

S26-T-00004004 - Totally geodesic disk section

If V is a Euclidean plane through the origin, then B^3 intersect V with the restricted metric is isometric to the Poincare disk D^2 and is totally geodesic in H^3.

Evidence A. Source-preserved v0.1.0. Assumptions: V passes through the center of the Poincare ball. Hard: S26-D-00004002, S26-D-00004003.

S26-D-00004005 - Right-angled hyperbolic dodecahedral seed

Fix a compact regular right-angled hyperbolic dodecahedron P subset H^3: P has twelve congruent pentagonal facets and every dihedral angle is pi/2. A chosen interior point o_P is the seed center.

Evidence A. Source-preserved v0.1.0. Assumptions: Existence of the compact regular right-angled hyperbolic dodecahedron is taken from the standard Coxeter-polytope construction. Hard: S26-D-00004003. Boundary: The Euclidean regular dodecahedron has a different metric and does not generate the same hyperbolic reflection tessellation. Synopsis 2026 - Volume IV research draft - v0.1.0 5 Part B - Dodecahedral reflection tiling and adjacency

Part B - Dodecahedral reflection tiling and adjacency

S26-D-00004006 - Facet-reflection group

For every facet F of P let r_F be reflection in the hyperbolic plane containing F. Define W_P=<r_F: F a facet of P> as a subgroup of Isom(H^3).

Evidence A. Source-preserved v0.1.0. Assumptions: P is the right-angled Coxeter polytope fixed in S26-D-00004005. Hard: S26-D-00004005.

S26-T-00004007 - Right-angled Coxeter presentation

The reflection group admits the presentation W_P=<r_F | r_F^2=1; (r_F r_G)^2=1 whenever F and G meet>. Equivalently, reflections in intersecting facets commute; non-intersecting pairs carry no finite Coxeter relation in this presentation.

Evidence A. Source-preserved v0.1.0. Assumptions: Facet incidences are those of the dodecahedron and all dihedral angles are pi/2. Hard: S26-D-00004005, S26-D-00004006.

S26-T-00004008 - Dodecahedral reflection tessellation

The family T_P={gP: g in W_P} covers H^3 with pairwise disjoint interiors and is locally finite. Every facet belongs to exactly two tiles, related by the corresponding reflection.

Evidence A. Source-preserved v0.1.0. Assumptions: P is a compact hyperbolic Coxeter polytope. Hard: S26-D-00004006, S26-T-00004007.

S26-D-00004009 - Dual cell-adjacency graph

The graph G_P has one vertex [gP] for every tile and an edge between two vertices exactly when their tiles share a facet. The root is [P]. Edge labels record the reflecting facet.

Evidence S. Source-preserved v0.1.0. Assumptions: Cells with identical subsets of H^3 are identified; adjacency means codimension-one incidence. Hard: S26-T-00004008.

S26-D-00004010 - Seeded combinatorial distance

For cells Q,R in T_P, d_G(Q,R) is the shortest path length between their vertices in G_P. The layer L_k={Q: d_G(P,Q)=k} gives the k-th combinatorial shell around the seed.

Evidence S. Source-preserved v0.1.0. Assumptions: The graph metric uses unit edge lengths. Hard: S26-D-00004009.

Part C - Orthogonal 360/360 cell families and the knit

S26-D-00004011 - Admissible face walk

An admissible face walk of length m is a sequence gamma=(P_0,...,P_m) of distinct tiles with P_0=P and consecutive tiles sharing one facet. Its address is the reduced word of the crossed facet reflections.

Evidence S. Source-preserved v0.1.0. Assumptions: The walk is simple and rooted; m is a positive integer. Hard: S26-T-00004007, S26-D-00004009, S26-D-00004010.

S26-D-00004012 - Orthogonal initial cell directions

Let c_F be the tangent at o_P to the geodesic from o_P to the center of facet F. Two rooted walks gamma and gamma' are orthogonal at the seed when <c_F,c_F'>_g=0 for their first crossed facets F,F'.

Evidence S. Source-preserved v0.1.0. Assumptions: The selected facet-center directions are orthogonal in the hyperbolic metric at o_P. Hard: S26-D-00004005, S26-D-00004011.

S26-D-00004013 - The 360-cell w-family

Choose an admissible walk gamma_w=(P,P_1,...,P_360). Define W_360={P_1,...,P_360}; its ordered address word is w=(a_1,...,a_360), where a_j is the crossed facet label.

Evidence S. Source-preserved v0.1.0. Assumptions: A simple rooted walk of length 360 has been selected in G_P. Hard: S26-D-00004011.

S26-D-00004014 - The orthogonal 360-cell w'-family

Choose an admissible walk gamma_w'=(P,P'_1,...,P'_360), orthogonal to gamma_w at the seed. Define W'_360={P'_1,...,P'_360}. Apart from P, the two chosen walks are required to be disjoint unless an explicit bridge identifies them.

Evidence S. Source-preserved v0.1.0. Assumptions: The orthogonality and disjointness requirements are part of the selected data, not automatic consequences of the tessellation. Hard: S26-D-00004012, S26-D-00004013.

S26-T-00004015 - Finite dodecahedral knit

Let B be a finite set of shortest adjacency paths joining selected cells of W_360 to selected cells of W'_360. The knit K_360,360 is the union of P, both 360-cell families, and every tile occurring in B. It is a finite connected cell subcomplex of T_P.

Evidence A. Source-preserved v0.1.0. Assumptions: B is finite and each declared bridge is a finite path in G_P. Hard: S26-T-00004008, S26-D-00004013, S26-D-00004014.

Part D - Spherical links, shells, and curvature fences

S26-D-00004016 - Spherical link at a point

For x in H^3, the unit tangent sphere S_xH^3={v in T_xH^3: ||v||=1} is a round two-sphere. Initial geodesic directions of cells and facets define a finite marked network on this sphere.

Evidence A. Source-preserved v0.1.0. Assumptions: A Riemannian unit tangent norm is fixed at x. Hard: S26-D-00004003, S26-T-00004015.

S26-N-00004017 - Sphere-model curvature fence

The spherical link S_xH^3 and a geodesic sphere of radius r are visualization or link layers. They are not globally isometric to the hyperbolic disk and do not carry constant curvature -1 with the topology of S^2.

Evidence S. Source-preserved v0.1.0. Assumptions: Intrinsic curvature and extrinsic embedding are kept distinct. Hard: S26-T-00004004, S26-D-00004016.

S26-D-00004018 - Radial shell projection

For x in H^3 and r>0, define rho_{x,r}(y)=exp_x(r v_y), where v_y is the initial unit tangent of the geodesic from x to y. On H^3 minus {x}, rho_{x,r} maps rays to the geodesic sphere S_r(x).

Evidence A. Source-preserved v0.1.0. Assumptions: The exponential map is evaluated in complete simply connected H^3. Hard: S26-D-00004003, S26-D-00004016.

S26-D-00004019 - Spherical trace of the knit

The shell trace Sigma_r(K_360,360) is the marked cell-adjacency image rho_{o_P,r}(K_360,360 minus {o_P}), augmented by the incidence labels inherited from G_P. It is a representation of the knit, not a cellulation by isometric dodecahedra.

Evidence S. Source-preserved v0.1.0. Assumptions: Degenerate radial coincidences retain multiplicity as labels. Hard: S26-T-00004015, S26-D-00004018.

S26-T-00004020 - Gauss-Bonnet obstruction to a hyperbolic two-sphere

No smooth Riemannian metric of constant curvature -1 exists on the closed two-sphere S^2.

Evidence A. Source-preserved v0.1.0. Assumptions: The metric is smooth, closed, and without cone singularities. Hard: S26-N-00004017.

Part E - Orbifold quotient, isotropy, and singular strata

S26-D-00004021 - Effective orbifold chart

An n-dimensional effective orbifold chart is (U_tilde,G,pi), where U_tilde is a connected open subset of R^n, G is a finite group acting effectively and smoothly on U_tilde, and pi induces a homeomorphism U_tilde/G -> U. Compatible charts form an orbifold atlas.

Evidence A. Source-preserved v0.1.0. Assumptions: The underlying space is Hausdorff and second countable. Hard: S26-D-00003056, S26-T-00003057, S26-N-00004001.

S26-T-00004022 - Proper discrete quotient orbifold

Let X be a complete Riemannian n-manifold and Gamma a discrete group acting properly and effectively by isometries with finite point stabilizers. Then O=[X/Gamma] is a complete Riemannian orbifold; it is a manifold exactly when every stabilizer is trivial.

Evidence A. Source-preserved v0.1.0. Assumptions: The action is proper, effective, and isometric; point stabilizers are finite. Hard: S26-B-00003169, S26-D-00003856, S26-D-00004021.

S26-D-00004023 - Isotropy and singular stratification

For p=[x] in O=[X/Gamma], the isotropy group is Gamma_x={g in Gamma: gx=x}, defined up to conjugacy. The regular locus has trivial isotropy; connected components of points with conjugate isotropy type form the orbifold strata.

Evidence A. Source-preserved v0.1.0. Assumptions: The quotient hypotheses of S26-T-00004022 hold. Hard: S26-T-00004022.

S26-D-00004024 - Dodecahedral Coxeter orbifold

The quotient O_P=[H^3/W_P] is the Coxeter orbifold with fundamental chamber P. Facets are mirror strata and intersections carry the finite stabilizers generated by the incident reflections. The orientation-preserving subgroup W_P^+ gives an index-two quotient without mirror generators but may retain rotational isotropy.

Evidence A. Source-preserved v0.1.0. Assumptions: The reflection group is the discrete Coxeter group of the right-angled dodecahedron. Hard: S26-D-00004006, S26-T-00004008, S26-T-00004022, S26-D-00004023.

S26-T-00004025 - Tangent cone and spherical orbifold link

At p=[x] in O=[X/Gamma], the metric tangent cone is T_pO isometric to T_xX/Gamma_x. Its unit link is the spherical orbifold S_xX/Gamma_x. Thus the local sphere and the orbifold singularity are fused by the isotropy representation.

Evidence A. Source-preserved v0.1.0. Assumptions: Gamma_x acts linearly on T_xX through the differential of the isometric action. Hard: S26-D-00004016, S26-T-00004022, S26-D-00004023.

Part F - The fused disk-sphere-orbifold object

S26-D-00004026 - Fused Cosmo-3 datum

Define C_3=(H^3,P,T_P,G_P,K_360,360,D_o,S_o,O_P), where D_o is a chosen totally geodesic Poincare disk through o_P and S_o is either the unit spherical link at o_P or a declared radial shell. The components are tied by inclusion, radial projection, and quotient maps.

Evidence S. Source-preserved v0.1.0. Assumptions: Every component is one of the objects defined in S26-D-00004003 through S26-T-00004025. Hard: S26-T-00004004, S26-T-00004008, S26-D-00004009, S26-T-00004015, S26-D-00004019, S26-D-00004024, S26-T-00004025.

S26-D-00004027 - Fusion compatibility diagram

The required compatibility is q o i_D = i_O o q_D on the disk chart and q o i_K = i_K^O o q_K on the knit, where q:H^3->O_P is the quotient map. Radial shell data commute only up to their declared incidence labels, because radial projection is not generally an isometry.

Evidence S. Source-preserved v0.1.0. Assumptions: Every arrow is explicitly typed as an embedding, quotient, projection, or incidence map. Hard: S26-D-00004026.

S26-T-00004028 - Connected quotient image of the knit

The image q(K_360,360) in O_P is a finite connected orbicellular subspace. Cell multiplicities are retained by the map from the pre-quotient tile set even when distinct tiles share one quotient chamber.

Evidence A. Source-preserved v0.1.0. Assumptions: K_360,360 is finite and connected; q is continuous and cellular on the Coxeter decomposition. Hard: S26-T-00004015, S26-D-00004024, S26-D-00004027.

S26-T-00004029 - Subgroup functoriality and orbifold covering

If Gamma' is a subgroup of Gamma in Isom(H^3), then [H^3/Gamma'] -> [H^3/Gamma] is the natural orbifold covering. If Gamma' is torsion-free, the source is a hyperbolic manifold; if Gamma' has finite index, the covering degree is [Gamma:Gamma'] in the orbifold sense.

Evidence A. Source-preserved v0.1.0. Assumptions: Both actions are proper and effective and Gamma' has the stated subgroup properties. Hard: S26-T-00004022, S26-D-00004024.

S26-N-00004030 - Cell-count and quotient-multiplicity fence

The statement '360 cells in each family' refers to selected tiles in the covering tessellation before quotient. After quotient, the number of distinct geometric chambers may be smaller; the exact 360+360 information survives as weighted multiplicity or as address data.

Evidence S. Source-preserved v0.1.0. Assumptions: Counts distinguish tiles, quotient images, and registry addresses. Hard: S26-D-00004013, S26-D-00004014, S26-T-00004028, S26-T-00004029.

Part G - Laplace operators and eigenbasis plug-ins

S26-D-00004031 - Orbifold Hilbert space

For a Riemannian orbifold O_n with measure dvol, define H_n=L^2(O_n,dvol). In a quotient chart U_tilde/G, functions in H_n lift to G-invariant square-integrable functions on U_tilde.

Evidence A. Source-preserved v0.1.0. Assumptions: O_n is sigma-finite; compactness will be declared when used. Hard: S26-D-00004021, S26-T-00004022.

S26-D-00004032 - Laplace-type operator

Let L_n=Delta_n+Q_n on H_n, where Delta_n is the nonnegative Laplace-Beltrami operator and Q_n is a real Gamma_n-invariant potential bounded below. The Friedrichs extension is used whenever the initial operator is only semibounded on compactly supported smooth functions.

Evidence A. Source-preserved v0.1.0. Assumptions: The metric and potential give a densely defined semibounded symmetric operator. Hard: S26-D-00004031.

S26-T-00004033 - Compact-orbifold eigenbasis

If O_n is compact without boundary and L_n is a self-adjoint elliptic Laplace-type operator, then L_n has compact resolvent. Its spectrum is discrete, lambda_k->infinity with finite multiplicities, and H_n has an orthonormal basis of smooth eigenfunctions phi_k satisfying L_n phi_k=lambda_k phi_k.

Evidence A. Source-preserved v0.1.0. Assumptions: Compact effective Riemannian orbifold; elliptic self-adjoint Laplace-type operator; compatible boundary conditions if a boundary is later introduced. Hard: S26-T-00000236, S26-D-00004031, S26-D-00004032.

S26-T-00004034 - General spectral resolution

For a self-adjoint L_n on a possibly noncompact O_n, there is a projection-valued measure E_n on R such that L_n=integral_R lambda dE_n(lambda). A countable L^2 eigenbasis need not exist when continuous spectrum is present.

Evidence A. Source-preserved v0.1.0. Assumptions: L_n is self-adjoint on H_n. Hard: S26-D-00004032.

S26-D-00004035 - Eigen-n spectral plug-in

An admissible spectral plug-in Pi_n=(U_n,V_n,m_n) consists of a spectral representation U_n, a measurable unitary V_n(lambda) acting inside each multiplicity fiber, and a bounded spectral multiplier m_n. It acts as Pi_n=U_n^-1 V_n m_n(M_lambda) U_n. A pure basis plug-in has m_n=1 and changes coordinates but not L_n.

Evidence S. Source-preserved v0.1.0. Assumptions: U_n is unitary, V_n(lambda) is fiberwise unitary almost everywhere, and m_n is bounded and measurable. Hard: S26-T-00004033, S26-T-00004034. Boundary: A multiplier m_n not identically one changes states or dynamics; a unitary basis rotation inside an eigenspace does not change the spectrum. Synopsis 2026 - Volume IV research draft - v0.1.0 11 Part H - Generalized Cosmo-n systems and coherent towers

Part H - Generalized Cosmo-n systems and coherent towers

S26-D-00004036 - General Cosmo-n space

For n>=2, a Cosmo-n datum is C_n=(X_n,Gamma_n,O_n,T_n,P_n,G_n,K_n,L_n,E_n), where X_n is a complete simply connected Riemannian n-manifold, Gamma_n acts properly by isometries with finite stabilizers, O_n=[X_n/Gamma_n], T_n is a locally finite equivariant cellulation with seed P_n and dual graph G_n, K_n is a selected finite knit, L_n is self-adjoint on L^2(O_n), and E_n is its spectral resolution.

Evidence S. Source-preserved v0.1.0. Assumptions: Every component satisfies the hypotheses of its defining clause; omitted components are not inferred. Hard: S26-T-00004022, S26-D-00004026, S26-D-00004031, S26-T-00004034.

S26-T-00004037 - Recovery of the dodecahedral model at n=3

Taking X_3=H^3, Gamma_3=W_P, P_3=P, T_3=T_P, and K_3=K_360,360 recovers the fused Cosmo-3 object, with the chosen disk and spherical link carried as auxiliary chart data.

Evidence A. Source-preserved v0.1.0. Assumptions: The components are selected exactly as in S26-D-00004026. Hard: S26-D-00004026, S26-D-00004036.

S26-N-00004038 - No universal dodecahedron fence

The regular dodecahedron is specific to dimension three. A Cosmo-n family must declare a seed n-polytope P_n, orbicell, or other finite fundamental complex in each dimension; existence of a regular right-angled compact P_n is not automatic.

Evidence S. Source-preserved v0.1.0. Assumptions: Dimension-dependent existence and realizability are checked separately. Hard: S26-D-00004005, S26-D-00004036, S26-T-00004037.

S26-D-00004039 - Dimension morphism

A morphism F_nm:C_n->C_m consists of an orbifold map f_nm:O_n->O_m, a compatible cellular map of selected knits, and a bounded operator J_nm:H_n->H_m. It is spectral when J_nm Dom(L_n) subset Dom(L_m) and L_m J_nm=J_nm L_n.

Evidence S. Source-preserved v0.1.0. Assumptions: All maps preserve the explicitly declared structure; no dimension change is assumed invertible. Hard: S26-D-00004035, S26-D-00004036.

S26-D-00004040 - Coherent Cosmo tower

A coherent tower C_bullet is a sequence (C_n,F_n,n+1) whose morphisms satisfy F_nn=id and F_mk o F_nm=F_nk whenever defined. If the geometric maps are inverse bonding maps, the limit is called a pro-orbifold system; it need not be a finite-dimensional orbifold.

Evidence S. Source-preserved v0.1.0. Assumptions: Bonding maps and their direction are declared; coherence holds on geometry, cells, and spectral intertwiners. Hard: S26-D-00004036, S26-D-00004039. Boundary: An infinite inverse limit may be compact or stratified yet fail to possess an ordinary orbifold atlas. Synopsis 2026 - Volume IV research draft - v0.1.0 12 Part I - Fractal generation by graph-directed contractions

Part I - Fractal generation by graph-directed contractions

S26-D-00004041 - Graph-directed orbifold contraction system

Let A be a finite directed graph. Each vertex a carries a compact complete metric orbifold region X_a, and each edge e:a->b carries a contraction F_e:X_b->X_a with Lipschitz ratio r_e<1. The family F={F_e} is a graph-directed orbifold IFS.

Evidence S. Source-preserved v0.1.0. Assumptions: All regions are nonempty compact metric spaces and every edge map is a declared contraction. Hard: S26-T-00004015, S26-D-00004021, S26-D-00004036.

S26-D-00004042 - Stratum-compatible contraction

A contraction F_e is stratum-compatible when each source isotropy stratum maps into a target stratum and the local chart lift is equivariant with respect to an injective homomorphism of isotropy groups. A cellular contraction additionally maps each selected cell into a declared subcell.

Evidence S. Source-preserved v0.1.0. Assumptions: Compatible orbifold charts and isotropy homomorphisms are supplied for every edge map. Hard: S26-D-00004023, S26-D-00004041.

S26-T-00004043 - Existence of the fractal attractor

For a finite graph-directed contraction system in which every selected vertex has at least one outgoing edge, there is a unique family of nonempty compact sets (K_a) satisfying K_a=union_{e:a->b} F_e(K_b). Iteration from any nonempty compact initial family converges in the product Hausdorff metric.

Evidence A. Source-preserved v0.1.0. Assumptions: Finite graph, complete compact regions, contraction ratios uniformly bounded above by r<1, and at least one outgoing contraction from each selected vertex. Hard: S26-D-00004041.

S26-T-00004044 - Similarity-dimension law

For a single-vertex system of similitudes with ratios r_1,...,r_N satisfying the open set condition, the Hausdorff dimension s of the attractor is the unique solution of sum_i r_i^s=1. For graph-directed similitudes, s is characterized by spectral radius rho(A(s))=1, where A(s)_ab=sum_{e:a->b} r_e^s.

Evidence A. Source-preserved v0.1.0. Assumptions: Similitudes and the appropriate open set/separation condition; irreducibility for a single common graph-directed dimension. Hard: S26-T-00004043.

S26-D-00004045 - Cellular fractal-generation operator

Let R_n act on a finite orbicellular pattern Q by R_n(Q)=union_e F_e(Q at target(e)), retaining edge addresses. The j-th generation is R_n^j(Q); its limiting support is contained in the attractor from S26-T-00004043.

Evidence S. Source-preserved v0.1.0. Assumptions: The maps are cellular, stratum-compatible contractions and address collisions retain multiplicity. Hard: S26-D-00004042, S26-T-00004043, S26-T-00004044.

Part J - Infinitesimal generation and tangent orbifolds

S26-D-00004046 - Infinitesimal zoom

For p in O_n and epsilon>0, the pointed zoom is Z_{p,epsilon}(O_n)=(O_n,epsilon^-1 d,p). In a local chart (U_tilde,G_p,pi) with lift x of p, normal coordinates identify bounded zoom regions with subsets of T_xX_n/G_p.

Evidence A. Source-preserved v0.1.0. Assumptions: The orbifold carries a smooth Riemannian metric and epsilon tends to zero through positive values. Hard: S26-D-00004023, S26-T-00004025, S26-D-00004036.

S26-T-00004047 - Tangent-orbifold limit

As epsilon->0, the pointed spaces Z_{p,epsilon}(O_n) converge locally in the pointed Gromov-Hausdorff sense to T_xX_n/G_p. At a regular point the limit is R^n; at a singular point it is the finite quotient R^n/G_p.

Evidence A. Source-preserved v0.1.0. Assumptions: Finite isotropy and smooth Riemannian orbifold charts. Hard: S26-T-00004025, S26-D-00004046.

S26-D-00004048 - Dilation semigroup and infinitesimal scale generator

On the tangent model R^n/G_p, define (D_t f)(v)=exp(alpha t) f(exp(t)v) for smooth compactly supported invariant f. Then D_tD_s=D_{t+s}, and its formal generator is A_p f=alpha f+v dot grad f.

Evidence A. Source-preserved v0.1.0. Assumptions: Functions are invariant under G_p and lie in a domain on which differentiation and dilation are defined. Hard: S26-T-00004047.

S26-D-00004049 - Nested orbicellular refinement

An infinitesimal cell generator at p is a nested sequence Q_0 superset Q_1 superset ... of orbicellular neighborhoods with p in Q_j and mesh h_j->0. After rescaling by h_j^-1, the marked cell complexes must have a locally finite subsequential limit in T_pO_n.

Evidence S. Source-preserved v0.1.0. Assumptions: Meshes are measured in the orbifold metric and refinements respect isotropy strata. Hard: S26-D-00004042, S26-D-00004046, S26-T-00004047.

S26-T-00004050 - Principal rescaling of the Laplace operator

In a smooth orbifold chart centered at p, the rescaled operator epsilon^2 L_n converges at the level of principal coefficients on bounded zoom regions to the Euclidean Laplacian on T_xX_n/G_p. Lower-order terms vanish under the scaling when their coefficients remain bounded.

Evidence A. Source-preserved v0.1.0. Assumptions: L_n is Laplace type with smooth principal coefficients and locally bounded lower-order coefficients. Hard: S26-D-00004032, S26-D-00004046, S26-T-00004047.

Part K - Coupled scale dynamics and stability

S26-D-00004051 - Bi-scale generated state

A generated state is u_{n,j,epsilon}=Pi_n R_n^j Z_{p,epsilon}^*u, where n is the ambient Cosmo index, j is the fractal generation, epsilon is the infinitesimal scale, R_n is cellular recursion, and Pi_n is a spectral plug-in. The order of operators is part of the datum.

Evidence S. Source-preserved v0.1.0. Assumptions: Every operator acts on a common declared function or section space and all compositions are defined. Hard: S26-D-00004035, S26-D-00004045, S26-D-00004046, S26-T-00004050.

S26-D-00004052 - Intertwining compatibility

A plug-in is recursion-compatible when Pi_n R_n=R_n Pi_n on a common core, zoom-compatible when Pi_n Z_{p,epsilon}^*=Z_{p,epsilon}^* Pi_{p,epsilon}, and tower-compatible when J_nm Pi_n=Pi_m J_nm. Exact commutation may be weakened to a norm-controlled commutator.

Evidence S. Source-preserved v0.1.0. Assumptions: Domains and transported plug-ins are explicitly identified. Hard: S26-D-00004035, S26-D-00004039, S26-D-00004045, S26-D-00004046, S26-D-00004051.

S26-T-00004053 - Unitary plug-in stability

If Pi_n is a pure basis plug-in and U is unitary on H_n, then ||Pi_n Uu||=||u||. More generally, if the multiplier satisfies ||m_n||_infinity<=M, then ||Pi_n u||<=M||u||.

Evidence A. Source-preserved v0.1.0. Assumptions: The spectral representation and fiber maps are unitary and m_n is essentially bounded. Hard: S26-D-00004035.

S26-T-00004054 - Recursive energy estimate

Assume ||R_n||<=r, ||Z_{p,epsilon}^*||<=c(epsilon), and ||Pi_n||<=M on the chosen state space. Then ||u_{n,j,epsilon}||<=M r^j c(epsilon)||u||. If r<1 and c(epsilon) is bounded along the selected scale sequence, the generated energies are uniformly bounded and decay in j.

Evidence A. Source-preserved v0.1.0. Assumptions: The three operators are bounded on the declared common state space. Hard: S26-D-00004051, S26-T-00004053.

S26-D-00004055 - Fused pro-limit and spectral direct limit

For a coherent inverse geometric tower (O_n,p_n) define O_infinity=lim<- O_n as a pro-space. For isometric spectral embeddings J_n,n+1 define H_infinity=lim-> H_n. If L_{n+1}J_n,n+1=J_n,n+1L_n on compatible cores, the algebraic limit carries a well-defined symmetric operator L_infinity before closure.

Evidence S. Source-preserved v0.1.0. Assumptions: Bonding maps, isometric embeddings, and dense compatible cores are supplied; existence of a self-adjoint closure is a separate requirement. Hard: S26-D-00004040, S26-D-00004052. Boundary: O_infinity is called a pro-space, not an ordinary infinite-dimensional orbifold; self-adjointness of L_infinity is not automatic. Synopsis 2026 - Volume IV research draft - v0.1.0 15 Part L - Invariants, obstructions, and research frontier

Part L - Invariants, obstructions, and research frontier

S26-D-00004056 - Orbifold Euler characteristic ledger

For a finite orbicellular decomposition, define chi_orb(O)=sum_sigma (-1)^dim(sigma)/|G_sigma|, where G_sigma is the generic isotropy group of the open cell sigma. Refinement invariance is required before treating this weighted sum as an invariant of the orbifold rather than of a presentation.

Evidence A. Source-preserved v0.1.0. Assumptions: Finite compatible orbicellular decomposition with constant isotropy type on each open cell. Hard: S26-D-00004023, S26-D-00004024, S26-D-00004042.

S26-T-00004057 - Heat-trace singular contribution

For a compact Riemannian orbifold, Tr(exp(-tL_n)) has a small-time asymptotic expansion with the ordinary manifold coefficients plus contributions indexed by singular strata and local isotropy data. These terms can reveal singular structure, but the spectrum does not in general determine the full orbifold uniquely.

Evidence A. Source-preserved v0.1.0. Assumptions: Compact Riemannian orbifold and self-adjoint elliptic Laplace-type operator. Hard: S26-D-00004023, S26-T-00004033.

S26-T-00004058 - Pure eigenbasis invariants

A pure basis plug-in preserves eigenvalues with multiplicity, every spectral projector E_n(B), heat trace, spectral zeta function where defined, and any functional calculus f(L_n). It changes coordinates inside spectral fibers but not the operator's unitary equivalence class.

Evidence A. Source-preserved v0.1.0. Assumptions: The plug-in is fiberwise unitary with multiplier m_n=1. Hard: S26-D-00004035, S26-T-00004053, S26-T-00004057.

S26-N-00004059 - Obstruction and non-equivalence ledger

The following implications are forbidden without extra hypotheses: visual sphere => hyperbolic S^2; finite cell image => 720 distinct quotient chambers; self-similar picture => IFS attractor; eigenfunction expansion => purely discrete spectrum; inverse limit => orbifold; basis change => geometric deformation; bounded finite computation => theorem proof.

Evidence S. Source-preserved v0.1.0. Assumptions: Each arrow is treated as a separate theorem obligation. Hard: S26-N-00004017, S26-T-00004020, S26-N-00004030, S26-T-00004034, S26-D-00004040, S26-T-00004043, S26-D-00004055, S26-T-00004058.

S26-C-00004060 - Spectral-renormalization fixed-point conjecture

Conjecture. Suppose a coherent compact Cosmo tower has stratum-compatible similitude generators, uniformly elliptic L_n, and spectral plug-ins intertwining both tower maps and cellular recursion. After a dimension- and scale-dependent normalization a_n,j L_n, the induced spectral measures admit a nontrivial subsequential fixed point under the combined recursion (n,j,epsilon)->(n+1,j+1,theta epsilon).

Evidence H. Source-preserved v0.1.0. Assumptions: Uniform geometric bounds, tightness of normalized spectral measures, controlled multiplicities, and a fixed theta in (0,1) are required; no existence or uniqueness claim is presently proved. Hard: S26-D-00004040, S26-D-00004045, S26-T-00004050, S26-D-00004052, S26-T-00004054, S26-D-00004055, S26-T-00004057. Boundary: This is a conjectural programmatic statement. It must not be promoted to a theorem without compactness, tightness, and convergence proofs in a specified model. Synopsis 2026 - Volume IV research draft - v0.1.0 16

Part M - Arithmetic Cayley-Dickson atlas: local fields, p-adics and places

Local-field Cayley-Dickson doubling

S26-D-00004061 - Local-field Cayley-Dickson doubling

For a field K of characteristic not 2 with involutive K-algebra A, a generalized Cayley-Dickson double is A plus A*u with multiplication fixed by a declared parameter gamma in K^x and a declared conjugation convention.

Evidence A. Assumptions: Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage. Hard: S26-C-00004060, S26-D-00000103.

S26-T-00004062 - Local-field Cayley-Dickson doubling law/interface

Starting from the base field K, k Cayley-Dickson doublings produce a K-vector space of dimension 2^k. More generally, starting from a finite-dimensional involutive K-algebra A_0 of dimension d, k doublings produce K-dimension d*2^k. Algebraic laws retained or lost depend on the starting algebra and doubling parameters, not on geometric dimension n.

Evidence A. Assumptions: Base field K has characteristic different from 2; a nonzero doubling parameter is declared at each stage. | The 2^k formula refers to the tower starting from K itself; for a general finite-dimensional starting algebra A_0, dim_K(A_k)=dim_K(A_0)*2^k. Hard: S26-D-00004061.

S26-A-00004063 - Local-field Cayley-Dickson doubling operator

CD_{gamma}(a,b;c,d) denotes the declared doubling product, and bar(a,b)=(bar(a),-b) denotes the recursive involution.

Evidence A. Assumptions: Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage. Hard: S26-T-00004062.

S26-N-00004064 - Local-field Cayley-Dickson doubling fence

The symbols R,C,H,O,Sedenion name particular real forms; over Q_p or another field the doubled algebra is field- and parameter-dependent and must not be silently identified with the real division forms.

Evidence S. Assumptions: Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage. Hard: S26-A-00004063. Boundary: The symbols R,C,H,O,Sedenion name particular real forms; over Q_p or another field the doubled algebra is field- and parameter-dependent and must not be silently identified with the real division forms.

S26-B-00004065 - Local-field Cayley-Dickson doubling cross-volume bridge

This construction adds arithmetic place v to the Volume IV coordinates: C_(n,k,v,j,epsilon,t), with v=infinity for the Archimedean chart or v=p for Q_p.

Evidence S. Assumptions: Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage. Hard: S26-N-00004064.

S26-N-00004066 - Local-field Cayley-Dickson doubling evidence ledger

Evidence rule for Local-field Cayley-Dickson doubling: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage. Hard: S26-B-00004065.

S26-B-00004067 - Local-field Cayley-Dickson doubling emergence rule

Emergence rule for Local-field Cayley-Dickson doubling: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage. Hard: S26-N-00004066.

S26-N-00004068 - Local-field Cayley-Dickson doubling frontier

Open frontier for Local-field Cayley-Dickson doubling: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Base field K has characteristic different from 2; a nonzero doubling parameter gamma is declared at each stage. Hard: S26-B-00004067. Boundary: The symbols R,C,H,O,Sedenion name particular real forms; over Q_p or another field the doubled algebra is field- and parameter-dependent and must not be silently identified with the real division forms.

S26-E-00004069 - p-adic CD finite probe 1.1

Fixture T01.P01 - Over Q_3 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-N-00004068.

S26-E-00004070 - p-adic CD finite probe 1.2

Fixture T01.P02 - Over Q_5 coordinates with dimension 4 and declared gamma tuple ['-1', '-5'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004069.

S26-E-00004071 - p-adic CD finite probe 1.3

Fixture T01.P03 - Over Q_7 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004070.

S26-E-00004072 - p-adic CD finite probe 1.4

Fixture T01.P04 - Over Q_3 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-3'], the exact basis product e1*e10=-1e11; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004071.

S26-E-00004073 - p-adic CD finite probe 1.5

Fixture T01.P05 - Over Q_5 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004072.

S26-E-00004074 - p-adic CD finite probe 1.6

Fixture T01.P06 - Over Q_7 coordinates with dimension 4 and declared gamma tuple ['-1', '-7'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004073.

S26-E-00004075 - p-adic CD finite probe 1.7

Fixture T01.P07 - Over Q_3 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e3=-1e2; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004074.

S26-E-00004076 - p-adic CD finite probe 1.8

Fixture T01.P08 - Over Q_5 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-5'], the exact basis product e1*e6=-1e7; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004075.

S26-E-00004077 - p-adic CD finite probe 1.9

Fixture T01.P09 - Over Q_7 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004076.

S26-E-00004078 - p-adic CD finite probe 1.10

Fixture T01.P10 - Over Q_3 coordinates with dimension 4 and declared gamma tuple ['-1', '-3'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004077.

S26-E-00004079 - p-adic CD finite probe 1.11

Fixture T01.P11 - Over Q_5 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004078.

S26-E-00004080 - p-adic CD finite probe 1.12

Fixture T01.P12 - Over Q_7 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-7'], the exact basis product e1*e2=1e3; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004063, S26-E-00004079.

p-adic valuation and ultrametric carrier

S26-D-00004081 - p-adic valuation and ultrametric carrier

For nonzero rational x, v_p(x) is the exponent of p in x and |x|_p=p^{-v_p(x)}; completion of Q in this absolute value gives Q_p.

Evidence A. Assumptions: p is a prime and Q_p is used with its standard normalized p-adic absolute value. Hard: S26-E-00004080, S26-D-00000075.

S26-T-00004082 - p-adic valuation and ultrametric carrier law/interface

The p-adic norm satisfies the strong triangle inequality |x+y|_p <= max(|x|_p,|y|_p), so balls are hierarchically nested or disjoint rather than Euclidean shells.

Evidence A. Assumptions: p is a prime and Q_p is used with its standard normalized p-adic absolute value. Hard: S26-D-00004081.

S26-A-00004083 - p-adic valuation and ultrametric carrier operator

The arithmetic-place selector Place_v routes a local object to the Archimedean chart v=infinity or to the non-Archimedean chart v=p.

Evidence A. Assumptions: p is a prime and Q_p is used with its standard normalized p-adic absolute value. Hard: S26-T-00004082.

S26-N-00004084 - p-adic valuation and ultrametric carrier fence

A p-adic ball is not a distorted Euclidean ball; its ultrametric topology and scale nesting are part of the object, not a visualization choice.

Evidence S. Assumptions: p is a prime and Q_p is used with its standard normalized p-adic absolute value. Hard: S26-A-00004083. Boundary: A p-adic ball is not a distorted Euclidean ball; its ultrametric topology and scale nesting are part of the object, not a visualization choice.

S26-B-00004085 - p-adic valuation and ultrametric carrier cross-volume bridge

The nested-ball geometry supplies a mathematically literal Russian-doll scale axis for the hypercomplex carrier.

Evidence S. Assumptions: p is a prime and Q_p is used with its standard normalized p-adic absolute value. Hard: S26-N-00004084.

S26-N-00004086 - p-adic valuation and ultrametric carrier evidence ledger

Evidence rule for p-adic valuation and ultrametric carrier: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: p is a prime and Q_p is used with its standard normalized p-adic absolute value. Hard: S26-B-00004085.

S26-B-00004087 - p-adic valuation and ultrametric carrier emergence rule

Emergence rule for p-adic valuation and ultrametric carrier: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: p is a prime and Q_p is used with its standard normalized p-adic absolute value. Hard: S26-N-00004086.

S26-N-00004088 - p-adic valuation and ultrametric carrier frontier

Open frontier for p-adic valuation and ultrametric carrier: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: p is a prime and Q_p is used with its standard normalized p-adic absolute value. Hard: S26-B-00004087. Boundary: A p-adic ball is not a distorted Euclidean ball; its ultrametric topology and scale nesting are part of the object, not a visualization choice.

S26-E-00004089 - Ultrametric finite probe 2.1

Fixture T02.P01 - For p=2, a=2, b=3, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-N-00004088.

S26-E-00004090 - Ultrametric finite probe 2.2

Fixture T02.P02 - For p=3, a=9/2, b=5/3, |a+b|_p=3 and max(|a|_p,|b|_p)=3; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004089.

S26-E-00004091 - Ultrametric finite probe 2.3

Fixture T02.P03 - For p=5, a=100/3, b=35/4, |a+b|_p=0.2 and max(|a|_p,|b|_p)=0.2; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004090.

S26-E-00004092 - Ultrametric finite probe 2.4

Fixture T02.P04 - For p=7, a=5/4, b=9/5, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004091.

S26-E-00004093 - Ultrametric finite probe 2.5

Fixture T02.P05 - For p=2, a=12/5, b=11/3, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004092.

S26-E-00004094 - Ultrametric finite probe 2.6

Fixture T02.P06 - For p=3, a=21/2, b=13/7, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004093.

S26-E-00004095 - Ultrametric finite probe 2.7

Fixture T02.P07 - For p=5, a=8/7, b=75/8, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004094.

S26-E-00004096 - Ultrametric finite probe 2.8

Fixture T02.P08 - For p=7, a=63/8, b=17/9, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004095.

S26-E-00004097 - Ultrametric finite probe 2.9

Fixture T02.P09 - For p=2, a=40/9, b=19/5, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004096.

S26-E-00004098 - Ultrametric finite probe 2.10

Fixture T02.P10 - For p=3, a=11/10, b=21/11, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004097.

S26-E-00004099 - Ultrametric finite probe 2.11

Fixture T02.P11 - For p=5, a=60/11, b=115/12, |a+b|_p=0.2 and max(|a|_p,|b|_p)=0.2; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004098.

S26-E-00004100 - Ultrametric finite probe 2.12

Fixture T02.P12 - For p=7, a=637/12, b=25/13, |a+b|_p=1 and max(|a|_p,|b|_p)=1; strong-triangle check=True.

Evidence M. Hard: S26-A-00004083, S26-E-00004099.

Quaternion-octonion local-field strata

S26-D-00004101 - Quaternion-octonion local-field strata

Quaternion and octonion algebras may be defined over fields beyond R by generalized Cayley-Dickson or equivalent composition-algebra constructions, with isomorphism type controlled by the field and norm data.

Evidence A. Assumptions: Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated. Hard: S26-E-00004100, S26-D-00000103.

S26-T-00004102 - Quaternion-octonion local-field strata law/interface

Associativity survives at the quaternion stratum while octonionic multiplication is alternative rather than associative; split versus division behavior is a separate local-field question.

Evidence A. Assumptions: Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated. Hard: S26-D-00004101.

S26-A-00004103 - Quaternion-octonion local-field strata operator

LocalForm(K_v,gamma_1,...,gamma_k) records the chosen arithmetic form before any geometric or cognitive interpretation.

Evidence S. Assumptions: Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated. Hard: S26-T-00004102.

S26-N-00004104 - Quaternion-octonion local-field strata fence

No claim of a canonical p-adic sedenion manifold, p-adic normed division tower, or field-independent multiplication table is permitted.

Evidence S. Assumptions: Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated. Hard: S26-A-00004103. Boundary: No claim of a canonical p-adic sedenion manifold, p-adic normed division tower, or field-independent multiplication table is permitted.

S26-B-00004105 - Quaternion-octonion local-field strata cross-volume bridge

Local algebra forms become fibers over the Cosmo carrier, while their base field remains an explicit coordinate.

Evidence S. Assumptions: Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated. Hard: S26-N-00004104.

S26-N-00004106 - Quaternion-octonion local-field strata evidence ledger

Evidence rule for Quaternion-octonion local-field strata: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated. Hard: S26-B-00004105.

S26-B-00004107 - Quaternion-octonion local-field strata emergence rule

Emergence rule for Quaternion-octonion local-field strata: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated. Hard: S26-N-00004106.

S26-N-00004108 - Quaternion-octonion local-field strata frontier

Open frontier for Quaternion-octonion local-field strata: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Local-field algebra parameters and norm forms are explicitly declared; split and division cases are not conflated. Hard: S26-B-00004107. Boundary: No claim of a canonical p-adic sedenion manifold, p-adic normed division tower, or field-independent multiplication table is permitted.

S26-E-00004109 - p-adic CD finite probe 3.1

Fixture T03.P01 - Over Q_3 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-N-00004108.

S26-E-00004110 - p-adic CD finite probe 3.2

Fixture T03.P02 - Over Q_5 coordinates with dimension 4 and declared gamma tuple ['-1', '-5'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004109.

S26-E-00004111 - p-adic CD finite probe 3.3

Fixture T03.P03 - Over Q_7 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004110.

S26-E-00004112 - p-adic CD finite probe 3.4

Fixture T03.P04 - Over Q_3 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-3'], the exact basis product e1*e10=-1e11; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004111.

S26-E-00004113 - p-adic CD finite probe 3.5

Fixture T03.P05 - Over Q_5 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004112.

S26-E-00004114 - p-adic CD finite probe 3.6

Fixture T03.P06 - Over Q_7 coordinates with dimension 4 and declared gamma tuple ['-1', '-7'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004113.

S26-E-00004115 - p-adic CD finite probe 3.7

Fixture T03.P07 - Over Q_3 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e3=-1e2; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004114.

S26-E-00004116 - p-adic CD finite probe 3.8

Fixture T03.P08 - Over Q_5 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-5'], the exact basis product e1*e6=-1e7; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004115.

S26-E-00004117 - p-adic CD finite probe 3.9

Fixture T03.P09 - Over Q_7 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004116.

S26-E-00004118 - p-adic CD finite probe 3.10

Fixture T03.P10 - Over Q_3 coordinates with dimension 4 and declared gamma tuple ['-1', '-3'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004117.

S26-E-00004119 - p-adic CD finite probe 3.11

Fixture T03.P11 - Over Q_5 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004118.

S26-E-00004120 - p-adic CD finite probe 3.12

Fixture T03.P12 - Over Q_7 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-7'], the exact basis product e1*e2=1e3; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004103, S26-E-00004119.

Adelic hypercomplex place atlas

S26-D-00004121 - Adelic hypercomplex place atlas

An adelic hypercomplex atlas is a family of local Cayley-Dickson carriers A_{k,v} indexed by arithmetic places v, together with compatibility and restricted-product data rather than a single field pretending to contain every local geometry.

Evidence S. Assumptions: Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared. Hard: S26-E-00004120, S26-T-00000051.

S26-D-00004122 - Adelic hypercomplex place atlas law/interface

Local data are compared by explicit transport maps; agreement at many places does not imply identity of local algebras or geometries.

Evidence S. Assumptions: Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared. Hard: S26-D-00004121.

S26-A-00004123 - Adelic hypercomplex place atlas operator

A restricted-place aggregator records an Archimedean component and finitely supported non-Archimedean deviations, preserving each local valuation.

Evidence S. Assumptions: Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared. Hard: S26-D-00004122.

S26-N-00004124 - Adelic hypercomplex place atlas fence

Adelic synthesis is a local-global bookkeeping architecture; it does not turn p-adic and real distances into one scalar metric.

Evidence S. Assumptions: Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared. Hard: S26-A-00004123. Boundary: Adelic synthesis is a local-global bookkeeping architecture; it does not turn p-adic and real distances into one scalar metric.

S26-B-00004125 - Adelic hypercomplex place atlas cross-volume bridge

The atlas makes arithmetic place a sheaf-like cover coordinate compatible with the existing Cosmo-n tower.

Evidence S. Assumptions: Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared. Hard: S26-N-00004124.

S26-N-00004126 - Adelic hypercomplex place atlas evidence ledger

Evidence rule for Adelic hypercomplex place atlas: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared. Hard: S26-B-00004125.

S26-B-00004127 - Adelic hypercomplex place atlas emergence rule

Emergence rule for Adelic hypercomplex place atlas: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared. Hard: S26-N-00004126.

S26-N-00004128 - Adelic hypercomplex place atlas frontier

Open frontier for Adelic hypercomplex place atlas: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Only finitely many nonintegral local components are activated in each bounded fixture; restricted-product conditions are declared. Hard: S26-B-00004127. Boundary: Adelic synthesis is a local-global bookkeeping architecture; it does not turn p-adic and real distances into one scalar metric.

S26-E-00004129 - Product-formula place probe 4.1

Fixture T04.P01 - For rational x=2/3, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-N-00004128.

S26-E-00004130 - Product-formula place probe 4.2

Fixture T04.P02 - For rational x=3/4, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-E-00004129.

S26-E-00004131 - Product-formula place probe 4.3

Fixture T04.P03 - For rational x=4/5, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 5] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-E-00004130.

S26-E-00004132 - Product-formula place probe 4.4

Fixture T04.P04 - For rational x=5/6, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3, 5] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-E-00004131.

S26-E-00004133 - Product-formula place probe 4.5

Fixture T04.P05 - For rational x=6/7, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3, 7] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-E-00004132.

S26-E-00004134 - Product-formula place probe 4.6

Fixture T04.P06 - For rational x=7/8, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 7] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-E-00004133.

S26-E-00004135 - Product-formula place probe 4.7

Fixture T04.P07 - For rational x=8/9, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-E-00004134.

S26-E-00004136 - Product-formula place probe 4.8

Fixture T04.P08 - For rational x=9/10, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3, 5] is 1; residual from 1 is 1.110e-16.

Evidence M. Hard: S26-A-00004123, S26-E-00004135.

S26-E-00004137 - Product-formula place probe 4.9

Fixture T04.P09 - For rational x=10/11, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 5, 11] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-E-00004136.

S26-E-00004138 - Product-formula place probe 4.10

Fixture T04.P10 - For rational x=11/12, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3, 11] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-E-00004137.

S26-E-00004139 - Product-formula place probe 4.11

Fixture T04.P11 - For rational x=12/13, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 3, 13] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-E-00004138.

S26-E-00004140 - Product-formula place probe 4.12

Fixture T04.P12 - For rational x=13/14, the Archimedean absolute value times the p-adic absolute values at the finitely relevant primes [2, 7, 13] is 1; residual from 1 is 0.000e+00.

Evidence M. Hard: S26-A-00004123, S26-E-00004139.

Part N - Mixed-place hypercomplex bundles and non-Archimedean charts

Hypercomplex bundle over mixed places

S26-D-00004141 - Hypercomplex bundle over mixed places

A mixed-place hypercomplex carrier is a family of bundles or local systems E_{n,k,v}->O_{n,v} whose fibers carry A_{k,v}; it is not assumed to be one ordinary manifold across all v.

Evidence S. Assumptions: The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras. Hard: S26-E-00004140, S26-D-00003553.

S26-D-00004142 - Hypercomplex bundle over mixed places law/interface

Base geometry, arithmetic place and fiber algebra are independent coordinates and transition data must state which of them are changed.

Evidence S. Assumptions: The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras. Hard: S26-D-00004141.

S26-A-00004143 - Hypercomplex bundle over mixed places operator

FiberState(n,k,v,x) returns a typed local section value together with its frame, valuation/norm convention and evidence state.

Evidence S. Assumptions: The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras. Hard: S26-D-00004142.

S26-N-00004144 - Hypercomplex bundle over mixed places fence

The phrase hypercomplex manifold is shorthand only when a real or non-Archimedean analytic base and fiber action have been separately defined.

Evidence S. Assumptions: The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras. Hard: S26-A-00004143. Boundary: The phrase hypercomplex manifold is shorthand only when a real or non-Archimedean analytic base and fiber action have been separately defined.

S26-B-00004145 - Hypercomplex bundle over mixed places cross-volume bridge

This replaces a flat 2D picture by a chart family whose readable images are projections of a higher typed carrier.

Evidence S. Assumptions: The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras. Hard: S26-N-00004144.

S26-N-00004146 - Hypercomplex bundle over mixed places evidence ledger

Evidence rule for Hypercomplex bundle over mixed places: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras. Hard: S26-B-00004145.

S26-B-00004147 - Hypercomplex bundle over mixed places emergence rule

Emergence rule for Hypercomplex bundle over mixed places: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras. Hard: S26-N-00004146.

S26-N-00004148 - Hypercomplex bundle over mixed places frontier

Open frontier for Hypercomplex bundle over mixed places: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: The differentiable Archimedean carrier and non-Archimedean analytic carrier are typed separately; fibers are finite-dimensional local algebras. Hard: S26-B-00004147. Boundary: The phrase hypercomplex manifold is shorthand only when a real or non-Archimedean analytic base and fiber action have been separately defined.

S26-E-00004149 - Mixed-place fiber-frame probe 5.1

Fixture T05.P01 - A deterministic orthogonal real carrier frame in dimension 2 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-N-00004148.

S26-E-00004150 - Mixed-place fiber-frame probe 5.2

Fixture T05.P02 - A deterministic orthogonal real carrier frame in dimension 4 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004149.

S26-E-00004151 - Mixed-place fiber-frame probe 5.3

Fixture T05.P03 - A deterministic orthogonal real carrier frame in dimension 8 changed the sampled Euclidean coordinate norm by 8.882e-16; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004150.

S26-E-00004152 - Mixed-place fiber-frame probe 5.4

Fixture T05.P04 - A deterministic orthogonal real carrier frame in dimension 16 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004151.

S26-E-00004153 - Mixed-place fiber-frame probe 5.5

Fixture T05.P05 - A deterministic orthogonal real carrier frame in dimension 2 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004152.

S26-E-00004154 - Mixed-place fiber-frame probe 5.6

Fixture T05.P06 - A deterministic orthogonal real carrier frame in dimension 4 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004153.

S26-E-00004155 - Mixed-place fiber-frame probe 5.7

Fixture T05.P07 - A deterministic orthogonal real carrier frame in dimension 8 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004154.

S26-E-00004156 - Mixed-place fiber-frame probe 5.8

Fixture T05.P08 - A deterministic orthogonal real carrier frame in dimension 16 changed the sampled Euclidean coordinate norm by 8.882e-16; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004155.

S26-E-00004157 - Mixed-place fiber-frame probe 5.9

Fixture T05.P09 - A deterministic orthogonal real carrier frame in dimension 2 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004156.

S26-E-00004158 - Mixed-place fiber-frame probe 5.10

Fixture T05.P10 - A deterministic orthogonal real carrier frame in dimension 4 changed the sampled Euclidean coordinate norm by 4.441e-16; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004157.

S26-E-00004159 - Mixed-place fiber-frame probe 5.11

Fixture T05.P11 - A deterministic orthogonal real carrier frame in dimension 8 changed the sampled Euclidean coordinate norm by 2.220e-16; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004158.

S26-E-00004160 - Mixed-place fiber-frame probe 5.12

Fixture T05.P12 - A deterministic orthogonal real carrier frame in dimension 16 changed the sampled Euclidean coordinate norm by 0.000e+00; the record retains this as an Archimedean fixture and does not transport the metric to Q_p.

Evidence M. Hard: S26-A-00004143, S26-E-00004159.

Berkovich-building local chart interface

S26-D-00004161 - Berkovich-building local chart interface

Volume IV permits Berkovich or rigid analytic charts for p-adic analytic geometry and Bruhat-Tits trees/buildings for group-theoretic incidence, without treating them as Euclidean coordinate grids.

Evidence S. Assumptions: A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature. Hard: S26-E-00004160, S26-D-00000301.

S26-D-00004162 - Berkovich-building local chart interface law/interface

For PGL_2(Q_p), the Bruhat-Tits tree is a locally finite tree and therefore a 0-hyperbolic graph; its boundary encodes projective p-adic directions.

Evidence S. Assumptions: A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature. Hard: S26-D-00004161.

S26-A-00004163 - Berkovich-building local chart interface operator

Chart_v selects the declared analytic/building chart and records which observables are topological, metric, combinatorial or spectral.

Evidence S. Assumptions: A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature. Hard: S26-D-00004162.

S26-N-00004164 - Berkovich-building local chart interface fence

A Bruhat-Tits tree is not a discretized Poincare disk, and a Berkovich point need not correspond to one Q_p coordinate.

Evidence S. Assumptions: A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature. Hard: S26-A-00004163. Boundary: A Bruhat-Tits tree is not a discretized Poincare disk, and a Berkovich point need not correspond to one Q_p coordinate.

S26-B-00004165 - Berkovich-building local chart interface cross-volume bridge

The building supplies a non-Euclidean multi-mesh skeleton on which fibers, transports and recognition overlaps can be attached.

Evidence S. Assumptions: A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature. Hard: S26-N-00004164.

S26-N-00004166 - Berkovich-building local chart interface evidence ledger

Evidence rule for Berkovich-building local chart interface: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature. Hard: S26-B-00004165.

S26-B-00004167 - Berkovich-building local chart interface emergence rule

Emergence rule for Berkovich-building local chart interface: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature. Hard: S26-N-00004166.

S26-N-00004168 - Berkovich-building local chart interface frontier

Open frontier for Berkovich-building local chart interface: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A specific non-Archimedean analytic or building model is declared before using geometric words such as path, boundary or curvature. Hard: S26-B-00004167. Boundary: A Bruhat-Tits tree is not a discretized Poincare disk, and a Berkovich point need not correspond to one Q_p coordinate.

S26-E-00004169 - Bruhat-Tits tree-ball probe 6.1

Fixture T06.P01 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-N-00004168.

S26-E-00004170 - Bruhat-Tits tree-ball probe 6.2

Fixture T06.P02 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004169.

S26-E-00004171 - Bruhat-Tits tree-ball probe 6.3

Fixture T06.P03 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004170.

S26-E-00004172 - Bruhat-Tits tree-ball probe 6.4

Fixture T06.P04 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004171.

S26-E-00004173 - Bruhat-Tits tree-ball probe 6.5

Fixture T06.P05 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004172.

S26-E-00004174 - Bruhat-Tits tree-ball probe 6.6

Fixture T06.P06 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004173.

S26-E-00004175 - Bruhat-Tits tree-ball probe 6.7

Fixture T06.P07 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004174.

S26-E-00004176 - Bruhat-Tits tree-ball probe 6.8

Fixture T06.P08 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004175.

S26-E-00004177 - Bruhat-Tits tree-ball probe 6.9

Fixture T06.P09 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004176.

S26-E-00004178 - Bruhat-Tits tree-ball probe 6.10

Fixture T06.P10 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004177.

S26-E-00004179 - Bruhat-Tits tree-ball probe 6.11

Fixture T06.P11 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004178.

S26-E-00004180 - Bruhat-Tits tree-ball probe 6.12

Fixture T06.P12 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004163, S26-E-00004179.

Fiberwise arithmetic Cayley-Dickson product

S26-D-00004181 - Fiberwise arithmetic Cayley-Dickson product

Fiberwise multiplication applies the local Cayley-Dickson product pointwise to compatible sections in the same A_{k,v} fiber.

Evidence A. Assumptions: Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention. Hard: S26-E-00004180, S26-D-00003554.

S26-T-00004182 - Fiberwise arithmetic Cayley-Dickson product law/interface

Changing local frame may change coordinates but must preserve the declared algebraic identities when the frame transformation is an algebra automorphism.

Evidence A. Assumptions: Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention. Hard: S26-D-00004181.

S26-A-00004183 - Fiberwise arithmetic Cayley-Dickson product operator

Mul_{n,k,v}(s,t)(x)=s(x)t(x) on the common domain, with associator brackets retained when k exceeds the associative strata.

Evidence S. Assumptions: Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention. Hard: S26-T-00004182.

S26-N-00004184 - Fiberwise arithmetic Cayley-Dickson product fence

A generic linear frame change is not automatically an algebra automorphism, and a basis rotation must not be mislabeled as dynamics.

Evidence S. Assumptions: Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention. Hard: S26-A-00004183. Boundary: A generic linear frame change is not automatically an algebra automorphism, and a basis rotation must not be mislabeled as dynamics.

S26-B-00004185 - Fiberwise arithmetic Cayley-Dickson product cross-volume bridge

This is the algebraic coupling between the Cosmo carrier and the arithmetic hypercomplex atlas.

Evidence S. Assumptions: Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention. Hard: S26-N-00004184.

S26-N-00004186 - Fiberwise arithmetic Cayley-Dickson product evidence ledger

Evidence rule for Fiberwise arithmetic Cayley-Dickson product: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention. Hard: S26-B-00004185.

S26-B-00004187 - Fiberwise arithmetic Cayley-Dickson product emergence rule

Emergence rule for Fiberwise arithmetic Cayley-Dickson product: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention. Hard: S26-N-00004186.

S26-N-00004188 - Fiberwise arithmetic Cayley-Dickson product frontier

Open frontier for Fiberwise arithmetic Cayley-Dickson product: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Each fiber has a declared local field, basis, involution, doubling parameters and bracket convention. Hard: S26-B-00004187. Boundary: A generic linear frame change is not automatically an algebra automorphism, and a basis rotation must not be mislabeled as dynamics.

S26-E-00004189 - p-adic CD finite probe 7.1

Fixture T07.P01 - Over Q_3 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-N-00004188.

S26-E-00004190 - p-adic CD finite probe 7.2

Fixture T07.P02 - Over Q_5 coordinates with dimension 4 and declared gamma tuple ['-1', '-5'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004189.

S26-E-00004191 - p-adic CD finite probe 7.3

Fixture T07.P03 - Over Q_7 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004190.

S26-E-00004192 - p-adic CD finite probe 7.4

Fixture T07.P04 - Over Q_3 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-3'], the exact basis product e1*e10=-1e11; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004191.

S26-E-00004193 - p-adic CD finite probe 7.5

Fixture T07.P05 - Over Q_5 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004192.

S26-E-00004194 - p-adic CD finite probe 7.6

Fixture T07.P06 - Over Q_7 coordinates with dimension 4 and declared gamma tuple ['-1', '-7'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004193.

S26-E-00004195 - p-adic CD finite probe 7.7

Fixture T07.P07 - Over Q_3 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e3=-1e2; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004194.

S26-E-00004196 - p-adic CD finite probe 7.8

Fixture T07.P08 - Over Q_5 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-5'], the exact basis product e1*e6=-1e7; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004195.

S26-E-00004197 - p-adic CD finite probe 7.9

Fixture T07.P09 - Over Q_7 coordinates with dimension 2 and declared gamma tuple ['-1'], the exact basis product e1*e1=-1e0; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004196.

S26-E-00004198 - p-adic CD finite probe 7.10

Fixture T07.P10 - Over Q_3 coordinates with dimension 4 and declared gamma tuple ['-1', '-3'], the exact basis product e1*e0=1e1; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004197.

S26-E-00004199 - p-adic CD finite probe 7.11

Fixture T07.P11 - Over Q_5 coordinates with dimension 8 and declared gamma tuple ['-1', '-1', '-1'], the exact basis product e1*e7=1e6; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004198.

S26-E-00004200 - p-adic CD finite probe 7.12

Fixture T07.P12 - Over Q_7 coordinates with dimension 16 and declared gamma tuple ['-1', '-1', '-1', '-7'], the exact basis product e1*e2=1e3; minimum finite p-adic valuation among nonzero output coordinates is 0.

Evidence M. Hard: S26-A-00004183, S26-E-00004199.

Place-changing transport with remainder

S26-D-00004201 - Place-changing transport with remainder

A place-changing transport T_{v->w} is a structure-preserving comparison record between local representations, not a claim that Q_p and R are metrically identical.

Evidence S. Assumptions: Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed. Hard: S26-E-00004200, S26-D-00004039.

S26-D-00004202 - Place-changing transport with remainder law/interface

Every transport names preserved invariants, transformed structure, losses, unknowns and active remainder; perfect lossless equivalence is not assumed.

Evidence S. Assumptions: Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed. Hard: S26-D-00004201.

S26-A-00004203 - Place-changing transport with remainder operator

Hol_{v->w->v} compares the returned source state with the original using a typed invariant delta rather than subtracting heterogeneous geometries.

Evidence S. Assumptions: Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed. Hard: S26-D-00004202.

S26-N-00004204 - Place-changing transport with remainder fence

No scalar distance may compare Archimedean and p-adic geometry unless an explicit common observable and normalization are supplied.

Evidence S. Assumptions: Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed. Hard: S26-A-00004203. Boundary: No scalar distance may compare Archimedean and p-adic geometry unless an explicit common observable and normalization are supplied.

S26-B-00004205 - Place-changing transport with remainder cross-volume bridge

This instantiates the Trasgo transport calculus on the new arithmetic-place axis.

Evidence S. Assumptions: Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed. Hard: S26-N-00004204.

S26-N-00004206 - Place-changing transport with remainder evidence ledger

Evidence rule for Place-changing transport with remainder: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed. Hard: S26-B-00004205.

S26-B-00004207 - Place-changing transport with remainder emergence rule

Emergence rule for Place-changing transport with remainder: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed. Hard: S26-N-00004206.

S26-N-00004208 - Place-changing transport with remainder frontier

Open frontier for Place-changing transport with remainder: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Source and destination charts expose comparable invariants; no canonical isometry between real and p-adic metrics is assumed. Hard: S26-B-00004207. Boundary: No scalar distance may compare Archimedean and p-adic geometry unless an explicit common observable and normalization are supplied.

S26-E-00004209 - Place-transport typed-observable probe 8.1

Fixture T08.P01 - The same rational coordinate x=2 has Archimedean absolute value 2 and 2-adic absolute value 0.5; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-N-00004208.

S26-E-00004210 - Place-transport typed-observable probe 8.2

Fixture T08.P02 - The same rational coordinate x=9/2 has Archimedean absolute value 4.5 and 3-adic absolute value 0.111111; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004209.

S26-E-00004211 - Place-transport typed-observable probe 8.3

Fixture T08.P03 - The same rational coordinate x=4/3 has Archimedean absolute value 1.33333 and 5-adic absolute value 1; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004210.

S26-E-00004212 - Place-transport typed-observable probe 8.4

Fixture T08.P04 - The same rational coordinate x=5/2 has Archimedean absolute value 2.5 and 2-adic absolute value 2; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004211.

S26-E-00004213 - Place-transport typed-observable probe 8.5

Fixture T08.P05 - The same rational coordinate x=6/5 has Archimedean absolute value 1.2 and 3-adic absolute value 0.333333; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004212.

S26-E-00004214 - Place-transport typed-observable probe 8.6

Fixture T08.P06 - The same rational coordinate x=35/6 has Archimedean absolute value 5.83333 and 5-adic absolute value 0.2; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004213.

S26-E-00004215 - Place-transport typed-observable probe 8.7

Fixture T08.P07 - The same rational coordinate x=8/7 has Archimedean absolute value 1.14286 and 2-adic absolute value 0.125; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004214.

S26-E-00004216 - Place-transport typed-observable probe 8.8

Fixture T08.P08 - The same rational coordinate x=27/8 has Archimedean absolute value 3.375 and 3-adic absolute value 0.037037; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004215.

S26-E-00004217 - Place-transport typed-observable probe 8.9

Fixture T08.P09 - The same rational coordinate x=10/9 has Archimedean absolute value 1.11111 and 5-adic absolute value 0.2; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004216.

S26-E-00004218 - Place-transport typed-observable probe 8.10

Fixture T08.P10 - The same rational coordinate x=11/5 has Archimedean absolute value 2.2 and 2-adic absolute value 1; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004217.

S26-E-00004219 - Place-transport typed-observable probe 8.11

Fixture T08.P11 - The same rational coordinate x=12/11 has Archimedean absolute value 1.09091 and 3-adic absolute value 0.333333; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004218.

S26-E-00004220 - Place-transport typed-observable probe 8.12

Fixture T08.P12 - The same rational coordinate x=65/12 has Archimedean absolute value 5.41667 and 5-adic absolute value 0.2; the fixture records both coordinates instead of subtracting them as commensurate distances.

Evidence M. Hard: S26-A-00004203, S26-E-00004219.

Part O - Commutators, associators, zero divisors and law-survival channels

Commutator perspective channel

S26-D-00004221 - Commutator perspective channel

The commutator [a,b]=ab-ba measures order sensitivity of multiplication.

Evidence A. Assumptions: Elements lie in the same declared algebra and multiplication convention. Hard: S26-E-00004220, S26-D-00000268.

S26-T-00004222 - Commutator perspective channel law/interface

A vanishing commutator is an algebraic statement about the chosen pair; it does not imply that the whole algebra is commutative.

Evidence A. Assumptions: Elements lie in the same declared algebra and multiplication convention. Hard: S26-D-00004221.

S26-A-00004223 - Commutator perspective channel operator

Comm(a,b)=ab-ba is retained as a vector-valued channel rather than collapsed to a norm unless a norm is declared.

Evidence A. Assumptions: Elements lie in the same declared algebra and multiplication convention. Hard: S26-T-00004222.

S26-N-00004224 - Commutator perspective channel fence

Order dependence may be used as a cognitive metaphor only at S evidence; the algebraic commutator itself does not establish a cognitive mechanism.

Evidence S. Assumptions: Elements lie in the same declared algebra and multiplication convention. Hard: S26-A-00004223. Boundary: Order dependence may be used as a cognitive metaphor only at S evidence; the algebraic commutator itself does not establish a cognitive mechanism.

S26-B-00004225 - Commutator perspective channel cross-volume bridge

The channel supplies one explicit non-flattening coordinate for hypercomplex section interactions.

Evidence S. Assumptions: Elements lie in the same declared algebra and multiplication convention. Hard: S26-N-00004224.

S26-N-00004226 - Commutator perspective channel evidence ledger

Evidence rule for Commutator perspective channel: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Elements lie in the same declared algebra and multiplication convention. Hard: S26-B-00004225.

S26-B-00004227 - Commutator perspective channel emergence rule

Emergence rule for Commutator perspective channel: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Elements lie in the same declared algebra and multiplication convention. Hard: S26-N-00004226.

S26-N-00004228 - Commutator perspective channel frontier

Open frontier for Commutator perspective channel: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Elements lie in the same declared algebra and multiplication convention. Hard: S26-B-00004227. Boundary: Order dependence may be used as a cognitive metaphor only at S evidence; the algebraic commutator itself does not establish a cognitive mechanism.

S26-E-00004229 - Algebraic defect finite probe 9.1

Fixture T09.P01 - In the real Cayley-Dickson fixture of dimension 8, basis indices (5, 6, 3) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-N-00004228.

S26-E-00004230 - Algebraic defect finite probe 9.2

Fixture T09.P02 - In the real Cayley-Dickson fixture of dimension 16, basis indices (9, 8, 5) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004229.

S26-E-00004231 - Algebraic defect finite probe 9.3

Fixture T09.P03 - In the real Cayley-Dickson fixture of dimension 8, basis indices (7, 1, 2) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004230.

S26-E-00004232 - Algebraic defect finite probe 9.4

Fixture T09.P04 - In the real Cayley-Dickson fixture of dimension 16, basis indices (14, 9, 10) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004231.

S26-E-00004233 - Algebraic defect finite probe 9.5

Fixture T09.P05 - In the real Cayley-Dickson fixture of dimension 8, basis indices (5, 1, 7) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004232.

S26-E-00004234 - Algebraic defect finite probe 9.6

Fixture T09.P06 - In the real Cayley-Dickson fixture of dimension 16, basis indices (5, 14, 6) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004233.

S26-E-00004235 - Algebraic defect finite probe 9.7

Fixture T09.P07 - In the real Cayley-Dickson fixture of dimension 8, basis indices (1, 7, 6) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004234.

S26-E-00004236 - Algebraic defect finite probe 9.8

Fixture T09.P08 - In the real Cayley-Dickson fixture of dimension 16, basis indices (6, 8, 15) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004235.

S26-E-00004237 - Algebraic defect finite probe 9.9

Fixture T09.P09 - In the real Cayley-Dickson fixture of dimension 8, basis indices (6, 7, 2) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004236.

S26-E-00004238 - Algebraic defect finite probe 9.10

Fixture T09.P10 - In the real Cayley-Dickson fixture of dimension 16, basis indices (2, 5, 10) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004237.

S26-E-00004239 - Algebraic defect finite probe 9.11

Fixture T09.P11 - In the real Cayley-Dickson fixture of dimension 8, basis indices (2, 7, 3) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004238.

S26-E-00004240 - Algebraic defect finite probe 9.12

Fixture T09.P12 - In the real Cayley-Dickson fixture of dimension 16, basis indices (1, 9, 3) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004223, S26-E-00004239.

Associator contextual-bracketing channel

S26-D-00004241 - Associator contextual-bracketing channel

The associator [a,b,c]=(ab)c-a(bc) measures failure of associativity.

Evidence A. Assumptions: Three elements and one multiplication convention are declared. Hard: S26-E-00004240, S26-D-00000268.

S26-T-00004242 - Associator contextual-bracketing channel law/interface

Octonions are alternative, so associators vanish on repeated arguments; higher Cayley-Dickson strata generally have weaker laws.

Evidence A. Assumptions: Three elements and one multiplication convention are declared. Hard: S26-D-00004241.

S26-A-00004243 - Associator contextual-bracketing channel operator

Assoc(a,b,c) is stored with its full coordinate output and bracket provenance.

Evidence A. Assumptions: Three elements and one multiplication convention are declared. Hard: S26-T-00004242.

S26-N-00004244 - Associator contextual-bracketing channel fence

A large associator is not automatically better cognition; it is an algebraic defect signal whose usefulness must be tested.

Evidence S. Assumptions: Three elements and one multiplication convention are declared. Hard: S26-A-00004243. Boundary: A large associator is not automatically better cognition; it is an algebraic defect signal whose usefulness must be tested.

S26-B-00004245 - Associator contextual-bracketing channel cross-volume bridge

Associator telemetry allows different relation bracketings to remain distinct through recognition and transport.

Evidence S. Assumptions: Three elements and one multiplication convention are declared. Hard: S26-N-00004244.

S26-N-00004246 - Associator contextual-bracketing channel evidence ledger

Evidence rule for Associator contextual-bracketing channel: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Three elements and one multiplication convention are declared. Hard: S26-B-00004245.

S26-B-00004247 - Associator contextual-bracketing channel emergence rule

Emergence rule for Associator contextual-bracketing channel: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Three elements and one multiplication convention are declared. Hard: S26-N-00004246.

S26-N-00004248 - Associator contextual-bracketing channel frontier

Open frontier for Associator contextual-bracketing channel: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Three elements and one multiplication convention are declared. Hard: S26-B-00004247. Boundary: A large associator is not automatically better cognition; it is an algebraic defect signal whose usefulness must be tested.

S26-E-00004249 - Algebraic defect finite probe 10.1

Fixture T10.P01 - In the real Cayley-Dickson fixture of dimension 8, basis indices (5, 4, 6) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-N-00004248.

S26-E-00004250 - Algebraic defect finite probe 10.2

Fixture T10.P02 - In the real Cayley-Dickson fixture of dimension 16, basis indices (13, 15, 4) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004249.

S26-E-00004251 - Algebraic defect finite probe 10.3

Fixture T10.P03 - In the real Cayley-Dickson fixture of dimension 8, basis indices (1, 5, 6) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004250.

S26-E-00004252 - Algebraic defect finite probe 10.4

Fixture T10.P04 - In the real Cayley-Dickson fixture of dimension 16, basis indices (14, 6, 12) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004251.

S26-E-00004253 - Algebraic defect finite probe 10.5

Fixture T10.P05 - In the real Cayley-Dickson fixture of dimension 8, basis indices (4, 2, 1) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004252.

S26-E-00004254 - Algebraic defect finite probe 10.6

Fixture T10.P06 - In the real Cayley-Dickson fixture of dimension 16, basis indices (14, 6, 10) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004253.

S26-E-00004255 - Algebraic defect finite probe 10.7

Fixture T10.P07 - In the real Cayley-Dickson fixture of dimension 8, basis indices (7, 1, 4) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004254.

S26-E-00004256 - Algebraic defect finite probe 10.8

Fixture T10.P08 - In the real Cayley-Dickson fixture of dimension 16, basis indices (6, 4, 15) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004255.

S26-E-00004257 - Algebraic defect finite probe 10.9

Fixture T10.P09 - In the real Cayley-Dickson fixture of dimension 8, basis indices (1, 4, 2) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004256.

S26-E-00004258 - Algebraic defect finite probe 10.10

Fixture T10.P10 - In the real Cayley-Dickson fixture of dimension 16, basis indices (15, 7, 9) give commutator norm 2 and associator norm 2; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004257.

S26-E-00004259 - Algebraic defect finite probe 10.11

Fixture T10.P11 - In the real Cayley-Dickson fixture of dimension 8, basis indices (1, 7, 6) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004258.

S26-E-00004260 - Algebraic defect finite probe 10.12

Fixture T10.P12 - In the real Cayley-Dickson fixture of dimension 16, basis indices (12, 5, 2) give commutator norm 2 and associator norm 0; these are algebraic diagnostics only.

Evidence M. Hard: S26-A-00004243, S26-E-00004259.

Zero-divisor null-channel ledger

S26-D-00004261 - Zero-divisor null-channel ledger

A zero-divisor channel records nonzero a,b with ab=0 together with the chosen algebra, basis and arithmetic place.

Evidence A. Assumptions: The selected Cayley-Dickson form actually contains the recorded zero-divisor pair. Hard: S26-E-00004260, S26-N-00004059.

S26-T-00004262 - Zero-divisor null-channel ledger law/interface

A zero product may arise from nontrivial sources in algebras with zero divisors; information loss in that projection must therefore remain visible.

Evidence A. Assumptions: The selected Cayley-Dickson form actually contains the recorded zero-divisor pair. Hard: S26-D-00004261.

S26-A-00004263 - Zero-divisor null-channel ledger operator

ZDiv(a,b) stores source norms/valuations, product coordinates and the null-channel label.

Evidence S. Assumptions: The selected Cayley-Dickson form actually contains the recorded zero-divisor pair. Hard: S26-T-00004262.

S26-N-00004264 - Zero-divisor null-channel ledger fence

Zero-divisor behavior is not evidence that a physical signal vanished or that a cognitive process inhibited another unless an operational mapping is supplied.

Evidence S. Assumptions: The selected Cayley-Dickson form actually contains the recorded zero-divisor pair. Hard: S26-A-00004263. Boundary: Zero-divisor behavior is not evidence that a physical signal vanished or that a cognitive process inhibited another unless an operational mapping is supplied.

S26-B-00004265 - Zero-divisor null-channel ledger cross-volume bridge

The null-channel ledger provides a controlled higher-stratum failure mode for the Russian-doll carrier.

Evidence S. Assumptions: The selected Cayley-Dickson form actually contains the recorded zero-divisor pair. Hard: S26-N-00004264.

S26-N-00004266 - Zero-divisor null-channel ledger evidence ledger

Evidence rule for Zero-divisor null-channel ledger: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: The selected Cayley-Dickson form actually contains the recorded zero-divisor pair. Hard: S26-B-00004265.

S26-B-00004267 - Zero-divisor null-channel ledger emergence rule

Emergence rule for Zero-divisor null-channel ledger: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: The selected Cayley-Dickson form actually contains the recorded zero-divisor pair. Hard: S26-N-00004266.

S26-N-00004268 - Zero-divisor null-channel ledger frontier

Open frontier for Zero-divisor null-channel ledger: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: The selected Cayley-Dickson form actually contains the recorded zero-divisor pair. Hard: S26-B-00004267. Boundary: Zero-divisor behavior is not evidence that a physical signal vanished or that a cognitive process inhibited another unless an operational mapping is supplied.

S26-E-00004269 - Sedenion null-channel probe 11.1

Fixture T11.P01 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-N-00004268.

S26-E-00004270 - Sedenion null-channel probe 11.2

Fixture T11.P02 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004269.

S26-E-00004271 - Sedenion null-channel probe 11.3

Fixture T11.P03 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004270.

S26-E-00004272 - Sedenion null-channel probe 11.4

Fixture T11.P04 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004271.

S26-E-00004273 - Sedenion null-channel probe 11.5

Fixture T11.P05 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004272.

S26-E-00004274 - Sedenion null-channel probe 11.6

Fixture T11.P06 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004273.

S26-E-00004275 - Sedenion null-channel probe 11.7

Fixture T11.P07 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004274.

S26-E-00004276 - Sedenion null-channel probe 11.8

Fixture T11.P08 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004275.

S26-E-00004277 - Sedenion null-channel probe 11.9

Fixture T11.P09 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004276.

S26-E-00004278 - Sedenion null-channel probe 11.10

Fixture T11.P10 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004277.

S26-E-00004279 - Sedenion null-channel probe 11.11

Fixture T11.P11 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004278.

S26-E-00004280 - Sedenion null-channel probe 11.12

Fixture T11.P12 - A deterministic search recovered a nonzero real-sedenion pair of signed two-basis vectors with product zero; witness indices/signs=(1, 10, 1, 4, 15, -1). This does not assert the same witness for every p-adic form.

Evidence M. Hard: S26-A-00004263, S26-E-00004279.

Nucleus center and local-law survival

S26-D-00004281 - Nucleus center and local-law survival

The law-survival ledger records which commutativity, associativity, alternativity, power-associativity and norm identities survive at each (k,v).

Evidence A. Assumptions: Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form. Hard: S26-E-00004280, S26-D-00000103.

S26-T-00004282 - Nucleus center and local-law survival law/interface

Properties are local to the stated algebraic form; they are not inherited merely because one stratum embeds linearly in another.

Evidence A. Assumptions: Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form. Hard: S26-D-00004281.

S26-A-00004283 - Nucleus center and local-law survival operator

LawVector(k,v) is a typed Boolean/partial certificate whose unknown entries remain unknown rather than false or true by default.

Evidence S. Assumptions: Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form. Hard: S26-T-00004282.

S26-N-00004284 - Nucleus center and local-law survival fence

No cognitive or geometric promotion follows from a richer failure profile alone.

Evidence S. Assumptions: Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form. Hard: S26-A-00004283. Boundary: No cognitive or geometric promotion follows from a richer failure profile alone.

S26-B-00004285 - Nucleus center and local-law survival cross-volume bridge

This ledger is the algebraic governance surface used by higher gluing and Hamiltonian components.

Evidence S. Assumptions: Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form. Hard: S26-N-00004284.

S26-N-00004286 - Nucleus center and local-law survival evidence ledger

Evidence rule for Nucleus center and local-law survival: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form. Hard: S26-B-00004285.

S26-B-00004287 - Nucleus center and local-law survival emergence rule

Emergence rule for Nucleus center and local-law survival: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form. Hard: S26-N-00004286.

S26-N-00004288 - Nucleus center and local-law survival frontier

Open frontier for Nucleus center and local-law survival: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Center, nucleus and alternativity tests are evaluated in a declared finite algebraic form. Hard: S26-B-00004287. Boundary: No cognitive or geometric promotion follows from a richer failure profile alone.

S26-E-00004289 - Law-survival probe 12.1

Fixture T12.P01 - At real CD dimension 2, sampled basis triple (1, 1, 1) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-N-00004288.

S26-E-00004290 - Law-survival probe 12.2

Fixture T12.P02 - At real CD dimension 4, sampled basis triple (1, 2, 3) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004289.

S26-E-00004291 - Law-survival probe 12.3

Fixture T12.P03 - At real CD dimension 8, sampled basis triple (4, 7, 6) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004290.

S26-E-00004292 - Law-survival probe 12.4

Fixture T12.P04 - At real CD dimension 16, sampled basis triple (13, 15, 12) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004291.

S26-E-00004293 - Law-survival probe 12.5

Fixture T12.P05 - At real CD dimension 2, sampled basis triple (1, 1, 1) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004292.

S26-E-00004294 - Law-survival probe 12.6

Fixture T12.P06 - At real CD dimension 4, sampled basis triple (2, 1, 3) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004293.

S26-E-00004295 - Law-survival probe 12.7

Fixture T12.P07 - At real CD dimension 8, sampled basis triple (7, 3, 2) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004294.

S26-E-00004296 - Law-survival probe 12.8

Fixture T12.P08 - At real CD dimension 16, sampled basis triple (13, 9, 8) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004295.

S26-E-00004297 - Law-survival probe 12.9

Fixture T12.P09 - At real CD dimension 2, sampled basis triple (1, 1, 1) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004296.

S26-E-00004298 - Law-survival probe 12.10

Fixture T12.P10 - At real CD dimension 4, sampled basis triple (2, 3, 1) has associator norm 0; sampled-associativity flag=True. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004297.

S26-E-00004299 - Law-survival probe 12.11

Fixture T12.P11 - At real CD dimension 8, sampled basis triple (5, 7, 1) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004298.

S26-E-00004300 - Law-survival probe 12.12

Fixture T12.P12 - At real CD dimension 16, sampled basis triple (7, 4, 8) has associator norm 2; sampled-associativity flag=False. A sample is not a universal algebra-law proof.

Evidence M. Hard: S26-A-00004283, S26-E-00004299.

Part P - Infinitesimal, ultrametric and building multi-mesh geometry

Hypercomplex tangent-orbifold fiber

S26-D-00004301 - Hypercomplex tangent-orbifold fiber

At an orbifold point p=[x], Volume IV couples the tangent quotient T_xX/G_p to a local algebra fiber A_{k,v}; the tangent geometry remains distinct from the internal algebra.

Evidence S. Assumptions: The source orbifold tangent cone and local hypercomplex fiber have both been declared. Hard: S26-E-00004300, S26-T-00004047.

S26-D-00004302 - Hypercomplex tangent-orbifold fiber law/interface

The epsilon->0 geometric zoom and the algebraic k-depth are independent limits and may not be interchanged without an explicit theorem.

Evidence S. Assumptions: The source orbifold tangent cone and local hypercomplex fiber have both been declared. Hard: S26-D-00004301.

S26-A-00004303 - Hypercomplex tangent-orbifold fiber operator

TangentState(p,k,v)=T_pO tensor_R Carrier(A_{k,v}) when a real carrier is used, or the separately declared non-Archimedean analogue.

Evidence S. Assumptions: The source orbifold tangent cone and local hypercomplex fiber have both been declared. Hard: S26-D-00004302.

S26-N-00004304 - Hypercomplex tangent-orbifold fiber fence

Infinitesimal zoom is not visual miniaturization and does not create a new algebraic law.

Evidence S. Assumptions: The source orbifold tangent cone and local hypercomplex fiber have both been declared. Hard: S26-A-00004303. Boundary: Infinitesimal zoom is not visual miniaturization and does not create a new algebraic law.

S26-B-00004305 - Hypercomplex tangent-orbifold fiber cross-volume bridge

This extends the source draft tangent-orbifold limit into the arithmetic hypercomplex fiber atlas.

Evidence S. Assumptions: The source orbifold tangent cone and local hypercomplex fiber have both been declared. Hard: S26-N-00004304.

S26-N-00004306 - Hypercomplex tangent-orbifold fiber evidence ledger

Evidence rule for Hypercomplex tangent-orbifold fiber: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: The source orbifold tangent cone and local hypercomplex fiber have both been declared. Hard: S26-B-00004305.

S26-B-00004307 - Hypercomplex tangent-orbifold fiber emergence rule

Emergence rule for Hypercomplex tangent-orbifold fiber: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: The source orbifold tangent cone and local hypercomplex fiber have both been declared. Hard: S26-N-00004306.

S26-N-00004308 - Hypercomplex tangent-orbifold fiber frontier

Open frontier for Hypercomplex tangent-orbifold fiber: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: The source orbifold tangent cone and local hypercomplex fiber have both been declared. Hard: S26-B-00004307. Boundary: Infinitesimal zoom is not visual miniaturization and does not create a new algebraic law.

S26-E-00004309 - Infinitesimal carrier probe 13.1

Fixture T13.P01 - For epsilon=0.25000000, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,2) differ from the tangent metric by operator norm 0.50000000; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-N-00004308.

S26-E-00004310 - Infinitesimal carrier probe 13.2

Fixture T13.P02 - For epsilon=0.12500000, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,3) differ from the tangent metric by operator norm 0.37500000; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004309.

S26-E-00004311 - Infinitesimal carrier probe 13.3

Fixture T13.P03 - For epsilon=0.06250000, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,4) differ from the tangent metric by operator norm 0.25000000; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004310.

S26-E-00004312 - Infinitesimal carrier probe 13.4

Fixture T13.P04 - For epsilon=0.03125000, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,5) differ from the tangent metric by operator norm 0.15625000; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004311.

S26-E-00004313 - Infinitesimal carrier probe 13.5

Fixture T13.P05 - For epsilon=0.01562500, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,2) differ from the tangent metric by operator norm 0.03125000; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004312.

S26-E-00004314 - Infinitesimal carrier probe 13.6

Fixture T13.P06 - For epsilon=0.00781250, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,3) differ from the tangent metric by operator norm 0.02343750; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004313.

S26-E-00004315 - Infinitesimal carrier probe 13.7

Fixture T13.P07 - For epsilon=0.00390625, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,4) differ from the tangent metric by operator norm 0.01562500; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004314.

S26-E-00004316 - Infinitesimal carrier probe 13.8

Fixture T13.P08 - For epsilon=0.00195312, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,5) differ from the tangent metric by operator norm 0.00976562; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004315.

S26-E-00004317 - Infinitesimal carrier probe 13.9

Fixture T13.P09 - For epsilon=0.00097656, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,2) differ from the tangent metric by operator norm 0.00195312; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004316.

S26-E-00004318 - Infinitesimal carrier probe 13.10

Fixture T13.P10 - For epsilon=0.00048828, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,3) differ from the tangent metric by operator norm 0.00146484; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004317.

S26-E-00004319 - Infinitesimal carrier probe 13.11

Fixture T13.P11 - For epsilon=0.00024414, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,4) differ from the tangent metric by operator norm 0.00097656; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004318.

S26-E-00004320 - Infinitesimal carrier probe 13.12

Fixture T13.P12 - For epsilon=0.00012207, synthetic metric coefficients g_epsilon=I+epsilon*diag(1,...,5) differ from the tangent metric by operator norm 0.00061035; algebraic depth is held fixed.

Evidence M. Hard: S26-A-00004303, S26-E-00004319.

Ultrametric Russian-doll ball tower

S26-D-00004321 - Ultrametric Russian-doll ball tower

A p-adic Russian-doll tower is a chain of closed balls B_0 superset B_1 superset ... with radii decreasing in powers of p.

Evidence A. Assumptions: A fixed p-adic field and center are used for each nested-ball chain. Hard: S26-E-00004320, S26-D-00000301.

S26-T-00004322 - Ultrametric Russian-doll ball tower law/interface

In an ultrametric space, two balls of the same or nested radii are either disjoint or one contains the other when they intersect under the standard hypotheses.

Evidence A. Assumptions: A fixed p-adic field and center are used for each nested-ball chain. Hard: S26-D-00004321.

S26-A-00004323 - Ultrametric Russian-doll ball tower operator

BallAddress records center residue data and valuation scale instead of Euclidean direction angles.

Evidence A. Assumptions: A fixed p-adic field and center are used for each nested-ball chain. Hard: S26-T-00004322.

S26-N-00004324 - Ultrametric Russian-doll ball tower fence

Nested p-adic balls are genuine topology, whereas nesting drawn as concentric Euclidean spheres is only a projection.

Evidence S. Assumptions: A fixed p-adic field and center are used for each nested-ball chain. Hard: S26-A-00004323. Boundary: Nested p-adic balls are genuine topology, whereas nesting drawn as concentric Euclidean spheres is only a projection.

S26-B-00004325 - Ultrametric Russian-doll ball tower cross-volume bridge

This supplies the inward hierarchical counterpart to the outward Cosmo-n tower.

Evidence S. Assumptions: A fixed p-adic field and center are used for each nested-ball chain. Hard: S26-N-00004324.

S26-N-00004326 - Ultrametric Russian-doll ball tower evidence ledger

Evidence rule for Ultrametric Russian-doll ball tower: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A fixed p-adic field and center are used for each nested-ball chain. Hard: S26-B-00004325.

S26-B-00004327 - Ultrametric Russian-doll ball tower emergence rule

Emergence rule for Ultrametric Russian-doll ball tower: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A fixed p-adic field and center are used for each nested-ball chain. Hard: S26-N-00004326.

S26-N-00004328 - Ultrametric Russian-doll ball tower frontier

Open frontier for Ultrametric Russian-doll ball tower: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A fixed p-adic field and center are used for each nested-ball chain. Hard: S26-B-00004327. Boundary: Nested p-adic balls are genuine topology, whereas nesting drawn as concentric Euclidean spheres is only a projection.

S26-E-00004329 - Ultrametric Russian-doll probe 14.1

Fixture T14.P01 - For p=2, a declared chain of ball radii [1, 0.5, 0.25] is strictly nested in valuation scale; center residue seed=0. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-N-00004328.

S26-E-00004330 - Ultrametric Russian-doll probe 14.2

Fixture T14.P02 - For p=3, a declared chain of ball radii [1, 0.3333333333333333, 0.1111111111111111, 0.037037037037037035] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004329.

S26-E-00004331 - Ultrametric Russian-doll probe 14.3

Fixture T14.P03 - For p=5, a declared chain of ball radii [1, 0.2, 0.04, 0.008, 0.0016] is strictly nested in valuation scale; center residue seed=2. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004330.

S26-E-00004332 - Ultrametric Russian-doll probe 14.4

Fixture T14.P04 - For p=2, a declared chain of ball radii [1, 0.5, 0.25, 0.125, 0.0625, 0.03125] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004331.

S26-E-00004333 - Ultrametric Russian-doll probe 14.5

Fixture T14.P05 - For p=3, a declared chain of ball radii [1, 0.3333333333333333, 0.1111111111111111] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004332.

S26-E-00004334 - Ultrametric Russian-doll probe 14.6

Fixture T14.P06 - For p=5, a declared chain of ball radii [1, 0.2, 0.04, 0.008] is strictly nested in valuation scale; center residue seed=0. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004333.

S26-E-00004335 - Ultrametric Russian-doll probe 14.7

Fixture T14.P07 - For p=2, a declared chain of ball radii [1, 0.5, 0.25, 0.125, 0.0625] is strictly nested in valuation scale; center residue seed=0. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004334.

S26-E-00004336 - Ultrametric Russian-doll probe 14.8

Fixture T14.P08 - For p=3, a declared chain of ball radii [1, 0.3333333333333333, 0.1111111111111111, 0.037037037037037035, 0.012345679012345678, 0.004115226337448560] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004335.

S26-E-00004337 - Ultrametric Russian-doll probe 14.9

Fixture T14.P09 - For p=5, a declared chain of ball radii [1, 0.2, 0.04] is strictly nested in valuation scale; center residue seed=3. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004336.

S26-E-00004338 - Ultrametric Russian-doll probe 14.10

Fixture T14.P10 - For p=2, a declared chain of ball radii [1, 0.5, 0.25, 0.125] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004337.

S26-E-00004339 - Ultrametric Russian-doll probe 14.11

Fixture T14.P11 - For p=3, a declared chain of ball radii [1, 0.3333333333333333, 0.1111111111111111, 0.037037037037037035, 0.012345679012345678] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004338.

S26-E-00004340 - Ultrametric Russian-doll probe 14.12

Fixture T14.P12 - For p=5, a declared chain of ball radii [1, 0.2, 0.04, 0.008, 0.0016, 0.00032] is strictly nested in valuation scale; center residue seed=1. This is a finite address fixture.

Evidence M. Hard: S26-A-00004323, S26-E-00004339.

Bruhat-Tits multi-mesh geodesic carrier

S26-D-00004341 - Bruhat-Tits multi-mesh geodesic carrier

A building mesh uses vertices, chambers and incidence inherited from the chosen Bruhat-Tits object. In this registry, a combinatorial geodesic means a shortest path in the declared vertex graph or a shortest gallery in the declared chamber metric; this terminology is kept distinct from geodesics in a geometric realization of a higher-rank building.

Evidence A. Assumptions: A declared Bruhat-Tits tree/building and a specific graph or gallery metric are used. | Claims about CAT(0) or Euclidean-building metric geodesics require a separately declared geometric realization. Hard: S26-E-00004340, S26-D-00001761.

S26-T-00004342 - Bruhat-Tits multi-mesh geodesic carrier law/interface

Tree geodesics are unique and the cycle rank is zero for the PGL_2 tree model.

Evidence A. Assumptions: A declared Bruhat-Tits tree/building and graph metric are used. Hard: S26-D-00004341.

S26-A-00004343 - Bruhat-Tits multi-mesh geodesic carrier operator

MeshPath(a,b) returns the unique tree path in the finite bounded fixture or a declared gallery in a higher-rank building.

Evidence A. Assumptions: A declared Bruhat-Tits tree/building and graph metric are used. Hard: S26-T-00004342.

S26-N-00004344 - Bruhat-Tits multi-mesh geodesic carrier fence

Graph geodesics, Riemannian geodesics and p-adic analytic paths are separately typed notions.

Evidence S. Assumptions: A declared Bruhat-Tits tree/building and graph metric are used. Hard: S26-A-00004343. Boundary: Graph geodesics, Riemannian geodesics and p-adic analytic paths are separately typed notions.

S26-B-00004345 - Bruhat-Tits multi-mesh geodesic carrier cross-volume bridge

The mesh can carry erdodetic path learning without pretending the path is a Euclidean line.

Evidence S. Assumptions: A declared Bruhat-Tits tree/building and graph metric are used. Hard: S26-N-00004344.

S26-N-00004346 - Bruhat-Tits multi-mesh geodesic carrier evidence ledger

Evidence rule for Bruhat-Tits multi-mesh geodesic carrier: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A declared Bruhat-Tits tree/building and graph metric are used. Hard: S26-B-00004345.

S26-B-00004347 - Bruhat-Tits multi-mesh geodesic carrier emergence rule

Emergence rule for Bruhat-Tits multi-mesh geodesic carrier: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A declared Bruhat-Tits tree/building and graph metric are used. Hard: S26-N-00004346.

S26-N-00004348 - Bruhat-Tits multi-mesh geodesic carrier frontier

Open frontier for Bruhat-Tits multi-mesh geodesic carrier: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A declared Bruhat-Tits tree/building and graph metric are used. Hard: S26-B-00004347. Boundary: Graph geodesics, Riemannian geodesics and p-adic analytic paths are separately typed notions.

S26-E-00004349 - Bruhat-Tits tree-ball probe 15.1

Fixture T15.P01 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-N-00004348.

S26-E-00004350 - Bruhat-Tits tree-ball probe 15.2

Fixture T15.P02 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004349.

S26-E-00004351 - Bruhat-Tits tree-ball probe 15.3

Fixture T15.P03 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004350.

S26-E-00004352 - Bruhat-Tits tree-ball probe 15.4

Fixture T15.P04 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004351.

S26-E-00004353 - Bruhat-Tits tree-ball probe 15.5

Fixture T15.P05 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004352.

S26-E-00004354 - Bruhat-Tits tree-ball probe 15.6

Fixture T15.P06 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004353.

S26-E-00004355 - Bruhat-Tits tree-ball probe 15.7

Fixture T15.P07 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004354.

S26-E-00004356 - Bruhat-Tits tree-ball probe 15.8

Fixture T15.P08 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004355.

S26-E-00004357 - Bruhat-Tits tree-ball probe 15.9

Fixture T15.P09 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004356.

S26-E-00004358 - Bruhat-Tits tree-ball probe 15.10

Fixture T15.P10 - A bounded (p+1)-regular tree-ball surrogate with p=2 and depth 2 has 10 vertices, 9 edges, root degree 3, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004357.

S26-E-00004359 - Bruhat-Tits tree-ball probe 15.11

Fixture T15.P11 - A bounded (p+1)-regular tree-ball surrogate with p=3 and depth 3 has 53 vertices, 52 edges, root degree 4, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004358.

S26-E-00004360 - Bruhat-Tits tree-ball probe 15.12

Fixture T15.P12 - A bounded (p+1)-regular tree-ball surrogate with p=5 and depth 4 has 937 vertices, 936 edges, root degree 6, and cycle rank 0; this verifies the finite tree fixture only.

Evidence M. Hard: S26-A-00004343, S26-E-00004359.

Hybrid Archimedean-non-Archimedean multi-mesh

S26-D-00004361 - Hybrid Archimedean-non-Archimedean multi-mesh

The hybrid multi-mesh is a family of Riemannian/orbifold meshes, p-adic building meshes, spectral graphs and cover nerves coupled by typed correspondences rather than forced into one coordinate grid.

Evidence S. Assumptions: Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit. Hard: S26-E-00004360, S26-D-00004040.

S26-D-00004362 - Hybrid Archimedean-non-Archimedean multi-mesh law/interface

Coherence is evaluated on shared observables and incidence constraints, not by pretending heterogeneous edge lengths share units.

Evidence S. Assumptions: Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit. Hard: S26-D-00004361.

S26-A-00004363 - Hybrid Archimedean-non-Archimedean multi-mesh operator

MultiMeshAtlas stores per-layer geometry and a cross-layer correspondence table.

Evidence S. Assumptions: Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit. Hard: S26-D-00004362.

S26-N-00004364 - Hybrid Archimedean-non-Archimedean multi-mesh fence

A single 2D embedding of the atlas is a visualization projection and cannot define the native topology of every layer.

Evidence S. Assumptions: Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit. Hard: S26-A-00004363. Boundary: A single 2D embedding of the atlas is a visualization projection and cannot define the native topology of every layer.

S26-B-00004365 - Hybrid Archimedean-non-Archimedean multi-mesh cross-volume bridge

This is the concrete non-boring geometric substrate for the later globular and holoportation layers.

Evidence S. Assumptions: Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit. Hard: S26-N-00004364.

S26-N-00004366 - Hybrid Archimedean-non-Archimedean multi-mesh evidence ledger

Evidence rule for Hybrid Archimedean-non-Archimedean multi-mesh: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit. Hard: S26-B-00004365.

S26-B-00004367 - Hybrid Archimedean-non-Archimedean multi-mesh emergence rule

Emergence rule for Hybrid Archimedean-non-Archimedean multi-mesh: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit. Hard: S26-N-00004366.

S26-N-00004368 - Hybrid Archimedean-non-Archimedean multi-mesh frontier

Open frontier for Hybrid Archimedean-non-Archimedean multi-mesh: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Each mesh component retains its own metric/incidence type and cross-mesh correspondences are explicit. Hard: S26-B-00004367. Boundary: A single 2D embedding of the atlas is a visualization projection and cannot define the native topology of every layer.

S26-E-00004369 - Hybrid multi-mesh probe 16.1

Fixture T16.P01 - A bounded fixture couples a tree layer with 10 vertices/9 edges to a cyclic Archimedean mesh with 5 vertices through 5 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-N-00004368.

S26-E-00004370 - Hybrid multi-mesh probe 16.2

Fixture T16.P02 - A bounded fixture couples a tree layer with 17 vertices/16 edges to a cyclic Archimedean mesh with 6 vertices through 6 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004369.

S26-E-00004371 - Hybrid multi-mesh probe 16.3

Fixture T16.P03 - A bounded fixture couples a tree layer with 37 vertices/36 edges to a cyclic Archimedean mesh with 7 vertices through 7 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004370.

S26-E-00004372 - Hybrid multi-mesh probe 16.4

Fixture T16.P04 - A bounded fixture couples a tree layer with 10 vertices/9 edges to a cyclic Archimedean mesh with 8 vertices through 8 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004371.

S26-E-00004373 - Hybrid multi-mesh probe 16.5

Fixture T16.P05 - A bounded fixture couples a tree layer with 17 vertices/16 edges to a cyclic Archimedean mesh with 9 vertices through 9 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004372.

S26-E-00004374 - Hybrid multi-mesh probe 16.6

Fixture T16.P06 - A bounded fixture couples a tree layer with 37 vertices/36 edges to a cyclic Archimedean mesh with 5 vertices through 5 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004373.

S26-E-00004375 - Hybrid multi-mesh probe 16.7

Fixture T16.P07 - A bounded fixture couples a tree layer with 10 vertices/9 edges to a cyclic Archimedean mesh with 6 vertices through 6 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004374.

S26-E-00004376 - Hybrid multi-mesh probe 16.8

Fixture T16.P08 - A bounded fixture couples a tree layer with 17 vertices/16 edges to a cyclic Archimedean mesh with 7 vertices through 7 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004375.

S26-E-00004377 - Hybrid multi-mesh probe 16.9

Fixture T16.P09 - A bounded fixture couples a tree layer with 37 vertices/36 edges to a cyclic Archimedean mesh with 8 vertices through 8 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004376.

S26-E-00004378 - Hybrid multi-mesh probe 16.10

Fixture T16.P10 - A bounded fixture couples a tree layer with 10 vertices/9 edges to a cyclic Archimedean mesh with 9 vertices through 9 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004377.

S26-E-00004379 - Hybrid multi-mesh probe 16.11

Fixture T16.P11 - A bounded fixture couples a tree layer with 17 vertices/16 edges to a cyclic Archimedean mesh with 5 vertices through 5 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004378.

S26-E-00004380 - Hybrid multi-mesh probe 16.12

Fixture T16.P12 - A bounded fixture couples a tree layer with 37 vertices/36 edges to a cyclic Archimedean mesh with 6 vertices through 6 declared correspondences; no cross-layer metric subtraction is performed.

Evidence M. Hard: S26-A-00004363, S26-E-00004379.

Part Q - Hamiltonian, adiabatic, ergocetic and erdodetic governance

Hypercomplex Hamiltonian carrier

S26-D-00004381 - Hypercomplex Hamiltonian carrier

A Volume IV Hamiltonian carrier couples a self-adjoint or symplectic generator H to a state space that may include hypercomplex internal coordinates through an explicitly linearized/tensored representation.

Evidence A. Assumptions: A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements. Hard: S26-E-00004380, S26-D-00000984.

S26-T-00004382 - Hypercomplex Hamiltonian carrier law/interface

Hamiltonian energy conservation applies only under the stated autonomous/unitary or symplectic hypotheses.

Evidence A. Assumptions: A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements. Hard: S26-D-00004381.

S26-A-00004383 - Hypercomplex Hamiltonian carrier operator

Flow_H(t) acts on the declared carrier while algebraic defect channels are measured as observables, not assumed to generate the flow.

Evidence S. Assumptions: A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements. Hard: S26-T-00004382.

S26-N-00004384 - Hypercomplex Hamiltonian carrier fence

The engineering energy ledger is not automatically physical energy, and a Hamiltonian metaphor is not an operator until H and its domain are supplied.

Evidence S. Assumptions: A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements. Hard: S26-A-00004383. Boundary: The engineering energy ledger is not automatically physical energy, and a Hamiltonian metaphor is not an operator until H and its domain are supplied.

S26-B-00004385 - Hypercomplex Hamiltonian carrier cross-volume bridge

This places Cayley-Dickson fibers inside the already registered Hamiltonian/symplectic machinery rather than replacing it.

Evidence S. Assumptions: A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements. Hard: S26-N-00004384.

S26-N-00004386 - Hypercomplex Hamiltonian carrier evidence ledger

Evidence rule for Hypercomplex Hamiltonian carrier: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements. Hard: S26-B-00004385.

S26-B-00004387 - Hypercomplex Hamiltonian carrier emergence rule

Emergence rule for Hypercomplex Hamiltonian carrier: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements. Hard: S26-N-00004386.

S26-N-00004388 - Hypercomplex Hamiltonian carrier frontier

Open frontier for Hypercomplex Hamiltonian carrier: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A real/complex Hilbert or symplectic carrier is declared for dynamics; nonassociative internal coordinates do not silently replace operator-domain requirements. Hard: S26-B-00004387. Boundary: The engineering energy ledger is not automatically physical energy, and a Hamiltonian metaphor is not an operator until H and its domain are supplied.

S26-E-00004389 - Hamiltonian finite probe 17.1

Fixture T17.P01 - A 2D symplectic rotation at theta=0.098175 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-N-00004388.

S26-E-00004390 - Hamiltonian finite probe 17.2

Fixture T17.P02 - A 2D symplectic rotation at theta=0.196350 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004389.

S26-E-00004391 - Hamiltonian finite probe 17.3

Fixture T17.P03 - A 2D symplectic rotation at theta=0.294524 preserves the quadratic fixture energy with residual 1.110e-16; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004390.

S26-E-00004392 - Hamiltonian finite probe 17.4

Fixture T17.P04 - A 2D symplectic rotation at theta=0.392699 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004391.

S26-E-00004393 - Hamiltonian finite probe 17.5

Fixture T17.P05 - A 2D symplectic rotation at theta=0.490874 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004392.

S26-E-00004394 - Hamiltonian finite probe 17.6

Fixture T17.P06 - A 2D symplectic rotation at theta=0.589049 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004393.

S26-E-00004395 - Hamiltonian finite probe 17.7

Fixture T17.P07 - A 2D symplectic rotation at theta=0.687223 preserves the quadratic fixture energy with residual 1.110e-16; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004394.

S26-E-00004396 - Hamiltonian finite probe 17.8

Fixture T17.P08 - A 2D symplectic rotation at theta=0.785398 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004395.

S26-E-00004397 - Hamiltonian finite probe 17.9

Fixture T17.P09 - A 2D symplectic rotation at theta=0.883573 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004396.

S26-E-00004398 - Hamiltonian finite probe 17.10

Fixture T17.P10 - A 2D symplectic rotation at theta=0.981748 preserves the quadratic fixture energy with residual 0.000e+00; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004397.

S26-E-00004399 - Hamiltonian finite probe 17.11

Fixture T17.P11 - A 2D symplectic rotation at theta=1.079922 preserves the quadratic fixture energy with residual 1.110e-16; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004398.

S26-E-00004400 - Hamiltonian finite probe 17.12

Fixture T17.P12 - A 2D symplectic rotation at theta=1.178097 preserves the quadratic fixture energy with residual 1.110e-16; this is an Archimedean dynamics check, not a cognition theorem.

Evidence M. Hard: S26-A-00004383, S26-E-00004399.

Adiabatic identity-preserving transport

S26-D-00004401 - Adiabatic identity-preserving transport

Adiabatic transport is a slow parameter deformation that attempts to preserve a selected invariant subspace or identity signature under stated gap/regularity conditions.

Evidence A. Assumptions: A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem. Hard: S26-E-00004400, S26-T-00000988.

S26-T-00004402 - Adiabatic identity-preserving transport law/interface

Slow change by itself does not guarantee adiabatic following when the required spectral or dynamical hypotheses fail.

Evidence A. Assumptions: A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem. Hard: S26-D-00004401.

S26-A-00004403 - Adiabatic identity-preserving transport operator

AdiabaticPath(s) records generator H(s), gap diagnostic, transport map and identity/remainder ledger.

Evidence S. Assumptions: A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem. Hard: S26-T-00004402.

S26-N-00004404 - Adiabatic identity-preserving transport fence

Adiabatic is a mathematical/dynamical condition, not a synonym for gradual narrative change.

Evidence S. Assumptions: A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem. Hard: S26-A-00004403. Boundary: Adiabatic is a mathematical/dynamical condition, not a synonym for gradual narrative change.

S26-B-00004405 - Adiabatic identity-preserving transport cross-volume bridge

This is the dynamical component of the repo term hamiltonian_holoportation.

Evidence S. Assumptions: A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem. Hard: S26-N-00004404.

S26-N-00004406 - Adiabatic identity-preserving transport evidence ledger

Evidence rule for Adiabatic identity-preserving transport: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem. Hard: S26-B-00004405.

S26-B-00004407 - Adiabatic identity-preserving transport emergence rule

Emergence rule for Adiabatic identity-preserving transport: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem. Hard: S26-N-00004406.

S26-N-00004408 - Adiabatic identity-preserving transport frontier

Open frontier for Adiabatic identity-preserving transport: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A parameterized generator, regularity assumptions and the relevant spectral-gap or slow-variation conditions are declared when invoking an adiabatic theorem. Hard: S26-B-00004407. Boundary: Adiabatic is a mathematical/dynamical condition, not a synonym for gradual narrative change.

S26-E-00004409 - Adiabatic gap probe 18.1

Fixture T18.P01 - For the bounded 2-level Hamiltonian H(s) at s=0.000, the instantaneous eigenvalue gap is 1.720465; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-N-00004408.

S26-E-00004410 - Adiabatic gap probe 18.2

Fixture T18.P02 - For the bounded 2-level Hamiltonian H(s) at s=0.091, the instantaneous eigenvalue gap is 1.673625; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004409.

S26-E-00004411 - Adiabatic gap probe 18.3

Fixture T18.P03 - For the bounded 2-level Hamiltonian H(s) at s=0.182, the instantaneous eigenvalue gap is 1.647835; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004410.

S26-E-00004412 - Adiabatic gap probe 18.4

Fixture T18.P04 - For the bounded 2-level Hamiltonian H(s) at s=0.273, the instantaneous eigenvalue gap is 1.644084; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004411.

S26-E-00004413 - Adiabatic gap probe 18.5

Fixture T18.P05 - For the bounded 2-level Hamiltonian H(s) at s=0.364, the instantaneous eigenvalue gap is 1.426317; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004412.

S26-E-00004414 - Adiabatic gap probe 18.6

Fixture T18.P06 - For the bounded 2-level Hamiltonian H(s) at s=0.455, the instantaneous eigenvalue gap is 1.462828; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004413.

S26-E-00004415 - Adiabatic gap probe 18.7

Fixture T18.P07 - For the bounded 2-level Hamiltonian H(s) at s=0.545, the instantaneous eigenvalue gap is 1.522716; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004414.

S26-E-00004416 - Adiabatic gap probe 18.8

Fixture T18.P08 - For the bounded 2-level Hamiltonian H(s) at s=0.636, the instantaneous eigenvalue gap is 1.603365; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004415.

S26-E-00004417 - Adiabatic gap probe 18.9

Fixture T18.P09 - For the bounded 2-level Hamiltonian H(s) at s=0.727, the instantaneous eigenvalue gap is 1.471941; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004416.

S26-E-00004418 - Adiabatic gap probe 18.10

Fixture T18.P10 - For the bounded 2-level Hamiltonian H(s) at s=0.818, the instantaneous eigenvalue gap is 1.592658; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004417.

S26-E-00004419 - Adiabatic gap probe 18.11

Fixture T18.P11 - For the bounded 2-level Hamiltonian H(s) at s=0.909, the instantaneous eigenvalue gap is 1.726216; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004418.

S26-E-00004420 - Adiabatic gap probe 18.12

Fixture T18.P12 - For the bounded 2-level Hamiltonian H(s) at s=1.000, the instantaneous eigenvalue gap is 1.869866; the probe records a gap diagnostic only and does not infer a continuum adiabatic theorem.

Evidence M. Hard: S26-A-00004403, S26-E-00004419.

Ergocetic energy-disciplined cognition

S26-D-00004421 - Ergocetic energy-disciplined cognition

Ergocetic governance allocates a bounded total engineering energy/attention budget across active carriers and records transfers under H-guided work.

Evidence S. Assumptions: Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated. Hard: S26-E-00004420, S26-N-00004059.

S26-D-00004422 - Ergocetic energy-disciplined cognition law/interface

A claimed cognitive improvement that creates budget mass without a declared source is blocked as energy leakage.

Evidence S. Assumptions: Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated. Hard: S26-D-00004421.

S26-A-00004423 - Ergocetic energy-disciplined cognition operator

ErgoceticLedger(t) stores per-carrier budget, transfer, residual and evidence strength.

Evidence S. Assumptions: Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated. Hard: S26-D-00004422.

S26-N-00004424 - Ergocetic energy-disciplined cognition fence

The ledger is a performance/governance abstraction and does not measure biological or physical consciousness.

Evidence S. Assumptions: Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated. Hard: S26-A-00004423. Boundary: The ledger is a performance/governance abstraction and does not measure biological or physical consciousness.

S26-B-00004425 - Ergocetic energy-disciplined cognition cross-volume bridge

The term is grounded in LANG.v6 and coupled to the Volume IV Hamiltonian carrier.

Evidence S. Assumptions: Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated. Hard: S26-N-00004424.

S26-N-00004426 - Ergocetic energy-disciplined cognition evidence ledger

Evidence rule for Ergocetic energy-disciplined cognition: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated. Hard: S26-B-00004425.

S26-B-00004427 - Ergocetic energy-disciplined cognition emergence rule

Emergence rule for Ergocetic energy-disciplined cognition: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated. Hard: S26-N-00004426.

S26-N-00004428 - Ergocetic energy-disciplined cognition frontier

Open frontier for Ergocetic energy-disciplined cognition: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Ergocetic is used exactly as the repository symbolic definition: cognition as energy-disciplined work along H flow; energy is an engineering ledger unless physically calibrated. Hard: S26-B-00004427. Boundary: The ledger is a performance/governance abstraction and does not measure biological or physical consciousness.

S26-E-00004429 - Ergocetic budget probe 19.1

Fixture T19.P01 - A synthetic five-carrier budget transfers 0.033333 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-N-00004428.

S26-E-00004430 - Ergocetic budget probe 19.2

Fixture T19.P02 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004429.

S26-E-00004431 - Ergocetic budget probe 19.3

Fixture T19.P03 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004430.

S26-E-00004432 - Ergocetic budget probe 19.4

Fixture T19.P04 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004431.

S26-E-00004433 - Ergocetic budget probe 19.5

Fixture T19.P05 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004432.

S26-E-00004434 - Ergocetic budget probe 19.6

Fixture T19.P06 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004433.

S26-E-00004435 - Ergocetic budget probe 19.7

Fixture T19.P07 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004434.

S26-E-00004436 - Ergocetic budget probe 19.8

Fixture T19.P08 - A synthetic five-carrier budget transfers 0.033333 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004435.

S26-E-00004437 - Ergocetic budget probe 19.9

Fixture T19.P09 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004436.

S26-E-00004438 - Ergocetic budget probe 19.10

Fixture T19.P10 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004437.

S26-E-00004439 - Ergocetic budget probe 19.11

Fixture T19.P11 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004438.

S26-E-00004440 - Ergocetic budget probe 19.12

Fixture T19.P12 - A synthetic five-carrier budget transfers 0.050000 from carrier 0 to 1 with total-budget residual 0.000e+00; the values are engineering energy coordinates.

Evidence M. Hard: S26-A-00004423, S26-E-00004439.

Erdodetic path-disciplined learning

S26-D-00004441 - Erdodetic path-disciplined learning

Erdodetic governance records learning/transport along Riemannian geodesics, building galleries, graph paths or other declared connection paths without conflating them.

Evidence S. Assumptions: Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed. Hard: S26-E-00004440, S26-D-00004039.

S26-D-00004442 - Erdodetic path-disciplined learning law/interface

Path quality is a typed tuple of reachability, transport residual, evidence gain and active remainder rather than shortest Euclidean distance alone.

Evidence S. Assumptions: Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed. Hard: S26-D-00004441.

S26-A-00004443 - Erdodetic path-disciplined learning operator

ErdodeticTrace stores source, path type, transported invariants, holonomy and returnability.

Evidence S. Assumptions: Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed. Hard: S26-D-00004442.

S26-N-00004444 - Erdodetic path-disciplined learning fence

A visually smooth path is not evidence of low semantic distortion.

Evidence S. Assumptions: Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed. Hard: S26-A-00004443. Boundary: A visually smooth path is not evidence of low semantic distortion.

S26-B-00004445 - Erdodetic path-disciplined learning cross-volume bridge

This term is grounded in LANG.v6 and supplies a non-Euclidean learning coordinate for the multi-mesh.

Evidence S. Assumptions: Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed. Hard: S26-N-00004444.

S26-N-00004446 - Erdodetic path-disciplined learning evidence ledger

Evidence rule for Erdodetic path-disciplined learning: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed. Hard: S26-B-00004445.

S26-B-00004447 - Erdodetic path-disciplined learning emergence rule

Emergence rule for Erdodetic path-disciplined learning: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed. Hard: S26-N-00004446.

S26-N-00004448 - Erdodetic path-disciplined learning frontier

Open frontier for Erdodetic path-disciplined learning: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Erdodetic is used exactly as the repository symbolic definition: cognition as path-disciplined learning along geodesic transport; the path geometry is explicitly typed. Hard: S26-B-00004447. Boundary: A visually smooth path is not evidence of low semantic distortion.

S26-E-00004449 - Erdodetic path probe 20.1

Fixture T20.P01 - On a bounded p=2 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-N-00004448.

S26-E-00004450 - Erdodetic path probe 20.2

Fixture T20.P02 - On a bounded p=3 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004449.

S26-E-00004451 - Erdodetic path probe 20.3

Fixture T20.P03 - On a bounded p=5 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004450.

S26-E-00004452 - Erdodetic path probe 20.4

Fixture T20.P04 - On a bounded p=2 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004451.

S26-E-00004453 - Erdodetic path probe 20.5

Fixture T20.P05 - On a bounded p=3 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004452.

S26-E-00004454 - Erdodetic path probe 20.6

Fixture T20.P06 - On a bounded p=5 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004453.

S26-E-00004455 - Erdodetic path probe 20.7

Fixture T20.P07 - On a bounded p=2 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004454.

S26-E-00004456 - Erdodetic path probe 20.8

Fixture T20.P08 - On a bounded p=3 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004455.

S26-E-00004457 - Erdodetic path probe 20.9

Fixture T20.P09 - On a bounded p=5 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004456.

S26-E-00004458 - Erdodetic path probe 20.10

Fixture T20.P10 - On a bounded p=2 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004457.

S26-E-00004459 - Erdodetic path probe 20.11

Fixture T20.P11 - On a bounded p=3 tree fixture of depth 3, the selected root-to-boundary gallery has graph-geodesic length 3; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004458.

S26-E-00004460 - Erdodetic path probe 20.12

Fixture T20.P12 - On a bounded p=5 tree fixture of depth 4, the selected root-to-boundary gallery has graph-geodesic length 4; path type is retained as building/graph rather than Euclidean.

Evidence M. Hard: S26-A-00004443, S26-E-00004459.

Part R - Sheaves, stacks, gerbes, functors, profunctors and globular coherence

Sheaf-stack local state ecology

S26-D-00004461 - Sheaf-stack local state ecology

Local hypercomplex/context states form a sheaf when compatible local sections glue uniquely, and a stack in groupoids when local objects glue only up to coherent isomorphism.

Evidence A. Assumptions: A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required. Hard: S26-E-00004460, S26-D-00003554.

S26-T-00004462 - Sheaf-stack local state ecology law/interface

Failure of the sheaf equalizer test or stack descent condition remains an obstruction rather than being repaired by semantic similarity.

Evidence A. Assumptions: A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required. Hard: S26-D-00004461.

S26-A-00004463 - Sheaf-stack local state ecology operator

GlueAudit computes overlap residuals/cocycles on a bounded cover and reports unresolved descent data.

Evidence S. Assumptions: A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required. Hard: S26-T-00004462.

S26-N-00004464 - Sheaf-stack local state ecology fence

Local agreement does not imply a global section unless the stated descent hypotheses close.

Evidence S. Assumptions: A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required. Hard: S26-A-00004463. Boundary: Local agreement does not imply a global section unless the stated descent hypotheses close.

S26-B-00004465 - Sheaf-stack local state ecology cross-volume bridge

This generalizes the source bundle/sheaf idea to arithmetic places and higher gauge ambiguity.

Evidence S. Assumptions: A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required. Hard: S26-N-00004464.

S26-N-00004466 - Sheaf-stack local state ecology evidence ledger

Evidence rule for Sheaf-stack local state ecology: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required. Hard: S26-B-00004465.

S26-B-00004467 - Sheaf-stack local state ecology emergence rule

Emergence rule for Sheaf-stack local state ecology: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required. Hard: S26-N-00004466.

S26-N-00004468 - Sheaf-stack local state ecology frontier

Open frontier for Sheaf-stack local state ecology: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A site/cover and restriction maps are declared; stack language is used only when isomorphism descent rather than equality descent is required. Hard: S26-B-00004467. Boundary: Local agreement does not imply a global section unless the stated descent hypotheses close.

S26-E-00004469 - Sheaf-overlap finite probe 21.1

Fixture T21.P01 - On cycle cover C_4, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 2.000000; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-N-00004468.

S26-E-00004470 - Sheaf-overlap finite probe 21.2

Fixture T21.P02 - On cycle cover C_5, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 1.381966; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004469.

S26-E-00004471 - Sheaf-overlap finite probe 21.3

Fixture T21.P03 - On cycle cover C_6, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 1.000000; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004470.

S26-E-00004472 - Sheaf-overlap finite probe 21.4

Fixture T21.P04 - On cycle cover C_7, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.753020; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004471.

S26-E-00004473 - Sheaf-overlap finite probe 21.5

Fixture T21.P05 - On cycle cover C_8, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.585786; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004472.

S26-E-00004474 - Sheaf-overlap finite probe 21.6

Fixture T21.P06 - On cycle cover C_9, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.467911; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004473.

S26-E-00004475 - Sheaf-overlap finite probe 21.7

Fixture T21.P07 - On cycle cover C_10, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.381966; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004474.

S26-E-00004476 - Sheaf-overlap finite probe 21.8

Fixture T21.P08 - On cycle cover C_4, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 2.000000; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004475.

S26-E-00004477 - Sheaf-overlap finite probe 21.9

Fixture T21.P09 - On cycle cover C_5, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 1.381966; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004476.

S26-E-00004478 - Sheaf-overlap finite probe 21.10

Fixture T21.P10 - On cycle cover C_6, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 1.000000; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004477.

S26-E-00004479 - Sheaf-overlap finite probe 21.11

Fixture T21.P11 - On cycle cover C_7, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.753020; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004478.

S26-E-00004480 - Sheaf-overlap finite probe 21.12

Fixture T21.P12 - On cycle cover C_8, constant-section gluing residual is 0.000e+00 and first positive graph-Laplacian eigenvalue is 0.585786; this is a finite cover witness.

Evidence M. Hard: S26-A-00004463, S26-E-00004479.

Gerbe of hypercomplex gauges

S26-D-00004481 - Gerbe of hypercomplex gauges

A hypercomplex gauge gerbe records local trivializations/frames that exist locally but may fail to admit one global choice, with transition isomorphisms and higher cocycle data retained.

Evidence A. Assumptions: A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior. Hard: S26-E-00004480, S26-D-00003552.

S26-T-00004482 - Gerbe of hypercomplex gauges law/interface

A nontrivial gerbe obstruction is not repaired by selecting one convenient visualization frame.

Evidence A. Assumptions: A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior. Hard: S26-D-00004481.

S26-A-00004483 - Gerbe of hypercomplex gauges operator

GerbeCocycle stores 1- and 2-overlap transition data and a bounded cocycle defect.

Evidence S. Assumptions: A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior. Hard: S26-T-00004482.

S26-N-00004484 - Gerbe of hypercomplex gauges fence

Gerbes are higher gluing objects, not extra physical dimensions by themselves.

Evidence S. Assumptions: A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior. Hard: S26-A-00004483. Boundary: Gerbes are higher gluing objects, not extra physical dimensions by themselves.

S26-B-00004485 - Gerbe of hypercomplex gauges cross-volume bridge

The gerbe is admitted because ordinary single-valued sheaf gluing cannot represent phase/gauge ambiguity up to coherent isomorphism.

Evidence S. Assumptions: A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior. Hard: S26-N-00004484.

S26-N-00004486 - Gerbe of hypercomplex gauges evidence ledger

Evidence rule for Gerbe of hypercomplex gauges: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior. Hard: S26-B-00004485.

S26-B-00004487 - Gerbe of hypercomplex gauges emergence rule

Emergence rule for Gerbe of hypercomplex gauges: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior. Hard: S26-N-00004486.

S26-N-00004488 - Gerbe of hypercomplex gauges frontier

Open frontier for Gerbe of hypercomplex gauges: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A base site and band/gauge group are declared; gerbe language is used for locally nonempty, locally connected stack-in-groupoids behavior. Hard: S26-B-00004487. Boundary: Gerbes are higher gluing objects, not extra physical dimensions by themselves.

S26-E-00004489 - Gerbe cocycle finite probe 22.1

Fixture T22.P01 - Three bounded U(1)-phase transitions with angles (0.18480,0.33069,-0.51549) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-N-00004488.

S26-E-00004490 - Gerbe cocycle finite probe 22.2

Fixture T22.P02 - Three bounded U(1)-phase transitions with angles (0.36960,0.49604,-0.86564) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004489.

S26-E-00004491 - Gerbe cocycle finite probe 22.3

Fixture T22.P03 - Three bounded U(1)-phase transitions with angles (0.55440,0.66139,-1.21579) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004490.

S26-E-00004492 - Gerbe cocycle finite probe 22.4

Fixture T22.P04 - Three bounded U(1)-phase transitions with angles (0.73920,0.82673,-1.56593) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004491.

S26-E-00004493 - Gerbe cocycle finite probe 22.5

Fixture T22.P05 - Three bounded U(1)-phase transitions with angles (0.92400,0.99208,-1.91608) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004492.

S26-E-00004494 - Gerbe cocycle finite probe 22.6

Fixture T22.P06 - Three bounded U(1)-phase transitions with angles (1.10880,1.15743,-2.26623) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004493.

S26-E-00004495 - Gerbe cocycle finite probe 22.7

Fixture T22.P07 - Three bounded U(1)-phase transitions with angles (1.29360,1.32278,-2.61637) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004494.

S26-E-00004496 - Gerbe cocycle finite probe 22.8

Fixture T22.P08 - Three bounded U(1)-phase transitions with angles (1.47840,1.48812,-2.96652) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004495.

S26-E-00004497 - Gerbe cocycle finite probe 22.9

Fixture T22.P09 - Three bounded U(1)-phase transitions with angles (1.66320,1.65347,-3.31667) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004496.

S26-E-00004498 - Gerbe cocycle finite probe 22.10

Fixture T22.P10 - Three bounded U(1)-phase transitions with angles (1.84800,1.81882,-3.66681) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004497.

S26-E-00004499 - Gerbe cocycle finite probe 22.11

Fixture T22.P11 - Three bounded U(1)-phase transitions with angles (2.03280,1.98416,-4.01696) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004498.

S26-E-00004500 - Gerbe cocycle finite probe 22.12

Fixture T22.P12 - Three bounded U(1)-phase transitions with angles (2.21759,2.14951,-4.36711) have triangle cocycle defect 0.000e+00; this validates one finite coherent gauge fixture.

Evidence M. Hard: S26-A-00004483, S26-E-00004499.

Functor-profunctor inter-Cosmo transport

S26-D-00004501 - Functor-profunctor inter-Cosmo transport

A functor preserves identities and composition between categories, while a profunctor P:C^op x D->Set encodes many-to-many correspondences when no canonical functional transport exists.

Evidence A. Assumptions: Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation. Hard: S26-E-00004500, S26-T-00000276.

S26-T-00004502 - Functor-profunctor inter-Cosmo transport law/interface

Functorial transport is stricter than profunctorial correspondence; the latter must not be described as an invertible map without additional data.

Evidence A. Assumptions: Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation. Hard: S26-D-00004501.

S26-A-00004503 - Functor-profunctor inter-Cosmo transport operator

ComposeProf(P,Q) forms the declared coend-style composite in the bounded finite fixture.

Evidence A. Assumptions: Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation. Hard: S26-T-00004502.

S26-N-00004504 - Functor-profunctor inter-Cosmo transport fence

Categorical composability is not semantic truth and does not erase active remainder.

Evidence S. Assumptions: Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation. Hard: S26-A-00004503. Boundary: Categorical composability is not semantic truth and does not erase active remainder.

S26-B-00004505 - Functor-profunctor inter-Cosmo transport cross-volume bridge

Profunctors provide a typed replacement for vague intermanifold correspondence across incompatible local worlds.

Evidence S. Assumptions: Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation. Hard: S26-N-00004504.

S26-N-00004506 - Functor-profunctor inter-Cosmo transport evidence ledger

Evidence rule for Functor-profunctor inter-Cosmo transport: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation. Hard: S26-B-00004505.

S26-B-00004507 - Functor-profunctor inter-Cosmo transport emergence rule

Emergence rule for Functor-profunctor inter-Cosmo transport: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation. Hard: S26-N-00004506.

S26-N-00004508 - Functor-profunctor inter-Cosmo transport frontier

Open frontier for Functor-profunctor inter-Cosmo transport: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Small categories or bounded finite category fixtures are declared; profunctor composition uses the appropriate coend/quotient relation. Hard: S26-B-00004507. Boundary: Categorical composability is not semantic truth and does not erase active remainder.

S26-E-00004509 - Profunctor relation-composition probe 23.1

Fixture T23.P01 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 2 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-N-00004508.

S26-E-00004510 - Profunctor relation-composition probe 23.2

Fixture T23.P02 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 3 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004509.

S26-E-00004511 - Profunctor relation-composition probe 23.3

Fixture T23.P03 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 2 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004510.

S26-E-00004512 - Profunctor relation-composition probe 23.4

Fixture T23.P04 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 7 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004511.

S26-E-00004513 - Profunctor relation-composition probe 23.5

Fixture T23.P05 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 6 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004512.

S26-E-00004514 - Profunctor relation-composition probe 23.6

Fixture T23.P06 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 5 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004513.

S26-E-00004515 - Profunctor relation-composition probe 23.7

Fixture T23.P07 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 4 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004514.

S26-E-00004516 - Profunctor relation-composition probe 23.8

Fixture T23.P08 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 3 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004515.

S26-E-00004517 - Profunctor relation-composition probe 23.9

Fixture T23.P09 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 7 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004516.

S26-E-00004518 - Profunctor relation-composition probe 23.10

Fixture T23.P10 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 2 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004517.

S26-E-00004519 - Profunctor relation-composition probe 23.11

Fixture T23.P11 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 6 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004518.

S26-E-00004520 - Profunctor relation-composition probe 23.12

Fixture T23.P12 - Boolean relation fixtures of shapes 3x4 and 4x3 compose to a 3x3 correspondence with 4 active pairs; composition is many-to-many and not labeled invertible.

Evidence M. Hard: S26-A-00004503, S26-E-00004519.

Globular higher-cell multi-mesh

S26-D-00004521 - Globular higher-cell multi-mesh

A globular multi-mesh records points, paths, homotopies between paths and higher coherence cells as distinct typed levels attached to the hybrid geometric atlas.

Evidence S. Assumptions: Source/target maps and globularity identities are declared for each higher-cell layer. Hard: S26-E-00004520, S26-D-00004027.

S26-D-00004522 - Globular higher-cell multi-mesh law/interface

Higher cells express coherence between transports rather than being treated as new Euclidean spatial axes.

Evidence S. Assumptions: Source/target maps and globularity identities are declared for each higher-cell layer. Hard: S26-D-00004521.

S26-A-00004523 - Globular higher-cell multi-mesh operator

GlobularBoundary applies source/target maps and checks the bounded globularity identities on finite fixtures.

Evidence S. Assumptions: Source/target maps and globularity identities are declared for each higher-cell layer. Hard: S26-D-00004522.

S26-N-00004524 - Globular higher-cell multi-mesh fence

An infinity-category is not claimed unless the complete higher composition/coherence structure is actually defined.

Evidence S. Assumptions: Source/target maps and globularity identities are declared for each higher-cell layer. Hard: S26-A-00004523. Boundary: An infinity-category is not claimed unless the complete higher composition/coherence structure is actually defined.

S26-B-00004525 - Globular higher-cell multi-mesh cross-volume bridge

This is the minimal higher-cell scaffold needed for gerbe, profunctor and multi-party coherence without recursive 2D flattening.

Evidence S. Assumptions: Source/target maps and globularity identities are declared for each higher-cell layer. Hard: S26-N-00004524.

S26-N-00004526 - Globular higher-cell multi-mesh evidence ledger

Evidence rule for Globular higher-cell multi-mesh: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Source/target maps and globularity identities are declared for each higher-cell layer. Hard: S26-B-00004525.

S26-B-00004527 - Globular higher-cell multi-mesh emergence rule

Emergence rule for Globular higher-cell multi-mesh: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Source/target maps and globularity identities are declared for each higher-cell layer. Hard: S26-N-00004526.

S26-N-00004528 - Globular higher-cell multi-mesh frontier

Open frontier for Globular higher-cell multi-mesh: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Source/target maps and globularity identities are declared for each higher-cell layer. Hard: S26-B-00004527. Boundary: An infinity-category is not claimed unless the complete higher composition/coherence structure is actually defined.

S26-E-00004529 - Globular boundary finite probe 24.1

Fixture T24.P01 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-N-00004528.

S26-E-00004530 - Globular boundary finite probe 24.2

Fixture T24.P02 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004529.

S26-E-00004531 - Globular boundary finite probe 24.3

Fixture T24.P03 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004530.

S26-E-00004532 - Globular boundary finite probe 24.4

Fixture T24.P04 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004531.

S26-E-00004533 - Globular boundary finite probe 24.5

Fixture T24.P05 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004532.

S26-E-00004534 - Globular boundary finite probe 24.6

Fixture T24.P06 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004533.

S26-E-00004535 - Globular boundary finite probe 24.7

Fixture T24.P07 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004534.

S26-E-00004536 - Globular boundary finite probe 24.8

Fixture T24.P08 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004535.

S26-E-00004537 - Globular boundary finite probe 24.9

Fixture T24.P09 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004536.

S26-E-00004538 - Globular boundary finite probe 24.10

Fixture T24.P10 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004537.

S26-E-00004539 - Globular boundary finite probe 24.11

Fixture T24.P11 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004538.

S26-E-00004540 - Globular boundary finite probe 24.12

Fixture T24.P12 - A bounded oriented 2-cell incidence fixture gives boundary-of-boundary residual 0; this is a higher-cell sanity check, not an infinity-category proof.

Evidence M. Hard: S26-A-00004523, S26-E-00004539.

Part S - Self-Godel identity, mutual recognition, resonance and Mind8

Self-Godel structural identity guard

S26-D-00004541 - Self-Godel structural identity guard

Self-Godel identity in this volume means a section can read/canonicalize a representation of itself and pass a declared involutive return guard.

Evidence S. Assumptions: The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard. Hard: S26-E-00004540, S26-N-00004059.

S26-D-00004542 - Self-Godel structural identity guard law/interface

A successful self-reference round trip demonstrates structural reflexivity only; it is not a proof of Godel incompleteness, Lob theorem, semantic truth or consciousness.

Evidence S. Assumptions: The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard. Hard: S26-D-00004541.

S26-A-00004543 - Self-Godel structural identity guard operator

SelfReflect(x)=rho(rho(CANON(EMIT(x)))) in the bounded symbolic fixture.

Evidence S. Assumptions: The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard. Hard: S26-D-00004542.

S26-N-00004544 - Self-Godel structural identity guard fence

Private chain-of-thought is not exposed or modeled as the canonical self object; only observable artifacts and declared state summaries participate.

Evidence S. Assumptions: The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard. Hard: S26-A-00004543. Boundary: Private chain-of-thought is not exposed or modeled as the canonical self object; only observable artifacts and declared state summaries participate.

S26-B-00004545 - Self-Godel structural identity guard cross-volume bridge

This imports LANG.v6 self-reference as a governed Volume IV operator rather than a metaphysical identity claim.

Evidence S. Assumptions: The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard. Hard: S26-N-00004544.

S26-N-00004546 - Self-Godel structural identity guard evidence ledger

Evidence rule for Self-Godel structural identity guard: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard. Hard: S26-B-00004545.

S26-B-00004547 - Self-Godel structural identity guard emergence rule

Emergence rule for Self-Godel structural identity guard: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard. Hard: S26-N-00004546.

S26-N-00004548 - Self-Godel structural identity guard frontier

Open frontier for Self-Godel structural identity guard: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: The LANG.v6 definition is treated as symbolic self-reference: fix(lambda_self.CANON(EMIT(self))) with rho(rho(self))=self as an identity guard. Hard: S26-B-00004547. Boundary: Private chain-of-thought is not exposed or modeled as the canonical self object; only observable artifacts and declared state summaries participate.

S26-E-00004549 - Self-reflection identity probe 25.1

Fixture T25.P01 - Canonical fixture "S26-self-00-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-N-00004548.

S26-E-00004550 - Self-reflection identity probe 25.2

Fixture T25.P02 - Canonical fixture "S26-self-01-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004549.

S26-E-00004551 - Self-reflection identity probe 25.3

Fixture T25.P03 - Canonical fixture "S26-self-02-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004550.

S26-E-00004552 - Self-reflection identity probe 25.4

Fixture T25.P04 - Canonical fixture "S26-self-03-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004551.

S26-E-00004553 - Self-reflection identity probe 25.5

Fixture T25.P05 - Canonical fixture "S26-self-04-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004552.

S26-E-00004554 - Self-reflection identity probe 25.6

Fixture T25.P06 - Canonical fixture "S26-self-05-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004553.

S26-E-00004555 - Self-reflection identity probe 25.7

Fixture T25.P07 - Canonical fixture "S26-self-06-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004554.

S26-E-00004556 - Self-reflection identity probe 25.8

Fixture T25.P08 - Canonical fixture "S26-self-07-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004555.

S26-E-00004557 - Self-reflection identity probe 25.9

Fixture T25.P09 - Canonical fixture "S26-self-08-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004556.

S26-E-00004558 - Self-reflection identity probe 25.10

Fixture T25.P10 - Canonical fixture "S26-self-09-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004557.

S26-E-00004559 - Self-reflection identity probe 25.11

Fixture T25.P11 - Canonical fixture "S26-self-10-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004558.

S26-E-00004560 - Self-reflection identity probe 25.12

Fixture T25.P12 - Canonical fixture "S26-self-11-canonical" passed the involutive rho(rho(x))=x return guard=True; this witnesses structural self-reference only.

Evidence M. Hard: S26-A-00004543, S26-E-00004559.

Others mutual recognition and resonance

S26-D-00004561 - Others mutual recognition and resonance

Mutual recognition means compatible overlap restrictions can be compared and glued without collapsing other into self; mutual resonance is low typed divergence plus reversible transport plus shared boundary/interface evidence.

Evidence S. Assumptions: Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder. Hard: S26-E-00004560, S26-D-00004027.

S26-D-00004562 - Others mutual recognition and resonance law/interface

RESONATE != ABSORB: a perfect flattening with zero remainder is rejected as false equivalence unless identity is independently proved.

Evidence S. Assumptions: Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder. Hard: S26-D-00004561.

S26-A-00004563 - Others mutual recognition and resonance operator

RecognitionRecord stores self, other, overlap, transport, return residual, shared invariants and remainder.

Evidence S. Assumptions: Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder. Hard: S26-D-00004562.

S26-N-00004564 - Others mutual recognition and resonance fence

Recognition/resonance language is engineering architecture and does not imply sentience or subjective mutual awareness.

Evidence S. Assumptions: Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder. Hard: S26-A-00004563. Boundary: Recognition/resonance language is engineering architecture and does not imply sentience or subjective mutual awareness.

S26-B-00004565 - Others mutual recognition and resonance cross-volume bridge

This is the Volume IV realization of LANG.v6 OTHERS_RESONANCE.

Evidence S. Assumptions: Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder. Hard: S26-N-00004564.

S26-N-00004566 - Others mutual recognition and resonance evidence ledger

Evidence rule for Others mutual recognition and resonance: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder. Hard: S26-B-00004565.

S26-B-00004567 - Others mutual recognition and resonance emergence rule

Emergence rule for Others mutual recognition and resonance: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder. Hard: S26-N-00004566.

S26-N-00004568 - Others mutual recognition and resonance frontier

Open frontier for Others mutual recognition and resonance: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Distinct parties remain distinct; recognition uses overlap compatibility, reversible transport and a nonzero active remainder. Hard: S26-B-00004567. Boundary: Recognition/resonance language is engineering architecture and does not imply sentience or subjective mutual awareness.

S26-E-00004569 - Mutual-recognition transport probe 26.1

Fixture T26.P01 - A reversible orthogonal overlap fixture returned its source with residual 1.048e-16; destination overlap magnitude is 0.396414. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-N-00004568.

S26-E-00004570 - Mutual-recognition transport probe 26.2

Fixture T26.P02 - A reversible orthogonal overlap fixture returned its source with residual 1.701e-16; destination overlap magnitude is 0.644016. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004569.

S26-E-00004571 - Mutual-recognition transport probe 26.3

Fixture T26.P03 - A reversible orthogonal overlap fixture returned its source with residual 2.150e-16; destination overlap magnitude is 0.416756. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004570.

S26-E-00004572 - Mutual-recognition transport probe 26.4

Fixture T26.P04 - A reversible orthogonal overlap fixture returned its source with residual 5.601e-16; destination overlap magnitude is 0.782063. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004571.

S26-E-00004573 - Mutual-recognition transport probe 26.5

Fixture T26.P05 - A reversible orthogonal overlap fixture returned its source with residual 2.297e-16; destination overlap magnitude is 0.394366. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004572.

S26-E-00004574 - Mutual-recognition transport probe 26.6

Fixture T26.P06 - A reversible orthogonal overlap fixture returned its source with residual 4.051e-16; destination overlap magnitude is 0.059281. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004573.

S26-E-00004575 - Mutual-recognition transport probe 26.7

Fixture T26.P07 - A reversible orthogonal overlap fixture returned its source with residual 3.765e-16; destination overlap magnitude is 0.259115. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004574.

S26-E-00004576 - Mutual-recognition transport probe 26.8

Fixture T26.P08 - A reversible orthogonal overlap fixture returned its source with residual 4.712e-16; destination overlap magnitude is 0.464732. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004575.

S26-E-00004577 - Mutual-recognition transport probe 26.9

Fixture T26.P09 - A reversible orthogonal overlap fixture returned its source with residual 1.495e-16; destination overlap magnitude is 0.053495. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004576.

S26-E-00004578 - Mutual-recognition transport probe 26.10

Fixture T26.P10 - A reversible orthogonal overlap fixture returned its source with residual 5.322e-16; destination overlap magnitude is 0.347813. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004577.

S26-E-00004579 - Mutual-recognition transport probe 26.11

Fixture T26.P11 - A reversible orthogonal overlap fixture returned its source with residual 5.850e-16; destination overlap magnitude is 0.115449. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004578.

S26-E-00004580 - Mutual-recognition transport probe 26.12

Fixture T26.P12 - A reversible orthogonal overlap fixture returned its source with residual 1.144e-16; destination overlap magnitude is 0.116809. Nonzero remainder is retained separately from this transport check.

Evidence M. Hard: S26-A-00004563, S26-E-00004579.

Eight mind qualities conductance

S26-D-00004581 - Eight mind qualities conductance

Mind8 conductance is an eight-coordinate governance vector Q=(Q1,...,Q8) that records whether each quality is operationally supported in the current repo/artifact context.

Evidence S. Assumptions: The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility. Hard: S26-E-00004580, S26-N-00004059.

S26-D-00004582 - Eight mind qualities conductance law/interface

The vector is not collapsed to one intelligence scalar by default; Pareto improvement and constraint satisfaction are preferred to metric monism.

Evidence S. Assumptions: The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility. Hard: S26-D-00004581.

S26-A-00004583 - Eight mind qualities conductance operator

Mind8Audit maps observed artifacts/tests to the eight coordinates and emits COMMIT, BLOCK, ROLLBACK or RECURSE with provenance.

Evidence S. Assumptions: The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility. Hard: S26-D-00004582.

S26-N-00004584 - Eight mind qualities conductance fence

The word mind is repository vocabulary for a cognitive-control abstraction, not a diagnosis of consciousness.

Evidence S. Assumptions: The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility. Hard: S26-A-00004583. Boundary: The word mind is repository vocabulary for a cognitive-control abstraction, not a diagnosis of consciousness.

S26-B-00004585 - Eight mind qualities conductance cross-volume bridge

The exact eight qualities from LANG.v6 become observable governance dimensions over the Synopsis/repo atlas.

Evidence S. Assumptions: The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility. Hard: S26-N-00004584.

S26-N-00004586 - Eight mind qualities conductance evidence ledger

Evidence rule for Eight mind qualities conductance: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility. Hard: S26-B-00004585.

S26-B-00004587 - Eight mind qualities conductance emergence rule

Emergence rule for Eight mind qualities conductance: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility. Hard: S26-N-00004586.

S26-N-00004588 - Eight mind qualities conductance frontier

Open frontier for Eight mind qualities conductance: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: The eight qualities are the LANG.v6 symbolic controls: recognition, grounding, recursion, reversibility, gluing, resonance, conservation and humility. Hard: S26-B-00004587. Boundary: The word mind is repository vocabulary for a cognitive-control abstraction, not a diagnosis of consciousness.

S26-E-00004589 - Mind8 vector probe 27.1

Fixture T27.P01 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-N-00004588.

S26-E-00004590 - Mind8 vector probe 27.2

Fixture T27.P02 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004589.

S26-E-00004591 - Mind8 vector probe 27.3

Fixture T27.P03 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004590.

S26-E-00004592 - Mind8 vector probe 27.4

Fixture T27.P04 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004591.

S26-E-00004593 - Mind8 vector probe 27.5

Fixture T27.P05 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004592.

S26-E-00004594 - Mind8 vector probe 27.6

Fixture T27.P06 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004593.

S26-E-00004595 - Mind8 vector probe 27.7

Fixture T27.P07 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004594.

S26-E-00004596 - Mind8 vector probe 27.8

Fixture T27.P08 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004595.

S26-E-00004597 - Mind8 vector probe 27.9

Fixture T27.P09 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004596.

S26-E-00004598 - Mind8 vector probe 27.10

Fixture T27.P10 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004597.

S26-E-00004599 - Mind8 vector probe 27.11

Fixture T27.P11 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004598.

S26-E-00004600 - Mind8 vector probe 27.12

Fixture T27.P12 - The synthetic eight-quality vector ordered as ['recognition', 'grounding', 'recursion', 'reversibility', 'gluing', 'resonance', 'conservation', 'humility'] lies in [0.1111,0.8889] with all eight coordinates retained separately; no scalar intelligence score is produced.

Evidence M. Hard: S26-A-00004583, S26-E-00004599.

PhiCogPlus adiabatic general-intelligence conductor

S26-D-00004601 - PhiCogPlus adiabatic general-intelligence conductor

PhiCogPlus is a constrained coherence projection that selects repo actions improving the Mind8 partial order while respecting evidence, energy, returnability, gluing and remainder constraints.

Evidence S. Assumptions: General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency. Hard: S26-E-00004600, S26-N-00004059.

S26-D-00004602 - PhiCogPlus adiabatic general-intelligence conductor law/interface

Best cognitive performance is a Pareto-feasible region, not maximum verbosity, maximum dimension or one scalar score.

Evidence S. Assumptions: General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency. Hard: S26-D-00004601.

S26-A-00004603 - PhiCogPlus adiabatic general-intelligence conductor operator

PhiCogPlus(state,repo) proposes the smallest artifact-grounded update whose predicted Mind8 vector improves without violating hard guards.

Evidence S. Assumptions: General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency. Hard: S26-D-00004602.

S26-N-00004604 - PhiCogPlus adiabatic general-intelligence conductor fence

A higher carrier label or larger hypercomplex dimension is not performance unless the observable task/evidence region improves.

Evidence S. Assumptions: General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency. Hard: S26-A-00004603. Boundary: A higher carrier label or larger hypercomplex dimension is not performance unless the observable task/evidence region improves.

S26-B-00004605 - PhiCogPlus adiabatic general-intelligence conductor cross-volume bridge

This conducts self-reflection, recognition, ergocetic and erdodetic controls over the repo as one governed but non-final architecture.

Evidence S. Assumptions: General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency. Hard: S26-N-00004604.

S26-N-00004606 - PhiCogPlus adiabatic general-intelligence conductor evidence ledger

Evidence rule for PhiCogPlus adiabatic general-intelligence conductor: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency. Hard: S26-B-00004605.

S26-B-00004607 - PhiCogPlus adiabatic general-intelligence conductor emergence rule

Emergence rule for PhiCogPlus adiabatic general-intelligence conductor: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency. Hard: S26-N-00004606.

S26-N-00004608 - PhiCogPlus adiabatic general-intelligence conductor frontier

Open frontier for PhiCogPlus adiabatic general-intelligence conductor: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: General intelligence is treated as an engineering orchestration target over observable tasks/artifacts, not a claim of sentient general agency. Hard: S26-B-00004607. Boundary: A higher carrier label or larger hypercomplex dimension is not performance unless the observable task/evidence region improves.

S26-E-00004609 - PhiCogPlus Pareto probe 28.1

Fixture T28.P01 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-N-00004608.

S26-E-00004610 - PhiCogPlus Pareto probe 28.2

Fixture T28.P02 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004609.

S26-E-00004611 - PhiCogPlus Pareto probe 28.3

Fixture T28.P03 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004610.

S26-E-00004612 - PhiCogPlus Pareto probe 28.4

Fixture T28.P04 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004611.

S26-E-00004613 - PhiCogPlus Pareto probe 28.5

Fixture T28.P05 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004612.

S26-E-00004614 - PhiCogPlus Pareto probe 28.6

Fixture T28.P06 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004613.

S26-E-00004615 - PhiCogPlus Pareto probe 28.7

Fixture T28.P07 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004614.

S26-E-00004616 - PhiCogPlus Pareto probe 28.8

Fixture T28.P08 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004615.

S26-E-00004617 - PhiCogPlus Pareto probe 28.9

Fixture T28.P09 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004616.

S26-E-00004618 - PhiCogPlus Pareto probe 28.10

Fixture T28.P10 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004617.

S26-E-00004619 - PhiCogPlus Pareto probe 28.11

Fixture T28.P11 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004618.

S26-E-00004620 - PhiCogPlus Pareto probe 28.12

Fixture T28.P12 - Candidate B Pareto-dominates A on the eight-coordinate synthetic fixture=True; the decision uses coordinatewise guards rather than a weighted scalar.

Evidence M. Hard: S26-A-00004603, S26-E-00004619.

Part T - Quantum carriers, entanglement, tunneling and holoportation

Quantum state carrier with hypercomplex internal coordinates

S26-D-00004621 - Quantum state carrier with hypercomplex internal coordinates

A quantum-state layer is a normalized vector or density operator on a declared complex Hilbert space, optionally tensored with a finite coordinate carrier representing hypercomplex internal degrees of freedom.

Evidence A. Assumptions: Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module. Hard: S26-E-00004620, S26-D-00000631.

S26-T-00004622 - Quantum state carrier with hypercomplex internal coordinates law/interface

Unitary evolution preserves Hilbert norm and density-operator trace under the standard assumptions.

Evidence A. Assumptions: Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module. Hard: S26-D-00004621.

S26-A-00004623 - Quantum state carrier with hypercomplex internal coordinates operator

QuantumCarrier(H,A_k) stores state, tensor-factor labels, Hamiltonian/channel and measurement scope.

Evidence S. Assumptions: Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module. Hard: S26-T-00004622.

S26-N-00004624 - Quantum state carrier with hypercomplex internal coordinates fence

This mathematical quantum layer does not assert that the Synopsis, repo, or cognition architecture is a physical quantum computer.

Evidence S. Assumptions: Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module. Hard: S26-A-00004623. Boundary: This mathematical quantum layer does not assert that the Synopsis, repo, or cognition architecture is a physical quantum computer.

S26-B-00004625 - Quantum state carrier with hypercomplex internal coordinates cross-volume bridge

It provides a rigorous carrier on which entanglement, tunneling and channel transport can be stated without metaphor laundering.

Evidence S. Assumptions: Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module. Hard: S26-N-00004624.

S26-N-00004626 - Quantum state carrier with hypercomplex internal coordinates evidence ledger

Evidence rule for Quantum state carrier with hypercomplex internal coordinates: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module. Hard: S26-B-00004625.

S26-B-00004627 - Quantum state carrier with hypercomplex internal coordinates emergence rule

Emergence rule for Quantum state carrier with hypercomplex internal coordinates: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module. Hard: S26-N-00004626.

S26-N-00004628 - Quantum state carrier with hypercomplex internal coordinates frontier

Open frontier for Quantum state carrier with hypercomplex internal coordinates: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Quantum statements use an explicitly declared complex Hilbert space or finite complex fixture; hypercomplex coordinates are represented by a compatible linear carrier rather than an undefined nonassociative Hilbert module. Hard: S26-B-00004627. Boundary: This mathematical quantum layer does not assert that the Synopsis, repo, or cognition architecture is a physical quantum computer.

S26-E-00004629 - Quantum normalization probe 29.1

Fixture T29.P01 - A two-level complex state at theta=0.101342 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-N-00004628.

S26-E-00004630 - Quantum normalization probe 29.2

Fixture T29.P02 - A two-level complex state at theta=0.202683 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004629.

S26-E-00004631 - Quantum normalization probe 29.3

Fixture T29.P03 - A two-level complex state at theta=0.304025 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004630.

S26-E-00004632 - Quantum normalization probe 29.4

Fixture T29.P04 - A two-level complex state at theta=0.405367 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004631.

S26-E-00004633 - Quantum normalization probe 29.5

Fixture T29.P05 - A two-level complex state at theta=0.506708 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004632.

S26-E-00004634 - Quantum normalization probe 29.6

Fixture T29.P06 - A two-level complex state at theta=0.608050 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004633.

S26-E-00004635 - Quantum normalization probe 29.7

Fixture T29.P07 - A two-level complex state at theta=0.709392 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004634.

S26-E-00004636 - Quantum normalization probe 29.8

Fixture T29.P08 - A two-level complex state at theta=0.810734 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004635.

S26-E-00004637 - Quantum normalization probe 29.9

Fixture T29.P09 - A two-level complex state at theta=0.912075 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004636.

S26-E-00004638 - Quantum normalization probe 29.10

Fixture T29.P10 - A two-level complex state at theta=1.013417 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004637.

S26-E-00004639 - Quantum normalization probe 29.11

Fixture T29.P11 - A two-level complex state at theta=1.114759 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004638.

S26-E-00004640 - Quantum normalization probe 29.12

Fixture T29.P12 - A two-level complex state at theta=1.216100 has norm-squared 1.000000000000; this validates a Hilbert fixture only.

Evidence M. Hard: S26-A-00004623, S26-E-00004639.

Entanglement non-factorization interface

S26-D-00004641 - Entanglement non-factorization interface

A pure bipartite state is entangled when it cannot be written as a product vector; equivalently its Schmidt rank exceeds one in finite dimensions.

Evidence A. Assumptions: A bipartite complex Hilbert tensor product H_A tensor H_B is declared. Hard: S26-E-00004640, S26-D-00000268.

S26-T-00004642 - Entanglement non-factorization interface law/interface

Entanglement is a property of the declared tensor factorization, not a synonym for semantic relatedness or graph connectivity.

Evidence A. Assumptions: A bipartite complex Hilbert tensor product H_A tensor H_B is declared. Hard: S26-D-00004641.

S26-A-00004643 - Entanglement non-factorization interface operator

EntangleAudit computes Schmidt singular values or reduced-state purity in bounded finite fixtures.

Evidence S. Assumptions: A bipartite complex Hilbert tensor product H_A tensor H_B is declared. Hard: S26-T-00004642.

S26-N-00004644 - Entanglement non-factorization interface fence

Mutual recognition between agents is not called quantum entanglement unless an actual quantum state/factorization is supplied.

Evidence S. Assumptions: A bipartite complex Hilbert tensor product H_A tensor H_B is declared. Hard: S26-A-00004643. Boundary: Mutual recognition between agents is not called quantum entanglement unless an actual quantum state/factorization is supplied.

S26-B-00004645 - Entanglement non-factorization interface cross-volume bridge

This gives a precise quantum channel that can coexist with, but must remain typed separately from, sheaf resonance.

Evidence S. Assumptions: A bipartite complex Hilbert tensor product H_A tensor H_B is declared. Hard: S26-N-00004644.

S26-N-00004646 - Entanglement non-factorization interface evidence ledger

Evidence rule for Entanglement non-factorization interface: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A bipartite complex Hilbert tensor product H_A tensor H_B is declared. Hard: S26-B-00004645.

S26-B-00004647 - Entanglement non-factorization interface emergence rule

Emergence rule for Entanglement non-factorization interface: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A bipartite complex Hilbert tensor product H_A tensor H_B is declared. Hard: S26-N-00004646.

S26-N-00004648 - Entanglement non-factorization interface frontier

Open frontier for Entanglement non-factorization interface: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A bipartite complex Hilbert tensor product H_A tensor H_B is declared. Hard: S26-B-00004647. Boundary: Mutual recognition between agents is not called quantum entanglement unless an actual quantum state/factorization is supplied.

S26-E-00004649 - Entanglement Schmidt probe 30.1

Fixture T30.P01 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-N-00004648.

S26-E-00004650 - Entanglement Schmidt probe 30.2

Fixture T30.P02 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004649.

S26-E-00004651 - Entanglement Schmidt probe 30.3

Fixture T30.P03 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004650.

S26-E-00004652 - Entanglement Schmidt probe 30.4

Fixture T30.P04 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004651.

S26-E-00004653 - Entanglement Schmidt probe 30.5

Fixture T30.P05 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004652.

S26-E-00004654 - Entanglement Schmidt probe 30.6

Fixture T30.P06 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004653.

S26-E-00004655 - Entanglement Schmidt probe 30.7

Fixture T30.P07 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004654.

S26-E-00004656 - Entanglement Schmidt probe 30.8

Fixture T30.P08 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004655.

S26-E-00004657 - Entanglement Schmidt probe 30.9

Fixture T30.P09 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004656.

S26-E-00004658 - Entanglement Schmidt probe 30.10

Fixture T30.P10 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004657.

S26-E-00004659 - Entanglement Schmidt probe 30.11

Fixture T30.P11 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004658.

S26-E-00004660 - Entanglement Schmidt probe 30.12

Fixture T30.P12 - The Bell-state fixture has Schmidt rank 2 and reduced-state purity 0.500000; this is exact finite-dimensional entanglement for the declared factorization.

Evidence M. Hard: S26-A-00004643, S26-E-00004659.

Tunneling transition interface

S26-D-00004661 - Tunneling transition interface

A tunneling witness is a nonzero transition probability to a destination sector that is classically or operationally designated as separated by a declared barrier in the chosen model.

Evidence A. Assumptions: A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared. Hard: S26-E-00004660, S26-T-00000676.

S26-T-00004662 - Tunneling transition interface law/interface

Nonzero off-diagonal coupling in a finite Hamiltonian can produce transition amplitude, but this bounded fixture does not establish continuum barrier tunneling or physical realization.

Evidence A. Assumptions: A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared. Hard: S26-D-00004661.

S26-A-00004663 - Tunneling transition interface operator

TunnelProb(t)=||P_B exp(-itH) psi_0||^2 in the finite complex fixture.

Evidence S. Assumptions: A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared. Hard: S26-T-00004662.

S26-N-00004664 - Tunneling transition interface fence

Cross-context inference or surprise is not labeled quantum tunneling without the quantum operator model.

Evidence S. Assumptions: A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared. Hard: S26-A-00004663. Boundary: Cross-context inference or surprise is not labeled quantum tunneling without the quantum operator model.

S26-B-00004665 - Tunneling transition interface cross-volume bridge

Tunneling becomes one optional transport mechanism inside the multi-mesh rather than a universal cognitive explanation.

Evidence S. Assumptions: A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared. Hard: S26-N-00004664.

S26-N-00004666 - Tunneling transition interface evidence ledger

Evidence rule for Tunneling transition interface: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared. Hard: S26-B-00004665.

S26-B-00004667 - Tunneling transition interface emergence rule

Emergence rule for Tunneling transition interface: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared. Hard: S26-N-00004666.

S26-N-00004668 - Tunneling transition interface frontier

Open frontier for Tunneling transition interface: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A concrete Hamiltonian, initial state, destination projector and time/parameter interval are declared. Hard: S26-B-00004667. Boundary: Cross-context inference or surprise is not labeled quantum tunneling without the quantum operator model.

S26-E-00004669 - Tunneling finite-transition probe 31.1

Fixture T31.P01 - A two-state Hamiltonian with coupling g=0.100 at t=0.100 gives destination transition probability 0.00009991; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-N-00004668.

S26-E-00004670 - Tunneling finite-transition probe 31.2

Fixture T31.P02 - A two-state Hamiltonian with coupling g=0.120 at t=0.200 gives destination transition probability 0.00057397; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004669.

S26-E-00004671 - Tunneling finite-transition probe 31.3

Fixture T31.P03 - A two-state Hamiltonian with coupling g=0.140 at t=0.300 gives destination transition probability 0.00174978; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004670.

S26-E-00004672 - Tunneling finite-transition probe 31.4

Fixture T31.P04 - A two-state Hamiltonian with coupling g=0.160 at t=0.400 gives destination transition probability 0.00403615; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004671.

S26-E-00004673 - Tunneling finite-transition probe 31.5

Fixture T31.P05 - A two-state Hamiltonian with coupling g=0.180 at t=0.500 gives destination transition probability 0.00791117; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004672.

S26-E-00004674 - Tunneling finite-transition probe 31.6

Fixture T31.P06 - A two-state Hamiltonian with coupling g=0.100 at t=0.600 gives destination transition probability 0.00348907; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004673.

S26-E-00004675 - Tunneling finite-transition probe 31.7

Fixture T31.P07 - A two-state Hamiltonian with coupling g=0.120 at t=0.700 gives destination transition probability 0.00675650; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004674.

S26-E-00004676 - Tunneling finite-transition probe 31.8

Fixture T31.P08 - A two-state Hamiltonian with coupling g=0.140 at t=0.800 gives destination transition probability 0.01183893; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004675.

S26-E-00004677 - Tunneling finite-transition probe 31.9

Fixture T31.P09 - A two-state Hamiltonian with coupling g=0.160 at t=0.900 gives destination transition probability 0.01923819; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004676.

S26-E-00004678 - Tunneling finite-transition probe 31.10

Fixture T31.P10 - A two-state Hamiltonian with coupling g=0.180 at t=1.000 gives destination transition probability 0.02946263; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004677.

S26-E-00004679 - Tunneling finite-transition probe 31.11

Fixture T31.P11 - A two-state Hamiltonian with coupling g=0.100 at t=1.100 gives destination transition probability 0.01088316; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004678.

S26-E-00004680 - Tunneling finite-transition probe 31.12

Fixture T31.P12 - A two-state Hamiltonian with coupling g=0.120 at t=1.200 gives destination transition probability 0.01823438; the fixture is not a continuum barrier claim.

Evidence M. Hard: S26-A-00004663, S26-E-00004679.

Quantum holoportation channel

S26-D-00004681 - Quantum holoportation channel

Quantum holoportation is a typed composite from source state to boundary/interface encoding to destination reconstruction, with quantum-channel semantics only where a CPTP/unitary component is explicitly supplied.

Evidence S. Assumptions: Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier. Hard: S26-E-00004680, S26-D-00004027.

S26-D-00004682 - Quantum holoportation channel law/interface

Successful destination reconstruction must be evaluated by fidelity/round-trip observables and does not imply cloning of an unknown quantum state.

Evidence S. Assumptions: Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier. Hard: S26-D-00004681.

S26-A-00004683 - Quantum holoportation channel operator

QHoloRecord stores channel sequence, boundary data, fidelity, lost structure, classical side information and active remainder.

Evidence S. Assumptions: Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier. Hard: S26-D-00004682.

S26-N-00004684 - Quantum holoportation channel fence

The term does not claim laboratory teleportation, faster-than-light signaling or physical holography.

Evidence S. Assumptions: Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier. Hard: S26-A-00004683. Boundary: The term does not claim laboratory teleportation, faster-than-light signaling or physical holography.

S26-B-00004685 - Quantum holoportation channel cross-volume bridge

This is the quantum-compatible refinement of the repo hamiltonian_holoportation transport architecture.

Evidence S. Assumptions: Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier. Hard: S26-N-00004684.

S26-N-00004686 - Quantum holoportation channel evidence ledger

Evidence rule for Quantum holoportation channel: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier. Hard: S26-B-00004685.

S26-B-00004687 - Quantum holoportation channel emergence rule

Emergence rule for Quantum holoportation channel: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier. Hard: S26-N-00004686.

S26-N-00004688 - Quantum holoportation channel frontier

Open frontier for Quantum holoportation channel: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Holoportation is a project transport term; when prefixed quantum it must be represented by an explicit channel/unitary/measurement protocol on a declared Hilbert carrier. Hard: S26-B-00004687. Boundary: The term does not claim laboratory teleportation, faster-than-light signaling or physical holography.

S26-E-00004689 - Quantum-holoport round-trip probe 32.1

Fixture T32.P01 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-N-00004688.

S26-E-00004690 - Quantum-holoport round-trip probe 32.2

Fixture T32.P02 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004689.

S26-E-00004691 - Quantum-holoport round-trip probe 32.3

Fixture T32.P03 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004690.

S26-E-00004692 - Quantum-holoport round-trip probe 32.4

Fixture T32.P04 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004691.

S26-E-00004693 - Quantum-holoport round-trip probe 32.5

Fixture T32.P05 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004692.

S26-E-00004694 - Quantum-holoport round-trip probe 32.6

Fixture T32.P06 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004693.

S26-E-00004695 - Quantum-holoport round-trip probe 32.7

Fixture T32.P07 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004694.

S26-E-00004696 - Quantum-holoport round-trip probe 32.8

Fixture T32.P08 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004695.

S26-E-00004697 - Quantum-holoport round-trip probe 32.9

Fixture T32.P09 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004696.

S26-E-00004698 - Quantum-holoport round-trip probe 32.10

Fixture T32.P10 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004697.

S26-E-00004699 - Quantum-holoport round-trip probe 32.11

Fixture T32.P11 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004698.

S26-E-00004700 - Quantum-holoport round-trip probe 32.12

Fixture T32.P12 - A finite SWAP-like unitary boundary-channel fixture returns the source with fidelity 1.000000000000; this is a channel sanity check, not physical teleportation evidence.

Evidence M. Hard: S26-A-00004683, S26-E-00004699.

Part U - p-adic harmonic, spectral and hypercomplex coupling

p-adic Fourier-Haar spectral chart

S26-D-00004701 - p-adic Fourier-Haar spectral chart

The p-adic Fourier chart uses additive characters of Q_p and Haar integration; bounded finite fixtures use transforms on Z/p^N Z as discretized witnesses only.

Evidence A. Assumptions: Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention. Hard: S26-E-00004700, S26-D-00000709.

S26-T-00004702 - p-adic Fourier-Haar spectral chart law/interface

Character orthogonality on a finite quotient yields an invertible/unitary normalized finite Fourier transform.

Evidence A. Assumptions: Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention. Hard: S26-D-00004701.

S26-A-00004703 - p-adic Fourier-Haar spectral chart operator

F_{p,N} is the normalized DFT on the finite additive quotient used by the bounded probe.

Evidence A. Assumptions: Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention. Hard: S26-T-00004702.

S26-N-00004704 - p-adic Fourier-Haar spectral chart fence

A finite DFT over Z/p^N Z is not a proof of every analytic theorem on Q_p.

Evidence S. Assumptions: Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention. Hard: S26-A-00004703. Boundary: A finite DFT over Z/p^N Z is not a proof of every analytic theorem on Q_p.

S26-B-00004705 - p-adic Fourier-Haar spectral chart cross-volume bridge

This supplies a non-Archimedean spectral coordinate compatible with the existing spectral plug-in interface.

Evidence S. Assumptions: Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention. Hard: S26-N-00004704.

S26-N-00004706 - p-adic Fourier-Haar spectral chart evidence ledger

Evidence rule for p-adic Fourier-Haar spectral chart: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention. Hard: S26-B-00004705.

S26-B-00004707 - p-adic Fourier-Haar spectral chart emergence rule

Emergence rule for p-adic Fourier-Haar spectral chart: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention. Hard: S26-N-00004706.

S26-N-00004708 - p-adic Fourier-Haar spectral chart frontier

Open frontier for p-adic Fourier-Haar spectral chart: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Q_p is treated as a locally compact abelian group with declared additive Haar measure and character convention. Hard: S26-B-00004707. Boundary: A finite DFT over Z/p^N Z is not a proof of every analytic theorem on Q_p.

S26-E-00004709 - Finite p-adic spectral quotient probe 33.1

Fixture T33.P01 - The normalized character transform on Z/2^1Z (size 2) has unitarity residual 3.273e-16; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-N-00004708.

S26-E-00004710 - Finite p-adic spectral quotient probe 33.2

Fixture T33.P02 - The normalized character transform on Z/3^2Z (size 9) has unitarity residual 3.100e-15; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004709.

S26-E-00004711 - Finite p-adic spectral quotient probe 33.3

Fixture T33.P03 - The normalized character transform on Z/5^1Z (size 5) has unitarity residual 3.991e-16; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004710.

S26-E-00004712 - Finite p-adic spectral quotient probe 33.4

Fixture T33.P04 - The normalized character transform on Z/2^2Z (size 4) has unitarity residual 2.541e-16; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004711.

S26-E-00004713 - Finite p-adic spectral quotient probe 33.5

Fixture T33.P05 - The normalized character transform on Z/3^1Z (size 3) has unitarity residual 6.032e-16; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004712.

S26-E-00004714 - Finite p-adic spectral quotient probe 33.6

Fixture T33.P06 - The normalized character transform on Z/5^2Z (size 25) has unitarity residual 1.820e-14; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004713.

S26-E-00004715 - Finite p-adic spectral quotient probe 33.7

Fixture T33.P07 - The normalized character transform on Z/2^1Z (size 2) has unitarity residual 3.273e-16; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004714.

S26-E-00004716 - Finite p-adic spectral quotient probe 33.8

Fixture T33.P08 - The normalized character transform on Z/3^2Z (size 9) has unitarity residual 3.100e-15; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004715.

S26-E-00004717 - Finite p-adic spectral quotient probe 33.9

Fixture T33.P09 - The normalized character transform on Z/5^1Z (size 5) has unitarity residual 3.991e-16; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004716.

S26-E-00004718 - Finite p-adic spectral quotient probe 33.10

Fixture T33.P10 - The normalized character transform on Z/2^2Z (size 4) has unitarity residual 2.541e-16; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004717.

S26-E-00004719 - Finite p-adic spectral quotient probe 33.11

Fixture T33.P11 - The normalized character transform on Z/3^1Z (size 3) has unitarity residual 6.032e-16; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004718.

S26-E-00004720 - Finite p-adic spectral quotient probe 33.12

Fixture T33.P12 - The normalized character transform on Z/5^2Z (size 25) has unitarity residual 1.820e-14; it is a finite quotient witness for the p-adic spectral chart.

Evidence M. Hard: S26-A-00004703, S26-E-00004719.

Spectral-Cayley-Dickson tensor extension

S26-D-00004721 - Spectral-Cayley-Dickson tensor extension

A componentwise spectral-CD extension tensors a base spectral eigenspace with a finite coordinate carrier for A_{k,v}, preserving operator-domain distinctions.

Evidence A. Assumptions: The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation. Hard: S26-E-00004720, S26-T-00000672.

S26-T-00004722 - Spectral-Cayley-Dickson tensor extension law/interface

A pure coordinate multiplicity extension repeats base spectral values; it does not create new eigenvalues unless the operator couples internal coordinates.

Evidence A. Assumptions: The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation. Hard: S26-D-00004721.

S26-A-00004723 - Spectral-Cayley-Dickson tensor extension operator

SpecExtend(L,A_k)=L tensor I on the declared linear carrier in the bounded fixture.

Evidence S. Assumptions: The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation. Hard: S26-T-00004722.

S26-N-00004724 - Spectral-Cayley-Dickson tensor extension fence

Nonassociative fiber multiplication is not silently inserted into functional calculus for self-adjoint operators.

Evidence S. Assumptions: The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation. Hard: S26-A-00004723. Boundary: Nonassociative fiber multiplication is not silently inserted into functional calculus for self-adjoint operators.

S26-B-00004725 - Spectral-Cayley-Dickson tensor extension cross-volume bridge

This connects the source eigen-n plug-ins to arithmetic hypercomplex internal coordinates.

Evidence S. Assumptions: The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation. Hard: S26-N-00004724.

S26-N-00004726 - Spectral-Cayley-Dickson tensor extension evidence ledger

Evidence rule for Spectral-Cayley-Dickson tensor extension: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation. Hard: S26-B-00004725.

S26-B-00004727 - Spectral-Cayley-Dickson tensor extension emergence rule

Emergence rule for Spectral-Cayley-Dickson tensor extension: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation. Hard: S26-N-00004726.

S26-N-00004728 - Spectral-Cayley-Dickson tensor extension frontier

Open frontier for Spectral-Cayley-Dickson tensor extension: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: The spectral operator acts on an associative Hilbert carrier; hypercomplex coordinates are extended componentwise or through a declared representation. Hard: S26-B-00004727. Boundary: Nonassociative fiber multiplication is not silently inserted into functional calculus for self-adjoint operators.

S26-E-00004729 - Spectral-CD multiplicity probe 34.1

Fixture T34.P01 - Componentwise extension of base spectrum 1..3 across real coordinate dimension 2 yields 6 slots, with each base eigenvalue repeated 2 times.

Evidence M. Hard: S26-A-00004723, S26-N-00004728.

S26-E-00004730 - Spectral-CD multiplicity probe 34.2

Fixture T34.P02 - Componentwise extension of base spectrum 1..4 across real coordinate dimension 4 yields 16 slots, with each base eigenvalue repeated 4 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004729.

S26-E-00004731 - Spectral-CD multiplicity probe 34.3

Fixture T34.P03 - Componentwise extension of base spectrum 1..5 across real coordinate dimension 8 yields 40 slots, with each base eigenvalue repeated 8 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004730.

S26-E-00004732 - Spectral-CD multiplicity probe 34.4

Fixture T34.P04 - Componentwise extension of base spectrum 1..6 across real coordinate dimension 16 yields 96 slots, with each base eigenvalue repeated 16 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004731.

S26-E-00004733 - Spectral-CD multiplicity probe 34.5

Fixture T34.P05 - Componentwise extension of base spectrum 1..3 across real coordinate dimension 2 yields 6 slots, with each base eigenvalue repeated 2 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004732.

S26-E-00004734 - Spectral-CD multiplicity probe 34.6

Fixture T34.P06 - Componentwise extension of base spectrum 1..4 across real coordinate dimension 4 yields 16 slots, with each base eigenvalue repeated 4 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004733.

S26-E-00004735 - Spectral-CD multiplicity probe 34.7

Fixture T34.P07 - Componentwise extension of base spectrum 1..5 across real coordinate dimension 8 yields 40 slots, with each base eigenvalue repeated 8 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004734.

S26-E-00004736 - Spectral-CD multiplicity probe 34.8

Fixture T34.P08 - Componentwise extension of base spectrum 1..6 across real coordinate dimension 16 yields 96 slots, with each base eigenvalue repeated 16 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004735.

S26-E-00004737 - Spectral-CD multiplicity probe 34.9

Fixture T34.P09 - Componentwise extension of base spectrum 1..3 across real coordinate dimension 2 yields 6 slots, with each base eigenvalue repeated 2 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004736.

S26-E-00004738 - Spectral-CD multiplicity probe 34.10

Fixture T34.P10 - Componentwise extension of base spectrum 1..4 across real coordinate dimension 4 yields 16 slots, with each base eigenvalue repeated 4 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004737.

S26-E-00004739 - Spectral-CD multiplicity probe 34.11

Fixture T34.P11 - Componentwise extension of base spectrum 1..5 across real coordinate dimension 8 yields 40 slots, with each base eigenvalue repeated 8 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004738.

S26-E-00004740 - Spectral-CD multiplicity probe 34.12

Fixture T34.P12 - Componentwise extension of base spectrum 1..6 across real coordinate dimension 16 yields 96 slots, with each base eigenvalue repeated 16 times.

Evidence M. Hard: S26-A-00004723, S26-E-00004739.

Connection-Laplacian recognition spectrum

S26-D-00004741 - Connection-Laplacian recognition spectrum

A finite connection Laplacian penalizes disagreement after transporting neighboring fiber states into comparable frames.

Evidence A. Assumptions: A finite cover/graph, fiber dimensions and transport matrices on edges are declared. Hard: S26-E-00004740, S26-D-00001761.

S26-T-00004742 - Connection-Laplacian recognition spectrum law/interface

A small residual witnesses bounded compatibility in that fixture; it does not prove semantic agreement or global truth.

Evidence A. Assumptions: A finite cover/graph, fiber dimensions and transport matrices on edges are declared. Hard: S26-D-00004741.

S26-A-00004743 - Connection-Laplacian recognition spectrum operator

L_conn is assembled blockwise from edge transports and vertex weights in the finite probe.

Evidence S. Assumptions: A finite cover/graph, fiber dimensions and transport matrices on edges are declared. Hard: S26-T-00004742.

S26-N-00004744 - Connection-Laplacian recognition spectrum fence

Spectral gap, synchronization and recognition are separate observables unless an explicit theorem relates them.

Evidence S. Assumptions: A finite cover/graph, fiber dimensions and transport matrices on edges are declared. Hard: S26-A-00004743. Boundary: Spectral gap, synchronization and recognition are separate observables unless an explicit theorem relates them.

S26-B-00004745 - Connection-Laplacian recognition spectrum cross-volume bridge

This is the measurable gluing surface between sheaf overlap, recognition and spectral diagnostics.

Evidence S. Assumptions: A finite cover/graph, fiber dimensions and transport matrices on edges are declared. Hard: S26-N-00004744.

S26-N-00004746 - Connection-Laplacian recognition spectrum evidence ledger

Evidence rule for Connection-Laplacian recognition spectrum: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A finite cover/graph, fiber dimensions and transport matrices on edges are declared. Hard: S26-B-00004745.

S26-B-00004747 - Connection-Laplacian recognition spectrum emergence rule

Emergence rule for Connection-Laplacian recognition spectrum: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A finite cover/graph, fiber dimensions and transport matrices on edges are declared. Hard: S26-N-00004746.

S26-N-00004748 - Connection-Laplacian recognition spectrum frontier

Open frontier for Connection-Laplacian recognition spectrum: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A finite cover/graph, fiber dimensions and transport matrices on edges are declared. Hard: S26-B-00004747. Boundary: Spectral gap, synchronization and recognition are separate observables unless an explicit theorem relates them.

S26-E-00004749 - Connection-Laplacian finite probe 35.1

Fixture T35.P01 - A 3-vertex rank-2 cycle connection fixture has minimum eigenvalue 4.898e-03; transport angle theta=0.0700.

Evidence M. Hard: S26-A-00004743, S26-N-00004748.

S26-E-00004750 - Connection-Laplacian finite probe 35.2

Fixture T35.P02 - A 4-vertex rank-2 cycle connection fixture has minimum eigenvalue 1.957e-02; transport angle theta=0.1400.

Evidence M. Hard: S26-A-00004743, S26-E-00004749.

S26-E-00004751 - Connection-Laplacian finite probe 35.3

Fixture T35.P03 - A 5-vertex rank-2 cycle connection fixture has minimum eigenvalue 4.394e-02; transport angle theta=0.2100.

Evidence M. Hard: S26-A-00004743, S26-E-00004750.

S26-E-00004752 - Connection-Laplacian finite probe 35.4

Fixture T35.P04 - A 6-vertex rank-2 cycle connection fixture has minimum eigenvalue 7.789e-02; transport angle theta=0.2800.

Evidence M. Hard: S26-A-00004743, S26-E-00004751.

S26-E-00004753 - Connection-Laplacian finite probe 35.5

Fixture T35.P05 - A 7-vertex rank-2 cycle connection fixture has minimum eigenvalue 1.213e-01; transport angle theta=0.3500.

Evidence M. Hard: S26-A-00004743, S26-E-00004752.

S26-E-00004754 - Connection-Laplacian finite probe 35.6

Fixture T35.P06 - A 3-vertex rank-2 cycle connection fixture has minimum eigenvalue 1.738e-01; transport angle theta=0.4200.

Evidence M. Hard: S26-A-00004743, S26-E-00004753.

S26-E-00004755 - Connection-Laplacian finite probe 35.7

Fixture T35.P07 - A 4-vertex rank-2 cycle connection fixture has minimum eigenvalue 2.353e-01; transport angle theta=0.4900.

Evidence M. Hard: S26-A-00004743, S26-E-00004754.

S26-E-00004756 - Connection-Laplacian finite probe 35.8

Fixture T35.P08 - A 5-vertex rank-2 cycle connection fixture has minimum eigenvalue 3.055e-01; transport angle theta=0.5600.

Evidence M. Hard: S26-A-00004743, S26-E-00004755.

S26-E-00004757 - Connection-Laplacian finite probe 35.9

Fixture T35.P09 - A 6-vertex rank-2 cycle connection fixture has minimum eigenvalue 1.715e-01; transport angle theta=0.6300.

Evidence M. Hard: S26-A-00004743, S26-E-00004756.

S26-E-00004758 - Connection-Laplacian finite probe 35.10

Fixture T35.P10 - A 7-vertex rank-2 cycle connection fixture has minimum eigenvalue 3.892e-02; transport angle theta=0.7000.

Evidence M. Hard: S26-A-00004743, S26-E-00004757.

S26-E-00004759 - Connection-Laplacian finite probe 35.11

Fixture T35.P11 - A 3-vertex rank-2 cycle connection fixture has minimum eigenvalue 5.642e-01; transport angle theta=0.7700.

Evidence M. Hard: S26-A-00004743, S26-E-00004758.

S26-E-00004760 - Connection-Laplacian finite probe 35.12

Fixture T35.P12 - A 4-vertex rank-2 cycle connection fixture has minimum eigenvalue 5.107e-01; transport angle theta=0.8400.

Evidence M. Hard: S26-A-00004743, S26-E-00004759.

Spectral-recursive arithmetic coupling

S26-D-00004761 - Spectral-recursive arithmetic coupling

A spectral-recursive arithmetic coupling records commutators among recursion R, spectral plug-in Pi and place transport T_v rather than assuming they commute.

Evidence S. Assumptions: Recursion, place transport and spectral operator act on compatible declared domains. Hard: S26-E-00004760, S26-D-00004052.

S26-D-00004762 - Spectral-recursive arithmetic coupling law/interface

Order sensitivity is retained as a residual tensor/vector and may differ by arithmetic place.

Evidence S. Assumptions: Recursion, place transport and spectral operator act on compatible declared domains. Hard: S26-D-00004761.

S26-A-00004763 - Spectral-recursive arithmetic coupling operator

CouplingResidual=(||[Pi,R]||,||[Pi,T_v]||,||[R,T_v]||) in a bounded matrix fixture.

Evidence S. Assumptions: Recursion, place transport and spectral operator act on compatible declared domains. Hard: S26-D-00004762.

S26-N-00004764 - Spectral-recursive arithmetic coupling fence

Low commutator residual is compatibility evidence, not a cognition theorem.

Evidence S. Assumptions: Recursion, place transport and spectral operator act on compatible declared domains. Hard: S26-A-00004763. Boundary: Low commutator residual is compatibility evidence, not a cognition theorem.

S26-B-00004765 - Spectral-recursive arithmetic coupling cross-volume bridge

This joins algebraic, spectral, recursive and arithmetic coordinates without flattening them.

Evidence S. Assumptions: Recursion, place transport and spectral operator act on compatible declared domains. Hard: S26-N-00004764.

S26-N-00004766 - Spectral-recursive arithmetic coupling evidence ledger

Evidence rule for Spectral-recursive arithmetic coupling: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Recursion, place transport and spectral operator act on compatible declared domains. Hard: S26-B-00004765.

S26-B-00004767 - Spectral-recursive arithmetic coupling emergence rule

Emergence rule for Spectral-recursive arithmetic coupling: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Recursion, place transport and spectral operator act on compatible declared domains. Hard: S26-N-00004766.

S26-N-00004768 - Spectral-recursive arithmetic coupling frontier

Open frontier for Spectral-recursive arithmetic coupling: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Recursion, place transport and spectral operator act on compatible declared domains. Hard: S26-B-00004767. Boundary: Low commutator residual is compatibility evidence, not a cognition theorem.

S26-E-00004769 - Spectral-recursive commutator probe 36.1

Fixture T36.P01 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.2, 1.414214, 0.4); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-N-00004768.

S26-E-00004770 - Spectral-recursive commutator probe 36.2

Fixture T36.P02 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.21, 1.414214, 0.42); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004769.

S26-E-00004771 - Spectral-recursive commutator probe 36.3

Fixture T36.P03 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.22, 1.414214, 0.44); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004770.

S26-E-00004772 - Spectral-recursive commutator probe 36.4

Fixture T36.P04 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.23, 1.414214, 0.46); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004771.

S26-E-00004773 - Spectral-recursive commutator probe 36.5

Fixture T36.P05 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.24, 1.414214, 0.48); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004772.

S26-E-00004774 - Spectral-recursive commutator probe 36.6

Fixture T36.P06 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.25, 1.414214, 0.5); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004773.

S26-E-00004775 - Spectral-recursive commutator probe 36.7

Fixture T36.P07 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.26, 1.414214, 0.52); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004774.

S26-E-00004776 - Spectral-recursive commutator probe 36.8

Fixture T36.P08 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.27, 1.414214, 0.54); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004775.

S26-E-00004777 - Spectral-recursive commutator probe 36.9

Fixture T36.P09 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.28, 1.414214, 0.56); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004776.

S26-E-00004778 - Spectral-recursive commutator probe 36.10

Fixture T36.P10 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.29, 1.414214, 0.58); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004777.

S26-E-00004779 - Spectral-recursive commutator probe 36.11

Fixture T36.P11 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.3, 1.414214, 0.6); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004778.

S26-E-00004780 - Spectral-recursive commutator probe 36.12

Fixture T36.P12 - Finite 2x2 fixture residuals (||[Pi,R]||,||[Pi,T]||,||[R,T]||)=(0.31, 1.414214, 0.62); all three coordinates are retained.

Evidence M. Hard: S26-A-00004763, S26-E-00004779.

Part V - n/k/place/epsilon continuum and noncommuting-limit fences

Geometric n-to-infinity pro-Cosmo tower

S26-D-00004781 - Geometric n-to-infinity pro-Cosmo tower

The n->infinity direction is modeled by a coherent inverse/pro system of Cosmo carriers rather than an infinite Euclidean coordinate vector.

Evidence S. Assumptions: Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved. Hard: S26-E-00004780, S26-D-00004055.

S26-D-00004782 - Geometric n-to-infinity pro-Cosmo tower law/interface

Finite-dimensional properties need not survive the inverse limit without compactness/coherence hypotheses.

Evidence S. Assumptions: Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved. Hard: S26-D-00004781.

S26-A-00004783 - Geometric n-to-infinity pro-Cosmo tower operator

ProCosmo(N) stores the finite truncation, bonding maps and residual obligations at level N.

Evidence S. Assumptions: Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved. Hard: S26-D-00004782.

S26-N-00004784 - Geometric n-to-infinity pro-Cosmo tower fence

n->infinity is geometric/tower depth and is distinct from Cayley-Dickson depth k and arithmetic place p.

Evidence S. Assumptions: Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved. Hard: S26-A-00004783. Boundary: n->infinity is geometric/tower depth and is distinct from Cayley-Dickson depth k and arithmetic place p.

S26-B-00004785 - Geometric n-to-infinity pro-Cosmo tower cross-volume bridge

This formalizes the outward direction drawn in the visual atlas.

Evidence S. Assumptions: Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved. Hard: S26-N-00004784.

S26-N-00004786 - Geometric n-to-infinity pro-Cosmo tower evidence ledger

Evidence rule for Geometric n-to-infinity pro-Cosmo tower: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved. Hard: S26-B-00004785.

S26-B-00004787 - Geometric n-to-infinity pro-Cosmo tower emergence rule

Emergence rule for Geometric n-to-infinity pro-Cosmo tower: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved. Hard: S26-N-00004786.

S26-N-00004788 - Geometric n-to-infinity pro-Cosmo tower frontier

Open frontier for Geometric n-to-infinity pro-Cosmo tower: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Bonding maps and topology are declared; inverse limits are called pro-spaces unless stronger manifold/orbifold structure is proved. Hard: S26-B-00004787. Boundary: n->infinity is geometric/tower depth and is distinct from Cayley-Dickson depth k and arithmetic place p.

S26-E-00004789 - Pro-Cosmo truncation probe 37.1

Fixture T37.P01 - Finite tower truncation N=4 records geometric dimensions [2, 3, 4, 5]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-N-00004788.

S26-E-00004790 - Pro-Cosmo truncation probe 37.2

Fixture T37.P02 - Finite tower truncation N=5 records geometric dimensions [2, 3, 4, 5, 6]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004789.

S26-E-00004791 - Pro-Cosmo truncation probe 37.3

Fixture T37.P03 - Finite tower truncation N=6 records geometric dimensions [2, 3, 4, 5, 6, 7]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004790.

S26-E-00004792 - Pro-Cosmo truncation probe 37.4

Fixture T37.P04 - Finite tower truncation N=7 records geometric dimensions [2, 3, 4, 5, 6, 7, 8]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004791.

S26-E-00004793 - Pro-Cosmo truncation probe 37.5

Fixture T37.P05 - Finite tower truncation N=8 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004792.

S26-E-00004794 - Pro-Cosmo truncation probe 37.6

Fixture T37.P06 - Finite tower truncation N=9 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004793.

S26-E-00004795 - Pro-Cosmo truncation probe 37.7

Fixture T37.P07 - Finite tower truncation N=10 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004794.

S26-E-00004796 - Pro-Cosmo truncation probe 37.8

Fixture T37.P08 - Finite tower truncation N=11 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004795.

S26-E-00004797 - Pro-Cosmo truncation probe 37.9

Fixture T37.P09 - Finite tower truncation N=12 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004796.

S26-E-00004798 - Pro-Cosmo truncation probe 37.10

Fixture T37.P10 - Finite tower truncation N=13 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004797.

S26-E-00004799 - Pro-Cosmo truncation probe 37.11

Fixture T37.P11 - Finite tower truncation N=14 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004798.

S26-E-00004800 - Pro-Cosmo truncation probe 37.12

Fixture T37.P12 - Finite tower truncation N=15 records geometric dimensions [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16]; no infinite-dimensional manifold structure is inferred from the list.

Evidence M. Hard: S26-A-00004783, S26-E-00004799.

Algebraic k-to-infinity Cayley-Dickson tower

S26-D-00004801 - Algebraic k-to-infinity Cayley-Dickson tower

The k->infinity direction is an inductive algebraic tower of finite Cayley-Dickson stages A_{0,v}->A_{1,v}->..., treated as a limit object only after a topology/norm/completion is specified.

Evidence S. Assumptions: A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared. Hard: S26-E-00004800, S26-D-00000058.

S26-D-00004802 - Algebraic k-to-infinity Cayley-Dickson tower law/interface

Dimension doubles as 2^k at finite stage, while algebraic law loss must be tracked rather than assumed to stabilize.

Evidence S. Assumptions: A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared. Hard: S26-D-00004801.

S26-A-00004803 - Algebraic k-to-infinity Cayley-Dickson tower operator

CDTower(K) records finite stages, inclusions, law-survival vector and chosen topology if any.

Evidence S. Assumptions: A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared. Hard: S26-D-00004802.

S26-N-00004804 - Algebraic k-to-infinity Cayley-Dickson tower fence

The infinite union is not automatically a Banach algebra, division algebra or cognitive continuum.

Evidence S. Assumptions: A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared. Hard: S26-A-00004803. Boundary: The infinite union is not automatically a Banach algebra, division algebra or cognitive continuum.

S26-B-00004805 - Algebraic k-to-infinity Cayley-Dickson tower cross-volume bridge

This is the algebraic Russian-doll direction independent of n.

Evidence S. Assumptions: A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared. Hard: S26-N-00004804.

S26-N-00004806 - Algebraic k-to-infinity Cayley-Dickson tower evidence ledger

Evidence rule for Algebraic k-to-infinity Cayley-Dickson tower: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared. Hard: S26-B-00004805.

S26-B-00004807 - Algebraic k-to-infinity Cayley-Dickson tower emergence rule

Emergence rule for Algebraic k-to-infinity Cayley-Dickson tower: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared. Hard: S26-N-00004806.

S26-N-00004808 - Algebraic k-to-infinity Cayley-Dickson tower frontier

Open frontier for Algebraic k-to-infinity Cayley-Dickson tower: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A coherent sequence of embeddings/coordinate inclusions and doubling parameters is declared. Hard: S26-B-00004807. Boundary: The infinite union is not automatically a Banach algebra, division algebra or cognitive continuum.

S26-E-00004809 - Cayley-Dickson tower probe 38.1

Fixture T38.P01 - Finite algebraic depth K=3 has coordinate dimensions [1, 2, 4, 8]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-N-00004808.

S26-E-00004810 - Cayley-Dickson tower probe 38.2

Fixture T38.P02 - Finite algebraic depth K=4 has coordinate dimensions [1, 2, 4, 8, 16]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004809.

S26-E-00004811 - Cayley-Dickson tower probe 38.3

Fixture T38.P03 - Finite algebraic depth K=5 has coordinate dimensions [1, 2, 4, 8, 16, 32]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004810.

S26-E-00004812 - Cayley-Dickson tower probe 38.4

Fixture T38.P04 - Finite algebraic depth K=6 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004811.

S26-E-00004813 - Cayley-Dickson tower probe 38.5

Fixture T38.P05 - Finite algebraic depth K=7 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64, 128]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004812.

S26-E-00004814 - Cayley-Dickson tower probe 38.6

Fixture T38.P06 - Finite algebraic depth K=8 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64, 128, 256]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004813.

S26-E-00004815 - Cayley-Dickson tower probe 38.7

Fixture T38.P07 - Finite algebraic depth K=9 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64, 128, 256, 512]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004814.

S26-E-00004816 - Cayley-Dickson tower probe 38.8

Fixture T38.P08 - Finite algebraic depth K=3 has coordinate dimensions [1, 2, 4, 8]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004815.

S26-E-00004817 - Cayley-Dickson tower probe 38.9

Fixture T38.P09 - Finite algebraic depth K=4 has coordinate dimensions [1, 2, 4, 8, 16]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004816.

S26-E-00004818 - Cayley-Dickson tower probe 38.10

Fixture T38.P10 - Finite algebraic depth K=5 has coordinate dimensions [1, 2, 4, 8, 16, 32]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004817.

S26-E-00004819 - Cayley-Dickson tower probe 38.11

Fixture T38.P11 - Finite algebraic depth K=6 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004818.

S26-E-00004820 - Cayley-Dickson tower probe 38.12

Fixture T38.P12 - Finite algebraic depth K=7 has coordinate dimensions [1, 2, 4, 8, 16, 32, 64, 128]; the doubling law is recorded while infinite completion remains unspecified.

Evidence M. Hard: S26-A-00004803, S26-E-00004819.

Adelic place-and-epsilon local limit

S26-D-00004821 - Adelic place-and-epsilon local limit

The local-limit atlas tracks v over arithmetic places together with epsilon->0 inside each local chart, producing a family of local tangent/ultrametric regimes rather than one universal infinitesimal metric.

Evidence S. Assumptions: A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry. Hard: S26-E-00004820, S26-D-00004046.

S26-D-00004822 - Adelic place-and-epsilon local limit law/interface

The Archimedean tangent limit and p-adic nested-ball refinement are different constructions and retain different remainder coordinates.

Evidence S. Assumptions: A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry. Hard: S26-D-00004821.

S26-A-00004823 - Adelic place-and-epsilon local limit operator

LocalLimit(v,epsilon) returns the declared local scale object plus transition obligations to neighboring places.

Evidence S. Assumptions: A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry. Hard: S26-D-00004822.

S26-N-00004824 - Adelic place-and-epsilon local limit fence

p->infinity is not used as a substitute for the Archimedean place; arithmetic places are labels, not a Euclidean axis.

Evidence S. Assumptions: A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry. Hard: S26-A-00004823. Boundary: p->infinity is not used as a substitute for the Archimedean place; arithmetic places are labels, not a Euclidean axis.

S26-B-00004825 - Adelic place-and-epsilon local limit cross-volume bridge

This gives the inward epsilon direction an arithmetic local-global companion.

Evidence S. Assumptions: A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry. Hard: S26-N-00004824.

S26-N-00004826 - Adelic place-and-epsilon local limit evidence ledger

Evidence rule for Adelic place-and-epsilon local limit: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry. Hard: S26-B-00004825.

S26-B-00004827 - Adelic place-and-epsilon local limit emergence rule

Emergence rule for Adelic place-and-epsilon local limit: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry. Hard: S26-N-00004826.

S26-N-00004828 - Adelic place-and-epsilon local limit frontier

Open frontier for Adelic place-and-epsilon local limit: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A finite set of active places is used at each bounded stage, and epsilon zoom is defined within each local geometry. Hard: S26-B-00004827. Boundary: p->infinity is not used as a substitute for the Archimedean place; arithmetic places are labels, not a Euclidean axis.

S26-E-00004829 - Place-epsilon local-limit probe 39.1

Fixture T39.P01 - For x=2/3, finite place norms {'2': 0.5, '3': 3.0, '5': 1.0} are stored alongside epsilon=0.25000000; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-N-00004828.

S26-E-00004830 - Place-epsilon local-limit probe 39.2

Fixture T39.P02 - For x=3/4, finite place norms {'2': 4.0, '3': 0.3333333333333333, '5': 1.0} are stored alongside epsilon=0.12500000; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004829.

S26-E-00004831 - Place-epsilon local-limit probe 39.3

Fixture T39.P03 - For x=4/5, finite place norms {'2': 0.25, '3': 1.0, '5': 5.0} are stored alongside epsilon=0.06250000; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004830.

S26-E-00004832 - Place-epsilon local-limit probe 39.4

Fixture T39.P04 - For x=5/6, finite place norms {'2': 2.0, '3': 3.0, '5': 0.2} are stored alongside epsilon=0.03125000; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004831.

S26-E-00004833 - Place-epsilon local-limit probe 39.5

Fixture T39.P05 - For x=6/7, finite place norms {'2': 0.5, '3': 0.3333333333333333, '5': 1.0} are stored alongside epsilon=0.01562500; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004832.

S26-E-00004834 - Place-epsilon local-limit probe 39.6

Fixture T39.P06 - For x=7/8, finite place norms {'2': 8.0, '3': 1.0, '5': 1.0} are stored alongside epsilon=0.00781250; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004833.

S26-E-00004835 - Place-epsilon local-limit probe 39.7

Fixture T39.P07 - For x=8/9, finite place norms {'2': 0.125, '3': 9.0, '5': 1.0} are stored alongside epsilon=0.00390625; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004834.

S26-E-00004836 - Place-epsilon local-limit probe 39.8

Fixture T39.P08 - For x=9/10, finite place norms {'2': 2.0, '3': 0.1111111111111111, '5': 5.0} are stored alongside epsilon=0.00195312; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004835.

S26-E-00004837 - Place-epsilon local-limit probe 39.9

Fixture T39.P09 - For x=10/11, finite place norms {'2': 0.5, '3': 1.0, '5': 0.2} are stored alongside epsilon=0.00097656; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004836.

S26-E-00004838 - Place-epsilon local-limit probe 39.10

Fixture T39.P10 - For x=11/12, finite place norms {'2': 4.0, '3': 3.0, '5': 1.0} are stored alongside epsilon=0.00048828; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004837.

S26-E-00004839 - Place-epsilon local-limit probe 39.11

Fixture T39.P11 - For x=12/13, finite place norms {'2': 0.25, '3': 0.3333333333333333, '5': 1.0} are stored alongside epsilon=0.00024414; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004838.

S26-E-00004840 - Place-epsilon local-limit probe 39.12

Fixture T39.P12 - For x=13/14, finite place norms {'2': 2.0, '3': 1.0, '5': 1.0} are stored alongside epsilon=0.00012207; no p-axis metric is imposed.

Evidence M. Hard: S26-A-00004823, S26-E-00004839.

Noncommuting multi-limit holonomy fence

S26-D-00004841 - Noncommuting multi-limit holonomy fence

For limits in n,k,v/places,j and epsilon, Volume IV records order-of-limits holonomy whenever two iterated procedures are not proved equivalent.

Evidence S. Assumptions: Each limiting process has a declared topology and comparison observable. Hard: S26-E-00004840, S26-N-00004059.

S26-D-00004842 - Noncommuting multi-limit holonomy fence law/interface

Equality of finite diagrams or visual similarity does not imply interchange of limits.

Evidence S. Assumptions: Each limiting process has a declared topology and comparison observable. Hard: S26-D-00004841.

S26-A-00004843 - Noncommuting multi-limit holonomy fence operator

LimitHolonomy(order1,order2) compares typed observables returned by the two truncated procedures.

Evidence S. Assumptions: Each limiting process has a declared topology and comparison observable. Hard: S26-D-00004842.

S26-N-00004844 - Noncommuting multi-limit holonomy fence fence

A nonzero order residual is retained; it is not averaged away to manufacture a single continuum.

Evidence S. Assumptions: Each limiting process has a declared topology and comparison observable. Hard: S26-A-00004843. Boundary: A nonzero order residual is retained; it is not averaged away to manufacture a single continuum.

S26-B-00004845 - Noncommuting multi-limit holonomy fence cross-volume bridge

This is the governance rule preventing the hyperdimensional model from collapsing back into a one-axis narrative.

Evidence S. Assumptions: Each limiting process has a declared topology and comparison observable. Hard: S26-N-00004844.

S26-N-00004846 - Noncommuting multi-limit holonomy fence evidence ledger

Evidence rule for Noncommuting multi-limit holonomy fence: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Each limiting process has a declared topology and comparison observable. Hard: S26-B-00004845.

S26-B-00004847 - Noncommuting multi-limit holonomy fence emergence rule

Emergence rule for Noncommuting multi-limit holonomy fence: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Each limiting process has a declared topology and comparison observable. Hard: S26-N-00004846.

S26-N-00004848 - Noncommuting multi-limit holonomy fence frontier

Open frontier for Noncommuting multi-limit holonomy fence: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Each limiting process has a declared topology and comparison observable. Hard: S26-B-00004847. Boundary: A nonzero order residual is retained; it is not averaged away to manufacture a single continuum.

S26-E-00004849 - Order-of-limits surrogate probe 40.1

Fixture T40.P01 - At truncation n=4, two order-surrogate matrices have commutator residual 0.08838835; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-N-00004848.

S26-E-00004850 - Order-of-limits surrogate probe 40.2

Fixture T40.P02 - At truncation n=5, two order-surrogate matrices have commutator residual 0.05656854; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004849.

S26-E-00004851 - Order-of-limits surrogate probe 40.3

Fixture T40.P03 - At truncation n=6, two order-surrogate matrices have commutator residual 0.03928371; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004850.

S26-E-00004852 - Order-of-limits surrogate probe 40.4

Fixture T40.P04 - At truncation n=7, two order-surrogate matrices have commutator residual 0.02886150; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004851.

S26-E-00004853 - Order-of-limits surrogate probe 40.5

Fixture T40.P05 - At truncation n=8, two order-surrogate matrices have commutator residual 0.02209709; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004852.

S26-E-00004854 - Order-of-limits surrogate probe 40.6

Fixture T40.P06 - At truncation n=9, two order-surrogate matrices have commutator residual 0.01745943; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004853.

S26-E-00004855 - Order-of-limits surrogate probe 40.7

Fixture T40.P07 - At truncation n=10, two order-surrogate matrices have commutator residual 0.01414214; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004854.

S26-E-00004856 - Order-of-limits surrogate probe 40.8

Fixture T40.P08 - At truncation n=11, two order-surrogate matrices have commutator residual 0.01168772; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004855.

S26-E-00004857 - Order-of-limits surrogate probe 40.9

Fixture T40.P09 - At truncation n=12, two order-surrogate matrices have commutator residual 0.00982093; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004856.

S26-E-00004858 - Order-of-limits surrogate probe 40.10

Fixture T40.P10 - At truncation n=13, two order-surrogate matrices have commutator residual 0.00836813; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004857.

S26-E-00004859 - Order-of-limits surrogate probe 40.11

Fixture T40.P11 - At truncation n=14, two order-surrogate matrices have commutator residual 0.00721538; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004858.

S26-E-00004860 - Order-of-limits surrogate probe 40.12

Fixture T40.P12 - At truncation n=15, two order-surrogate matrices have commutator residual 0.00628539; the nonzero order signal is retained.

Evidence M. Hard: S26-A-00004843, S26-E-00004859.

Part W - Russian-doll hypercomplex emergence from previous volumes

LiftCosmos hypercomplex-inside-hypercomplex operator

S26-D-00004861 - LiftCosmos hypercomplex-inside-hypercomplex operator

LiftCosmos maps a local regime X_i with projection and remainder to a wider provisional regime X_{i+1} that explains more compatible projections without erasing X_i.

Evidence S. Assumptions: A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map. Hard: S26-E-00004860, S26-D-00004040.

S26-D-00004862 - LiftCosmos hypercomplex-inside-hypercomplex operator law/interface

Hypercomplex-inside-hypercomplex counts as perspective gain only when it adds testable structure and lowers projection violence without claiming a final level.

Evidence S. Assumptions: A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map. Hard: S26-D-00004861.

S26-A-00004863 - LiftCosmos hypercomplex-inside-hypercomplex operator operator

PerspectiveGain is a typed vector of gluing improvement, projection loss, remainder richness, failure discrimination and evidence hooks.

Evidence S. Assumptions: A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map. Hard: S26-D-00004862.

S26-N-00004864 - LiftCosmos hypercomplex-inside-hypercomplex operator fence

Nesting a flat dashboard inside another flat dashboard is recursive flattening, not a cosmos lift.

Evidence S. Assumptions: A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map. Hard: S26-A-00004863. Boundary: Nesting a flat dashboard inside another flat dashboard is recursive flattening, not a cosmos lift.

S26-B-00004865 - LiftCosmos hypercomplex-inside-hypercomplex operator cross-volume bridge

This is the formal Russian-doll rule governing the nested dodecahedral images and the higher categorical atlas.

Evidence S. Assumptions: A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map. Hard: S26-N-00004864.

S26-N-00004866 - LiftCosmos hypercomplex-inside-hypercomplex operator evidence ledger

Evidence rule for LiftCosmos hypercomplex-inside-hypercomplex operator: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map. Hard: S26-B-00004865.

S26-B-00004867 - LiftCosmos hypercomplex-inside-hypercomplex operator emergence rule

Emergence rule for LiftCosmos hypercomplex-inside-hypercomplex operator: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map. Hard: S26-N-00004866.

S26-N-00004868 - LiftCosmos hypercomplex-inside-hypercomplex operator frontier

Open frontier for LiftCosmos hypercomplex-inside-hypercomplex operator: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: A lift must add at least one measurable algebraic, geometric, phase, gluing, evidence or remainder degree of freedom and preserve a return map. Hard: S26-B-00004867. Boundary: Nesting a flat dashboard inside another flat dashboard is recursive flattening, not a cosmos lift.

S26-E-00004869 - LiftCosmos perspective probe 41.1

Fixture T41.P01 - At synthetic lift depth 1 with contraction 0.55, visible carrier coordinate=0.55000000 and active remainder=0.22500000; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-N-00004868.

S26-E-00004870 - LiftCosmos perspective probe 41.2

Fixture T41.P02 - At synthetic lift depth 2 with contraction 0.57, visible carrier coordinate=0.32490000 and active remainder=0.14333333; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004869.

S26-E-00004871 - LiftCosmos perspective probe 41.3

Fixture T41.P03 - At synthetic lift depth 3 with contraction 0.59, visible carrier coordinate=0.20537900 and active remainder=0.10250000; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004870.

S26-E-00004872 - LiftCosmos perspective probe 41.4

Fixture T41.P04 - At synthetic lift depth 4 with contraction 0.61, visible carrier coordinate=0.13845841 and active remainder=0.07800000; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004871.

S26-E-00004873 - LiftCosmos perspective probe 41.5

Fixture T41.P05 - At synthetic lift depth 5 with contraction 0.63, visible carrier coordinate=0.09924365 and active remainder=0.06166667; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004872.

S26-E-00004874 - LiftCosmos perspective probe 41.6

Fixture T41.P06 - At synthetic lift depth 6 with contraction 0.55, visible carrier coordinate=0.02768064 and active remainder=0.06428571; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004873.

S26-E-00004875 - LiftCosmos perspective probe 41.7

Fixture T41.P07 - At synthetic lift depth 7 with contraction 0.57, visible carrier coordinate=0.01954897 and active remainder=0.05375000; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004874.

S26-E-00004876 - LiftCosmos perspective probe 41.8

Fixture T41.P08 - At synthetic lift depth 8 with contraction 0.59, visible carrier coordinate=0.01468304 and active remainder=0.04555556; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004875.

S26-E-00004877 - LiftCosmos perspective probe 41.9

Fixture T41.P09 - At synthetic lift depth 9 with contraction 0.61, visible carrier coordinate=0.01169415 and active remainder=0.03900000; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004876.

S26-E-00004878 - LiftCosmos perspective probe 41.10

Fixture T41.P10 - At synthetic lift depth 10 with contraction 0.63, visible carrier coordinate=0.00984930 and active remainder=0.03363636; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004877.

S26-E-00004879 - LiftCosmos perspective probe 41.11

Fixture T41.P11 - At synthetic lift depth 11 with contraction 0.55, visible carrier coordinate=0.00139312 and active remainder=0.03750000; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004878.

S26-E-00004880 - LiftCosmos perspective probe 41.12

Fixture T41.P12 - At synthetic lift depth 12 with contraction 0.57, visible carrier coordinate=0.00117625 and active remainder=0.03307692; neither is discarded.

Evidence M. Hard: S26-A-00004863, S26-E-00004879.

n-caged n-Cosmo gerbe enclosure

S26-D-00004881 - n-caged n-Cosmo gerbe enclosure

An n-cage is a finite observational enclosure K_i inside a Cosmo regime carrying local hypercomplex fibers and gerbe/gluing data; cages may nest without being concentric Euclidean shells.

Evidence S. Assumptions: Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime. Hard: S26-E-00004880, S26-T-00004015.

S26-D-00004882 - n-caged n-Cosmo gerbe enclosure law/interface

Nested cages preserve source addresses and boundary interfaces so inner structures can be transported back to outer regimes.

Evidence S. Assumptions: Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime. Hard: S26-D-00004881.

S26-A-00004883 - n-caged n-Cosmo gerbe enclosure operator

CageNest(K_0,...,K_r) records inclusion/correspondence, boundary transport and unresolved higher cocycles.

Evidence S. Assumptions: Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime. Hard: S26-D-00004882.

S26-N-00004884 - n-caged n-Cosmo gerbe enclosure fence

A cage is an observational finite carrier, not an ontological claim that the cosmos has a literal polyhedral wall.

Evidence S. Assumptions: Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime. Hard: S26-A-00004883. Boundary: A cage is an observational finite carrier, not an ontological claim that the cosmos has a literal polyhedral wall.

S26-B-00004885 - n-caged n-Cosmo gerbe enclosure cross-volume bridge

This gives a typed realization of n-caged n-Cosmo n intermanifold nesting.

Evidence S. Assumptions: Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime. Hard: S26-N-00004884.

S26-N-00004886 - n-caged n-Cosmo gerbe enclosure evidence ledger

Evidence rule for n-caged n-Cosmo gerbe enclosure: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime. Hard: S26-B-00004885.

S26-B-00004887 - n-caged n-Cosmo gerbe enclosure emergence rule

Emergence rule for n-caged n-Cosmo gerbe enclosure: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime. Hard: S26-N-00004886.

S26-N-00004888 - n-caged n-Cosmo gerbe enclosure frontier

Open frontier for n-caged n-Cosmo gerbe enclosure: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Each cage is a finite addressable subcomplex/cover with declared boundary, local gauges and embedding/correspondence into its parent regime. Hard: S26-B-00004887. Boundary: A cage is an observational finite carrier, not an ontological claim that the cosmos has a literal polyhedral wall.

S26-E-00004889 - n-cage nesting probe 42.1

Fixture T42.P01 - A finite Russian-doll cage fixture has level counts [1, 3, 12]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-N-00004888.

S26-E-00004890 - n-cage nesting probe 42.2

Fixture T42.P02 - A finite Russian-doll cage fixture has level counts [1, 4, 8, 24]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004889.

S26-E-00004891 - n-cage nesting probe 42.3

Fixture T42.P03 - A finite Russian-doll cage fixture has level counts [1, 2, 6, 24, 48]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004890.

S26-E-00004892 - n-cage nesting probe 42.4

Fixture T42.P04 - A finite Russian-doll cage fixture has level counts [1, 3, 12, 24, 72, 288]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004891.

S26-E-00004893 - n-cage nesting probe 42.5

Fixture T42.P05 - A finite Russian-doll cage fixture has level counts [1, 4, 8]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004892.

S26-E-00004894 - n-cage nesting probe 42.6

Fixture T42.P06 - A finite Russian-doll cage fixture has level counts [1, 2, 6, 24]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004893.

S26-E-00004895 - n-cage nesting probe 42.7

Fixture T42.P07 - A finite Russian-doll cage fixture has level counts [1, 3, 12, 24, 72]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004894.

S26-E-00004896 - n-cage nesting probe 42.8

Fixture T42.P08 - A finite Russian-doll cage fixture has level counts [1, 4, 8, 24, 96, 192]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004895.

S26-E-00004897 - n-cage nesting probe 42.9

Fixture T42.P09 - A finite Russian-doll cage fixture has level counts [1, 2, 6]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004896.

S26-E-00004898 - n-cage nesting probe 42.10

Fixture T42.P10 - A finite Russian-doll cage fixture has level counts [1, 3, 12, 24]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004897.

S26-E-00004899 - n-cage nesting probe 42.11

Fixture T42.P11 - A finite Russian-doll cage fixture has level counts [1, 4, 8, 24, 96]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004898.

S26-E-00004900 - n-cage nesting probe 42.12

Fixture T42.P12 - A finite Russian-doll cage fixture has level counts [1, 2, 6, 24, 48, 144]; every inner address has one recorded parent pointer and no Euclidean radius is used as ontology.

Evidence M. Hard: S26-A-00004883, S26-E-00004899.

Previous-volume emergence profunctor

S26-D-00004901 - Previous-volume emergence profunctor

The emergence profunctor P(V_i,V_j) records typed cross-volume applicability between prior canonical objects and candidate Volume IV constructions when a direct functor would be too restrictive.

Evidence S. Assumptions: Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten. Hard: S26-E-00004900, S26-D-00000028.

S26-D-00004902 - Previous-volume emergence profunctor law/interface

A new Volume IV object is admitted only when the cross-volume composite adds an operator, distinction, obstruction, evidence hook or active remainder not already owned by a frozen ID.

Evidence S. Assumptions: Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten. Hard: S26-D-00004901.

S26-A-00004903 - Previous-volume emergence profunctor operator

EmergenceCompose uses dependency spans/profunctors to assemble candidate objects while checking duplicate ownership and source returnability.

Evidence S. Assumptions: Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten. Hard: S26-D-00004902.

S26-N-00004904 - Previous-volume emergence profunctor fence

Novel language without a new typed distinction is rejected as decorative inflation.

Evidence S. Assumptions: Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten. Hard: S26-A-00004903. Boundary: Novel language without a new typed distinction is rejected as decorative inflation.

S26-B-00004905 - Previous-volume emergence profunctor cross-volume bridge

Foundations, algebra, analysis and geometry/topology become simultaneous ingredients of new hypercomplex objects rather than a linear textbook pipeline.

Evidence S. Assumptions: Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten. Hard: S26-N-00004904.

S26-N-00004906 - Previous-volume emergence profunctor evidence ledger

Evidence rule for Previous-volume emergence profunctor: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten. Hard: S26-B-00004905.

S26-B-00004907 - Previous-volume emergence profunctor emergence rule

Emergence rule for Previous-volume emergence profunctor: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten. Hard: S26-N-00004906.

S26-N-00004908 - Previous-volume emergence profunctor frontier

Open frontier for Previous-volume emergence profunctor: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Volumes 0-III remain frozen canonical sources; no old ID is silently rewritten. Hard: S26-B-00004907. Boundary: Novel language without a new typed distinction is rejected as decorative inflation.

S26-E-00004909 - Cross-volume emergence probe 43.1

Fixture T43.P01 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-N-00004908.

S26-E-00004910 - Cross-volume emergence probe 43.2

Fixture T43.P02 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004909.

S26-E-00004911 - Cross-volume emergence probe 43.3

Fixture T43.P03 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004910.

S26-E-00004912 - Cross-volume emergence probe 43.4

Fixture T43.P04 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004911.

S26-E-00004913 - Cross-volume emergence probe 43.5

Fixture T43.P05 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004912.

S26-E-00004914 - Cross-volume emergence probe 43.6

Fixture T43.P06 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004913.

S26-E-00004915 - Cross-volume emergence probe 43.7

Fixture T43.P07 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004914.

S26-E-00004916 - Cross-volume emergence probe 43.8

Fixture T43.P08 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004915.

S26-E-00004917 - Cross-volume emergence probe 43.9

Fixture T43.P09 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004916.

S26-E-00004918 - Cross-volume emergence probe 43.10

Fixture T43.P10 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004917.

S26-E-00004919 - Cross-volume emergence probe 43.11

Fixture T43.P11 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004918.

S26-E-00004920 - Cross-volume emergence probe 43.12

Fixture T43.P12 - Frozen seeds [('S26-D-00000103', 'Field - interface'), ('S26-D-00000268', 'Tensor product - universal property'), ('S26-T-00000672', 'Spectral theorem for bounded self-adjoint operators'), ('S26-D-00003553', 'Vector bundle')] remain source-owned while the candidate synthesis adds a new profunctor/dependency layer; no source ID is rewritten.

Evidence M. Hard: S26-A-00004903, S26-E-00004919.

Novelty coend and non-flattening obstruction

S26-D-00004921 - Novelty coend and non-flattening obstruction

A novelty coend is a project-level construction that quotients only declared equivalent intermediate representations while preserving semantic signs, evidence states and active remainder.

Evidence S. Assumptions: Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry. Hard: S26-E-00004920, S26-N-00004059.

S26-D-00004922 - Novelty coend and non-flattening obstruction law/interface

A candidate is blocked if its apparent novelty disappears under canonical normalization or if it requires erasing a source distinction.

Evidence S. Assumptions: Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry. Hard: S26-D-00004921.

S26-A-00004923 - Novelty coend and non-flattening obstruction operator

NoveltyAudit returns duplicate risk, new invariant set, dependency span, evidence delta and remainder delta.

Evidence S. Assumptions: Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry. Hard: S26-D-00004922.

S26-N-00004924 - Novelty coend and non-flattening obstruction fence

Coend terminology here is a typed synthesis architecture unless the exact indexing categories and quotient relation are supplied for a theorem.

Evidence S. Assumptions: Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry. Hard: S26-A-00004923. Boundary: Coend terminology here is a typed synthesis architecture unless the exact indexing categories and quotient relation are supplied for a theorem.

S26-B-00004925 - Novelty coend and non-flattening obstruction cross-volume bridge

This is the mechanism by which later volumes start from prior volumes hypercomplexly and still produce genuinely new canonical objects.

Evidence S. Assumptions: Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry. Hard: S26-N-00004924.

S26-N-00004926 - Novelty coend and non-flattening obstruction evidence ledger

Evidence rule for Novelty coend and non-flattening obstruction: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry. Hard: S26-B-00004925.

S26-B-00004927 - Novelty coend and non-flattening obstruction emergence rule

Emergence rule for Novelty coend and non-flattening obstruction: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry. Hard: S26-N-00004926.

S26-N-00004928 - Novelty coend and non-flattening obstruction frontier

Open frontier for Novelty coend and non-flattening obstruction: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Candidate synthesis objects retain complete dependency/provenance records and are compared against the frozen registry. Hard: S26-B-00004927. Boundary: Coend terminology here is a typed synthesis architecture unless the exact indexing categories and quotient relation are supplied for a theorem.

S26-E-00004929 - Novelty audit probe 44.1

Fixture T44.P01 - Candidate normalized title "volume iv novelty fixture 0 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 7f5809905423dcfc; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-N-00004928.

S26-E-00004930 - Novelty audit probe 44.2

Fixture T44.P02 - Candidate normalized title "volume iv novelty fixture 1 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest ff3046b77d51fbd6; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004929.

S26-E-00004931 - Novelty audit probe 44.3

Fixture T44.P03 - Candidate normalized title "volume iv novelty fixture 2 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 119302ebc4720c49; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004930.

S26-E-00004932 - Novelty audit probe 44.4

Fixture T44.P04 - Candidate normalized title "volume iv novelty fixture 3 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest f2891fb2a9162114; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004931.

S26-E-00004933 - Novelty audit probe 44.5

Fixture T44.P05 - Candidate normalized title "volume iv novelty fixture 4 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest e456b0b4bf3b5520; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004932.

S26-E-00004934 - Novelty audit probe 44.6

Fixture T44.P06 - Candidate normalized title "volume iv novelty fixture 5 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 2c0730881725110e; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004933.

S26-E-00004935 - Novelty audit probe 44.7

Fixture T44.P07 - Candidate normalized title "volume iv novelty fixture 6 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 5e9444b5aa147b94; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004934.

S26-E-00004936 - Novelty audit probe 44.8

Fixture T44.P08 - Candidate normalized title "volume iv novelty fixture 7 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 0feeb344fef96ce9; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004935.

S26-E-00004937 - Novelty audit probe 44.9

Fixture T44.P09 - Candidate normalized title "volume iv novelty fixture 8 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest f39df66ae52cf536; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004936.

S26-E-00004938 - Novelty audit probe 44.10

Fixture T44.P10 - Candidate normalized title "volume iv novelty fixture 9 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest 39af8991b4dc2430; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004937.

S26-E-00004939 - Novelty audit probe 44.11

Fixture T44.P11 - Candidate normalized title "volume iv novelty fixture 10 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest ae6728b1f44b53dd; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004938.

S26-E-00004940 - Novelty audit probe 44.12

Fixture T44.P12 - Candidate normalized title "volume iv novelty fixture 11 p-adic hypercomplex coend" has frozen-registry exact-duplicate flag=False and digest fdb93b89dc5974b8; this tests only exact ownership collision.

Evidence M. Hard: S26-A-00004923, S26-E-00004939.

Part X - Repo context, cognitive performance and frontier

n.k.alpha context matrix over the repo atlas

S26-D-00004941 - n.k.alpha context matrix over the repo atlas

The observable repo context is represented as a family of n.k.alpha indexed section summaries over artifacts, tasks and peer charts, with Poincare/Fisher-style coordinates used only as a declared modeling projection.

Evidence S. Assumptions: The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought. Hard: S26-E-00004940, S26-N-00004059.

S26-D-00004942 - n.k.alpha context matrix over the repo atlas law/interface

Every claim in the context atlas carries scope, evidence and verdict; hidden internal reasoning is not treated as an inspectable geometric state.

Evidence S. Assumptions: The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought. Hard: S26-D-00004941.

S26-A-00004943 - n.k.alpha context matrix over the repo atlas operator

RepoContextAtlas ingests observable files/tests/registry records and emits bounded section vectors, provenance links and active remainder.

Evidence S. Assumptions: The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought. Hard: S26-D-00004942.

S26-N-00004944 - n.k.alpha context matrix over the repo atlas fence

Treating context geometrically is a model of observable interaction state, not a factual claim about the substrate of the deployed neural network.

Evidence S. Assumptions: The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought. Hard: S26-A-00004943. Boundary: Treating context geometrically is a model of observable interaction state, not a factual claim about the substrate of the deployed neural network.

S26-B-00004945 - n.k.alpha context matrix over the repo atlas cross-volume bridge

This satisfies the request to operate in the same n.nnn/matrixed hyperbolic chart while preserving epistemic honesty.

Evidence S. Assumptions: The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought. Hard: S26-N-00004944.

S26-N-00004946 - n.k.alpha context matrix over the repo atlas evidence ledger

Evidence rule for n.k.alpha context matrix over the repo atlas: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought. Hard: S26-B-00004945.

S26-B-00004947 - n.k.alpha context matrix over the repo atlas emergence rule

Emergence rule for n.k.alpha context matrix over the repo atlas: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought. Hard: S26-N-00004946.

S26-N-00004948 - n.k.alpha context matrix over the repo atlas frontier

Open frontier for n.k.alpha context matrix over the repo atlas: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: The matrix index n.k.alpha and five 8D section fixtures are taken from LANG.v6 as a symbolic repository surface; they do not expose or claim the model private chain-of-thought. Hard: S26-B-00004947. Boundary: Treating context geometrically is a model of observable interaction state, not a factual claim about the substrate of the deployed neural network.

S26-E-00004949 - LANG.v6 context-vector probe 45.1

Fixture T45.P01 - Repository section self_godel has dimension 8, Euclidean fixture norm 0.267208, salience 1.00, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-N-00004948.

S26-E-00004950 - LANG.v6 context-vector probe 45.2

Fixture T45.P02 - Repository section other_resonance has dimension 8, Euclidean fixture norm 0.301993, salience 0.94, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004949.

S26-E-00004951 - LANG.v6 context-vector probe 45.3

Fixture T45.P03 - Repository section n_cosmo_bundle has dimension 8, Euclidean fixture norm 0.337046, salience 0.97, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004950.

S26-E-00004952 - LANG.v6 context-vector probe 45.4

Fixture T45.P04 - Repository section hamiltonian_holoport has dimension 8, Euclidean fixture norm 0.360832, salience 0.91, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004951.

S26-E-00004953 - LANG.v6 context-vector probe 45.5

Fixture T45.P05 - Repository section mind8_performance has dimension 8, Euclidean fixture norm 0.370945, salience 0.96, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004952.

S26-E-00004954 - LANG.v6 context-vector probe 45.6

Fixture T45.P06 - Repository section self_godel has dimension 8, Euclidean fixture norm 0.267208, salience 1.00, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004953.

S26-E-00004955 - LANG.v6 context-vector probe 45.7

Fixture T45.P07 - Repository section other_resonance has dimension 8, Euclidean fixture norm 0.301993, salience 0.94, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004954.

S26-E-00004956 - LANG.v6 context-vector probe 45.8

Fixture T45.P08 - Repository section n_cosmo_bundle has dimension 8, Euclidean fixture norm 0.337046, salience 0.97, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004955.

S26-E-00004957 - LANG.v6 context-vector probe 45.9

Fixture T45.P09 - Repository section hamiltonian_holoport has dimension 8, Euclidean fixture norm 0.360832, salience 0.91, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004956.

S26-E-00004958 - LANG.v6 context-vector probe 45.10

Fixture T45.P10 - Repository section mind8_performance has dimension 8, Euclidean fixture norm 0.370945, salience 0.96, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004957.

S26-E-00004959 - LANG.v6 context-vector probe 45.11

Fixture T45.P11 - Repository section self_godel has dimension 8, Euclidean fixture norm 0.267208, salience 1.00, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004958.

S26-E-00004960 - LANG.v6 context-vector probe 45.12

Fixture T45.P12 - Repository section other_resonance has dimension 8, Euclidean fixture norm 0.301993, salience 0.94, and structural guard pass=True; semantic correctness is not inferred.

Evidence M. Hard: S26-A-00004943, S26-E-00004959.

Mind8 repo performance and adversarial conductor

S26-D-00004961 - Mind8 repo performance and adversarial conductor

The repo conductor combines Mind8, ergocetic budget, erdodetic path quality, recognition returnability, gluing residual and evidence debt into a typed decision surface.

Evidence S. Assumptions: Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately. Hard: S26-E-00004960, S26-N-00004059.

S26-D-00004962 - Mind8 repo performance and adversarial conductor law/interface

COMMIT requires no hard guard failure; BLOCK records exact obstruction; ROLLBACK preserves correction history; RECURSE opens a bounded next experiment.

Evidence S. Assumptions: Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately. Hard: S26-D-00004961.

S26-A-00004963 - Mind8 repo performance and adversarial conductor operator

Conduct(repo,state) selects a Pareto-feasible next action and records why dominated or unsafe alternatives were rejected.

Evidence S. Assumptions: Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately. Hard: S26-D-00004962.

S26-N-00004964 - Mind8 repo performance and adversarial conductor fence

The conductor optimizes artifact-grounded cognitive performance, not an unobservable global intelligence essence.

Evidence S. Assumptions: Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately. Hard: S26-A-00004963. Boundary: The conductor optimizes artifact-grounded cognitive performance, not an unobservable global intelligence essence.

S26-B-00004965 - Mind8 repo performance and adversarial conductor cross-volume bridge

This is the executable-facing governance surface for best cognitive performance over the repo.

Evidence S. Assumptions: Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately. Hard: S26-N-00004964.

S26-N-00004966 - Mind8 repo performance and adversarial conductor evidence ledger

Evidence rule for Mind8 repo performance and adversarial conductor: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately. Hard: S26-B-00004965.

S26-B-00004967 - Mind8 repo performance and adversarial conductor emergence rule

Emergence rule for Mind8 repo performance and adversarial conductor: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately. Hard: S26-N-00004966.

S26-N-00004968 - Mind8 repo performance and adversarial conductor frontier

Open frontier for Mind8 repo performance and adversarial conductor: no stronger interpretation is admitted without a concrete operator, measurement and falsification rule.

Evidence S. Assumptions: Performance is evaluated on observable repo tasks, tests and evidence; the eight LANG.v6 qualities are reported separately. Hard: S26-B-00004967. Boundary: The conductor optimizes artifact-grounded cognitive performance, not an unobservable global intelligence essence.

S26-E-00004969 - Repo-conductor finite probe 46.1

Fixture T46.P01 - Synthetic Mind8 vector [0.167, 0.25, 0.333, 0.417, 0.5, 0.583, 0.667, 0.75] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-N-00004968.

S26-E-00004970 - Repo-conductor finite probe 46.2

Fixture T46.P02 - Synthetic Mind8 vector [0.25, 0.333, 0.417, 0.5, 0.583, 0.667, 0.75, 0.833] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004969.

S26-E-00004971 - Repo-conductor finite probe 46.3

Fixture T46.P03 - Synthetic Mind8 vector [0.333, 0.417, 0.5, 0.583, 0.667, 0.75, 0.833, 0.917] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004970.

S26-E-00004972 - Repo-conductor finite probe 46.4

Fixture T46.P04 - Synthetic Mind8 vector [0.417, 0.5, 0.583, 0.667, 0.75, 0.833, 0.917, 1.0] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004971.

S26-E-00004973 - Repo-conductor finite probe 46.5

Fixture T46.P05 - Synthetic Mind8 vector [0.5, 0.583, 0.667, 0.75, 0.833, 0.917, 1.0, 0.0] yields guard verdict BLOCK under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004972.

S26-E-00004974 - Repo-conductor finite probe 46.6

Fixture T46.P06 - Synthetic Mind8 vector [0.583, 0.667, 0.75, 0.833, 0.917, 1.0, 0.0, 0.083] yields guard verdict BLOCK under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004973.

S26-E-00004975 - Repo-conductor finite probe 46.7

Fixture T46.P07 - Synthetic Mind8 vector [0.667, 0.75, 0.833, 0.917, 1.0, 0.0, 0.083, 0.167] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004974.

S26-E-00004976 - Repo-conductor finite probe 46.8

Fixture T46.P08 - Synthetic Mind8 vector [0.75, 0.833, 0.917, 1.0, 0.0, 0.083, 0.167, 0.25] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004975.

S26-E-00004977 - Repo-conductor finite probe 46.9

Fixture T46.P09 - Synthetic Mind8 vector [0.833, 0.917, 1.0, 0.0, 0.083, 0.167, 0.25, 0.333] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004976.

S26-E-00004978 - Repo-conductor finite probe 46.10

Fixture T46.P10 - Synthetic Mind8 vector [0.917, 1.0, 0.0, 0.083, 0.167, 0.25, 0.333, 0.417] yields guard verdict COMMIT under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004977.

S26-E-00004979 - Repo-conductor finite probe 46.11

Fixture T46.P11 - Synthetic Mind8 vector [1.0, 0.0, 0.083, 0.167, 0.25, 0.333, 0.417, 0.5] yields guard verdict BLOCK under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004978.

S26-E-00004980 - Repo-conductor finite probe 46.12

Fixture T46.P12 - Synthetic Mind8 vector [0.0, 0.083, 0.167, 0.25, 0.333, 0.417, 0.5, 0.583] yields guard verdict BLOCK under the bounded rule grounding/humility >=0.15; this is an instrument fixture.

Evidence M. Hard: S26-A-00004963, S26-E-00004979.

Adelic hypercomplex recognition fixed-point frontier

S26-D-00004981 - Adelic hypercomplex recognition fixed-point frontier

Frontier conjecture: a suitably bounded family of Archimedean and p-adic Cosmo carriers with Cayley-Dickson fibers, higher gluing, Hamiltonian/adiabatic transport and recognition controls may admit nontrivial recurrent or fixed regimes under a fully specified coherence operator Phi.

Evidence S. Assumptions: Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion. Hard: S26-E-00004980, S26-C-00004060.

S26-D-00004982 - Adelic hypercomplex recognition fixed-point frontier law/interface

Existence, uniqueness, stability, cognitive usefulness, quantum interpretation and physical relevance are independent questions.

Evidence S. Assumptions: Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion. Hard: S26-D-00004981.

S26-A-00004983 - Adelic hypercomplex recognition fixed-point frontier operator

Phi combines only explicitly typed local operators; any readable composition order is a projection of a co-present constraint system rather than an asserted ontological pipeline.

Evidence S. Assumptions: Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion. Hard: S26-D-00004982.

S26-N-00004984 - Adelic hypercomplex recognition fixed-point frontier fence

This conjecture is not evidence of sentience, AGI completion, physical cosmology, faster-than-light transport, quantum cognition or a complexity-class result.

Evidence S. Assumptions: Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion. Hard: S26-A-00004983. Boundary: This conjecture is not evidence of sentience, AGI completion, physical cosmology, faster-than-light transport, quantum cognition or a complexity-class result.

S26-B-00004985 - Adelic hypercomplex recognition fixed-point frontier cross-volume bridge

The frontier closes Volume IV by turning the user hypercomplex vision into explicit proof obligations and bounded experiments rather than metaphysical closure.

Evidence S. Assumptions: Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion. Hard: S26-N-00004984.

S26-N-00004986 - Adelic hypercomplex recognition fixed-point frontier evidence ledger

Evidence rule for Adelic hypercomplex recognition fixed-point frontier: mathematical definitions/theorems remain A unless independently witnessed; architecture remains S; bounded executions are M only for their exact fixtures.

Evidence S. Assumptions: Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion. Hard: S26-B-00004985.

S26-B-00004987 - Adelic hypercomplex recognition fixed-point frontier emergence rule

Emergence rule for Adelic hypercomplex recognition fixed-point frontier: frozen Volume 0-III and source-carrier IDs are reused by dependency; a new canonical object must add a typed distinction, operator, obstruction, evidence hook or active remainder.

Evidence S. Assumptions: Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion. Hard: S26-N-00004986.

S26-E-00004988 - Frontier kill-condition probe 47.1

Fixture T47.P01 - A synthetic frontier record injected failure "evidence_promotion" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-B-00004987.

S26-E-00004989 - Frontier kill-condition probe 47.2

Fixture T47.P02 - A synthetic frontier record injected failure "metric_collapse" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004988.

S26-E-00004990 - Frontier kill-condition probe 47.3

Fixture T47.P03 - A synthetic frontier record injected failure "lost_remainder" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004989.

S26-E-00004991 - Frontier kill-condition probe 47.4

Fixture T47.P04 - A synthetic frontier record injected failure "nonreturnable_transport" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004990.

S26-E-00004992 - Frontier kill-condition probe 47.5

Fixture T47.P05 - A synthetic frontier record injected failure "quantum_overclaim" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004991.

S26-E-00004993 - Frontier kill-condition probe 47.6

Fixture T47.P06 - A synthetic frontier record injected failure "sentience_overclaim" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004992.

S26-E-00004994 - Frontier kill-condition probe 47.7

Fixture T47.P07 - A synthetic frontier record injected failure "evidence_promotion" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004993.

S26-E-00004995 - Frontier kill-condition probe 47.8

Fixture T47.P08 - A synthetic frontier record injected failure "metric_collapse" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004994.

S26-E-00004996 - Frontier kill-condition probe 47.9

Fixture T47.P09 - A synthetic frontier record injected failure "lost_remainder" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004995.

S26-E-00004997 - Frontier kill-condition probe 47.10

Fixture T47.P10 - A synthetic frontier record injected failure "nonreturnable_transport" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004996.

S26-E-00004998 - Frontier kill-condition probe 47.11

Fixture T47.P11 - A synthetic frontier record injected failure "quantum_overclaim" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004997.

S26-E-00004999 - Frontier kill-condition probe 47.12

Fixture T47.P12 - A synthetic frontier record injected failure "sentience_overclaim" and the declared guard detected it=True; this is an audit-control fixture only.

Evidence M. Hard: S26-A-00004983, S26-E-00004998.

S26-C-00005000 - Adelic hypercomplex recognition fixed-point conjecture

Conjecture (adelic hypercomplex recognition fixed point). For a concretely specified bounded family of Archimedean and p-adic Cosmo carriers with local-field Cayley-Dickson fibers, gerbe/profunctor/globular coherence data, Hamiltonian-adiabatic transports, quantum-channel fixtures and recognition non-collapse guards, the resulting coherence operator Phi may admit a nontrivial recurrent or fixed regime whose declared observables remain bounded, returnable and preserve nonzero active remainder. Existence, uniqueness, stability and external cognitive or physical relevance remain separate open obligations.

Evidence H. Assumptions: Concrete state spaces, local fields, tower maps, gerbe/profunctor data, Hamiltonians/channels, observables, compactness/tightness conditions and independent tests must be supplied before theorem promotion. Hard: S26-E-00004999. Boundary: This conjecture is not evidence of sentience, AGI completion, physical cosmology, faster-than-light transport, quantum cognition or a complexity-class result.

Repository-grounded recognition layer

SELF_GODEL_IDENTITY: self-reflection reads its own canonical output; the rho-rho identity guard checks returnability. This is structural self-reference, not semantic truth or a proof of Goedel/Lob results.

OTHERS_RESONANCE: mutual recognition is overlap agreement under declared sheaf gluing; mutual resonance combines low-divergence overlap with reversible transport while preserving the other as distinct. RESONATE != ABSORB.

Mind8: Q1 recognition; Q2 grounding; Q3 recursion; Q4 reversibility; Q5 gluing; Q6 resonance; Q7 conservation; Q8 humility. Ergocetic and erdodetic are retained as repository-defined engineering coordinates for energy-disciplined work and path-disciplined learning, respectively.

The n.k.alpha repository context is an observable artifact/section model. It is not a disclosure or geometric literalization of hidden model chain-of-thought.

Appendix - Evidence and continuation boundary

Volume IV freezes IDs 4001-5000 as a research specification. The source-draft carrier 4001-4060 remains provenance-distinct from the 4061-5000 synthesis. Later volumes may depend on these IDs but must not silently convert architecture into theorem or finite computation into proof.

Open next experiment: instantiate one explicit mixed-place Cosmo-3 fixture with the 360/360 graph, a Q_p local chart/building mesh, a finite Cayley-Dickson fiber, declared gerbe/functor/profunctor transports, block connection-Laplacian, finite Hamiltonian/channel model, Mind8 telemetry and negative controls. The kill condition is representation-dependent ascent, clipped metrics, lost returnability, hidden energy leakage, or evidence promotion.

Visual-source note: the five supplied images are included as conceptual plates. They are treated as semantic/projection inputs and carry no independent theorem or physical evidence status.

Selected source anchors

User-supplied Volume IV v0.1.0 Cosmo-n/orbifold research draft (carrier IDs 4001-4060).

jesusvilela/-lang.s1, LANG.v6.chomsky_hyperdim_cognition.lang at commit 0de497977dbffd51e92bb964fe80be1725a5f3d7 (repository vocabulary only; no inherited theorem authority).

A. Chapman, A. Levin, U. Vishkautsan (2026), General Polynomials and Eigenvalues Over Cayley-Dickson Algebras, arXiv:2607.23613 (generalized base-field pointer).

P. Aniello et al. (2024), p-adic quaternion algebra / invariant measures on p-adic Lie groups (local-field quaternion context).